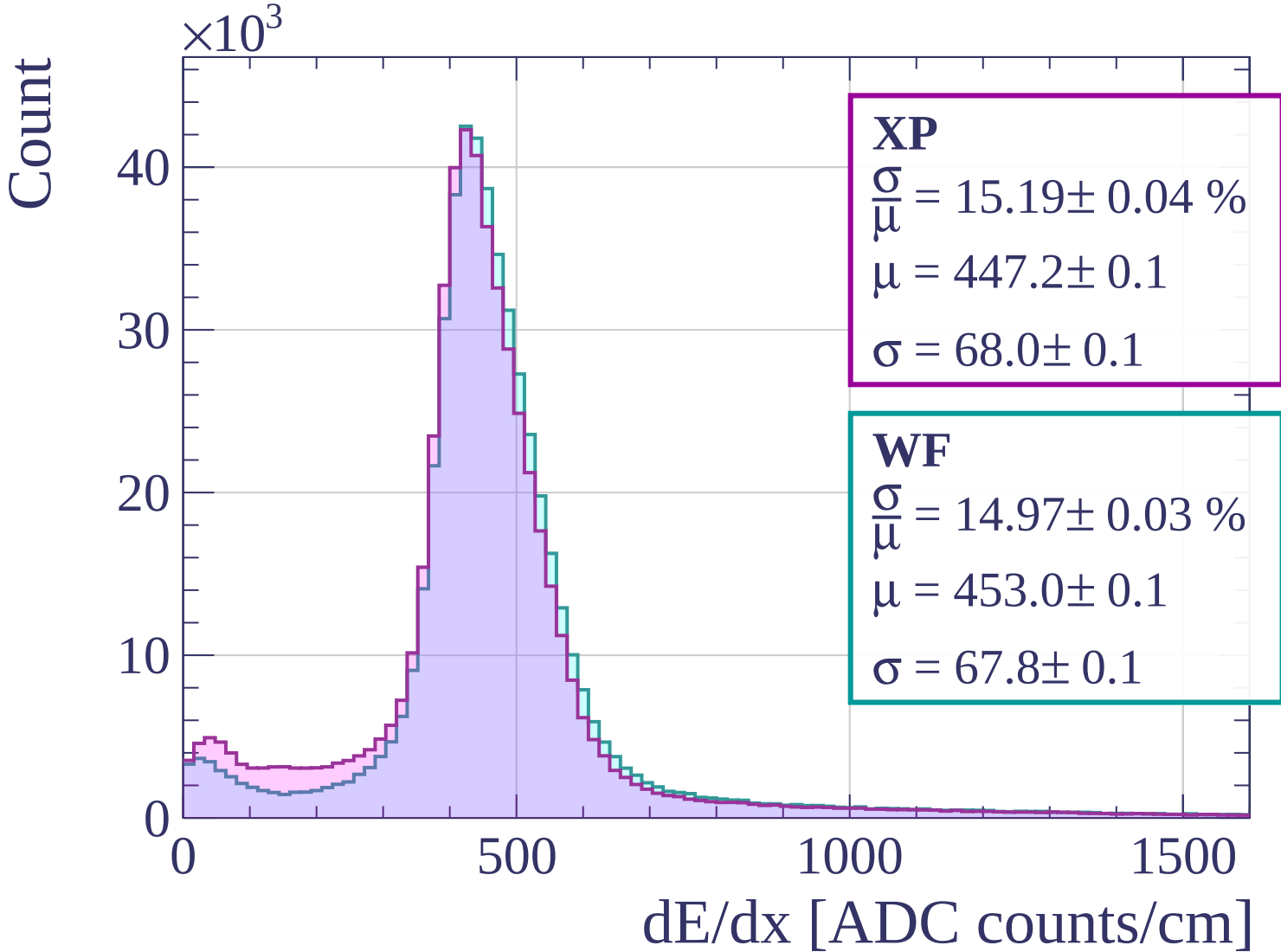
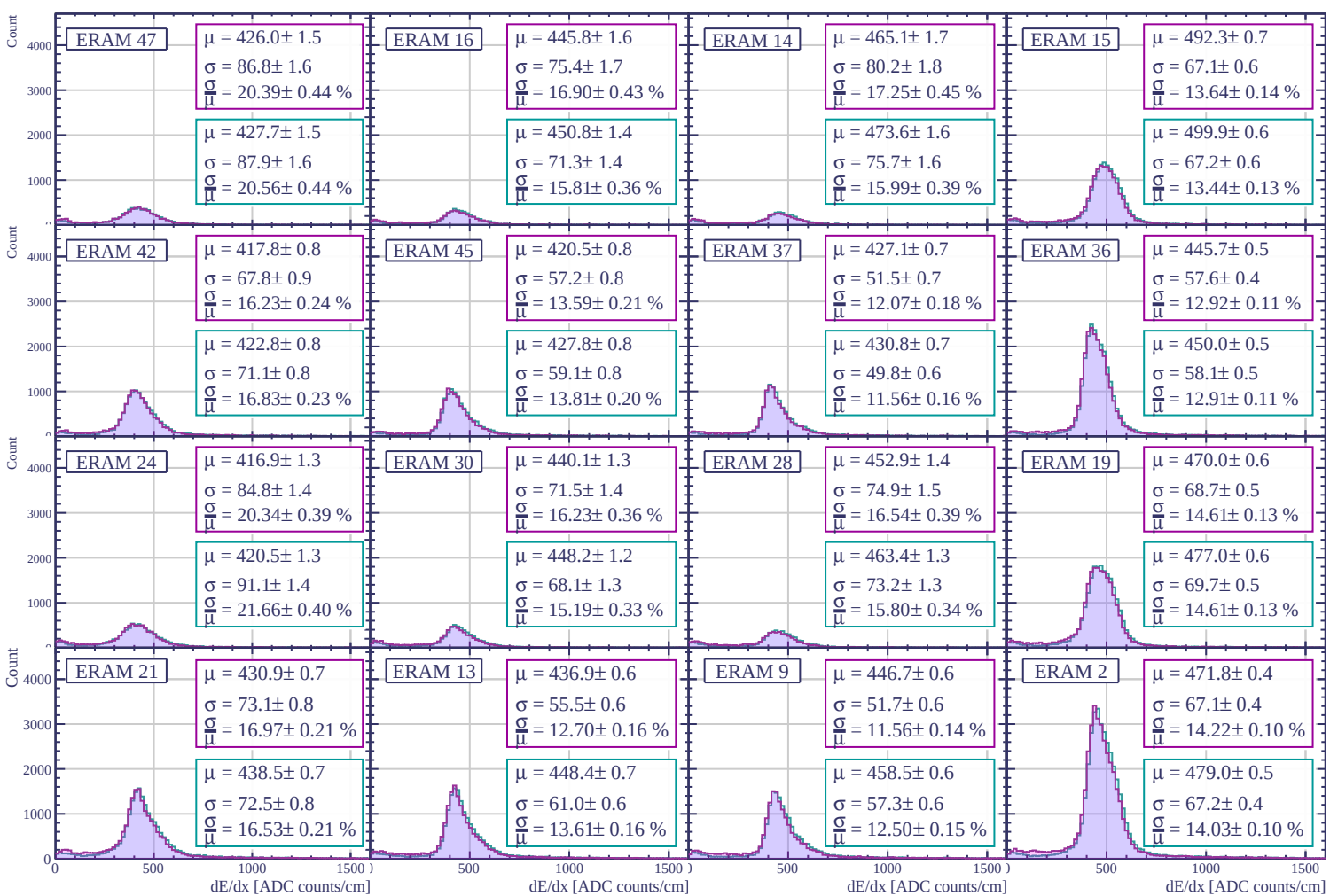
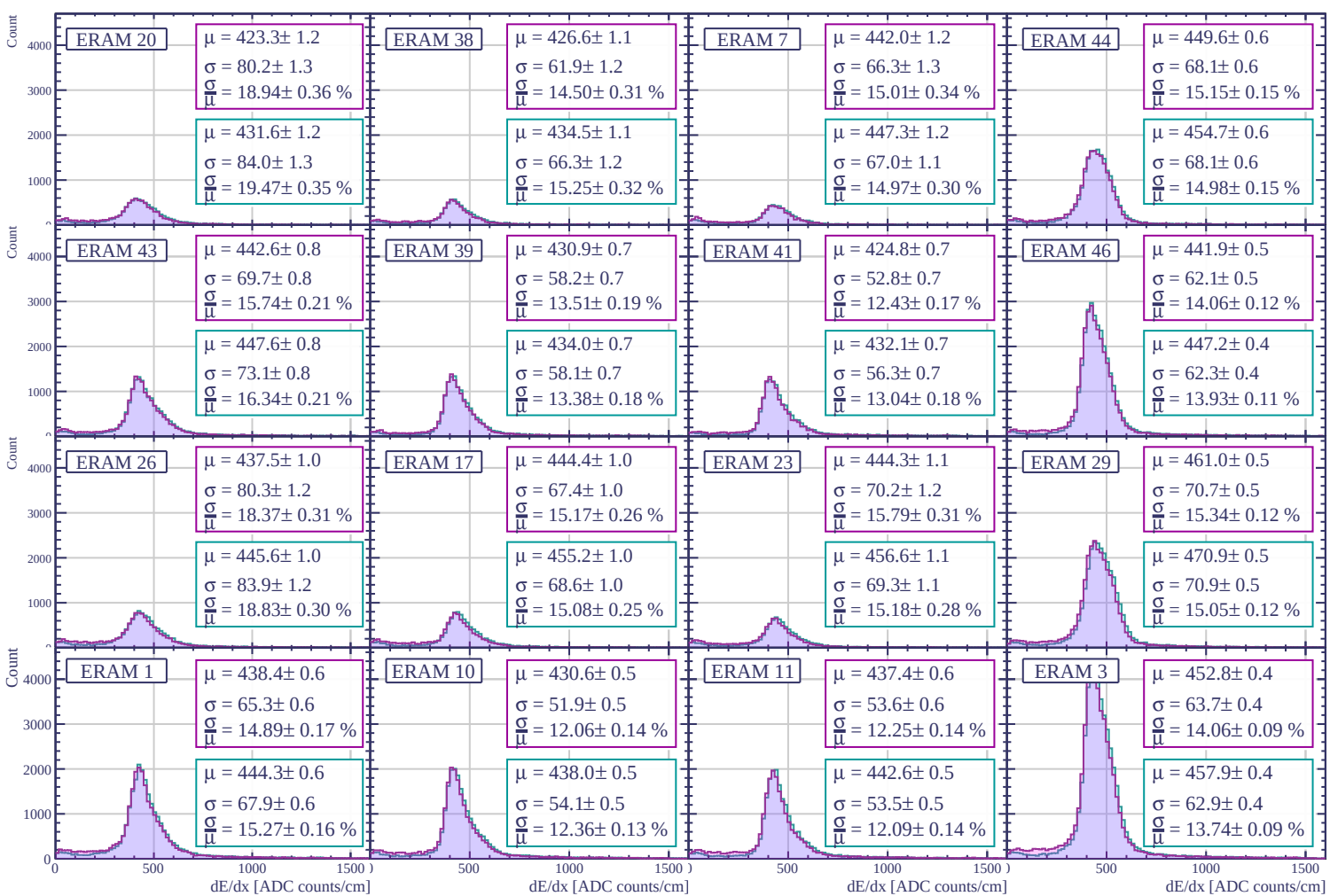
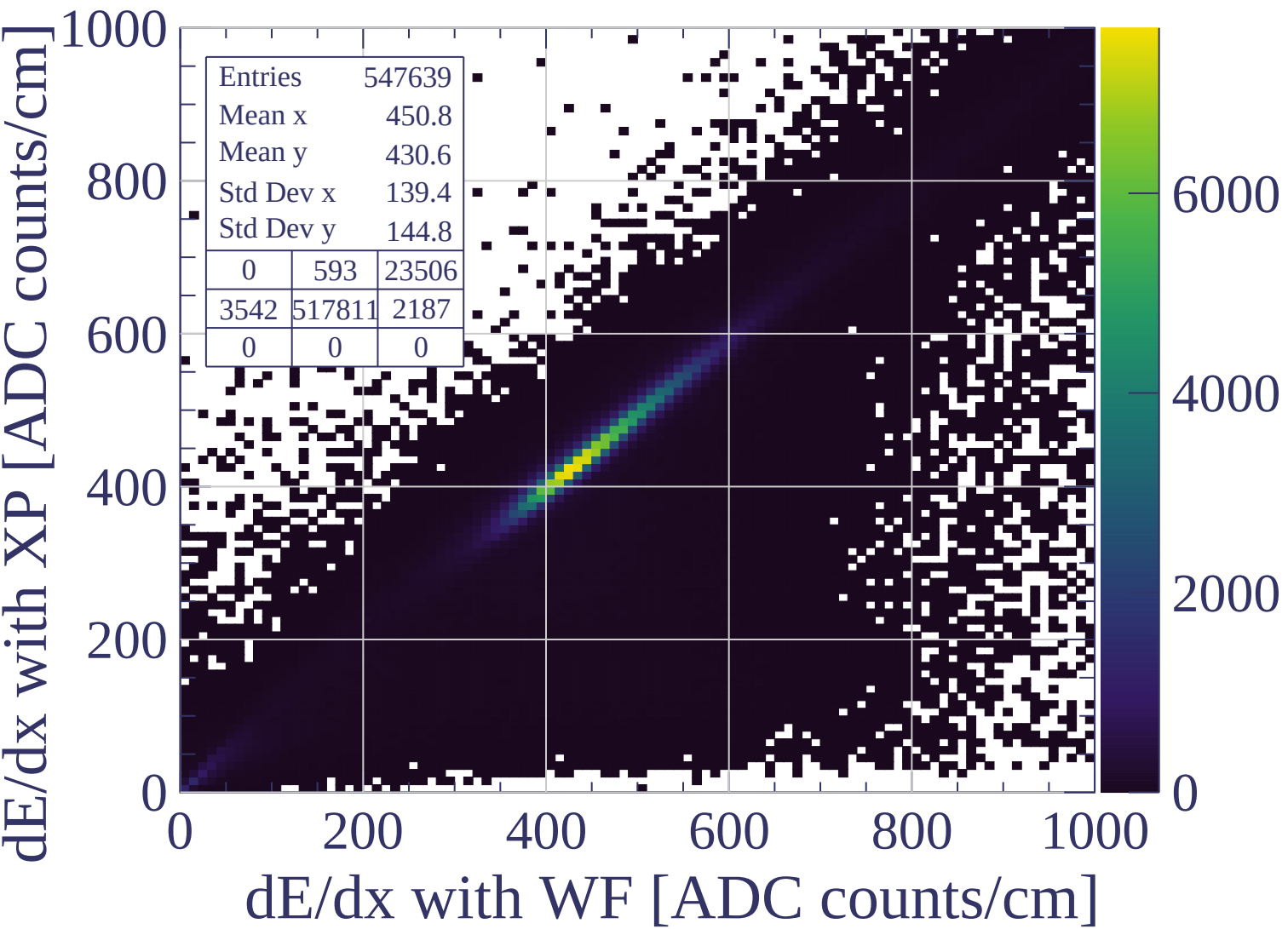
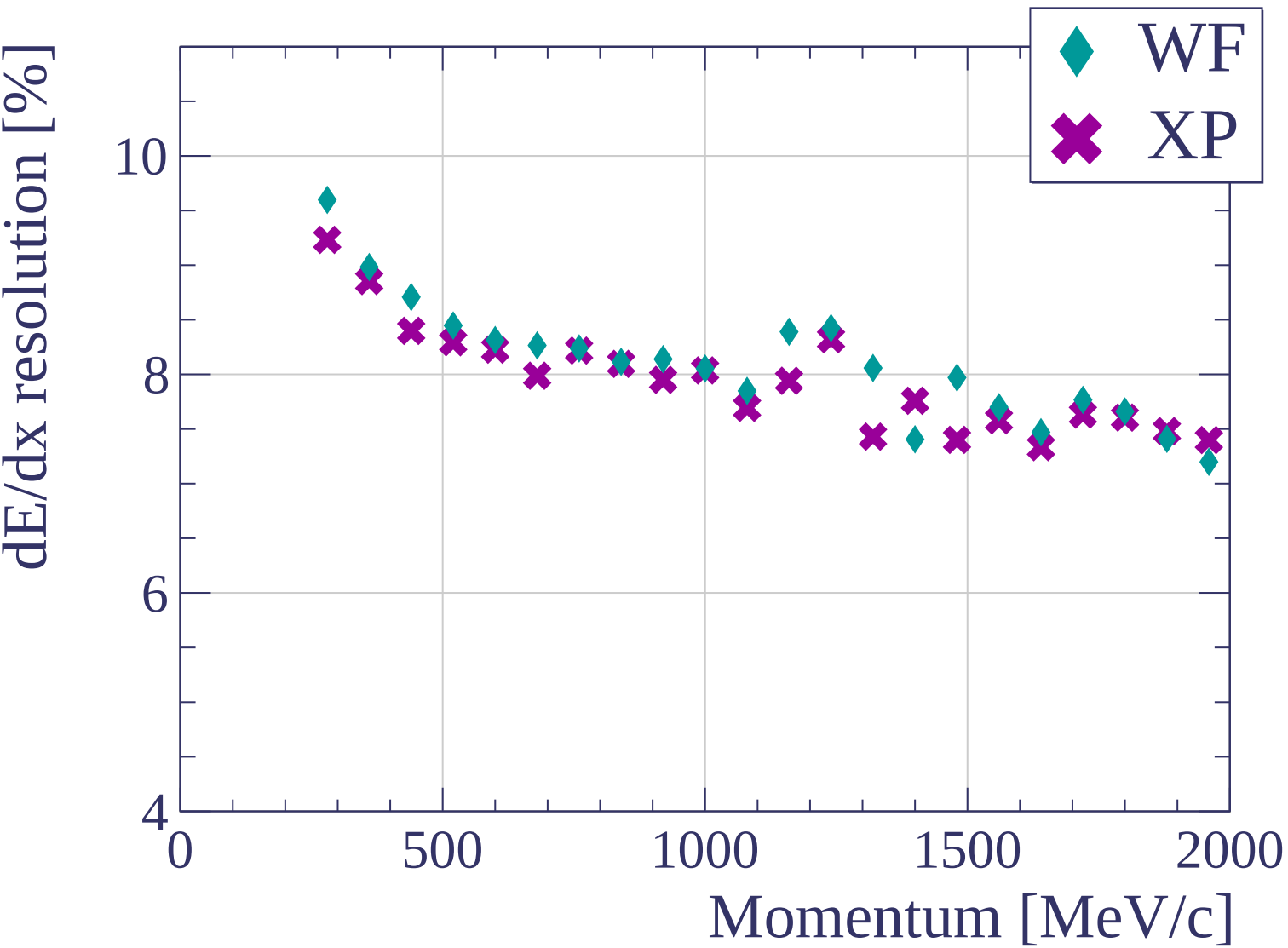


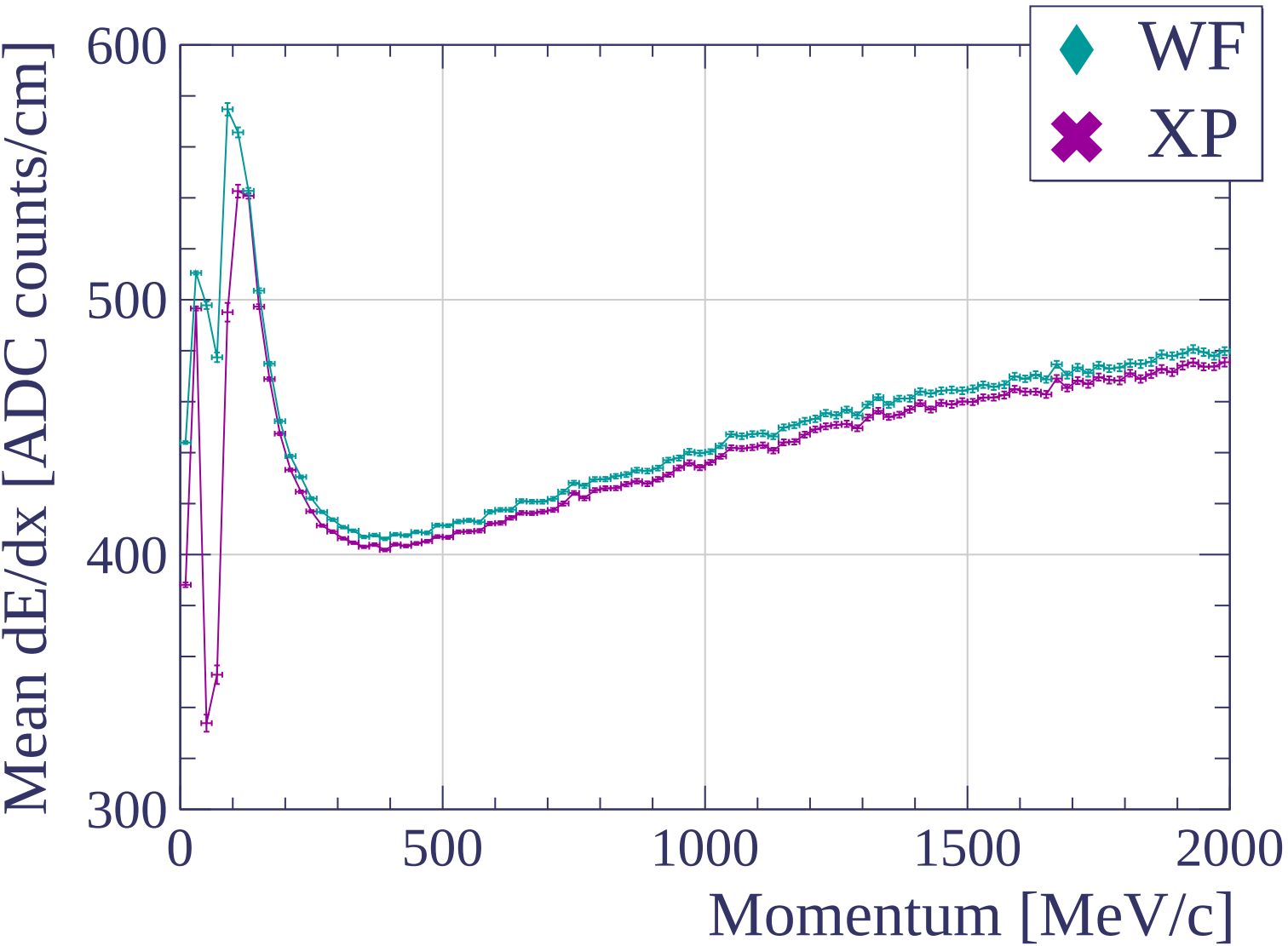


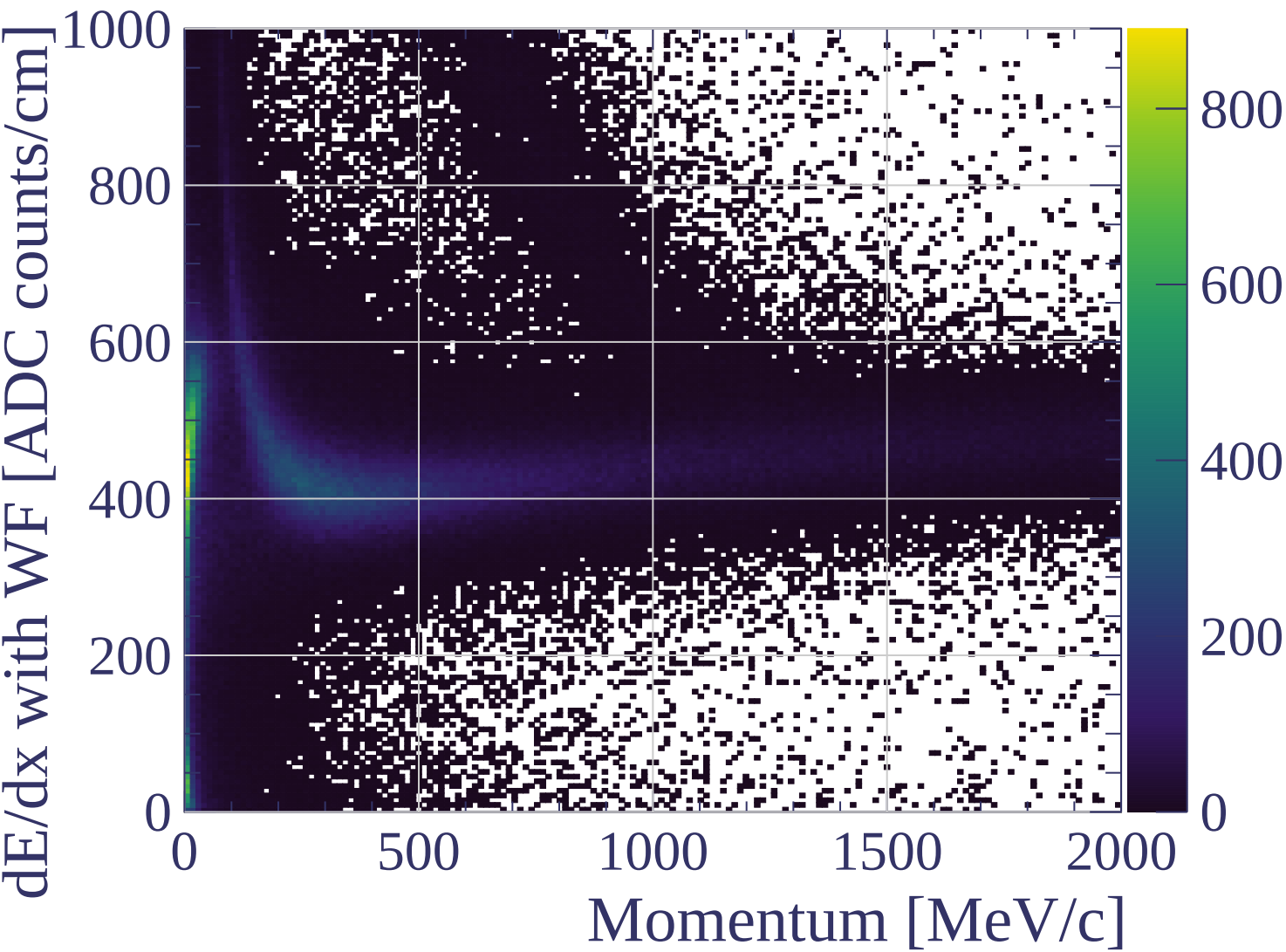


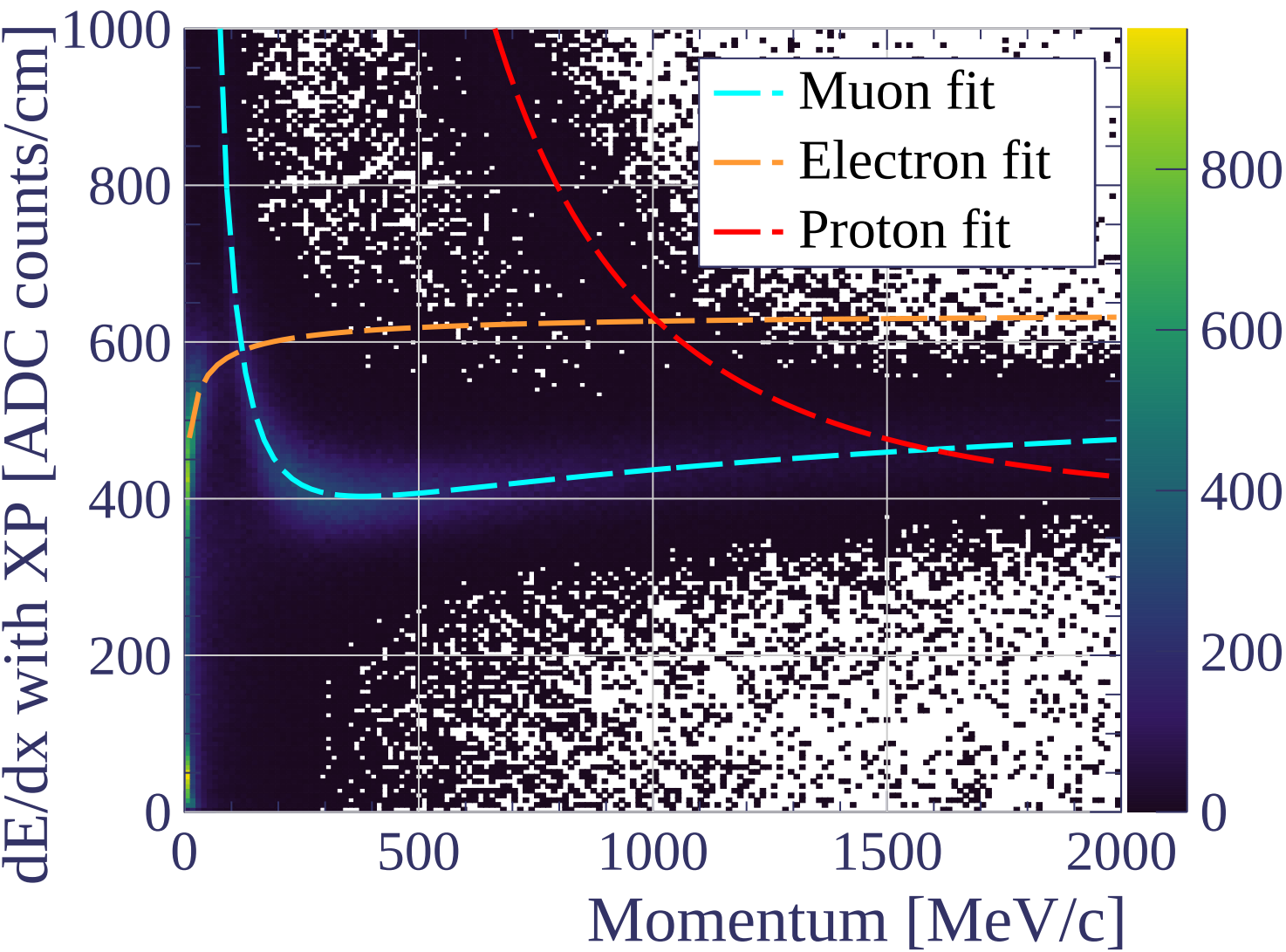




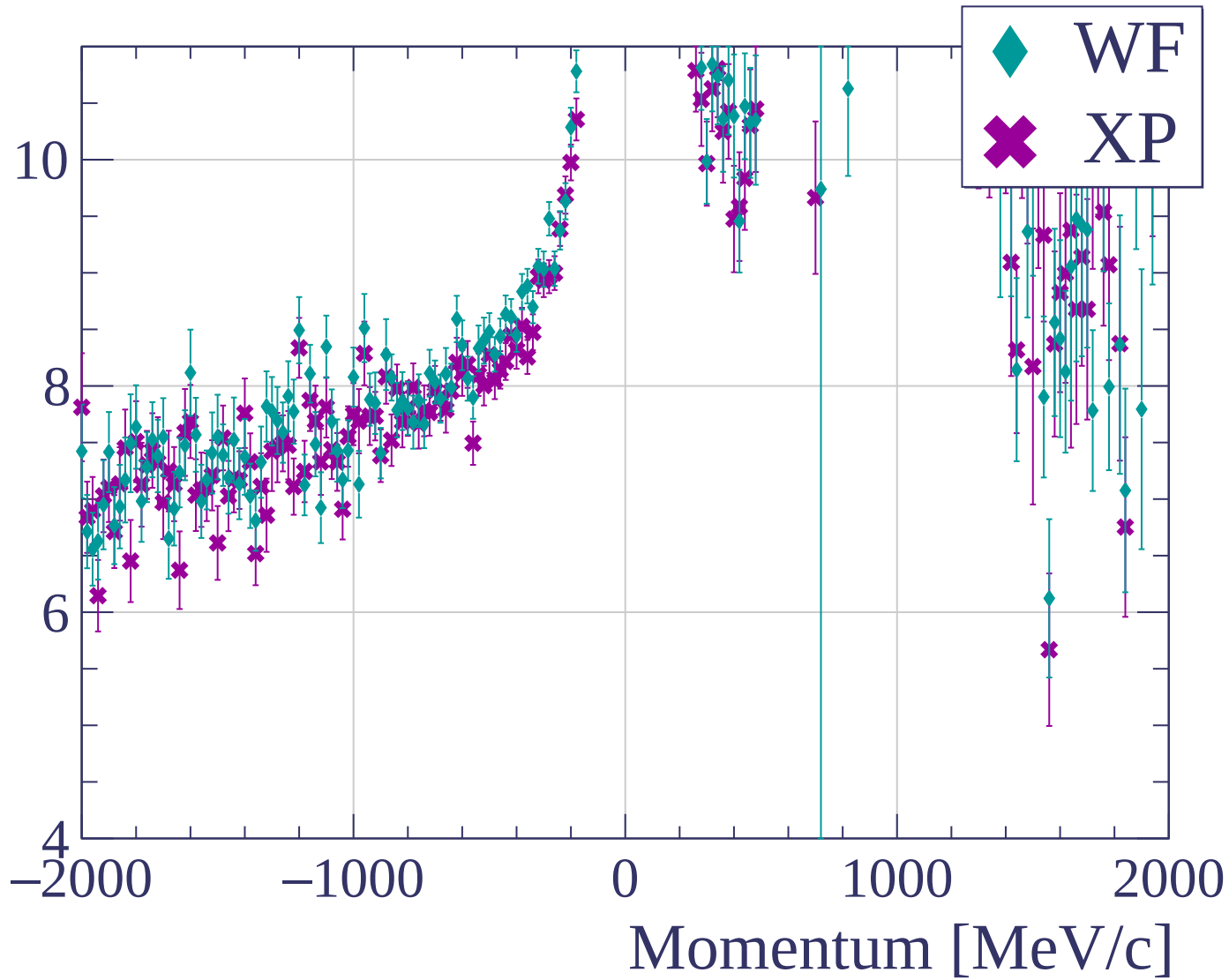


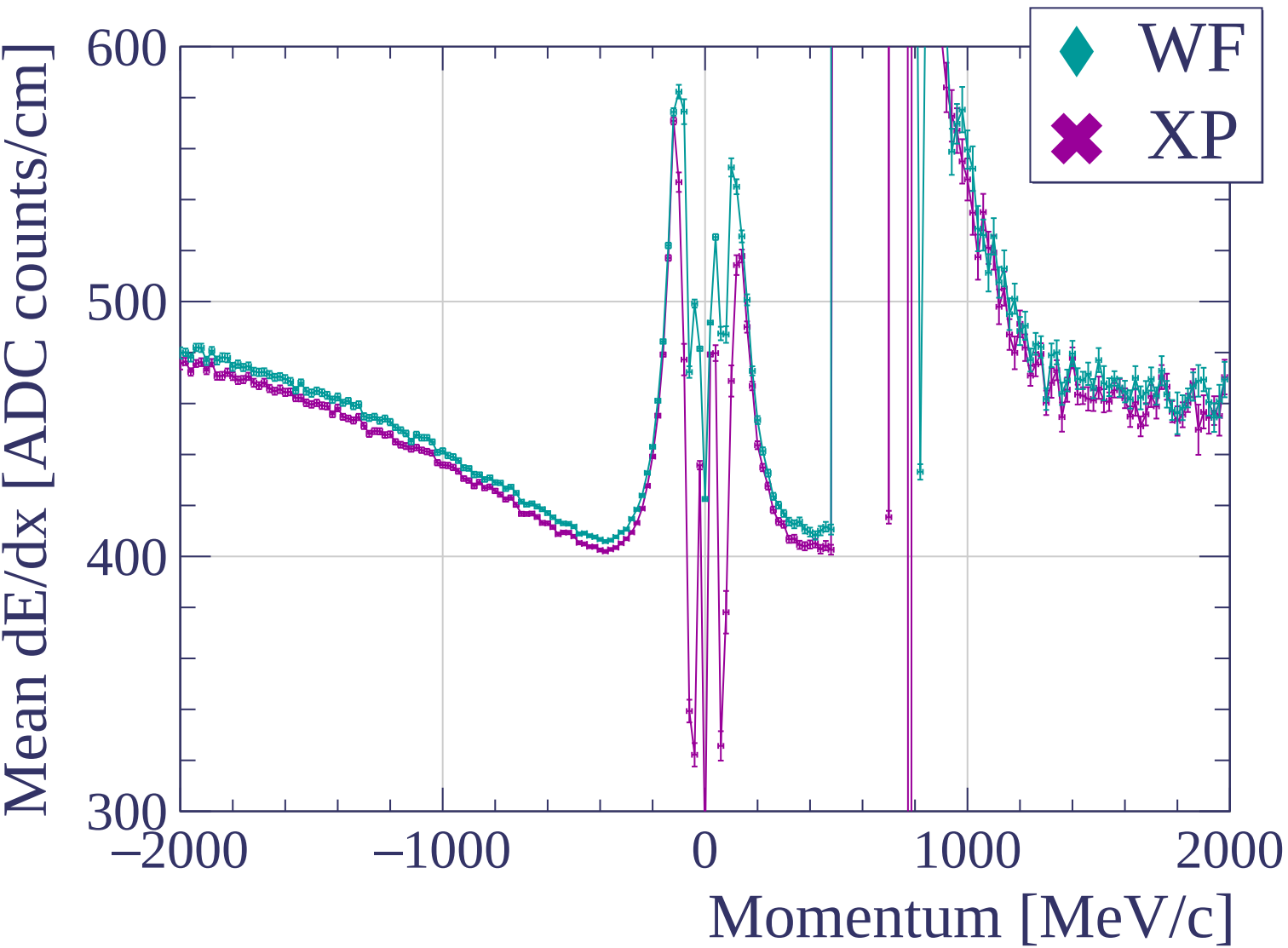


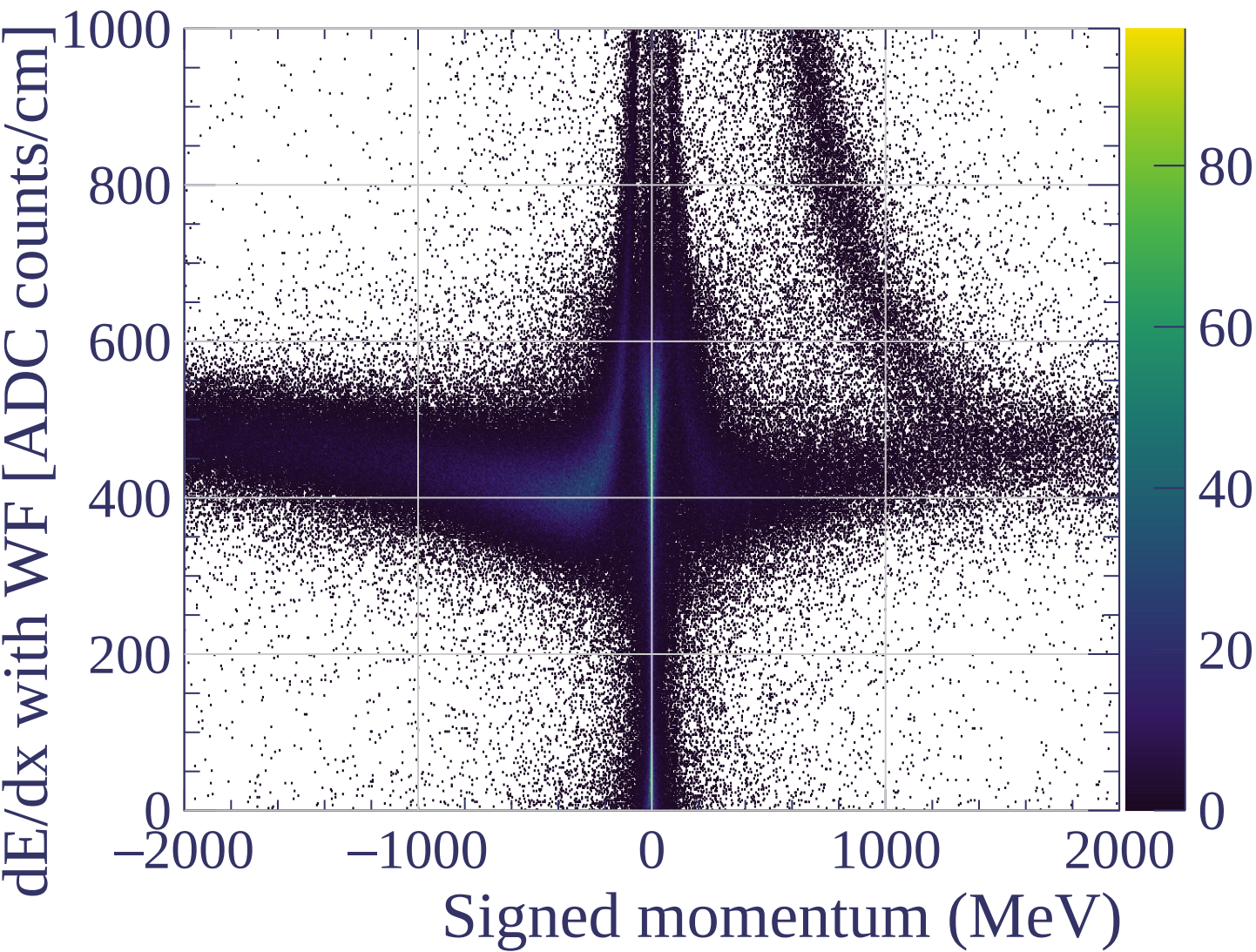


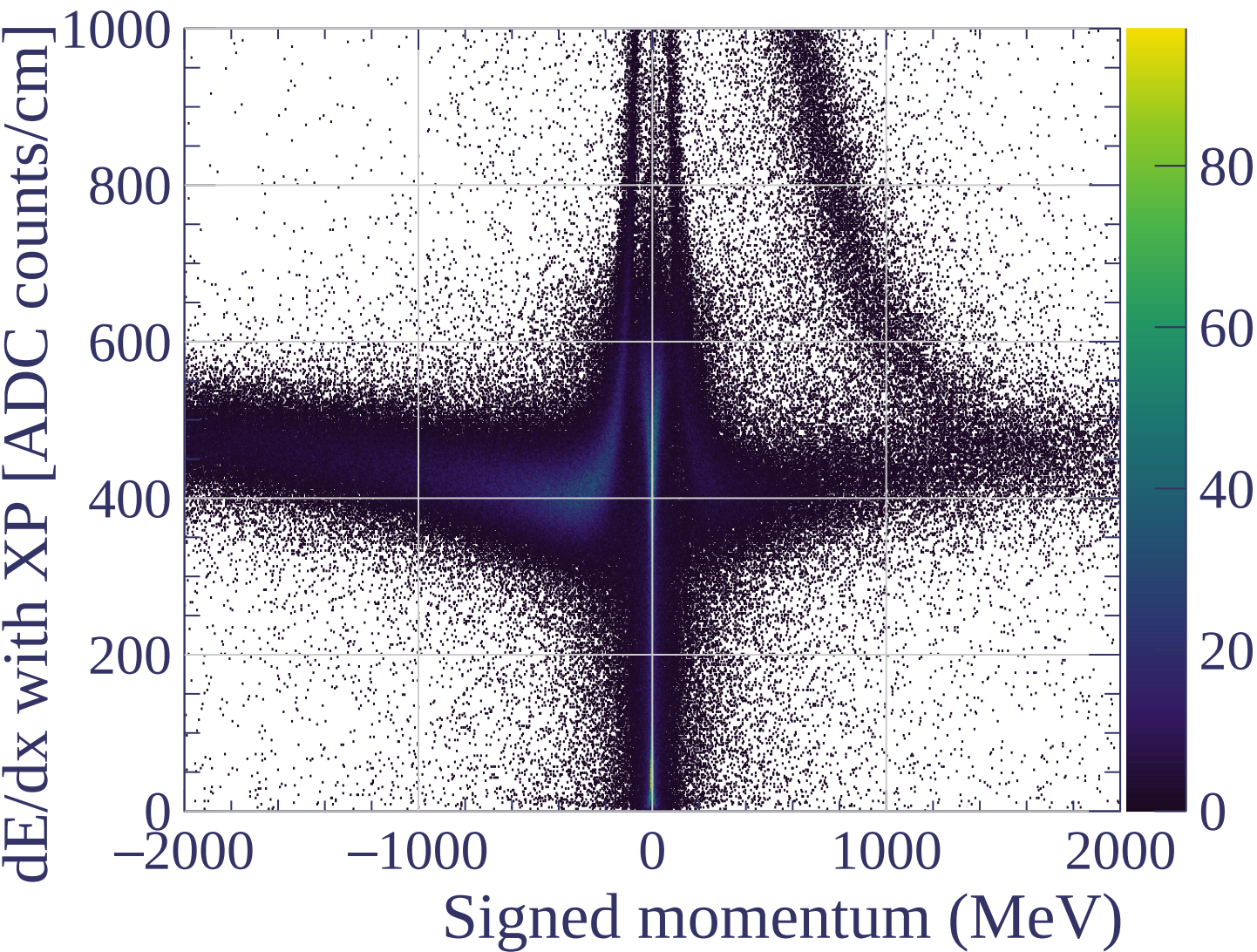


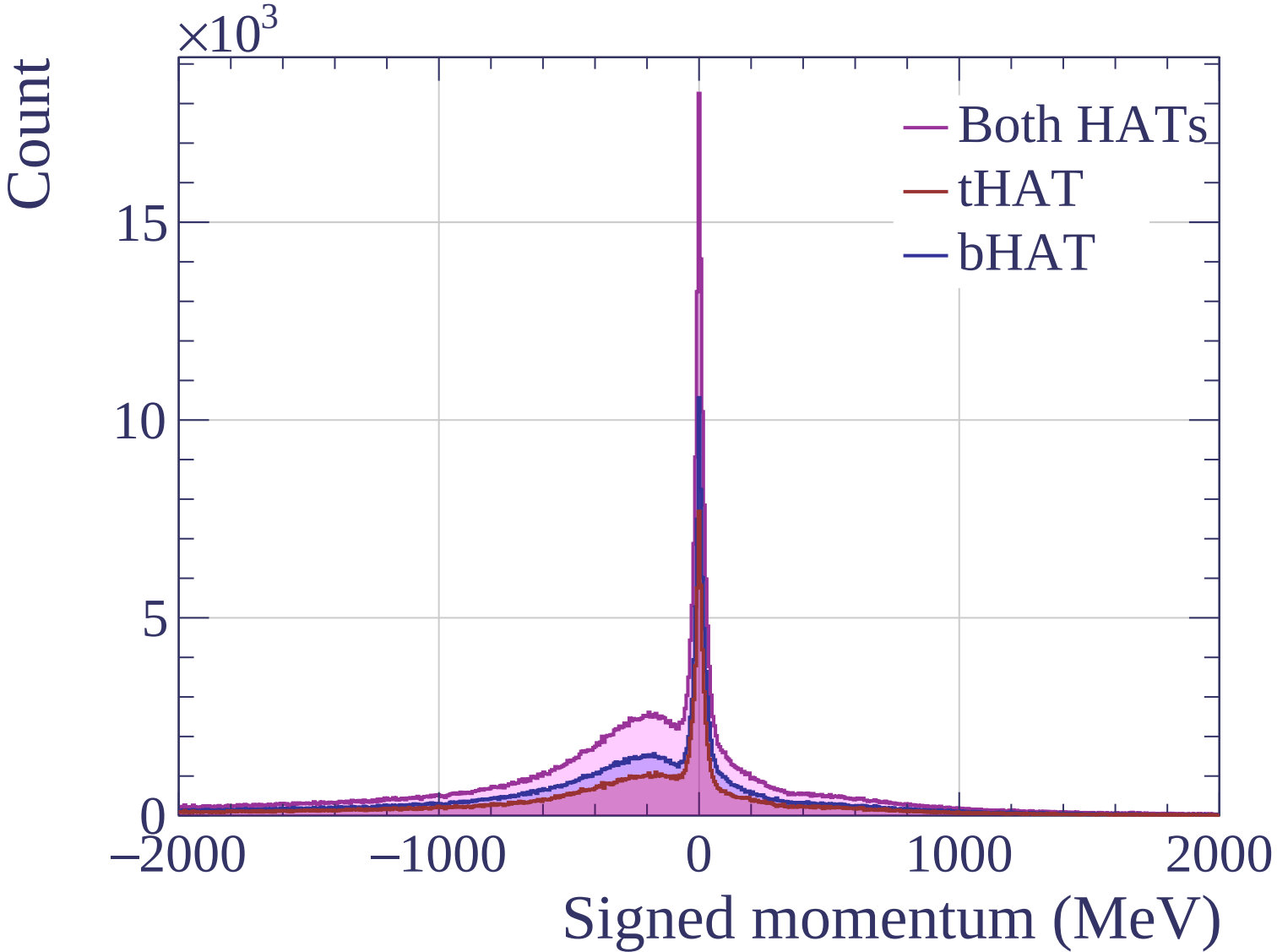
dE/dx resolution [%]

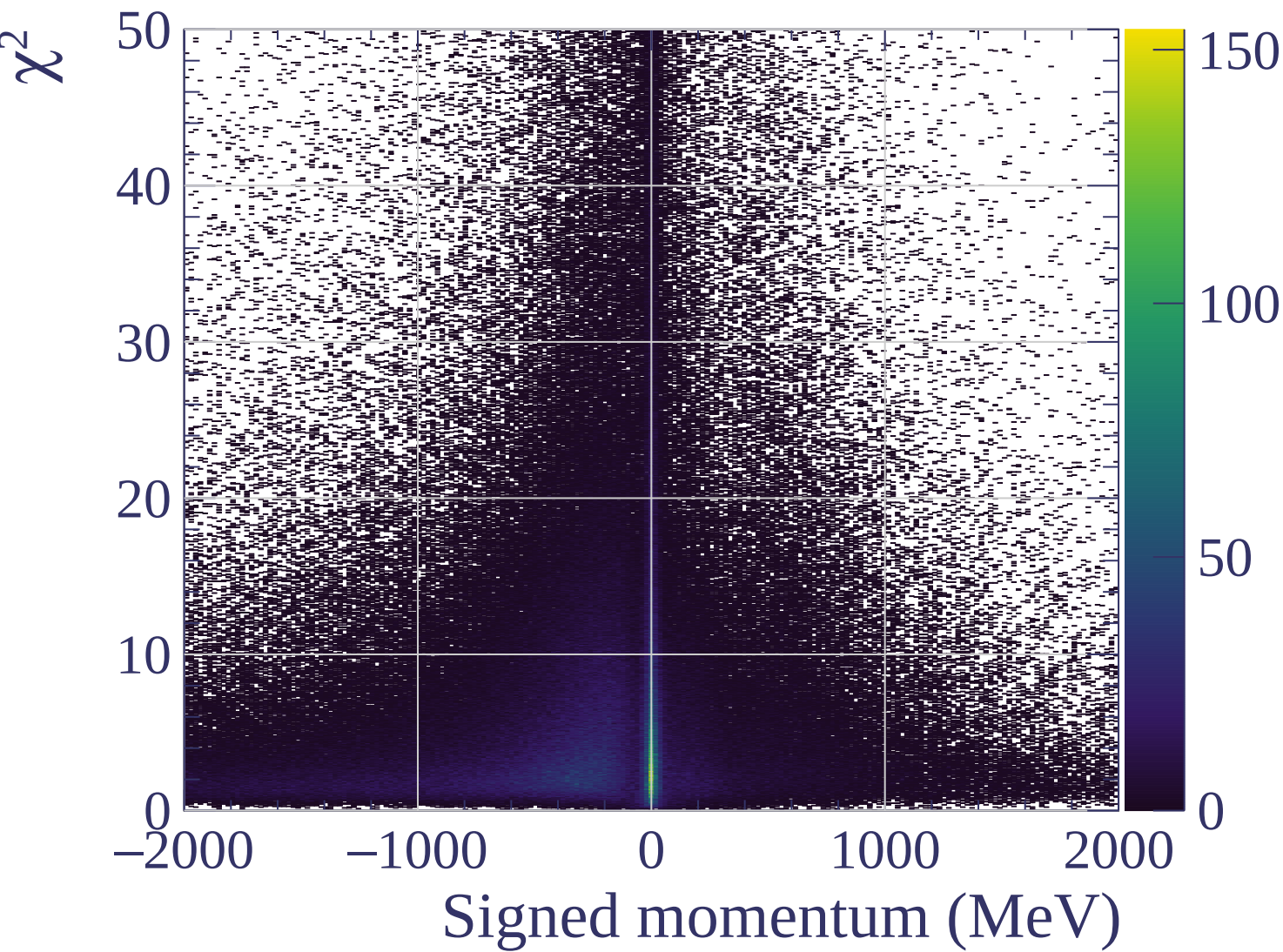


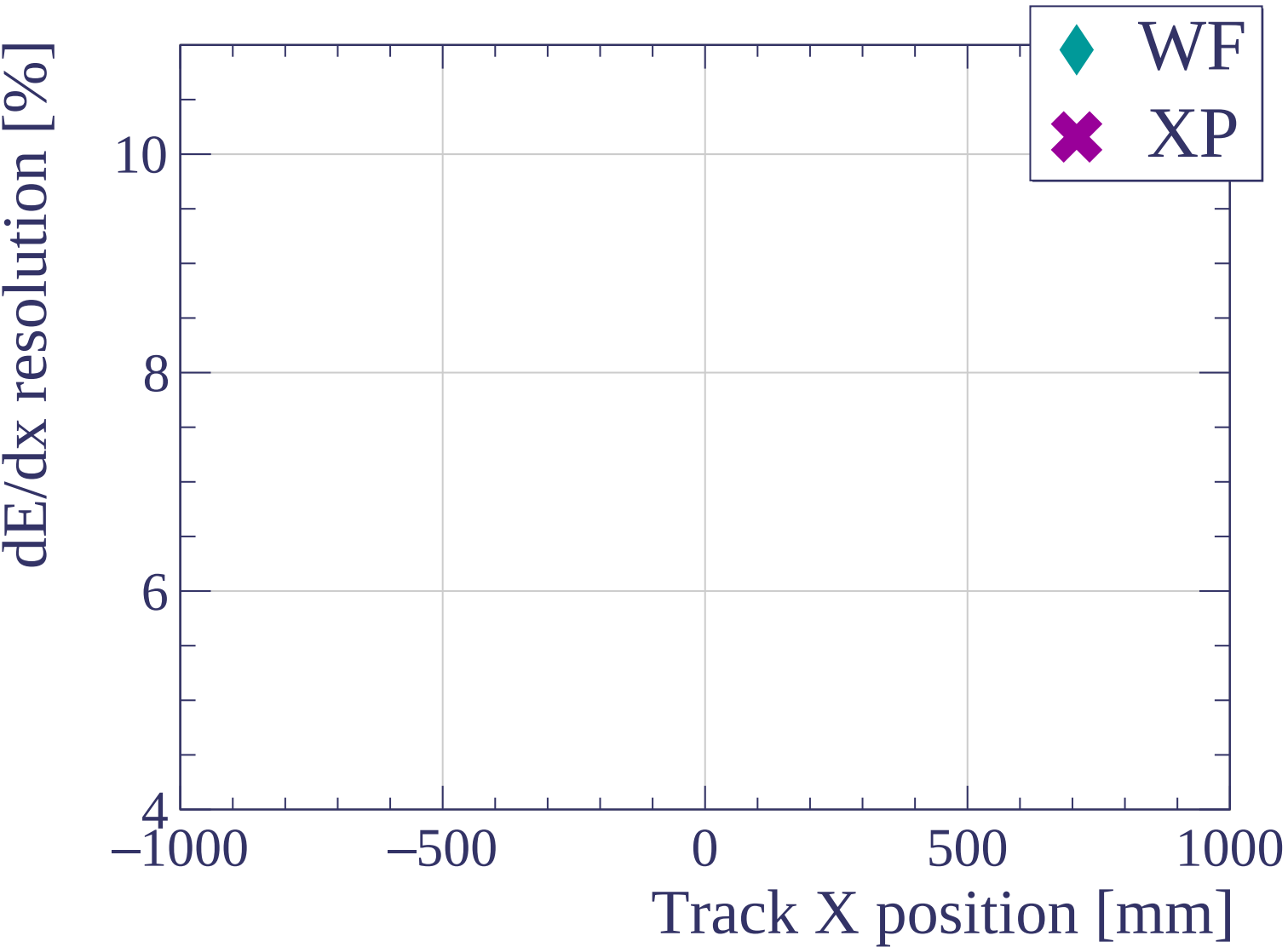


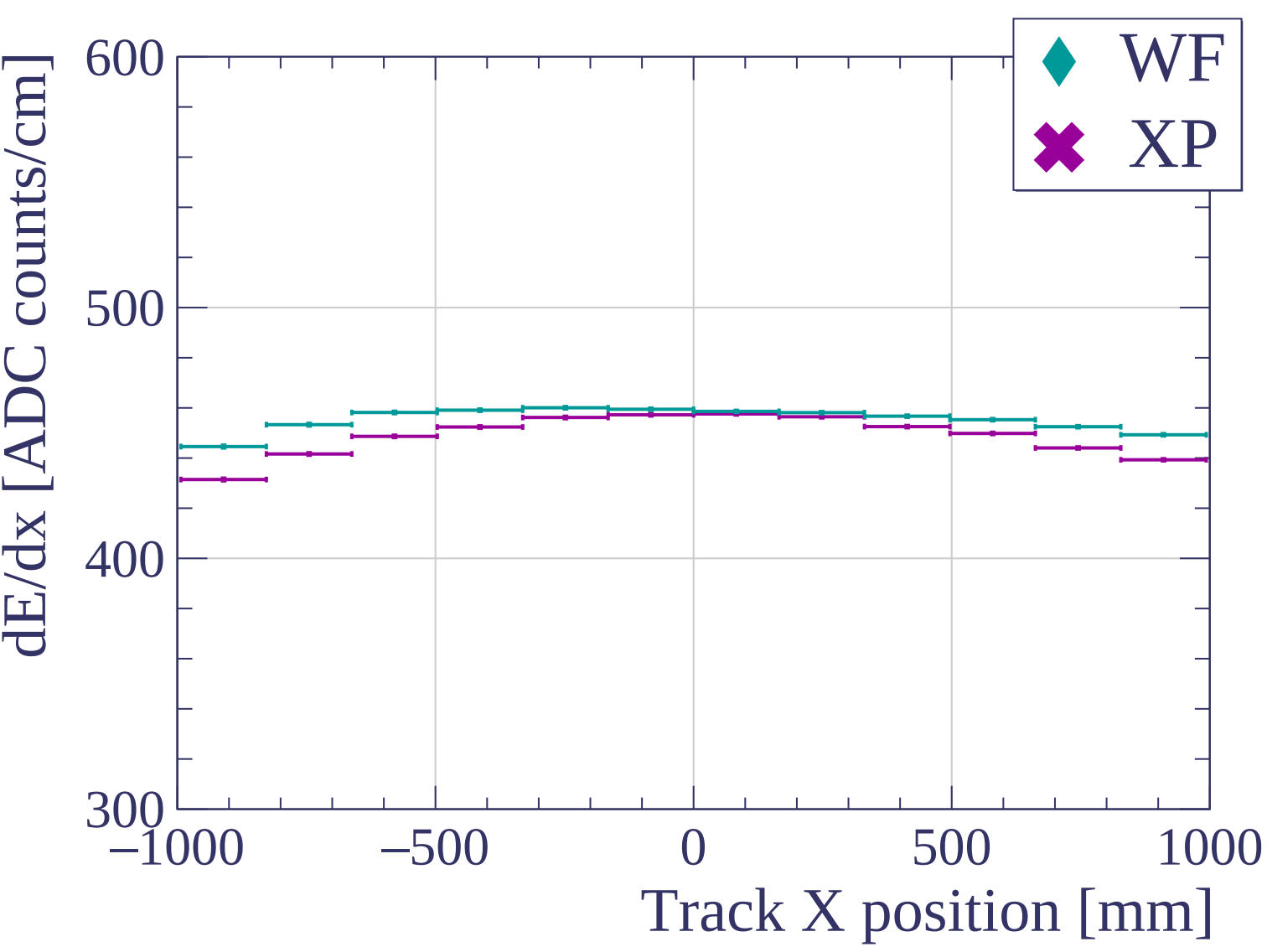


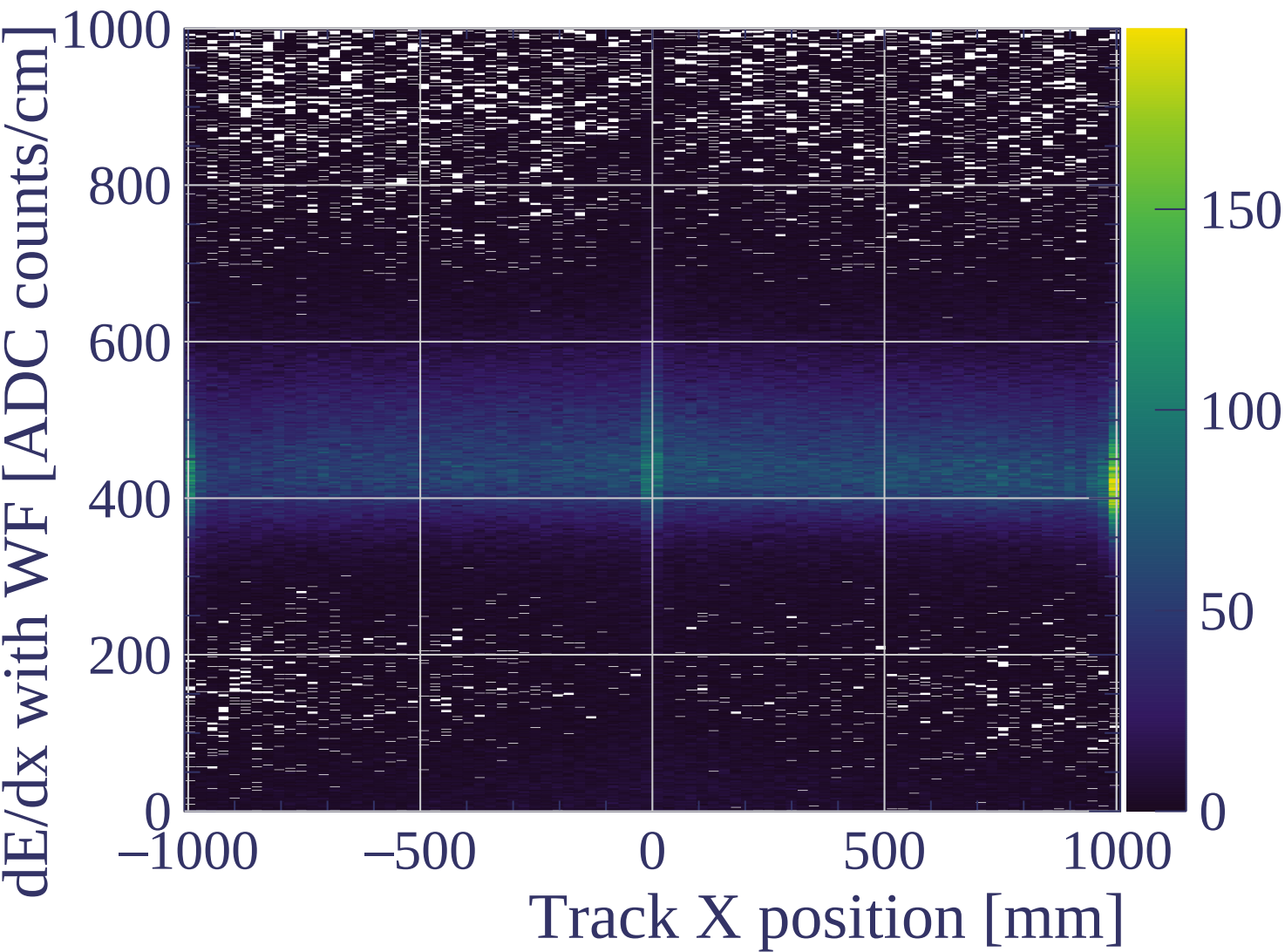


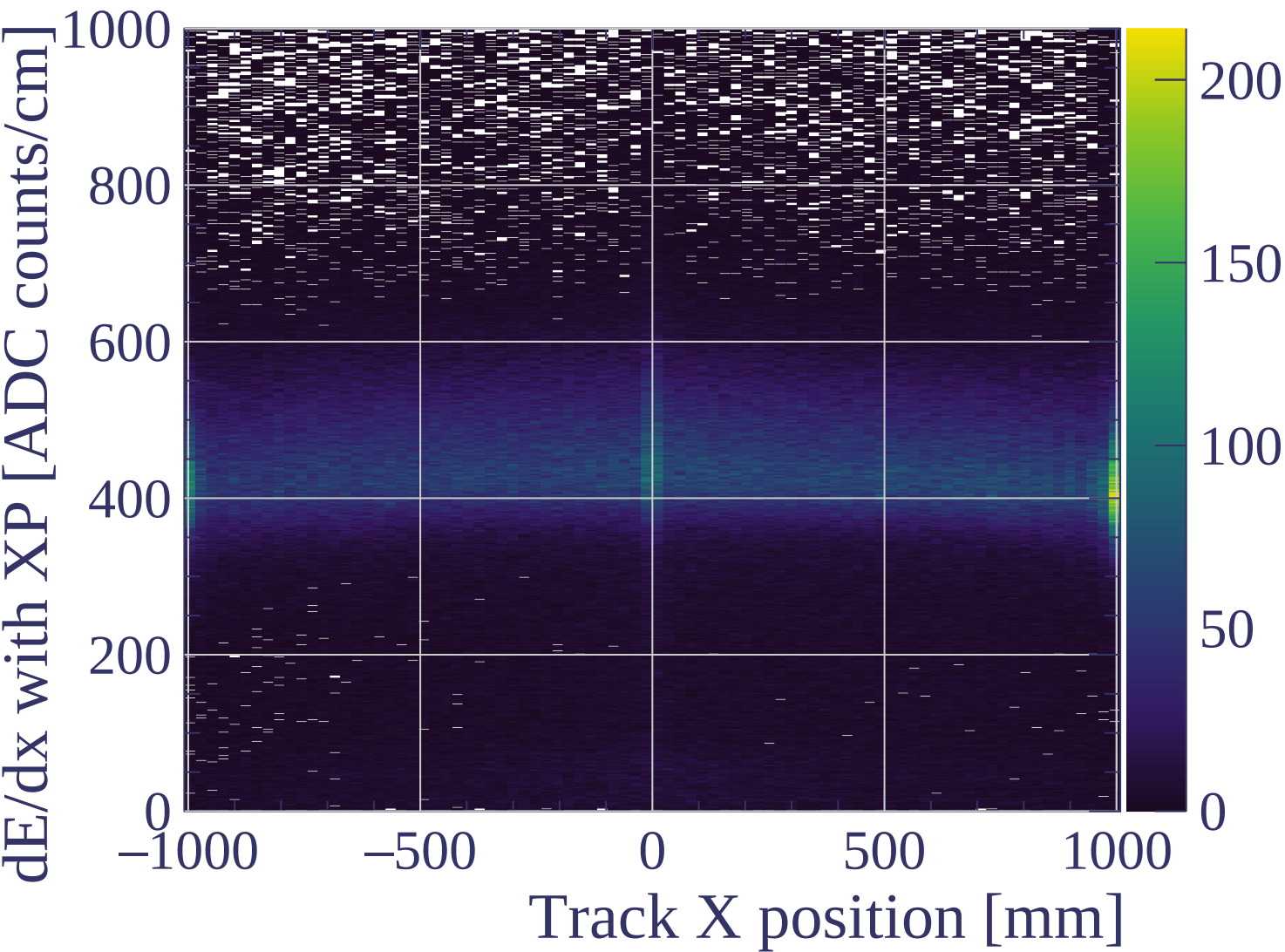




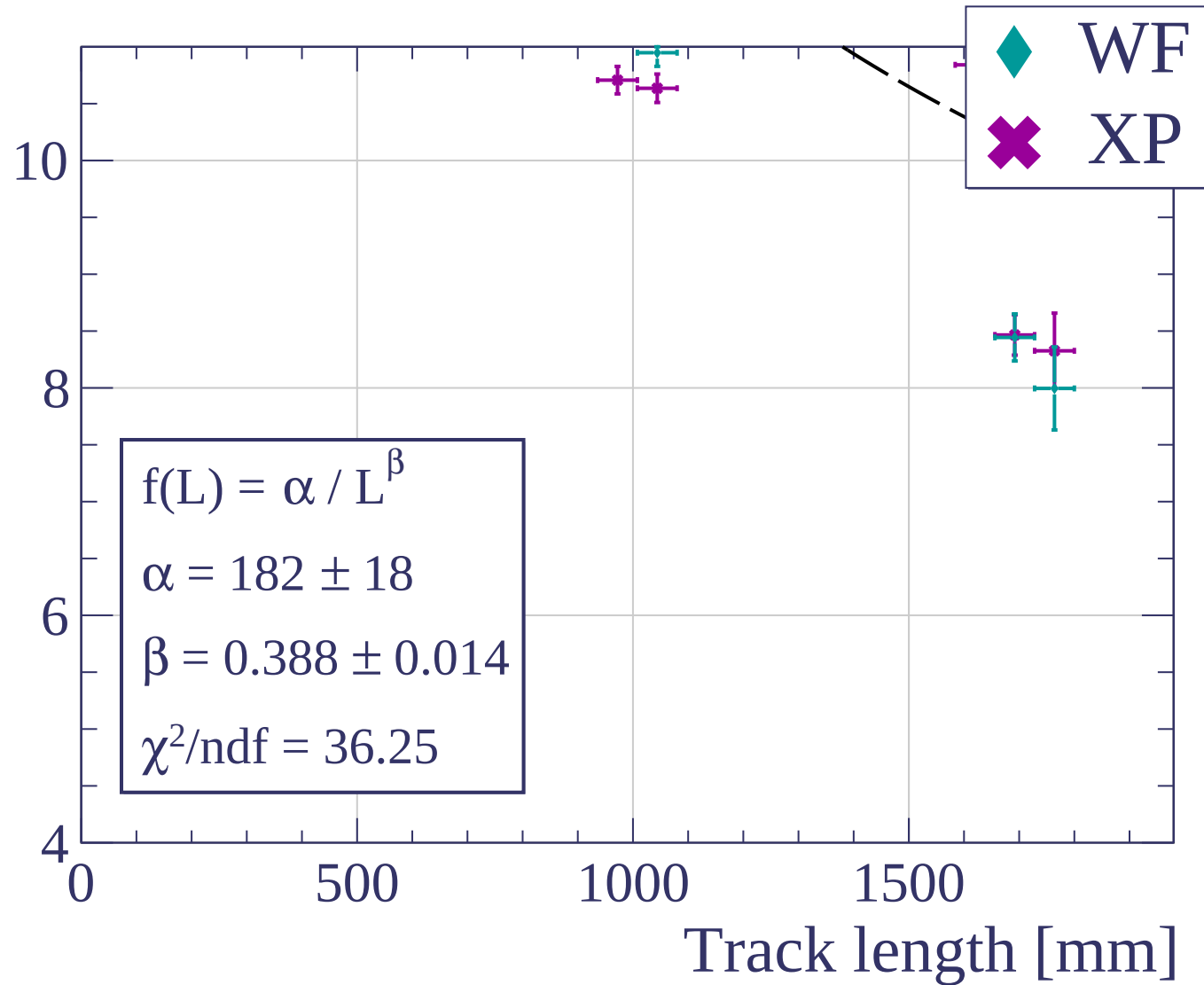


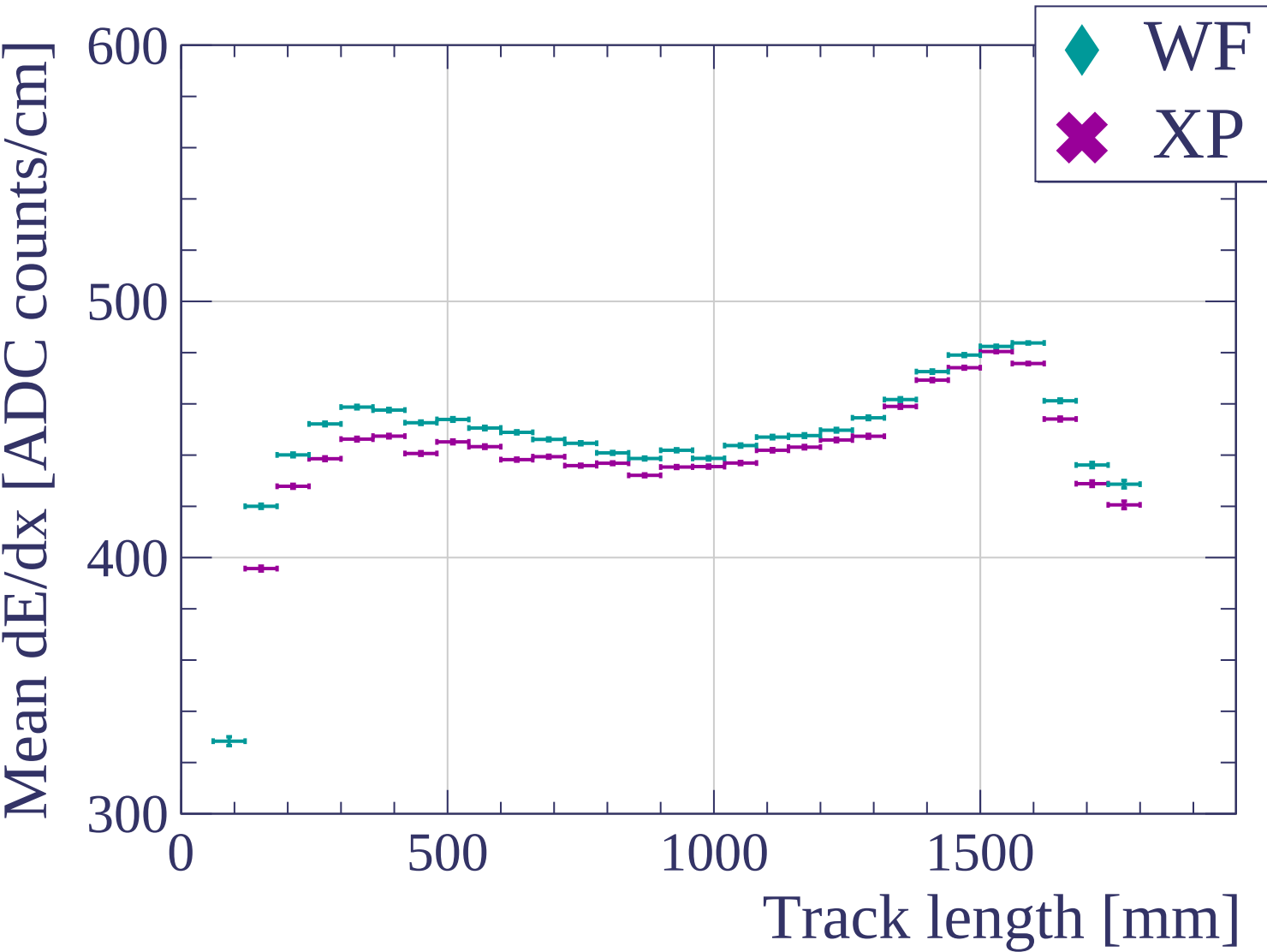


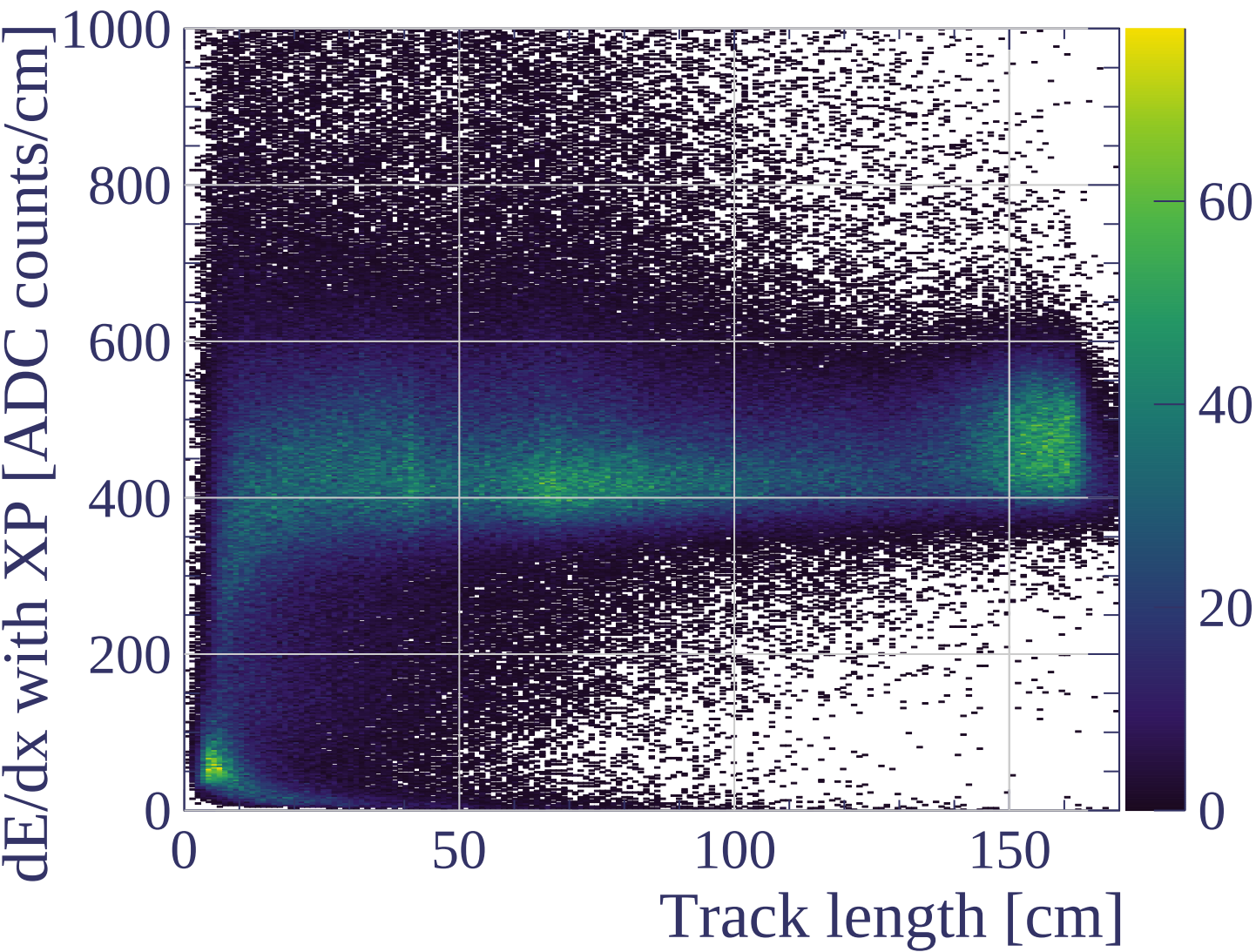


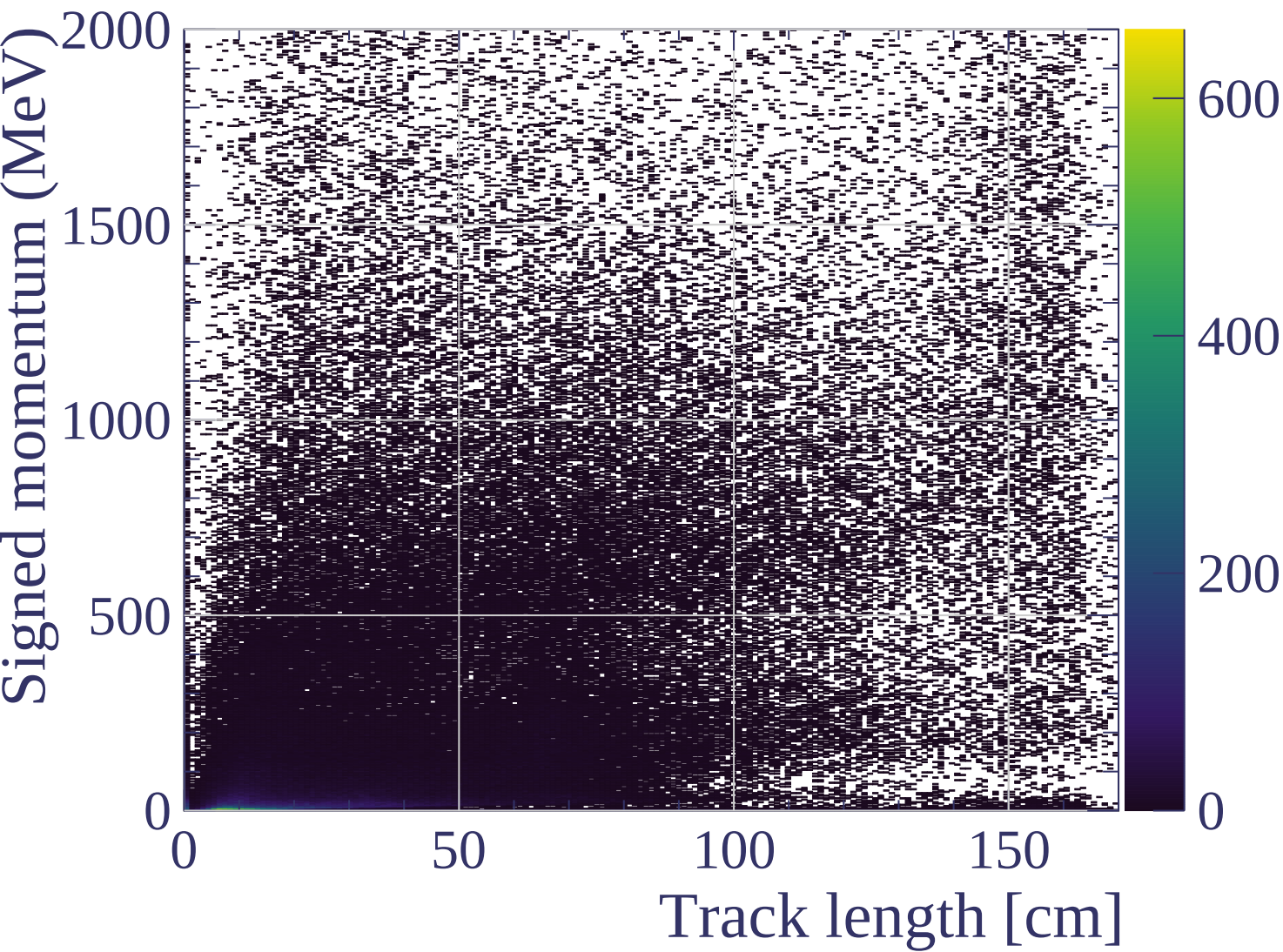


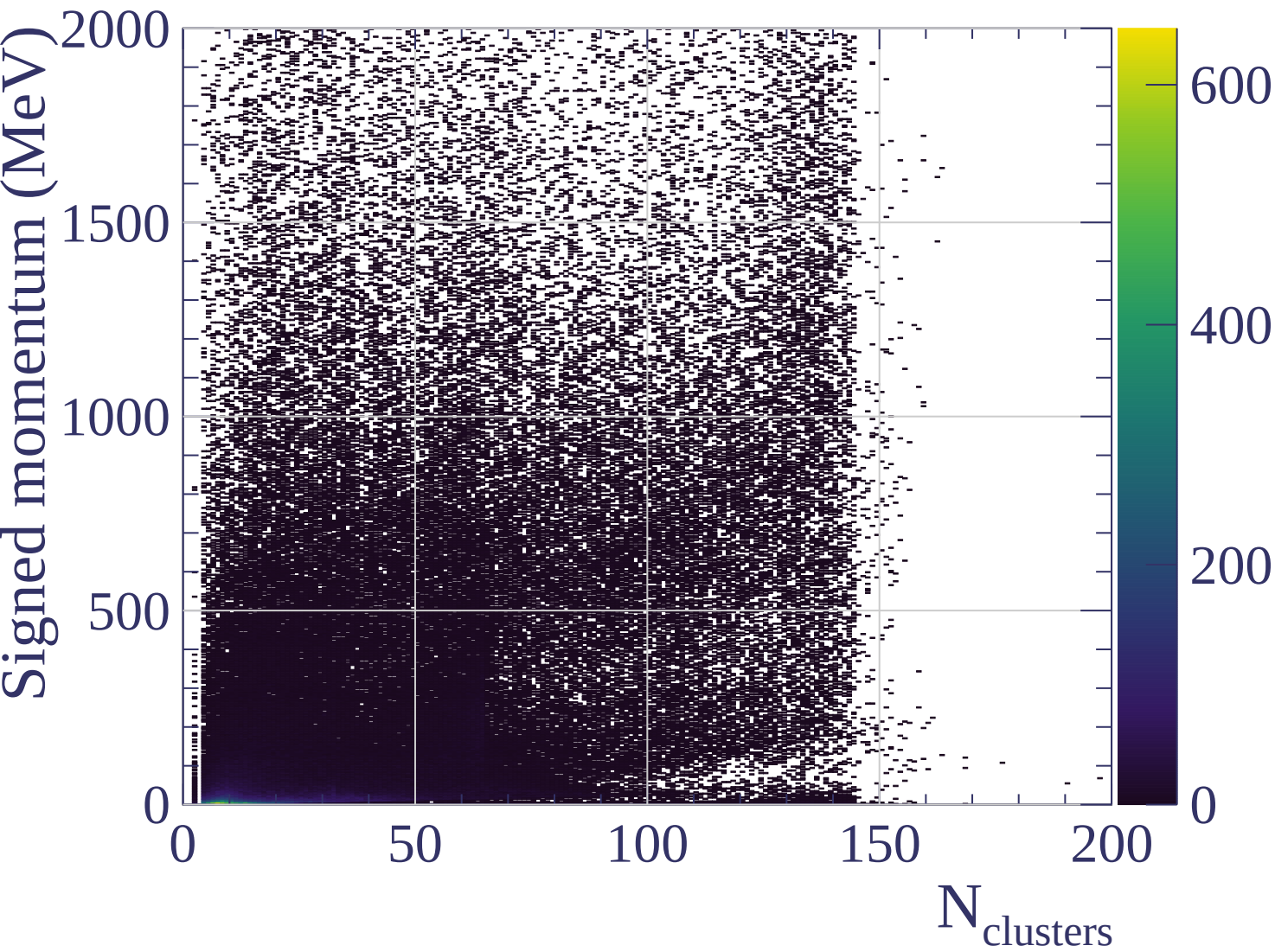
dE/dx resolution [%]

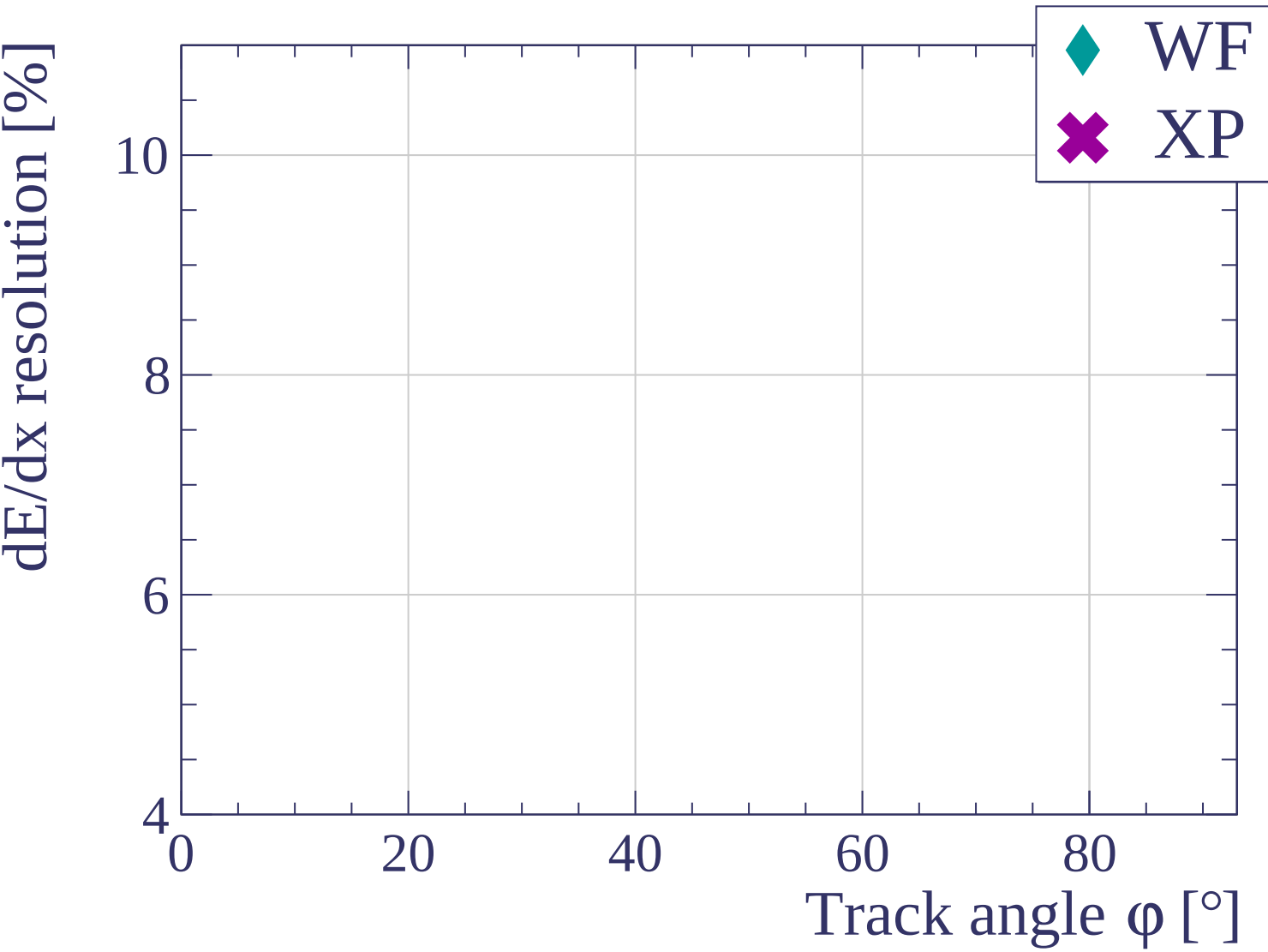


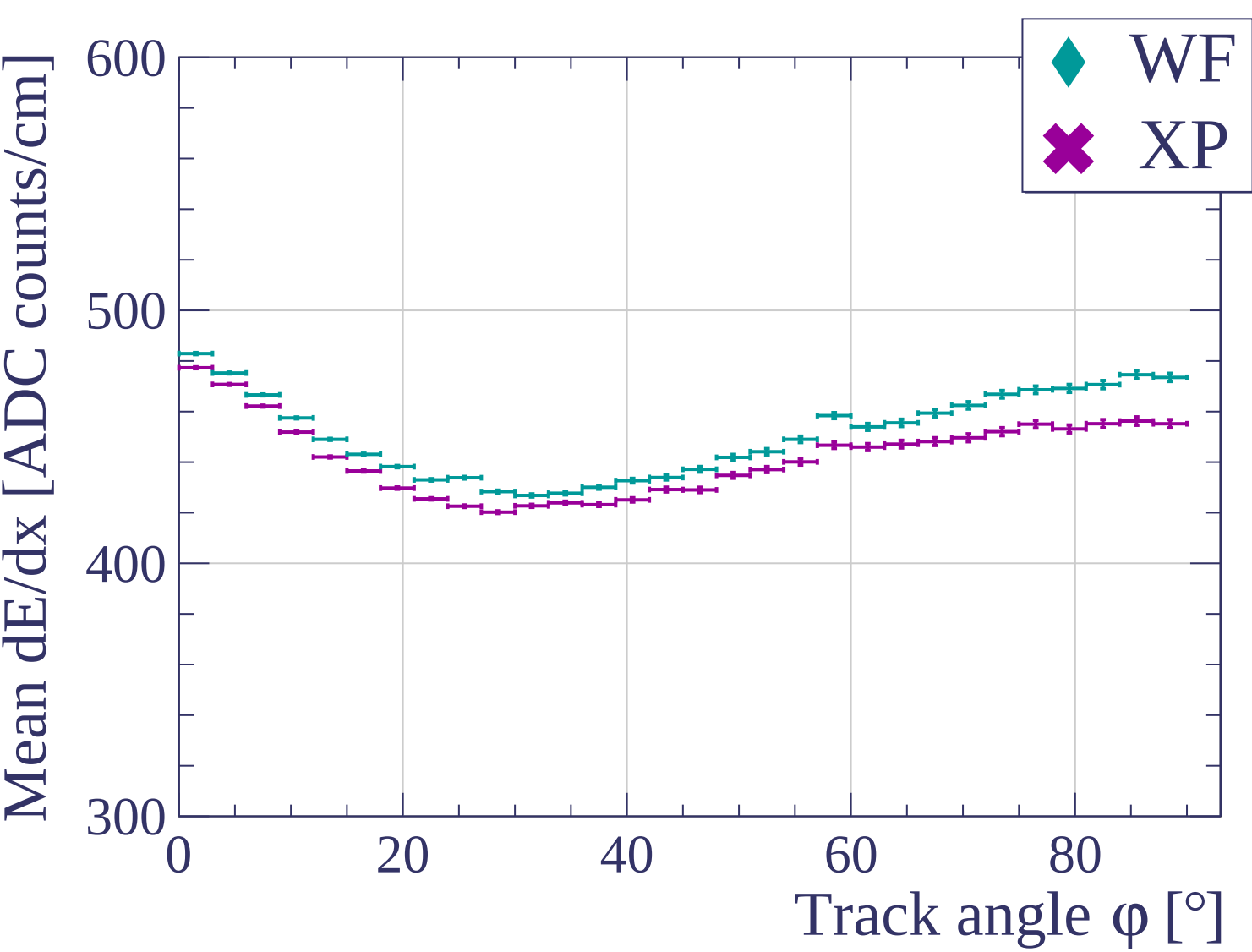


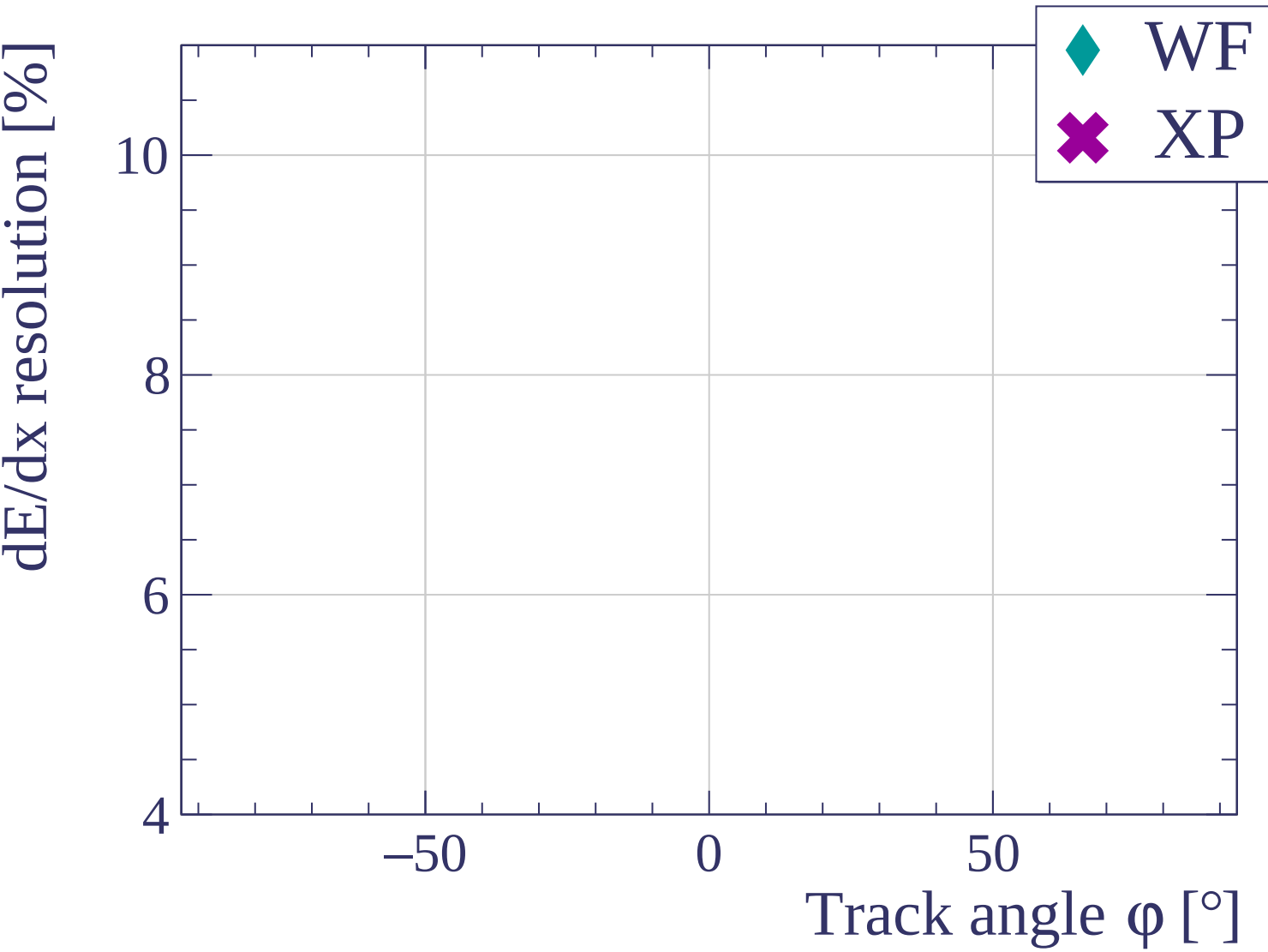


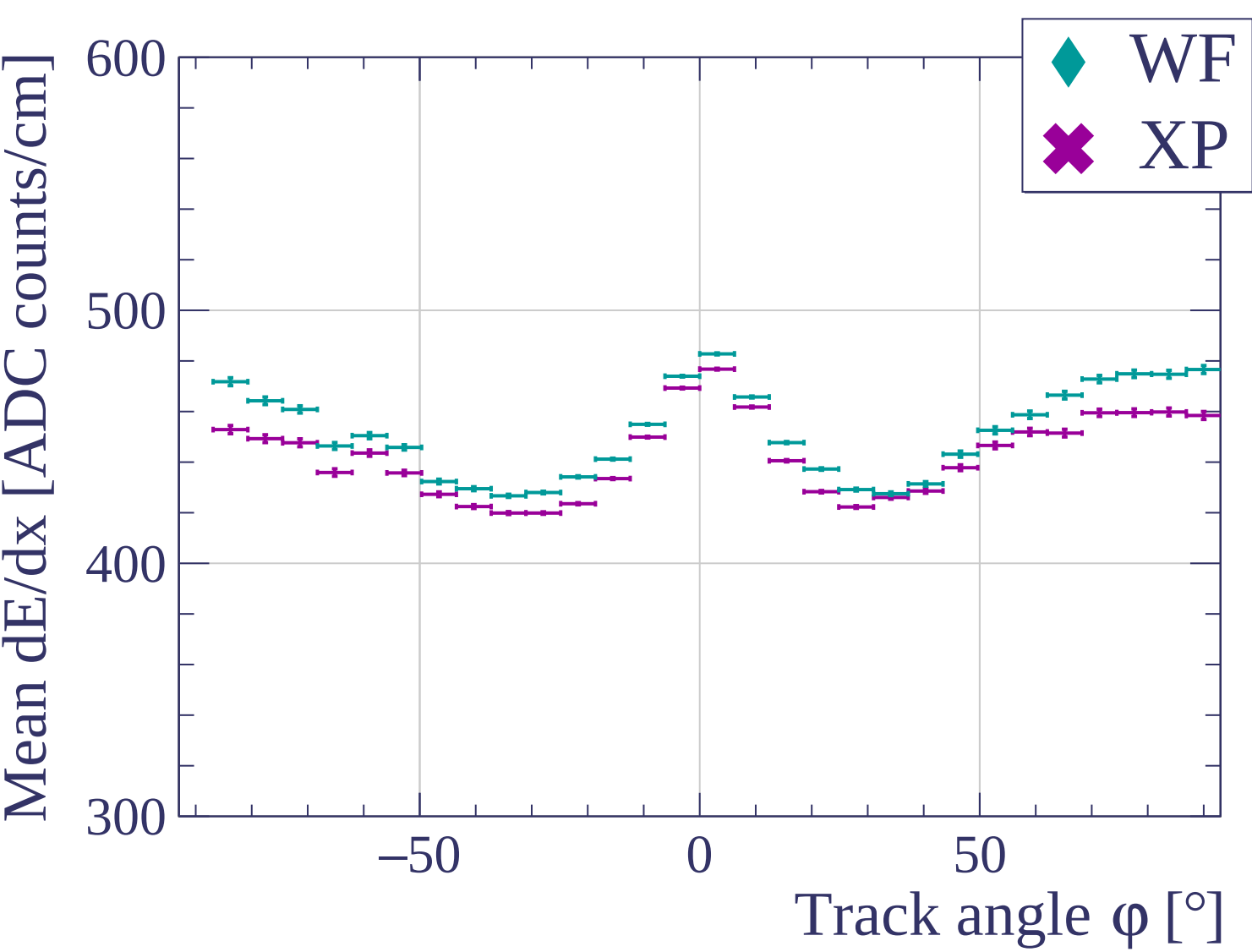


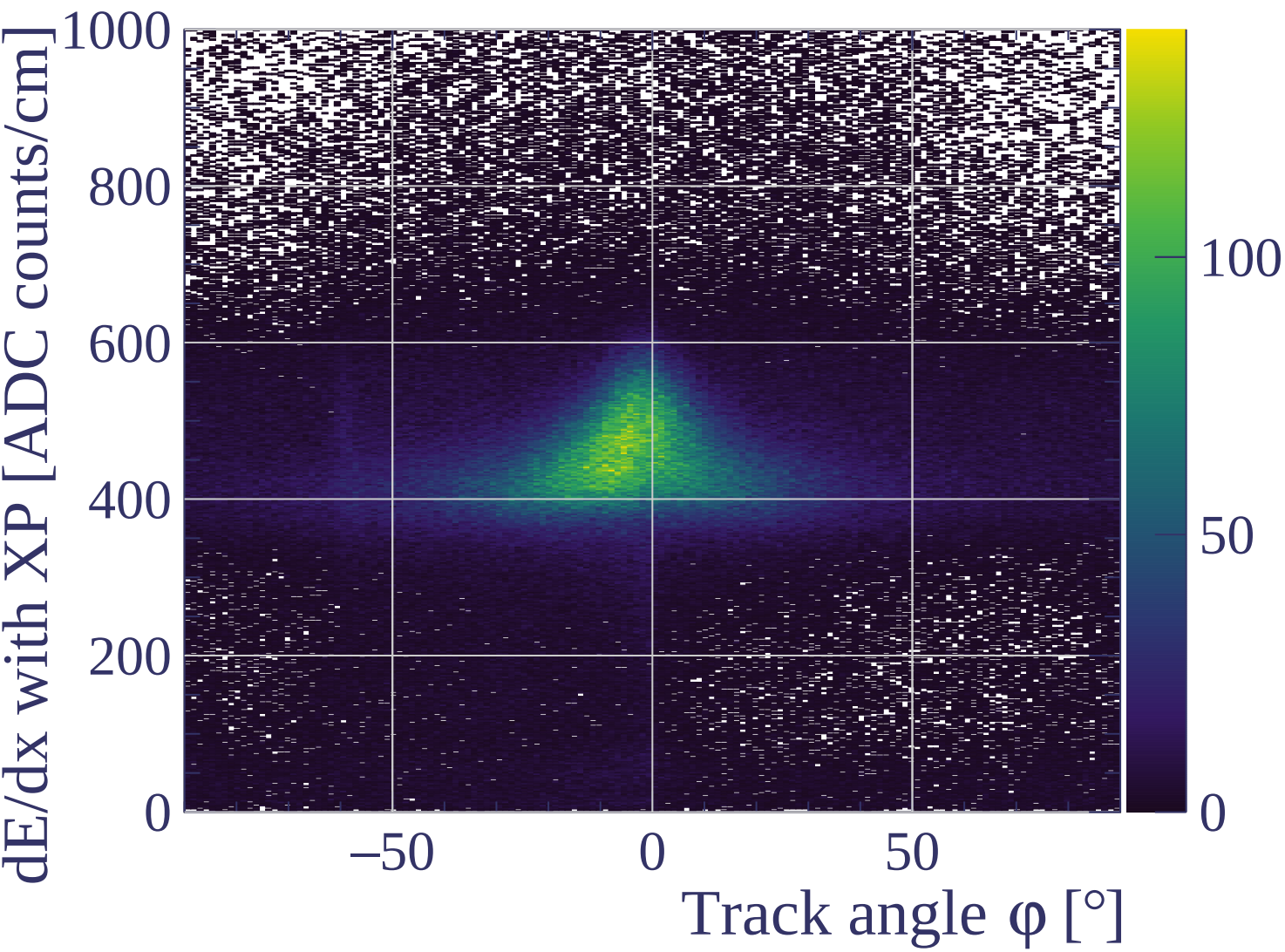




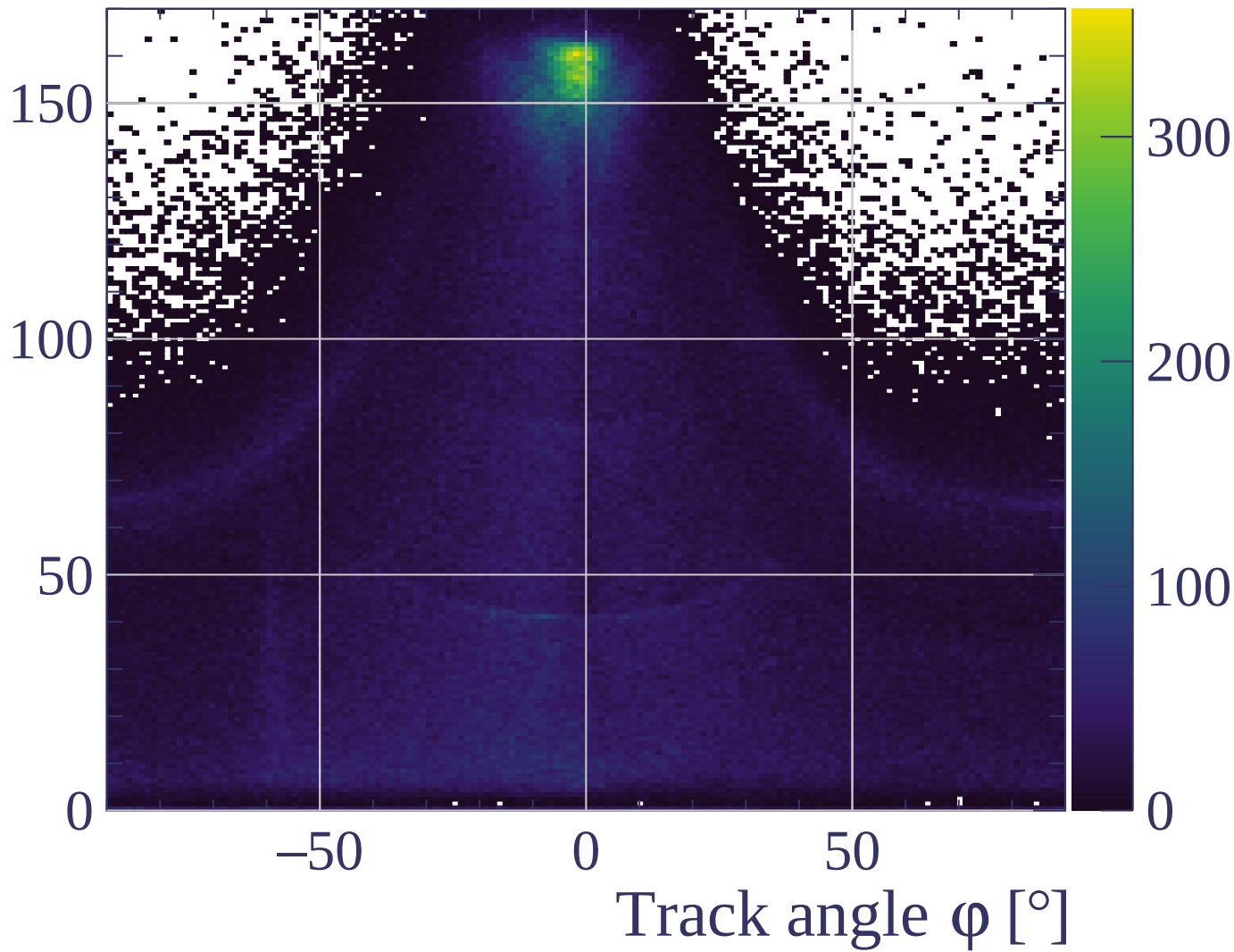


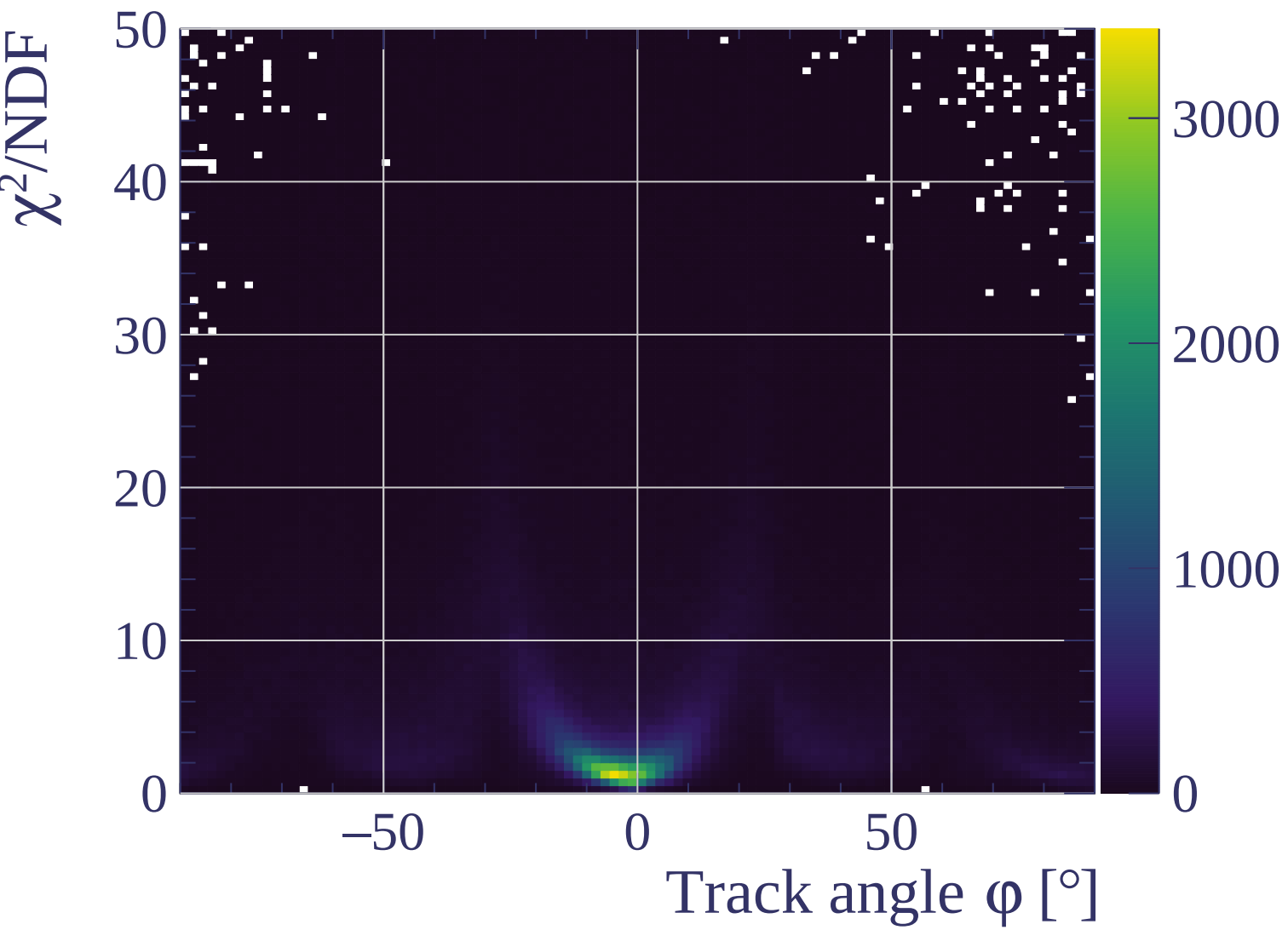


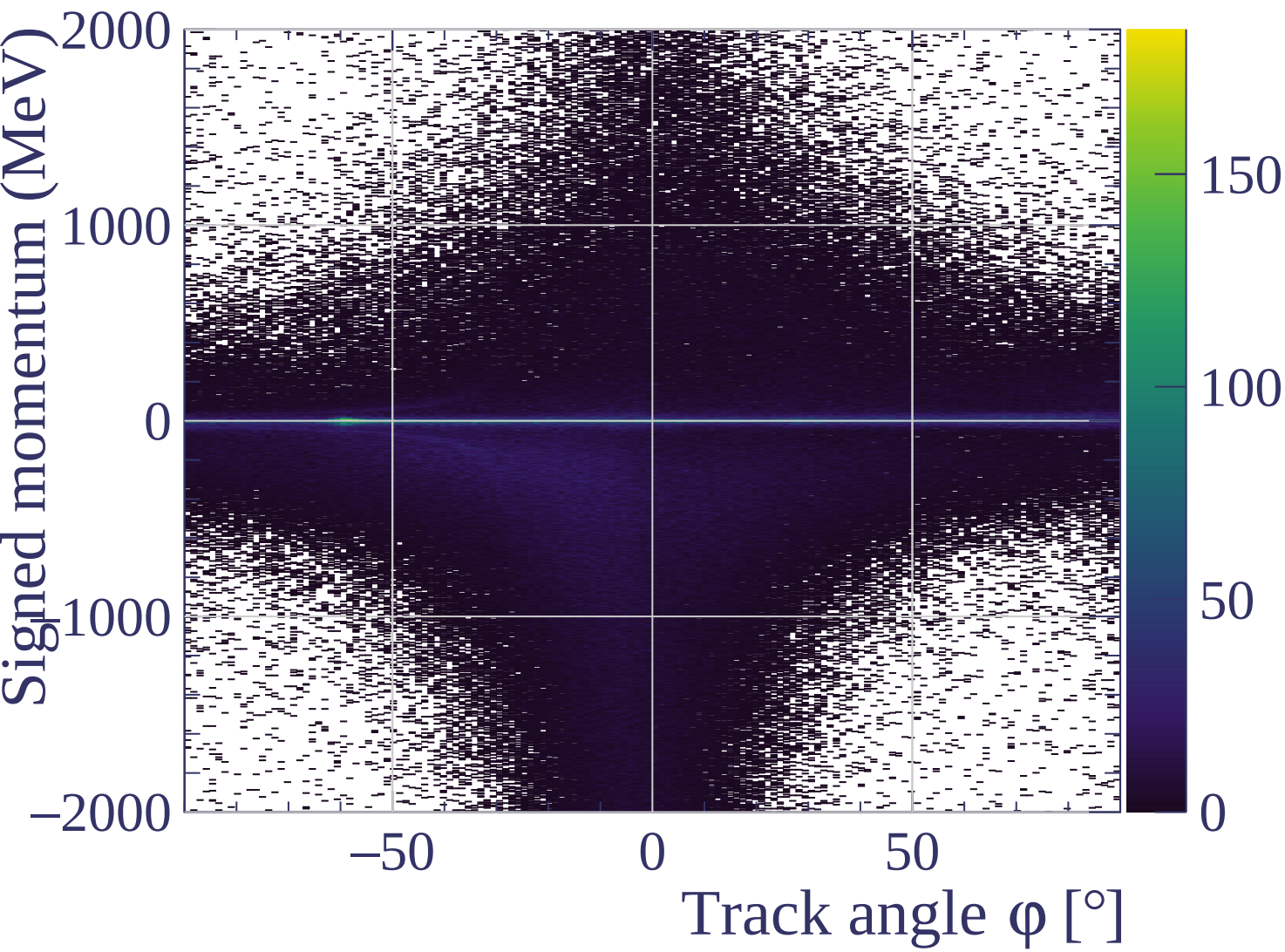


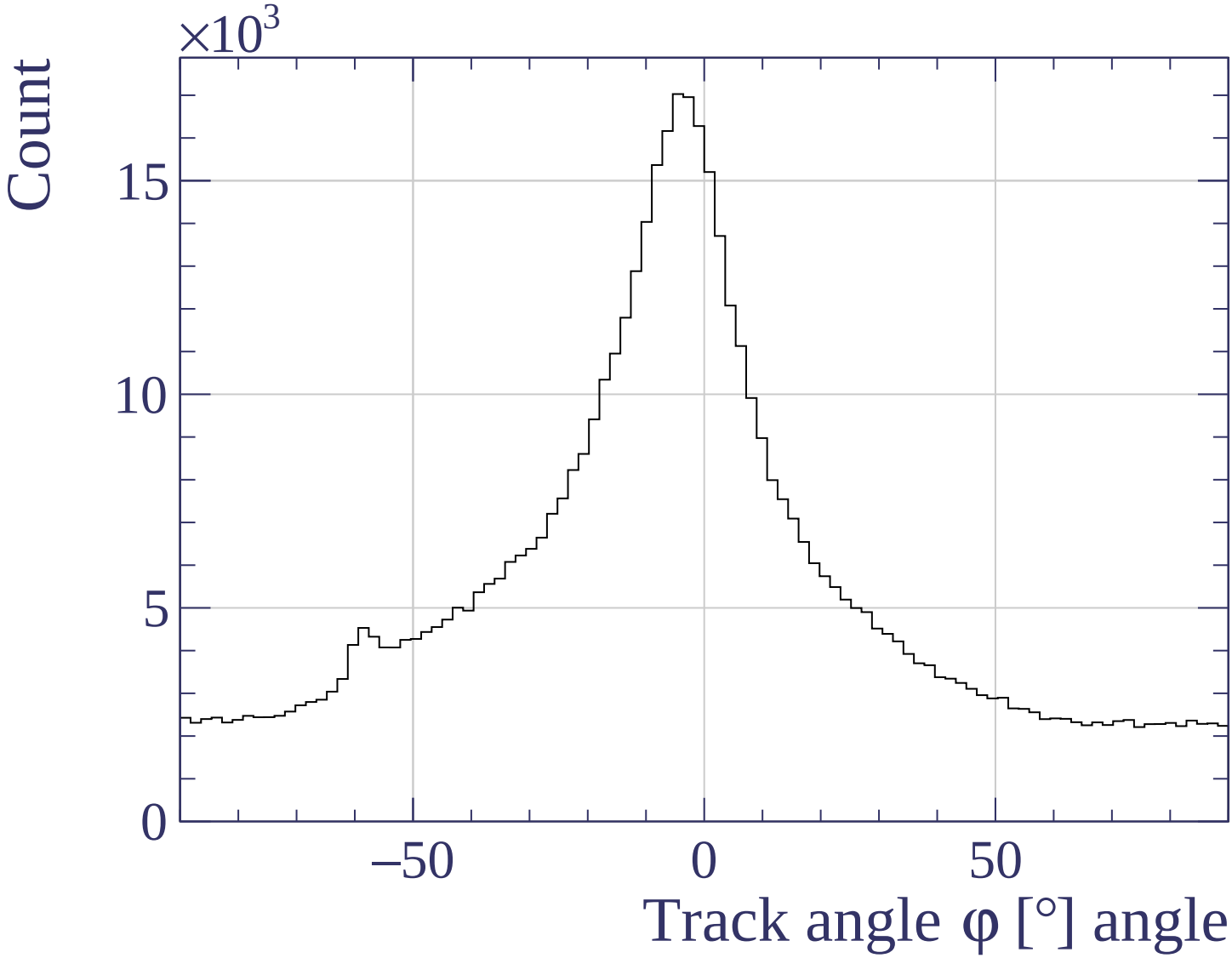


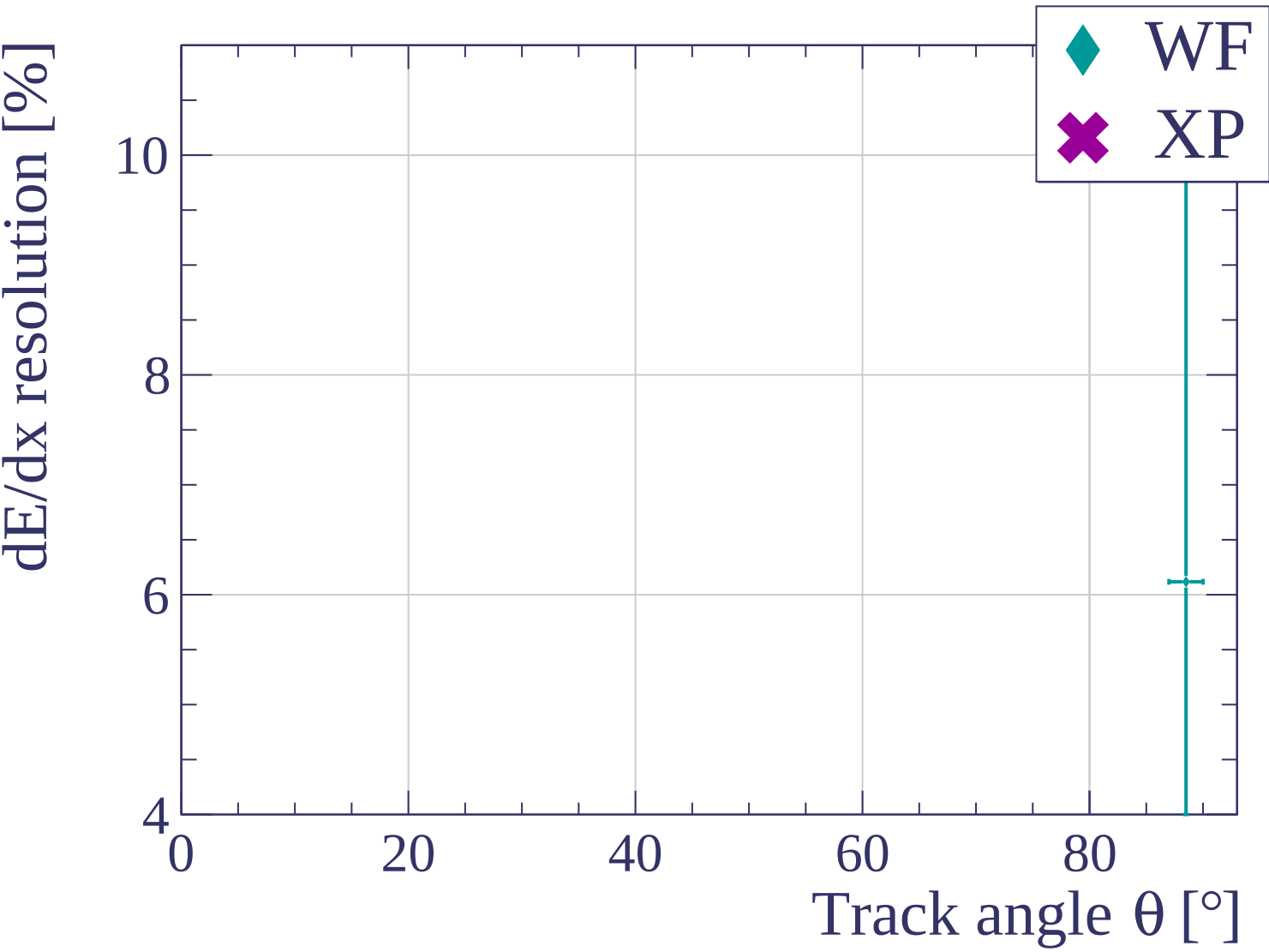
Track length [cm]

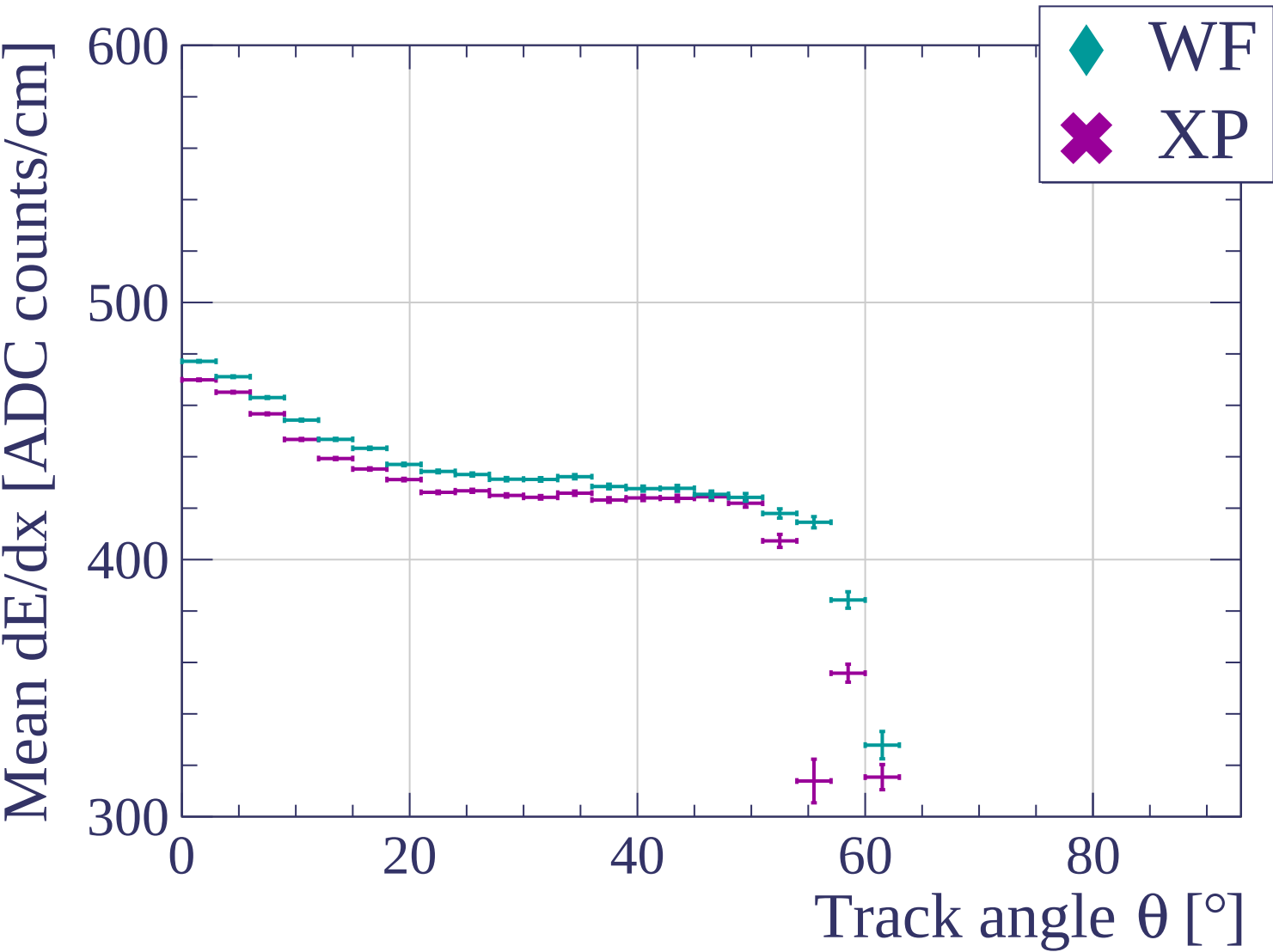


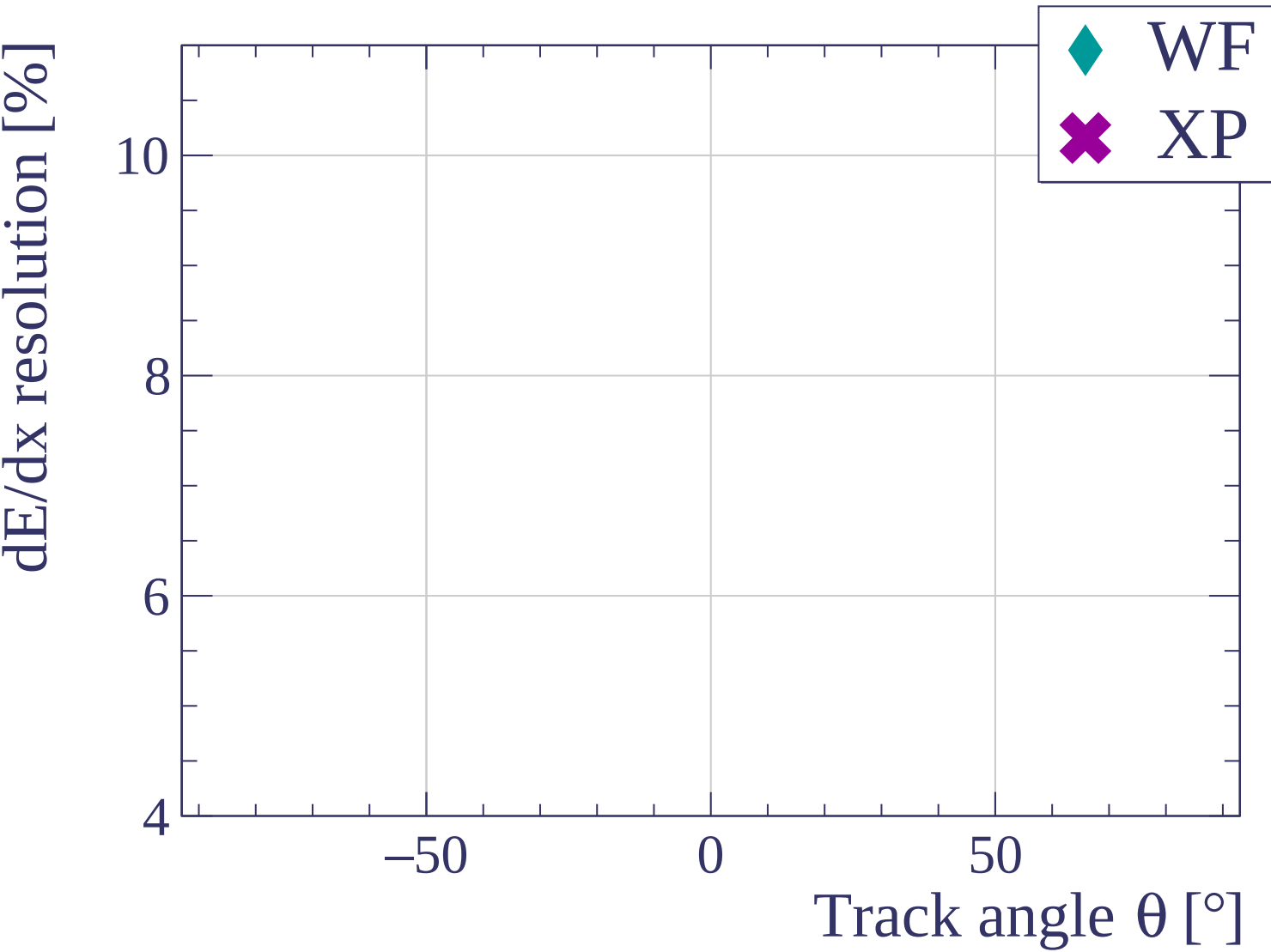


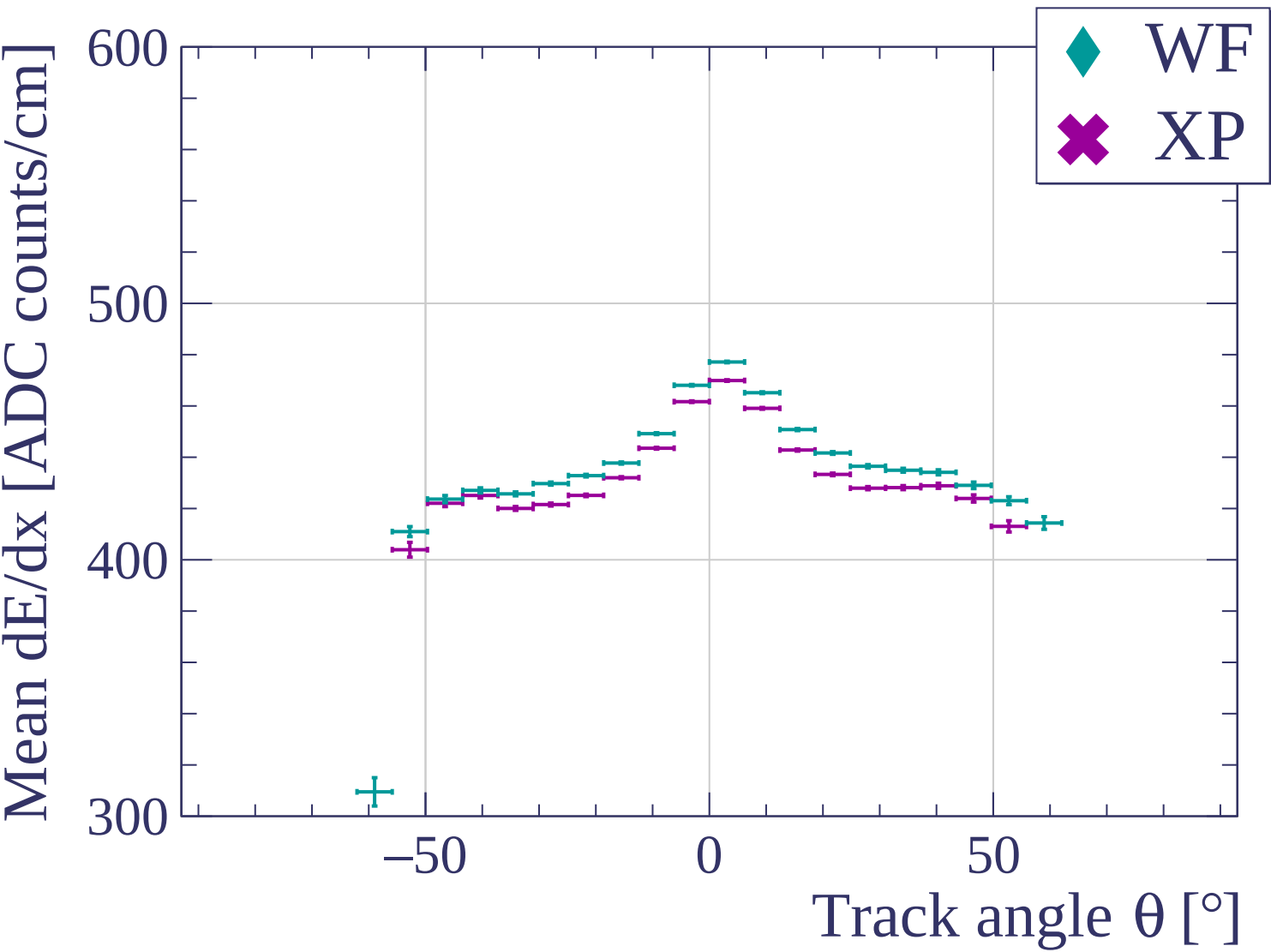


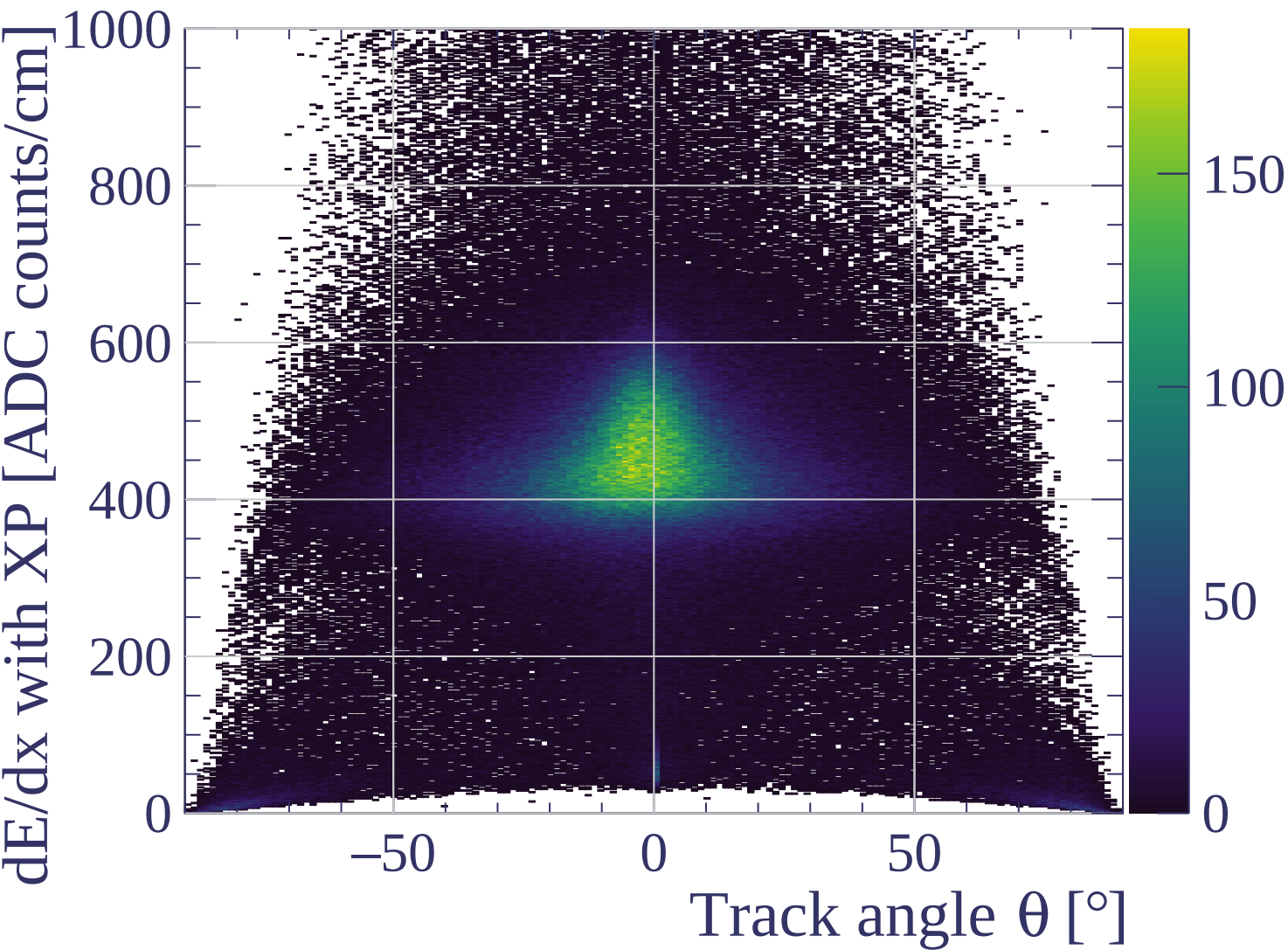




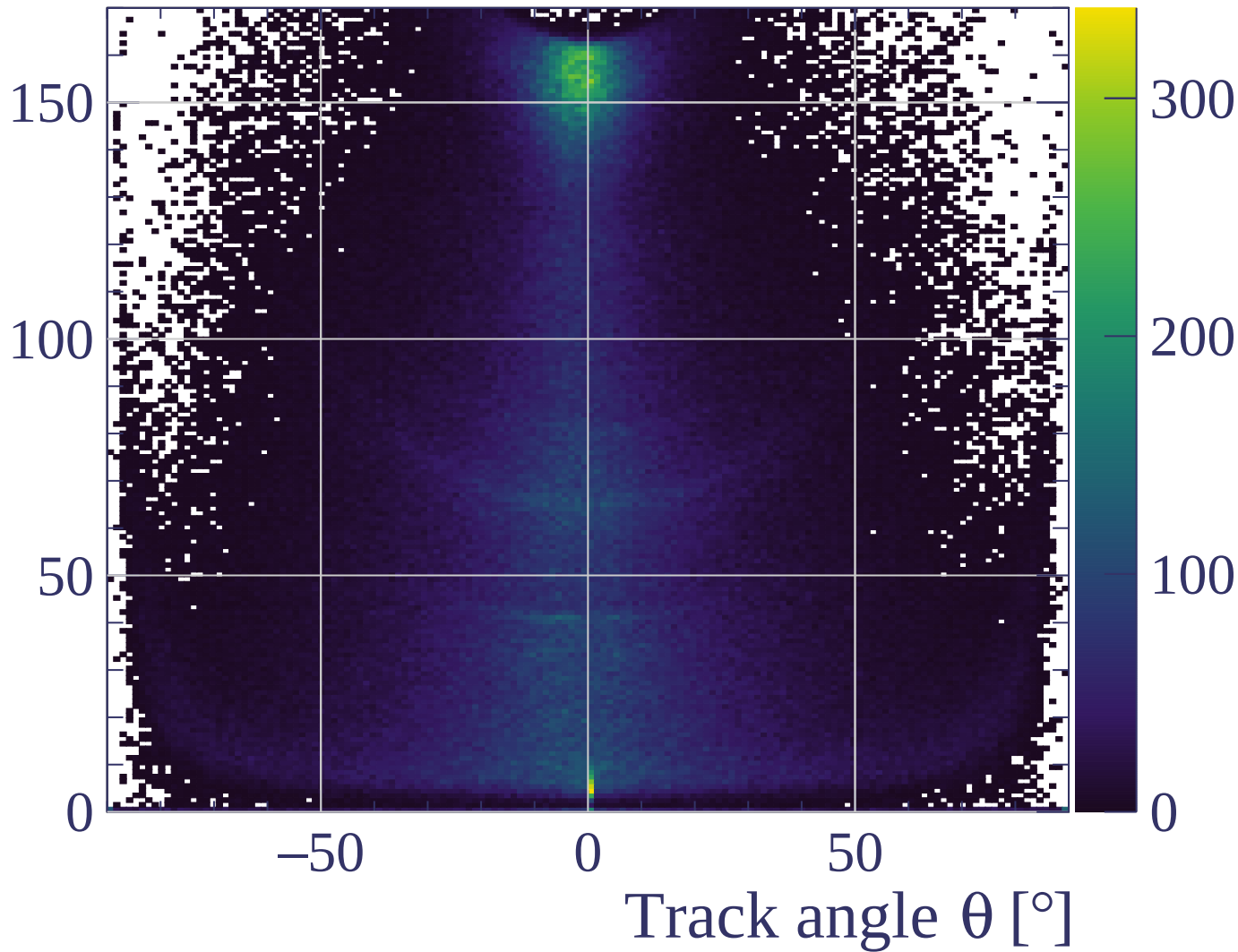


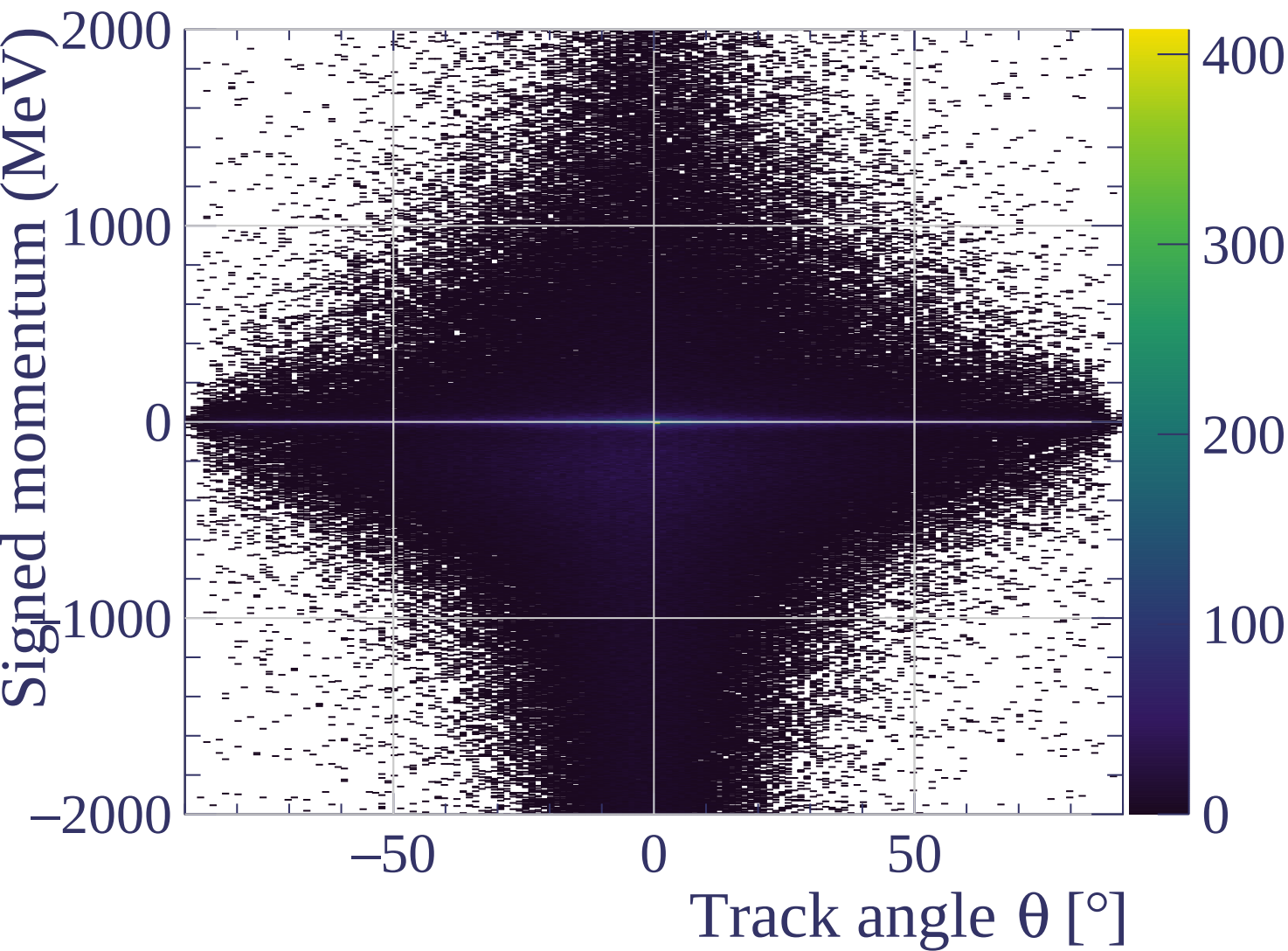


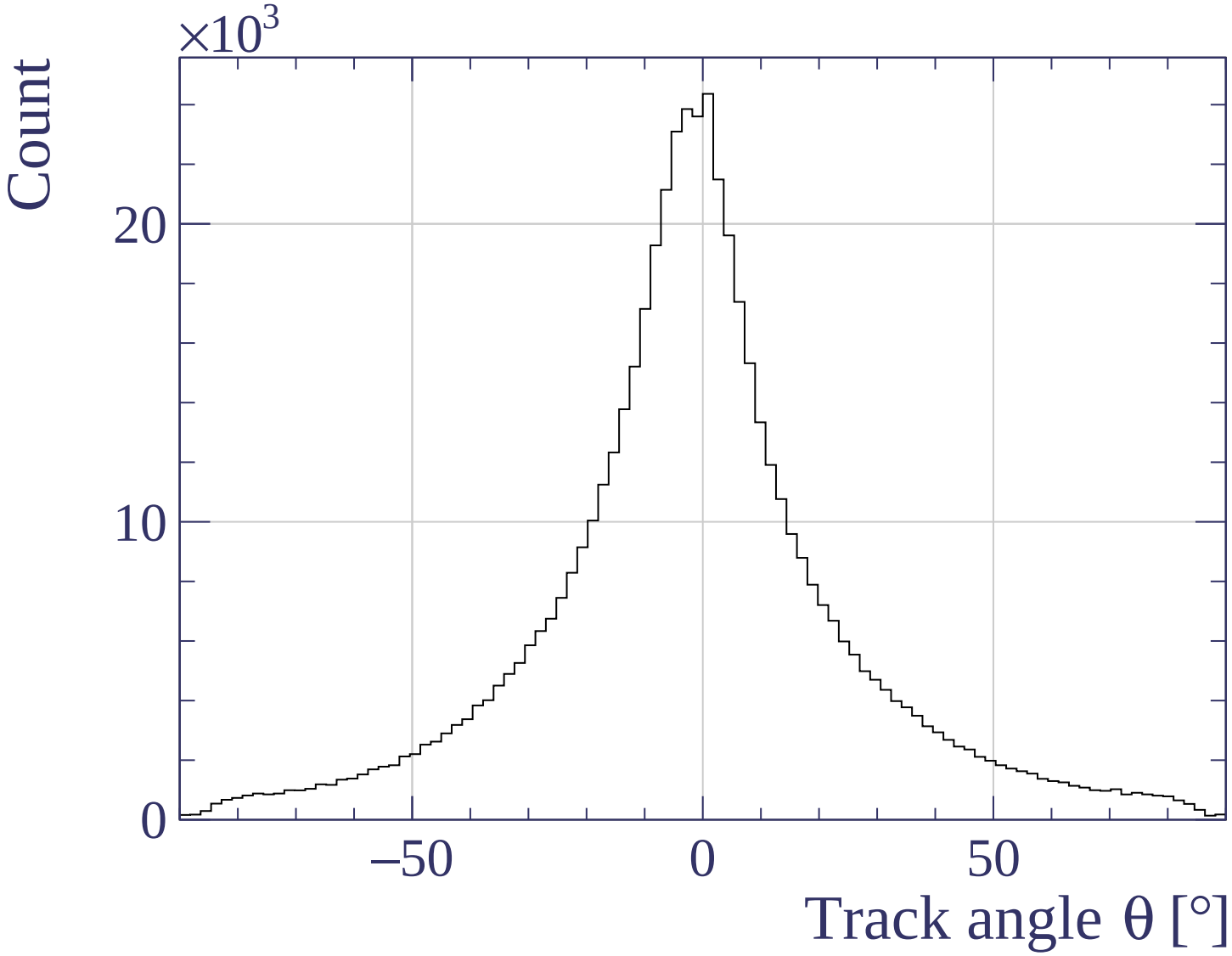




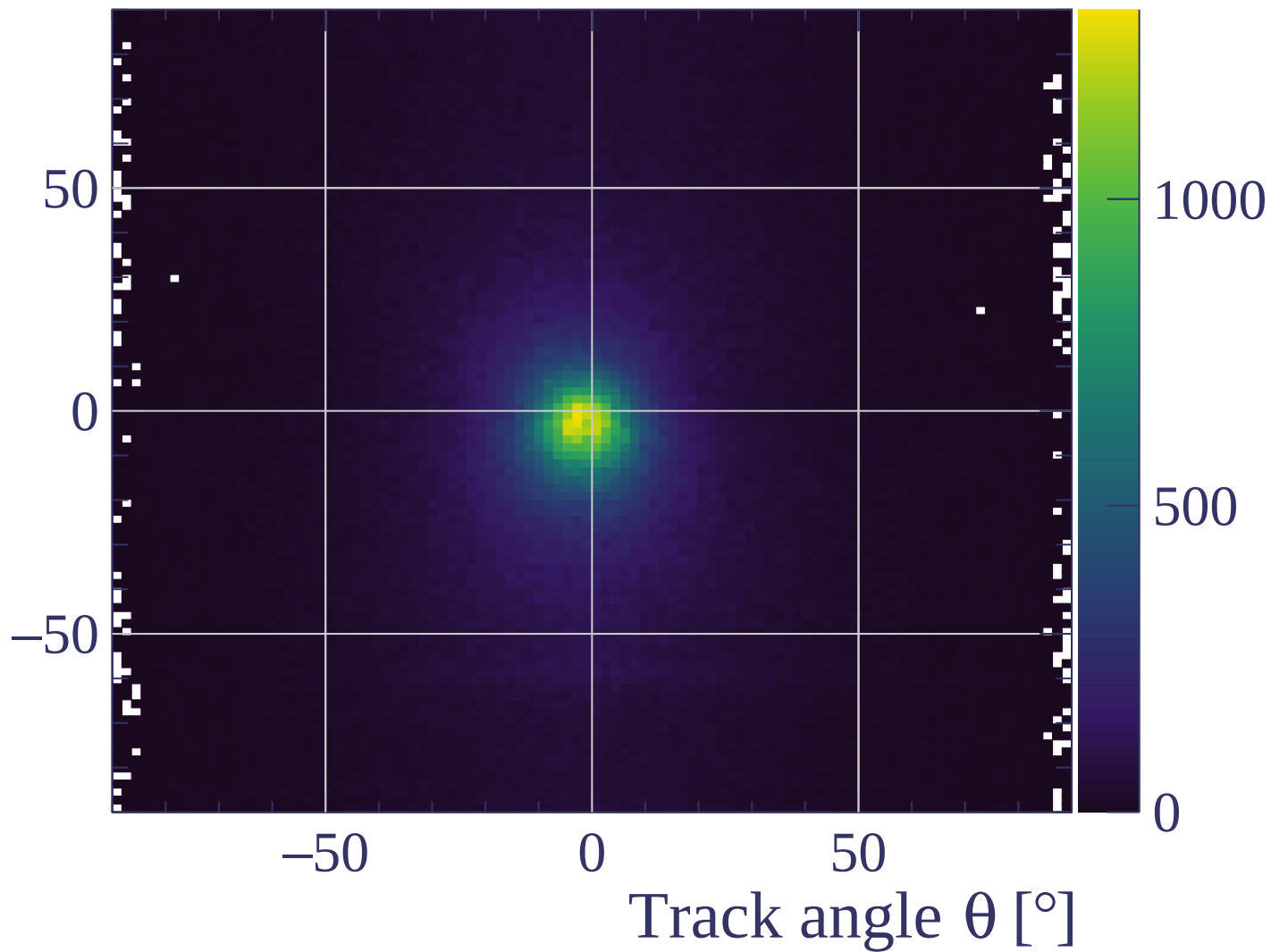
Track length [cm]



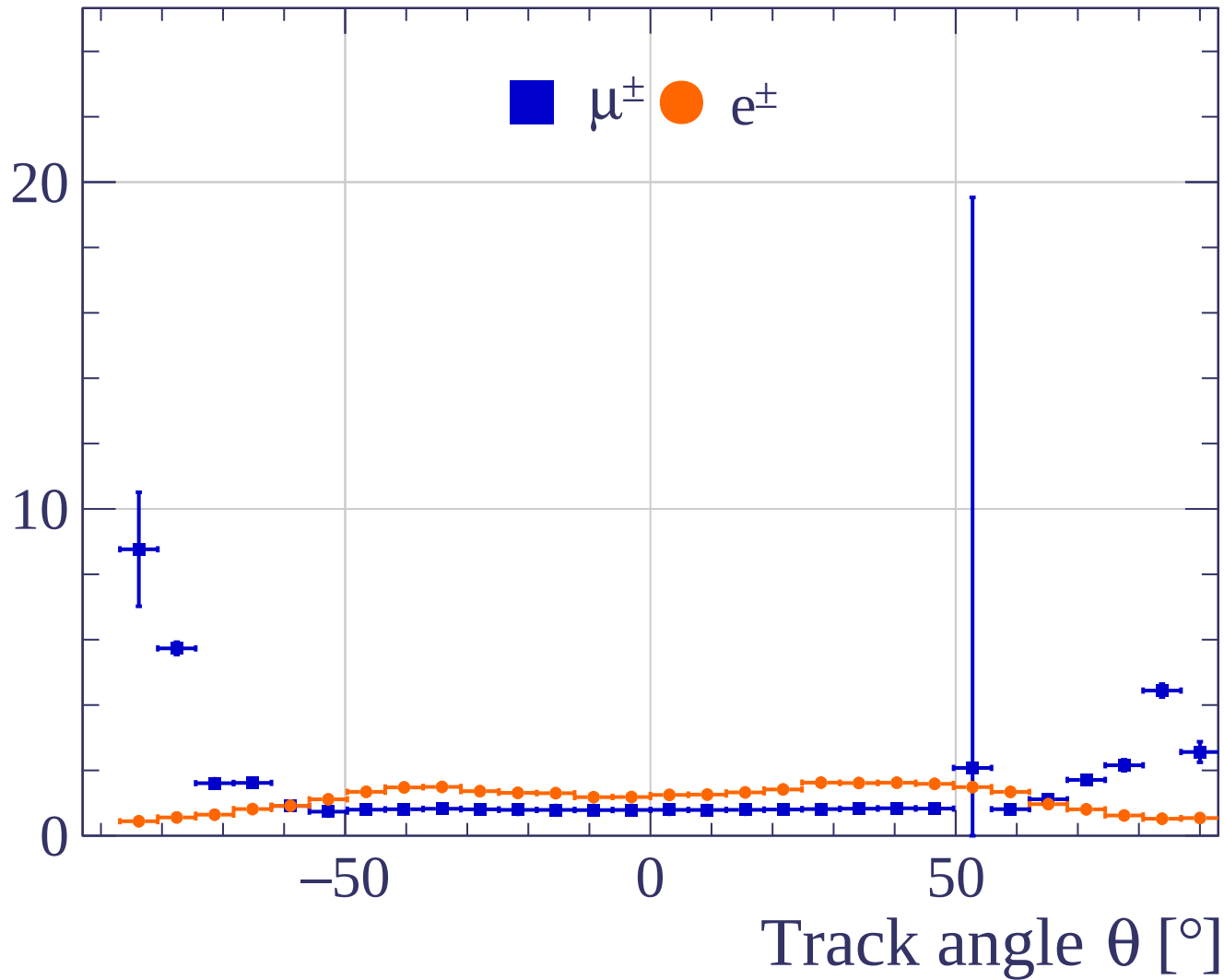




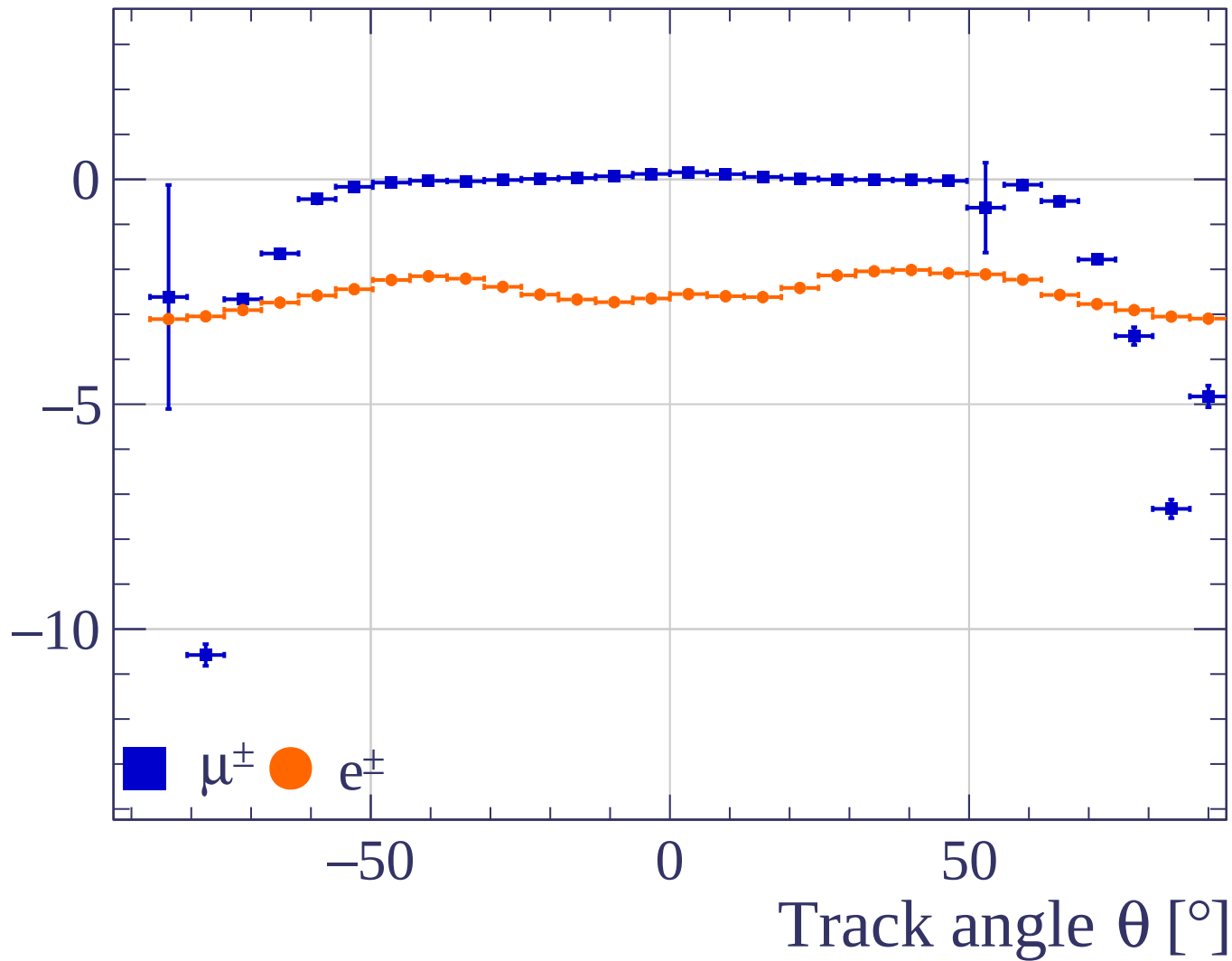
Track angle ϕ [°]



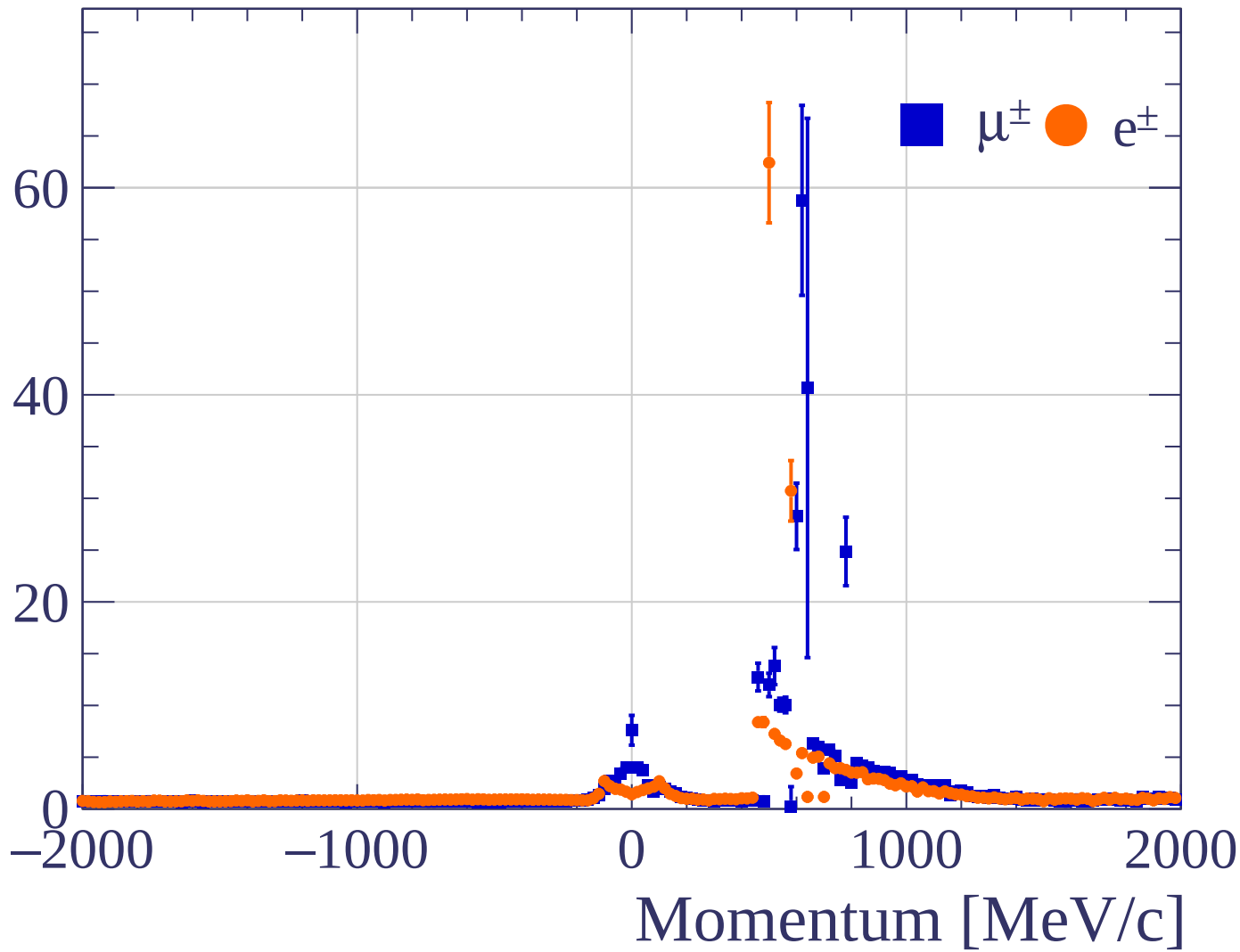
Pulls standard deviation



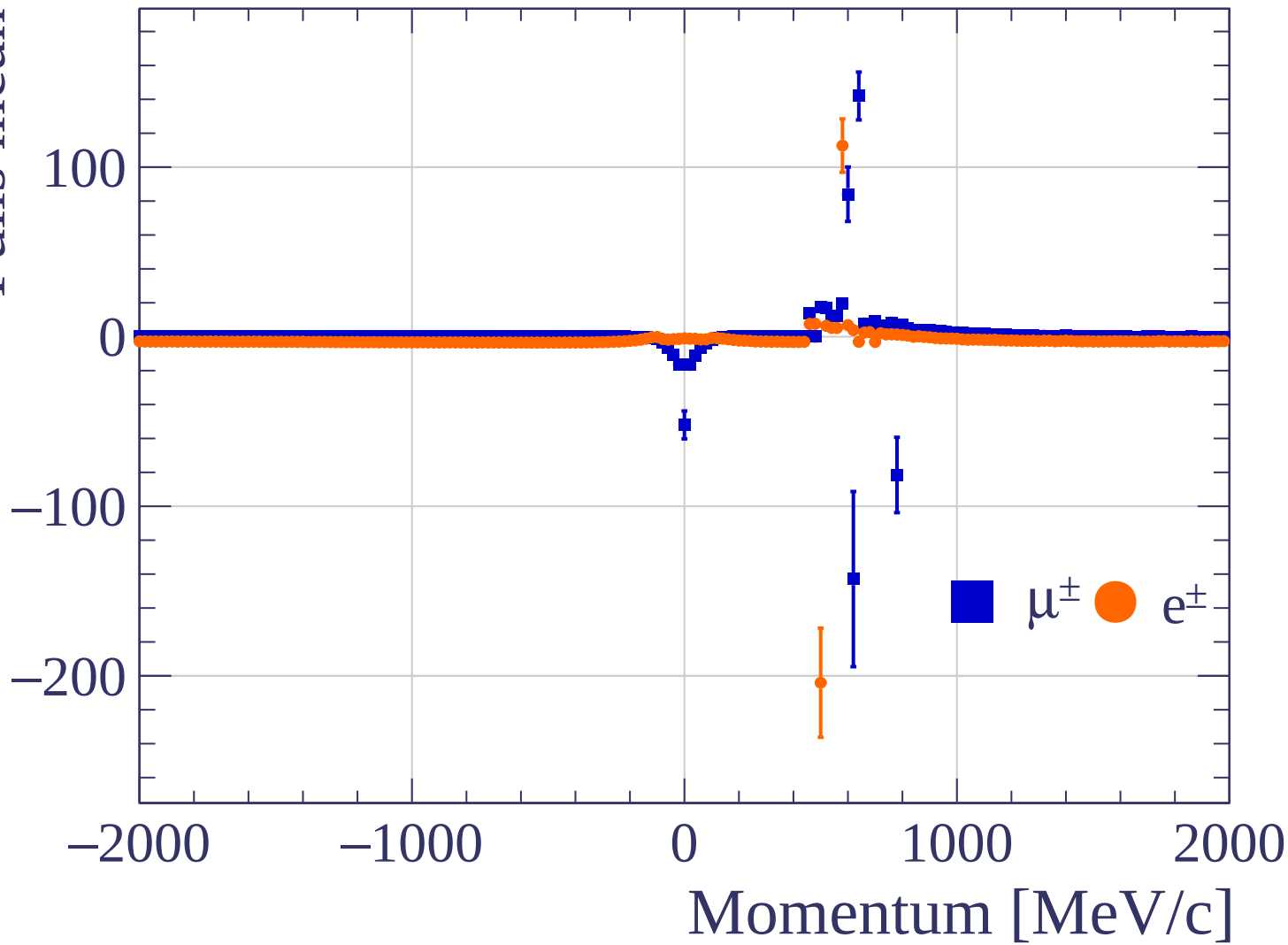
Pulls mean

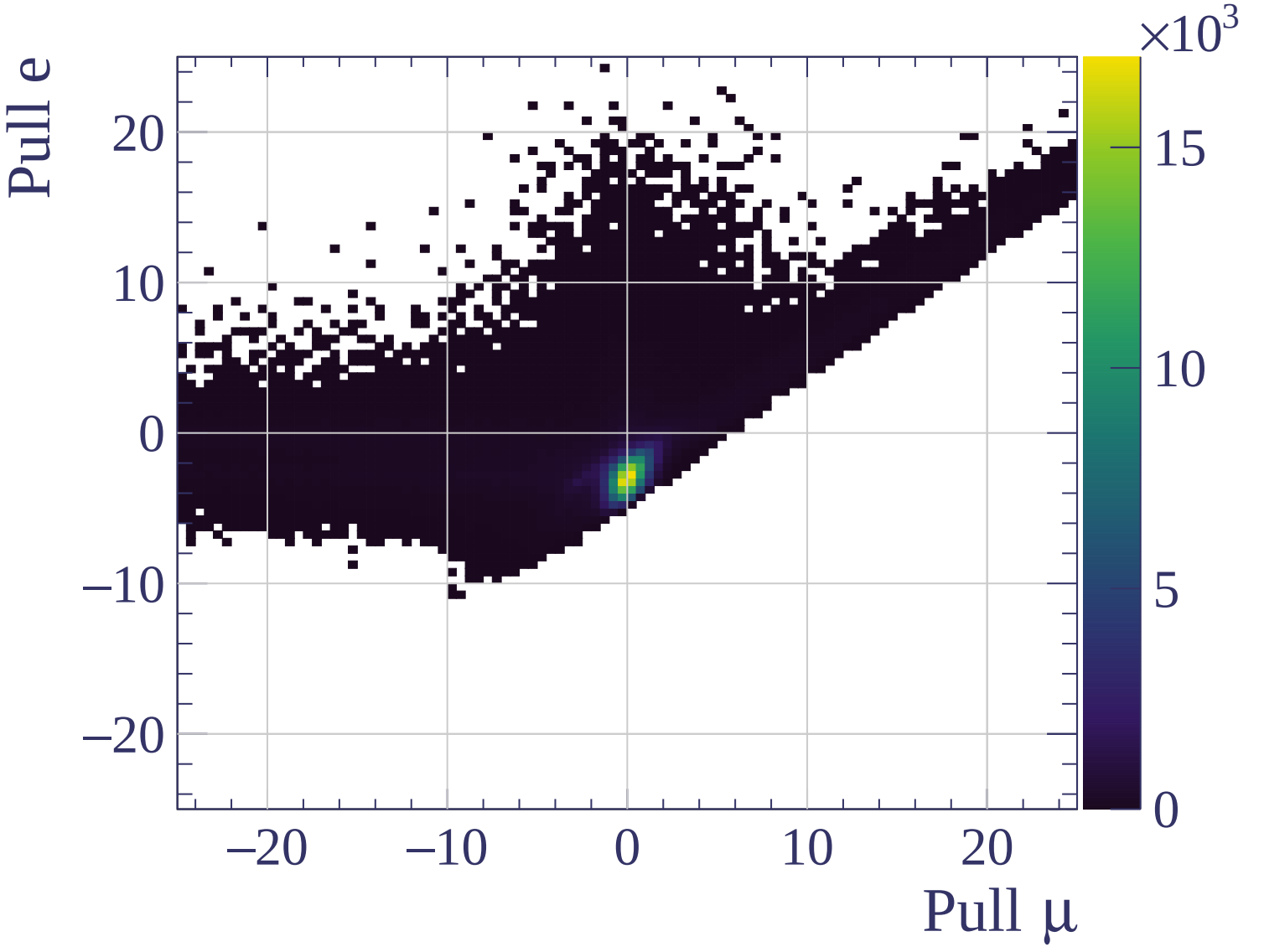


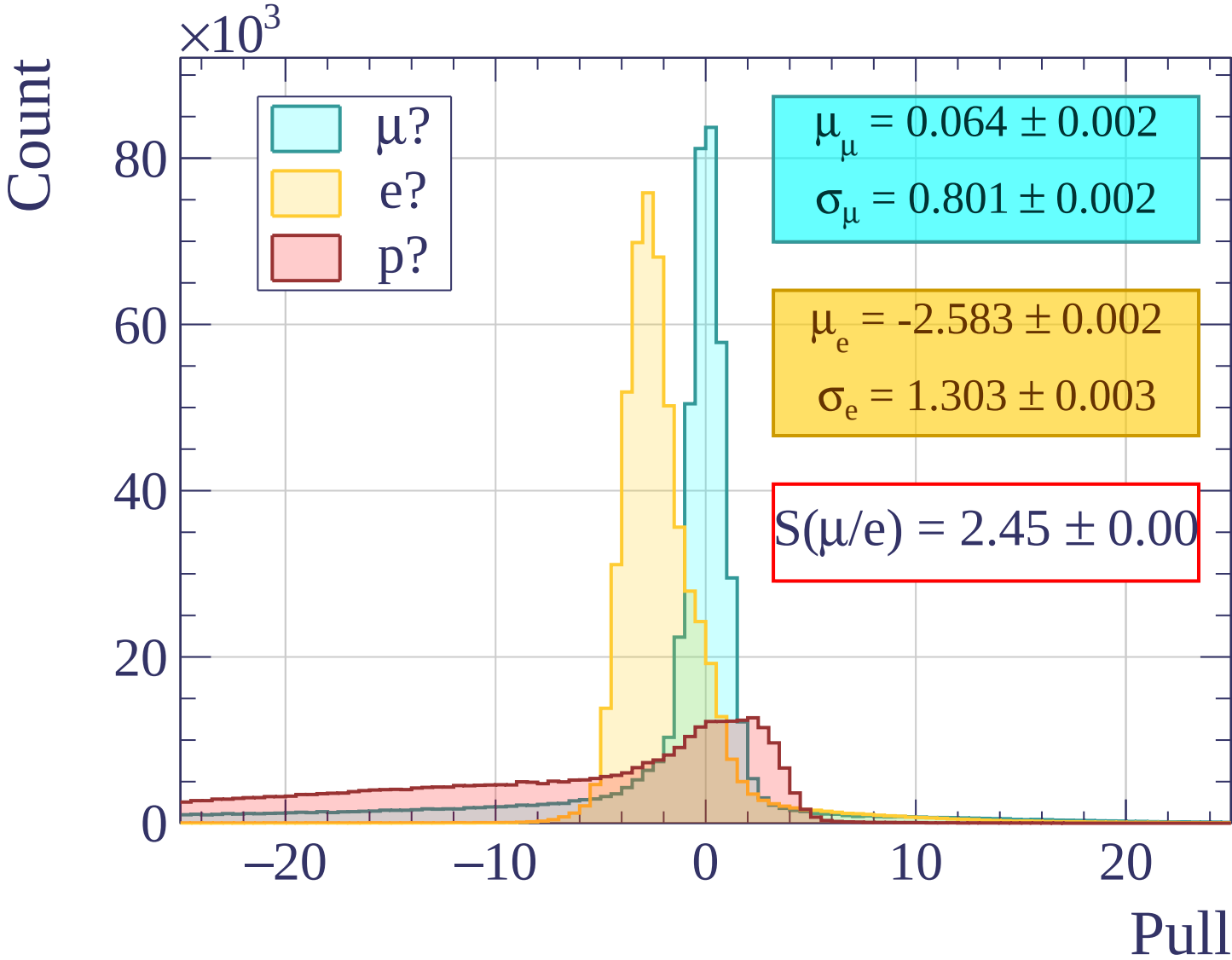
Pulls standard deviation

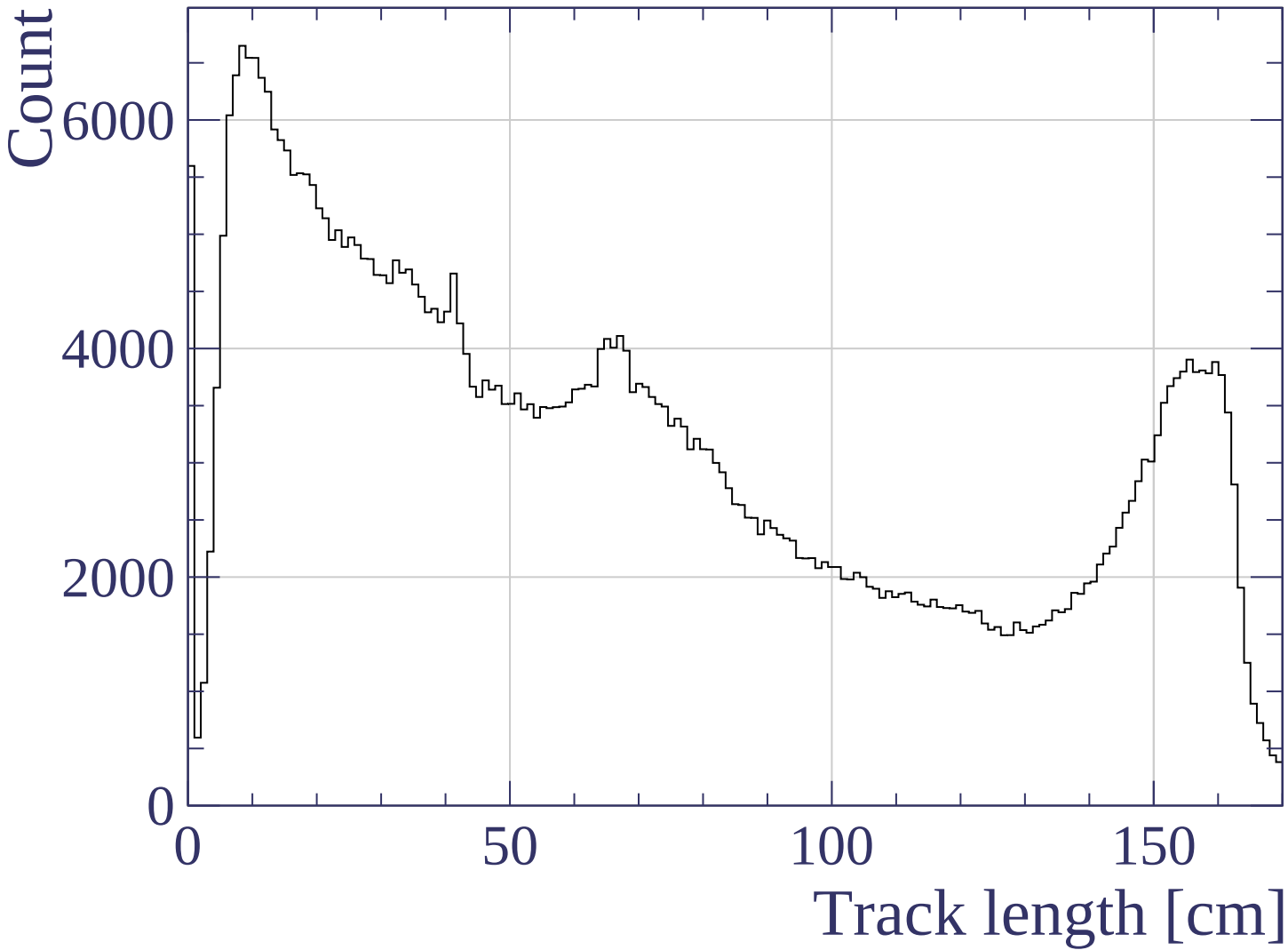


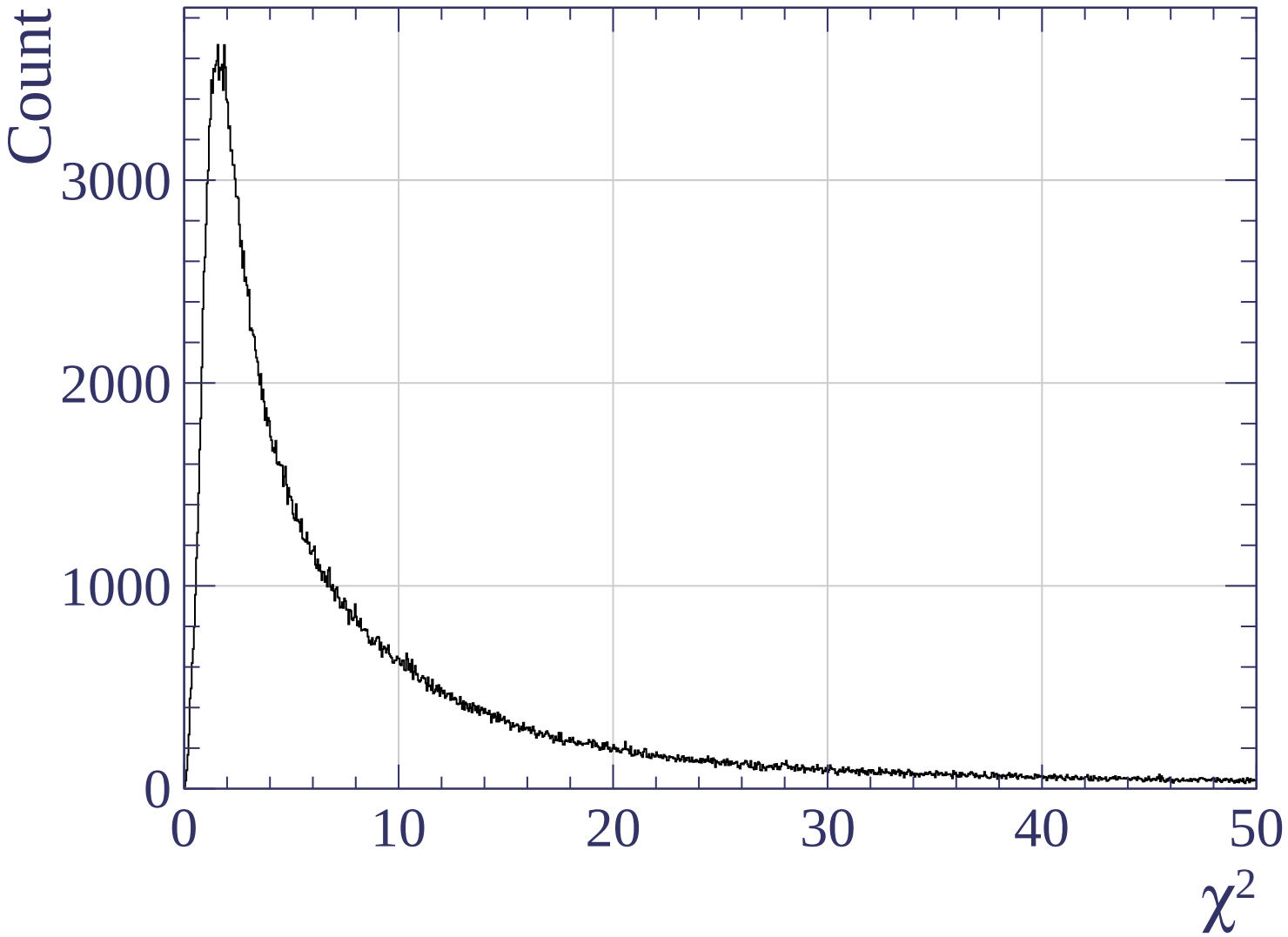
Pulls mean

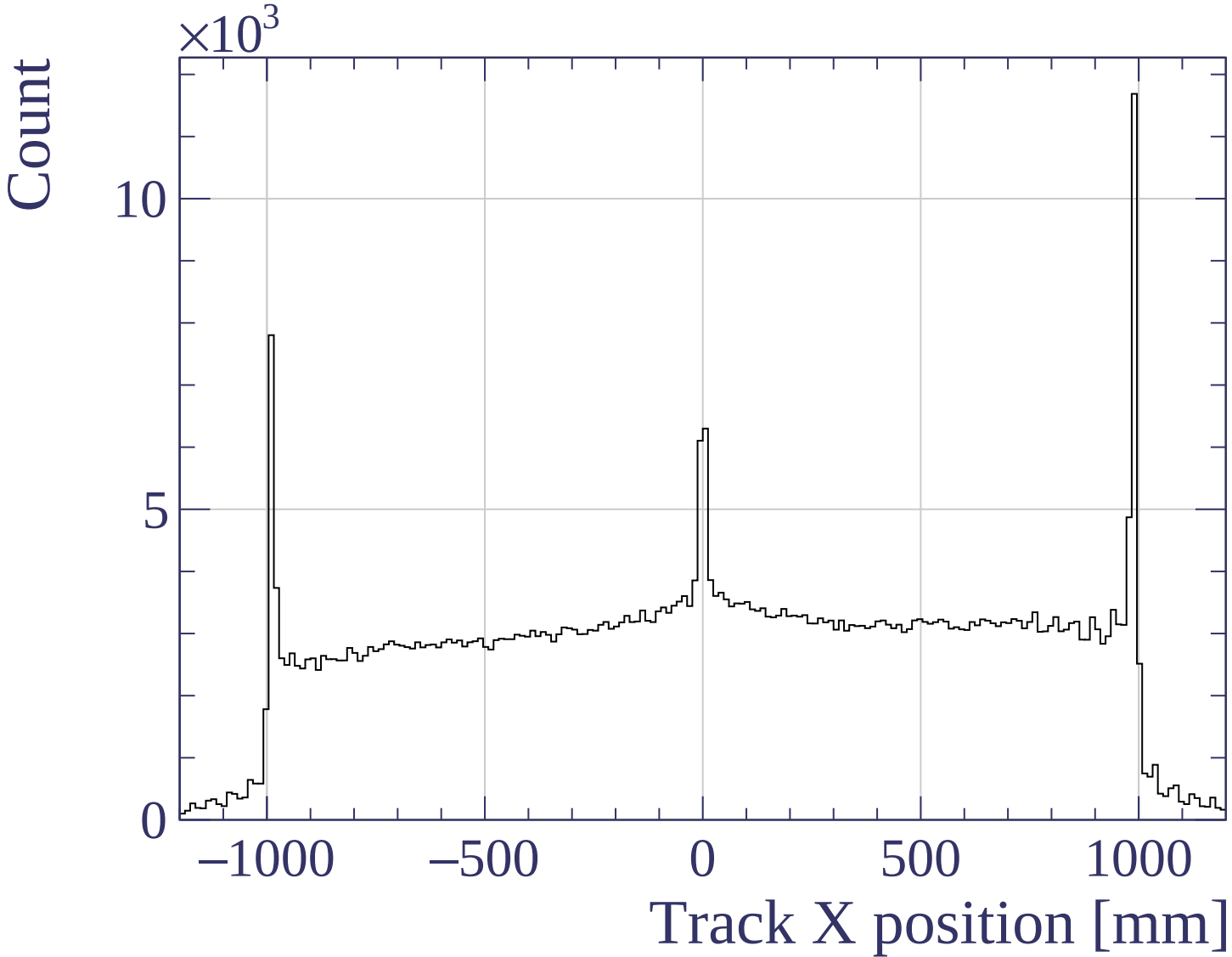


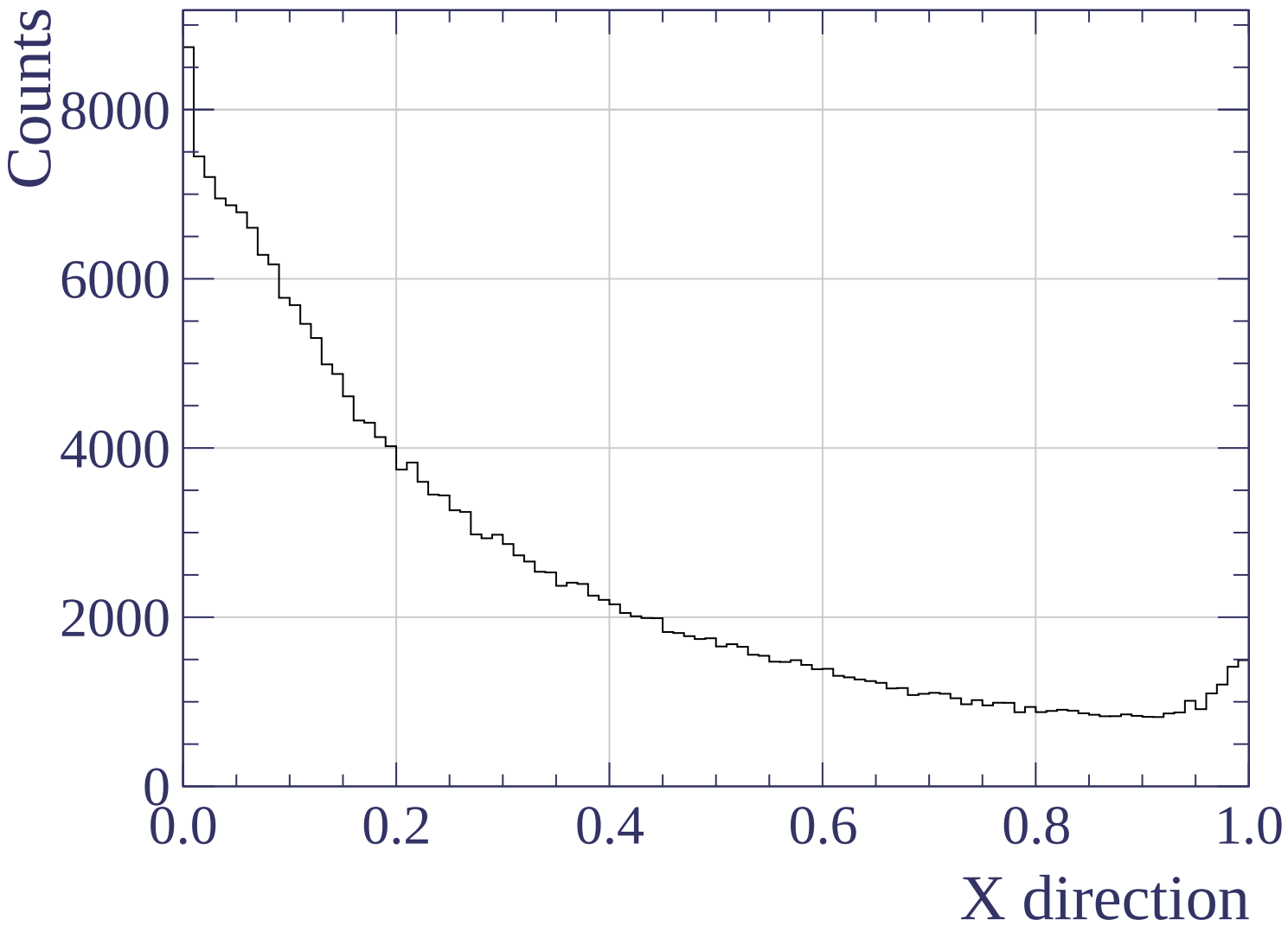


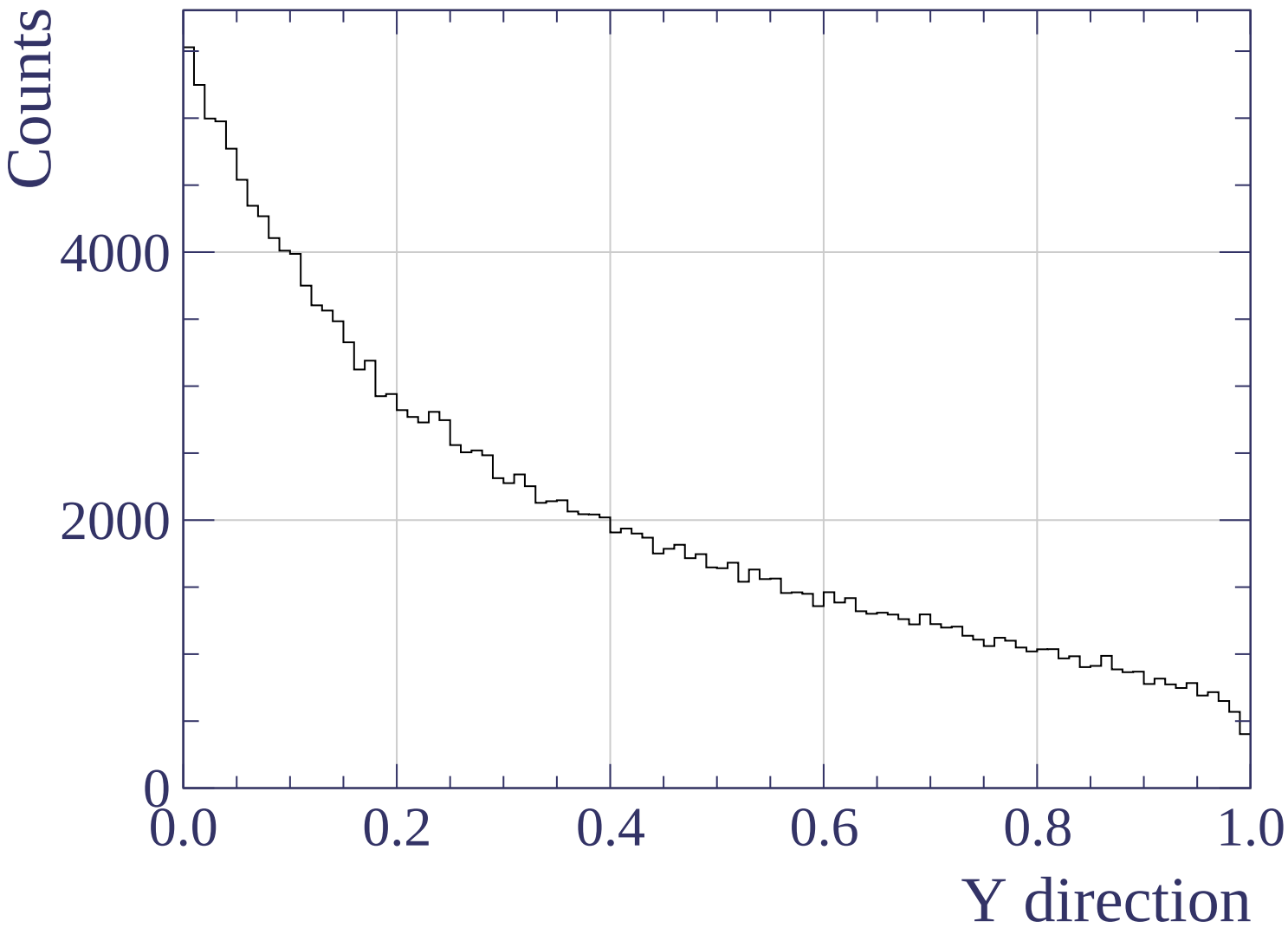


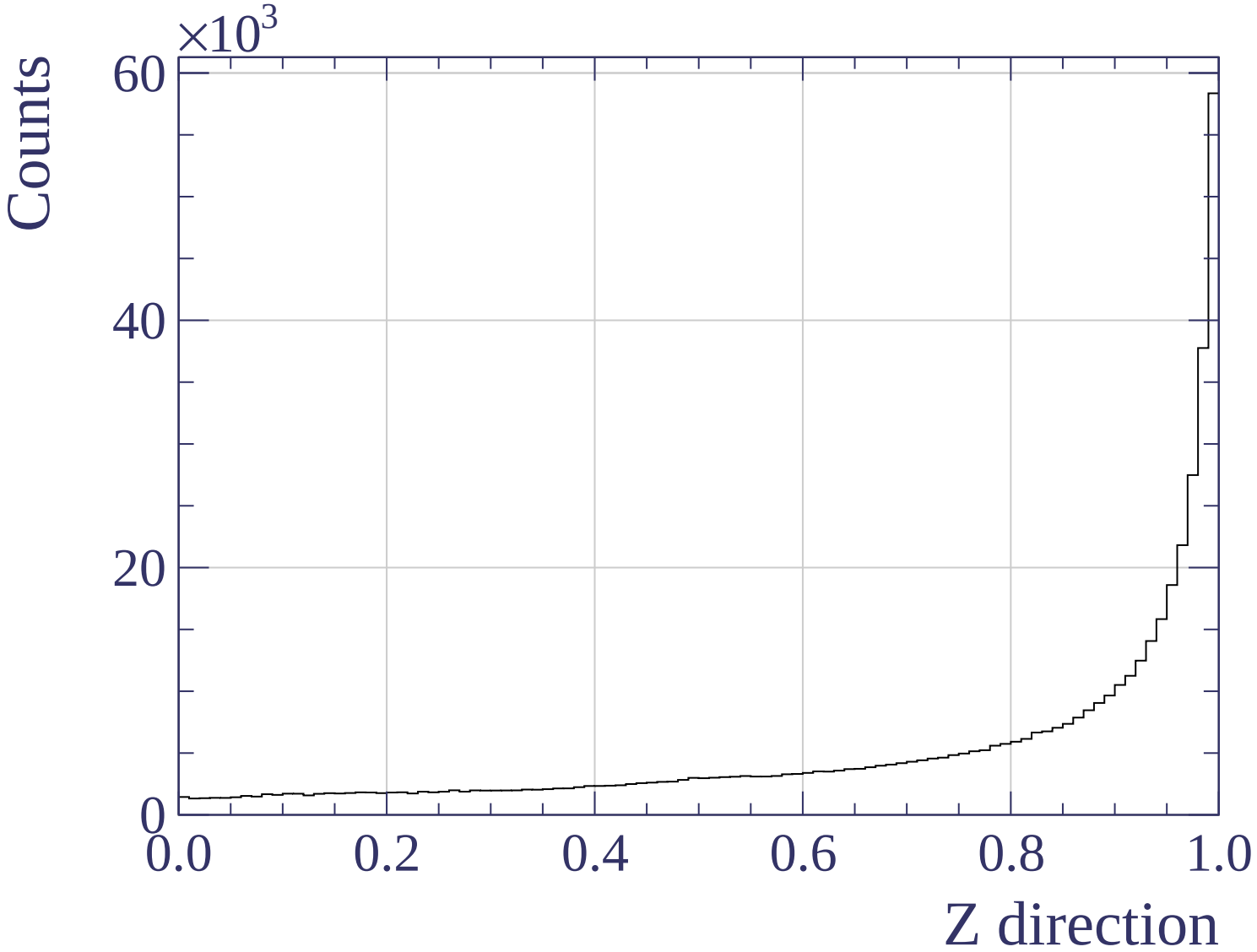


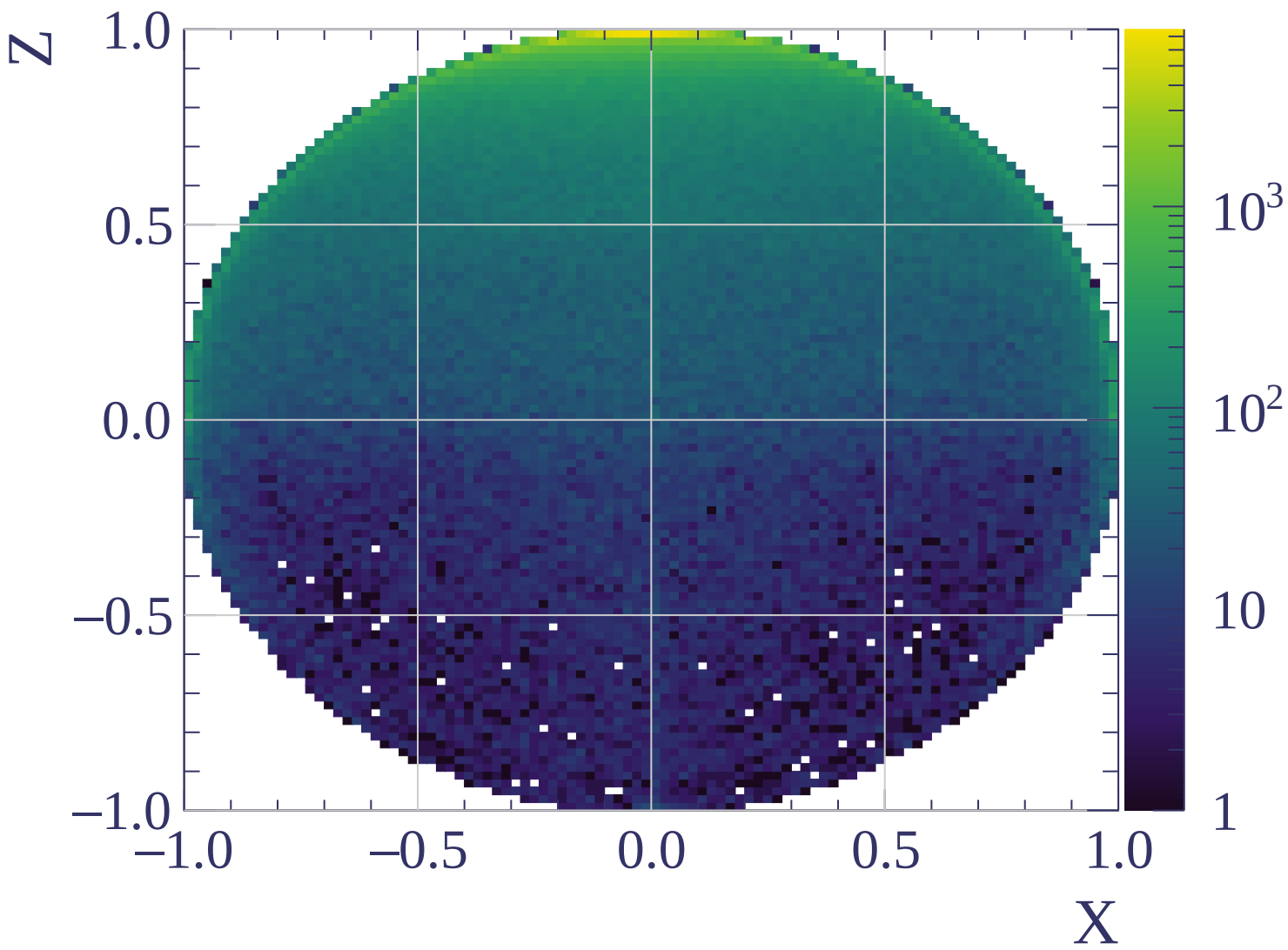


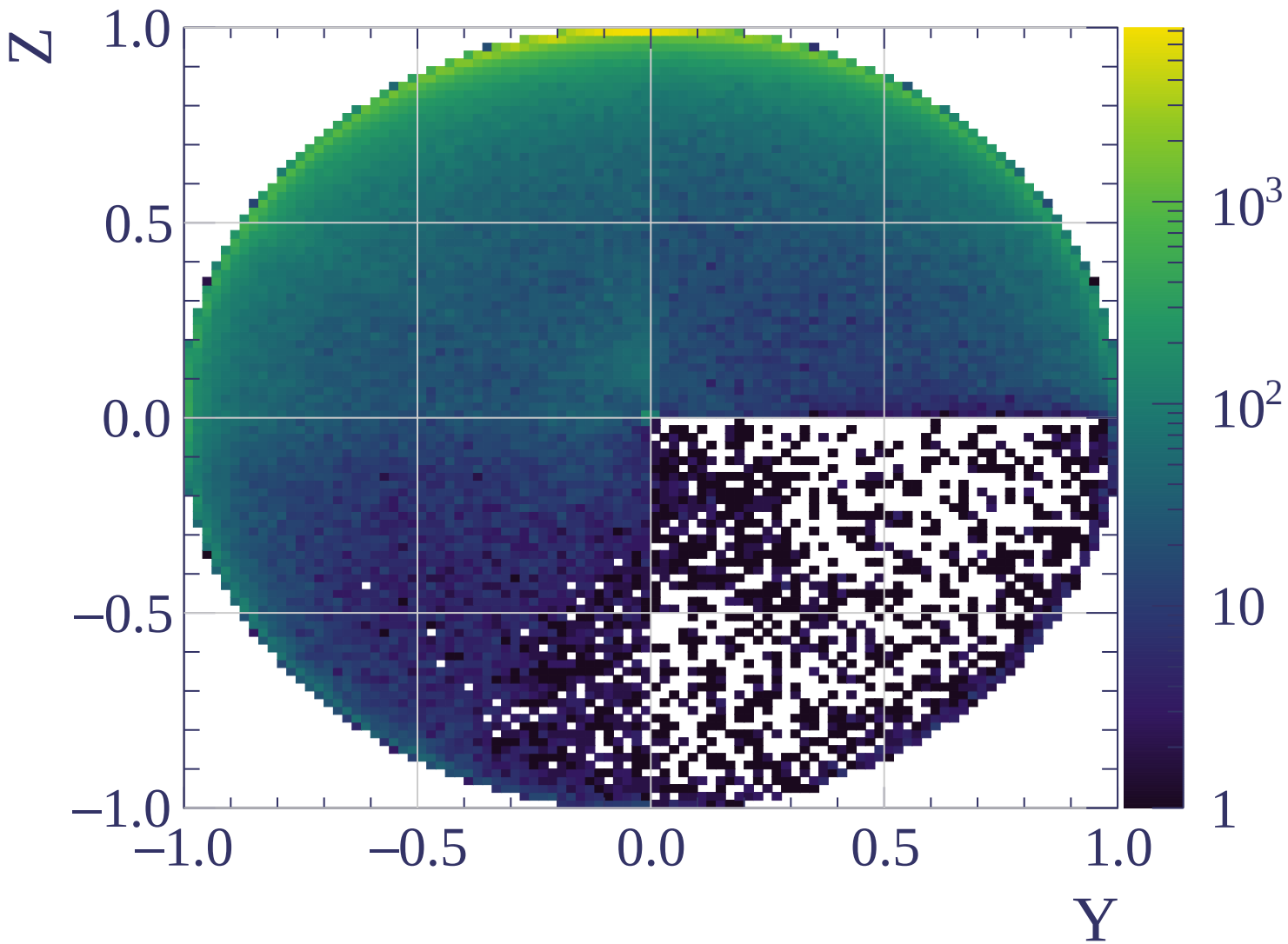


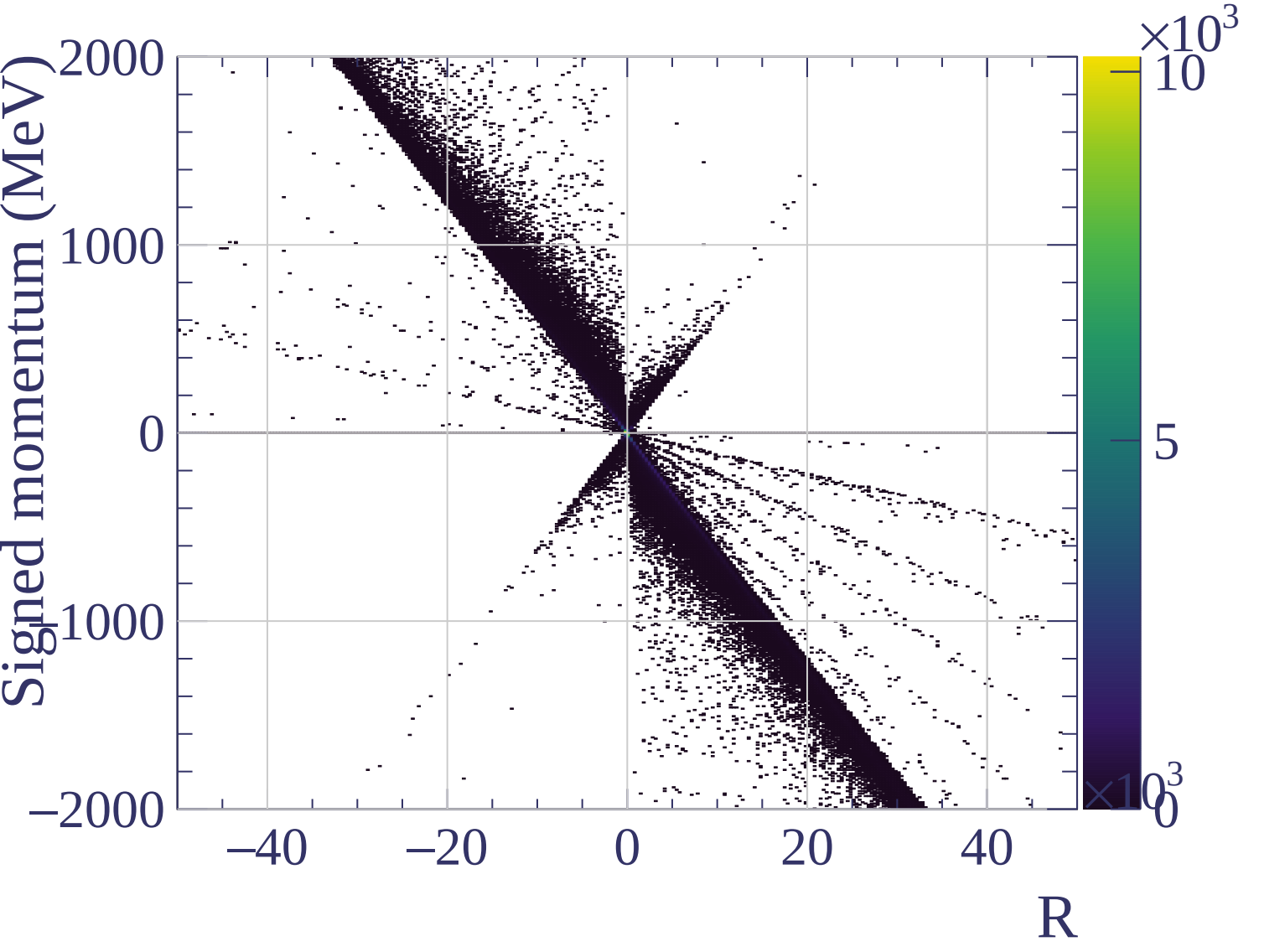


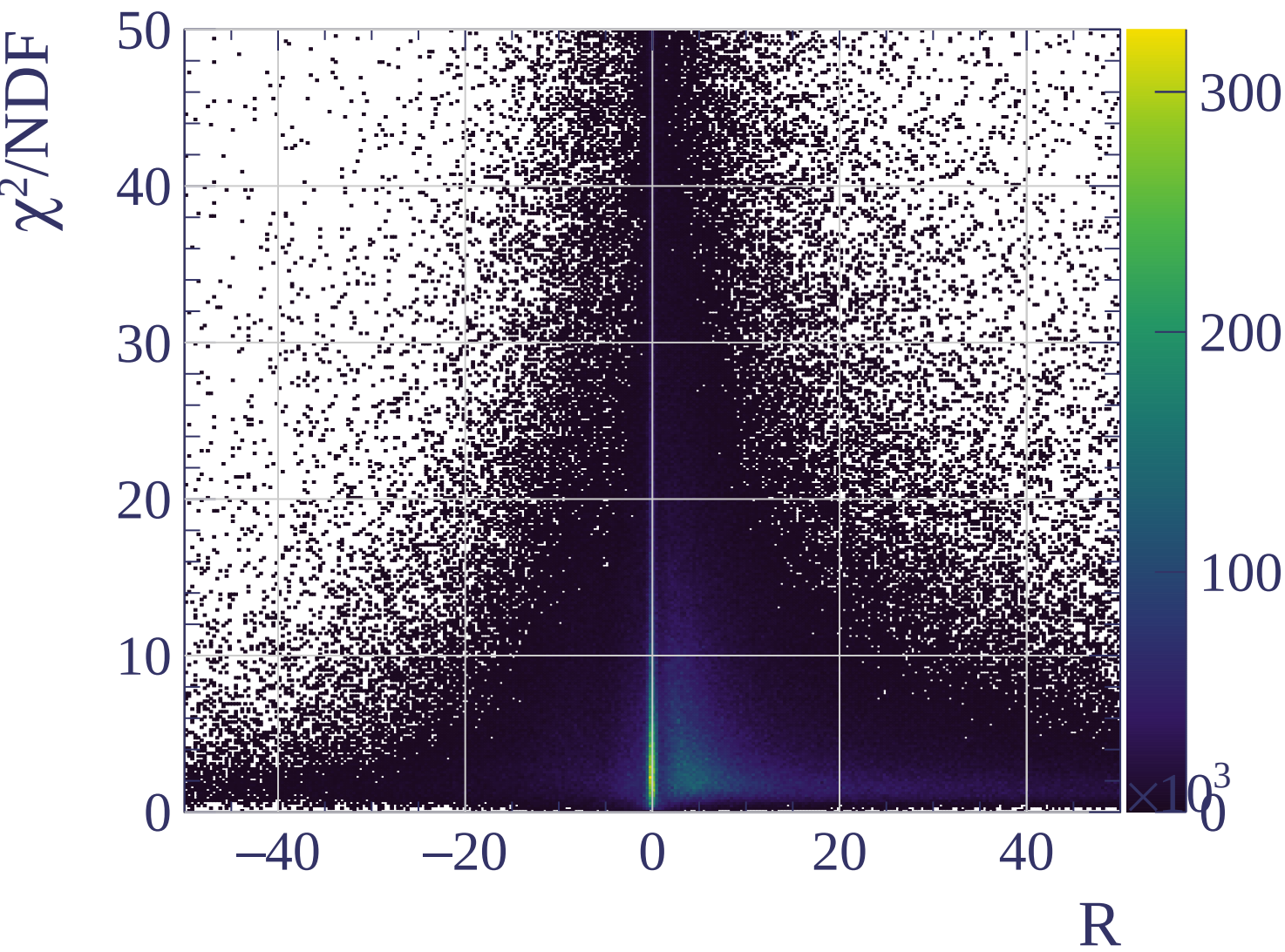


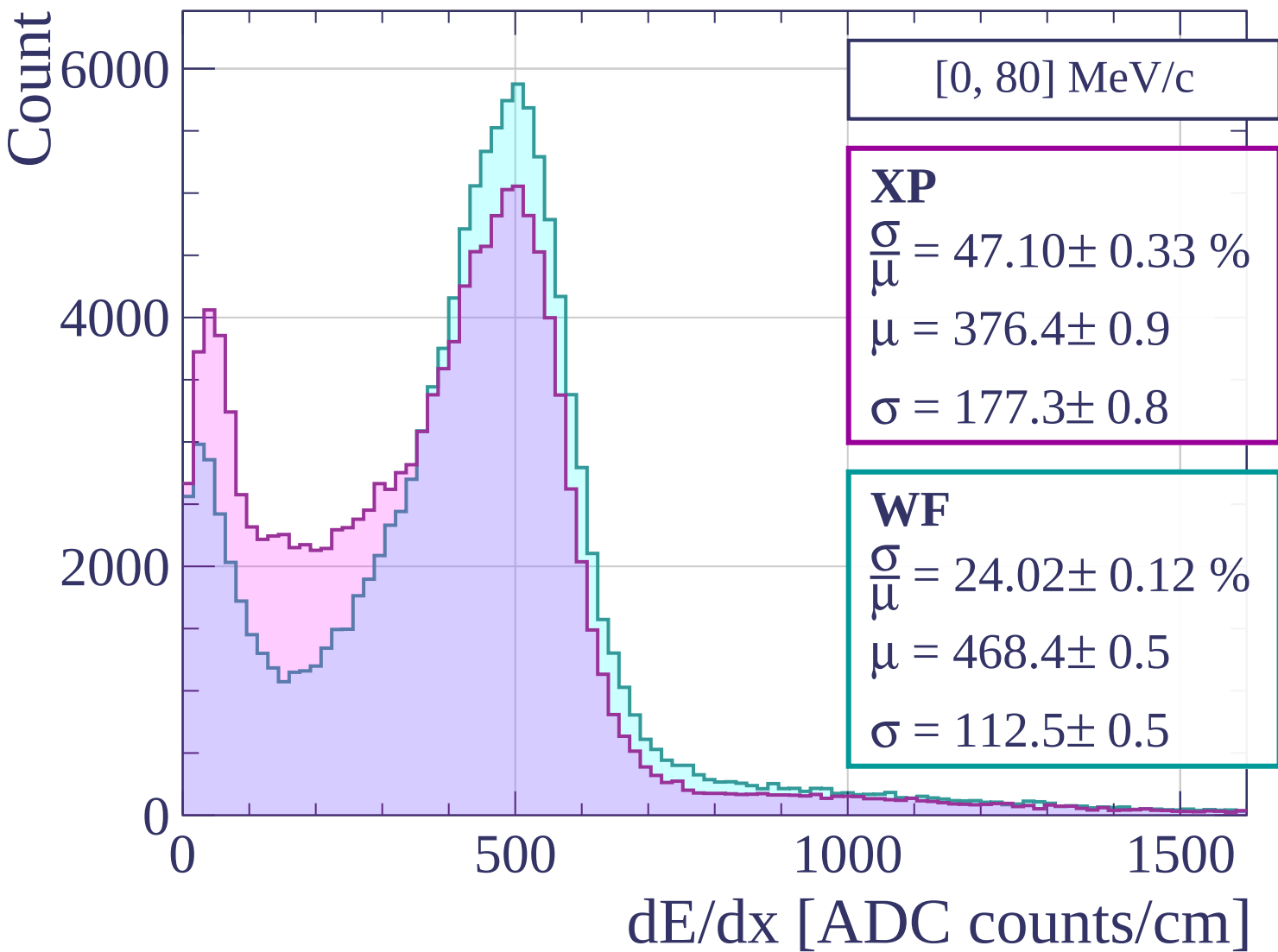


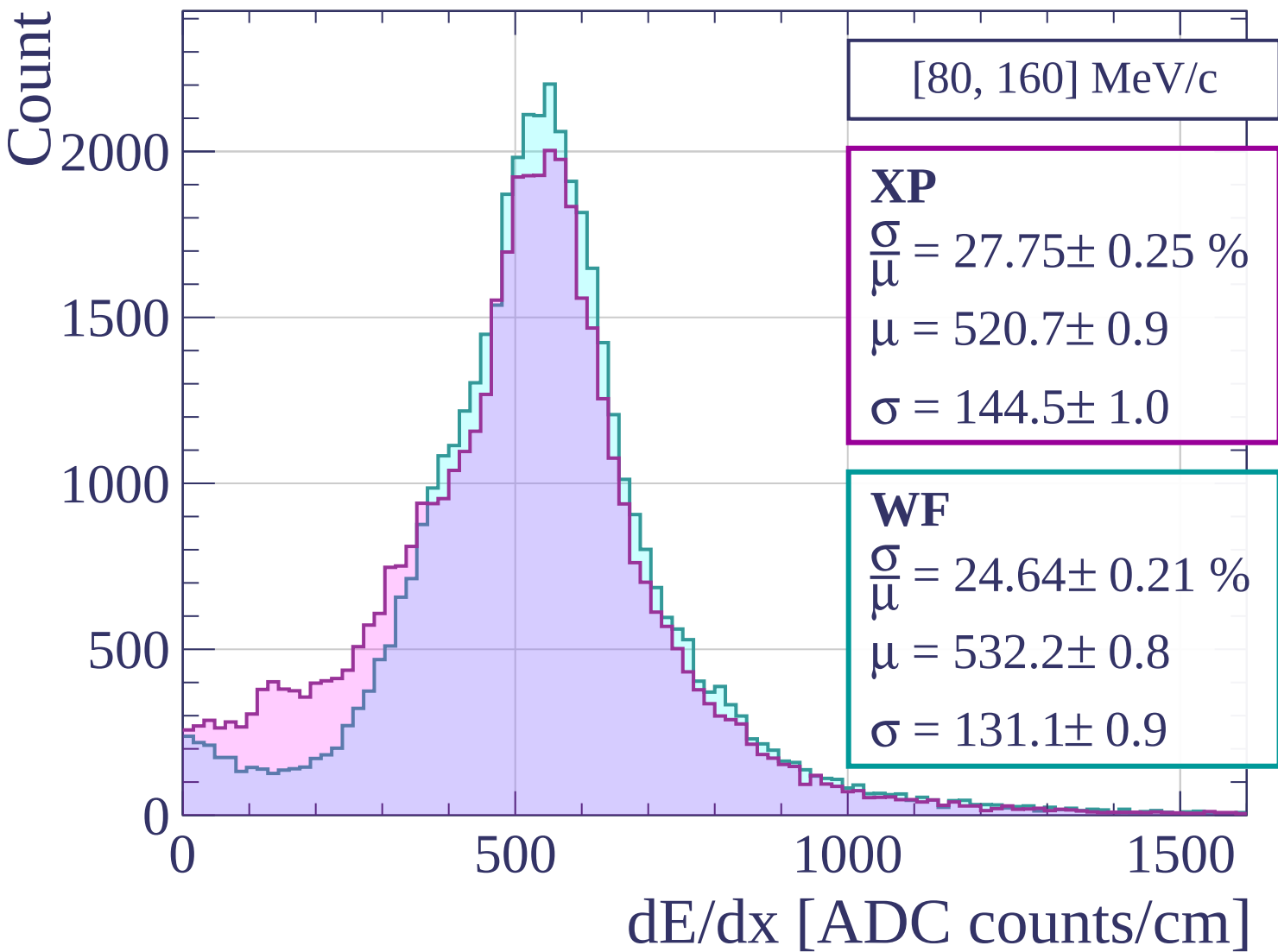


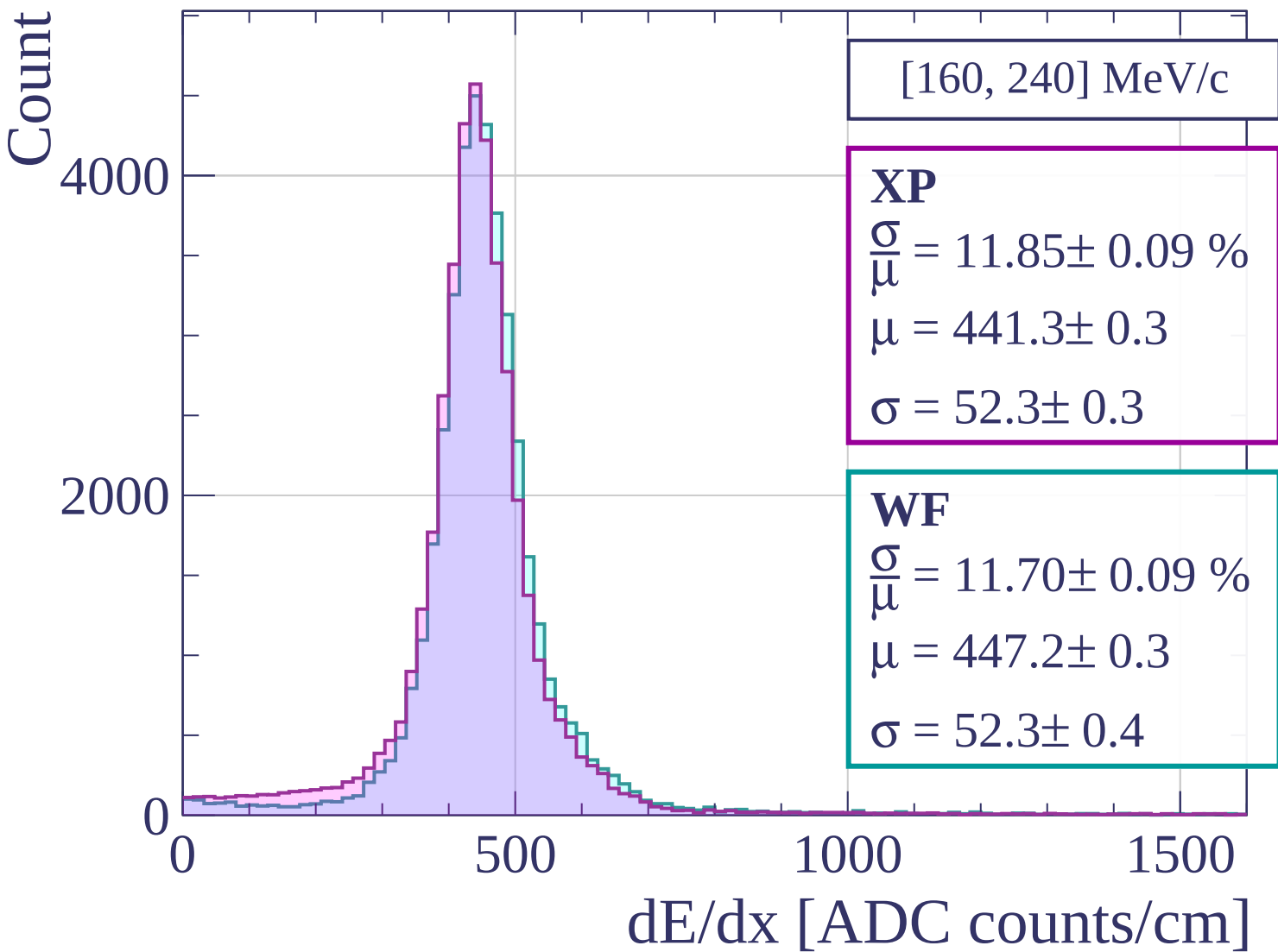




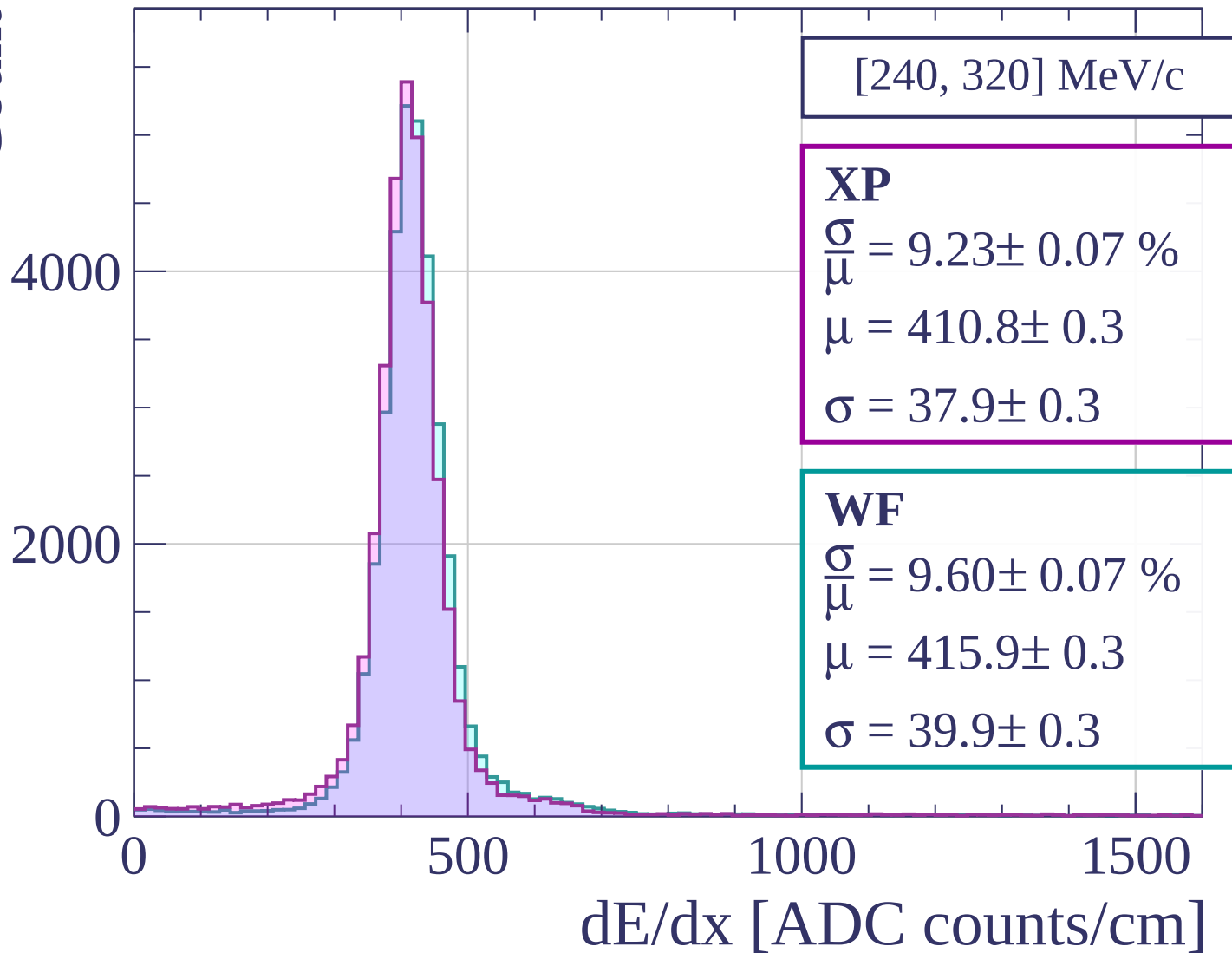


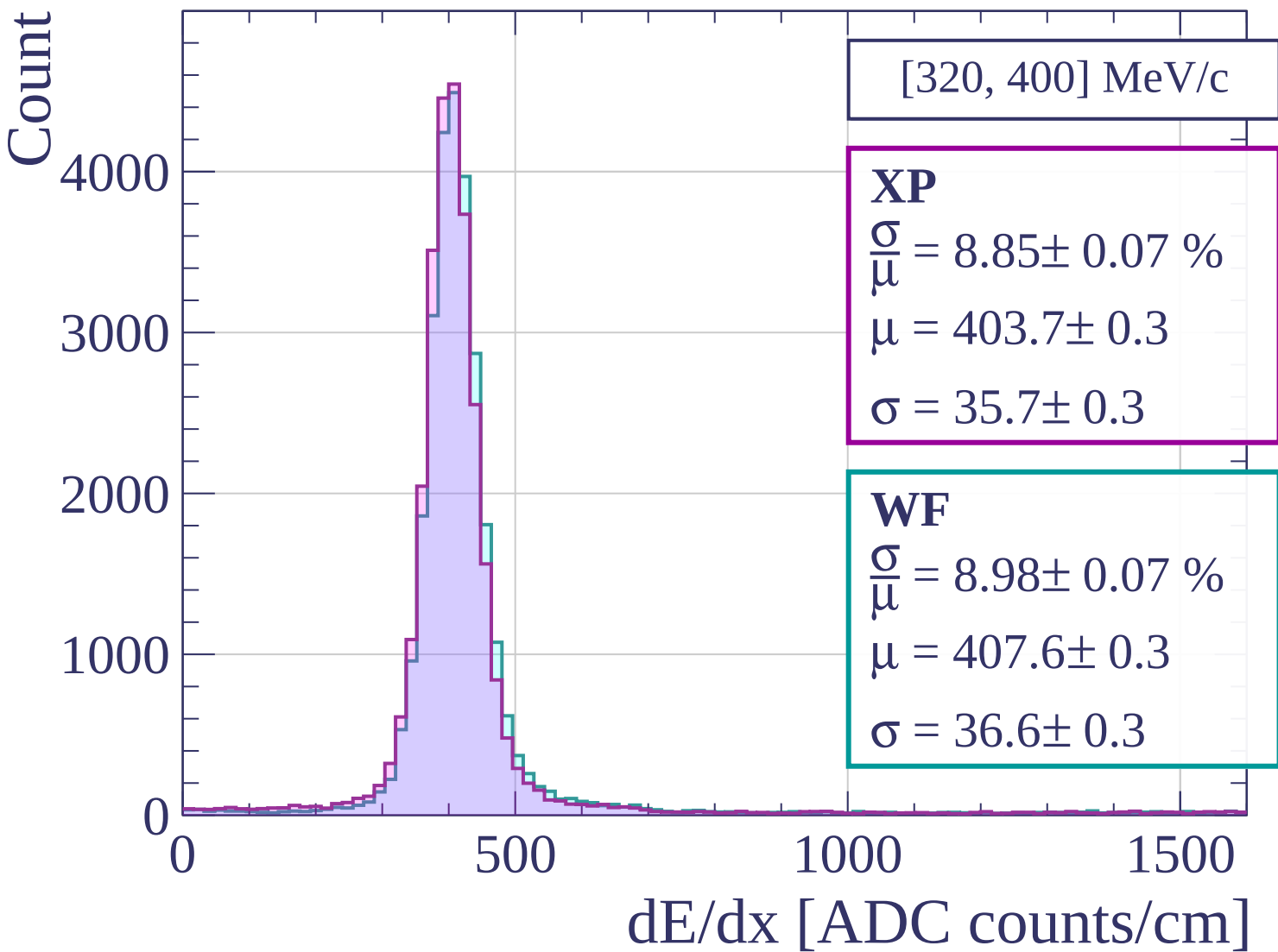


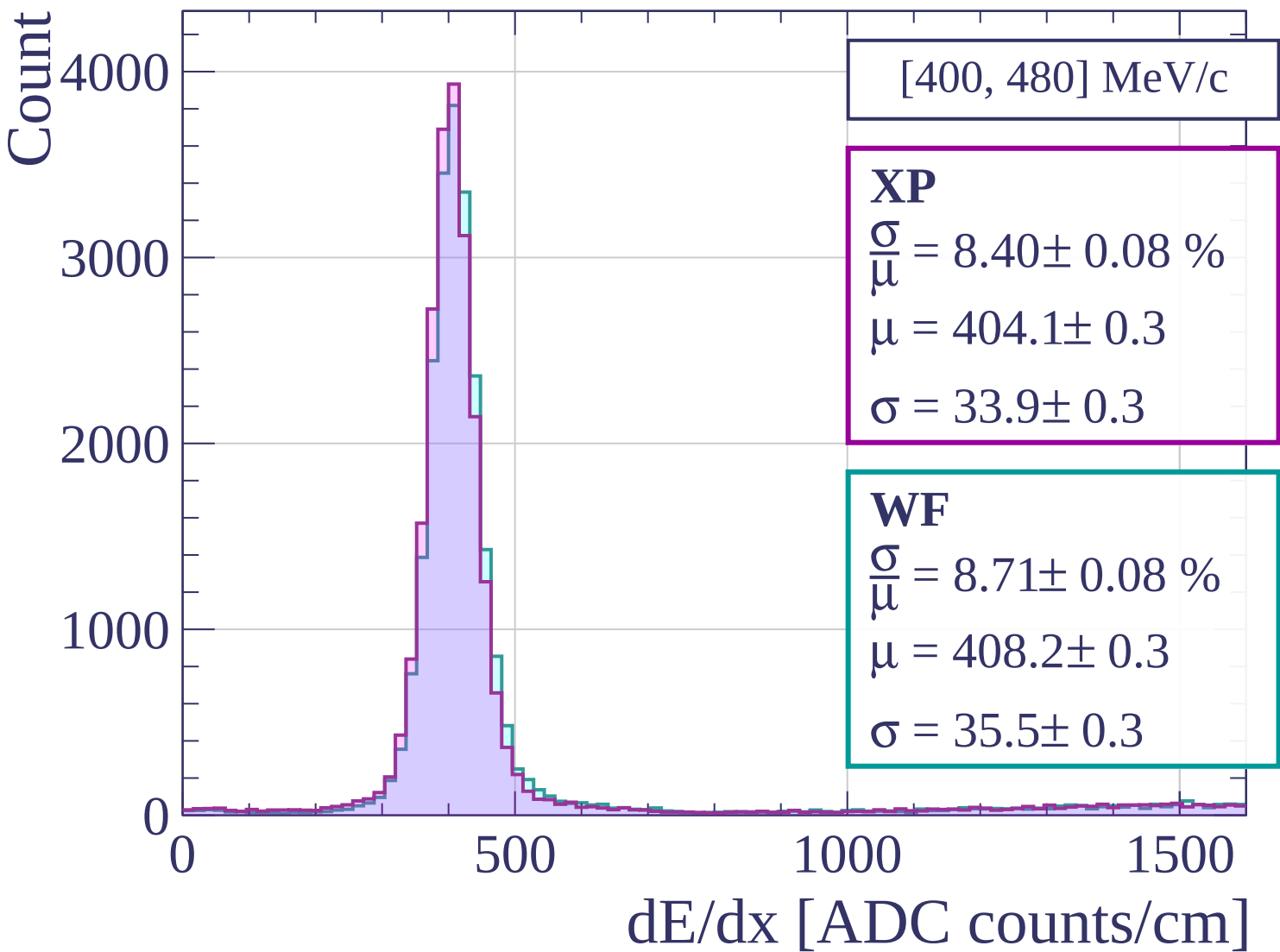


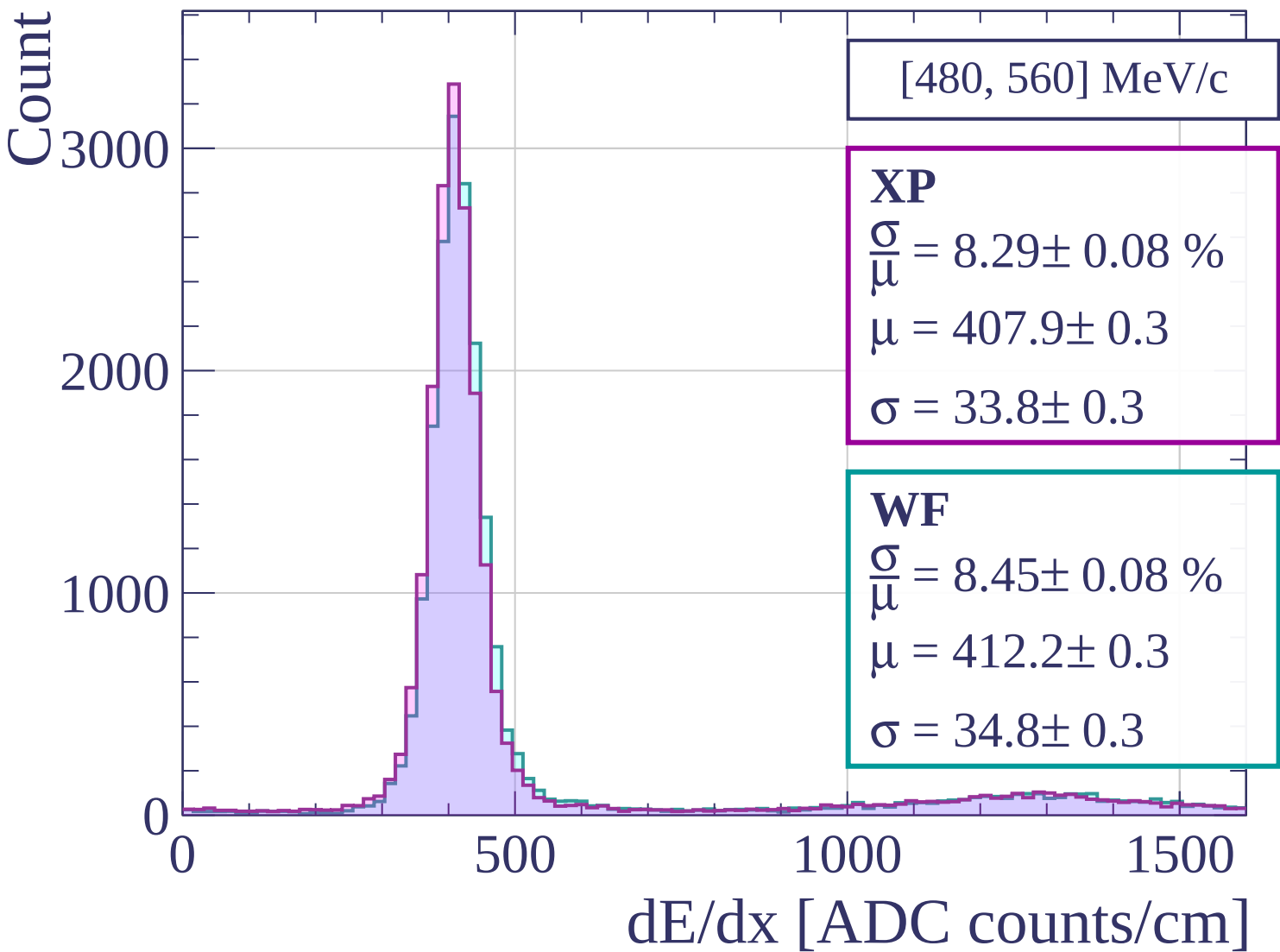


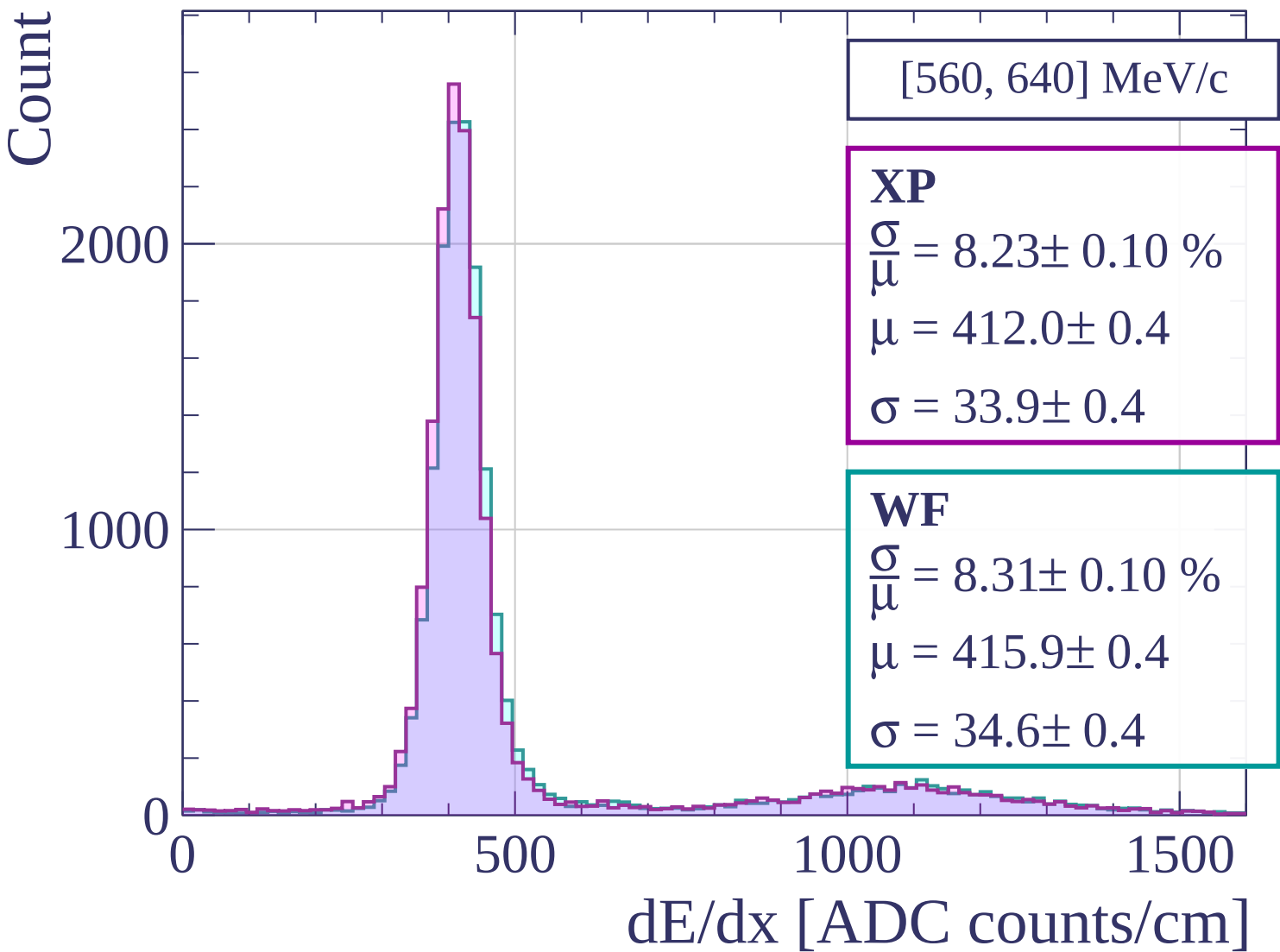
Count

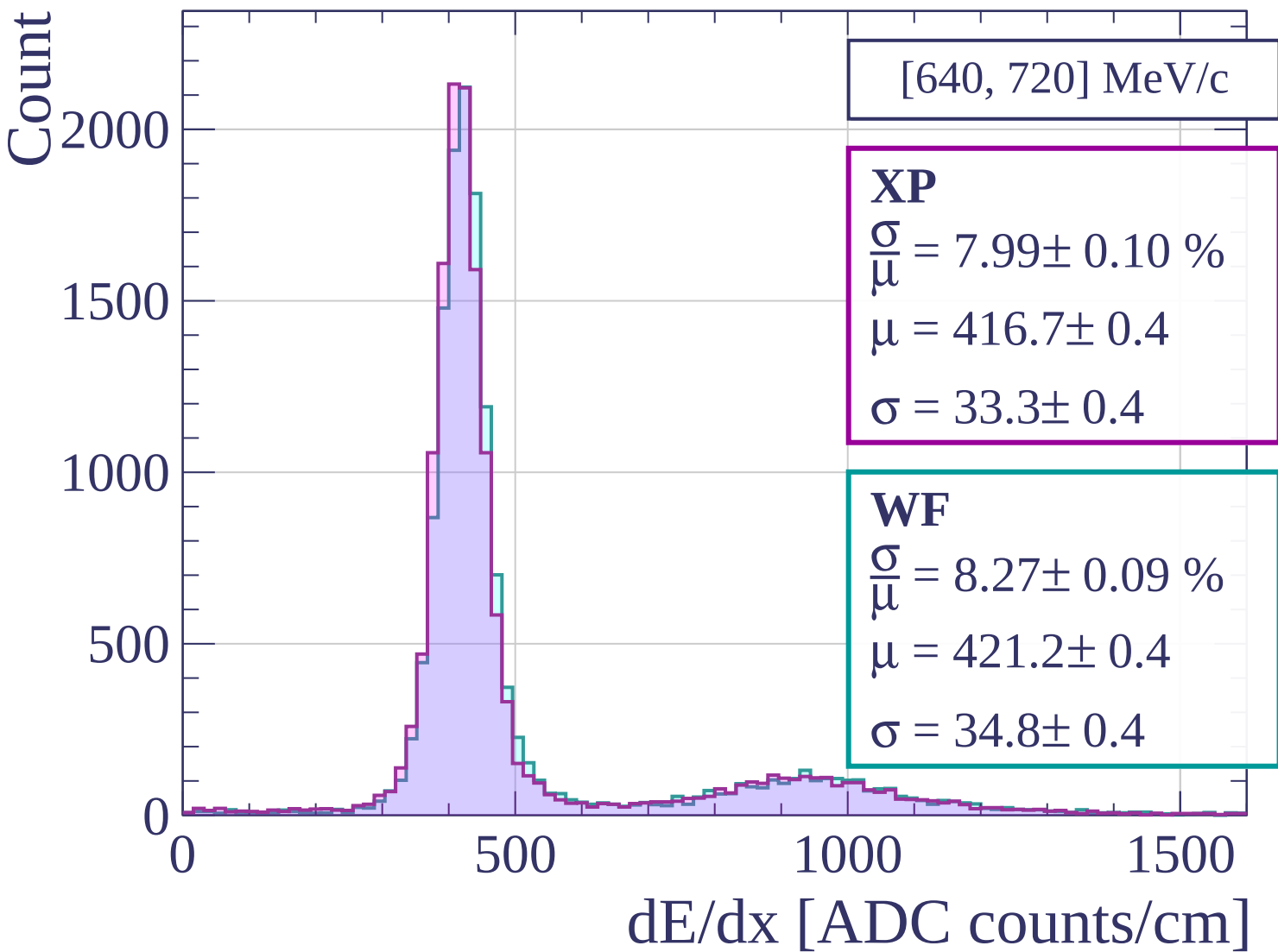


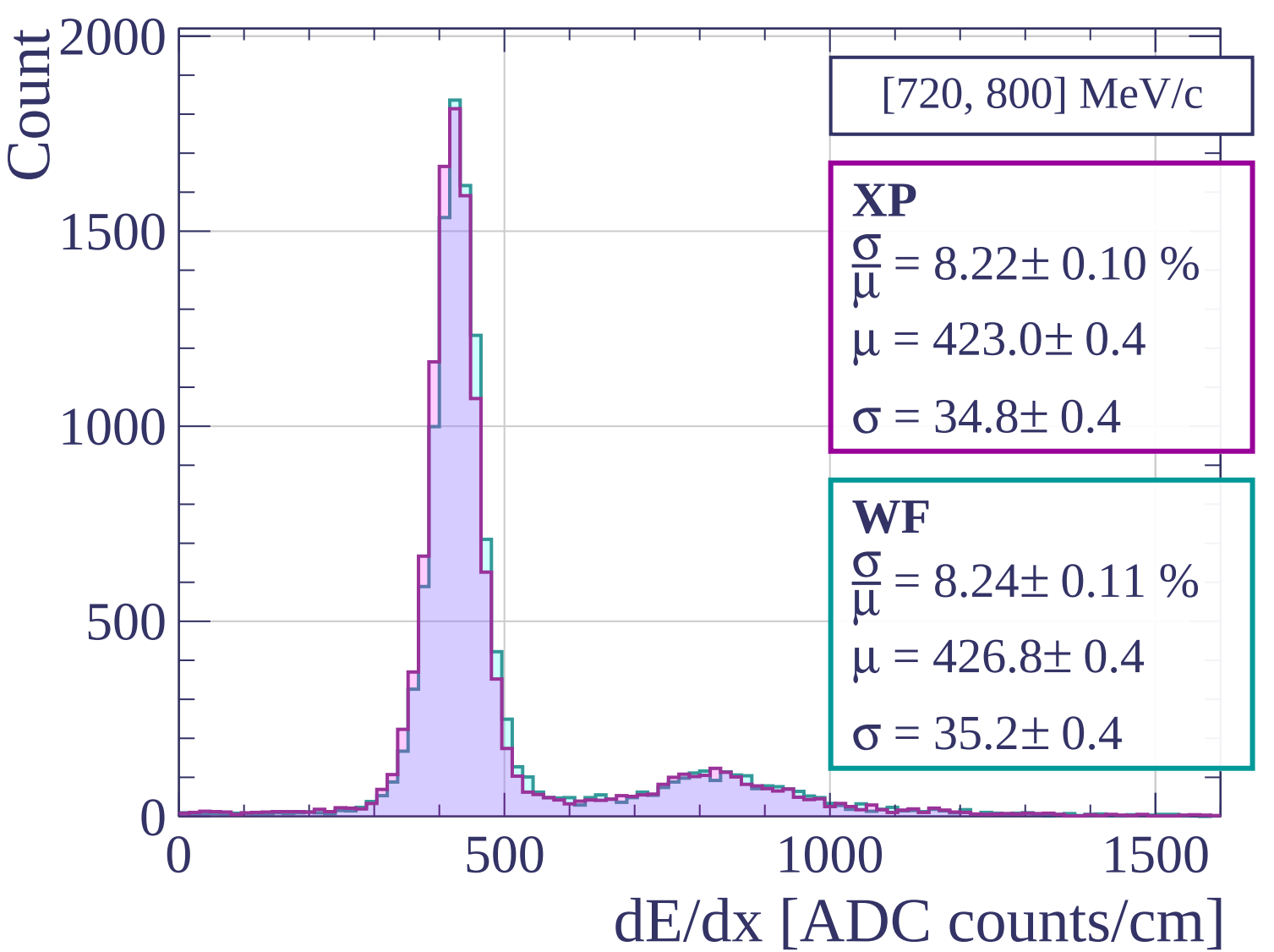


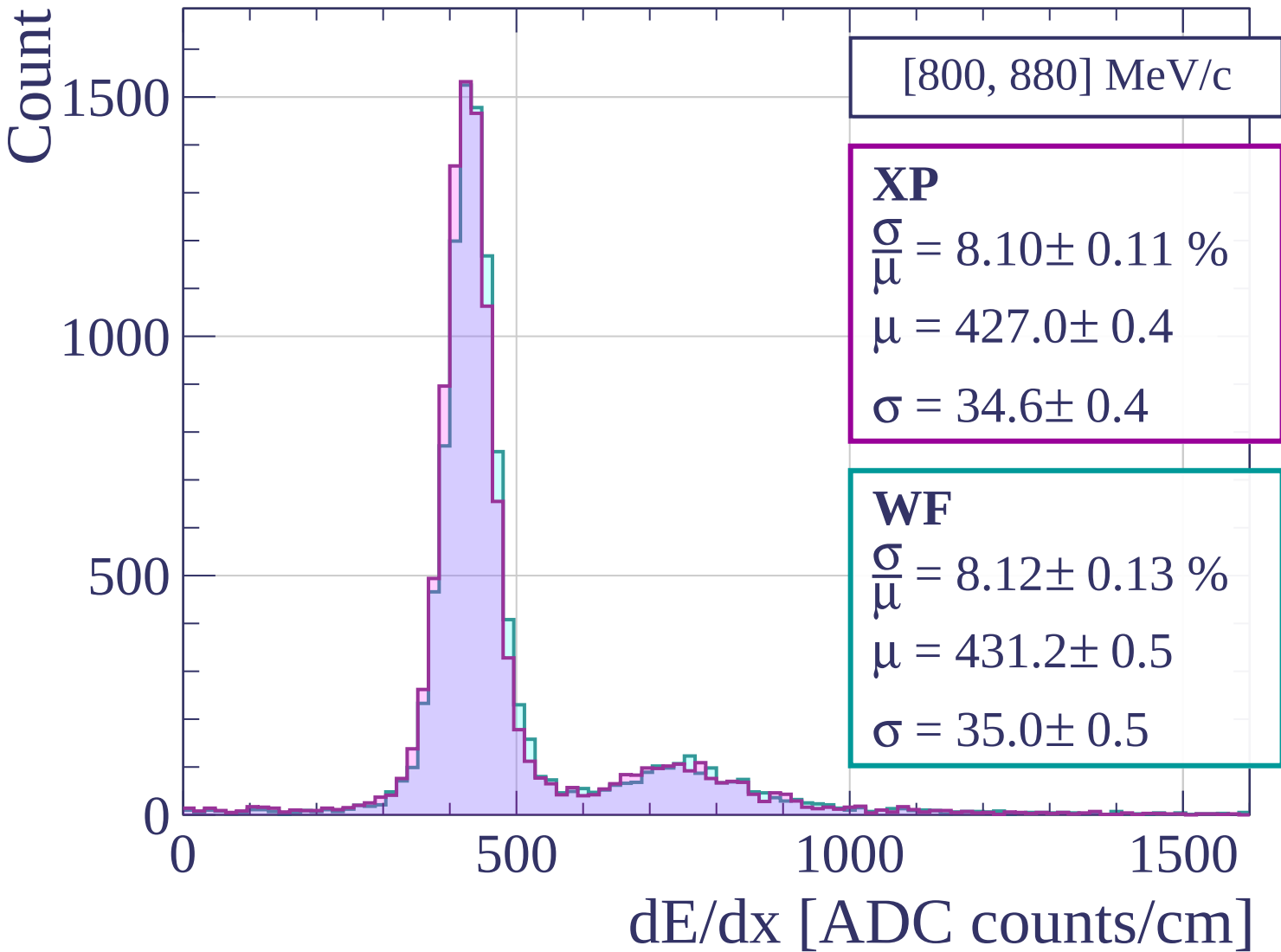


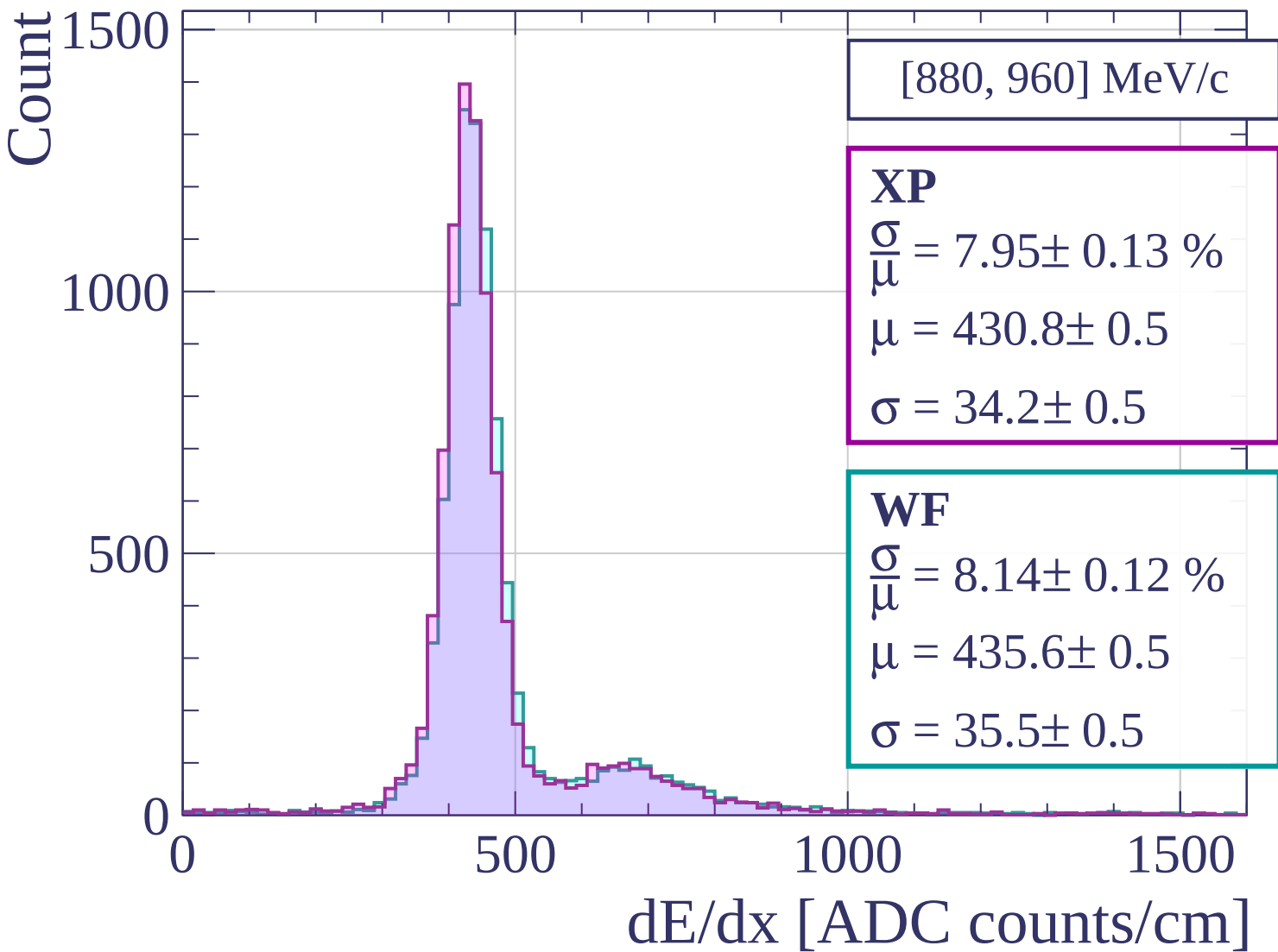


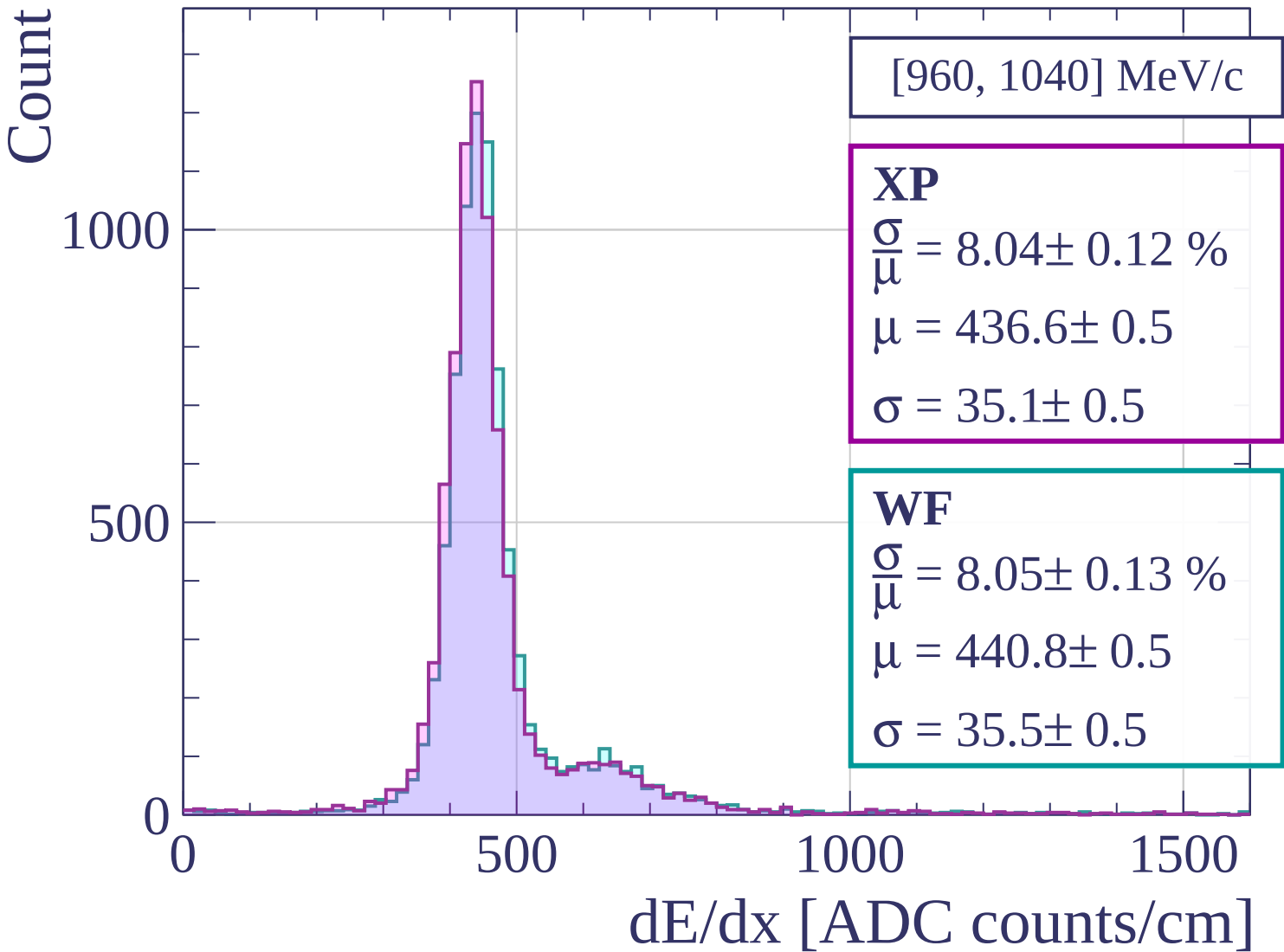


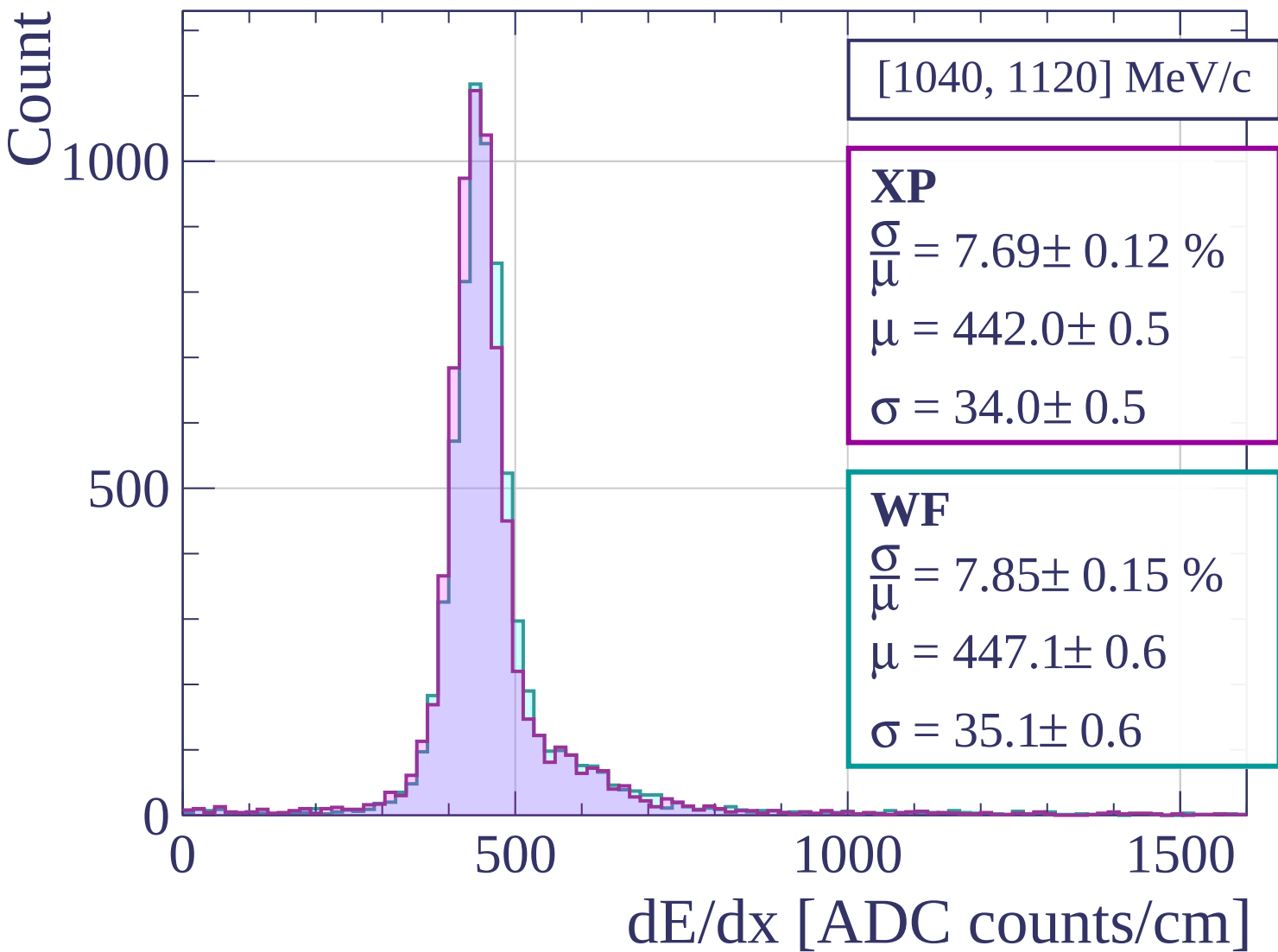


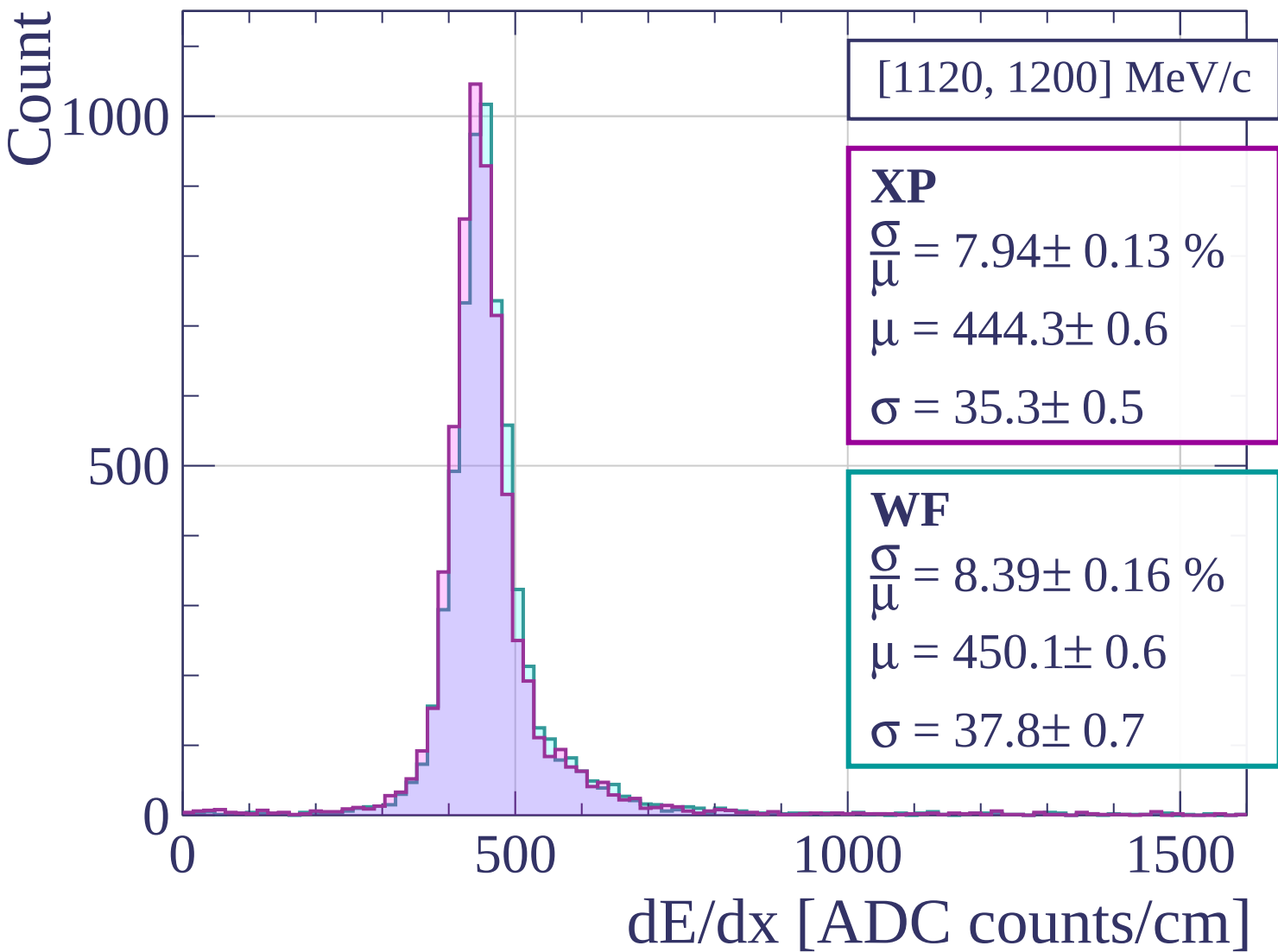












Count

[1200, 1280] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.32 \pm 0.15 \%$$

$$\mu = 450.3 \pm 0.6$$

$$\sigma = 37.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.42 \pm 0.16 \%$$

$$\mu = 455.0 \pm 0.6$$

$$\sigma = 38.3 \pm 0.7$$

800

600

400

200

0

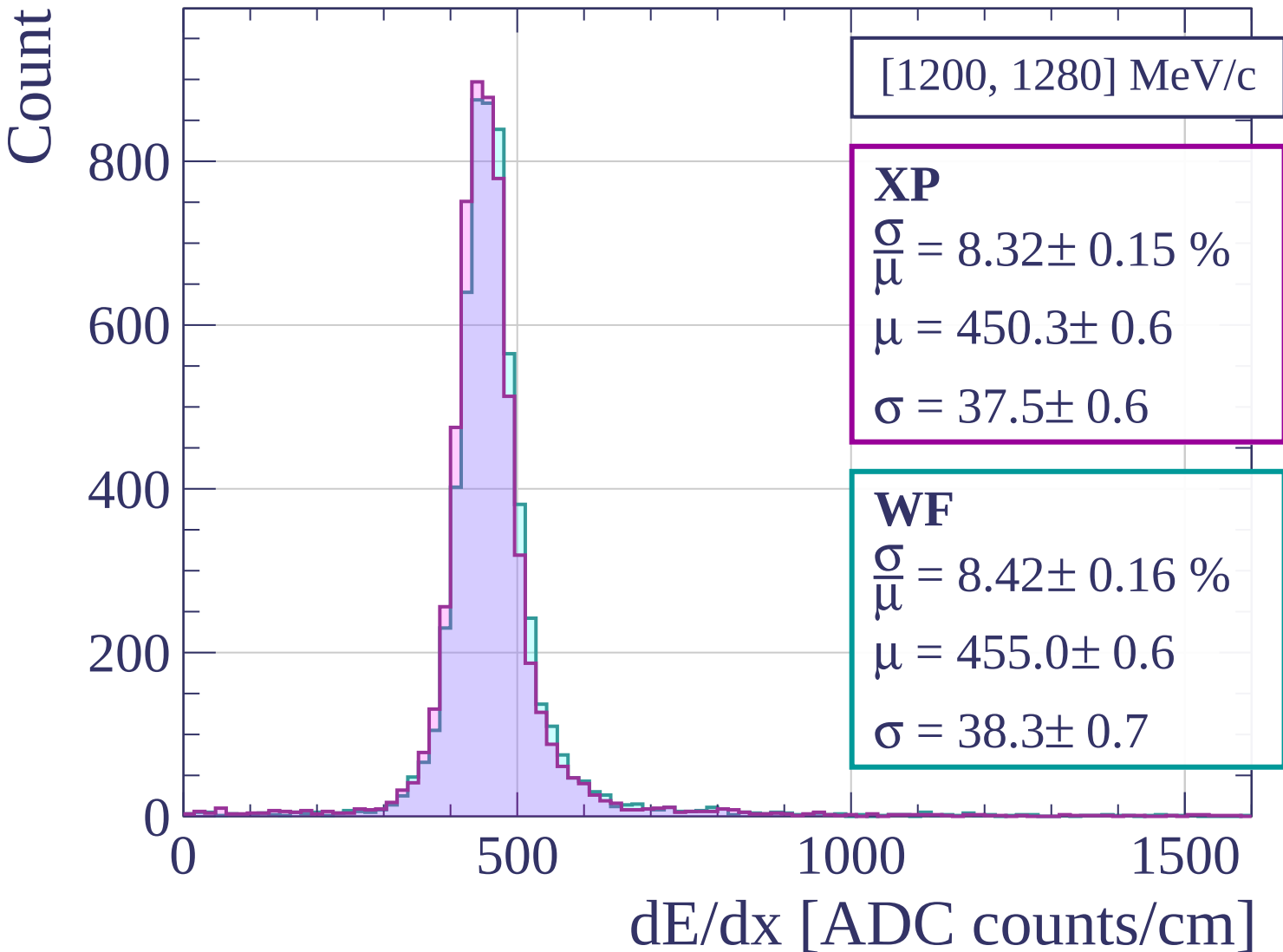
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1280, 1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.15 \%$$

$$\mu = 453.1 \pm 0.6$$

$$\sigma = 33.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.06 \pm 0.15 \%$$

$$\mu = 458.4 \pm 0.6$$

$$\sigma = 36.9 \pm 0.7$$

800

600

400

200

0

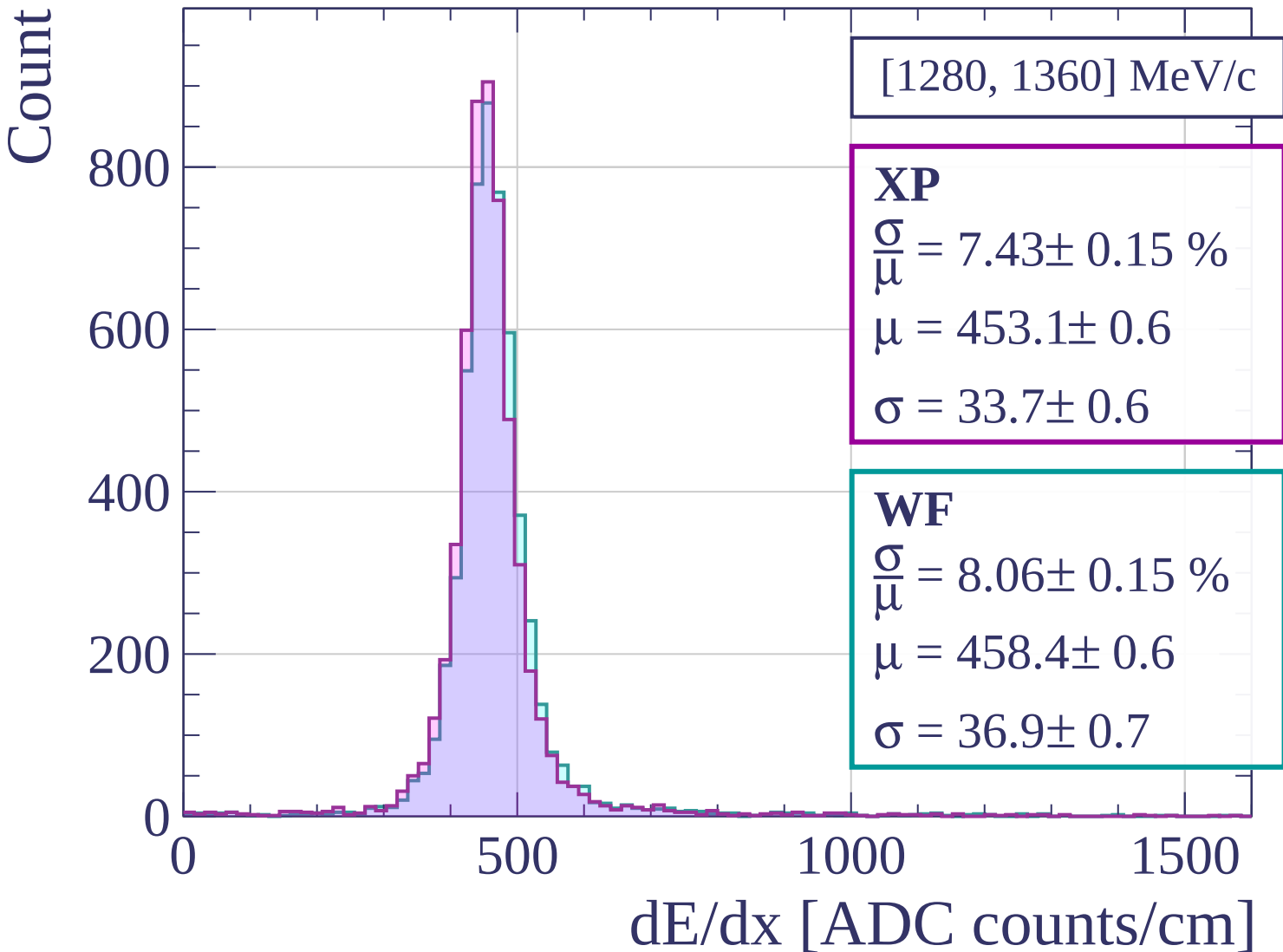
0

500

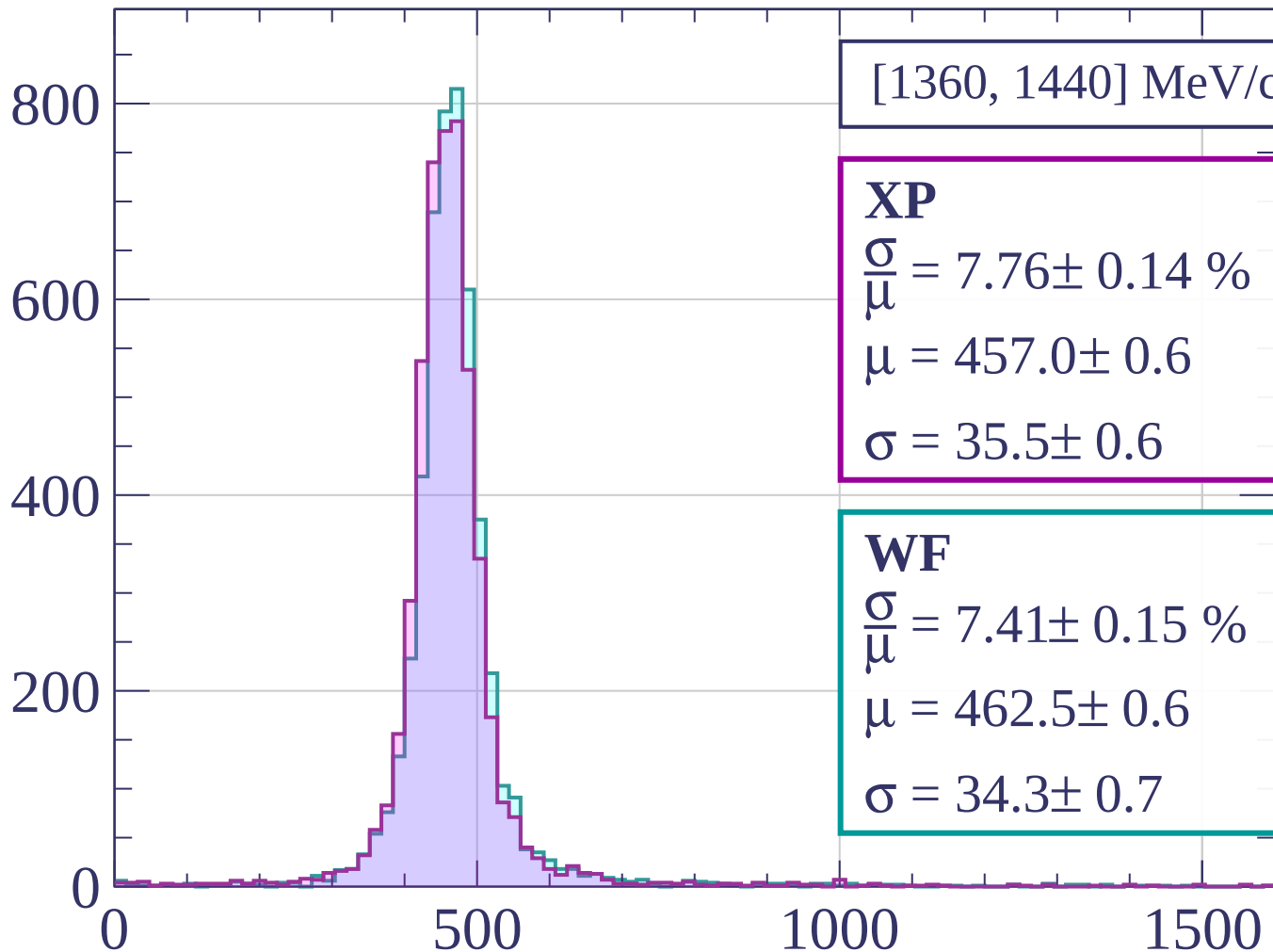
1000

1500

dE/dx [ADC counts/cm]



Count



[1360, 1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.76 \pm 0.14 \%$$

$$\mu = 457.0 \pm 0.6$$

$$\sigma = 35.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.41 \pm 0.15 \%$$

$$\mu = 462.5 \pm 0.6$$

$$\sigma = 34.3 \pm 0.7$$

dE/dx [ADC counts/cm]

Count

[1440, 1520] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.16 \%$$

$$\mu = 459.7 \pm 0.6$$

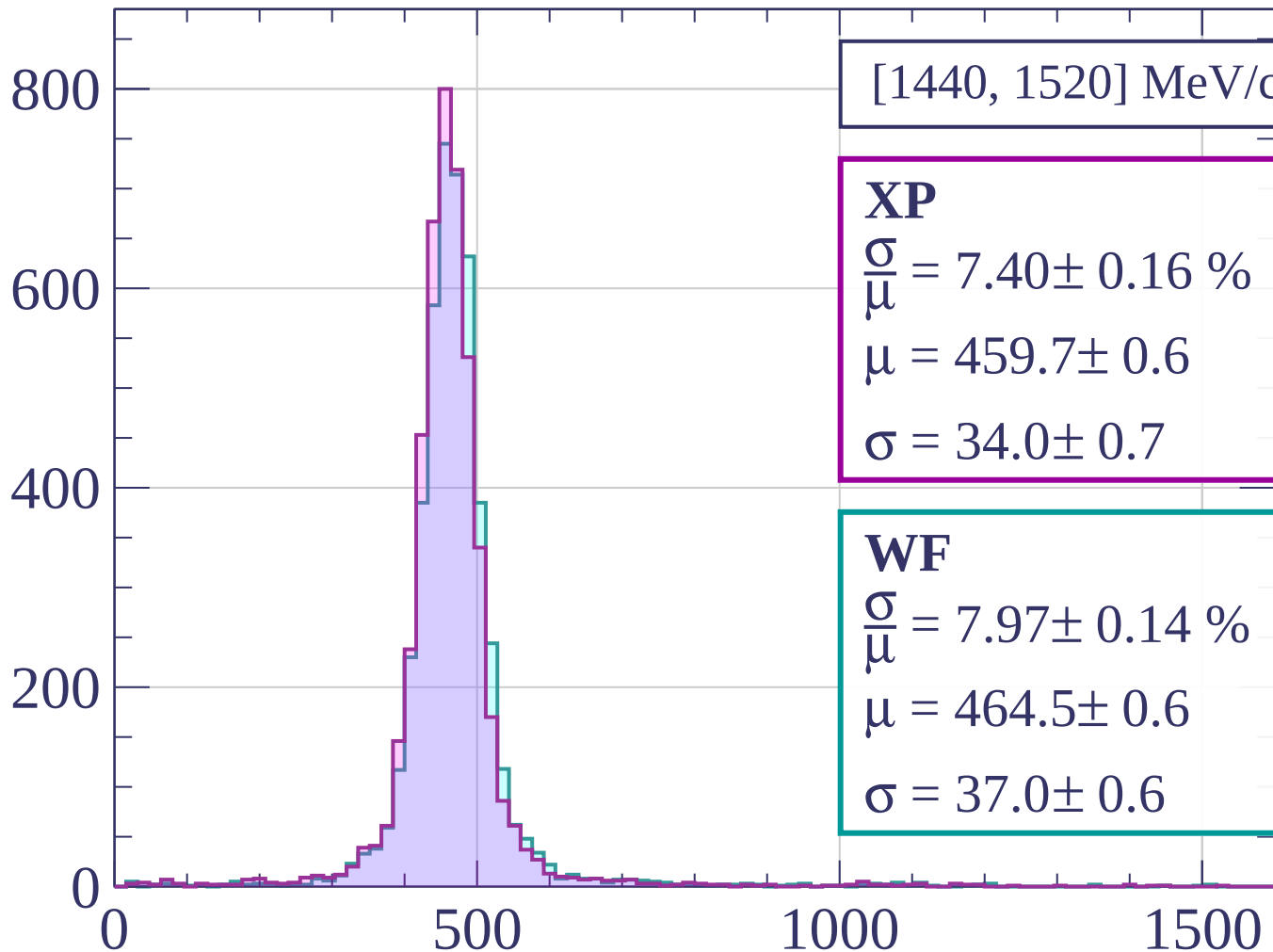
$$\sigma = 34.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.97 \pm 0.14 \%$$

$$\mu = 464.5 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$



dE/dx [ADC counts/cm]

Count

[1520, 1600] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.58 \pm 0.17 \%$$

$$\mu = 462.9 \pm 0.7$$

$$\sigma = 35.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.70 \pm 0.15 \%$$

$$\mu = 467.1 \pm 0.7$$

$$\sigma = 36.0 \pm 0.7$$

600

400

200

0

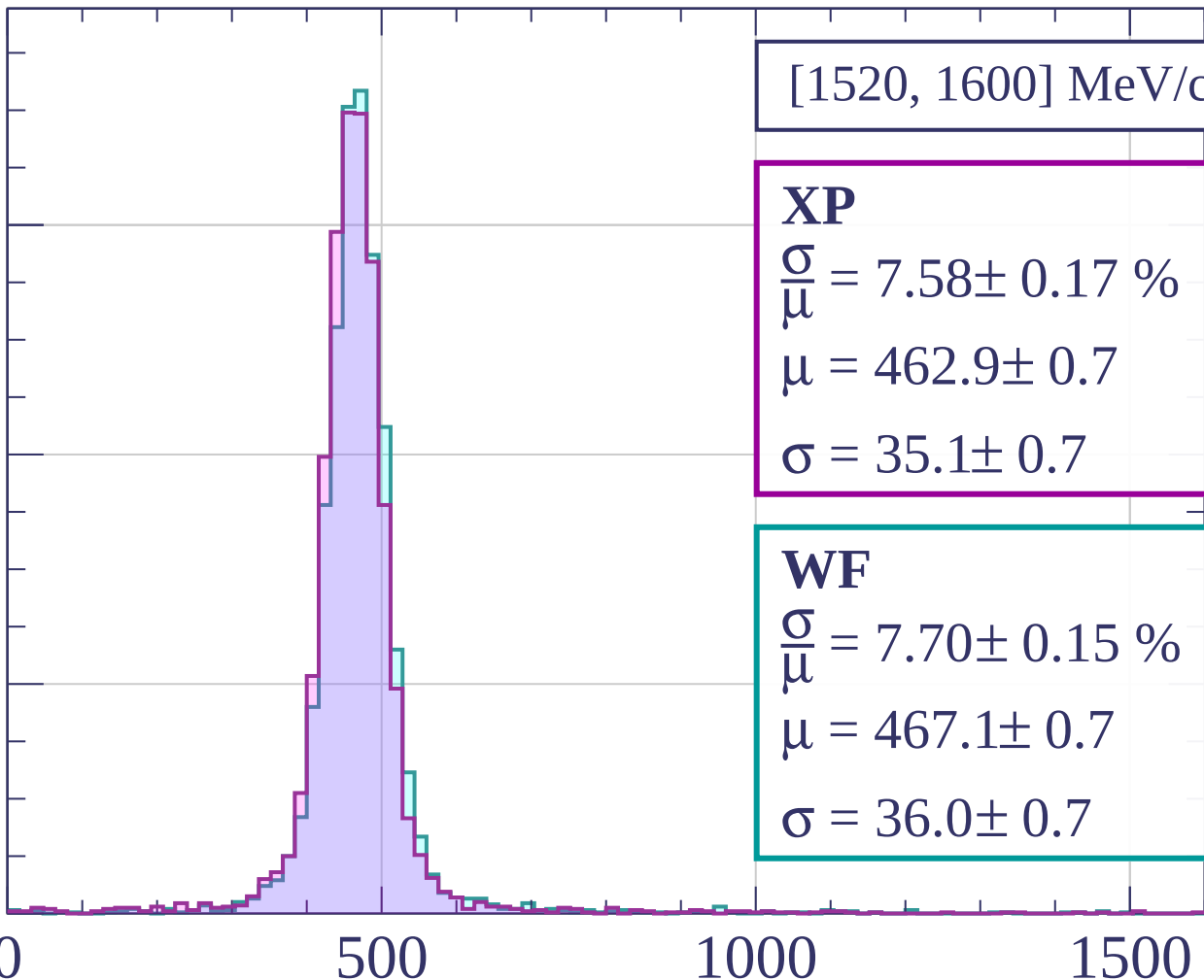
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1600, 1680] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.33 \pm 0.17 \%$$

$$\mu = 464.7 \pm 0.7$$

$$\sigma = 34.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.47 \pm 0.15 \%$$

$$\mu = 470.6 \pm 0.7$$

$$\sigma = 35.2 \pm 0.7$$

600

400

200

0

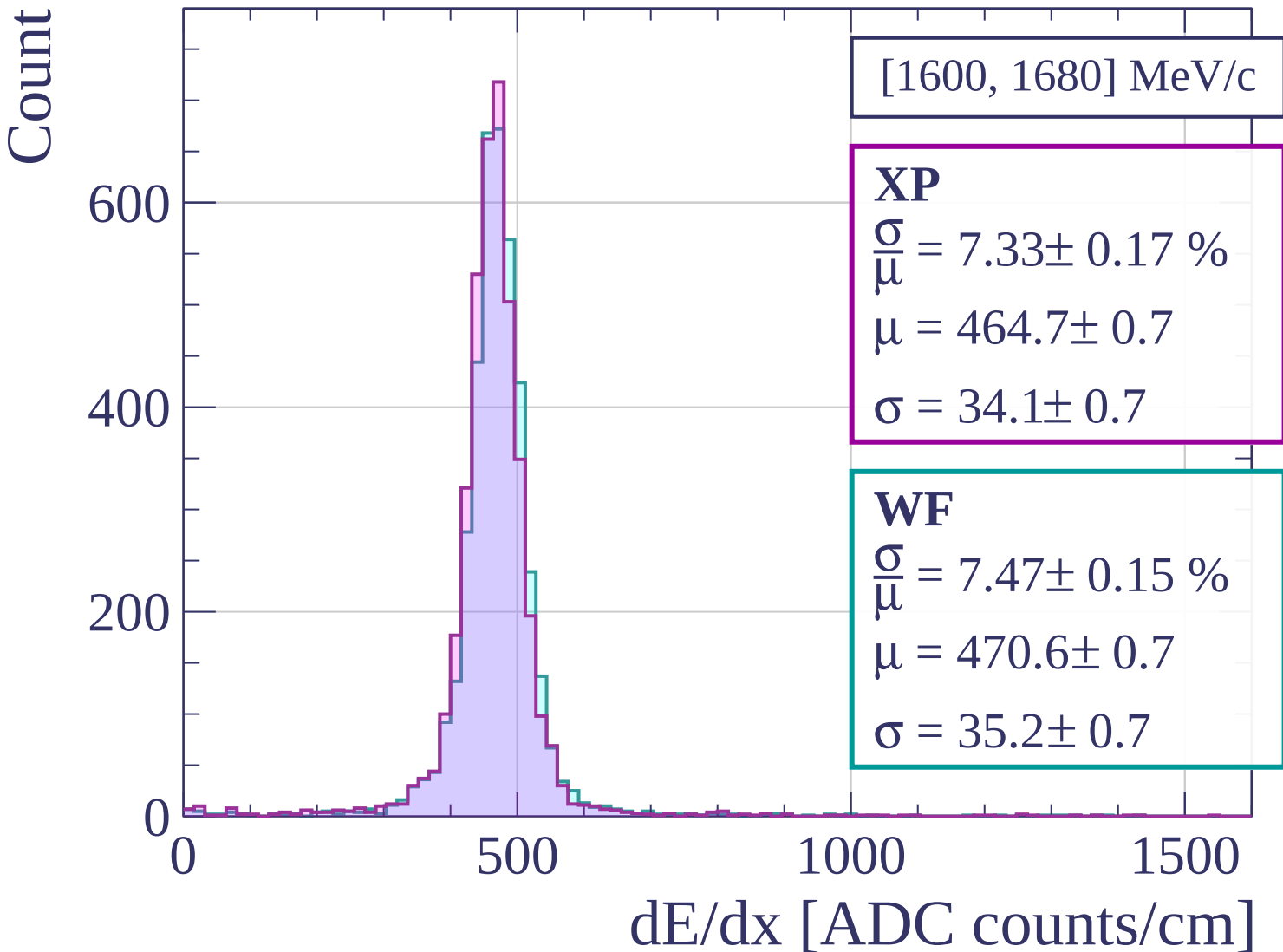
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1680, 1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.63 \pm 0.16 \%$$

$$\mu = 467.6 \pm 0.7$$

$$\sigma = 35.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.77 \pm 0.17 \%$$

$$\mu = 472.5 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

600

400

200

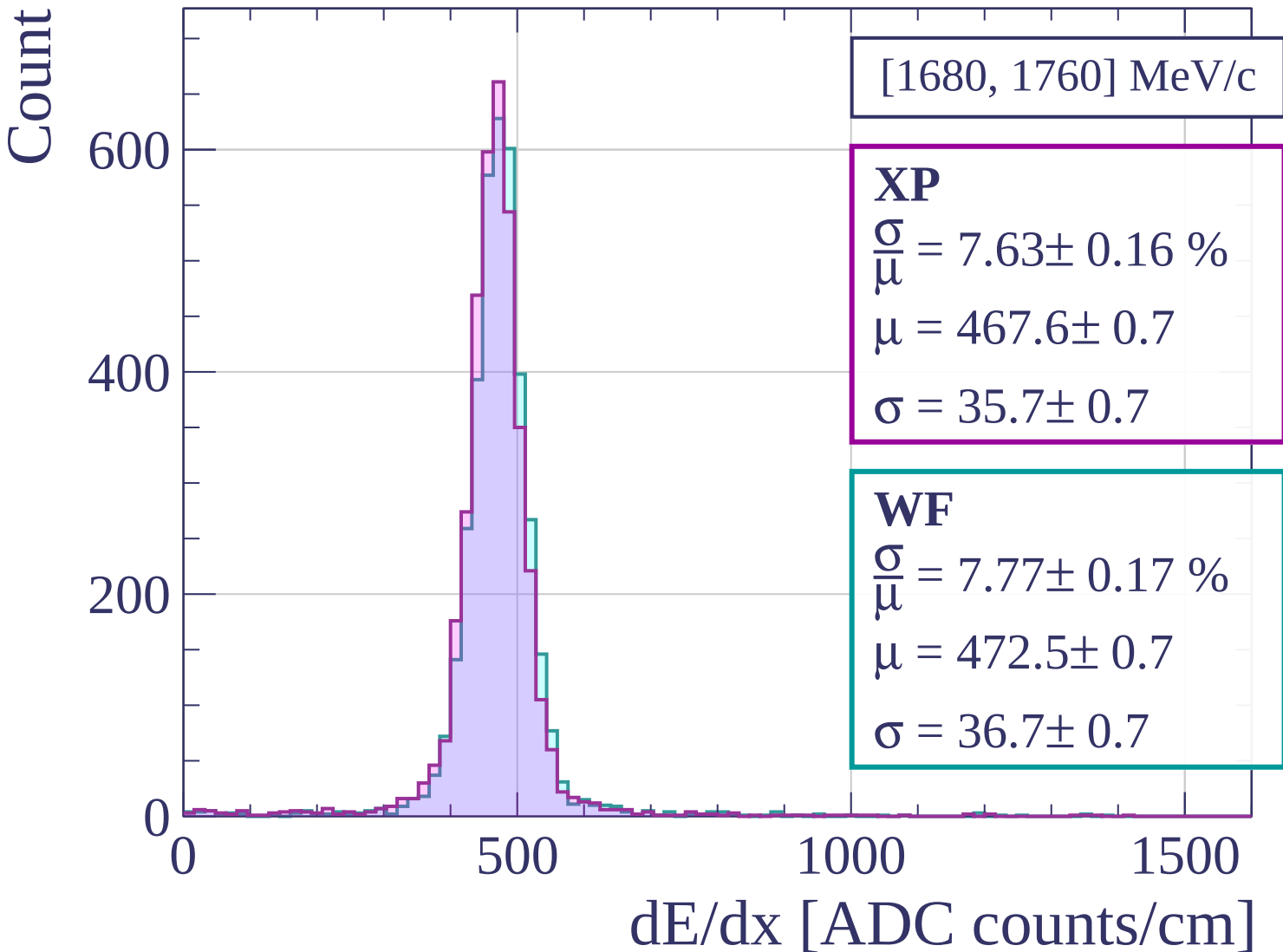
0

500

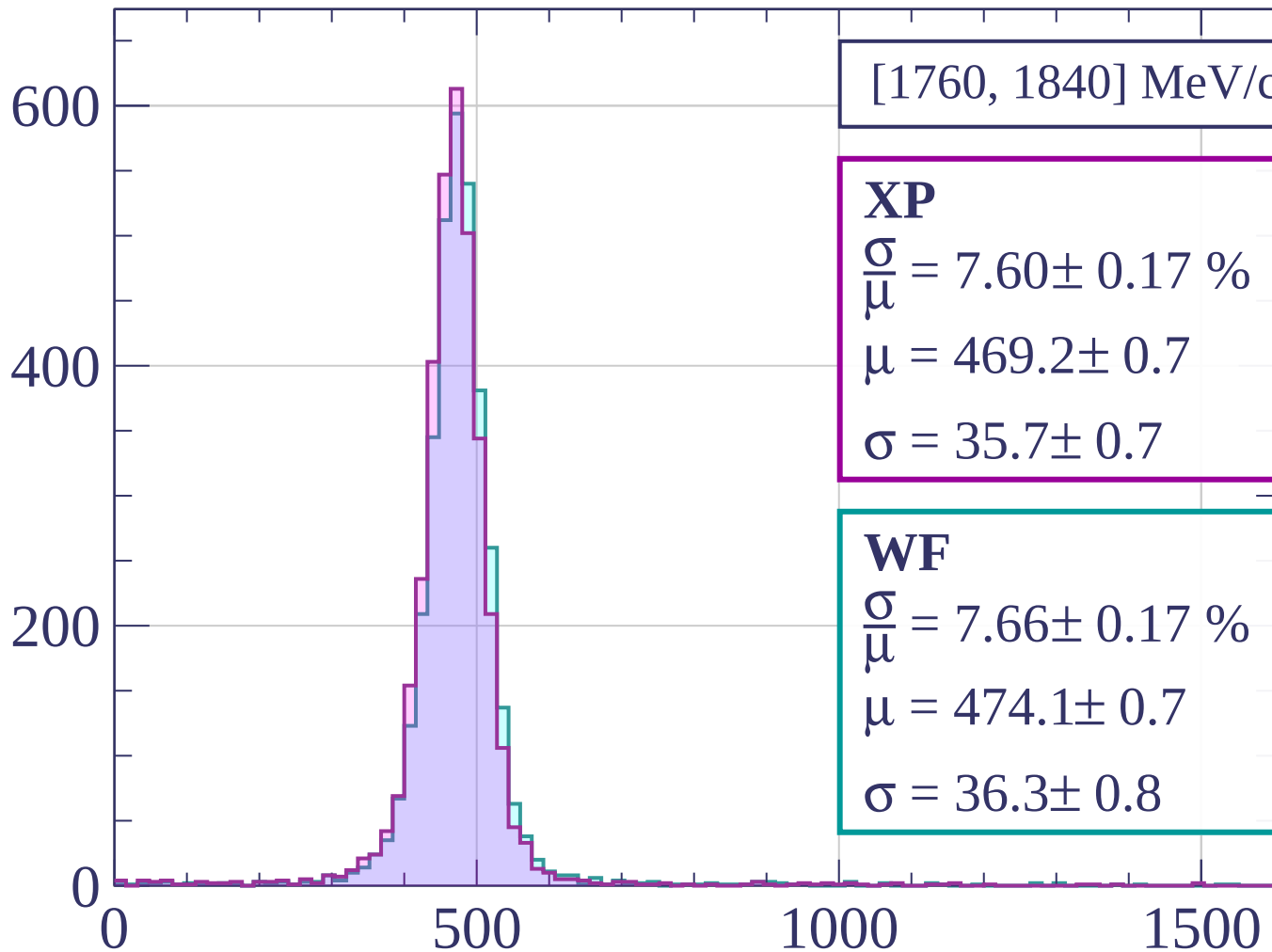
1000

1500

dE/dx [ADC counts/cm]



Count



[1760, 1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.17 \%$$

$$\mu = 469.2 \pm 0.7$$

$$\sigma = 35.7 \pm 0.7$$

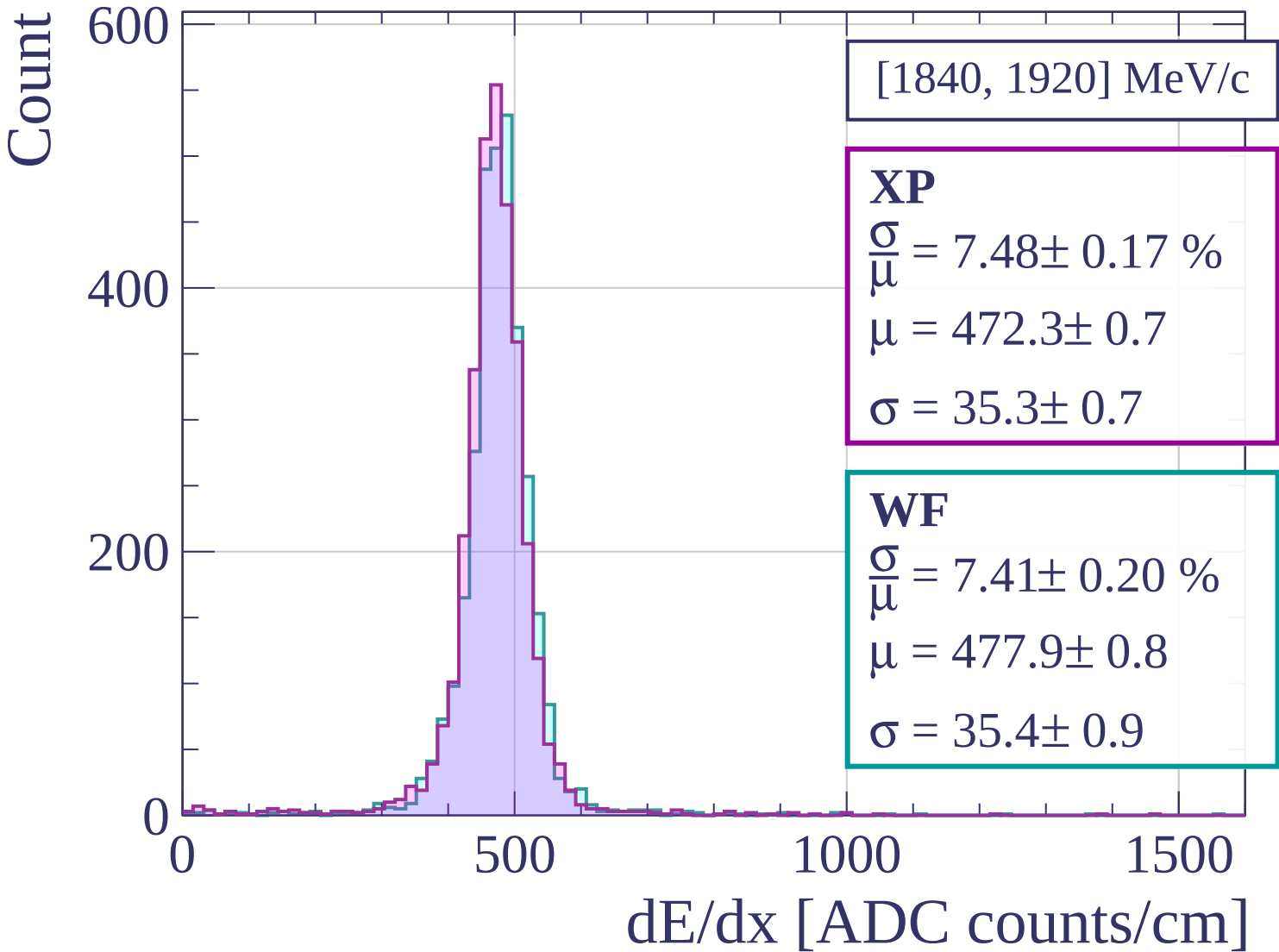
WF

$$\frac{\sigma}{\mu} = 7.66 \pm 0.17 \%$$

$$\mu = 474.1 \pm 0.7$$

$$\sigma = 36.3 \pm 0.8$$

dE/dx [ADC counts/cm]



Count

[1920, 2000] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.17 \%$$

$$\mu = 474.1 \pm 0.7$$

$$\sigma = 35.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.20 \pm 0.19 \%$$

$$\mu = 479.4 \pm 0.8$$

$$\sigma = 34.5 \pm 0.8$$

400

200

0

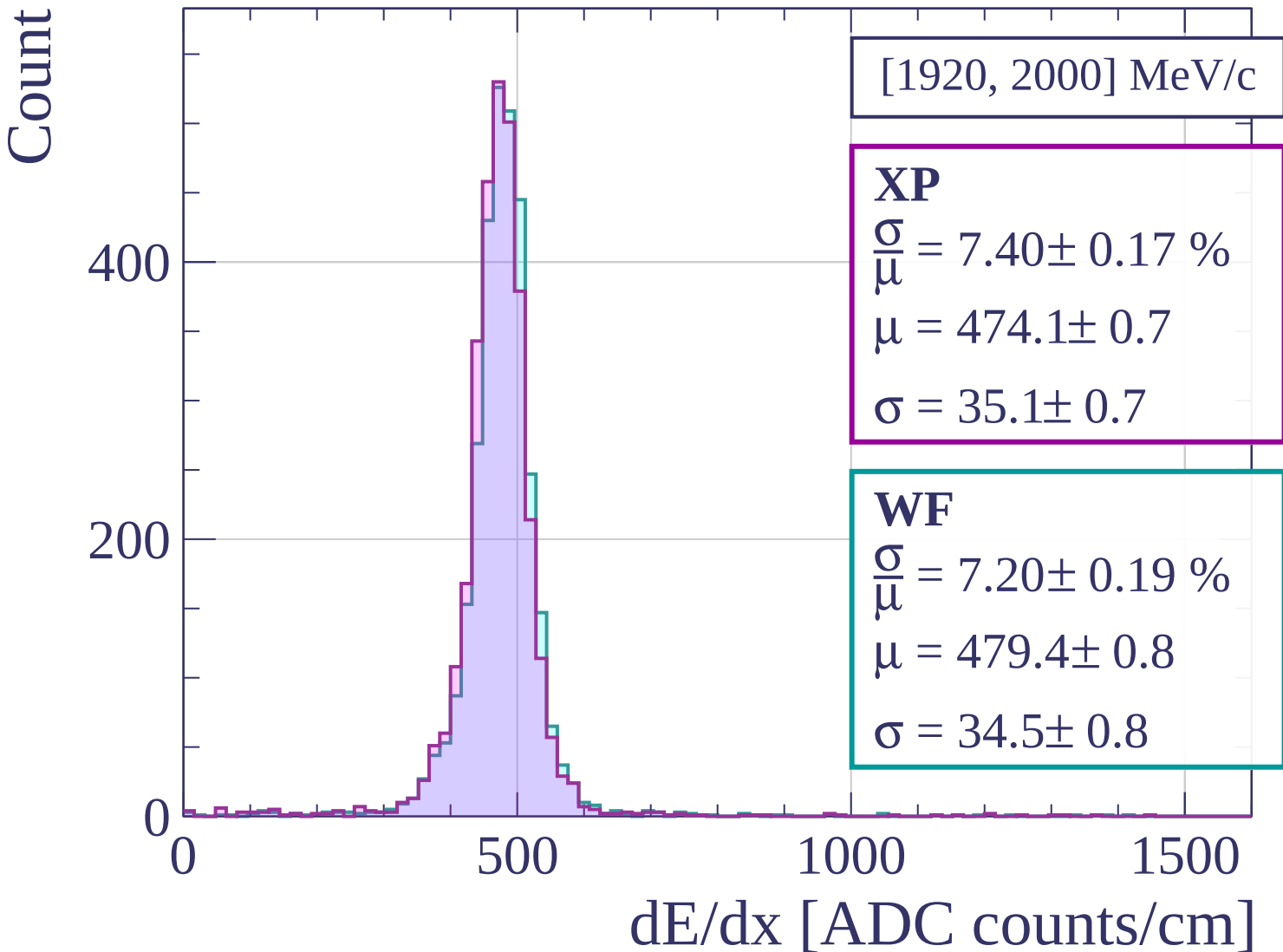
0

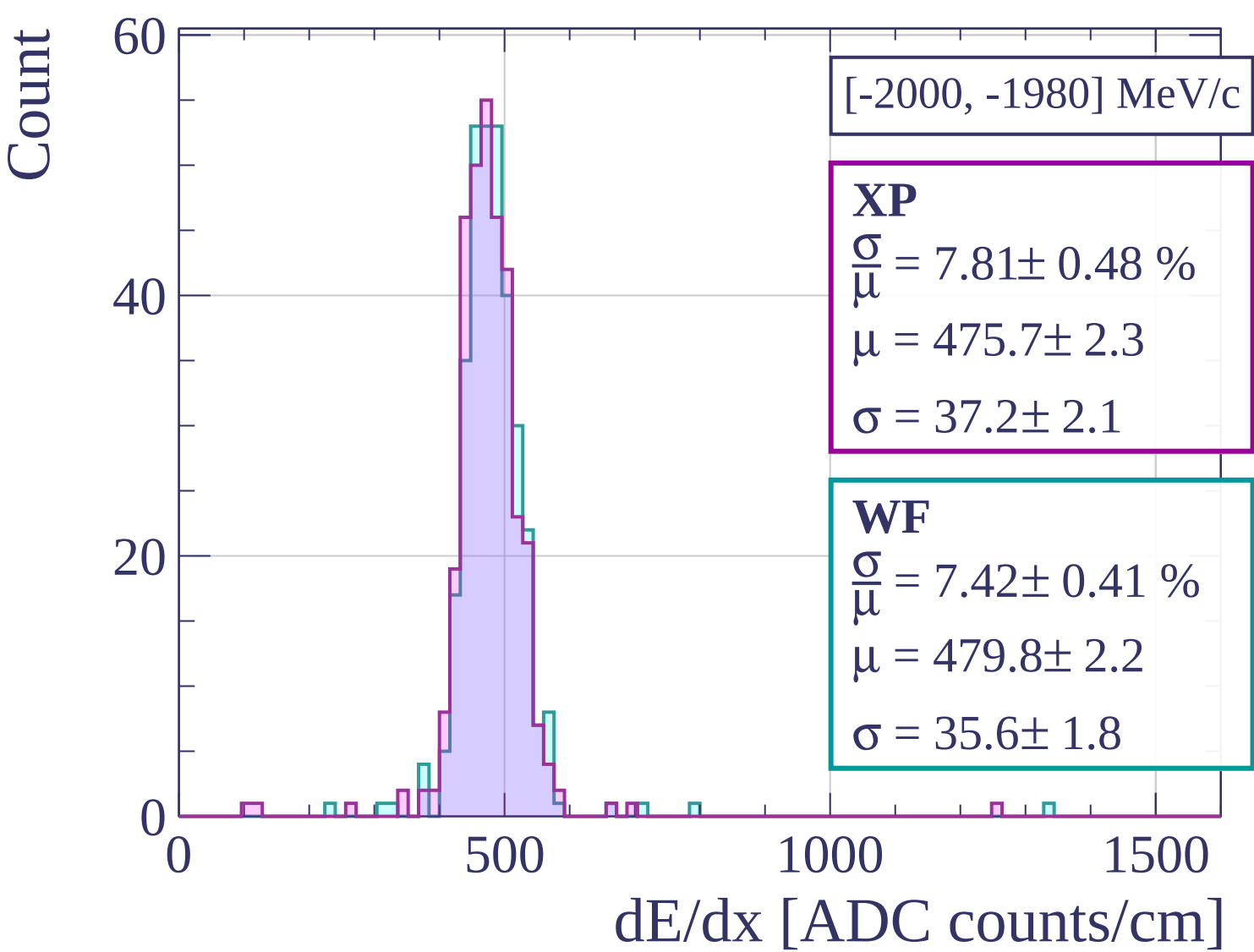
500

1000

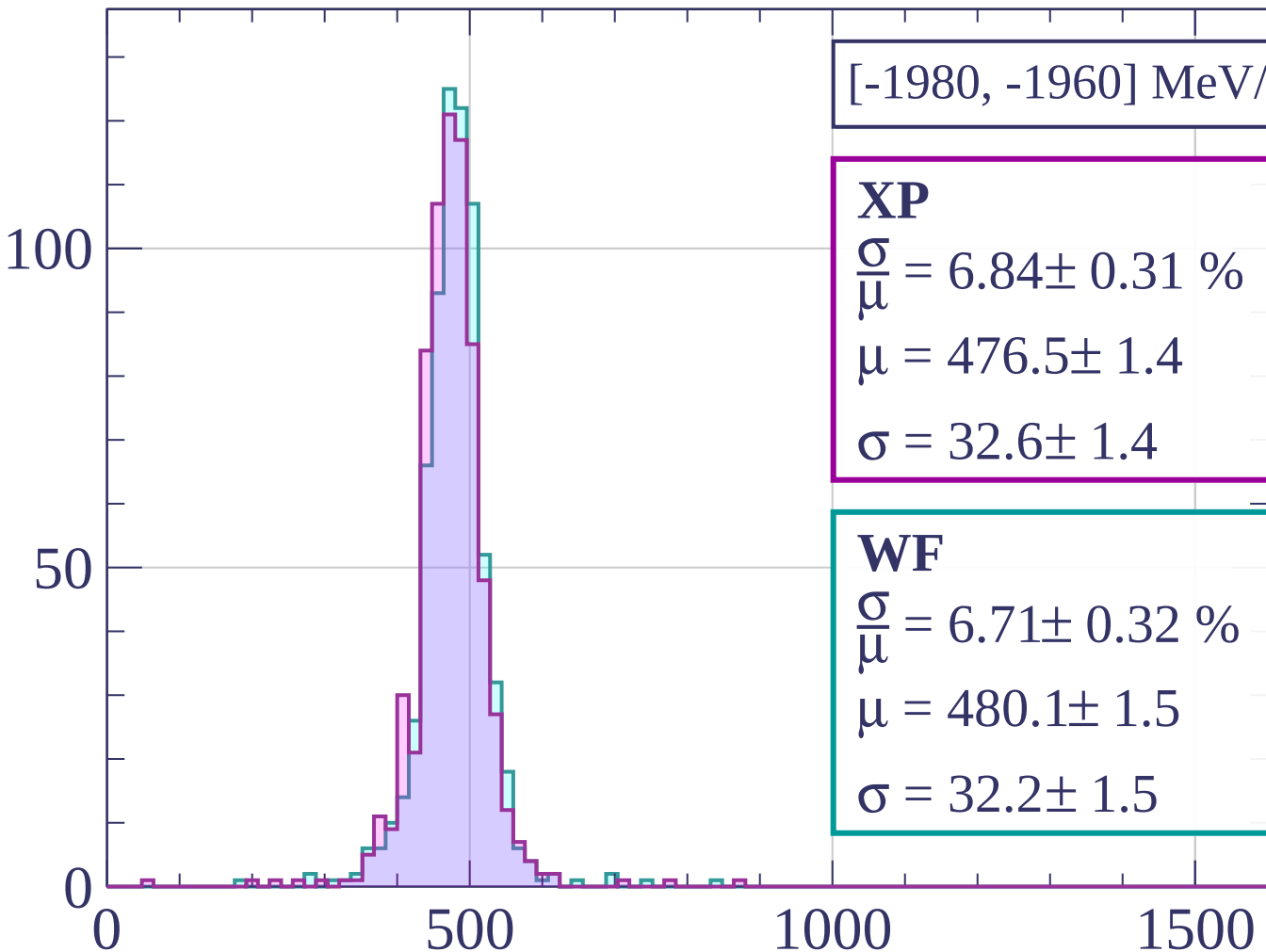
1500

dE/dx [ADC counts/cm]





Count



[-1980, -1960] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 6.84 \pm 0.31 \%$$

$$\mu = 476.5 \pm 1.4$$

$$\sigma = 32.6 \pm 1.4$$

WF

$$\frac{\sigma}{\bar{\mu}} = 6.71 \pm 0.32 \%$$

$$\mu = 480.1 \pm 1.5$$

$$\sigma = 32.2 \pm 1.5$$

dE/dx [ADC counts/cm]

Count

[-1960, -1940] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.90 \pm 0.30 \%$$

$$\mu = 472.4 \pm 1.5$$

$$\sigma = 32.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.56 \pm 0.32 \%$$

$$\mu = 478.0 \pm 1.5$$

$$\sigma = 31.3 \pm 1.4$$

100

50

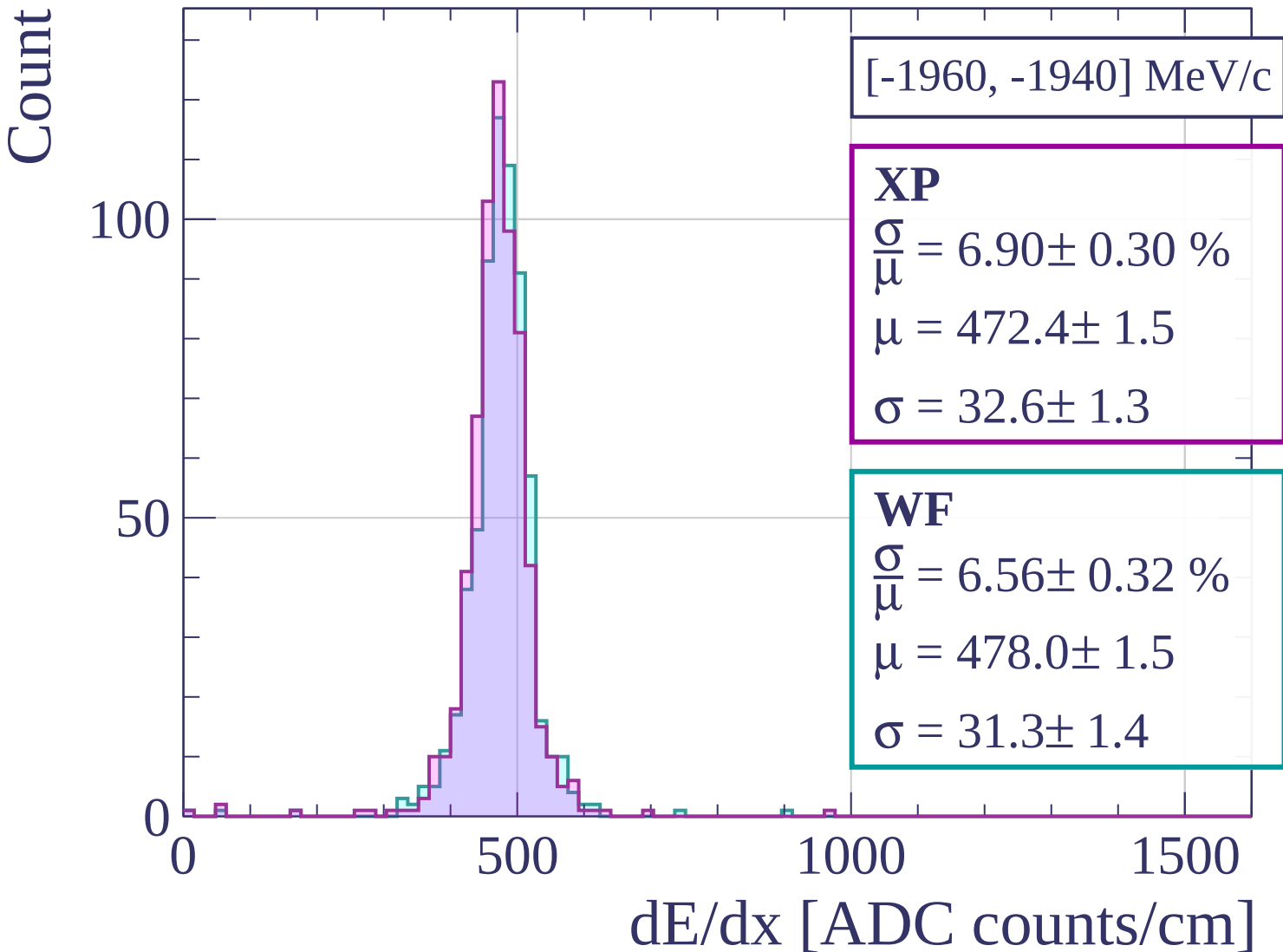
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1940, -1920] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.15 \pm 0.32 \%$$

$$\mu = 475.7 \pm 1.4$$

$$\sigma = 29.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.63 \pm 0.34 \%$$

$$\mu = 482.0 \pm 1.5$$

$$\sigma = 31.9 \pm 1.5$$

100

50

0

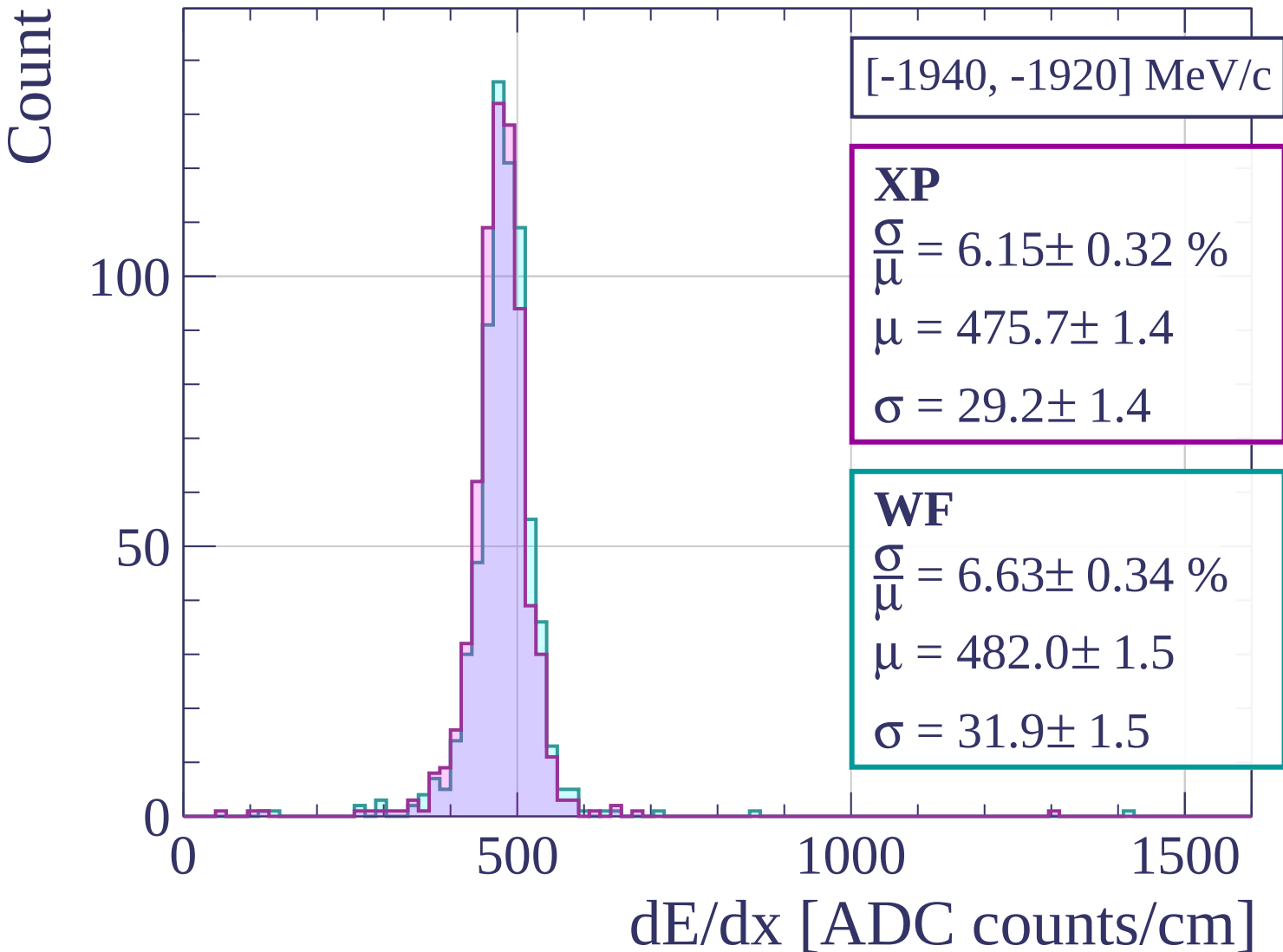
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1920, -1900] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.03 \pm 0.32 \%$$

$$\mu = 476.2 \pm 1.6$$

$$\sigma = 33.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.95 \pm 0.39 \%$$

$$\mu = 481.8 \pm 1.7$$

$$\sigma = 33.5 \pm 1.8$$

100

50

0

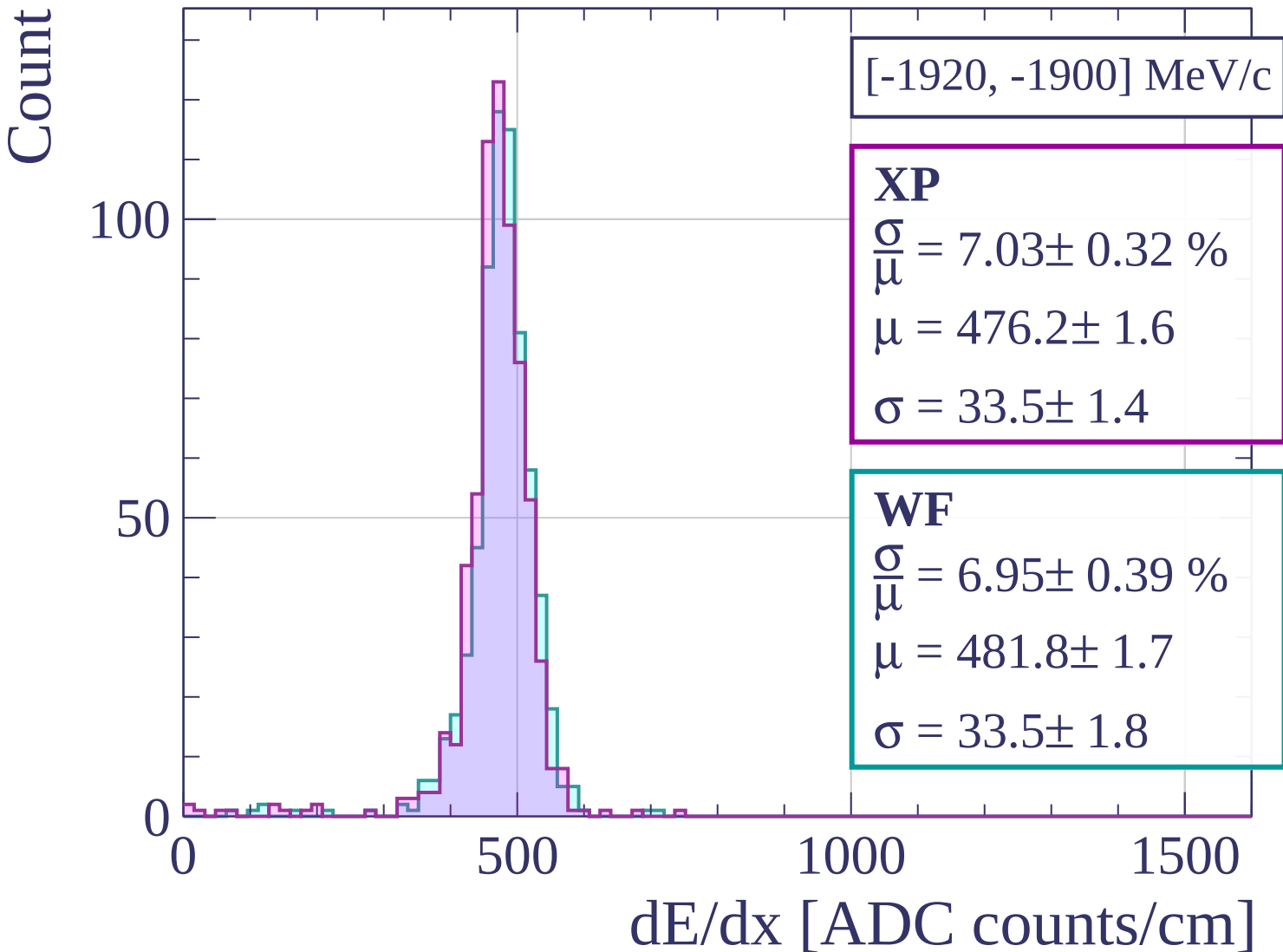
0

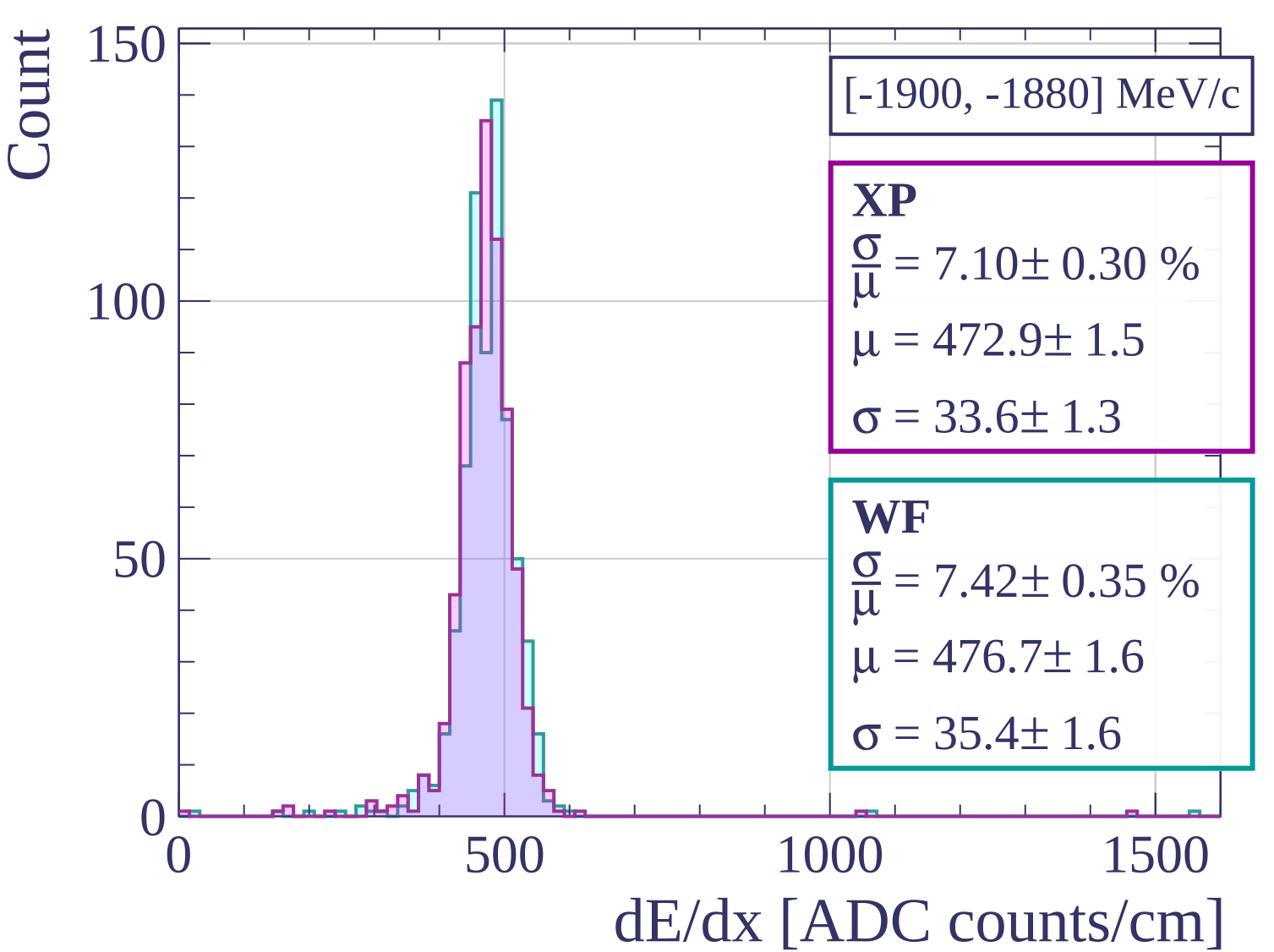
500

1000

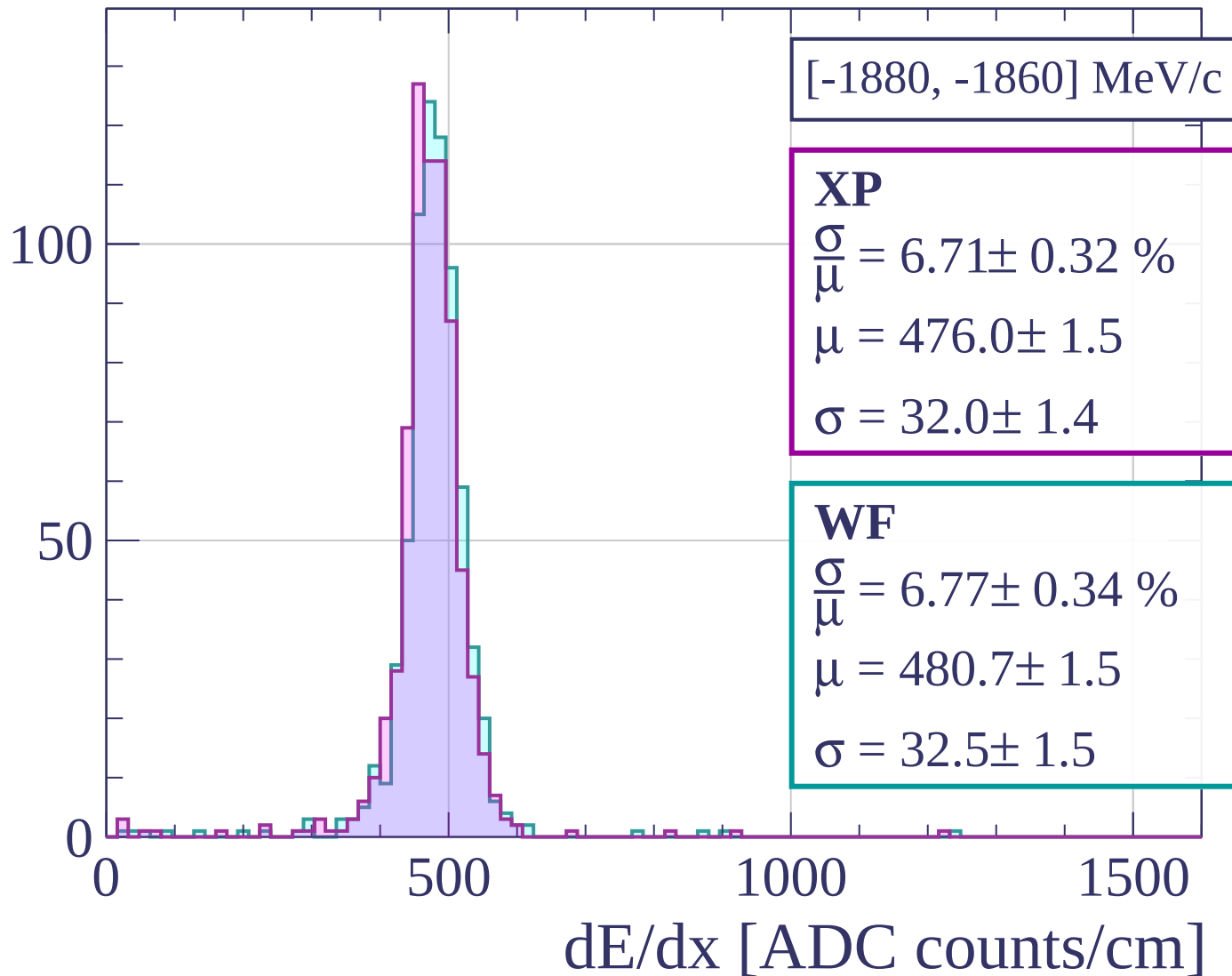
1500

dE/dx [ADC counts/cm]





Count



Count

[-1860, -1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.13 \pm 0.32 \%$$

$$\mu = 470.8 \pm 1.5$$

$$\sigma = 33.6 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.93 \pm 0.37 \%$$

$$\mu = 476.9 \pm 1.6$$

$$\sigma = 33.1 \pm 1.6$$

100

50

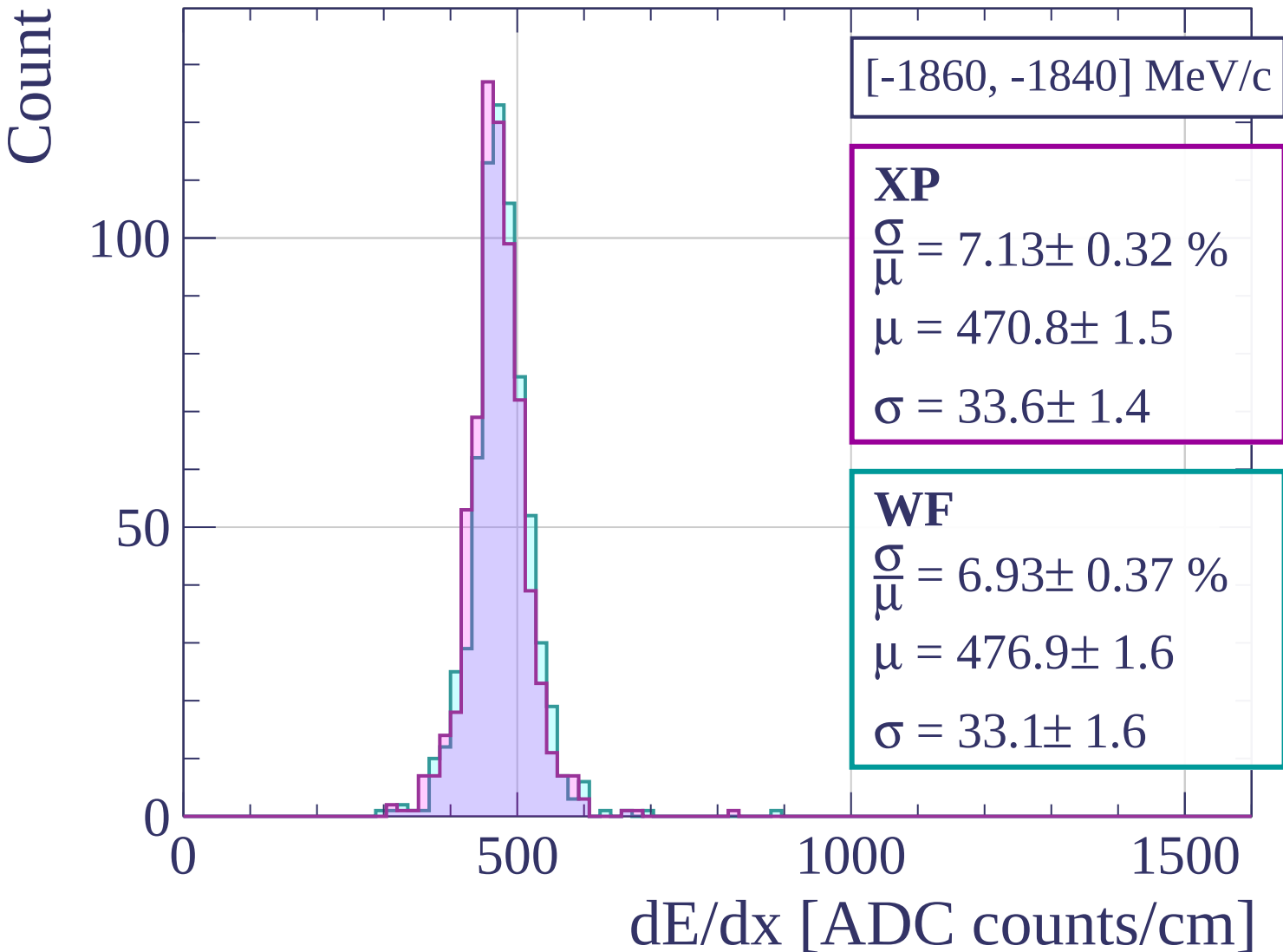
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1840, -1820] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.45 \pm 0.34 \%$$

$$\mu = 470.8 \pm 1.6$$

$$\sigma = 35.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.17 \pm 0.38 \%$$

$$\mu = 478.1 \pm 1.7$$

$$\sigma = 34.3 \pm 1.7$$

100

50

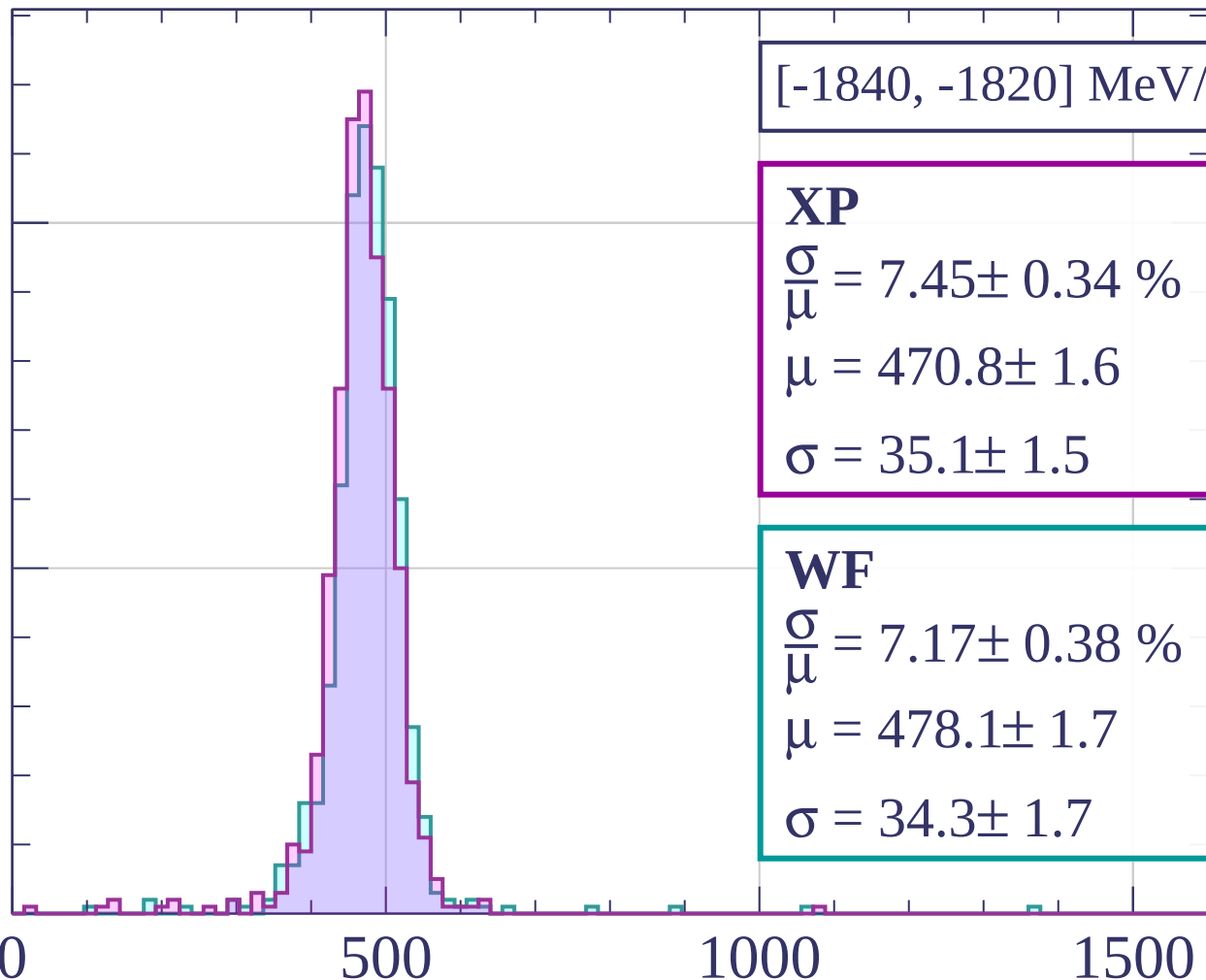
0

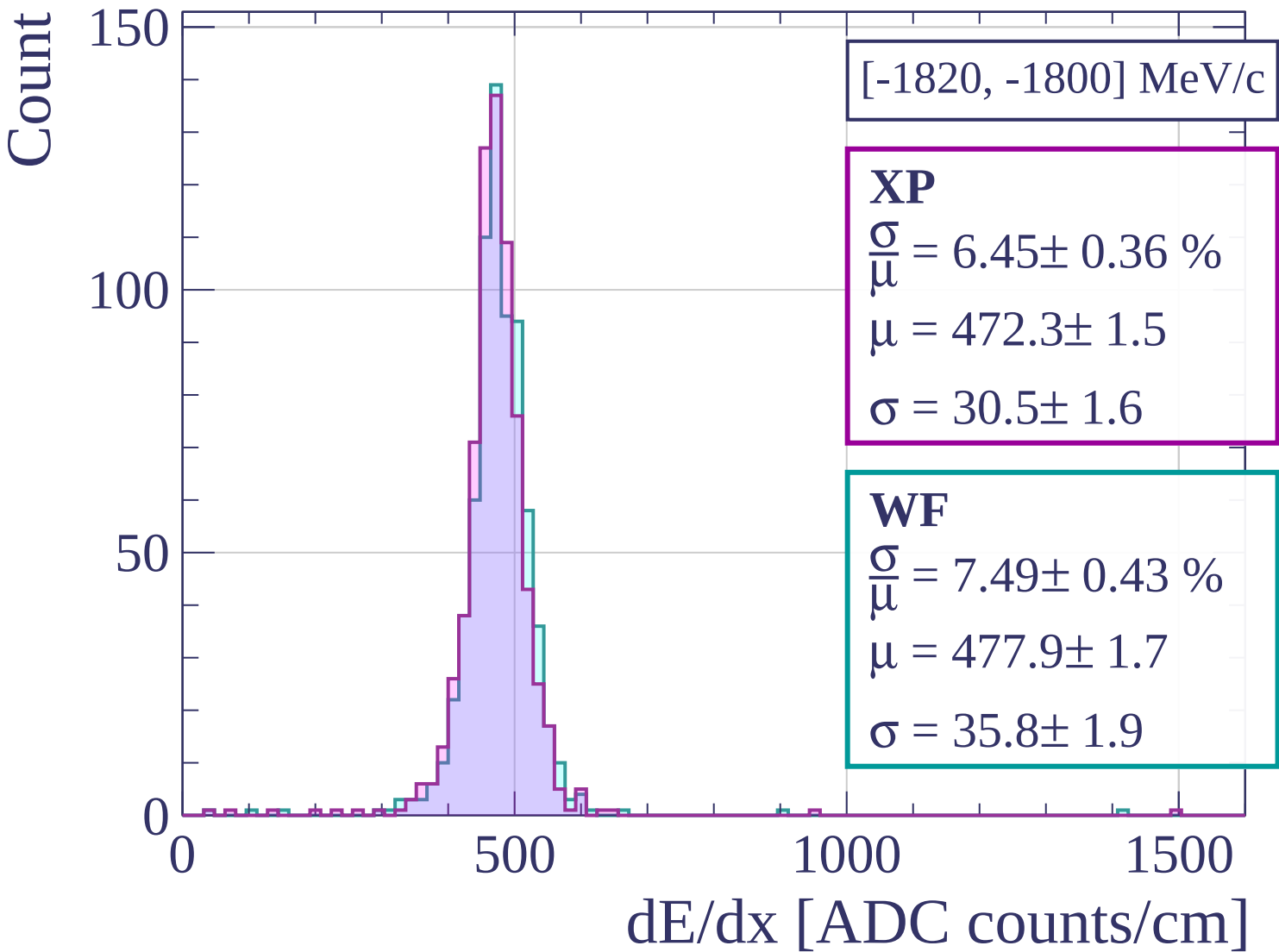
500

1000

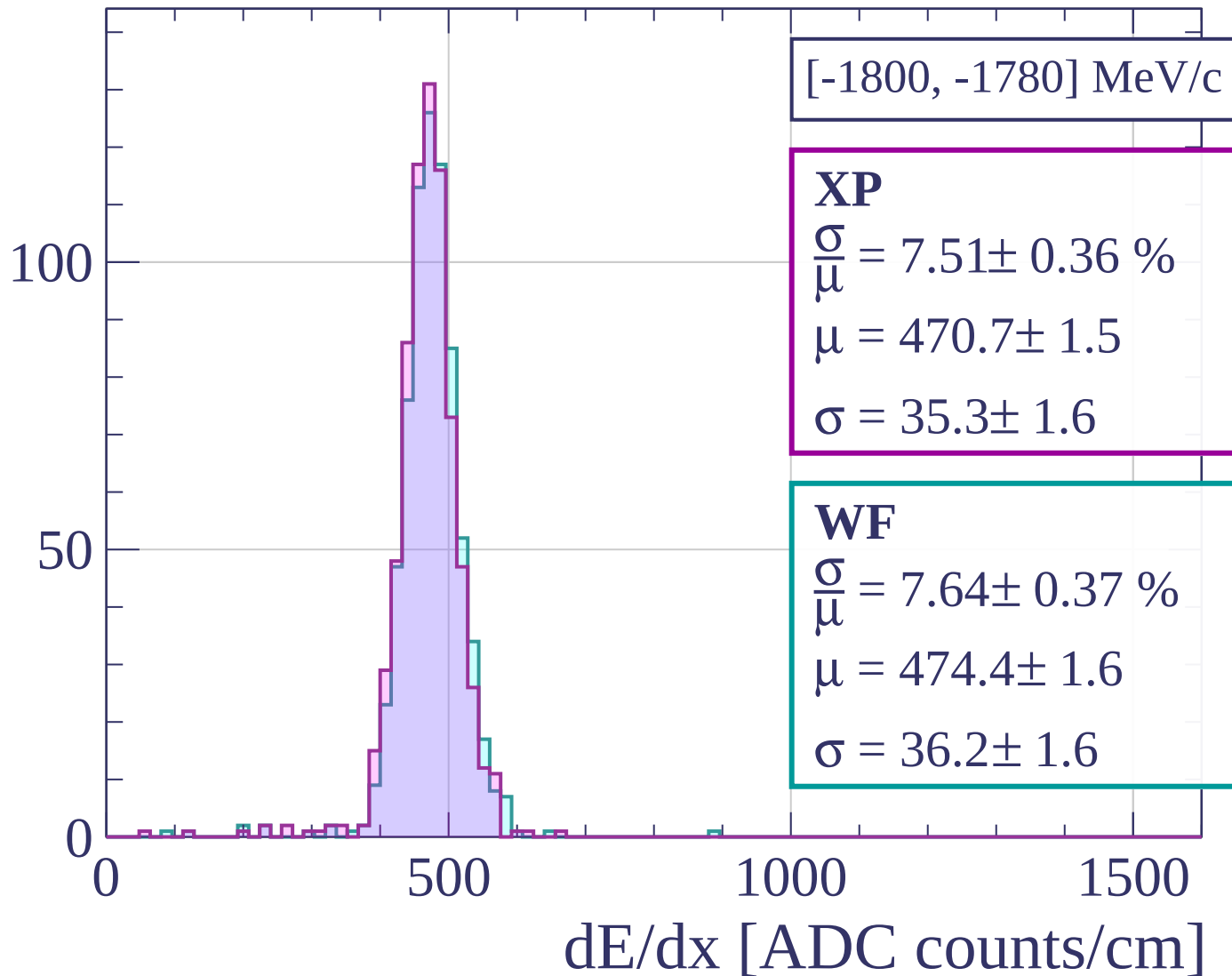
1500

dE/dx [ADC counts/cm]

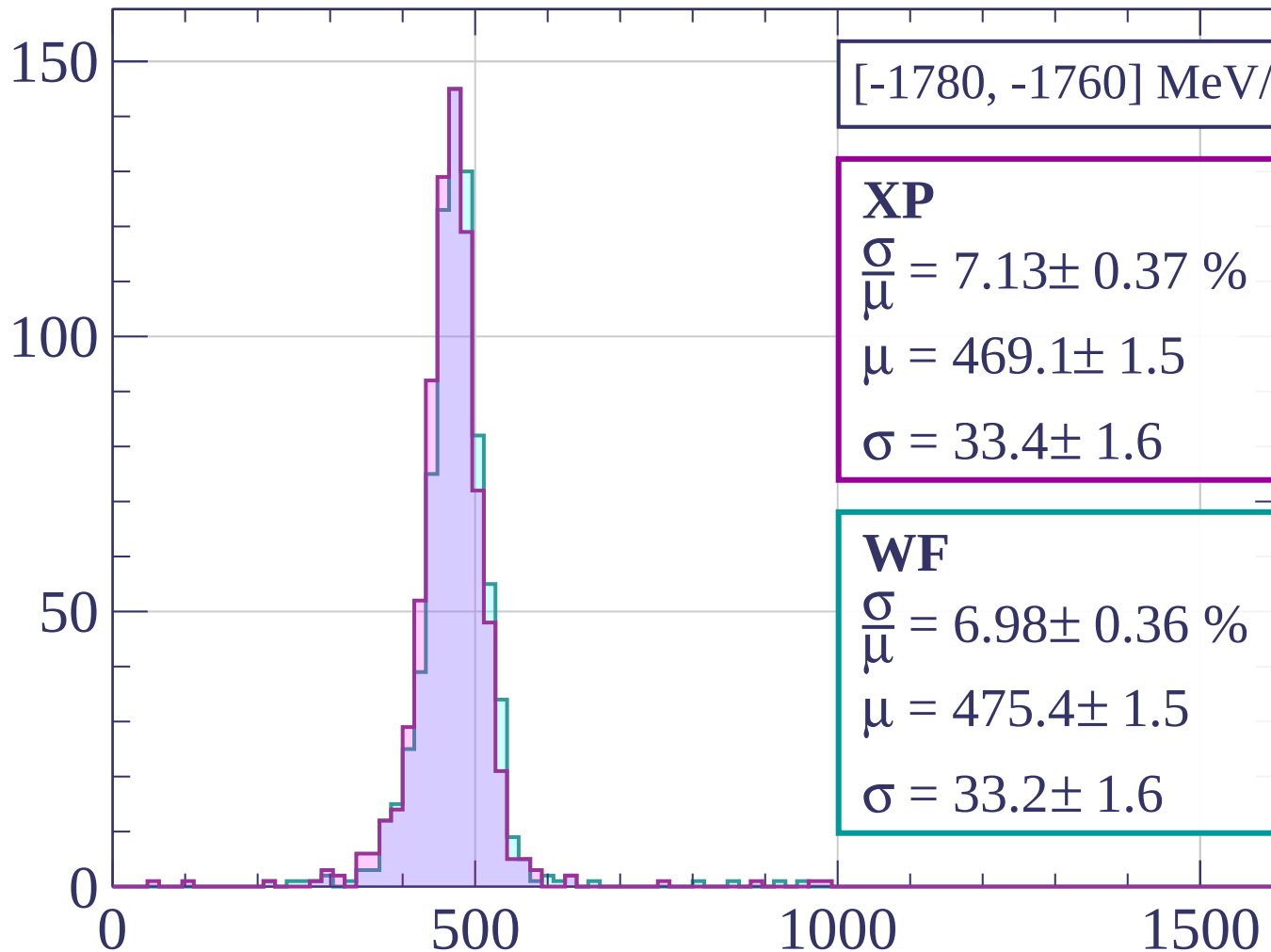




Count

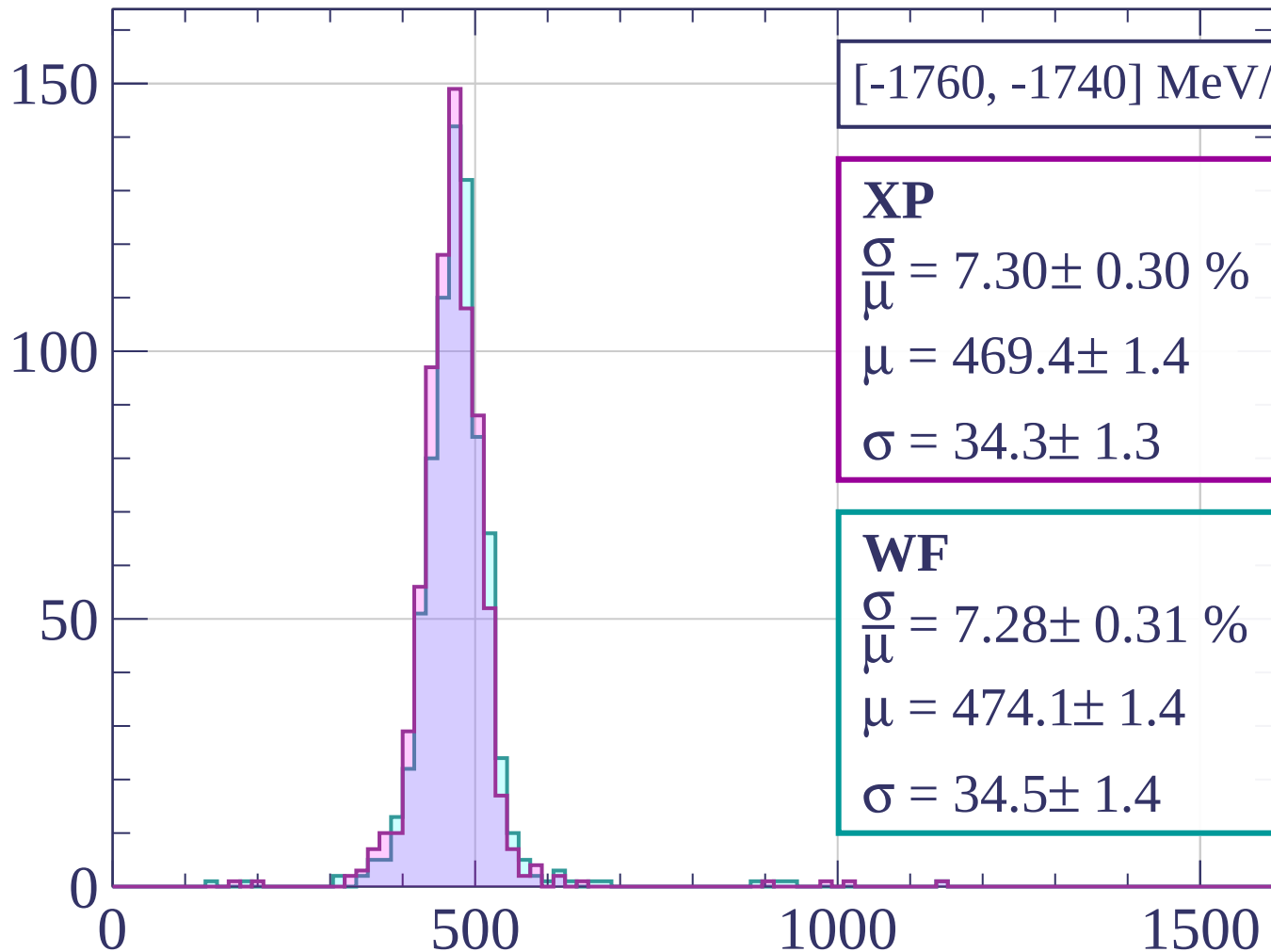


Count



dE/dx [ADC counts/cm]

Count



[-1760, -1740] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.30 \pm 0.30 \%$$

$$\mu = 469.4 \pm 1.4$$

$$\sigma = 34.3 \pm 1.3$$

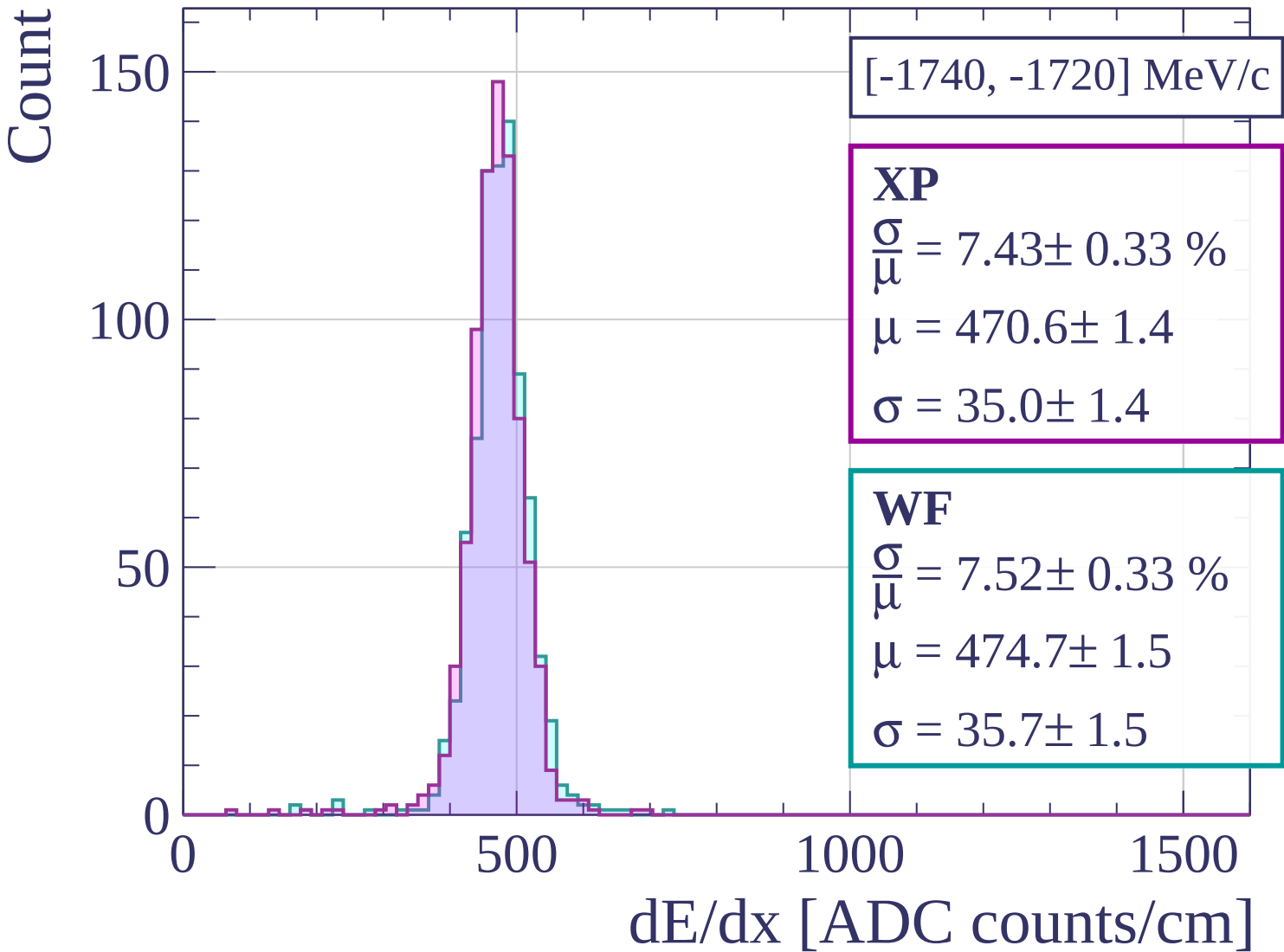
WF

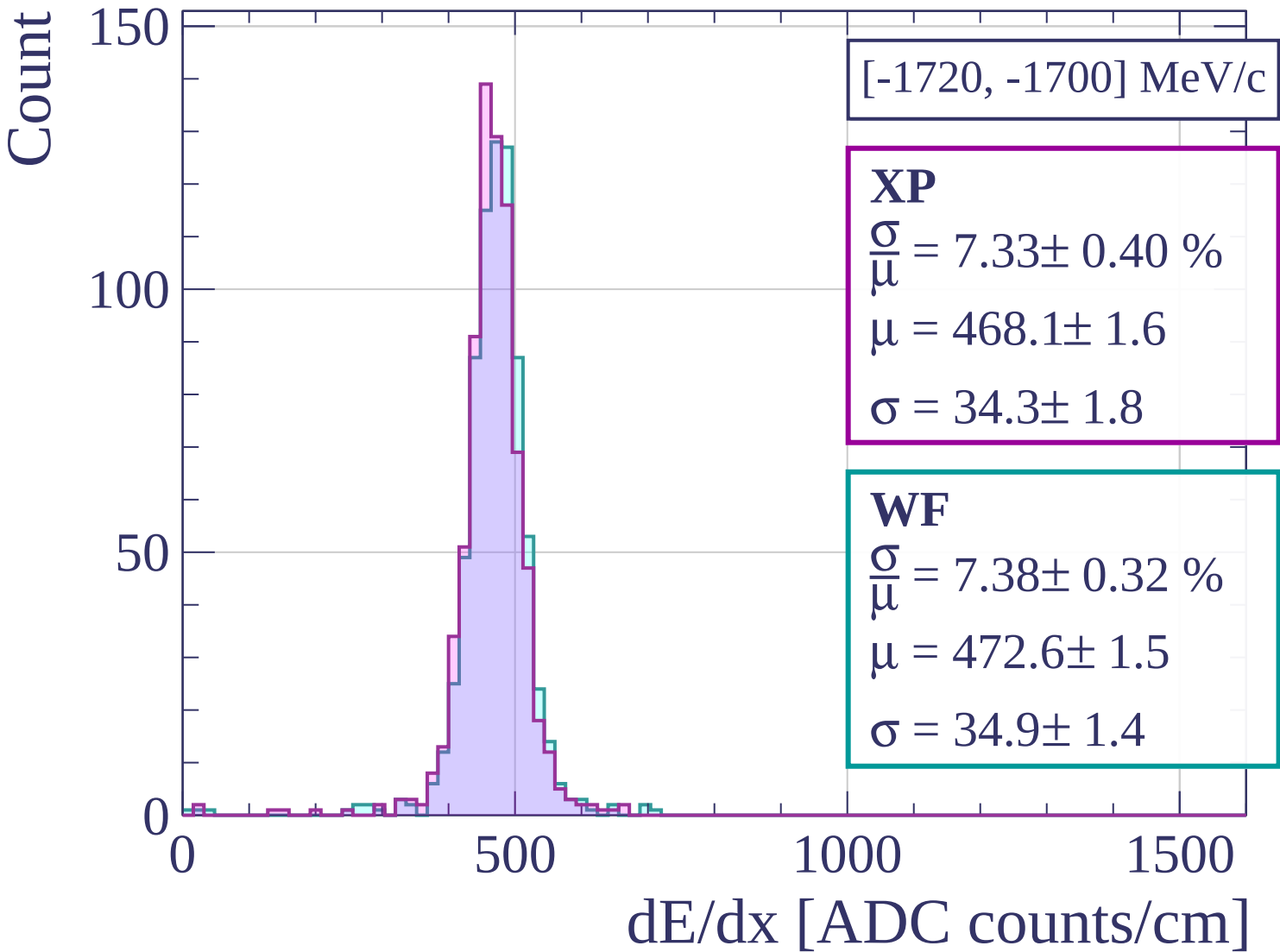
$$\frac{\sigma}{\mu} = 7.28 \pm 0.31 \%$$

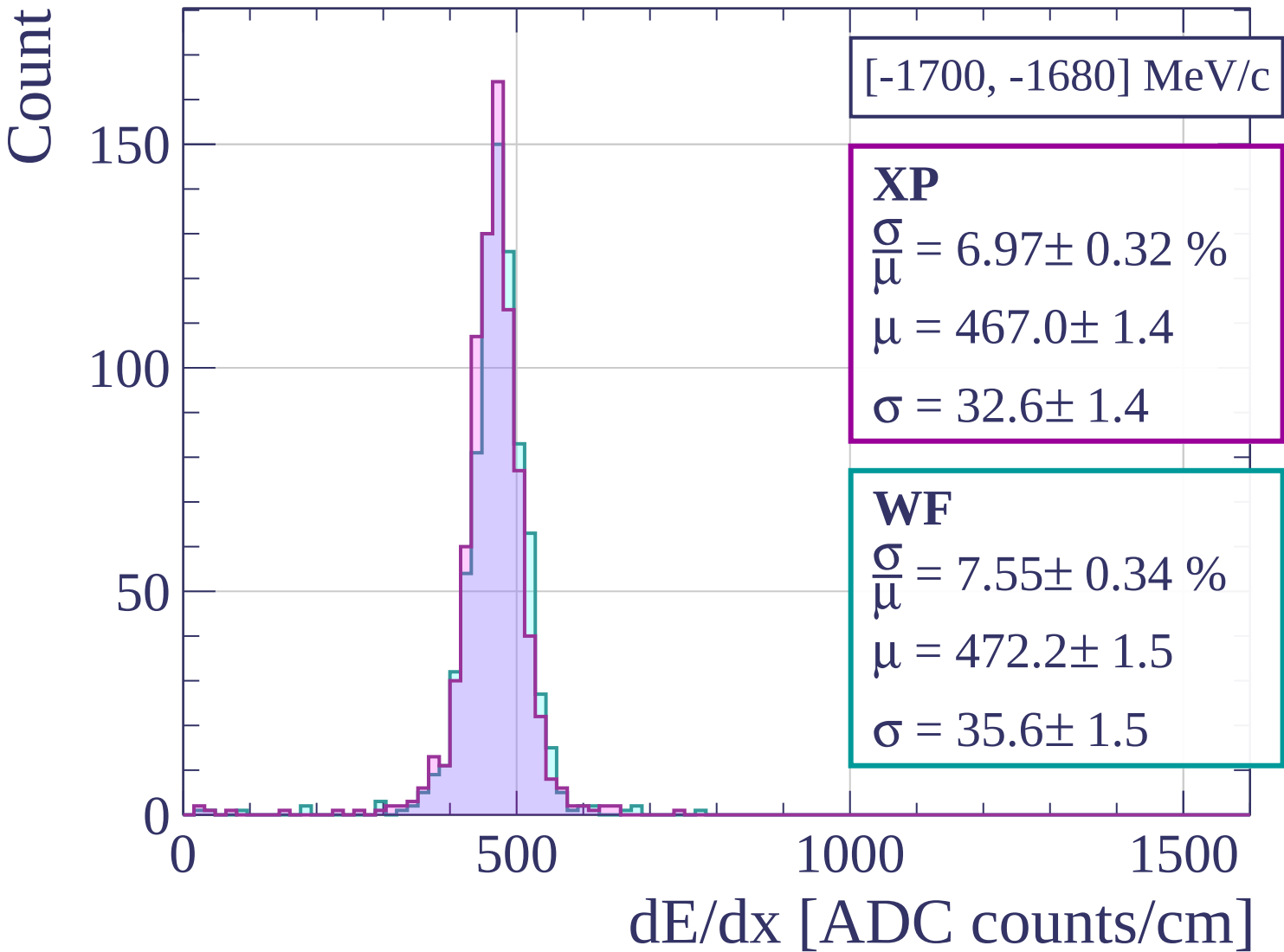
$$\mu = 474.1 \pm 1.4$$

$$\sigma = 34.5 \pm 1.4$$

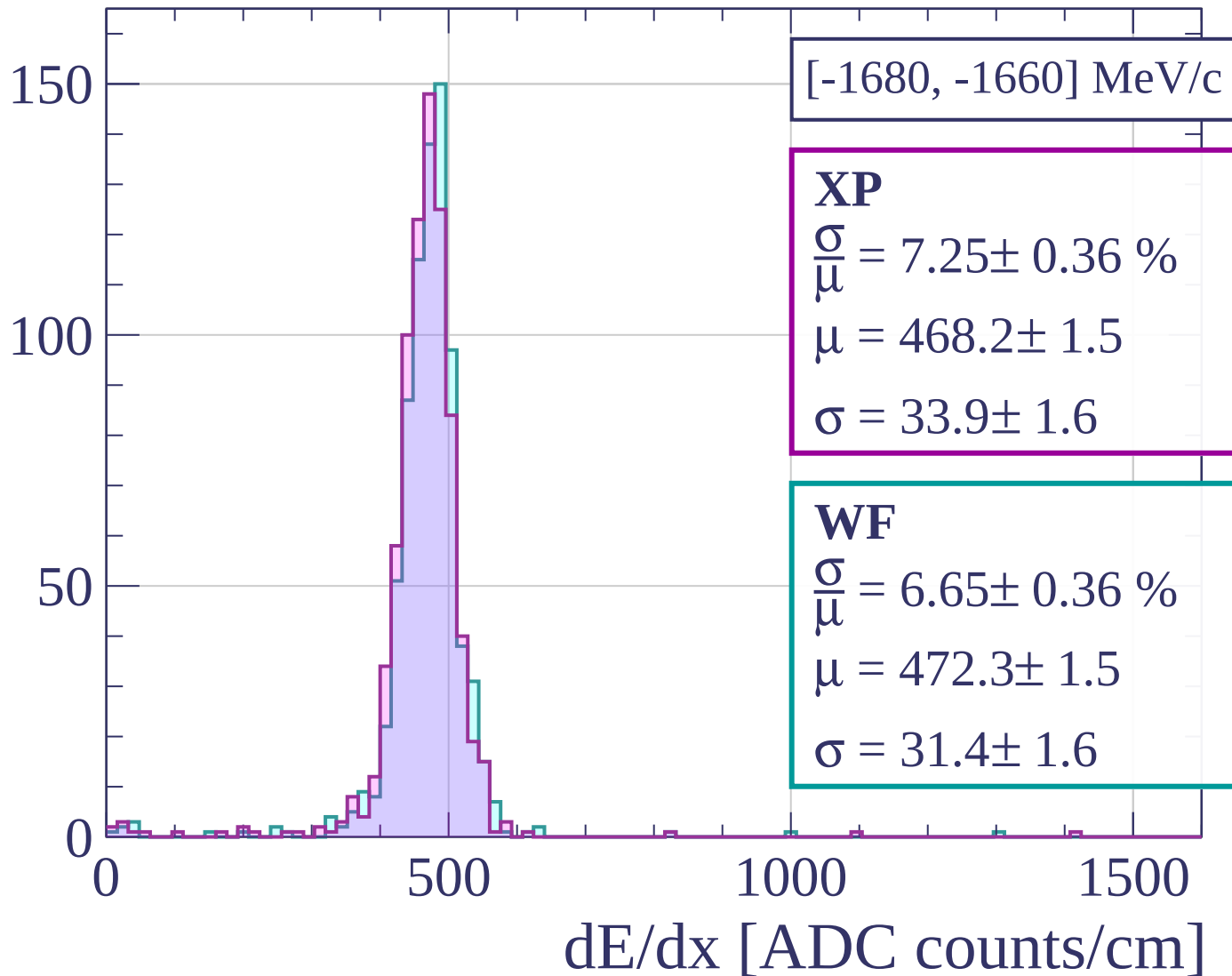
dE/dx [ADC counts/cm]







Count



Count

[-1660, -1640] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.13 \pm 0.33 \%$$

$$\mu = 465.9 \pm 1.4$$

$$\sigma = 33.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.91 \pm 0.33 \%$$

$$\mu = 471.3 \pm 1.5$$

$$\sigma = 32.6 \pm 1.4$$

150

100

50

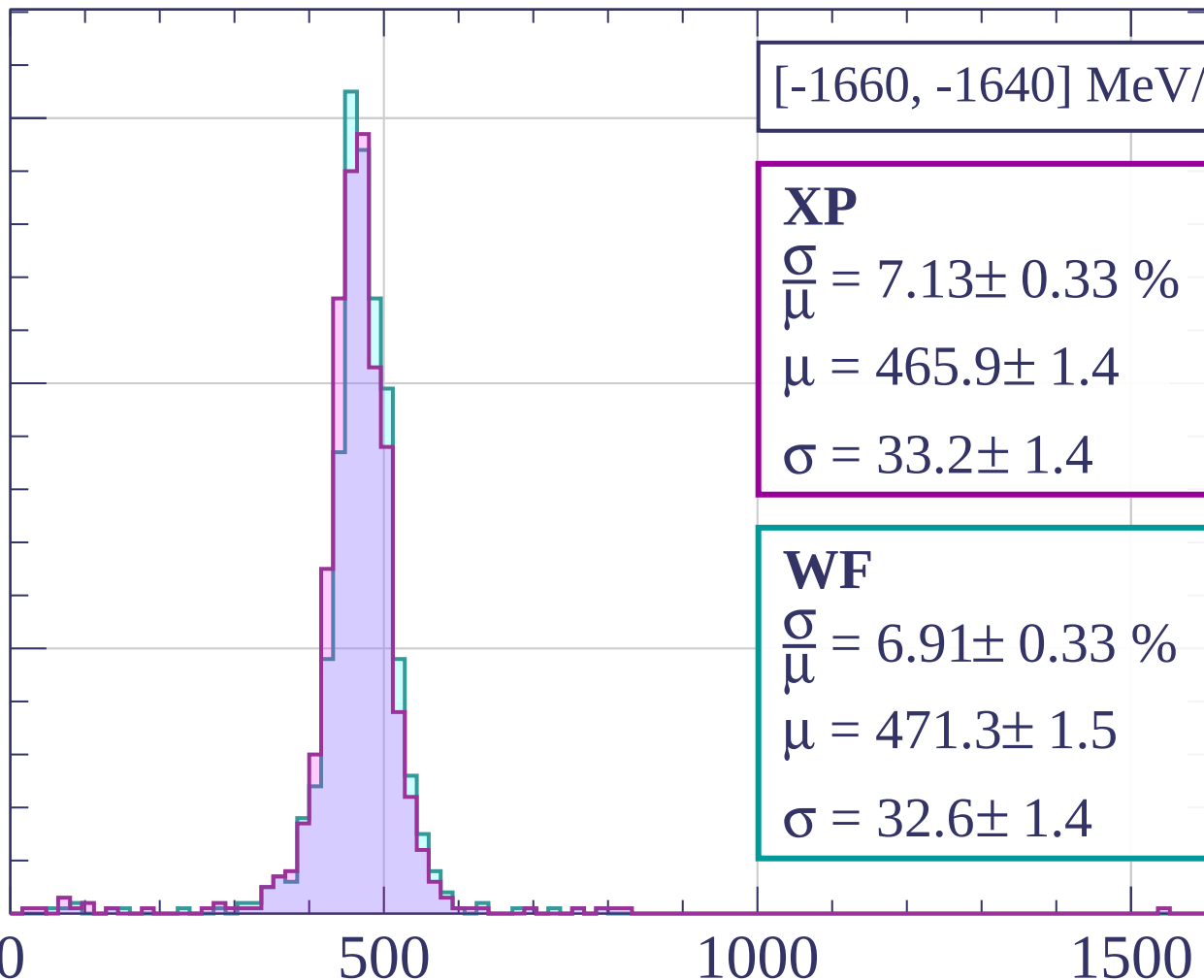
0

500

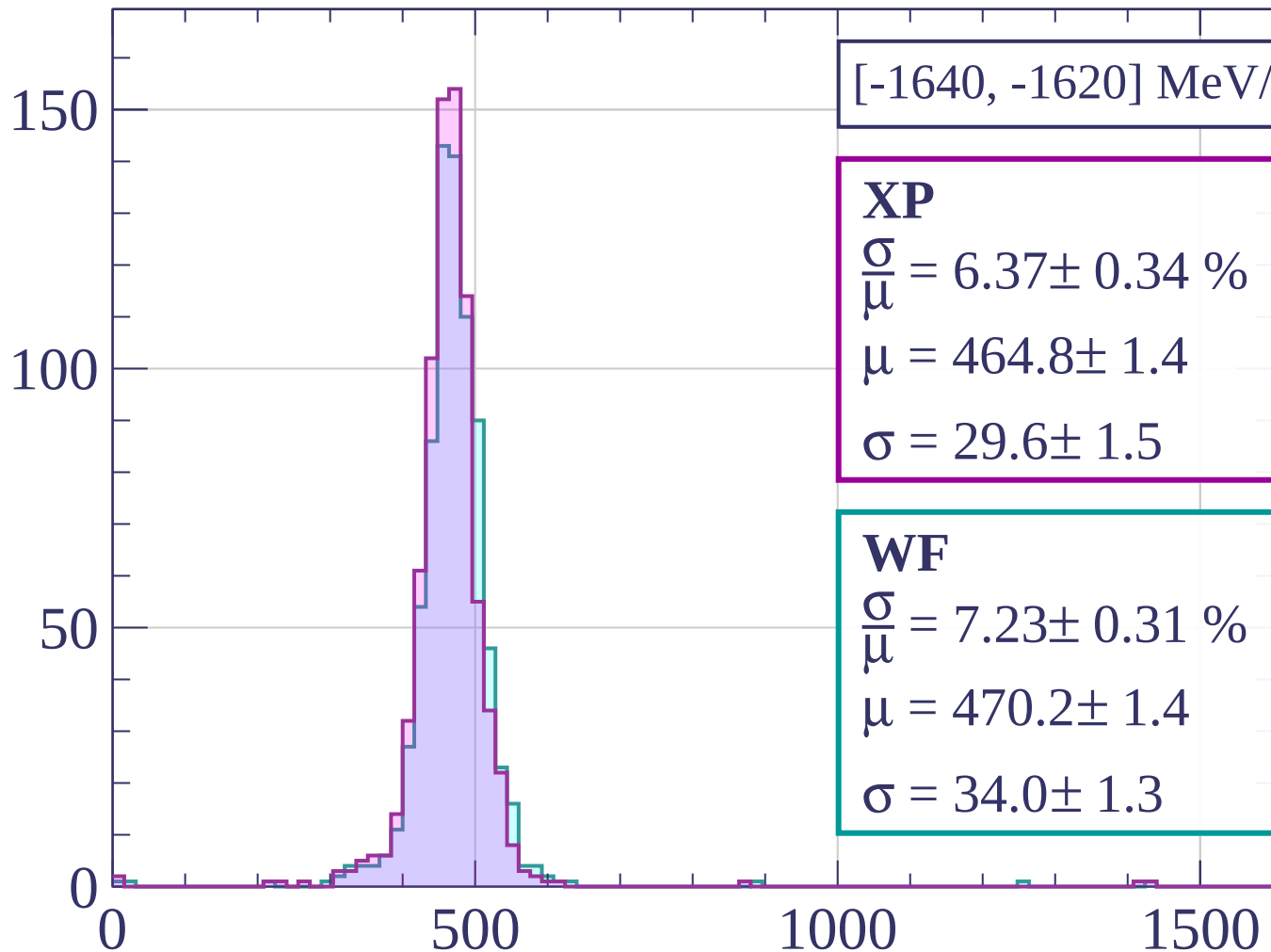
1000

1500

dE/dx [ADC counts/cm]



Count



[-1640, -1620] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 6.37 \pm 0.34 \%$$

$$\mu = 464.8 \pm 1.4$$

$$\sigma = 29.6 \pm 1.5$$

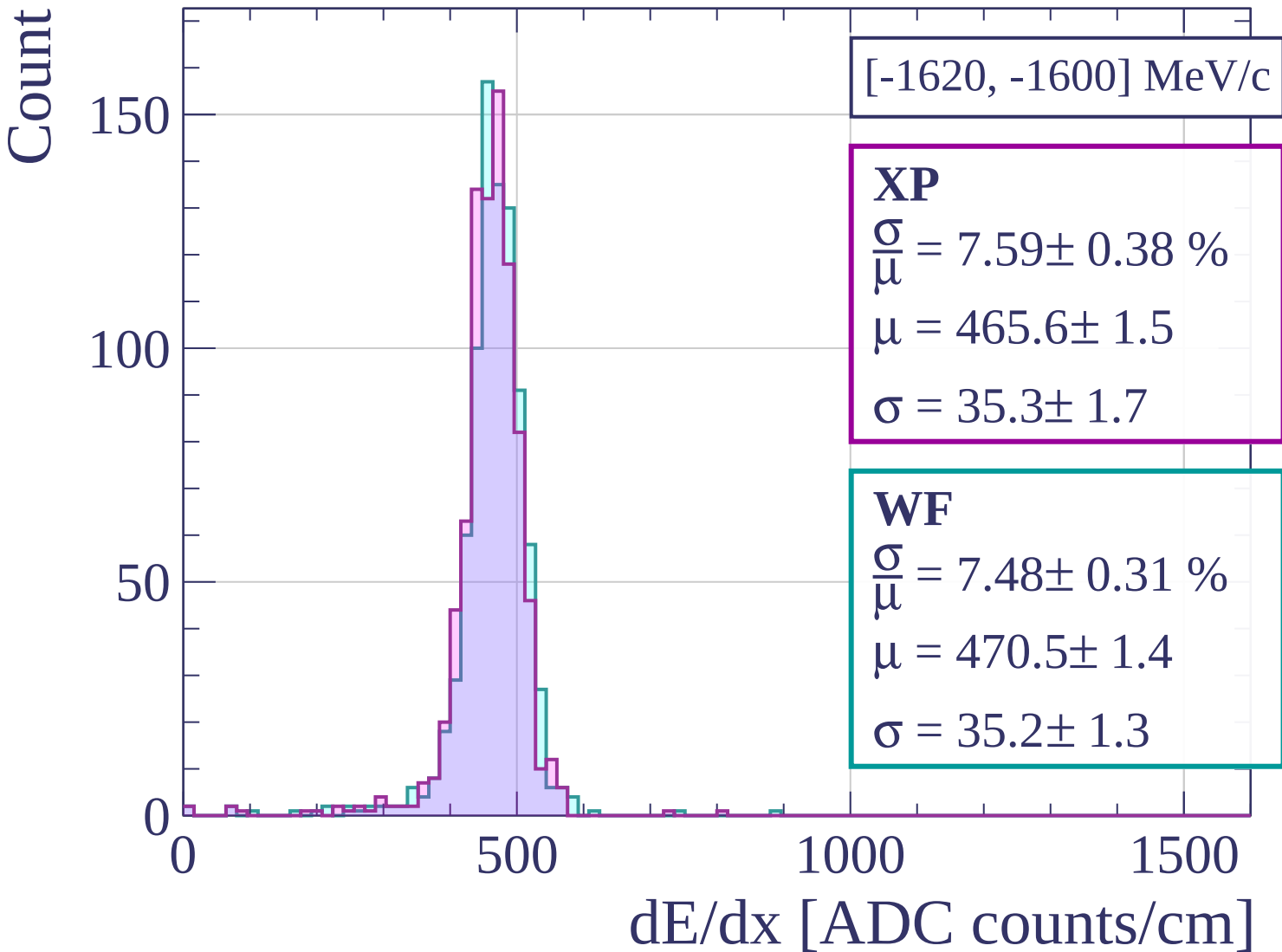
WF

$$\frac{\sigma}{\bar{\mu}} = 7.23 \pm 0.31 \%$$

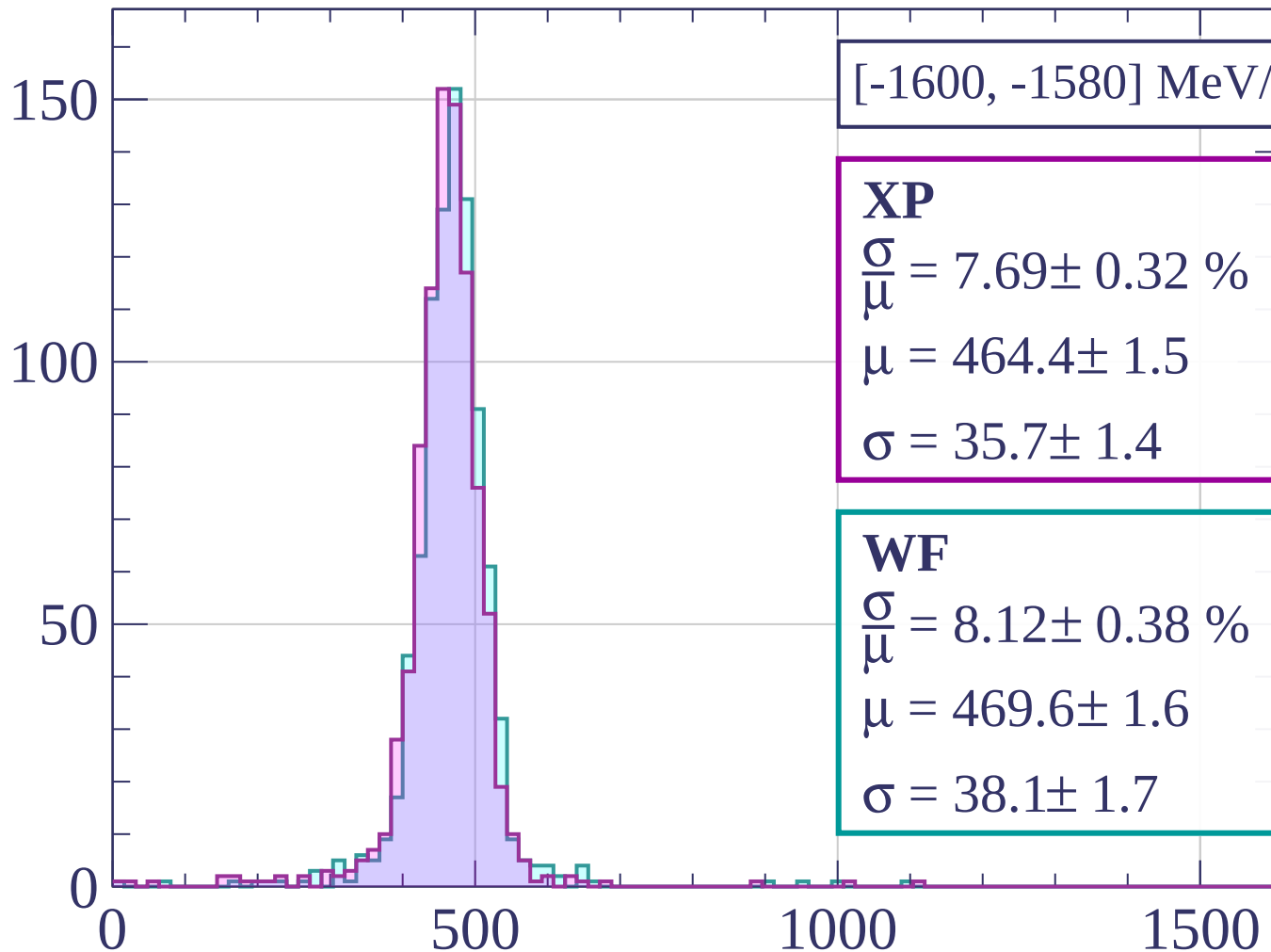
$$\mu = 470.2 \pm 1.4$$

$$\sigma = 34.0 \pm 1.3$$

dE/dx [ADC counts/cm]



Count



$[-1600, -1580] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 7.69 \pm 0.32 \%$$

$$\mu = 464.4 \pm 1.5$$

$$\sigma = 35.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.12 \pm 0.38 \%$$

$$\mu = 469.6 \pm 1.6$$

$$\sigma = 38.1 \pm 1.7$$

dE/dx [ADC counts/cm]

Count

[-1580, -1560] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.03 \pm 0.32 \%$$

$$\mu = 464.5 \pm 1.4$$

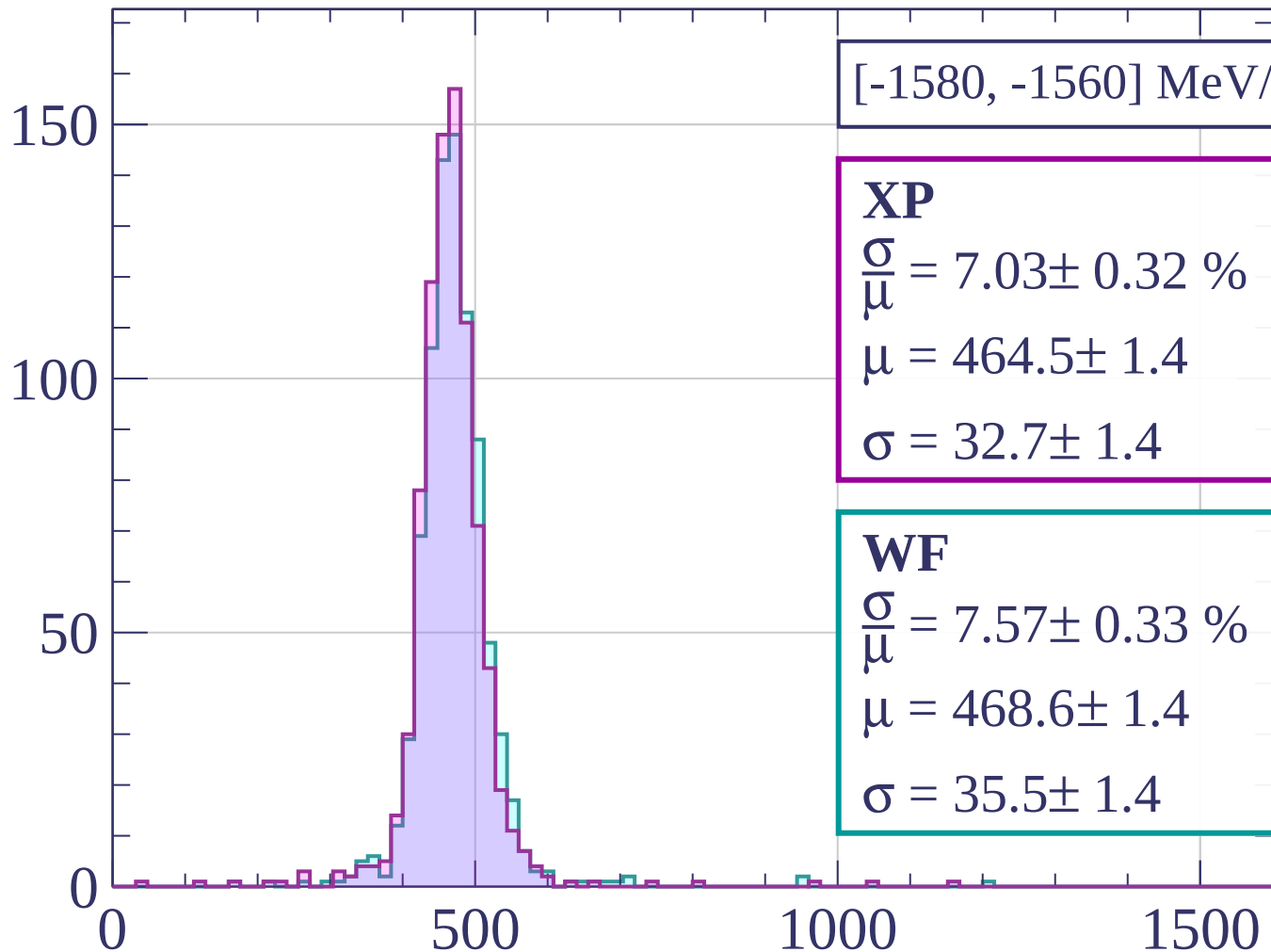
$$\sigma = 32.7 \pm 1.4$$

WF

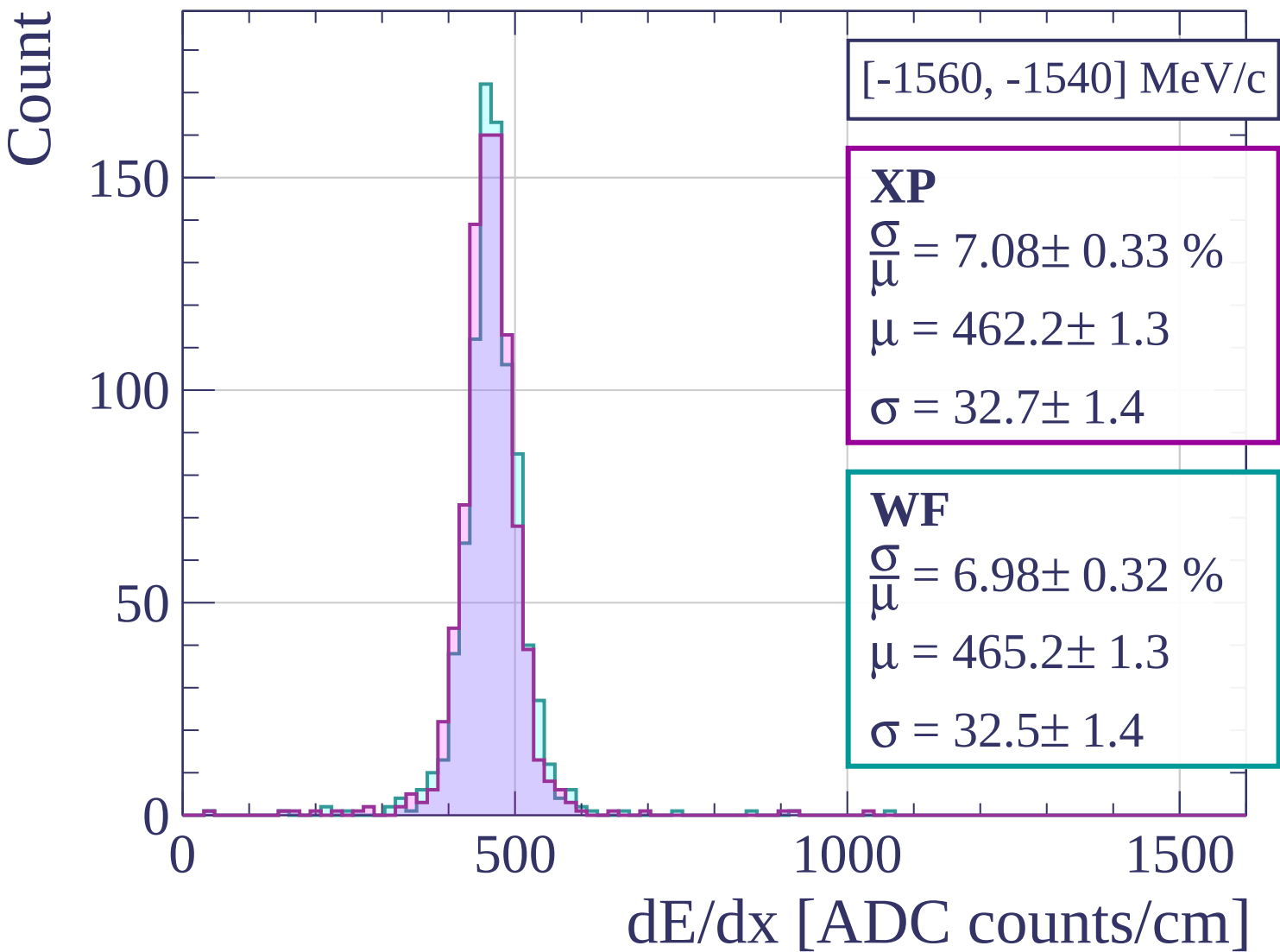
$$\frac{\sigma}{\mu} = 7.57 \pm 0.33 \%$$

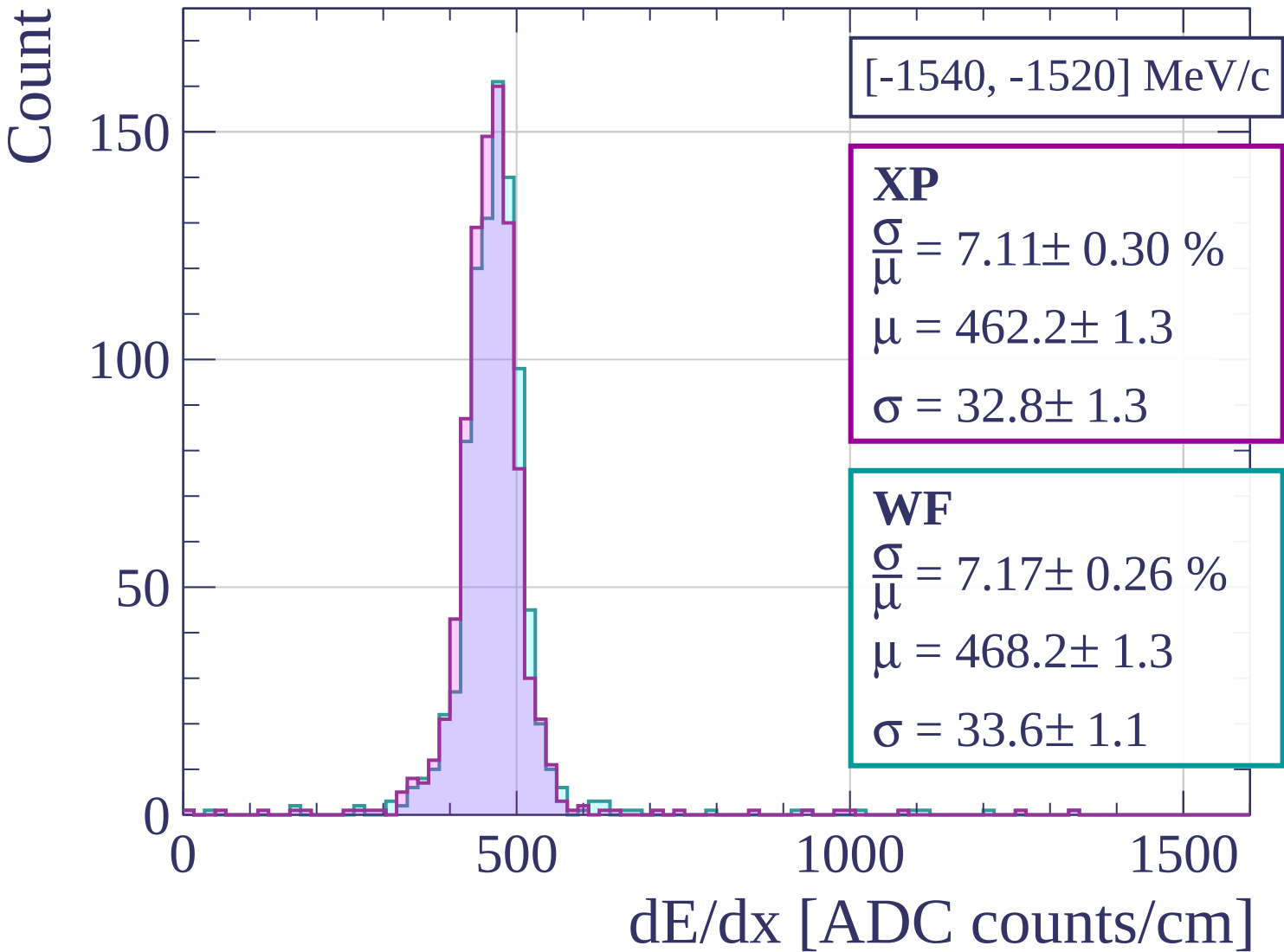
$$\mu = 468.6 \pm 1.4$$

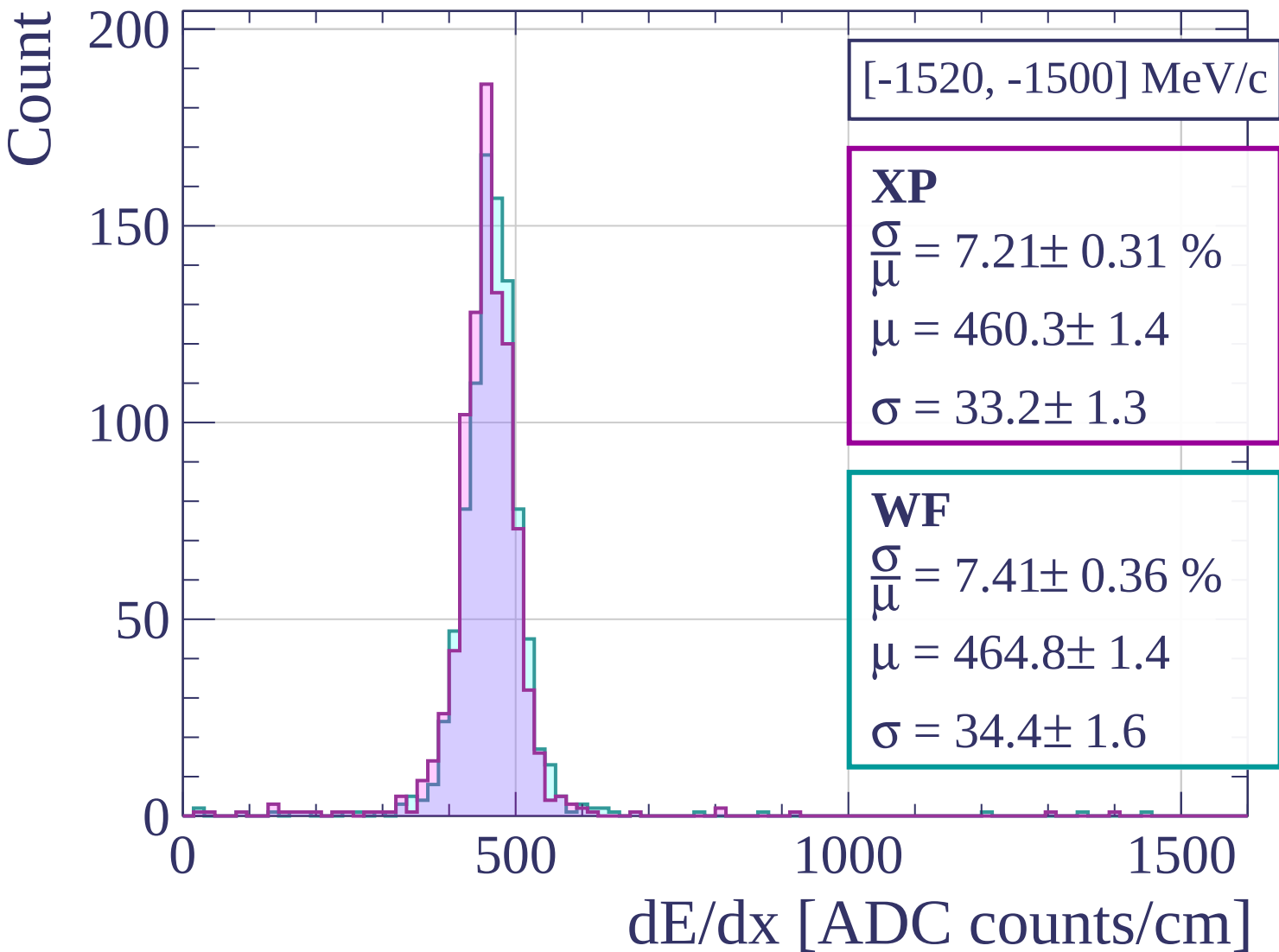
$$\sigma = 35.5 \pm 1.4$$



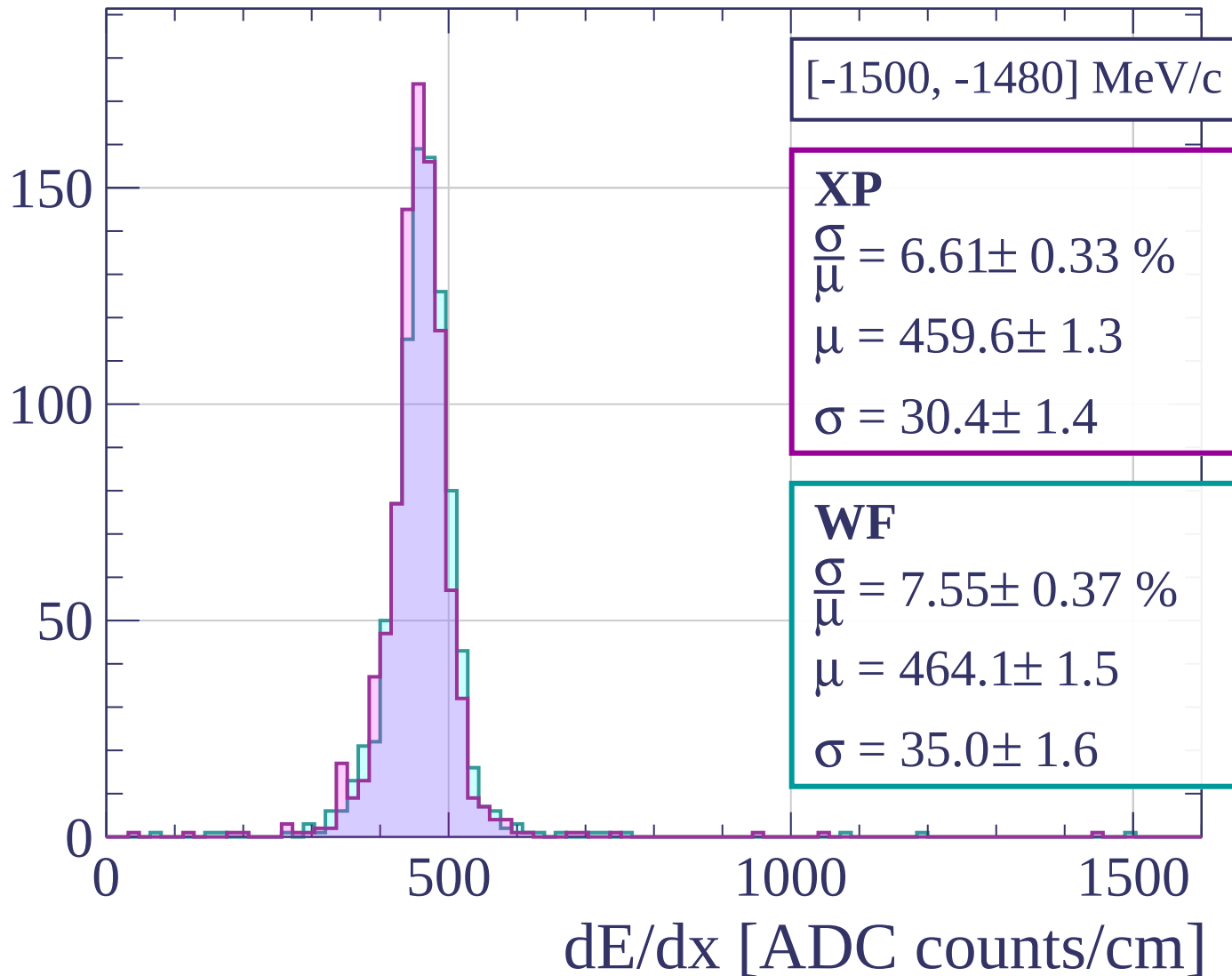
dE/dx [ADC counts/cm]







Count



Count

[-1480, -1460] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.54 \pm 0.29 \%$$

$$\mu = 460.2 \pm 1.3$$

$$\sigma = 34.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.27 \%$$

$$\mu = 465.0 \pm 1.3$$

$$\sigma = 34.4 \pm 1.2$$

150

100

50

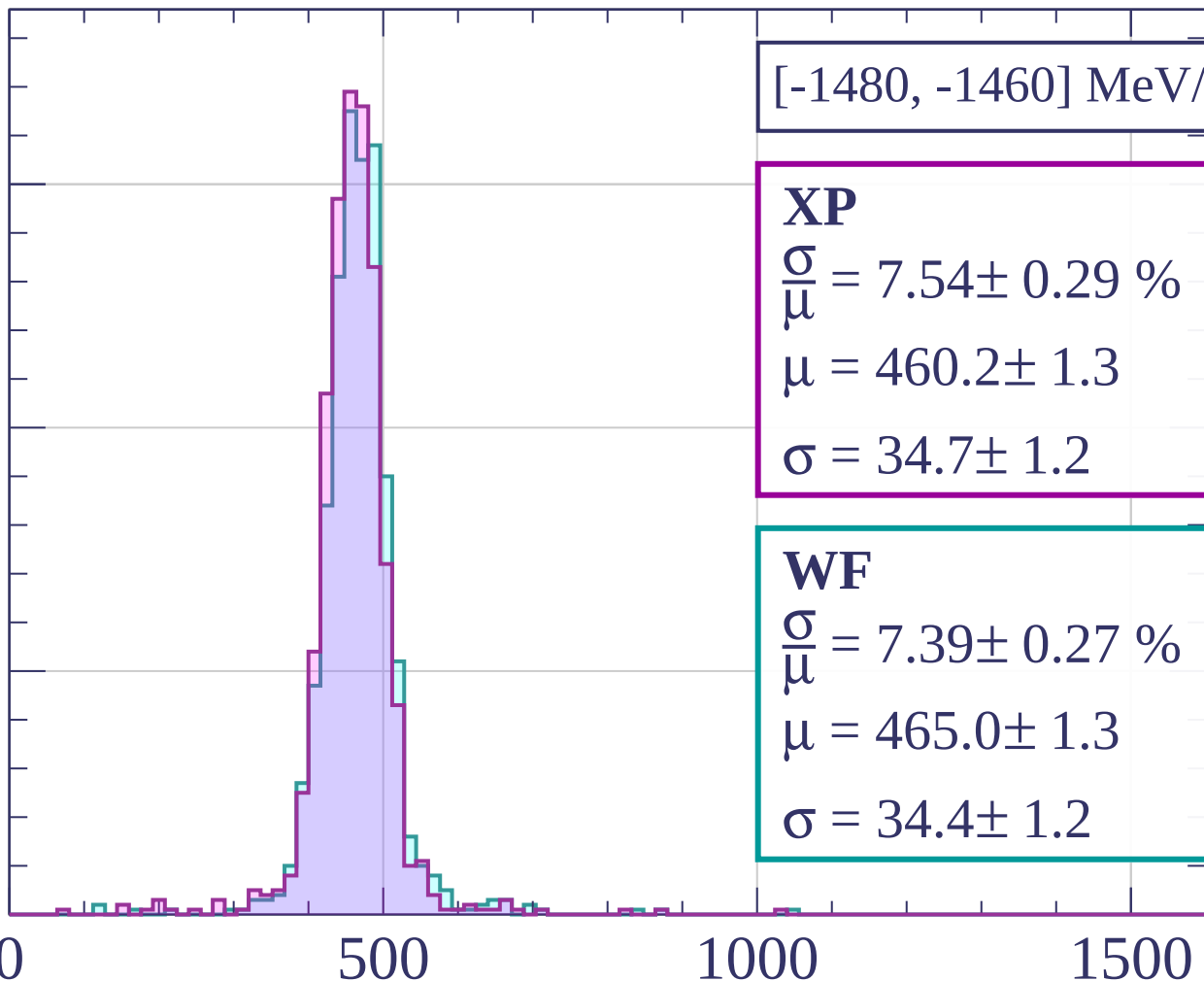
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1460, -1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.03 \pm 0.31 \%$$

$$\mu = 459.2 \pm 1.3$$

$$\sigma = 32.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.19 \pm 0.32 \%$$

$$\mu = 464.2 \pm 1.4$$

$$\sigma = 33.4 \pm 1.4$$

150

100

50

0

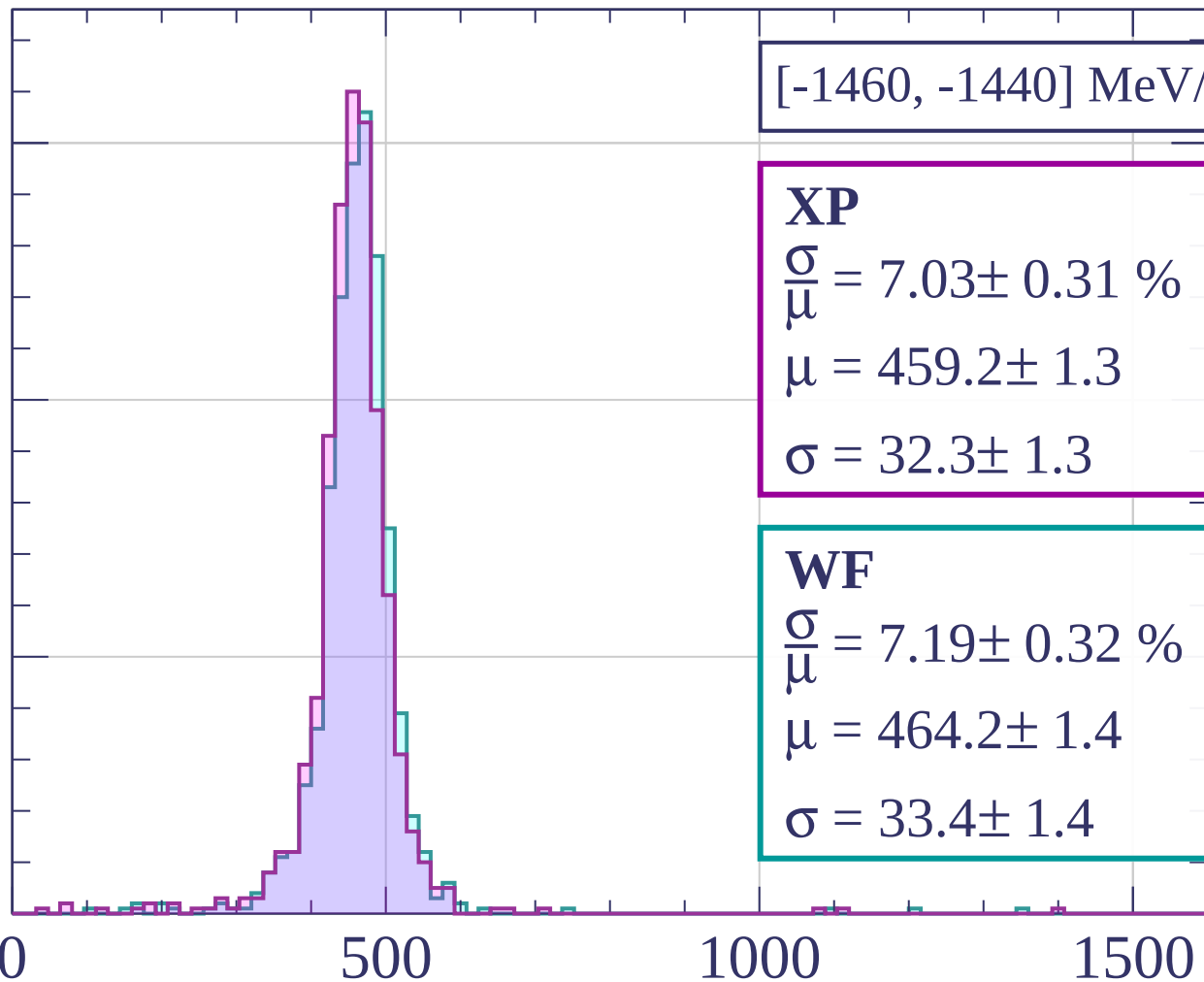
0

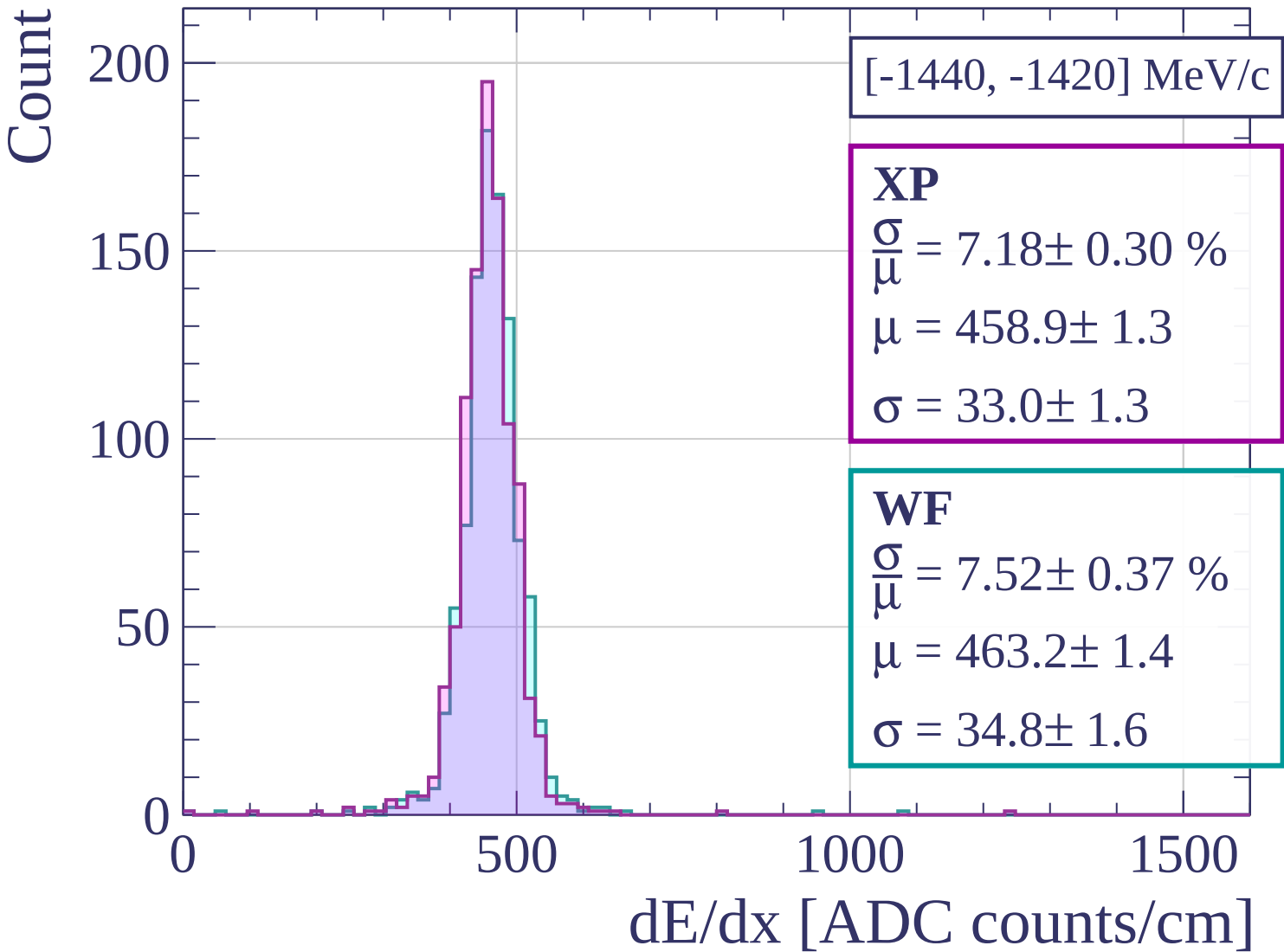
500

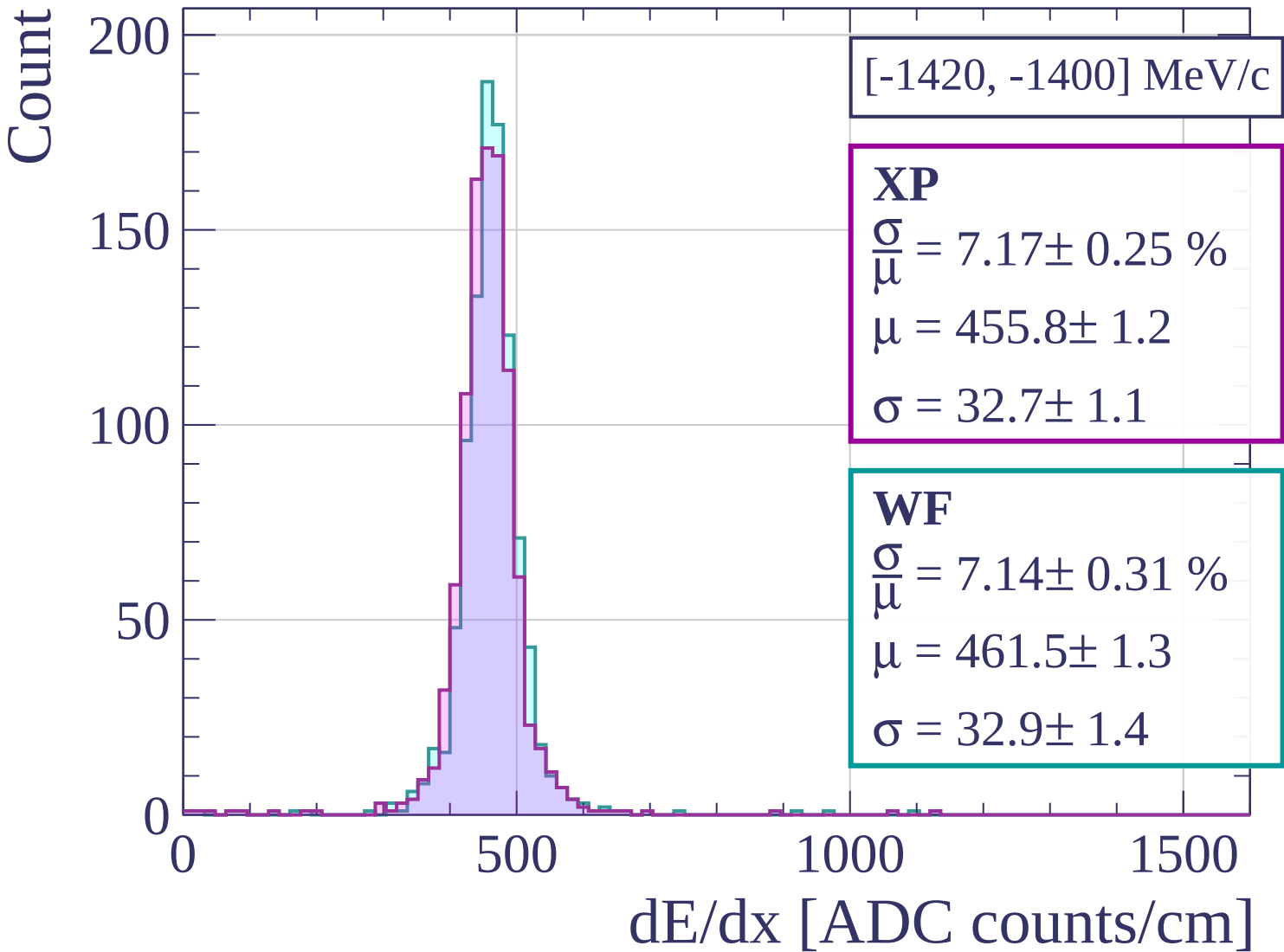
1000

1500

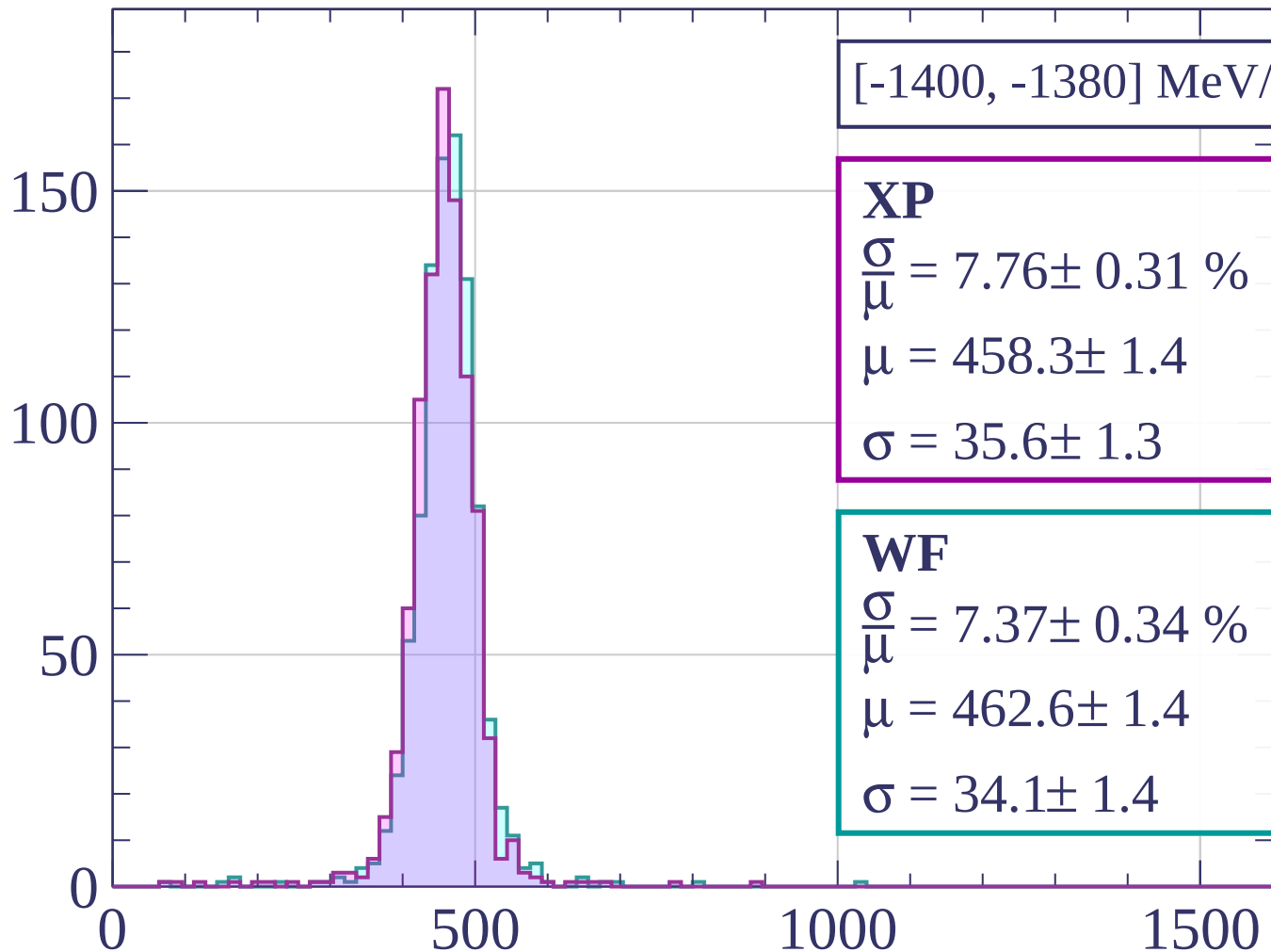
dE/dx [ADC counts/cm]







Count



[-1400, -1380] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.76 \pm 0.31 \%$$

$$\mu = 458.3 \pm 1.4$$

$$\sigma = 35.6 \pm 1.3$$

WF

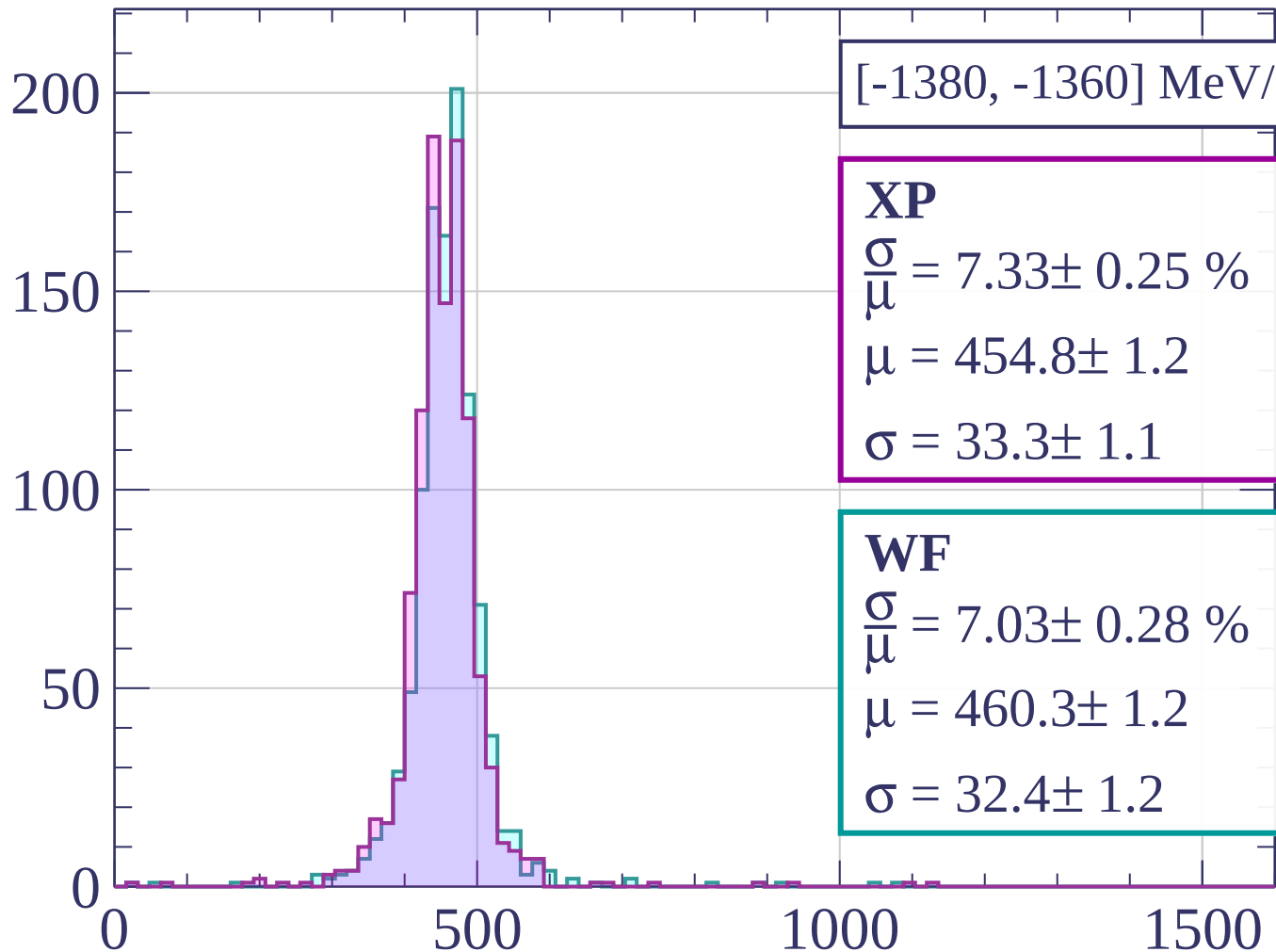
$$\frac{\sigma}{\mu} = 7.37 \pm 0.34 \%$$

$$\mu = 462.6 \pm 1.4$$

$$\sigma = 34.1 \pm 1.4$$

dE/dx [ADC counts/cm]

Count



[-1380, -1360] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 7.33 \pm 0.25 \%$$

$$\mu = 454.8 \pm 1.2$$

$$\sigma = 33.3 \pm 1.1$$

WF

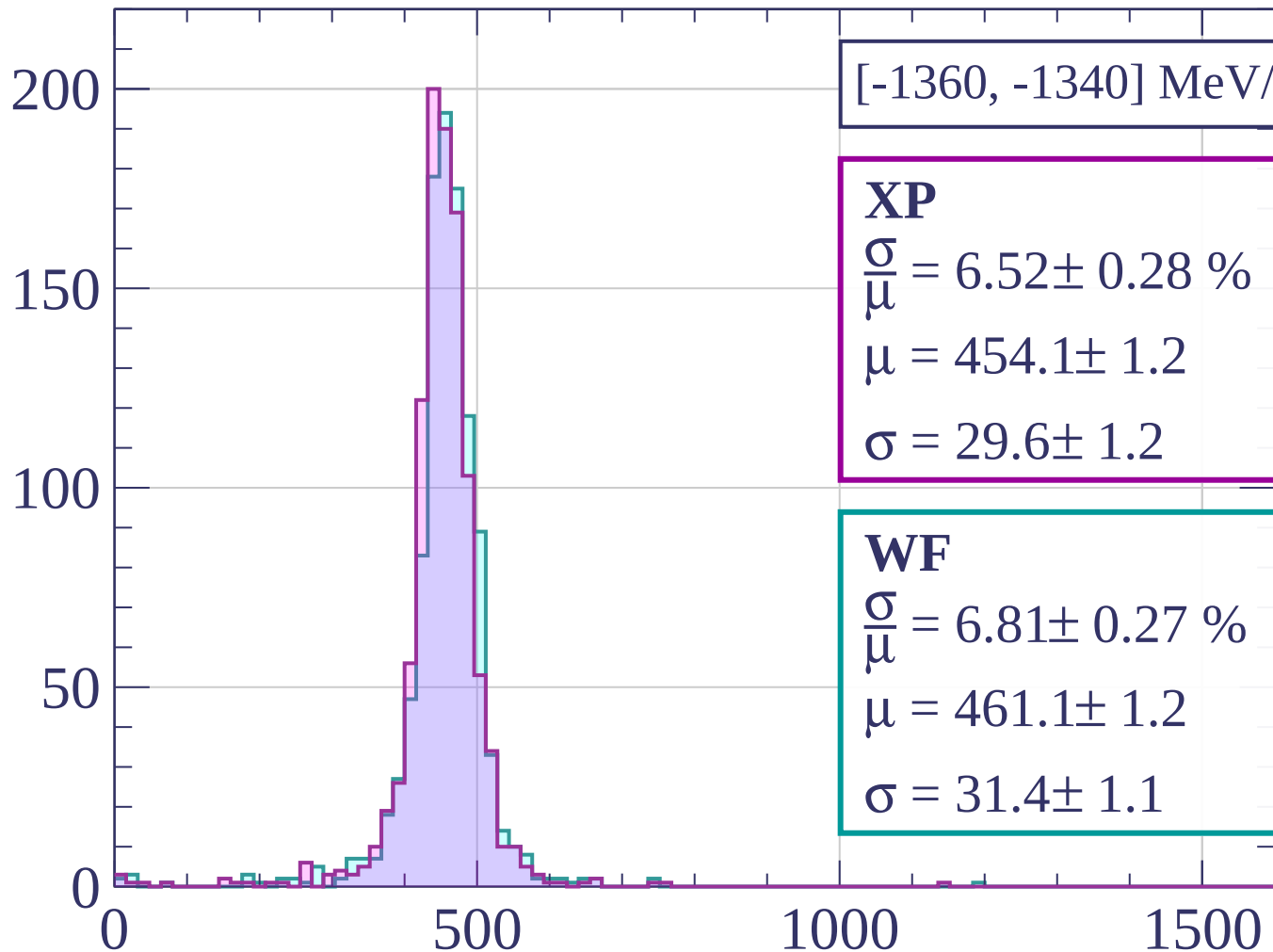
$$\frac{\sigma}{\bar{\mu}} = 7.03 \pm 0.28 \%$$

$$\mu = 460.3 \pm 1.2$$

$$\sigma = 32.4 \pm 1.2$$

dE/dx [ADC counts/cm]

Count



[-1360, -1340] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.52 \pm 0.28 \%$$

$$\mu = 454.1 \pm 1.2$$

$$\sigma = 29.6 \pm 1.2$$

WF

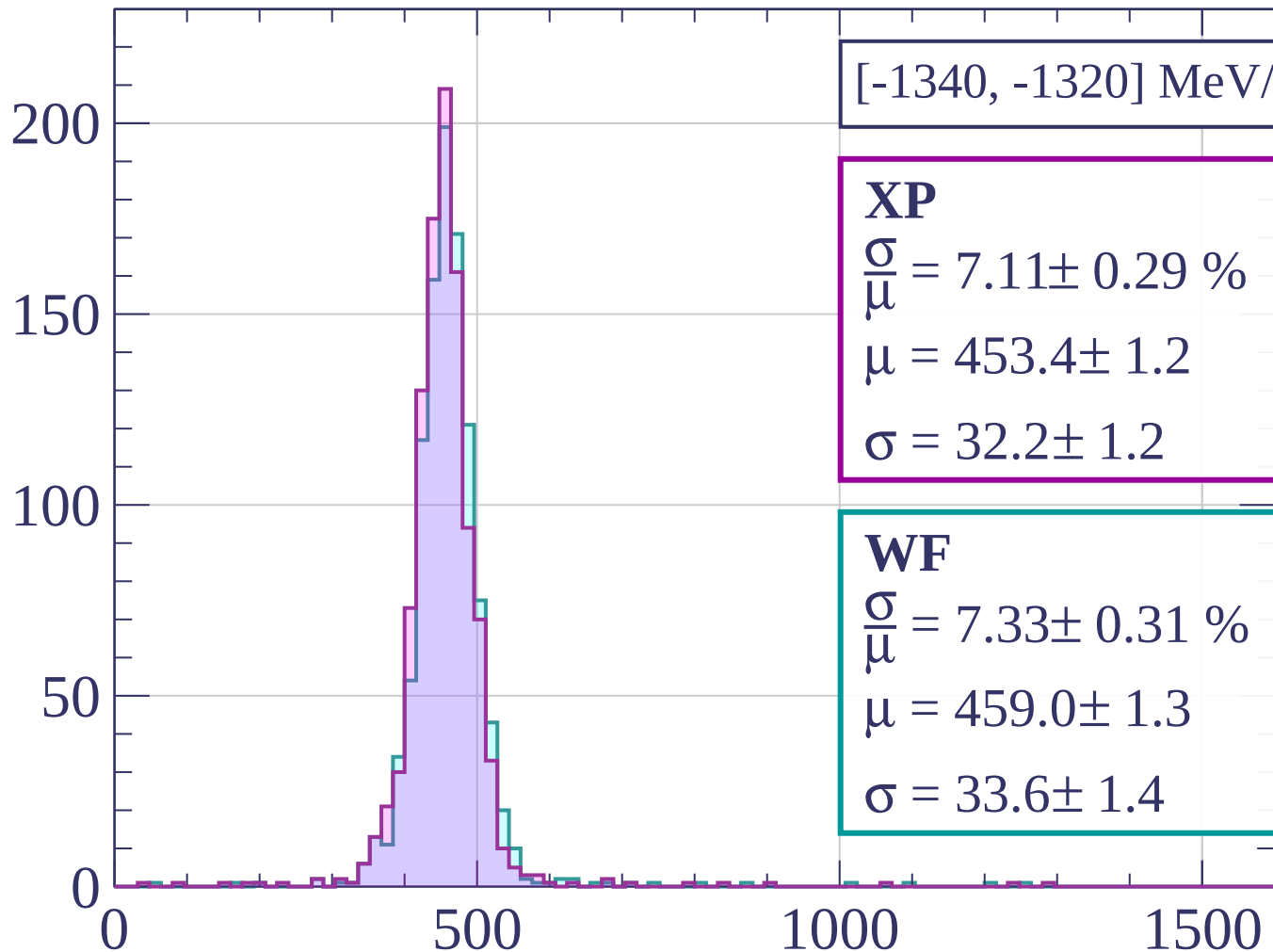
$$\frac{\sigma}{\mu} = 6.81 \pm 0.27 \%$$

$$\mu = 461.1 \pm 1.2$$

$$\sigma = 31.4 \pm 1.1$$

dE/dx [ADC counts/cm]

Count



[-1340, -1320] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.11 \pm 0.29 \%$$

$$\mu = 453.4 \pm 1.2$$

$$\sigma = 32.2 \pm 1.2$$

WF

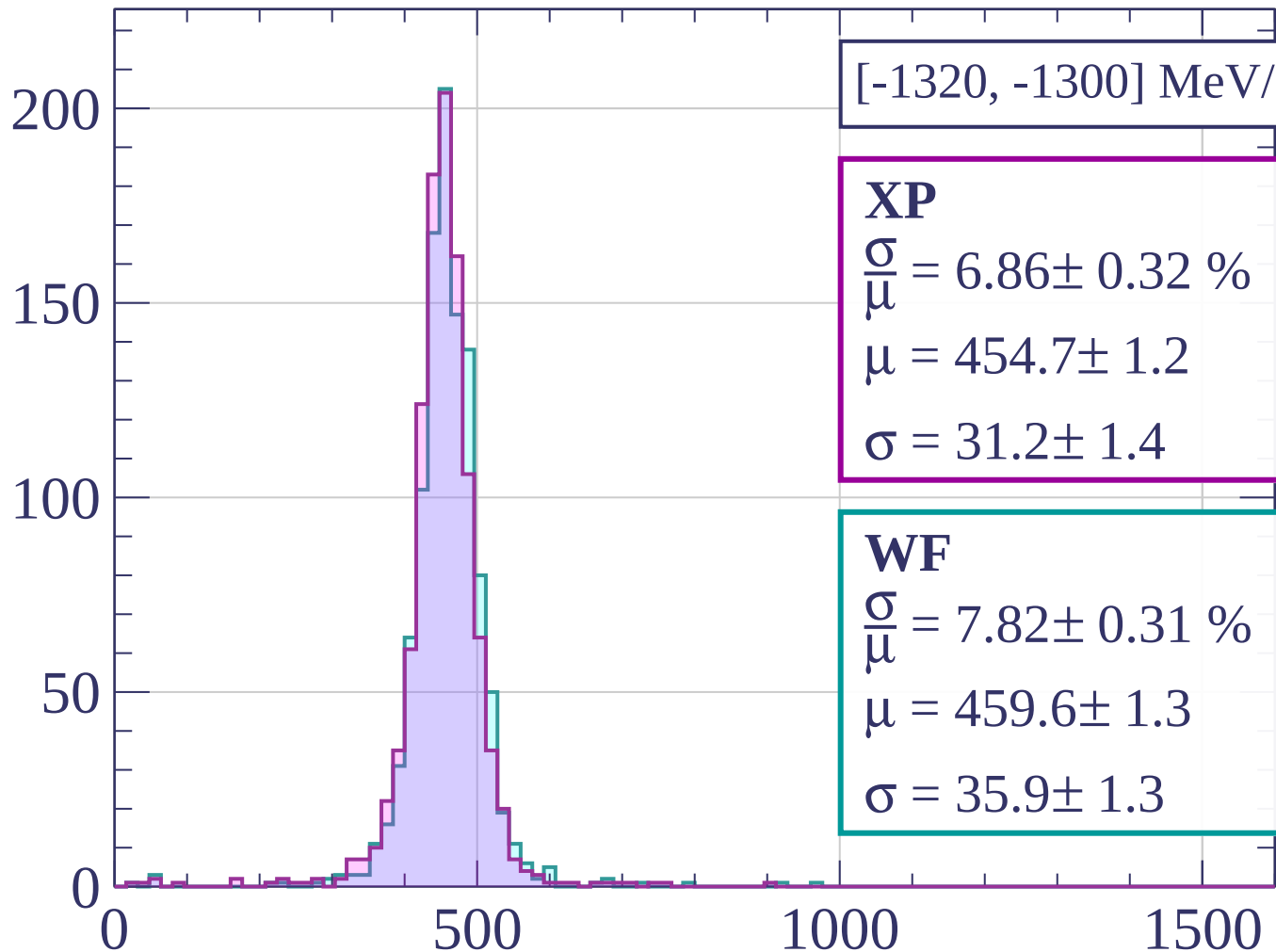
$$\frac{\sigma}{\mu} = 7.33 \pm 0.31 \%$$

$$\mu = 459.0 \pm 1.3$$

$$\sigma = 33.6 \pm 1.4$$

dE/dx [ADC counts/cm]

Count



[-1320, -1300] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.86 \pm 0.32 \%$$

$$\mu = 454.7 \pm 1.2$$

$$\sigma = 31.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.82 \pm 0.31 \%$$

$$\mu = 459.6 \pm 1.3$$

$$\sigma = 35.9 \pm 1.3$$

dE/dx [ADC counts/cm]

Count

[-1300, -1280] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.42 \pm 0.35 \%$$

$$\mu = 451.3 \pm 1.3$$

$$\sigma = 33.5 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.78 \pm 0.30 \%$$

$$\mu = 455.0 \pm 1.3$$

$$\sigma = 35.4 \pm 1.3$$

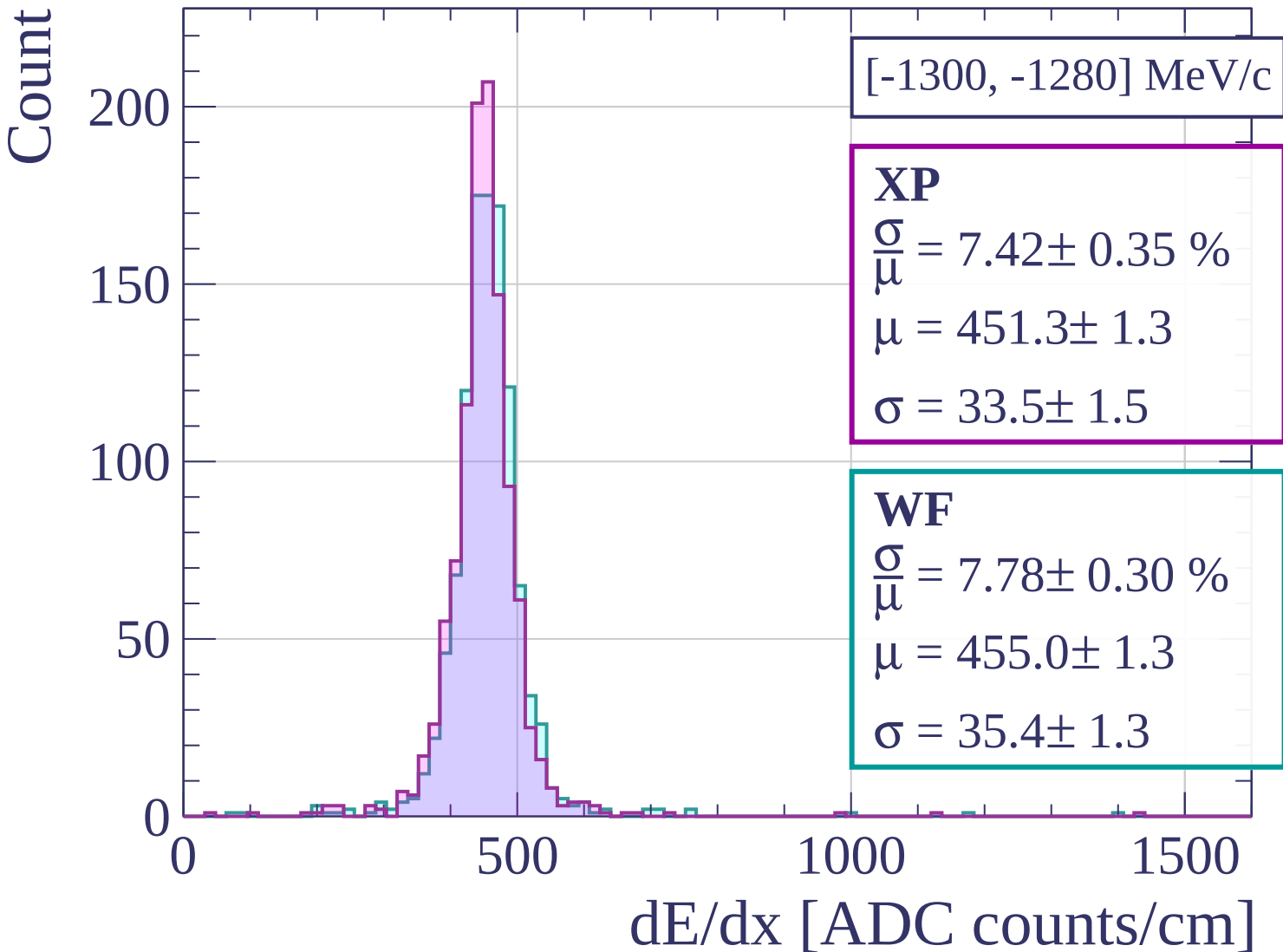
0

500

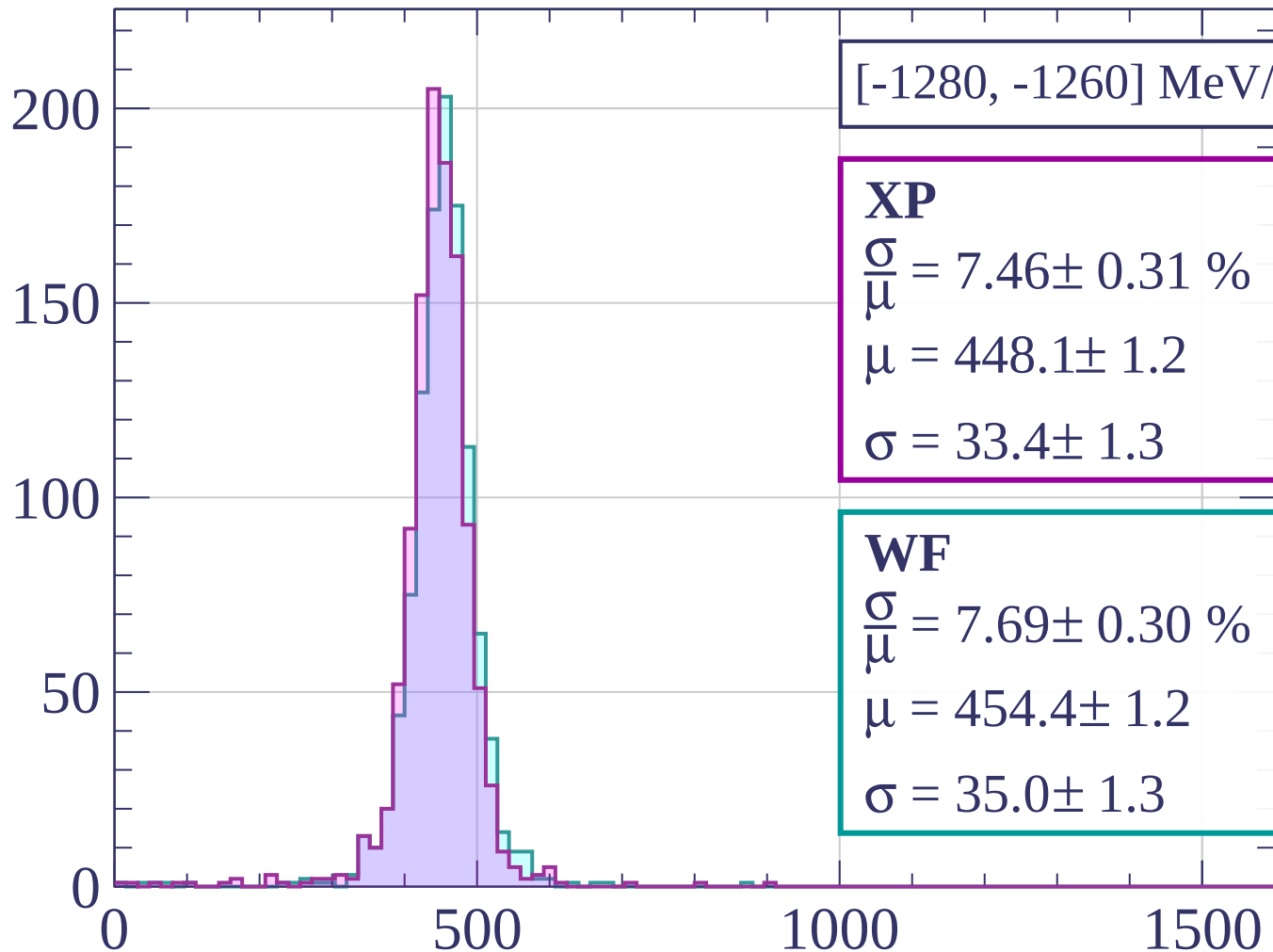
1000

1500

dE/dx [ADC counts/cm]



Count



[-1280, -1260] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.46 \pm 0.31 \%$$

$$\mu = 448.1 \pm 1.2$$

$$\sigma = 33.4 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.69 \pm 0.30 \%$$

$$\mu = 454.4 \pm 1.2$$

$$\sigma = 35.0 \pm 1.3$$

dE/dx [ADC counts/cm]

Count

[-1260, -1240] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.49 \pm 0.25 \%$$

$$\mu = 449.2 \pm 1.2$$

$$\sigma = 33.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.59 \pm 0.27 \%$$

$$\mu = 454.8 \pm 1.2$$

$$\sigma = 34.5 \pm 1.1$$

200

150

100

50

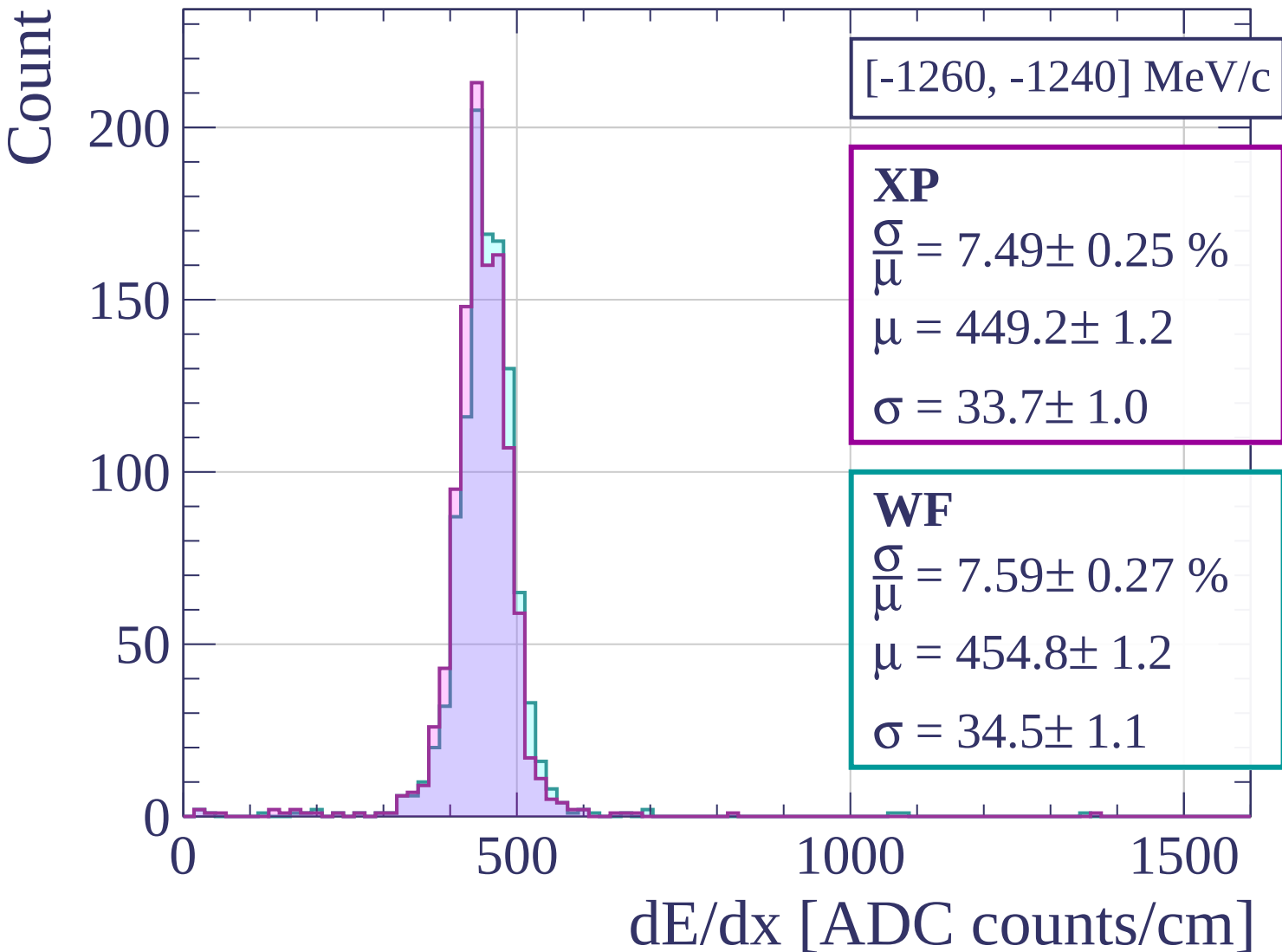
0

500

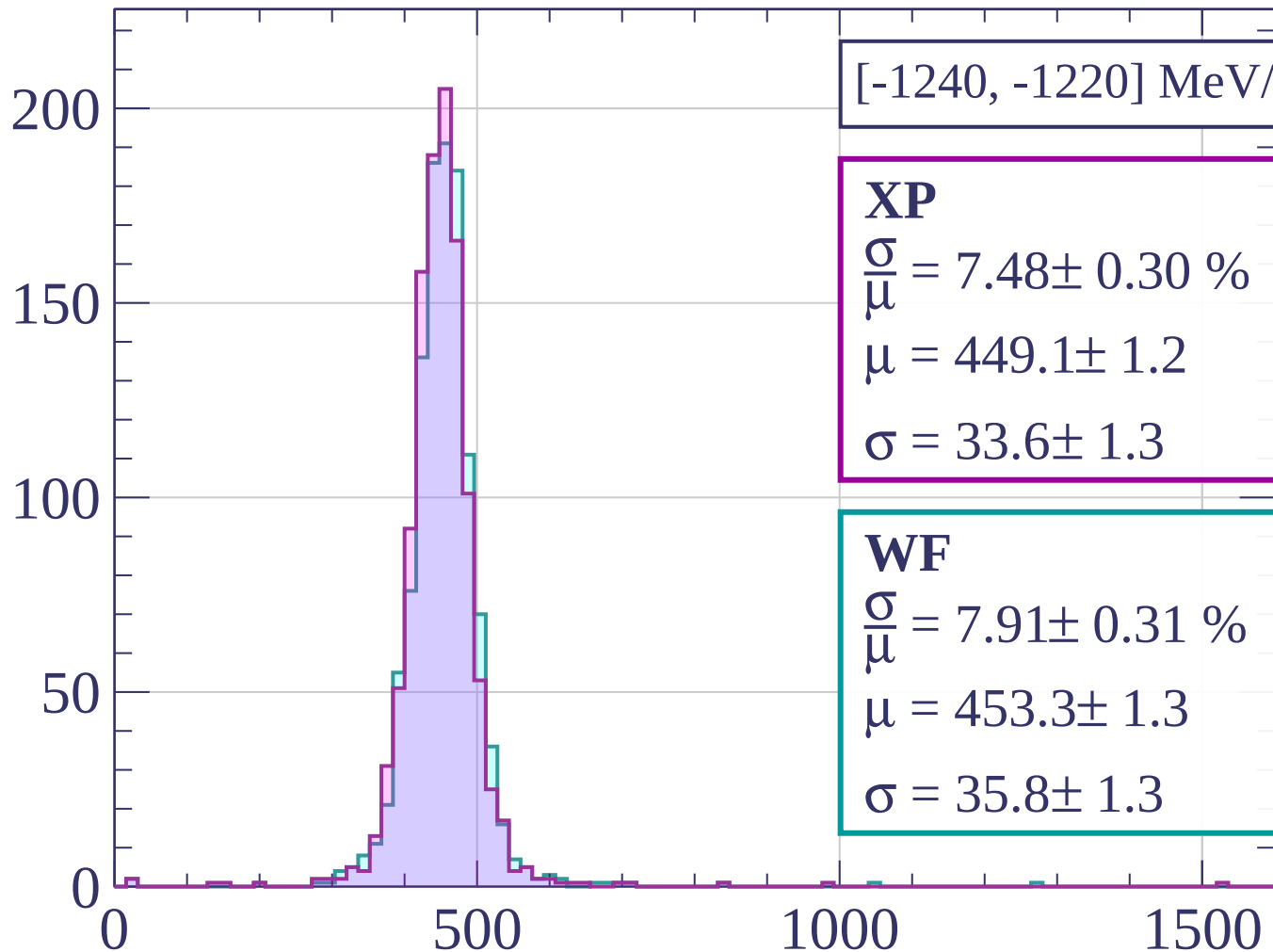
1000

1500

dE/dx [ADC counts/cm]



Count



[-1240, -1220] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.48 \pm 0.30 \%$$

$$\mu = 449.1 \pm 1.2$$

$$\sigma = 33.6 \pm 1.3$$

WF

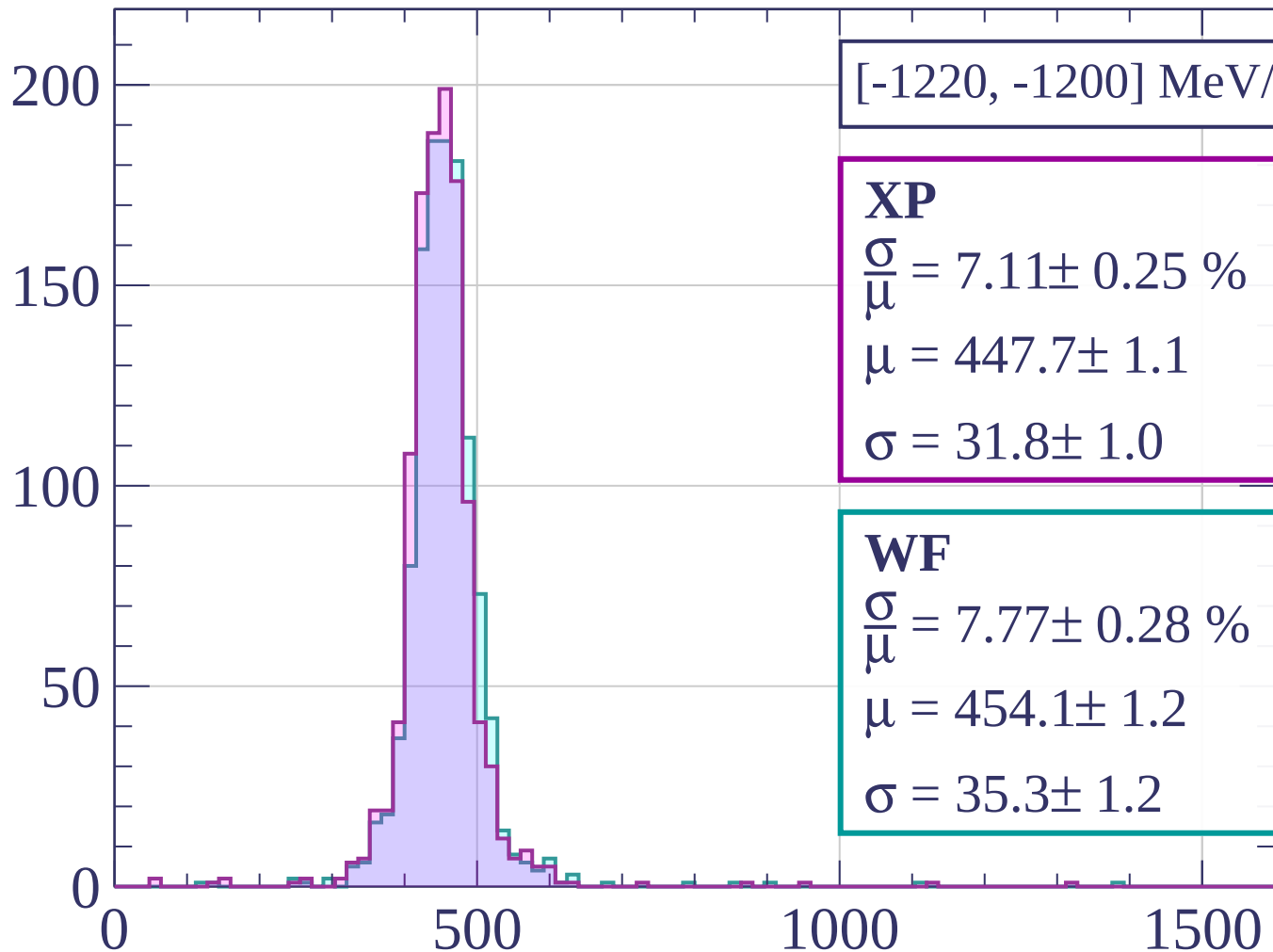
$$\frac{\sigma}{\mu} = 7.91 \pm 0.31 \%$$

$$\mu = 453.3 \pm 1.3$$

$$\sigma = 35.8 \pm 1.3$$

dE/dx [ADC counts/cm]

Count



[-1220, -1200] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.11 \pm 0.25 \%$$

$$\mu = 447.7 \pm 1.1$$

$$\sigma = 31.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.77 \pm 0.28 \%$$

$$\mu = 454.1 \pm 1.2$$

$$\sigma = 35.3 \pm 1.2$$

dE/dx [ADC counts/cm]

Count

[-1200, -1180] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.34 \pm 0.27 \%$$

$$\mu = 447.9 \pm 1.2$$

$$\sigma = 37.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.49 \pm 0.29 \%$$

$$\mu = 452.7 \pm 1.3$$

$$\sigma = 38.4 \pm 1.2$$

200

150

100

50

0

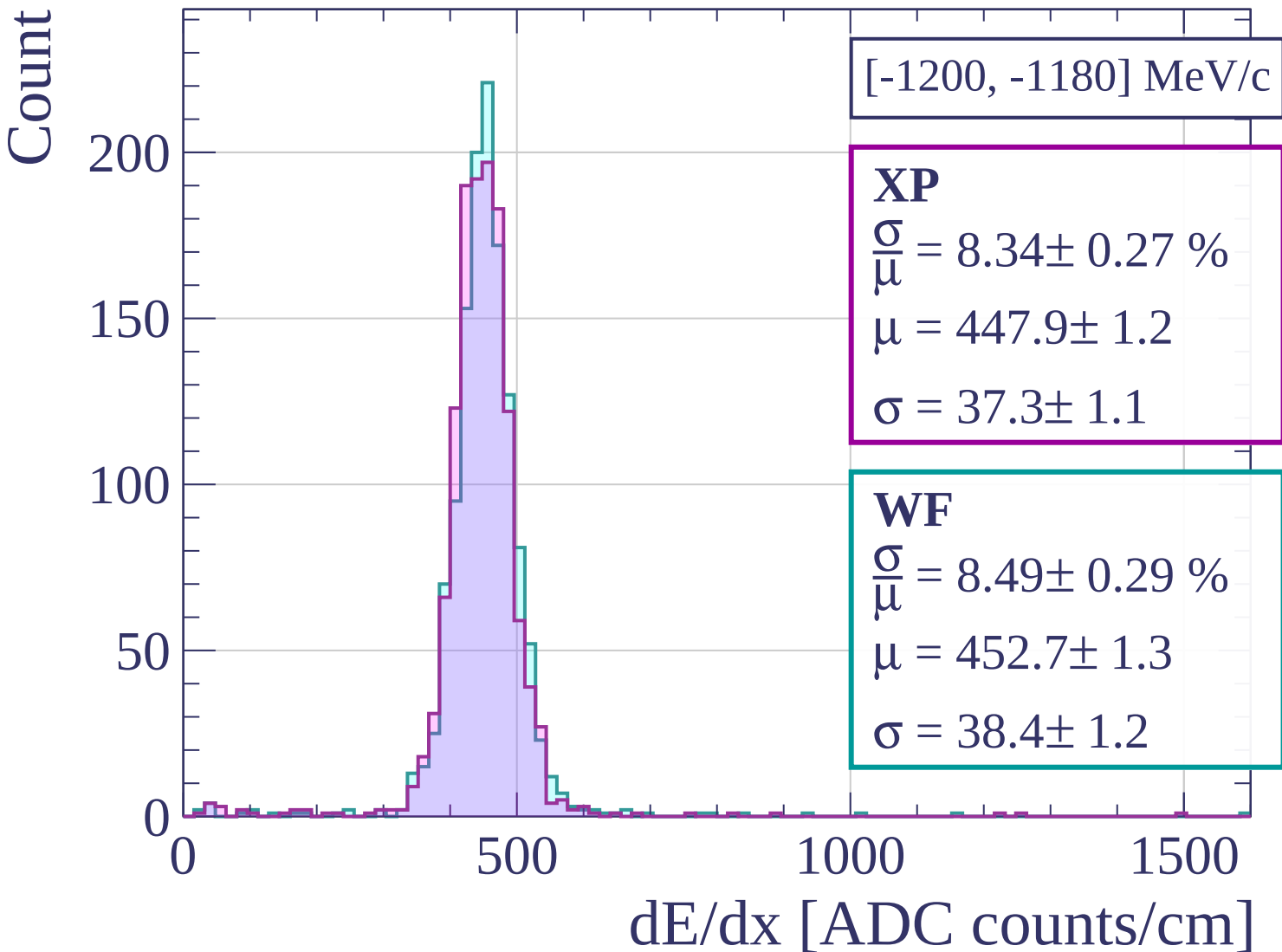
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1180, -1160] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.24 \pm 0.27 \%$$

$$\mu = 445.0 \pm 1.1$$

$$\sigma = 32.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.12 \pm 0.27 \%$$

$$\mu = 450.6 \pm 1.1$$

$$\sigma = 32.1 \pm 1.1$$

200

100

0

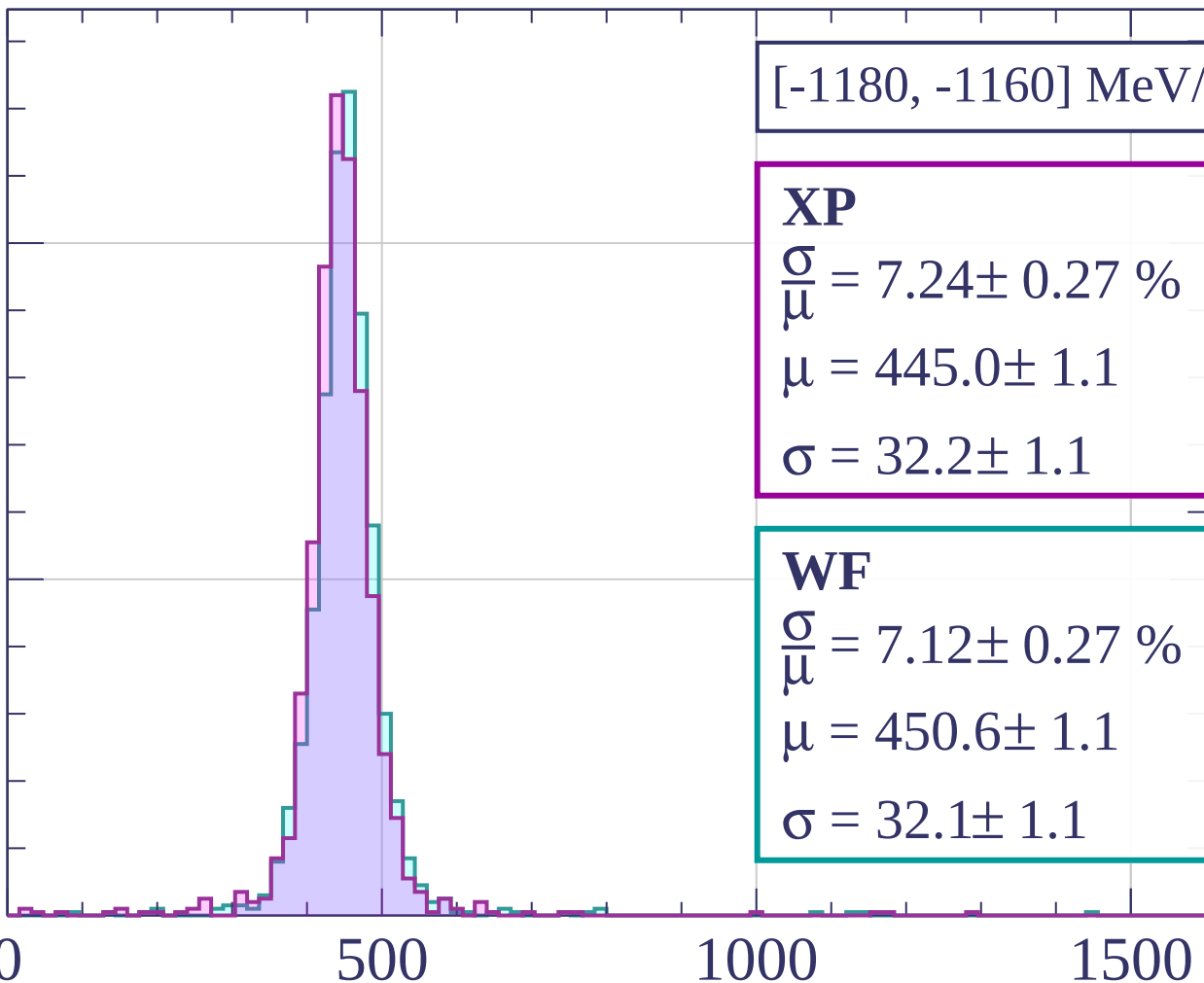
0

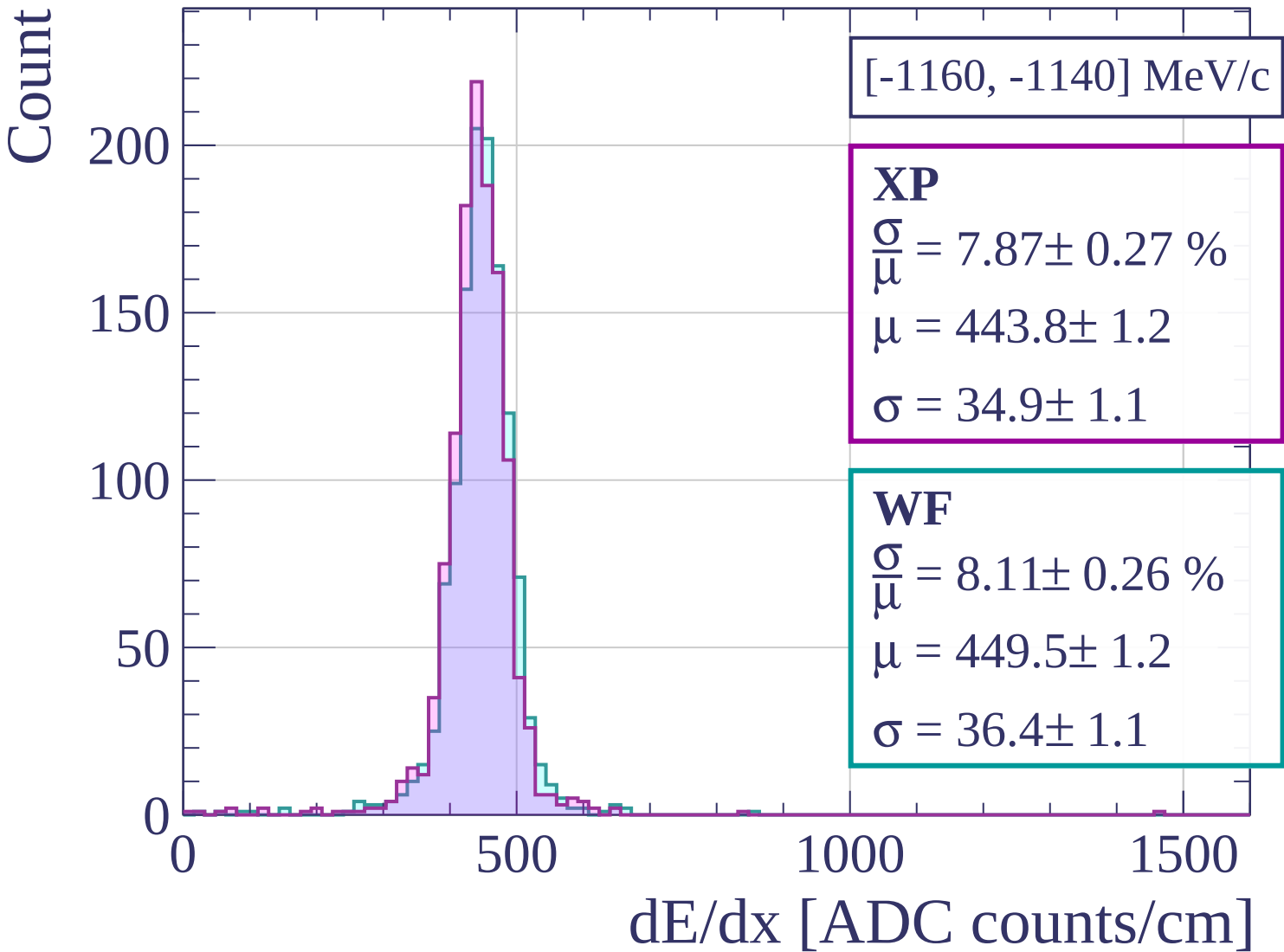
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-1140, -1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.69 \pm 0.31 \%$$

$$\mu = 443.3 \pm 1.2$$

$$\sigma = 34.1 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.48 \pm 0.28 \%$$

$$\mu = 448.3 \pm 1.2$$

$$\sigma = 33.6 \pm 1.2$$

200

100

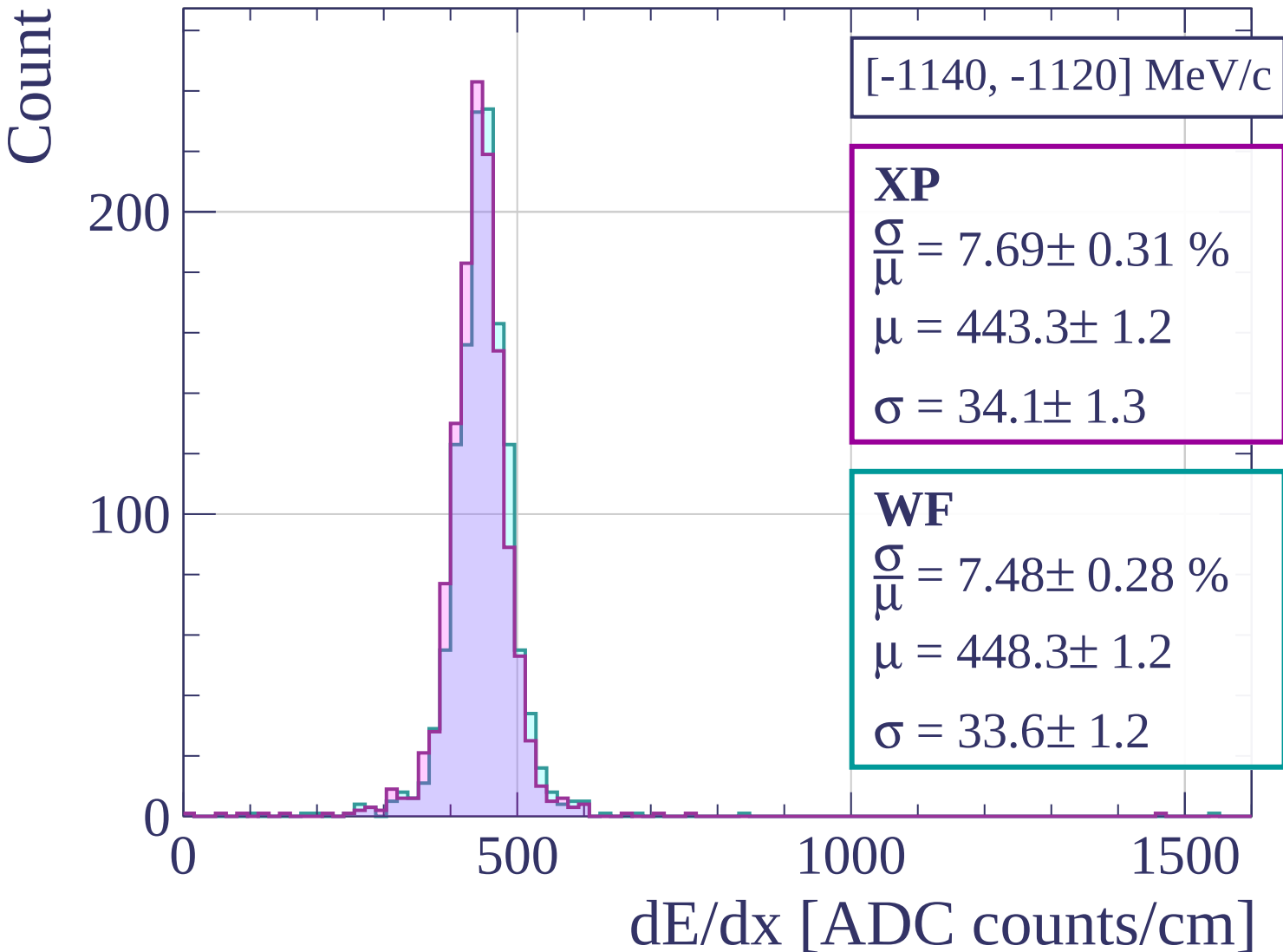
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1120, -1100] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.33 \pm 0.29 \%$$

$$\mu = 442.2 \pm 1.1$$

$$\sigma = 32.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.92 \pm 0.31 \%$$

$$\mu = 445.1 \pm 1.1$$

$$\sigma = 30.8 \pm 1.3$$

200

100

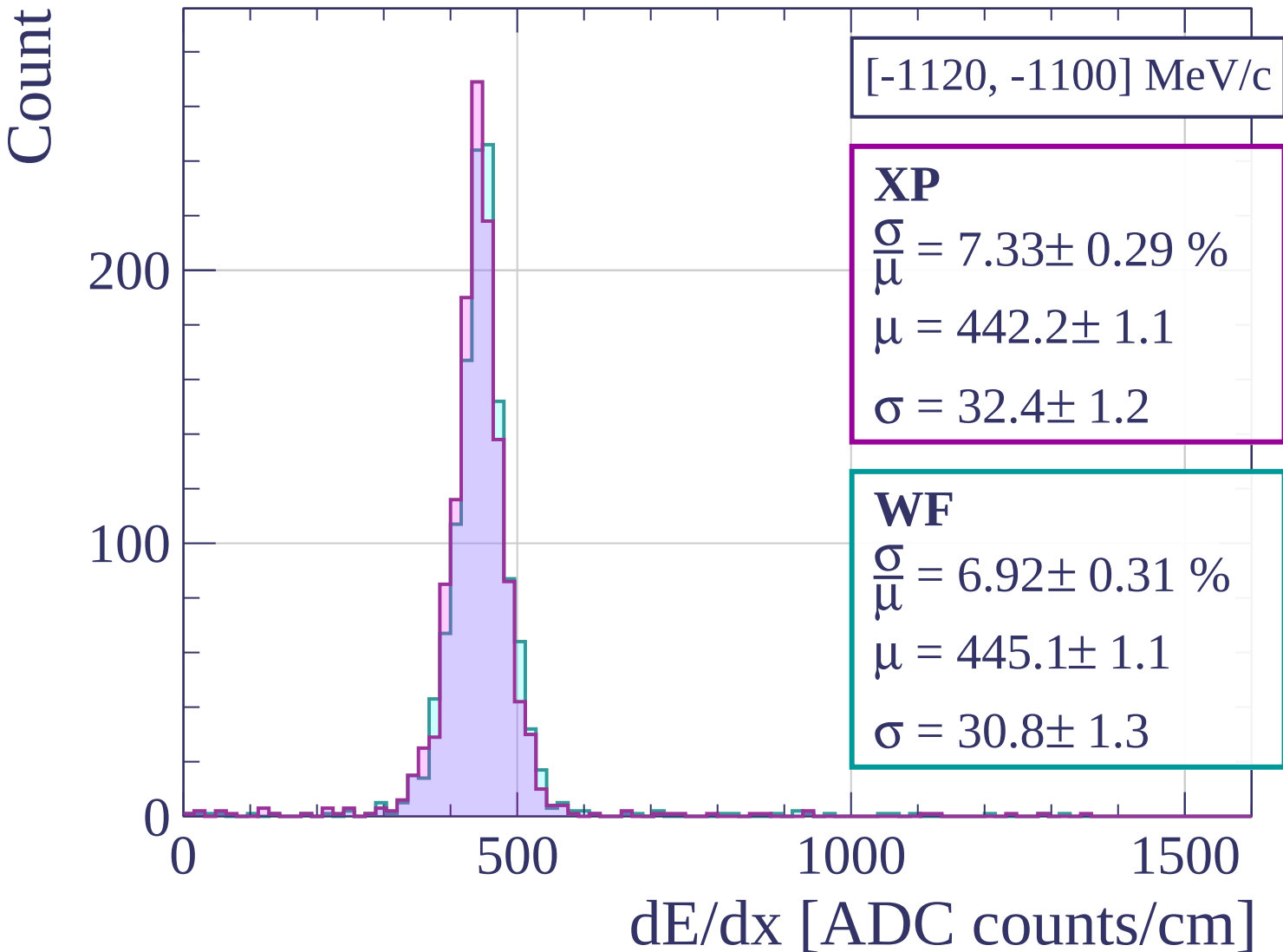
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1100, -1080] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.81 \pm 0.27 \%$$

$$\mu = 442.7 \pm 1.1$$

$$\sigma = 34.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.35 \pm 0.28 \%$$

$$\mu = 447.8 \pm 1.2$$

$$\sigma = 37.4 \pm 1.1$$

200

100

0

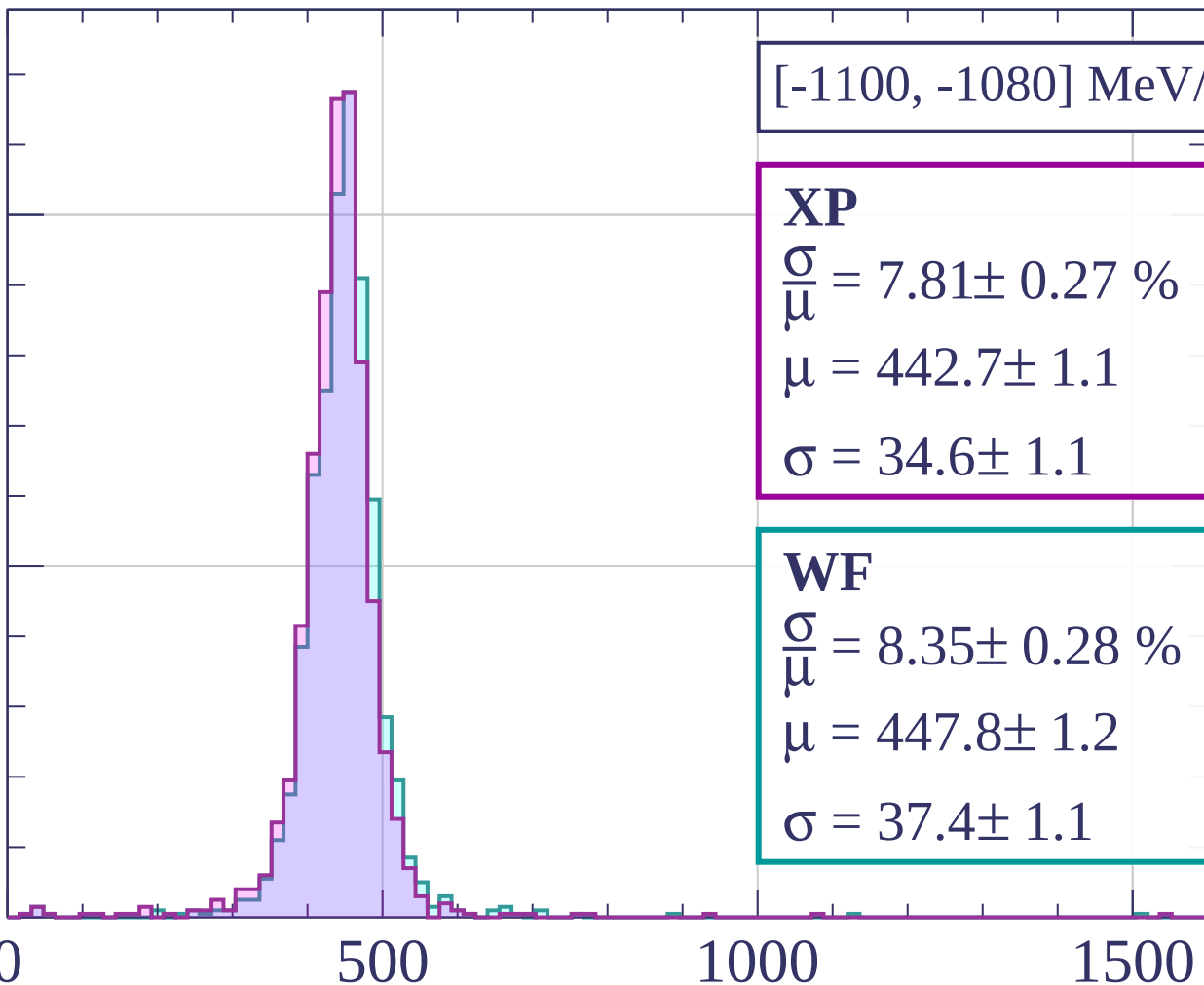
0

500

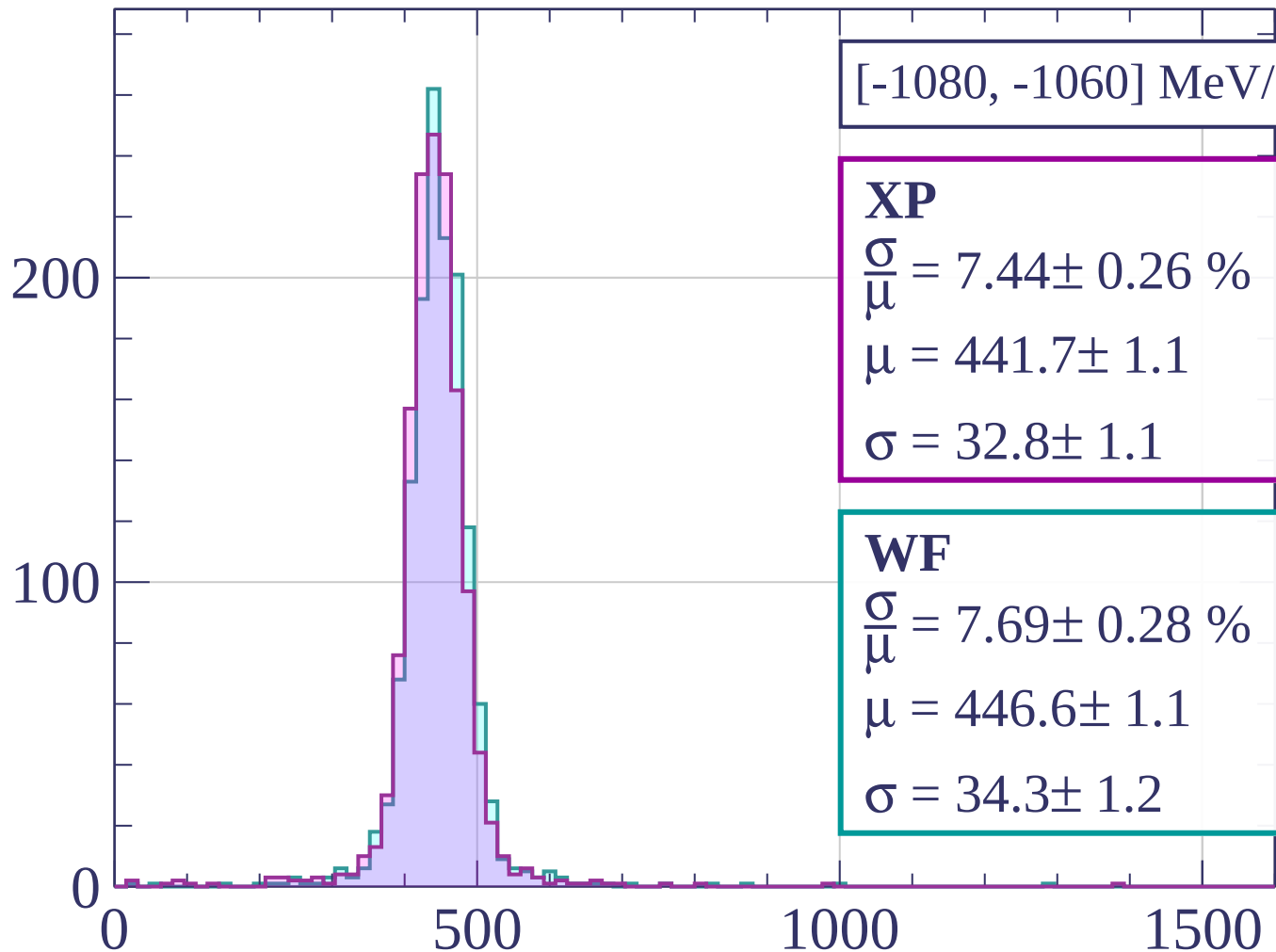
1000

1500

dE/dx [ADC counts/cm]



Count



[-1080, -1060] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.44 \pm 0.26 \%$$

$$\mu = 441.7 \pm 1.1$$

$$\sigma = 32.8 \pm 1.1$$

WF

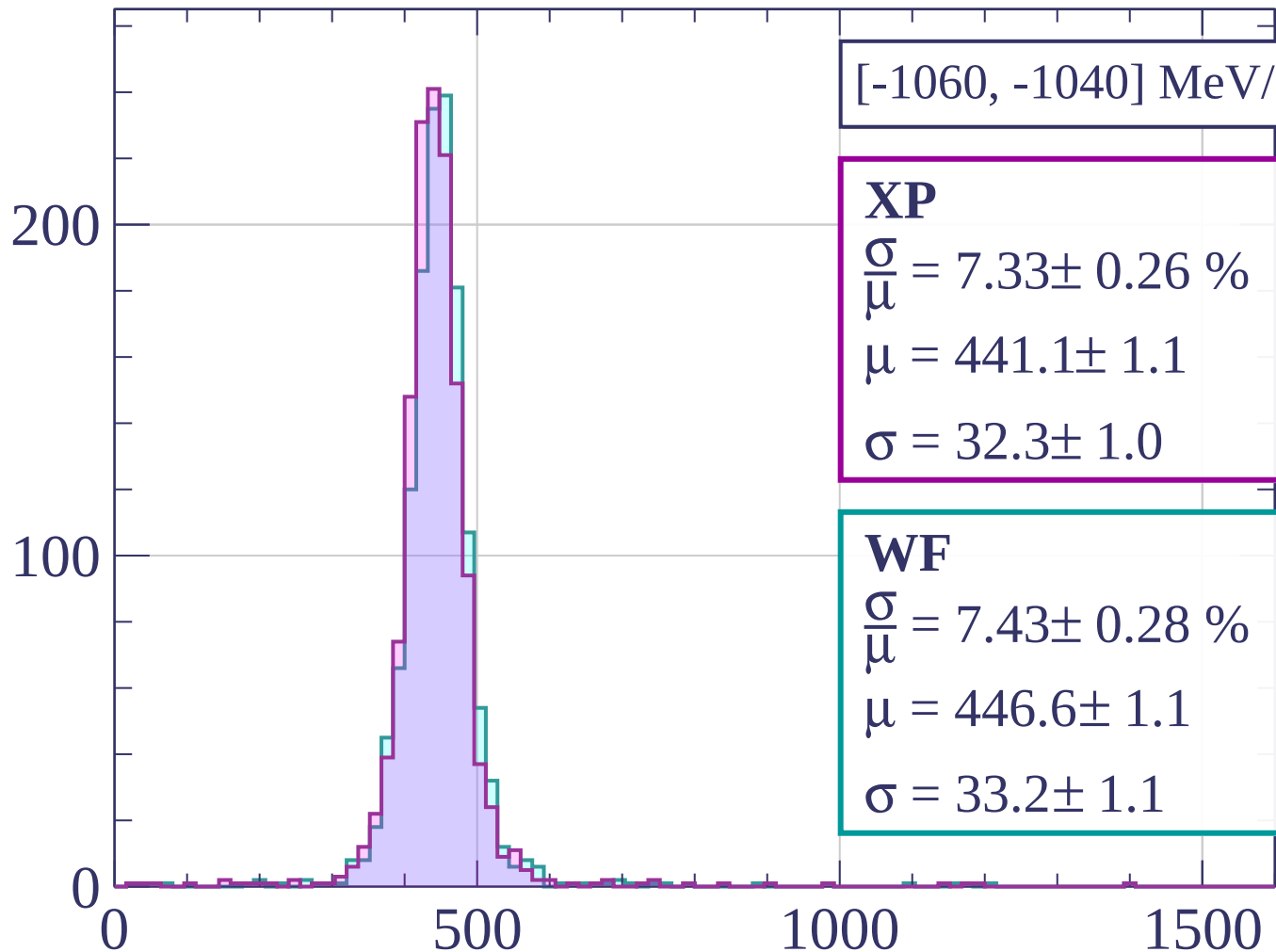
$$\frac{\sigma}{\mu} = 7.69 \pm 0.28 \%$$

$$\mu = 446.6 \pm 1.1$$

$$\sigma = 34.3 \pm 1.2$$

dE/dx [ADC counts/cm]

Count



$[-1060, -1040]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 7.33 \pm 0.26 \%$$

$$\mu = 441.1 \pm 1.1$$

$$\sigma = 32.3 \pm 1.0$$

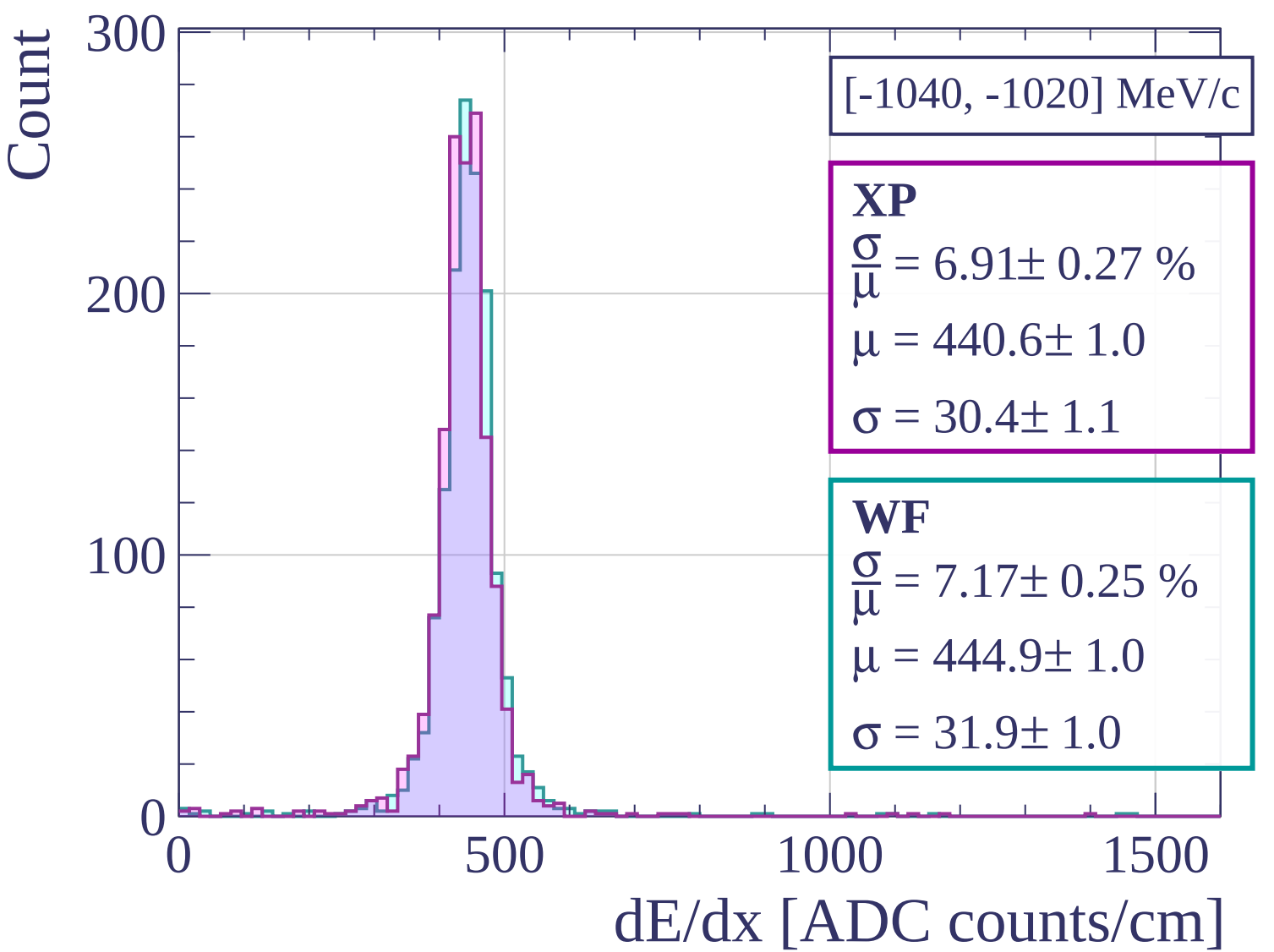
WF

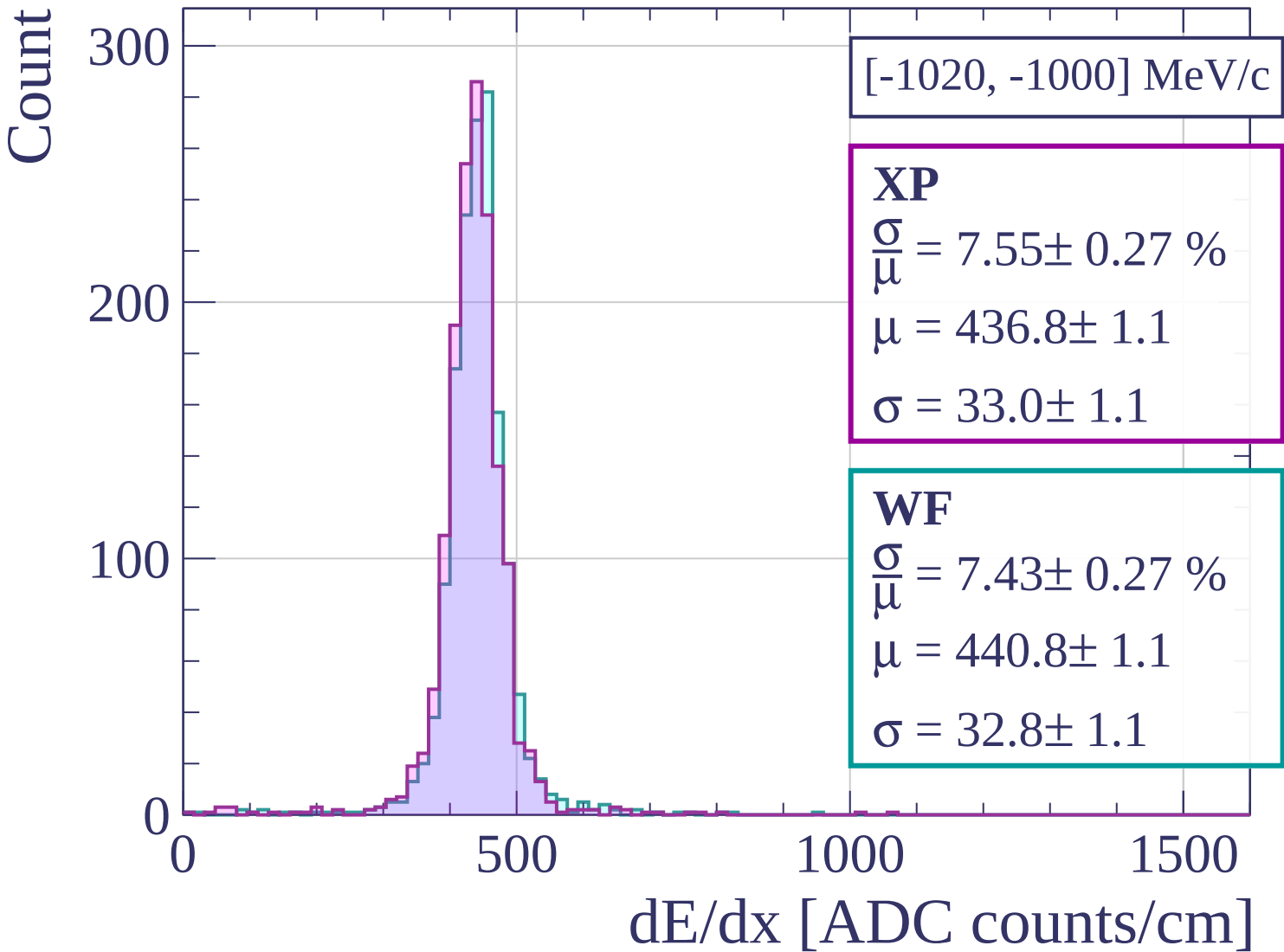
$$\frac{\sigma}{\mu} = 7.43 \pm 0.28 \%$$

$$\mu = 446.6 \pm 1.1$$

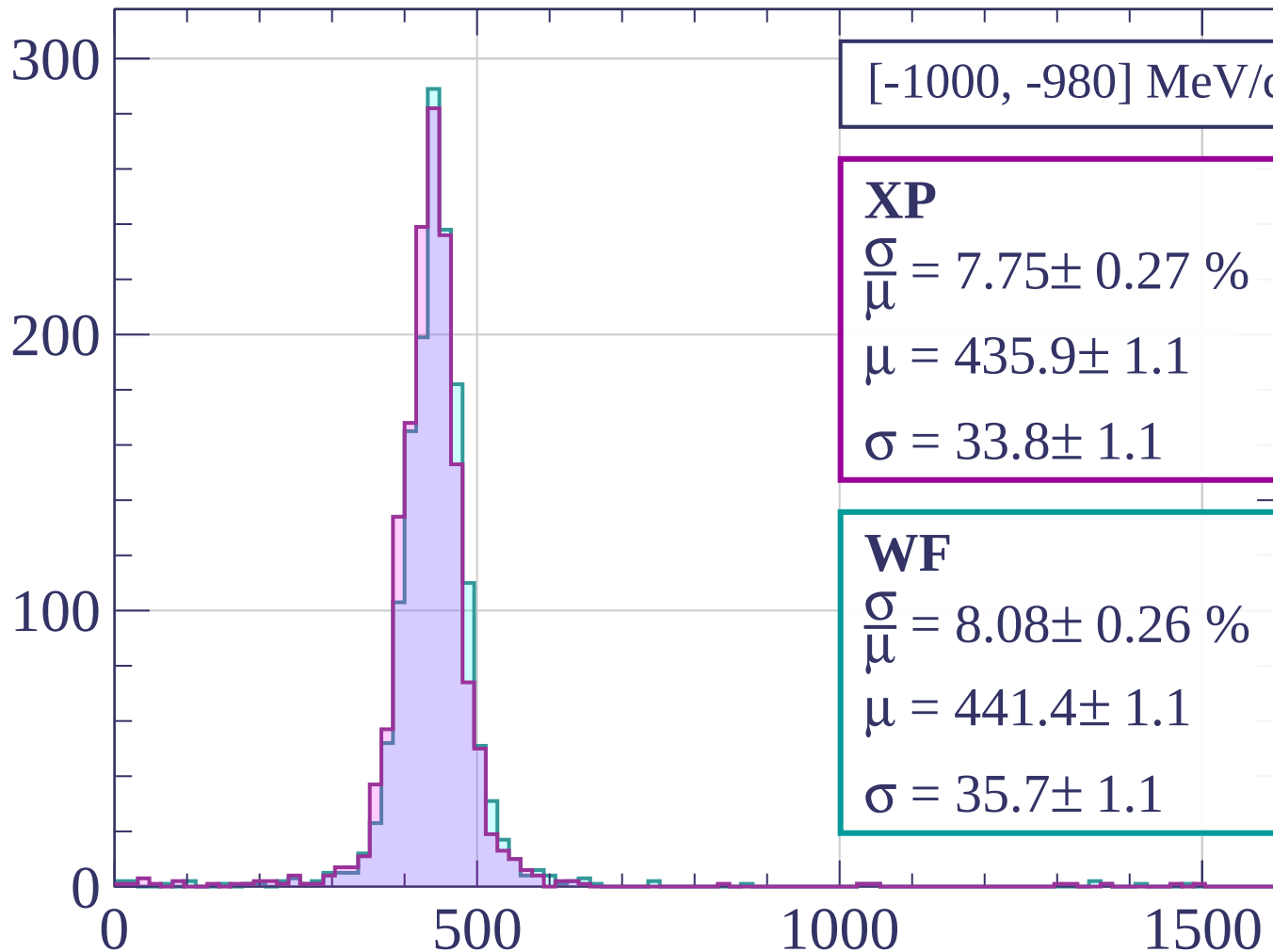
$$\sigma = 33.2 \pm 1.1$$

dE/dx [ADC counts/cm]





Count



dE/dx [ADC counts/cm]

Count

[-980, -960] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.69 \pm 0.28 \%$$

$$\mu = 435.7 \pm 1.1$$

$$\sigma = 33.5 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.13 \pm 0.29 \%$$

$$\mu = 439.6 \pm 1.1$$

$$\sigma = 31.3 \pm 1.2$$

200

100

0

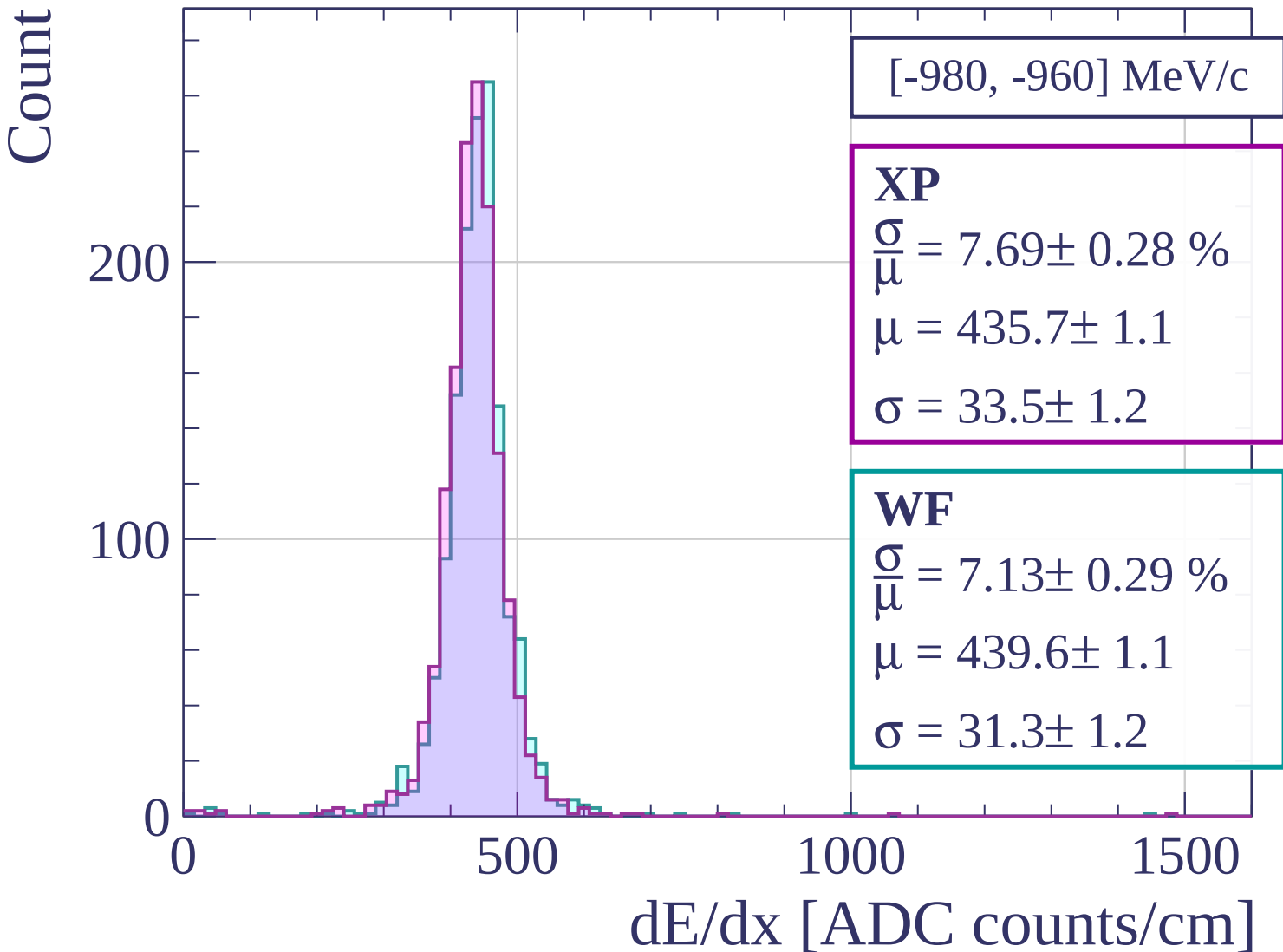
0

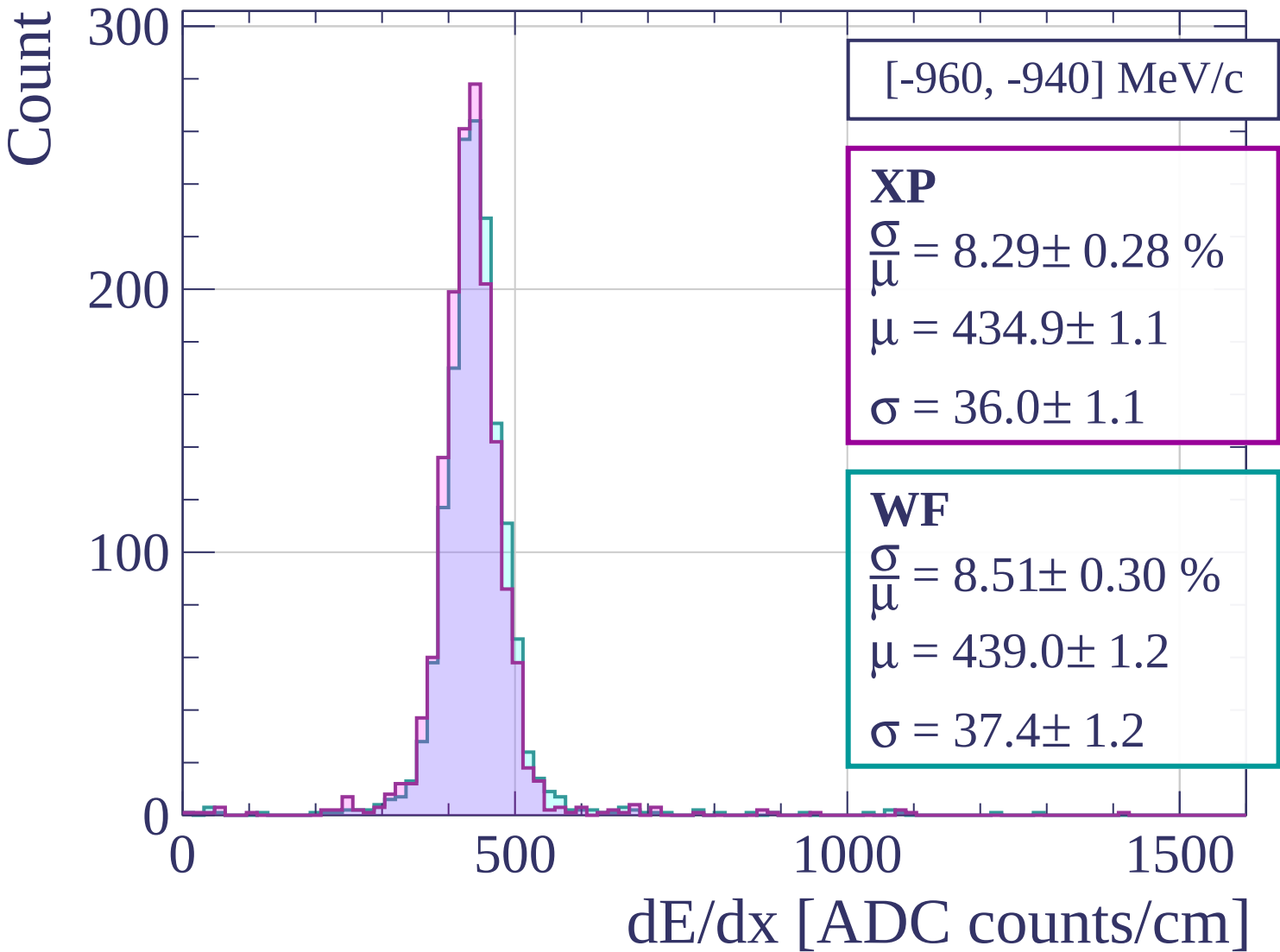
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-940, -920] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.73 \pm 0.26 \%$$

$$\mu = 433.4 \pm 1.0$$

$$\sigma = 33.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.23 \%$$

$$\mu = 437.6 \pm 1.0$$

$$\sigma = 34.5 \pm 0.9$$

300

200

100

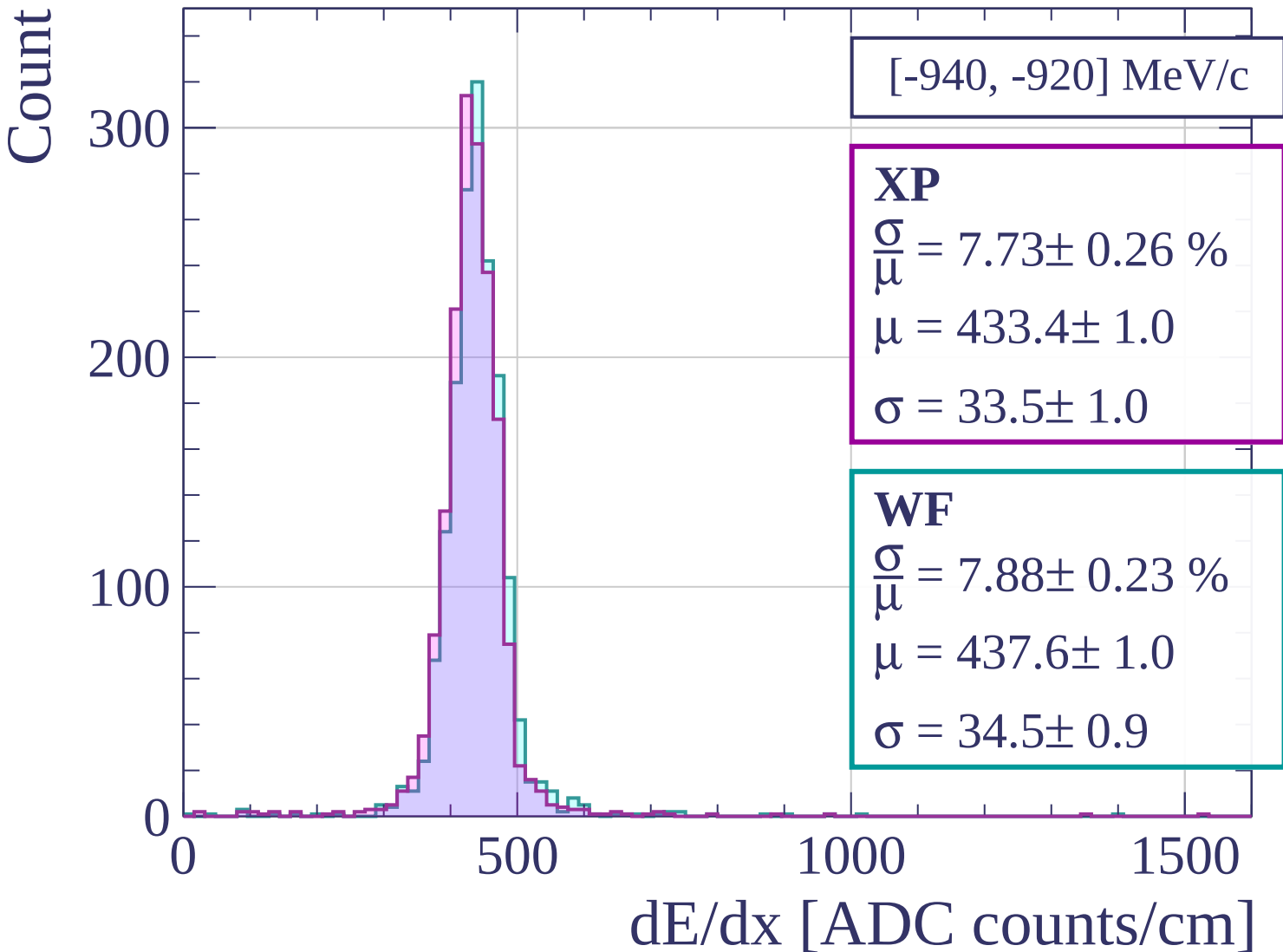
0

500

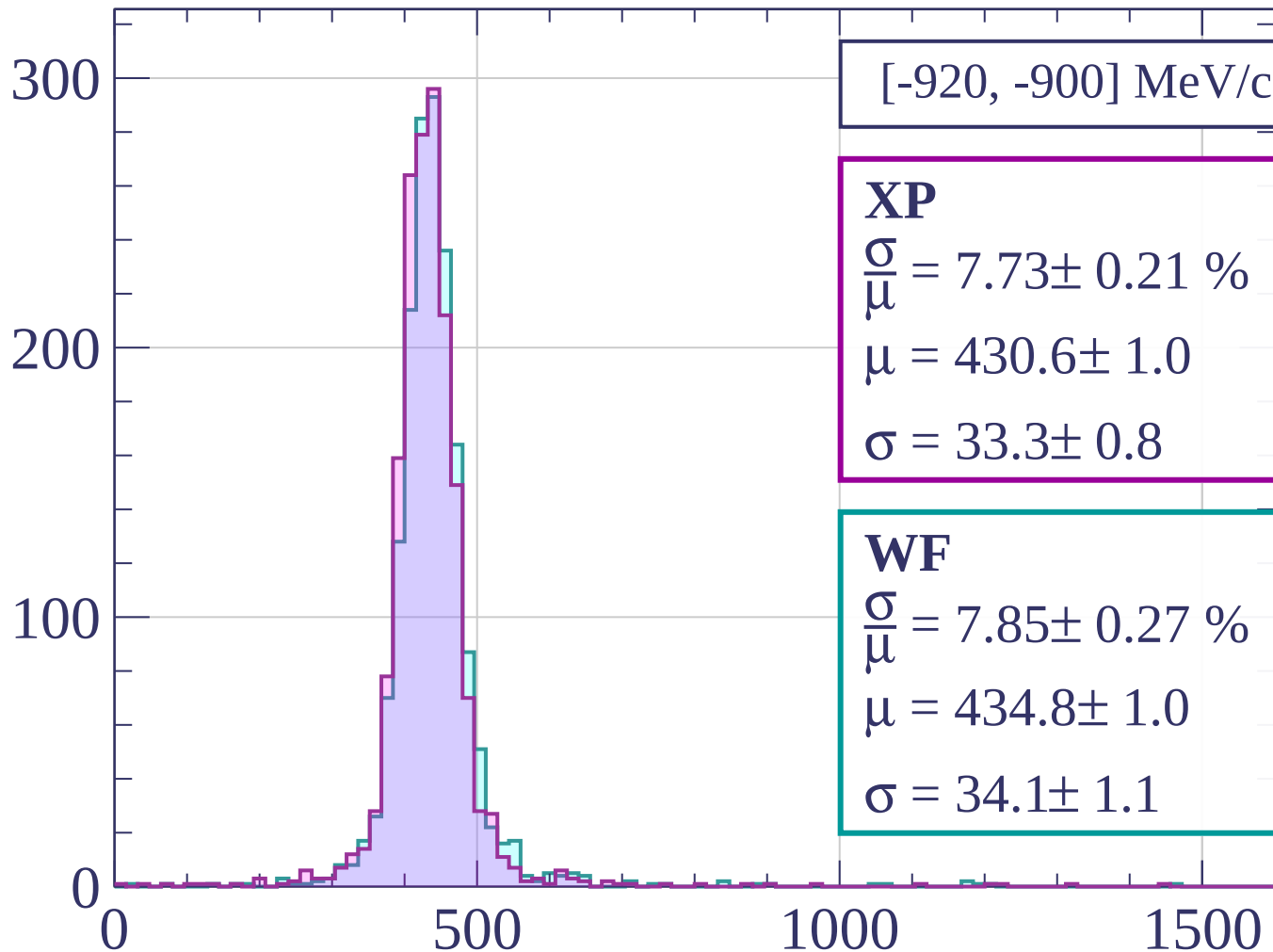
1000

1500

dE/dx [ADC counts/cm]



Count



[-920, -900] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.73 \pm 0.21 \%$$

$$\mu = 430.6 \pm 1.0$$

$$\sigma = 33.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.27 \%$$

$$\mu = 434.8 \pm 1.0$$

$$\sigma = 34.1 \pm 1.1$$

dE/dx [ADC counts/cm]

Count

[-900, -880] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.38 \pm 0.23 \%$$

$$\mu = 429.8 \pm 0.9$$

$$\sigma = 31.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.41 \pm 0.22 \%$$

$$\mu = 434.5 \pm 0.9$$

$$\sigma = 32.2 \pm 0.9$$

300

200

100

0

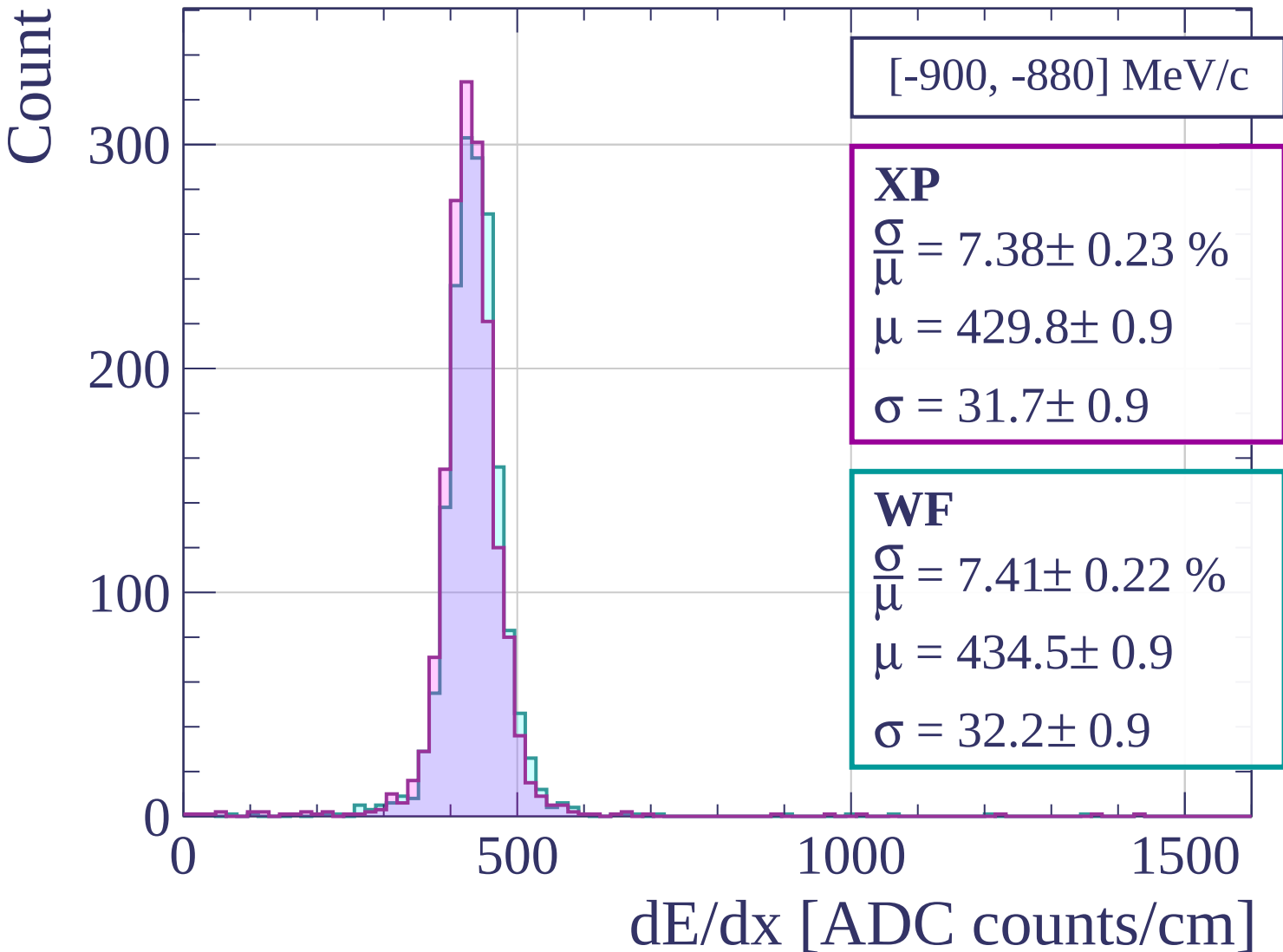
0

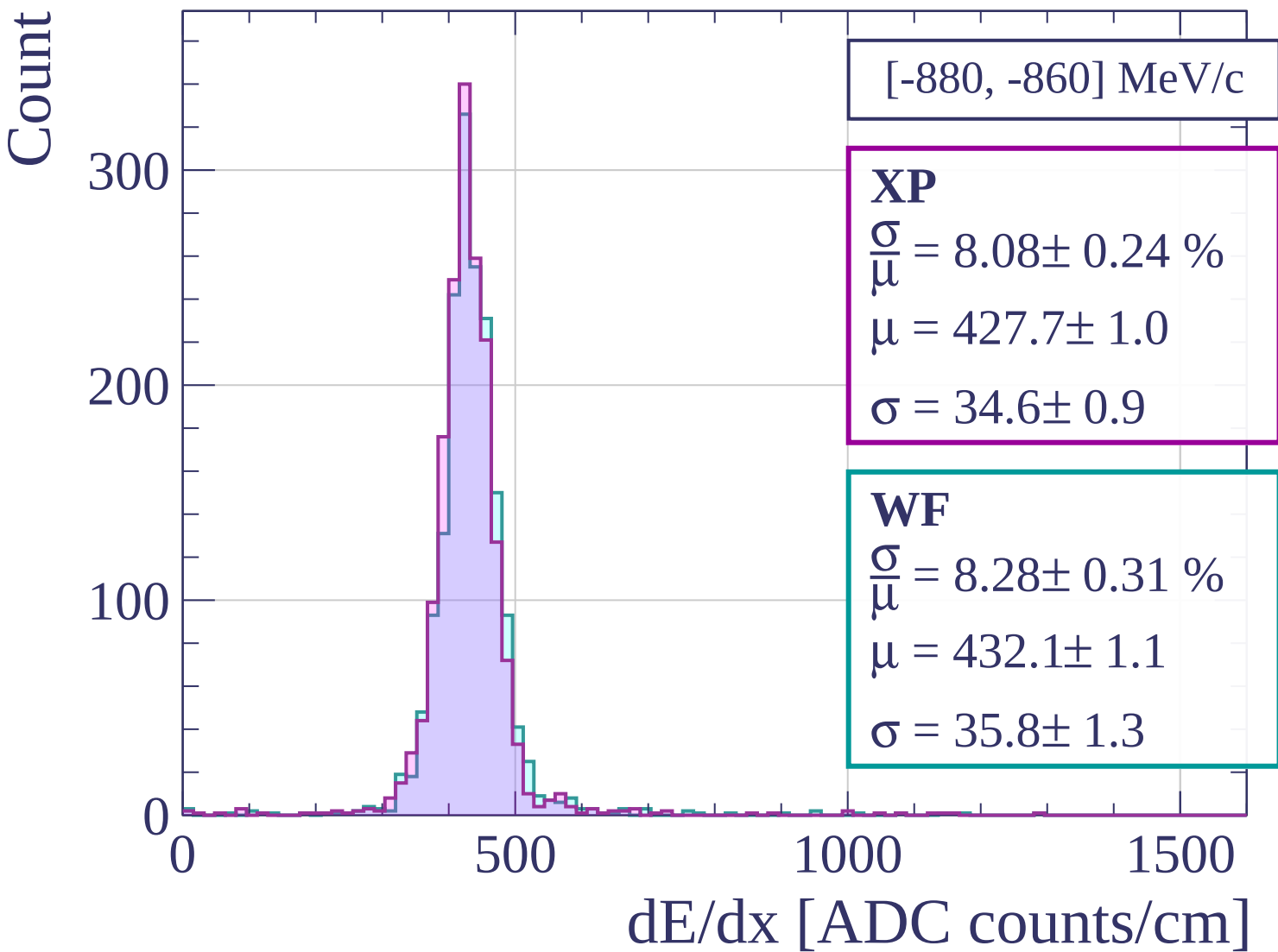
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-860, -840] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.52 \pm 0.23 \%$$

$$\mu = 429.2 \pm 0.9$$

$$\sigma = 32.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.20 \%$$

$$\mu = 432.1 \pm 0.9$$

$$\sigma = 34.9 \pm 0.8$$

300

200

100

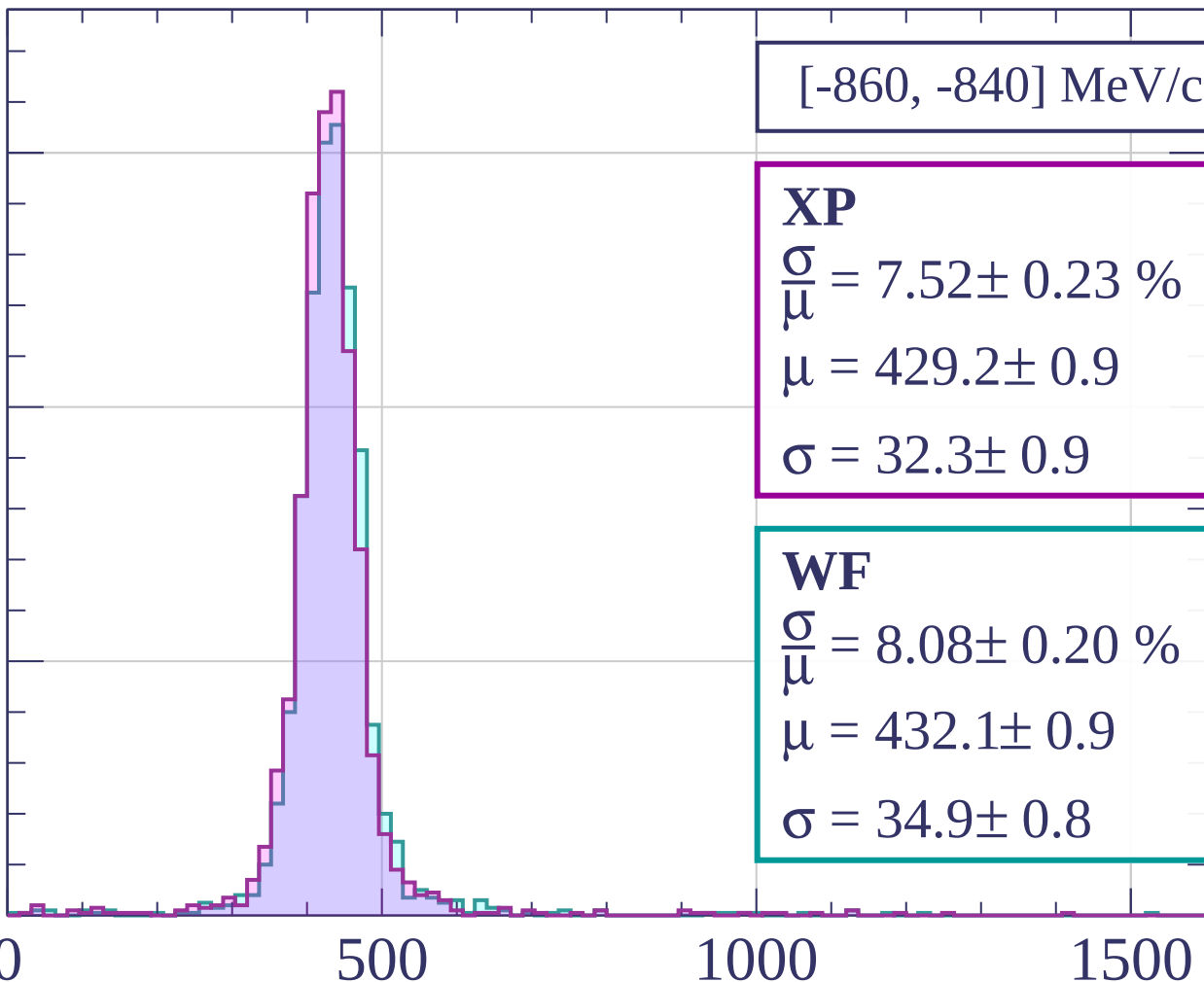
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-840, -820] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.98 \pm 0.21 \%$$

$$\mu = 426.8 \pm 0.9$$

$$\sigma = 34.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.79 \pm 0.24 \%$$

$$\mu = 430.2 \pm 0.9$$

$$\sigma = 33.5 \pm 1.0$$

300

200

100

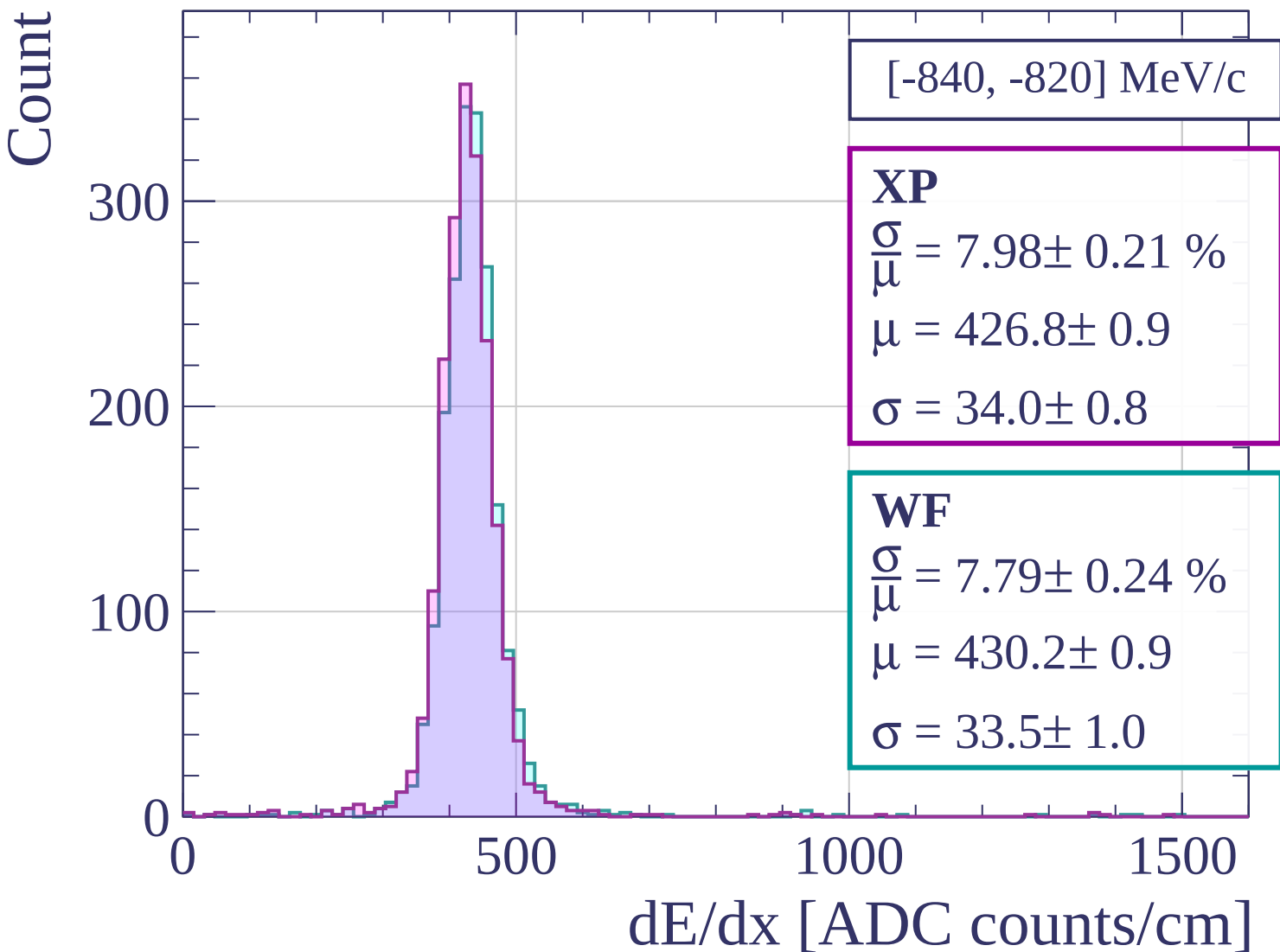
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-820, -800] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.68 \pm 0.23 \%$$

$$\mu = 427.3 \pm 0.9$$

$$\sigma = 32.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.87 \pm 0.26 \%$$

$$\mu = 430.9 \pm 0.9$$

$$\sigma = 33.9 \pm 1.0$$

300

200

100

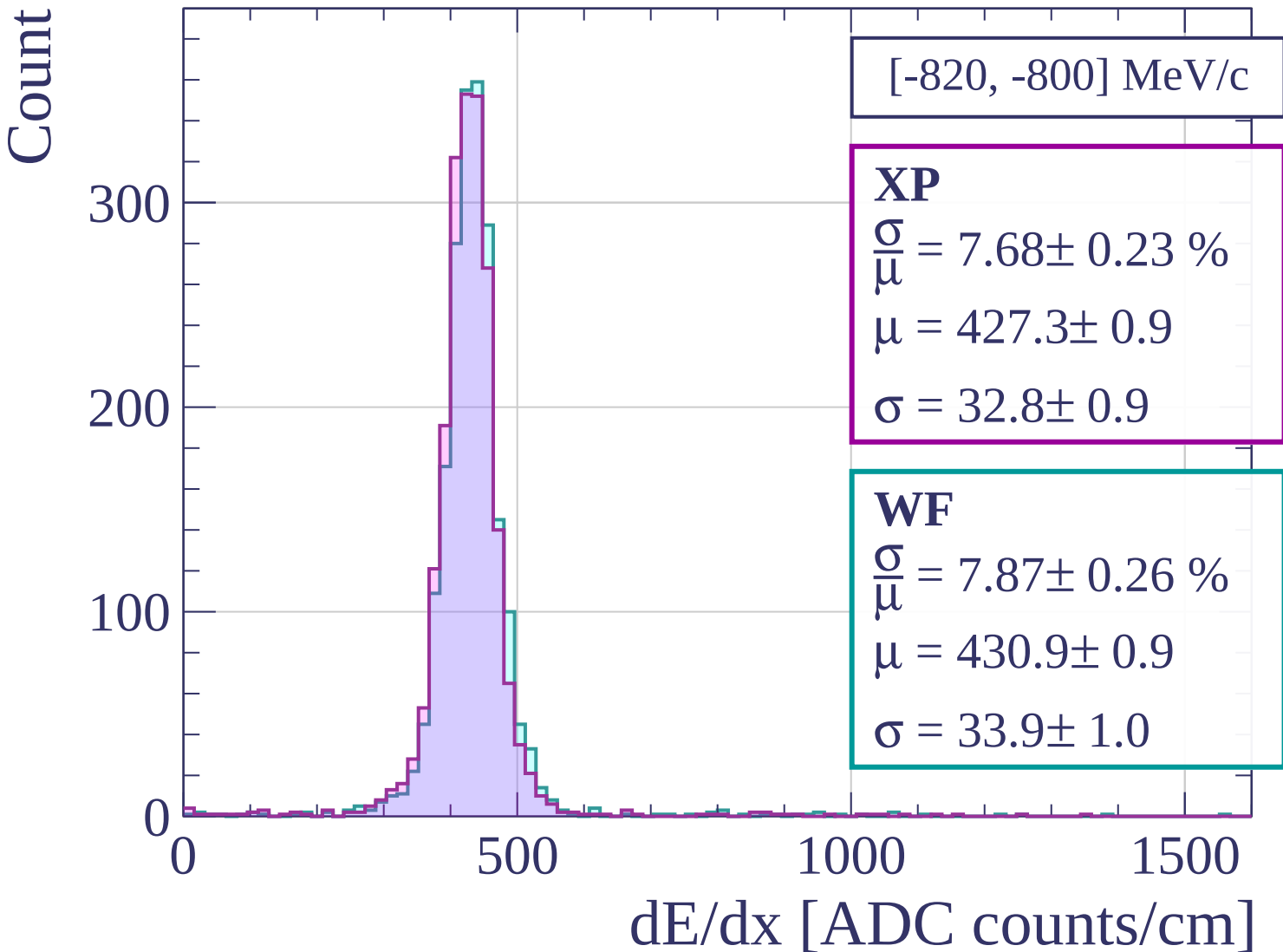
0

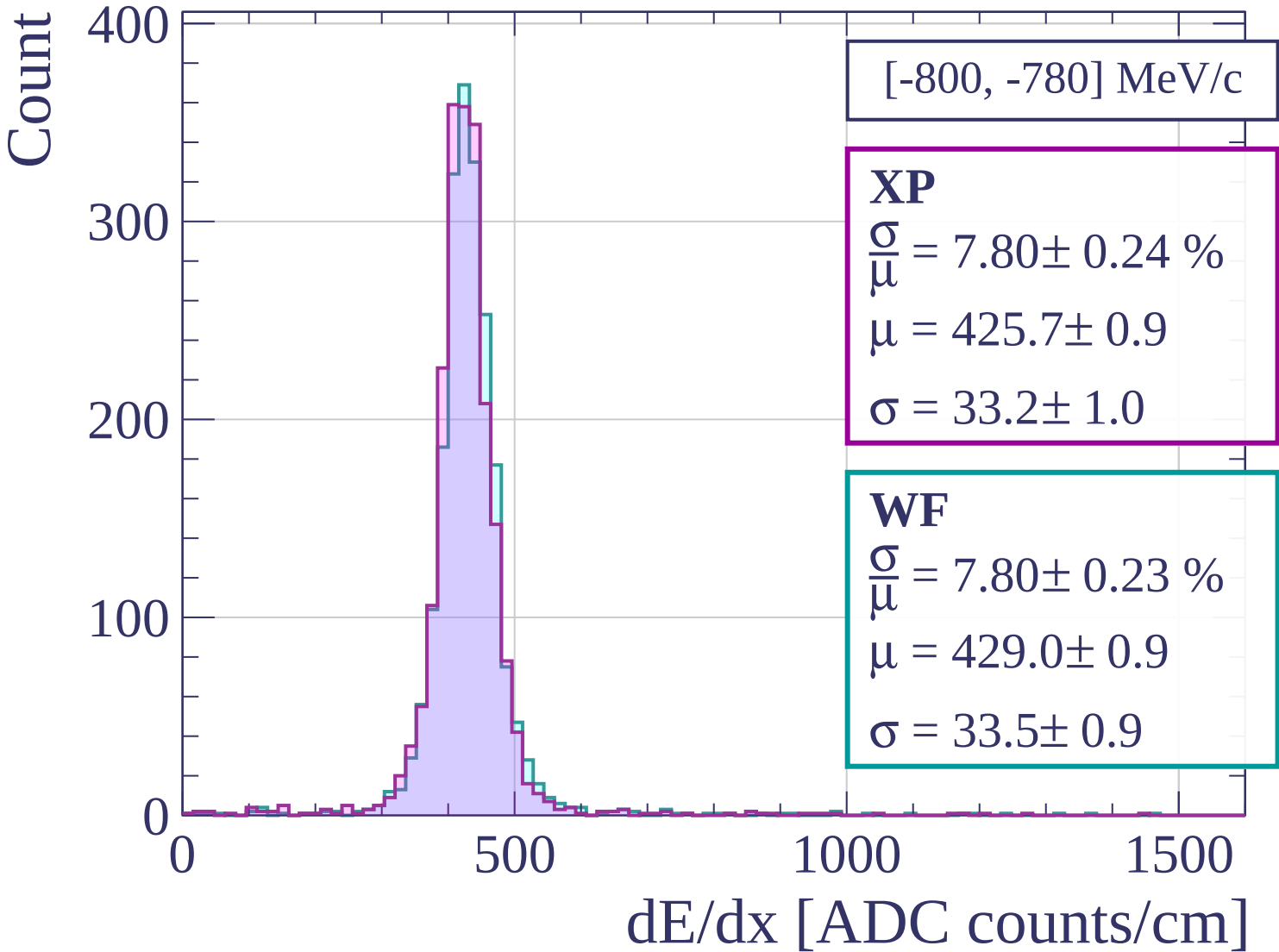
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-780, -760] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.99 \pm 0.21 \%$$

$$\mu = 424.3 \pm 0.8$$

$$\sigma = 33.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.68 \pm 0.23 \%$$

$$\mu = 428.8 \pm 0.9$$

$$\sigma = 32.9 \pm 0.9$$

400

300

200

100

0

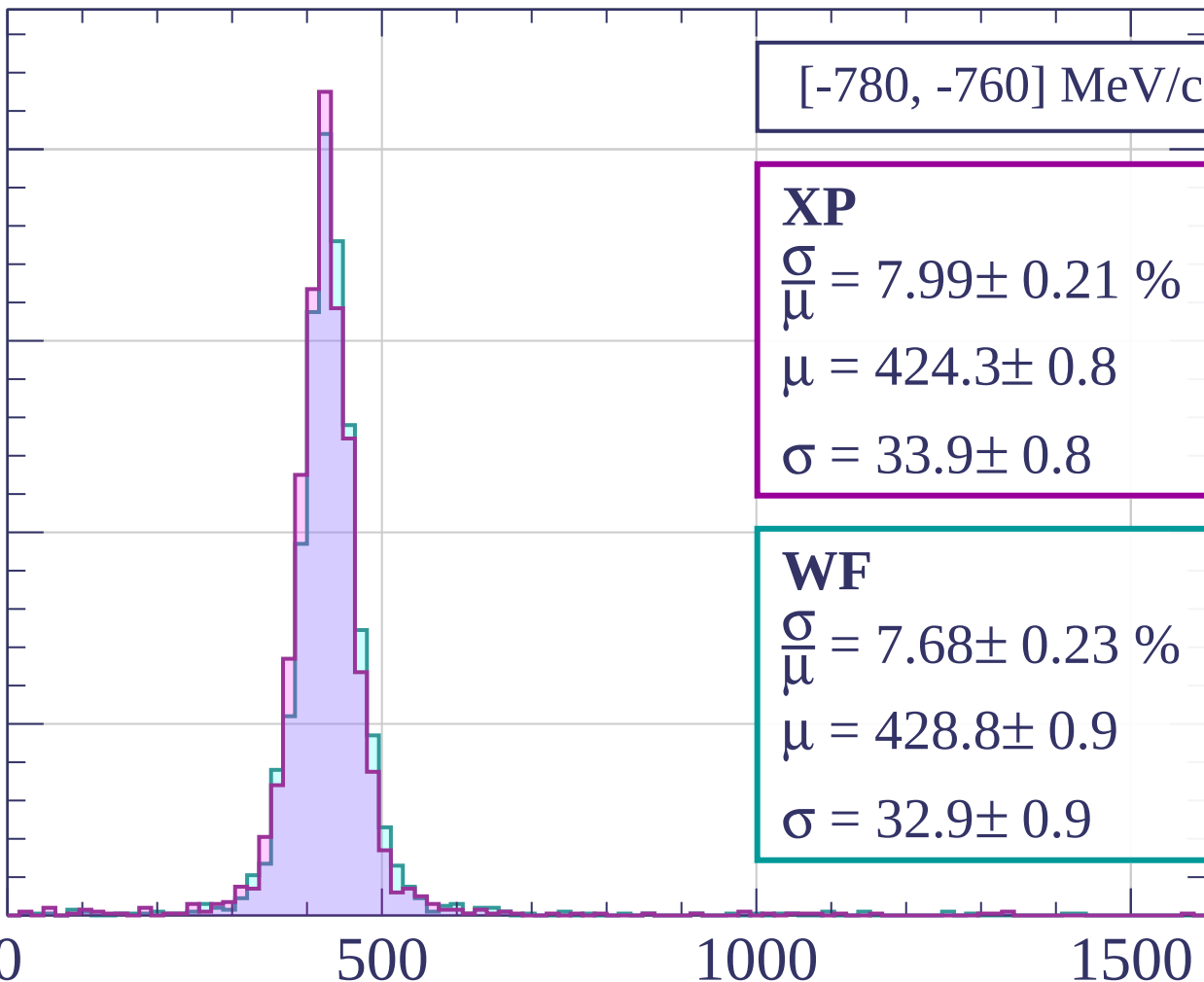
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-760, -740] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.67 \pm 0.22 \%$$

$$\mu = 422.4 \pm 0.9$$

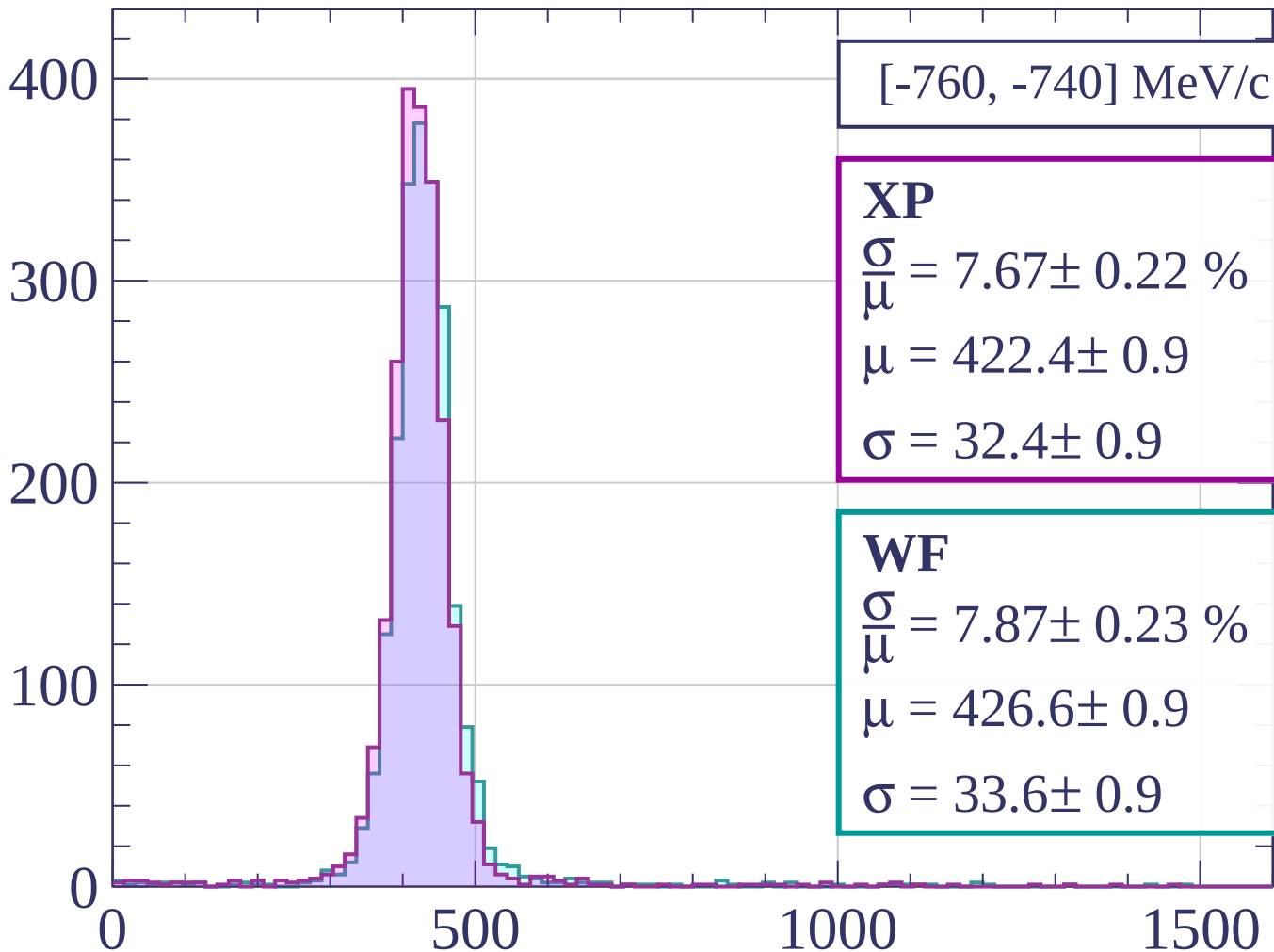
$$\sigma = 32.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.87 \pm 0.23 \%$$

$$\mu = 426.6 \pm 0.9$$

$$\sigma = 33.6 \pm 0.9$$



dE/dx [ADC counts/cm]

Count

[-740, -720] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.23 \%$$

$$\mu = 423.1 \pm 0.9$$

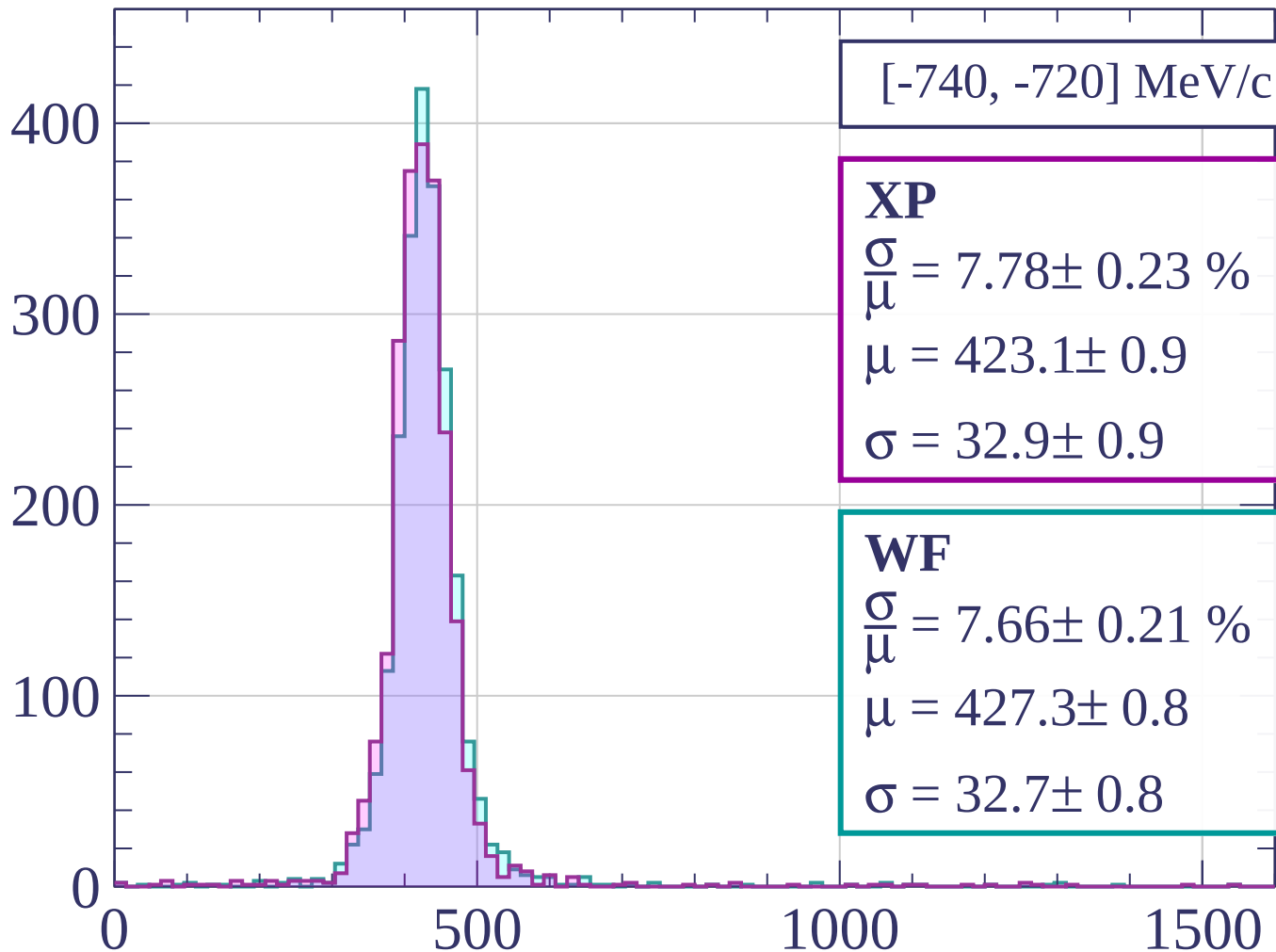
$$\sigma = 32.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.66 \pm 0.21 \%$$

$$\mu = 427.3 \pm 0.8$$

$$\sigma = 32.7 \pm 0.8$$



dE/dx [ADC counts/cm]

Count

[-720, -700] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.76 \pm 0.20 \%$$

$$\mu = 420.3 \pm 0.8$$

$$\sigma = 32.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.11 \pm 0.21 \%$$

$$\mu = 424.9 \pm 0.8$$

$$\sigma = 34.5 \pm 0.8$$

400

200

0

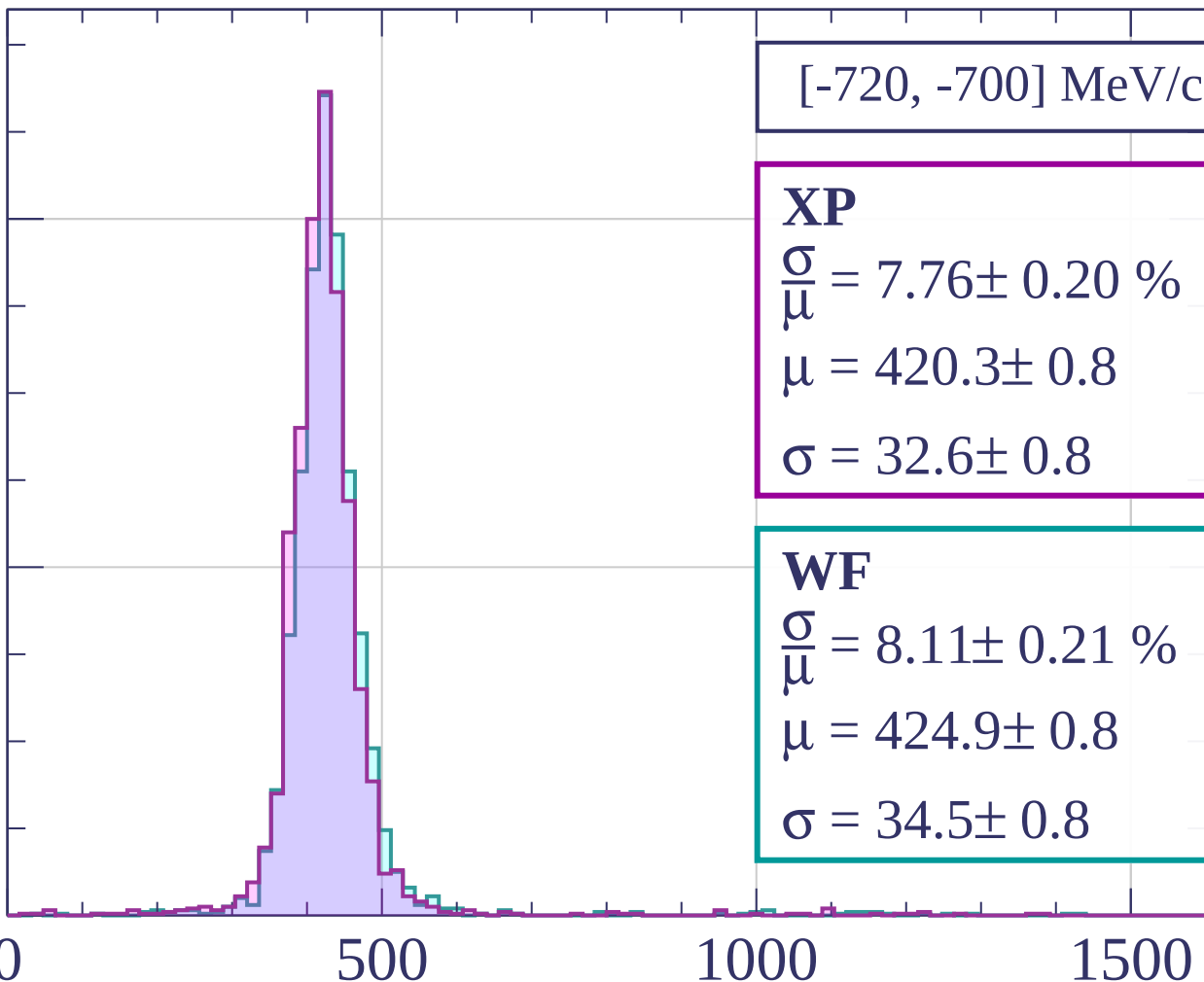
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-700, -680] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.96 \pm 0.22 \%$$

$$\mu = 416.7 \pm 0.8$$

$$\sigma = 33.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.04 \pm 0.19 \%$$

$$\mu = 421.5 \pm 0.8$$

$$\sigma = 33.9 \pm 0.7$$

400

300

200

100

0

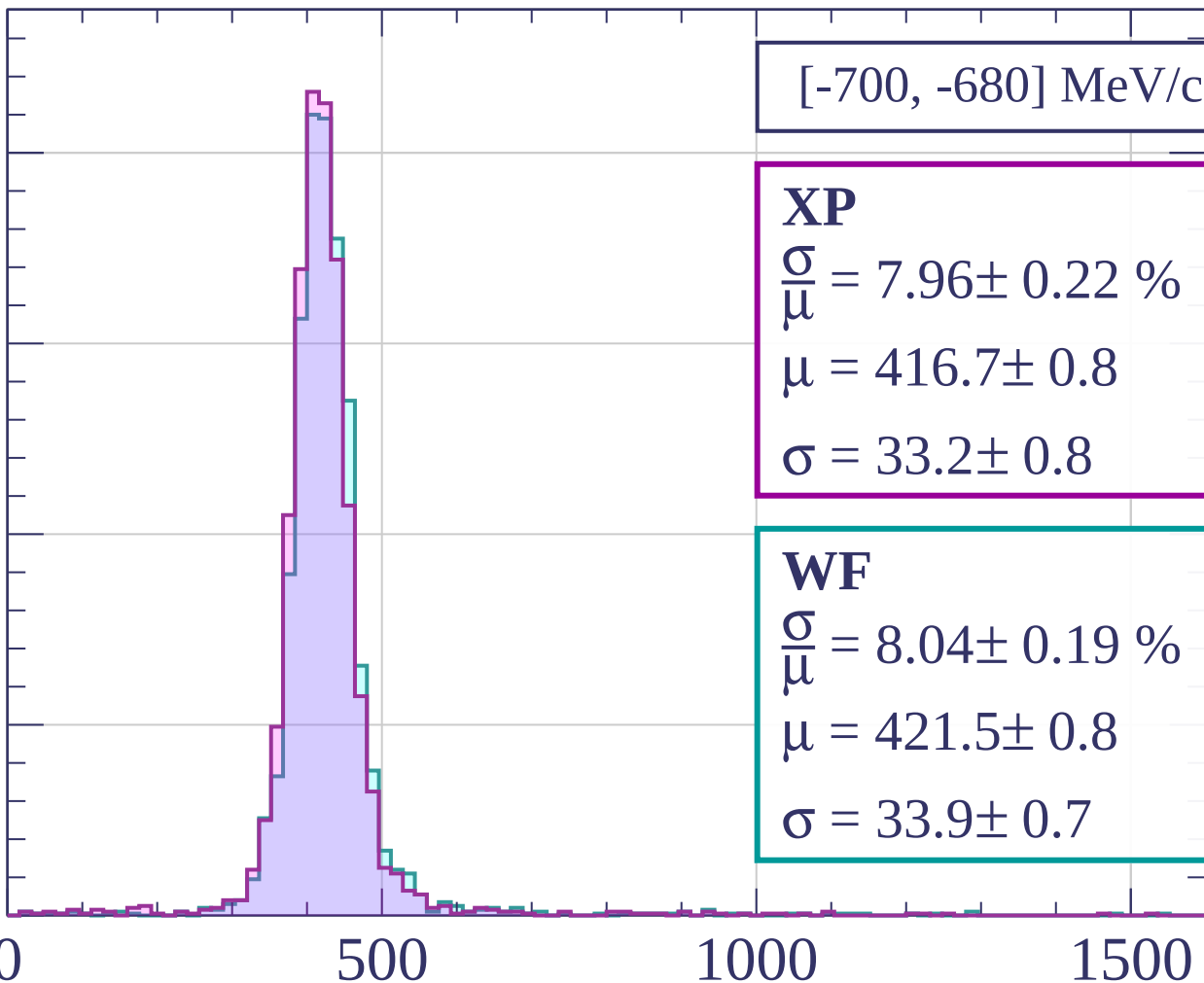
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-680, -660] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.90 \pm 0.20 \%$$

$$\mu = 416.7 \pm 0.8$$

$$\sigma = 32.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.89 \pm 0.21 \%$$

$$\mu = 420.3 \pm 0.8$$

$$\sigma = 33.1 \pm 0.8$$

400

200

0

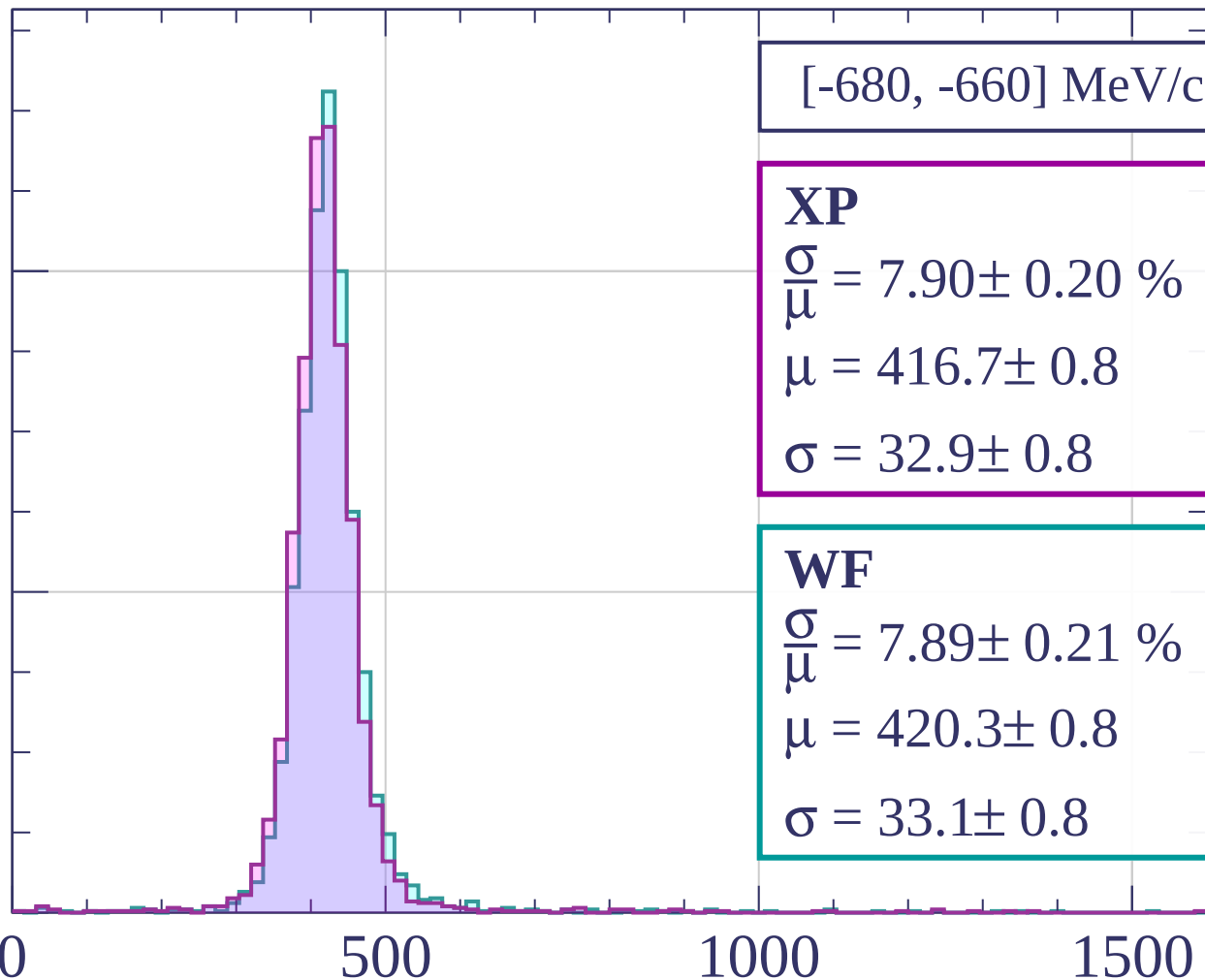
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-660, -640] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.79 \pm 0.20 \%$$

$$\mu = 416.8 \pm 0.8$$

$$\sigma = 32.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.11 \pm 0.23 \%$$

$$\mu = 420.8 \pm 0.8$$

$$\sigma = 34.1 \pm 0.9$$

400

200

0

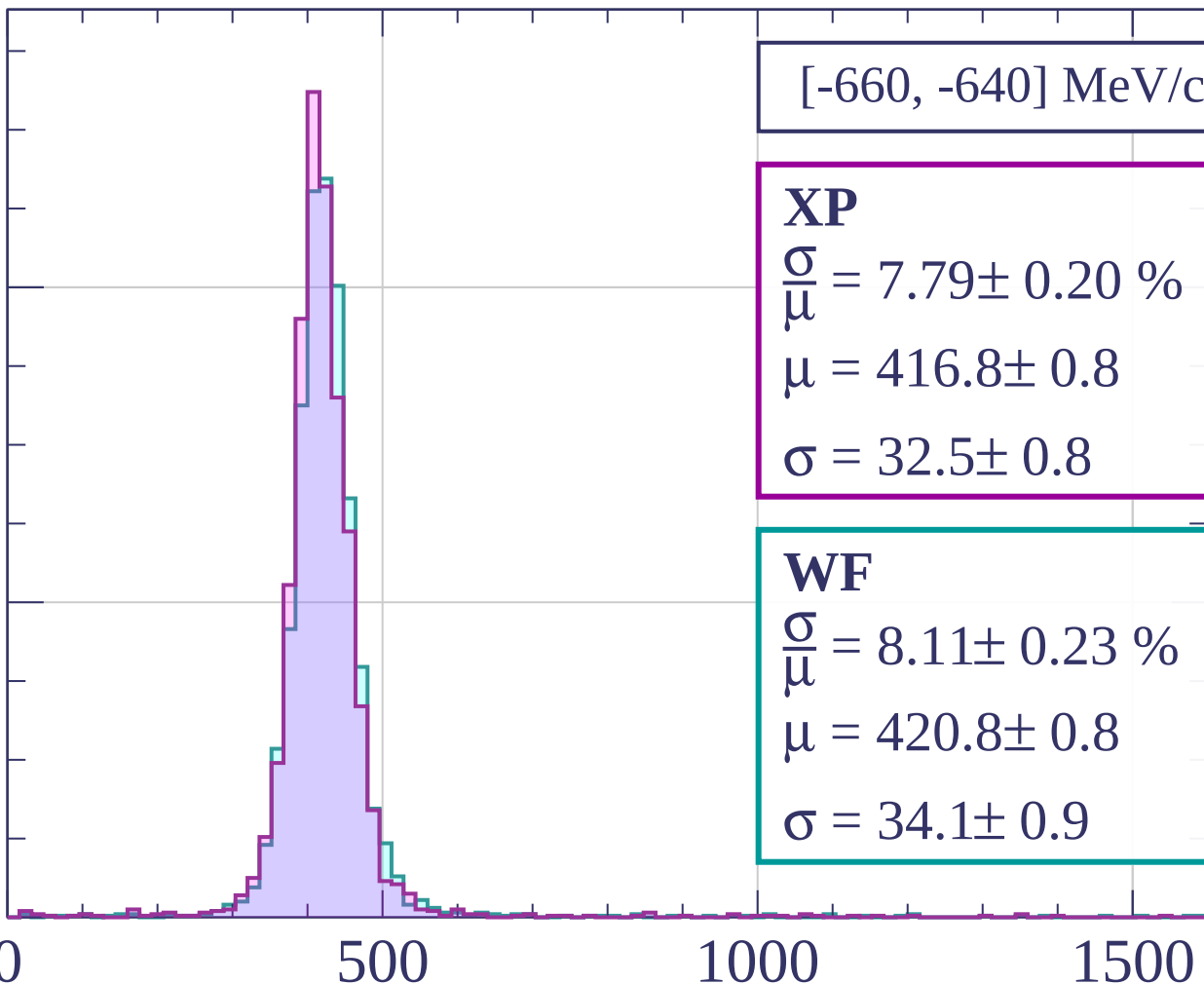
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-640, -620] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.95 \pm 0.20 \%$$

$$\mu = 415.5 \pm 0.8$$

$$\sigma = 33.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.99 \pm 0.21 \%$$

$$\mu = 419.6 \pm 0.8$$

$$\sigma = 33.5 \pm 0.8$$

400

200

0

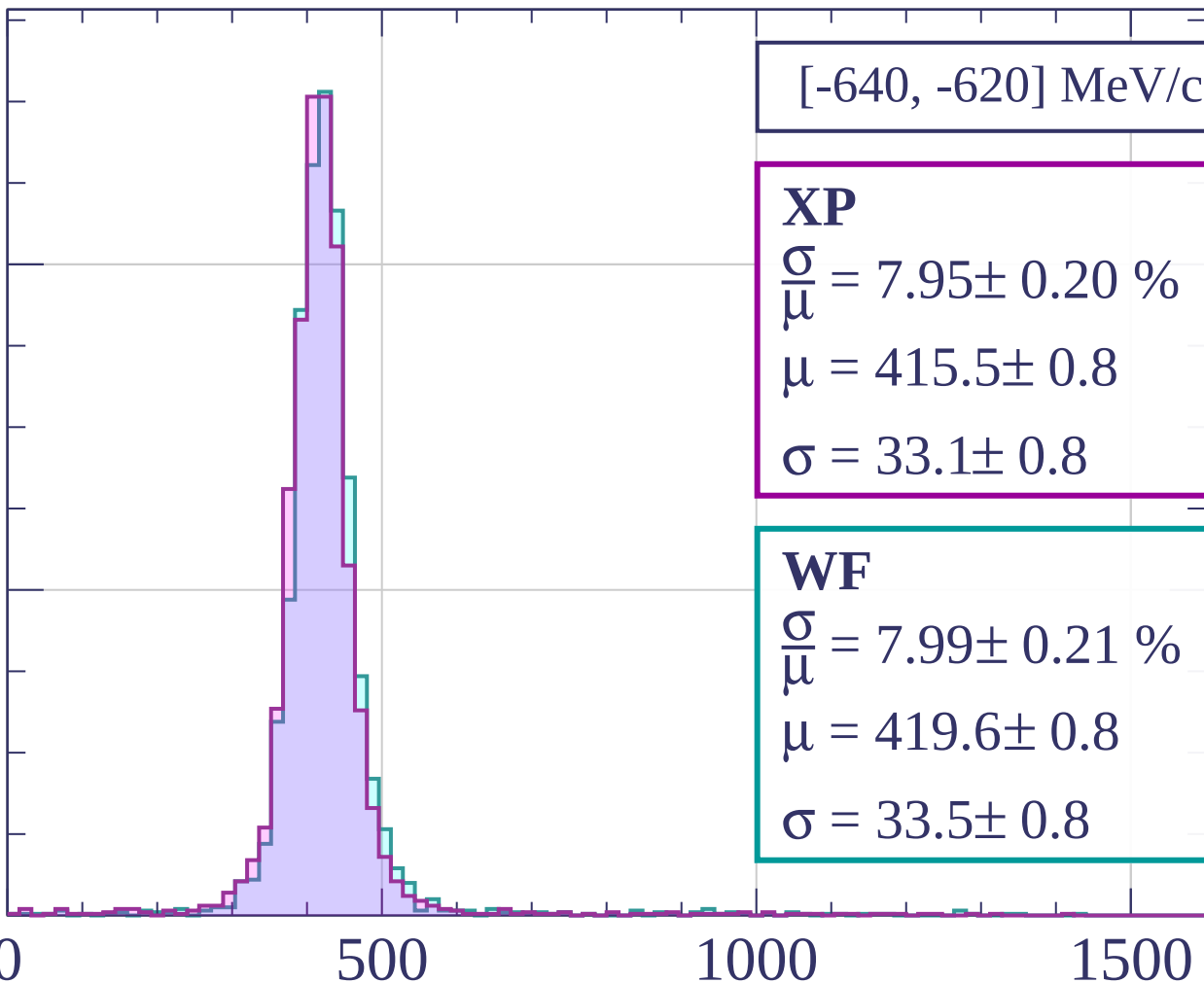
0

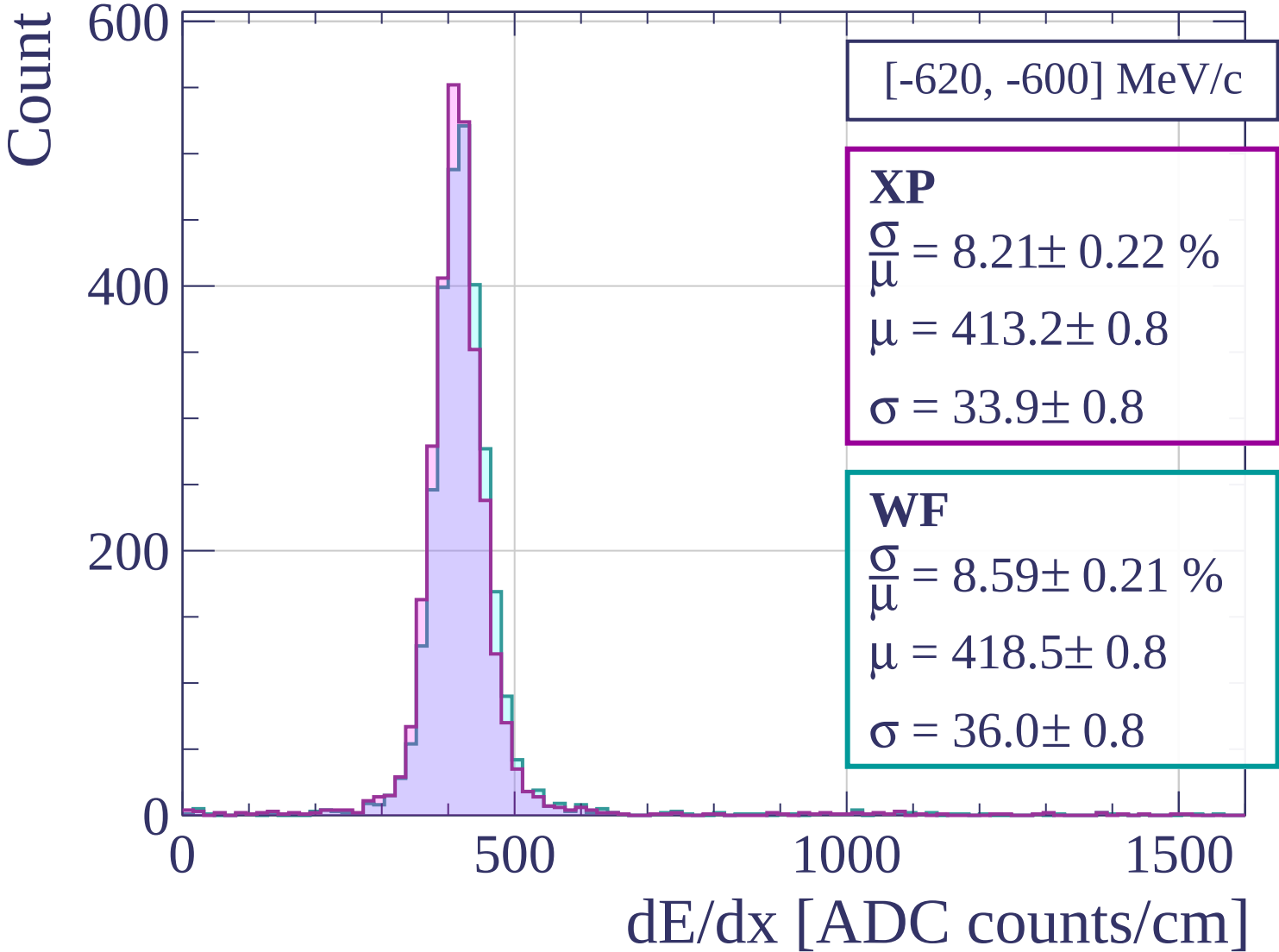
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-600, -580] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.11 \pm 0.20 \%$$

$$\mu = 413.0 \pm 0.7$$

$$\sigma = 33.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.36 \pm 0.22 \%$$

$$\mu = 417.0 \pm 0.8$$

$$\sigma = 34.9 \pm 0.9$$

400

200

0

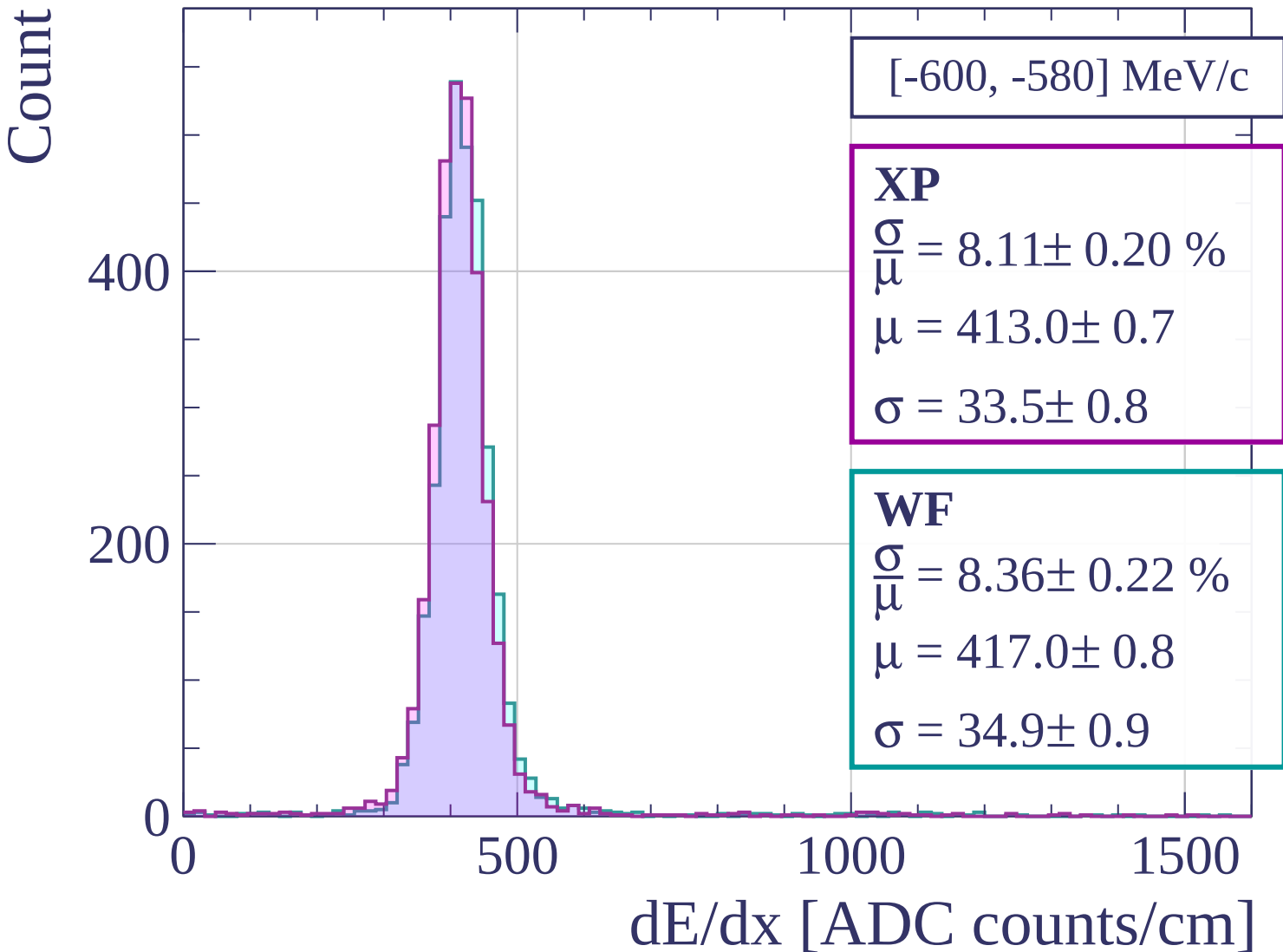
0

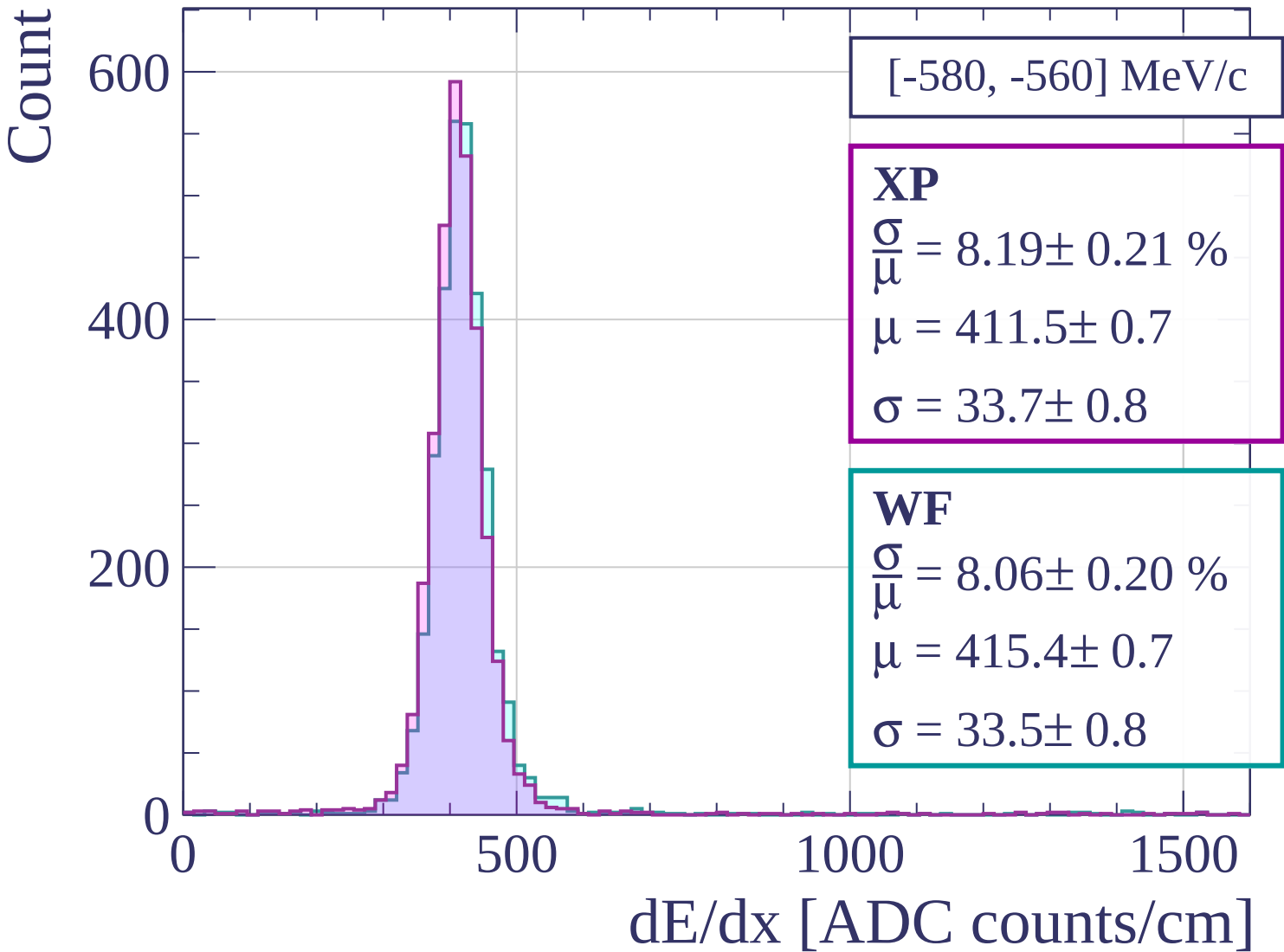
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-560, -540] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.49 \pm 0.19 \%$$

$$\mu = 408.7 \pm 0.7$$

$$\sigma = 30.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.19 \%$$

$$\mu = 413.7 \pm 0.7$$

$$\sigma = 32.7 \pm 0.7$$

600

400

200

0

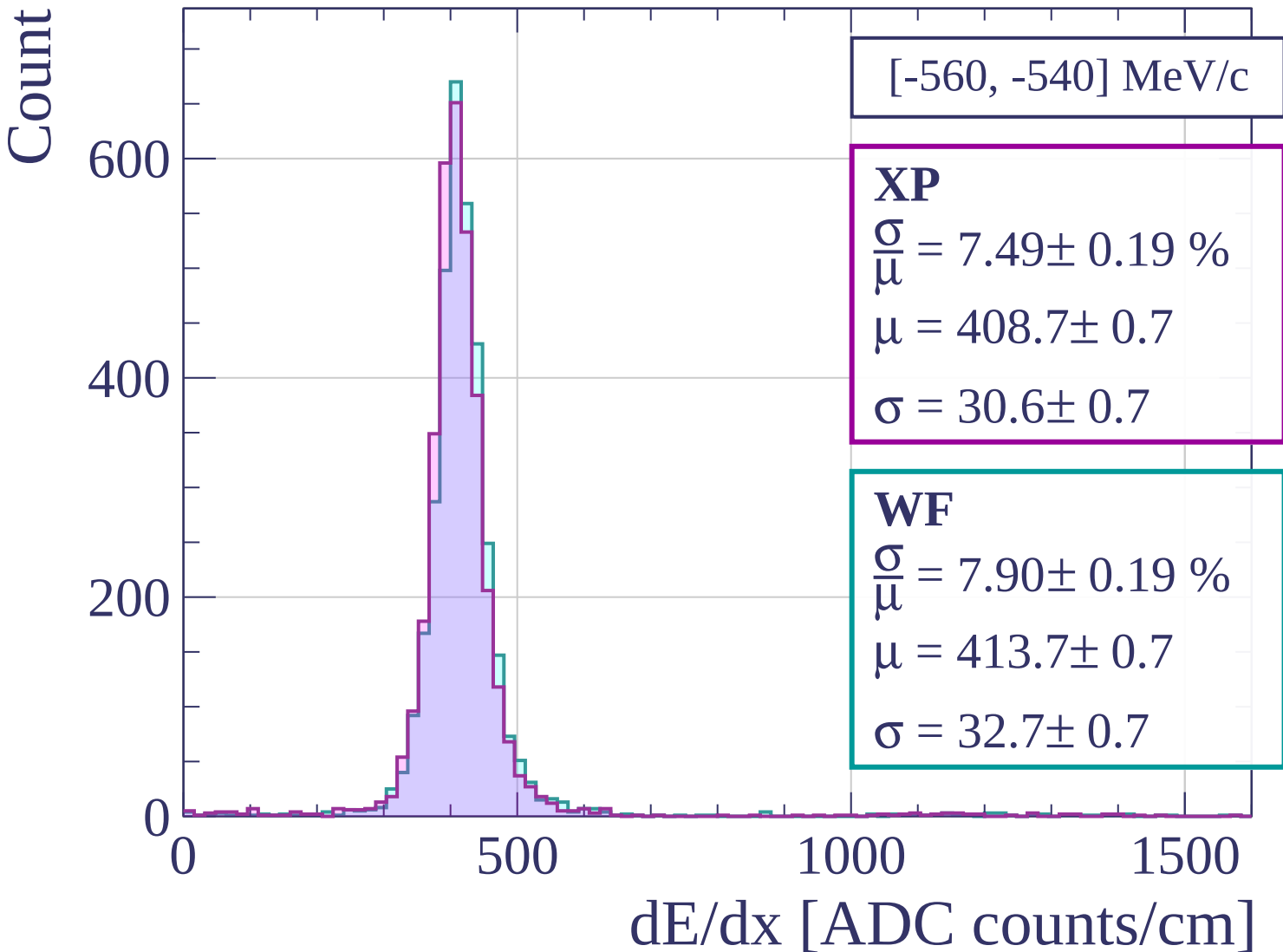
0

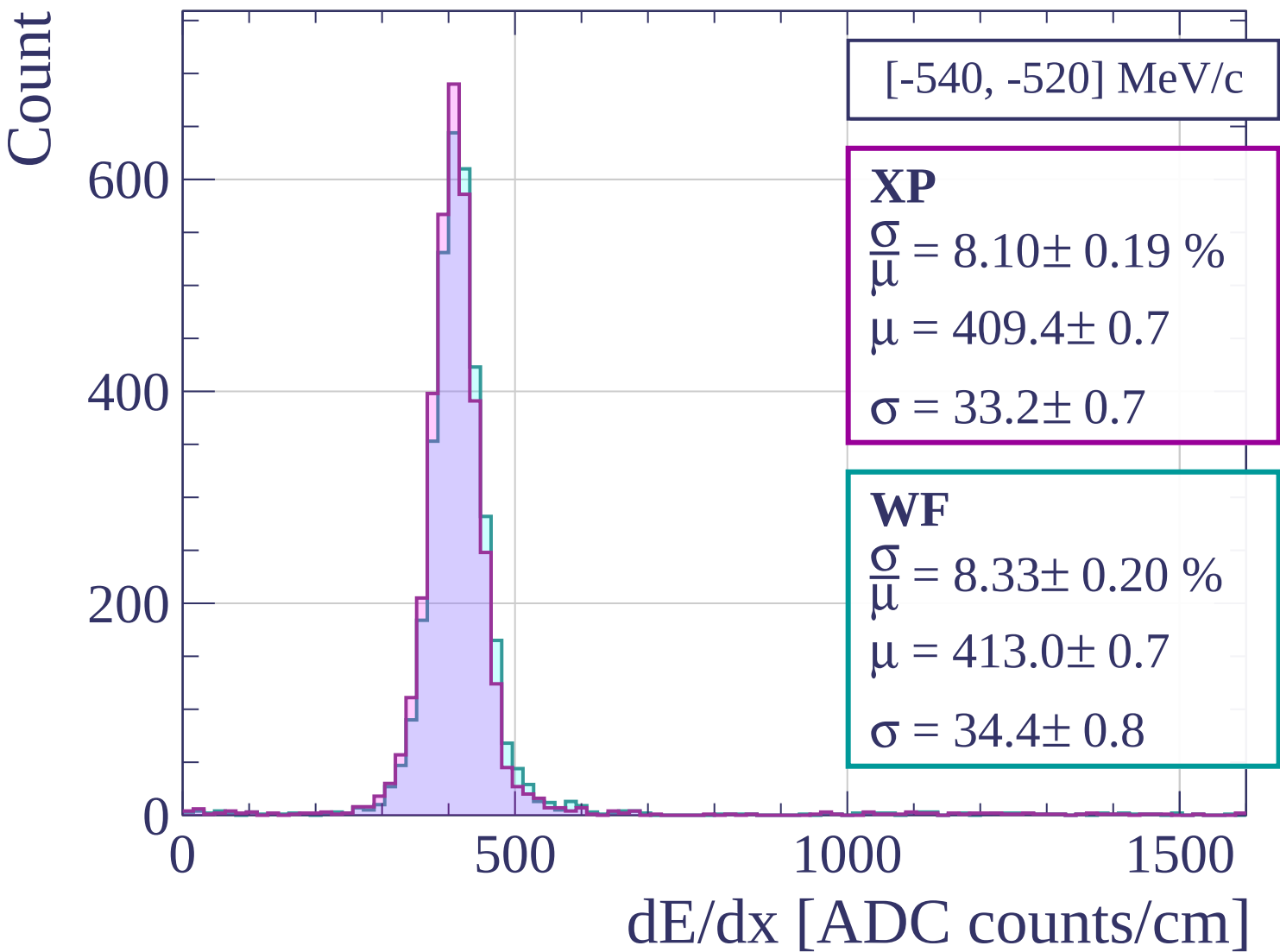
500

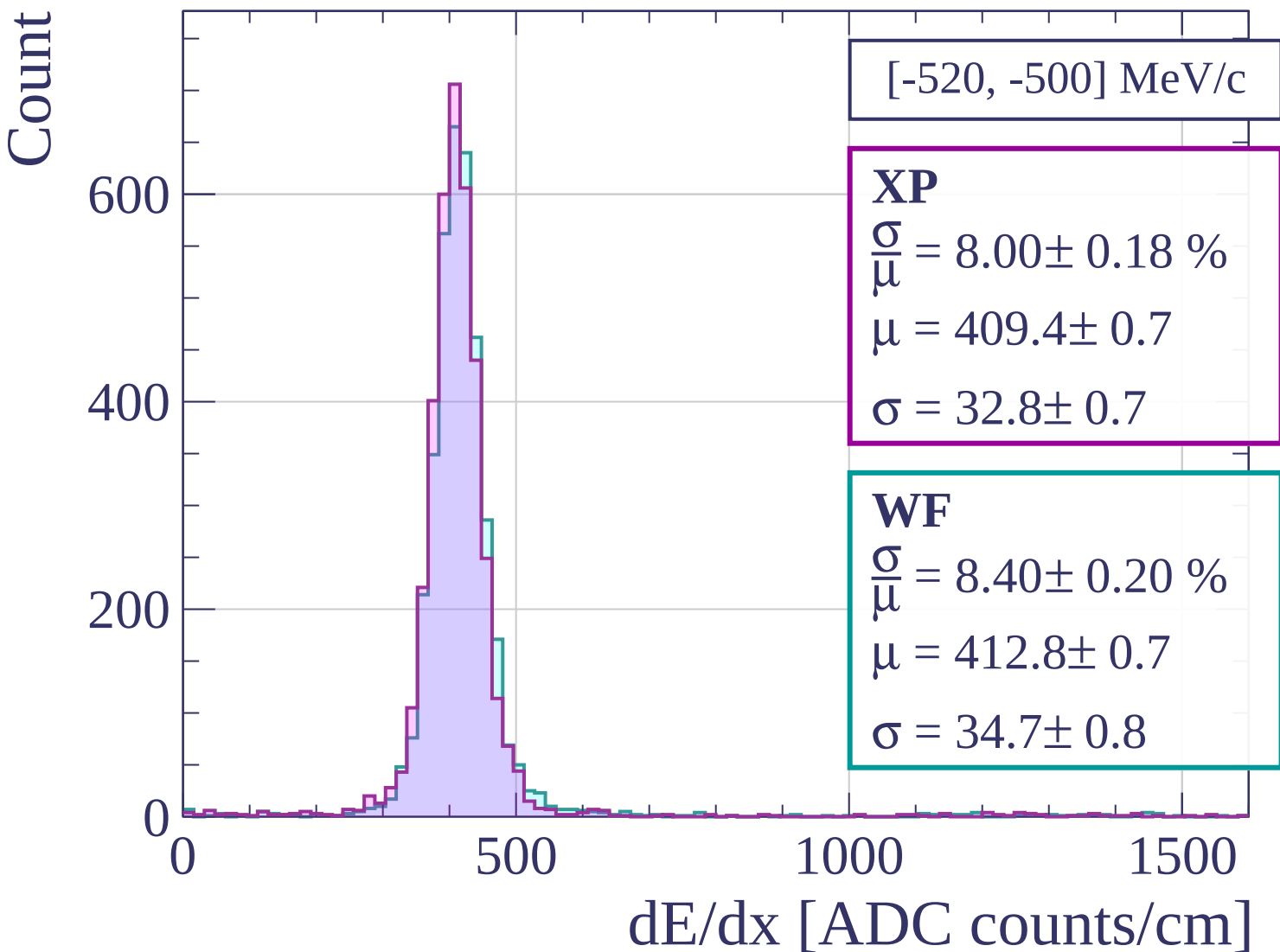
1000

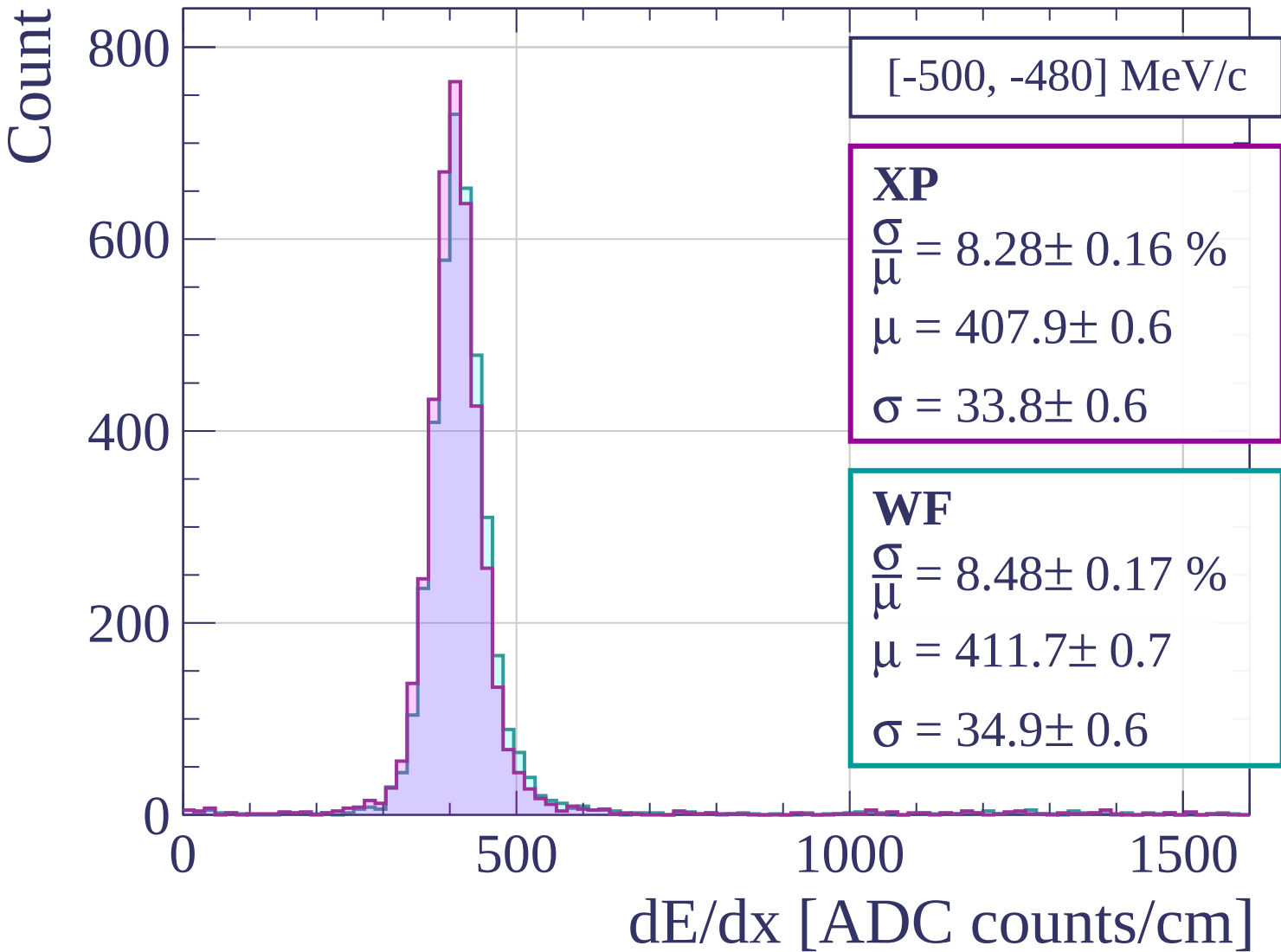
1500

dE/dx [ADC counts/cm]









Count

[-480, -460] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.17 \%$$

$$\mu = 405.3 \pm 0.6$$

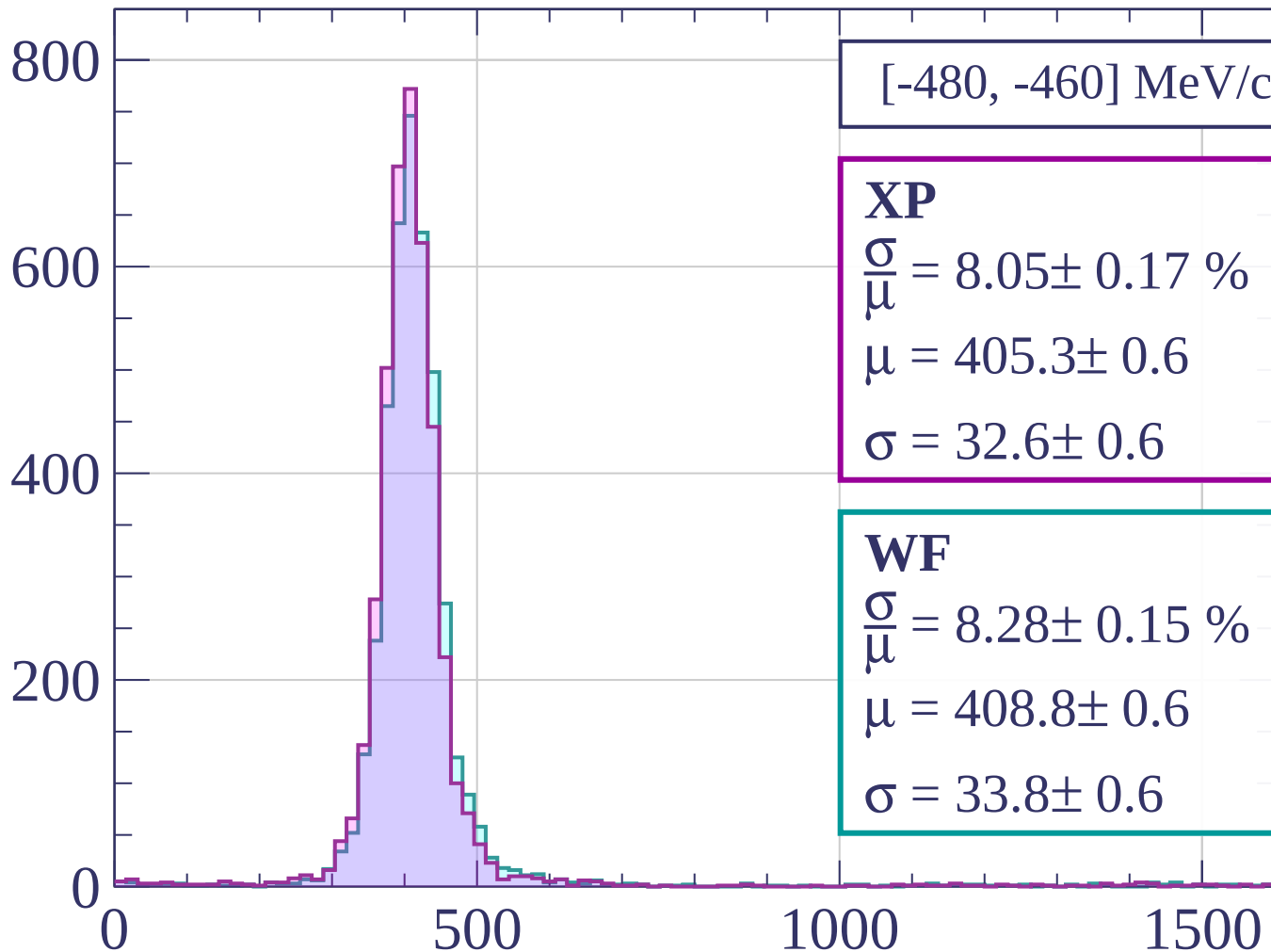
$$\sigma = 32.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.28 \pm 0.15 \%$$

$$\mu = 408.8 \pm 0.6$$

$$\sigma = 33.8 \pm 0.6$$



dE/dx [ADC counts/cm]

Count

[-460, -440] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.17 \%$$

$$\mu = 404.9 \pm 0.6$$

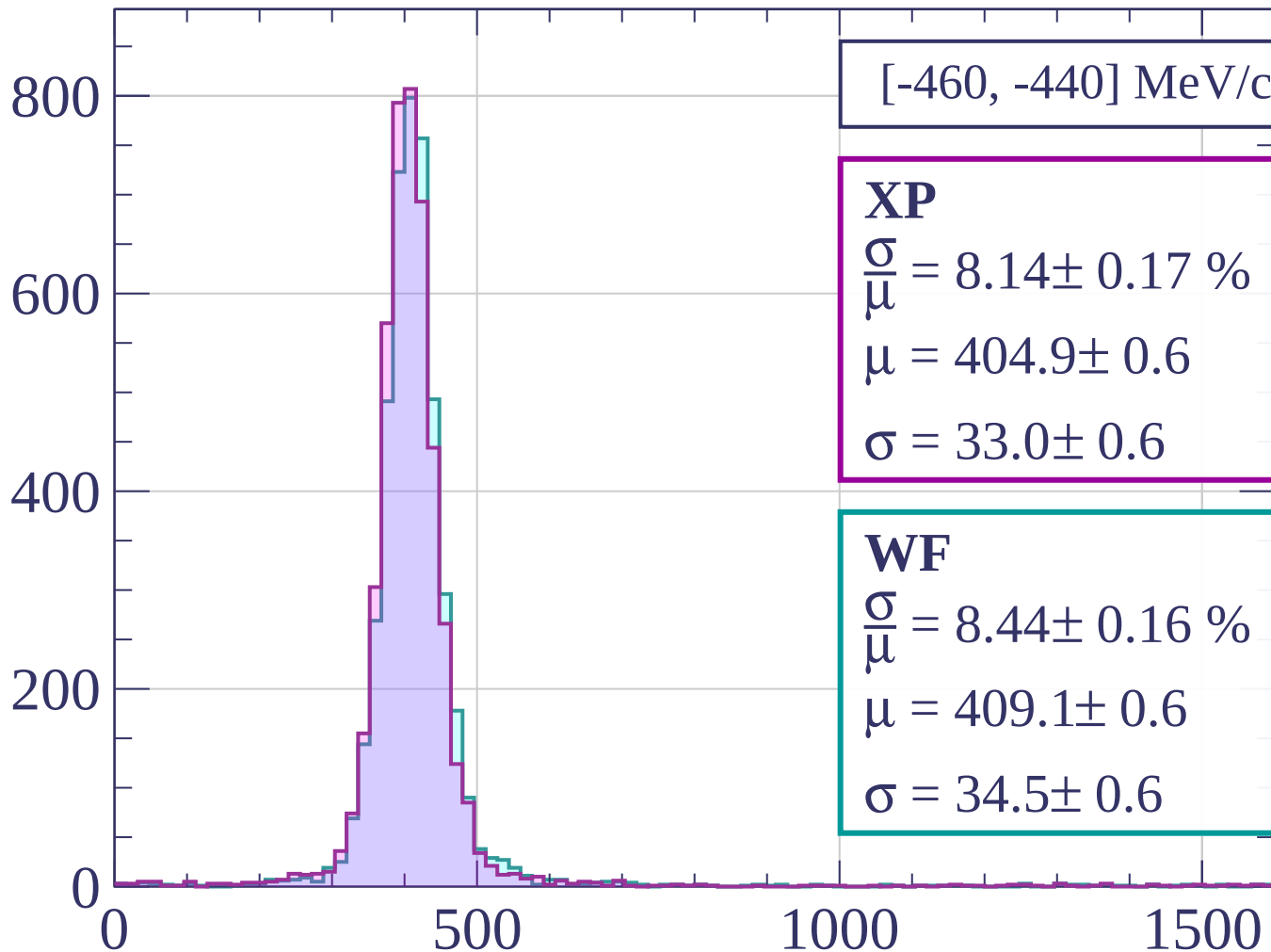
$$\sigma = 33.0 \pm 0.6$$

WF

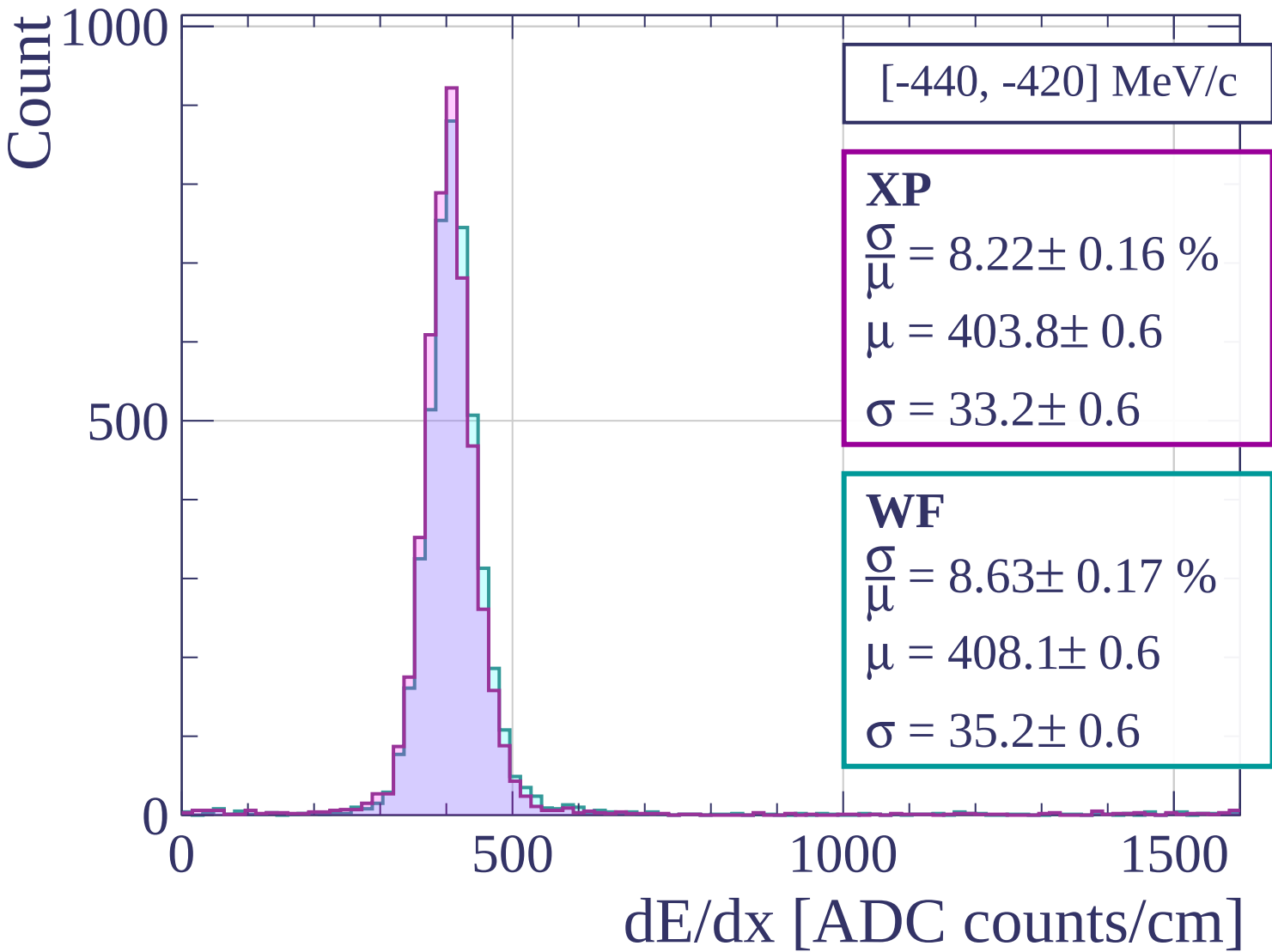
$$\frac{\sigma}{\mu} = 8.44 \pm 0.16 \%$$

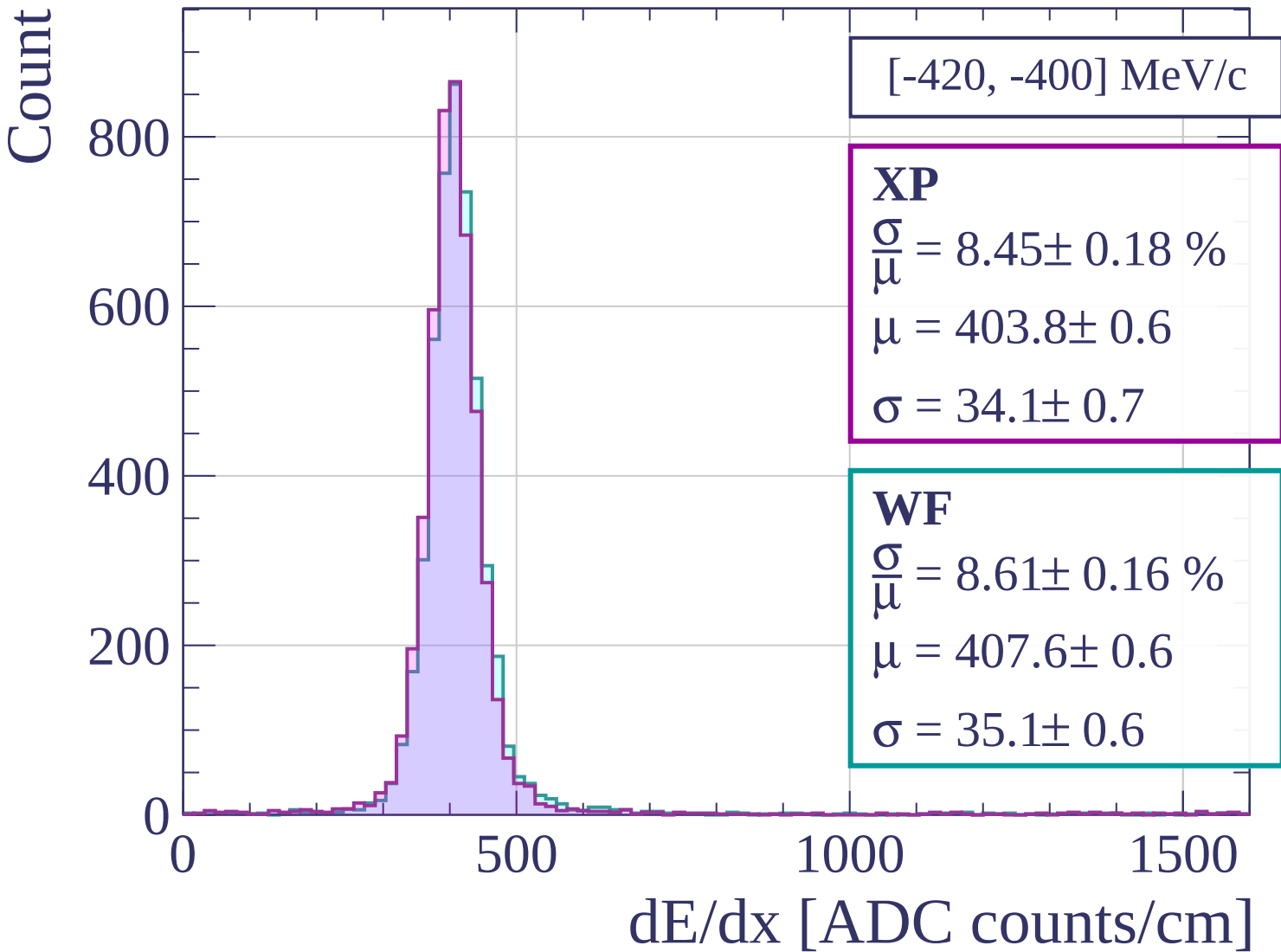
$$\mu = 409.1 \pm 0.6$$

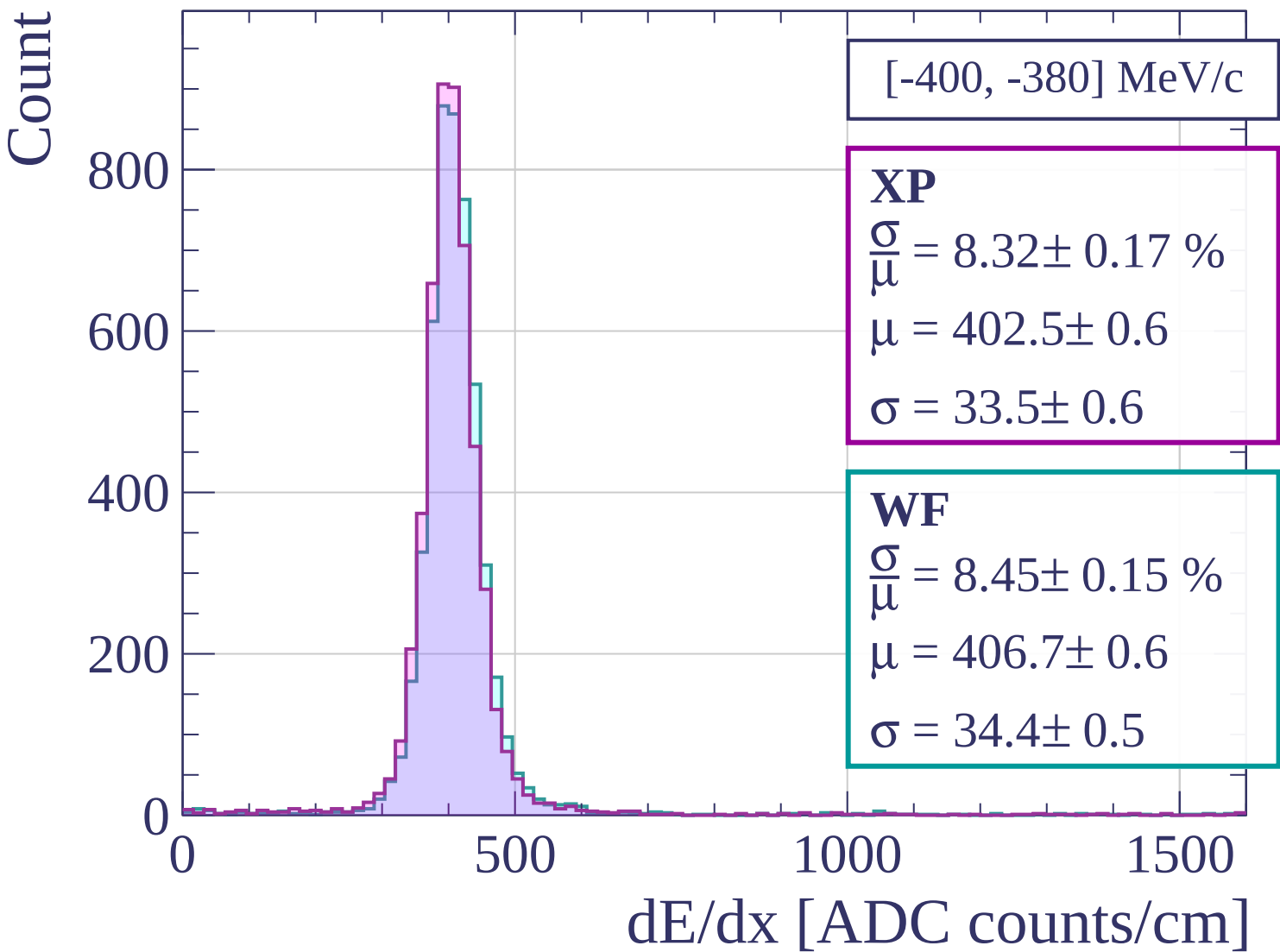
$$\sigma = 34.5 \pm 0.6$$

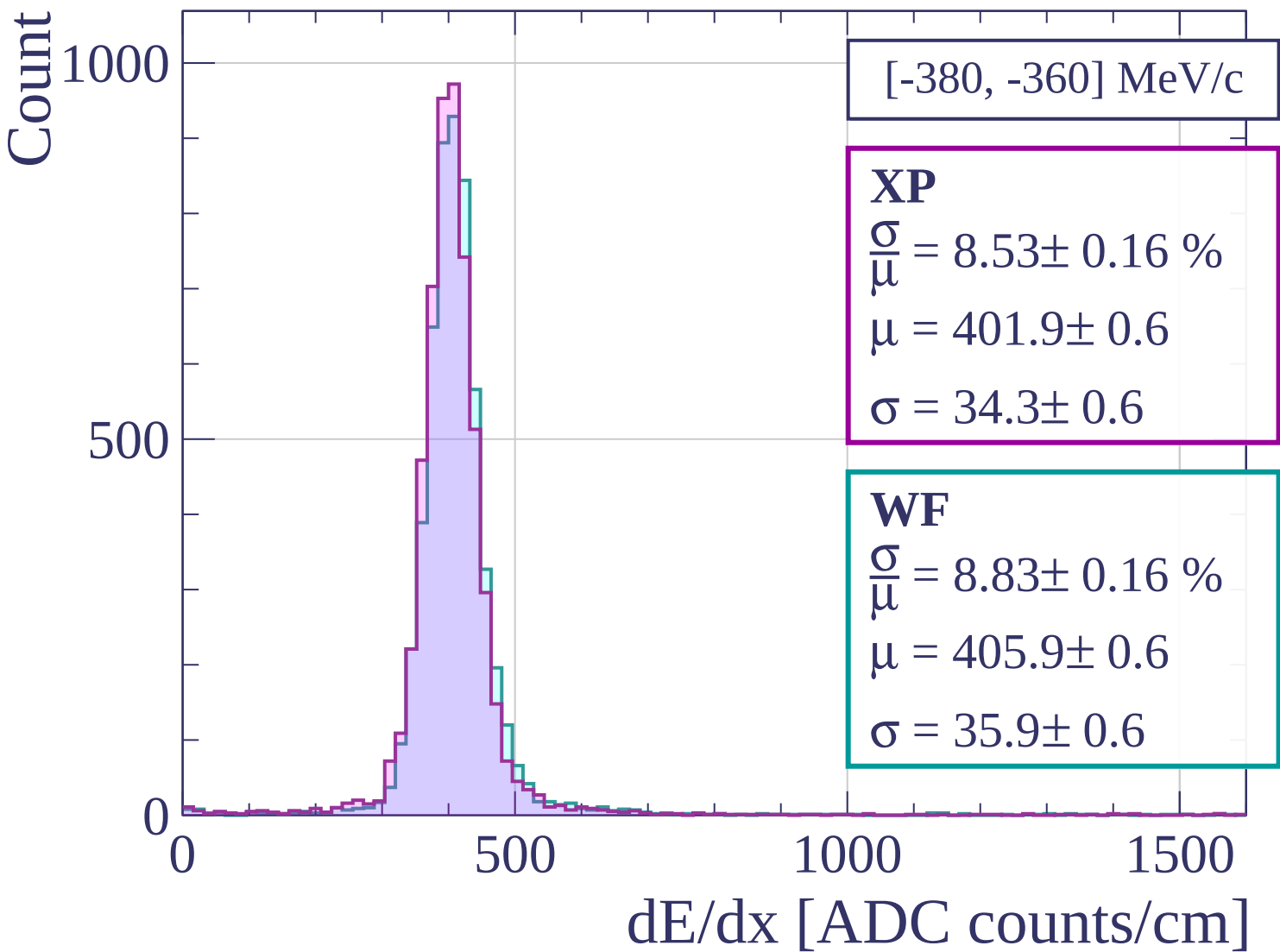


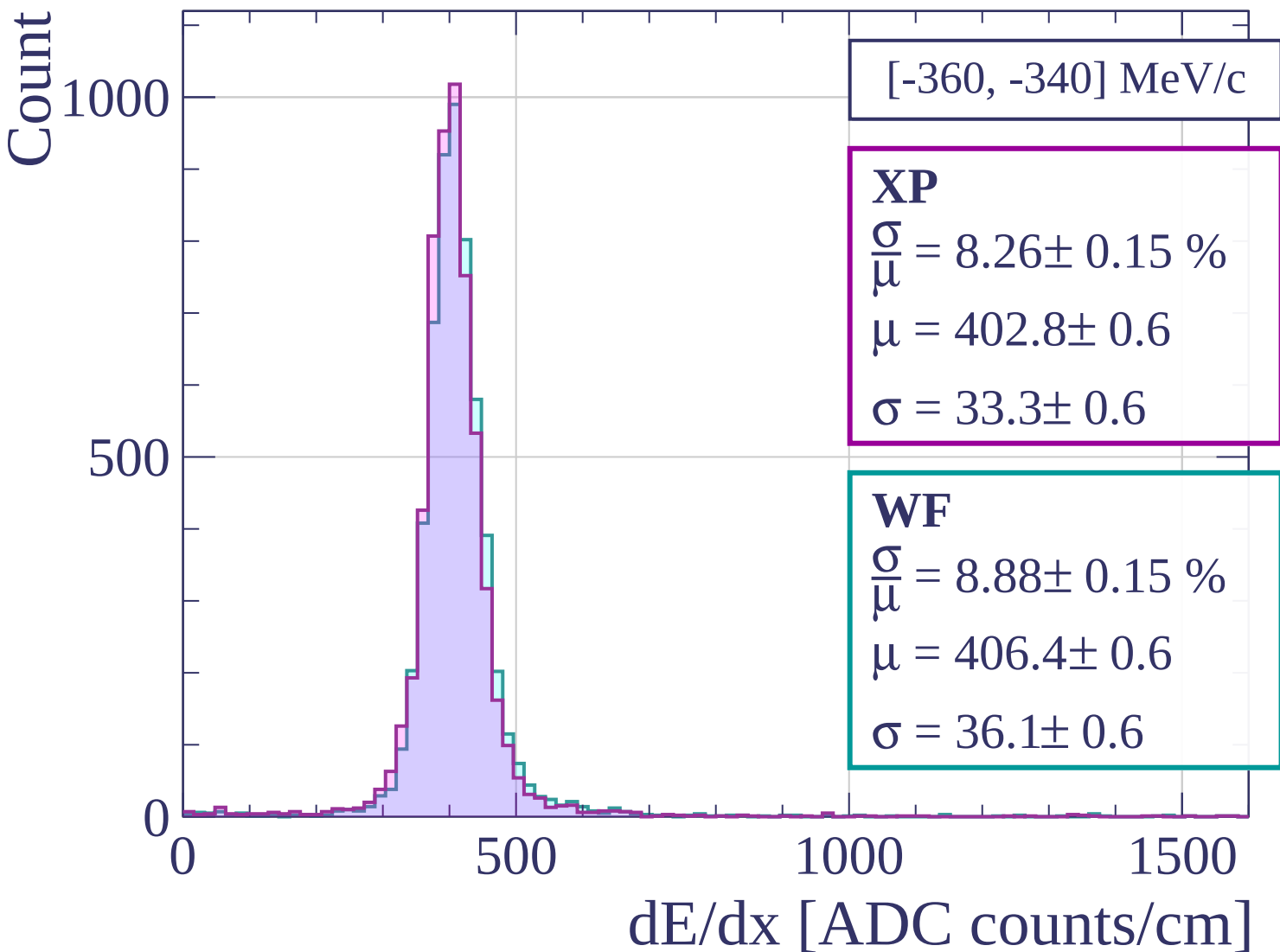
dE/dx [ADC counts/cm]

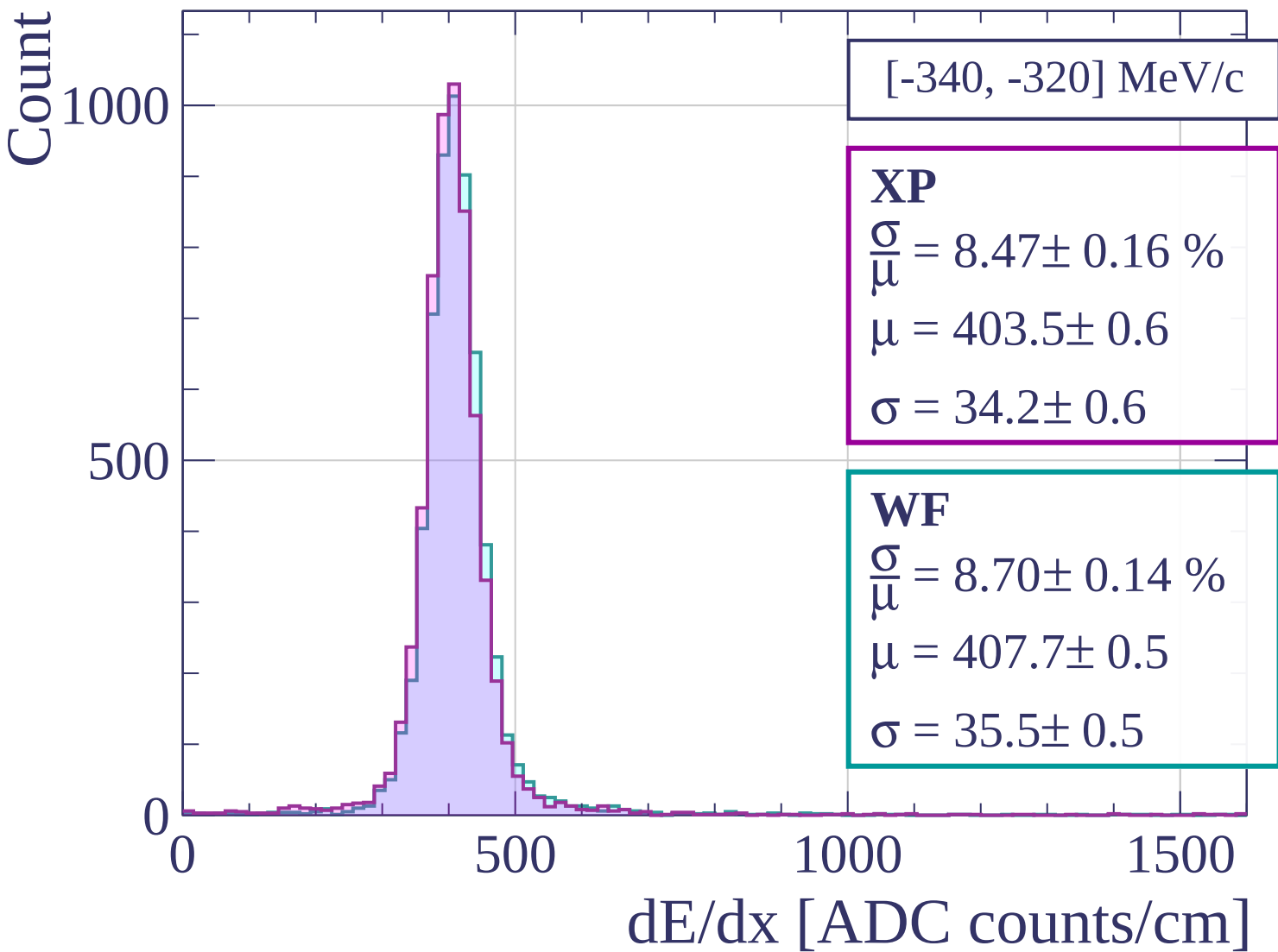


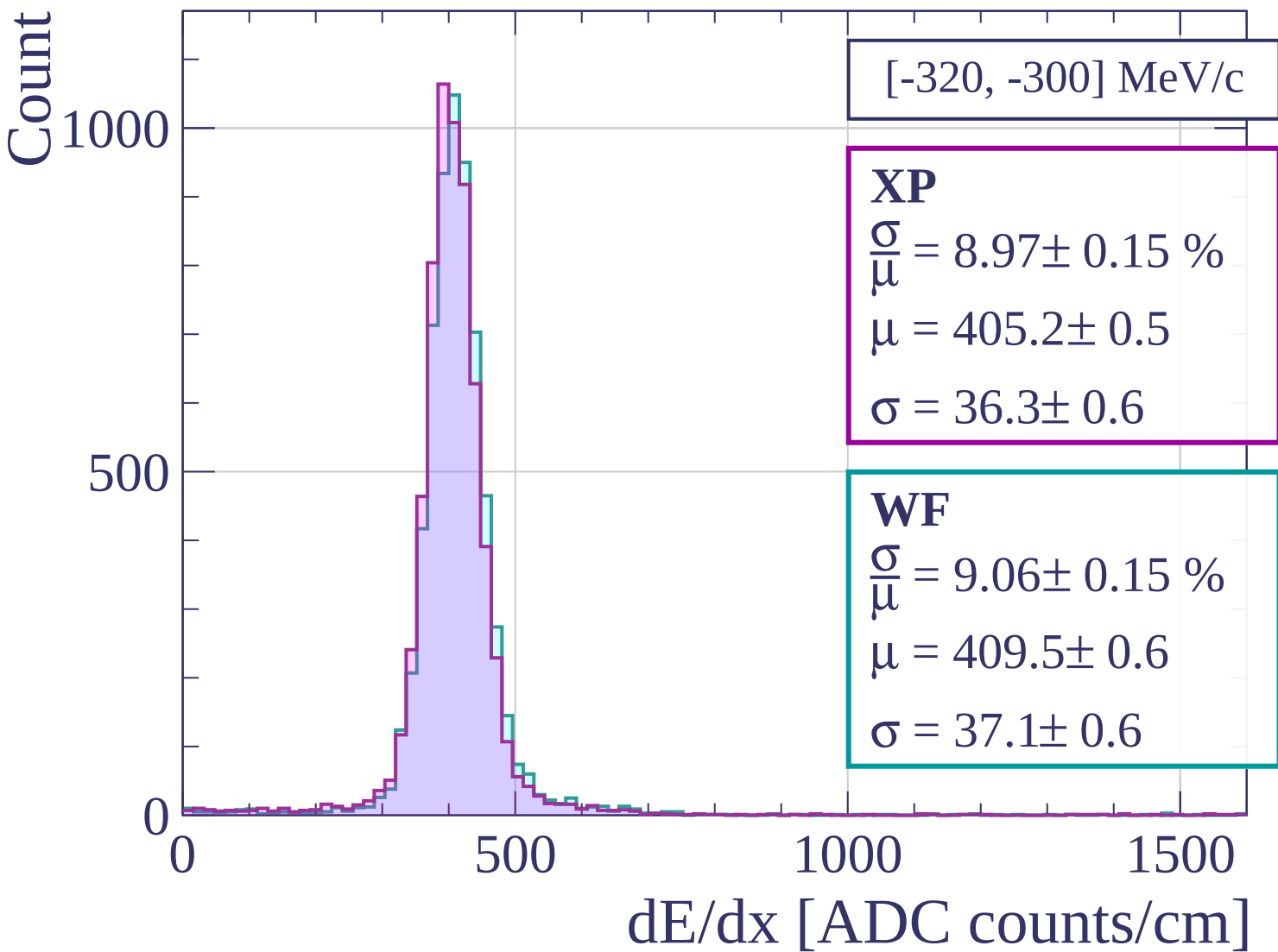


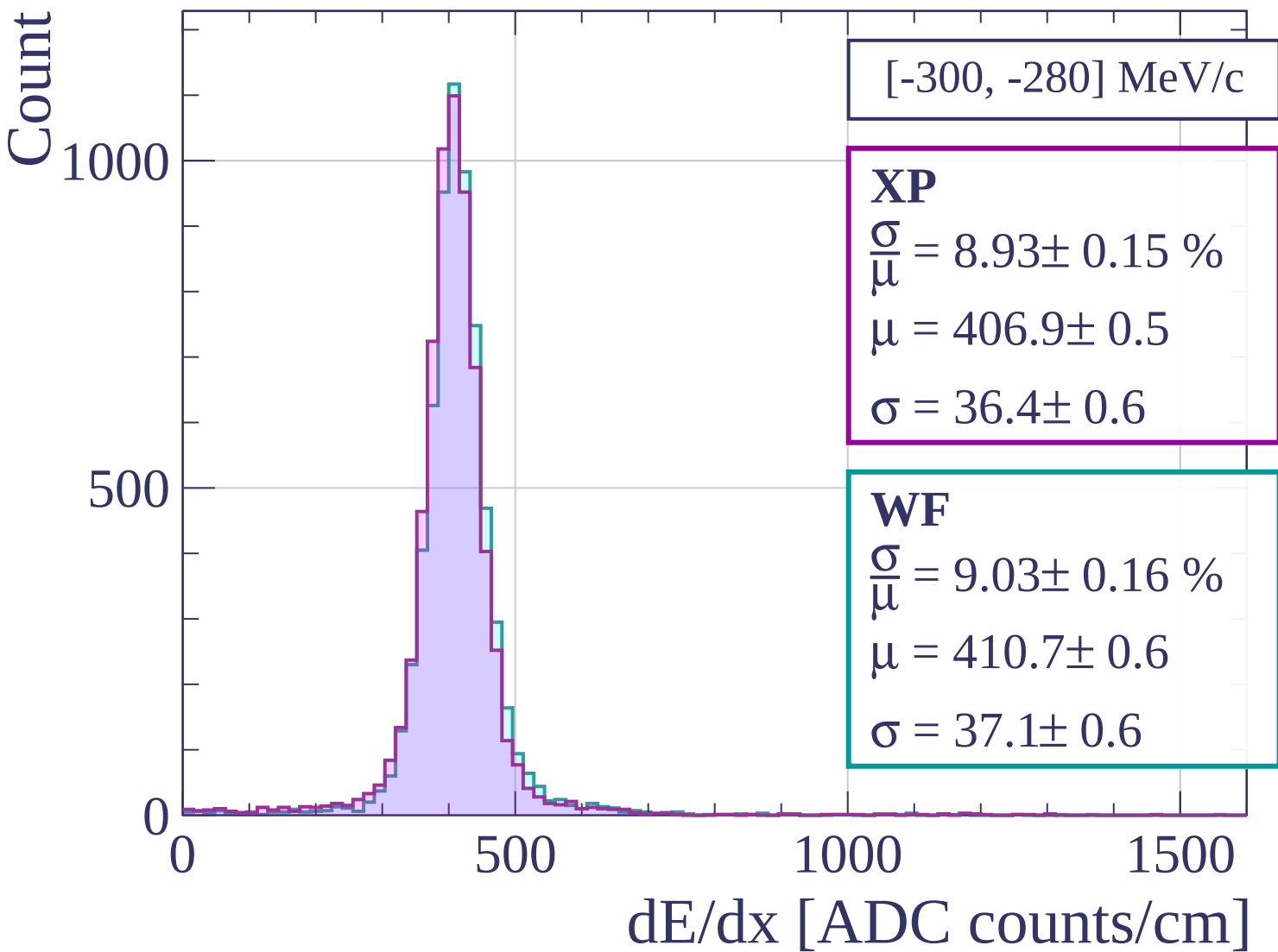


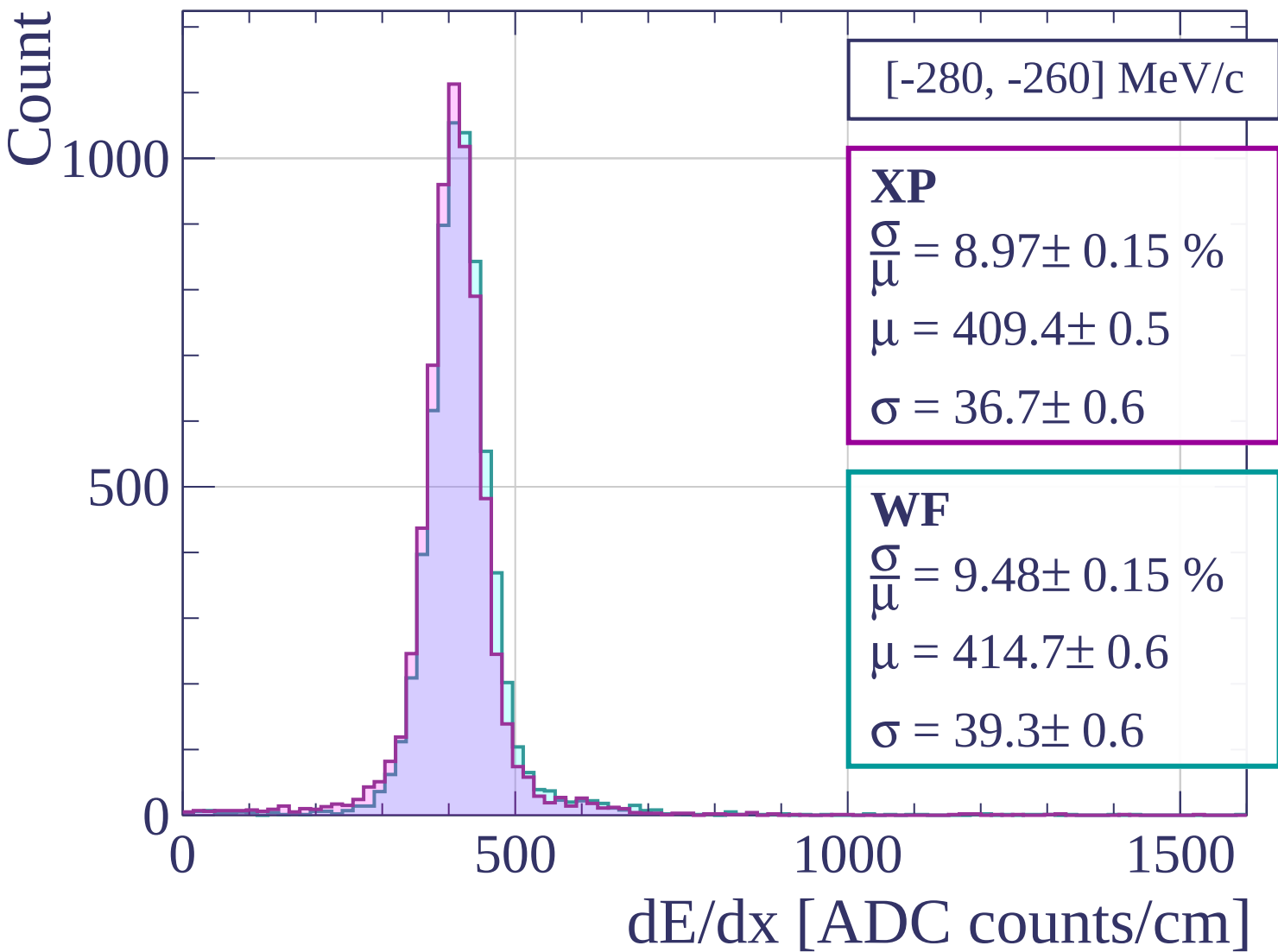


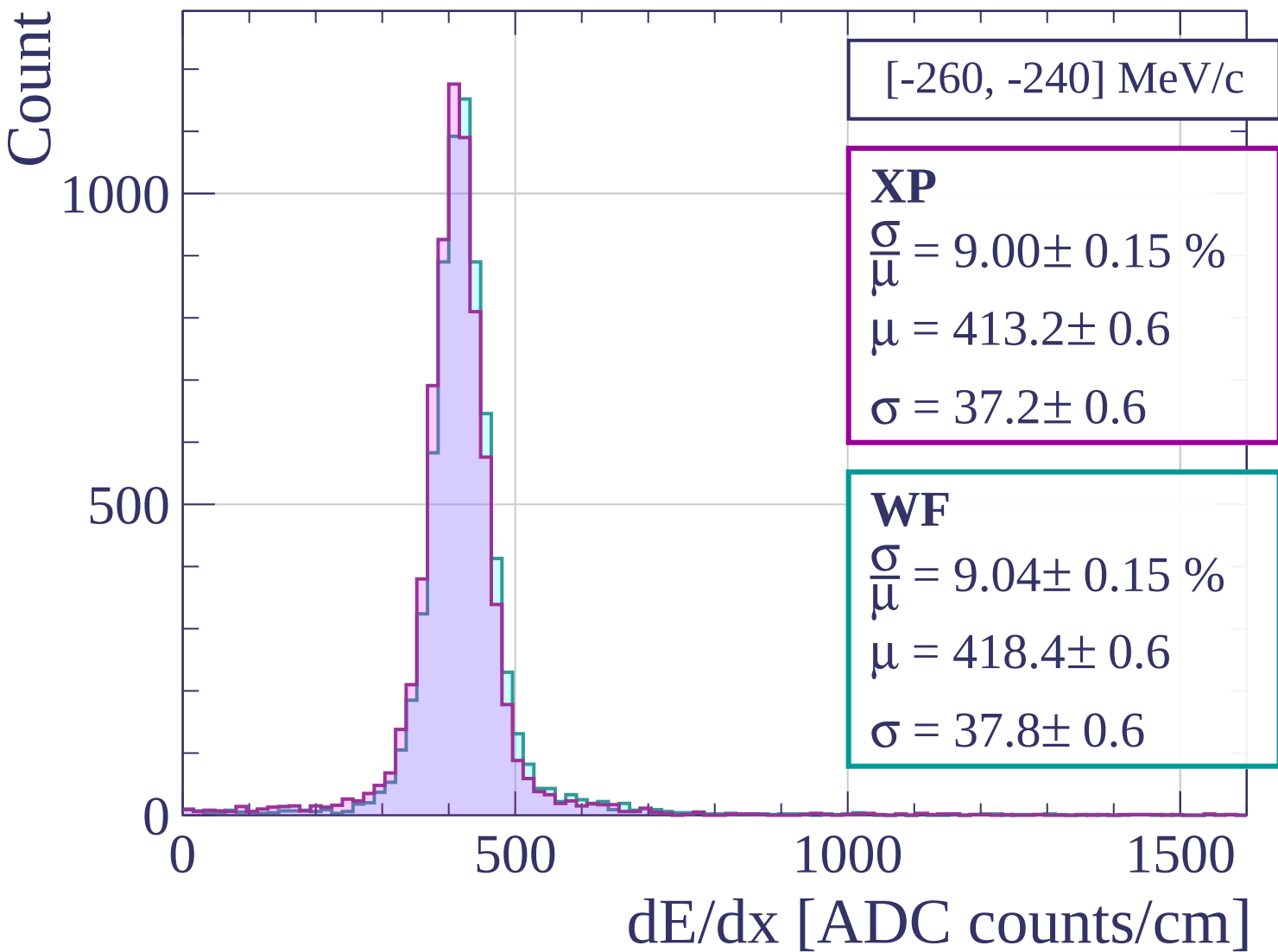


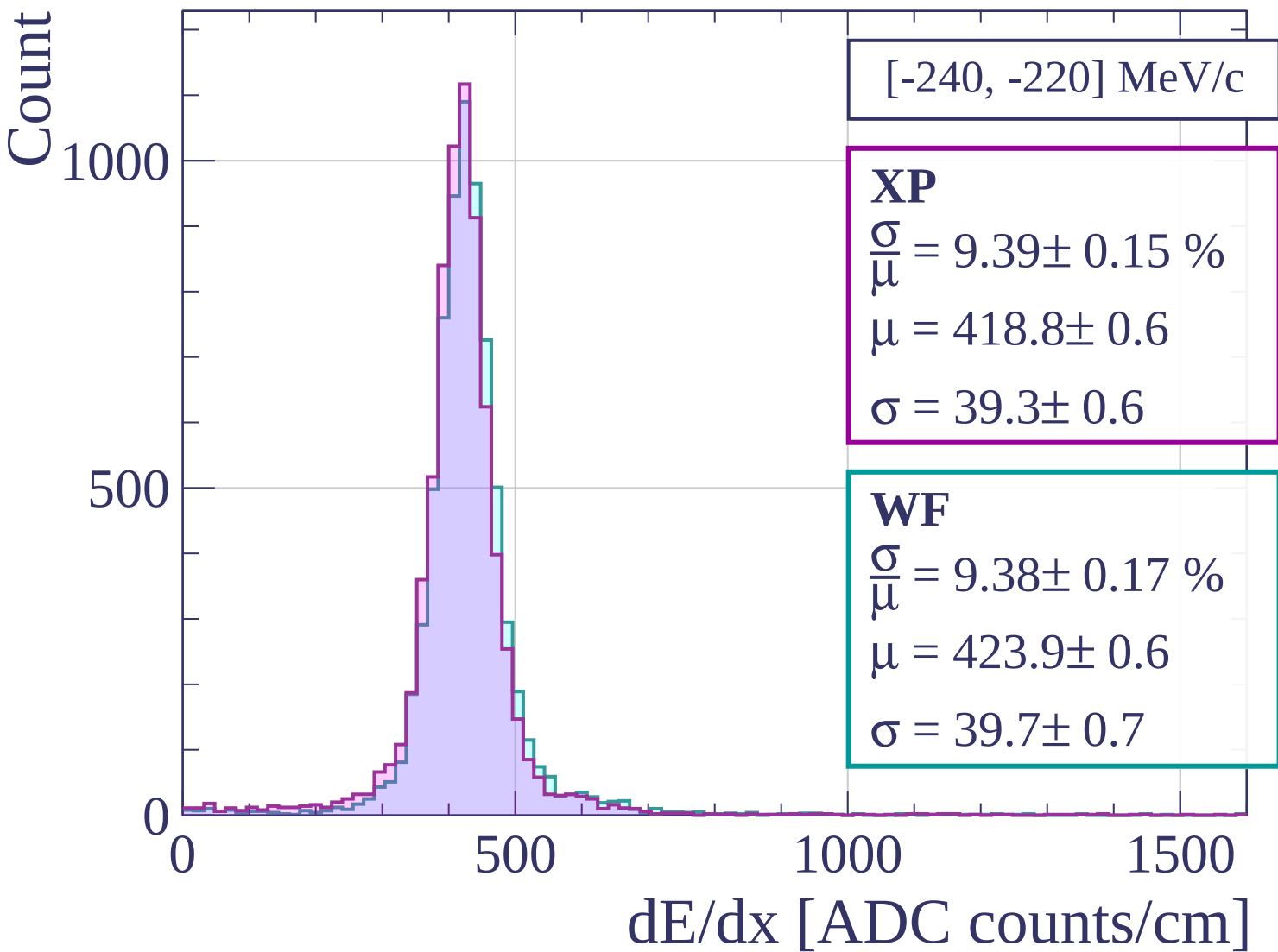


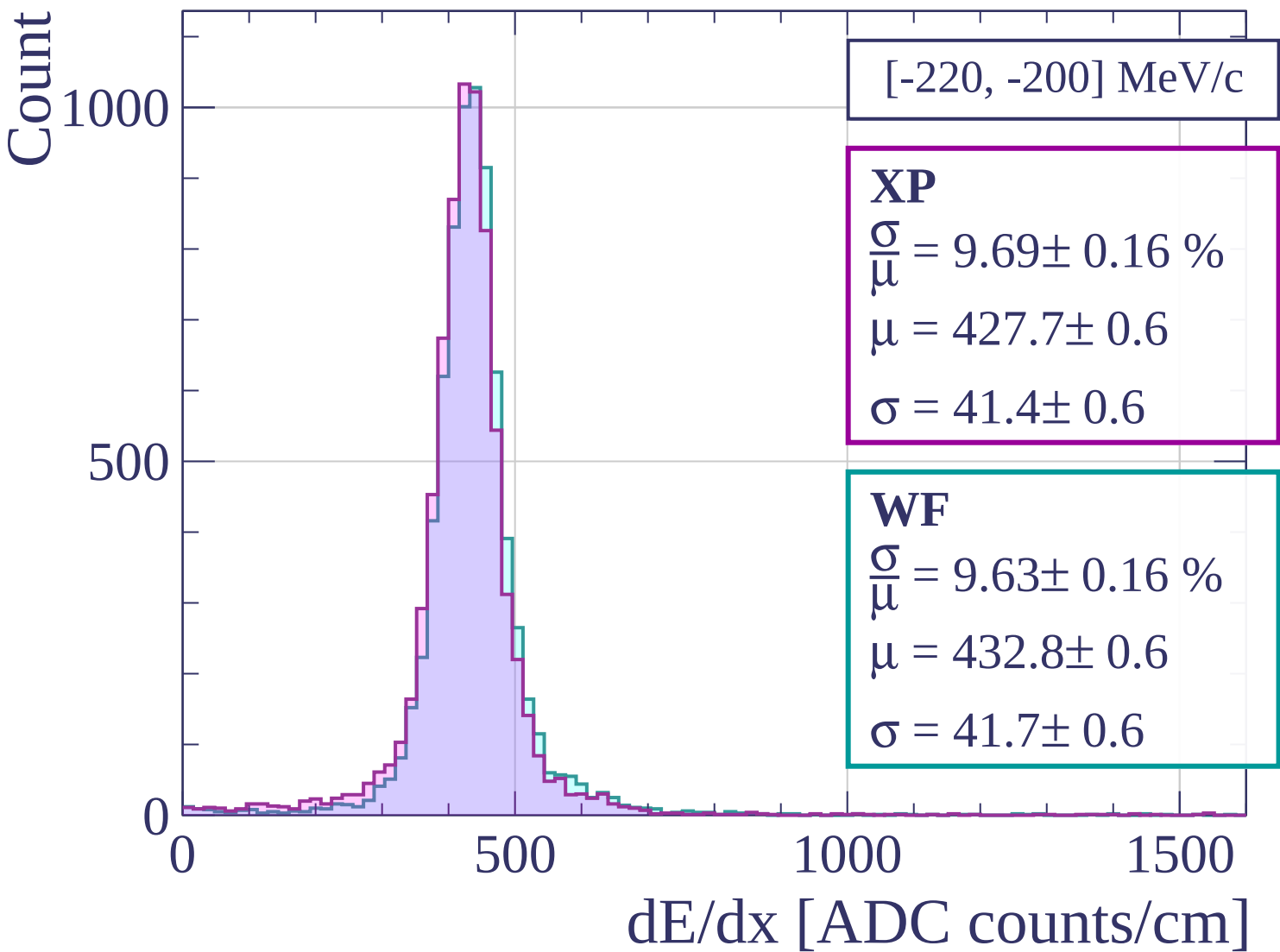


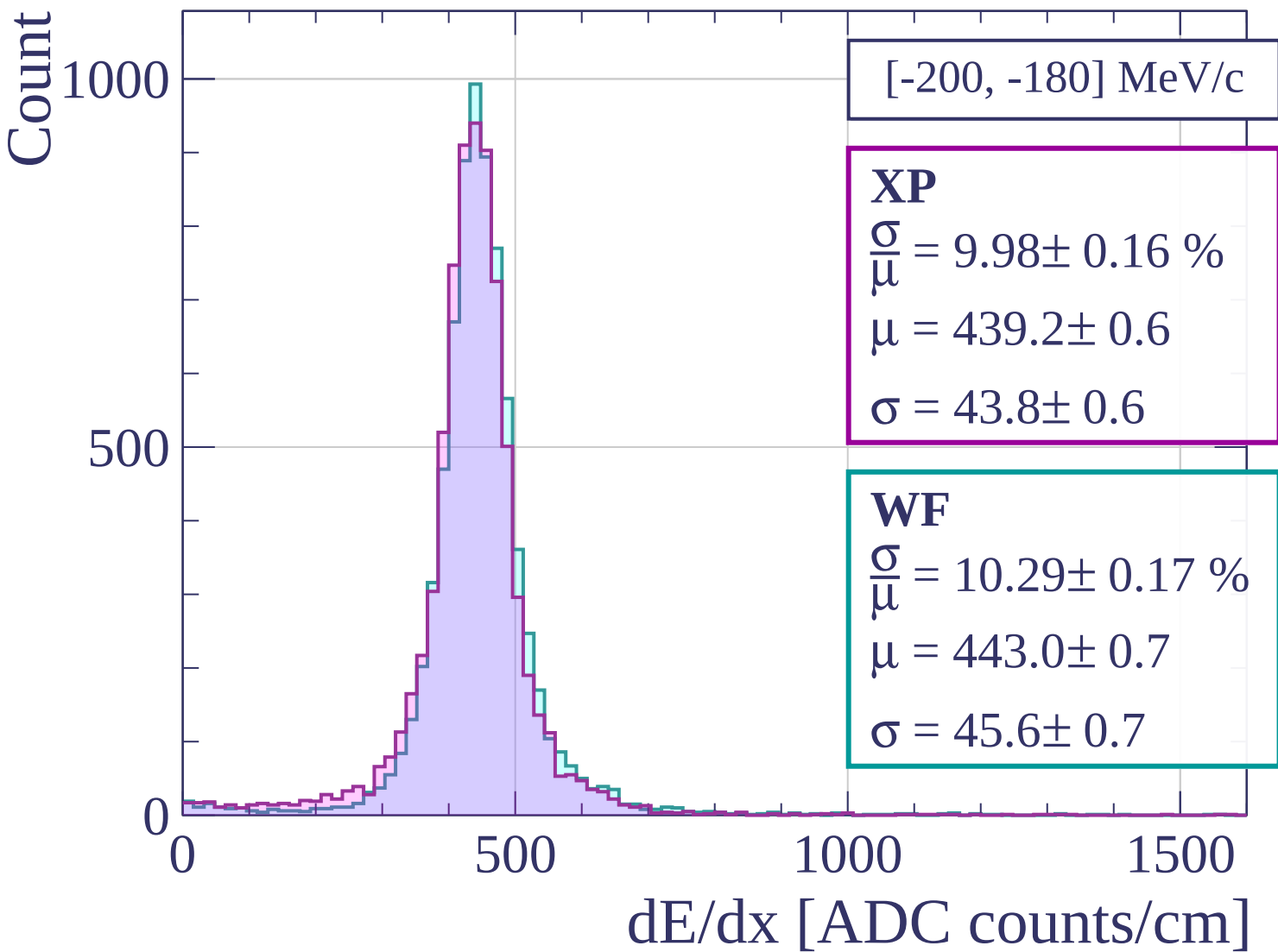












Count

[-180, -160] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.36 \pm 0.19 \%$$

$$\mu = 455.2 \pm 0.7$$

$$\sigma = 47.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.78 \pm 0.19 \%$$

$$\mu = 461.1 \pm 0.7$$

$$\sigma = 49.7 \pm 0.8$$

800

600

400

200

0

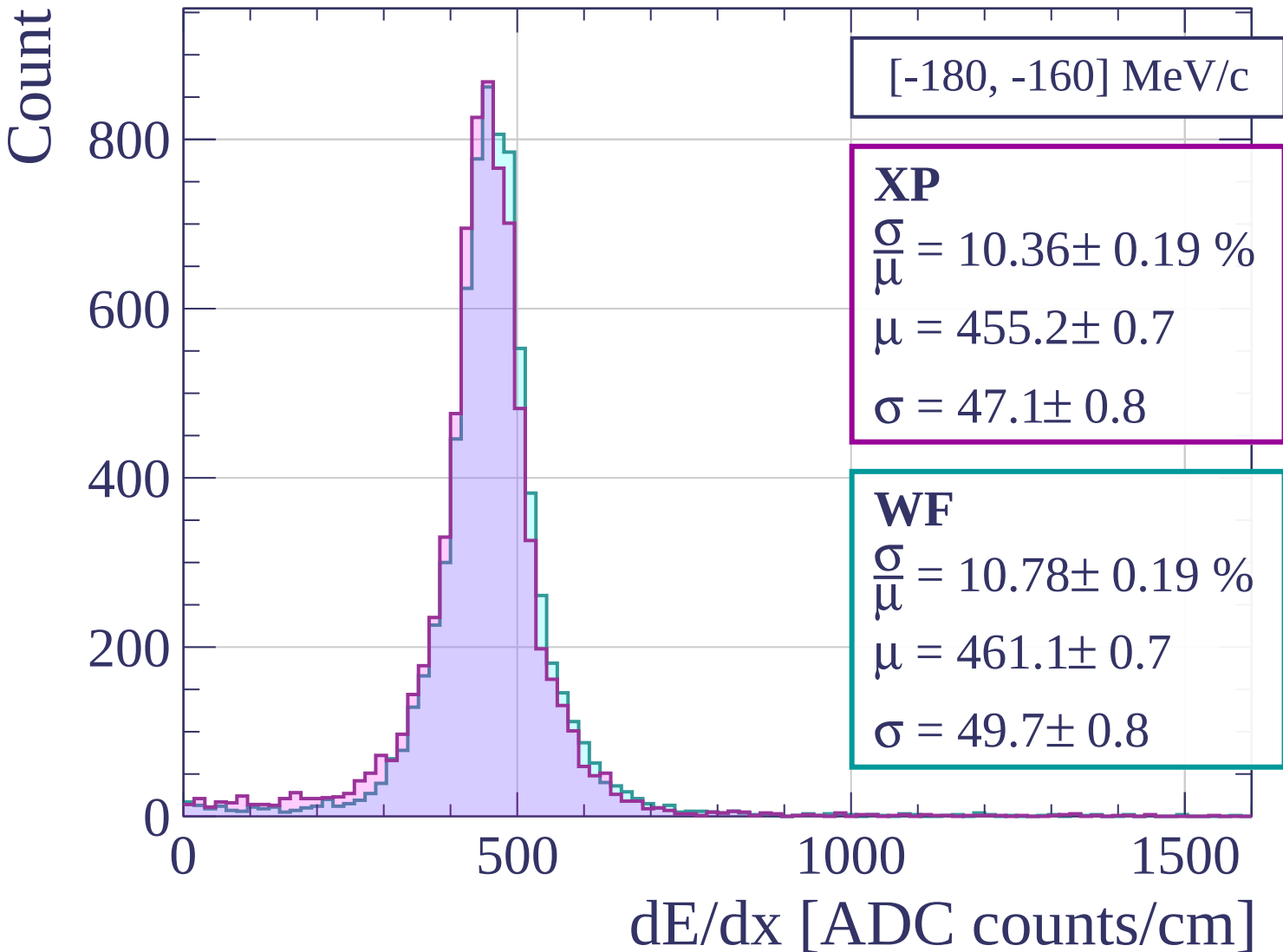
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-160, -140] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.71 \pm 0.21 \%$$

$$\mu = 479.2 \pm 0.8$$

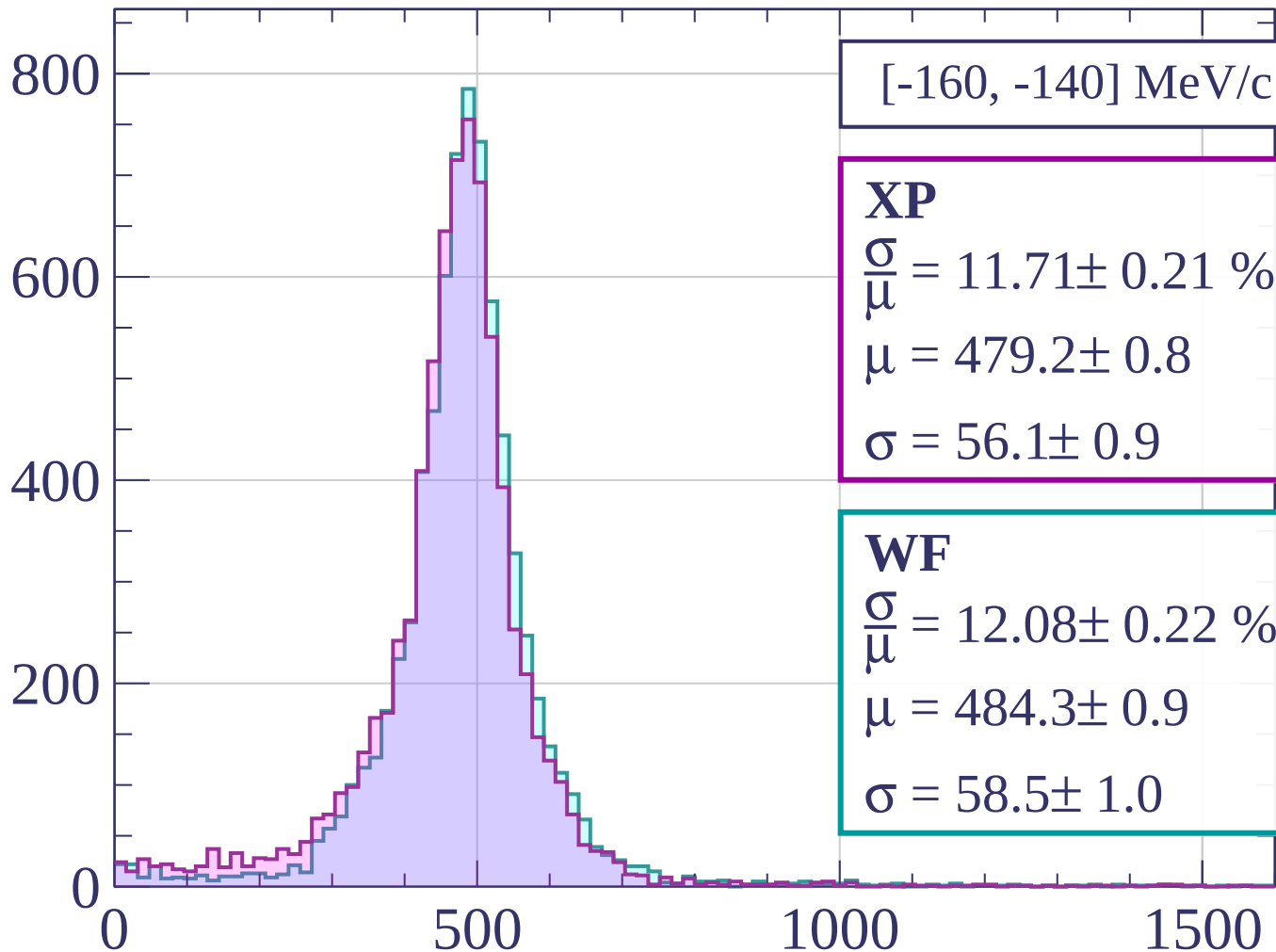
$$\sigma = 56.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 12.08 \pm 0.22 \%$$

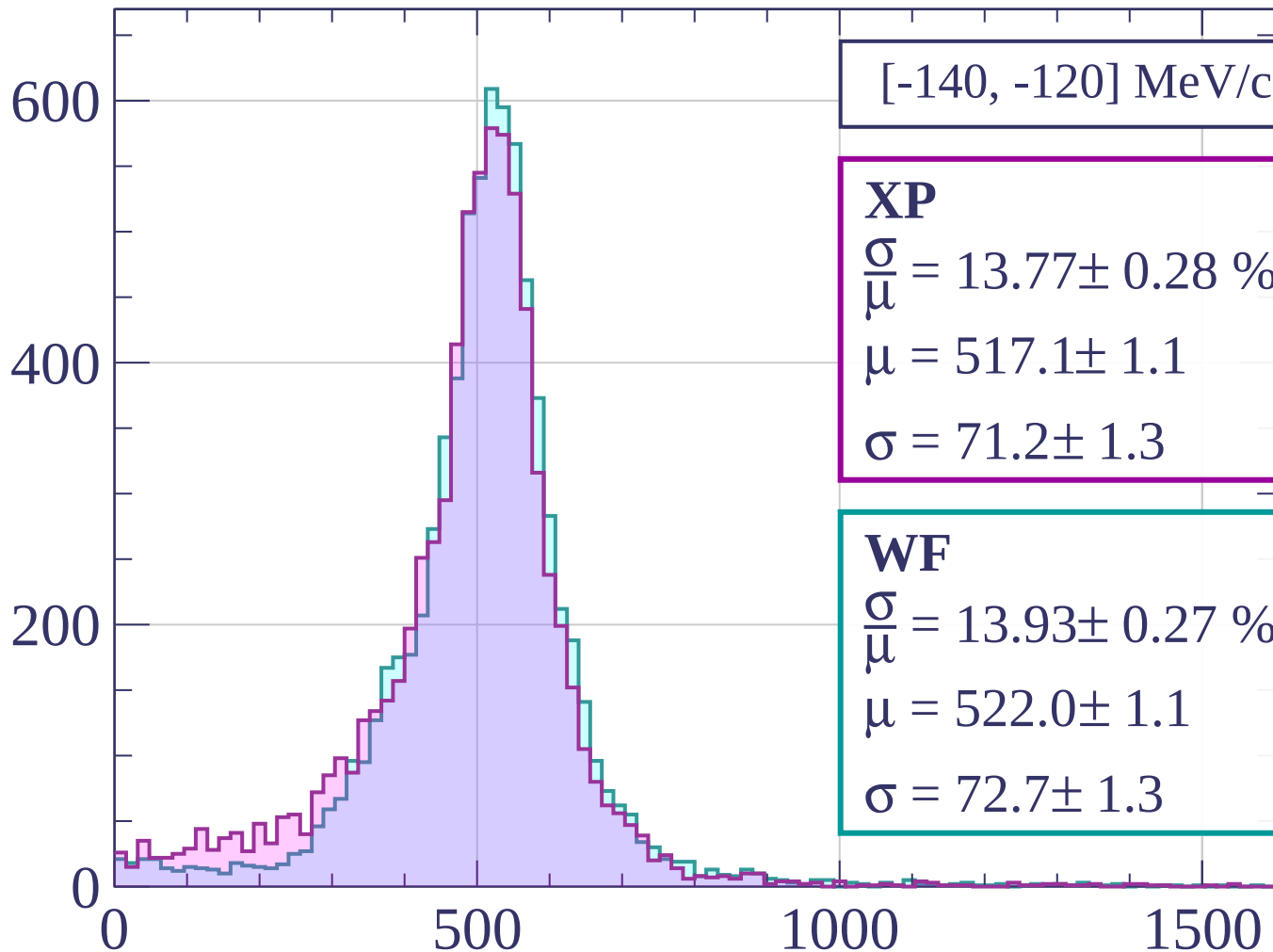
$$\mu = 484.3 \pm 0.9$$

$$\sigma = 58.5 \pm 1.0$$



dE/dx [ADC counts/cm]

Count



[-140, -120] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.77 \pm 0.28 \%$$

$$\mu = 517.1 \pm 1.1$$

$$\sigma = 71.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 13.93 \pm 0.27 \%$$

$$\mu = 522.0 \pm 1.1$$

$$\sigma = 72.7 \pm 1.3$$

dE/dx [ADC counts/cm]

Count

[-120, -100] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.80 \pm 0.32 \%$$

$$\mu = 570.9 \pm 1.4$$

$$\sigma = 84.5 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 15.55 \pm 0.33 \%$$

$$\mu = 574.4 \pm 1.5$$

$$\sigma = 89.3 \pm 1.7$$

400

200

0

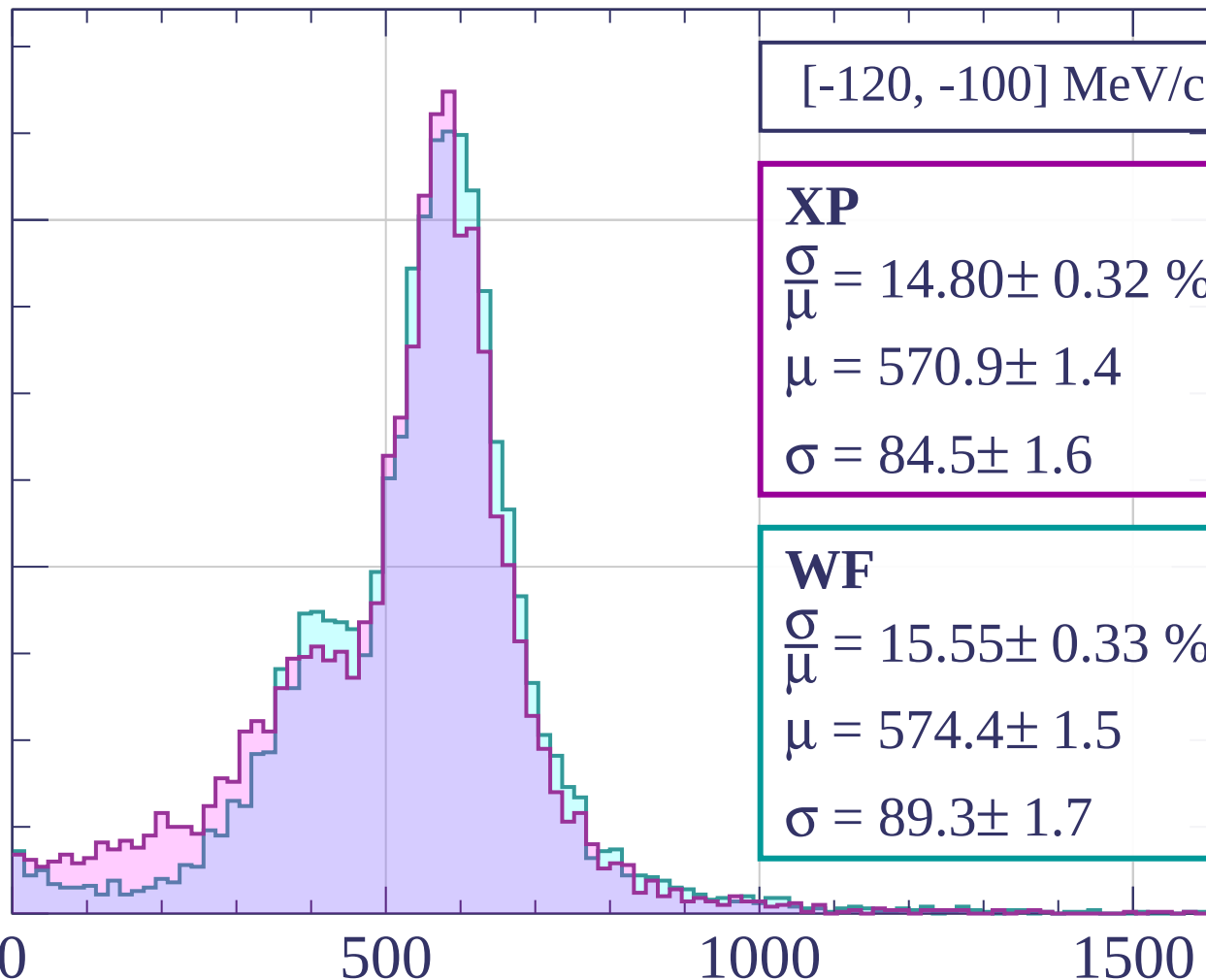
0

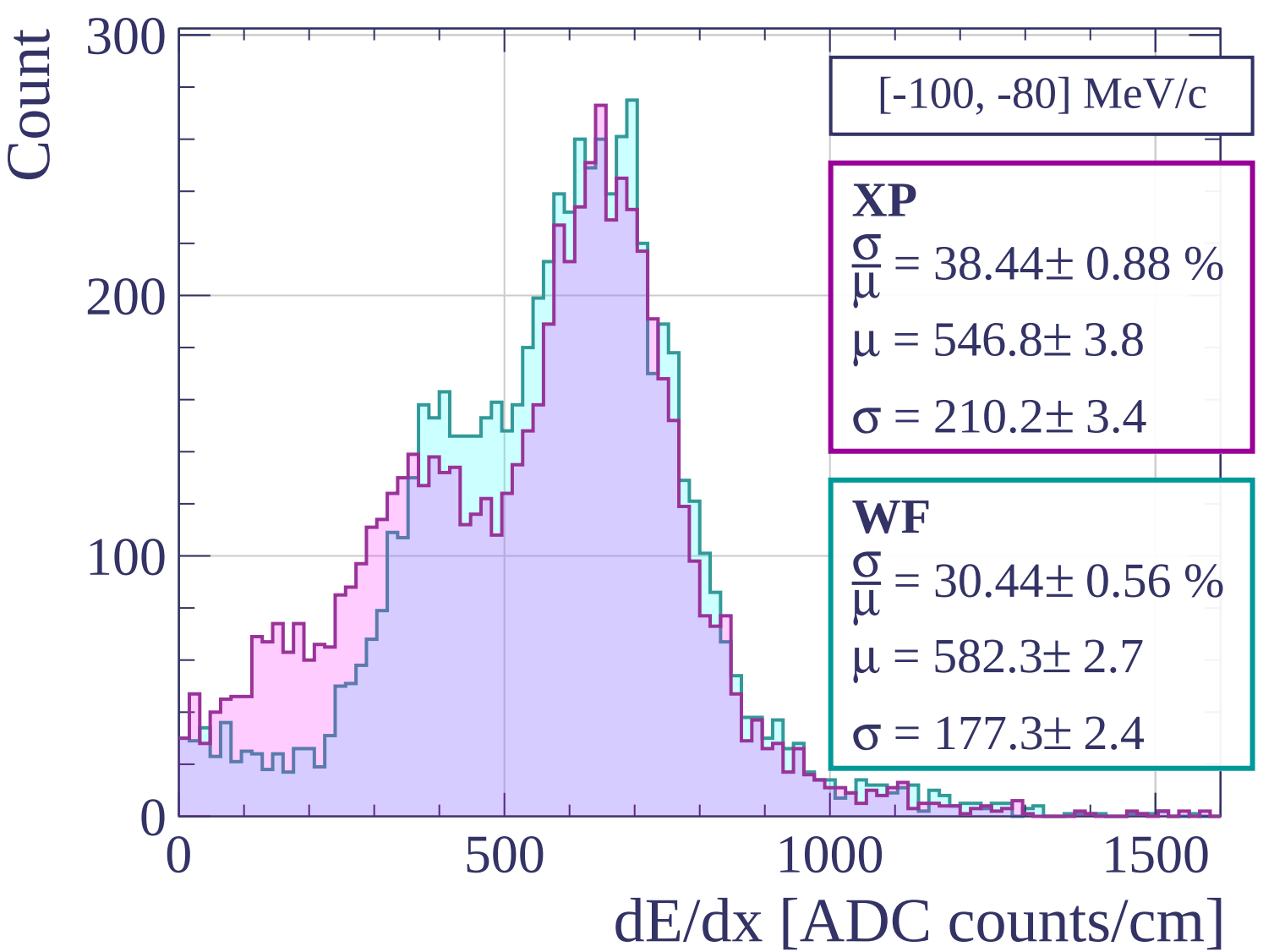
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-80, -60] MeV/c

XP

$$\frac{\sigma}{\mu} = 74.63 \pm 2.21 \%$$

$$\mu = 477.2 \pm 6.2$$

$$\sigma = 356.2 \pm 6.0$$

WF

$$\frac{\sigma}{\mu} = 47.90 \pm 1.32 \%$$

$$\mu = 574.5 \pm 4.9$$

$$\sigma = 275.2 \pm 5.2$$

200

150

100

50

0

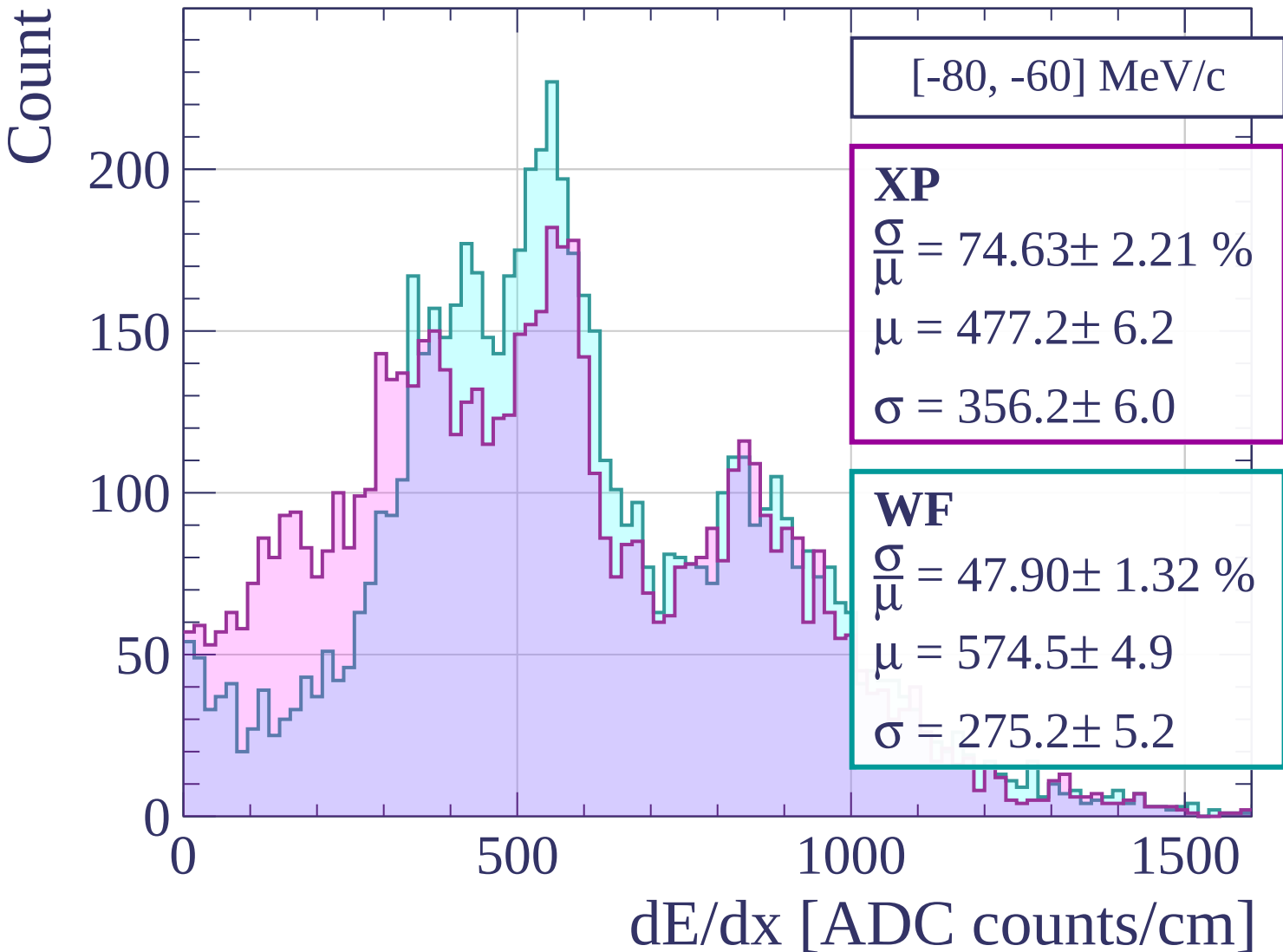
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-60, -40] MeV/c

XP

$$\frac{\sigma}{\mu} = 64.21 \pm 1.86 \%$$

$$\mu = 339.3 \pm 4.5$$

$$\sigma = 217.9 \pm 3.5$$

WF

$$\frac{\sigma}{\mu} = 29.47 \pm 0.60 \%$$

$$\mu = 472.3 \pm 2.3$$

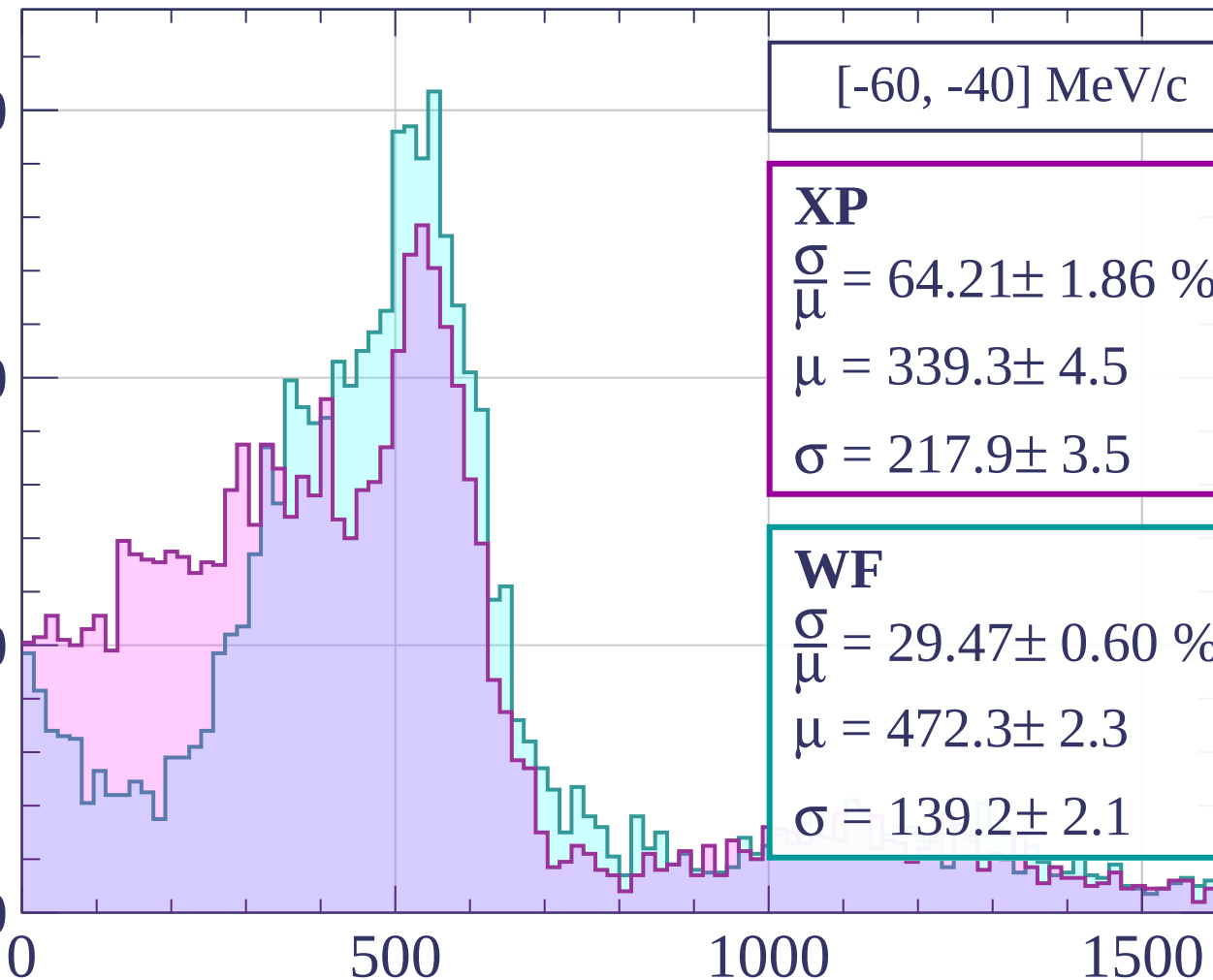
$$\sigma = 139.2 \pm 2.1$$

300

200

100

0



dE/dx [ADC counts/cm]

Count

[-40, -20] MeV/c

XP

$$\frac{\sigma}{\mu} = 70.84 \pm 2.08 \%$$

$$\mu = 322.2 \pm 4.6$$

$$\sigma = 228.2 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 21.41 \pm 0.38 \%$$

$$\mu = 499.2 \pm 1.6$$

$$\sigma = 106.9 \pm 1.6$$

600

400

200

0

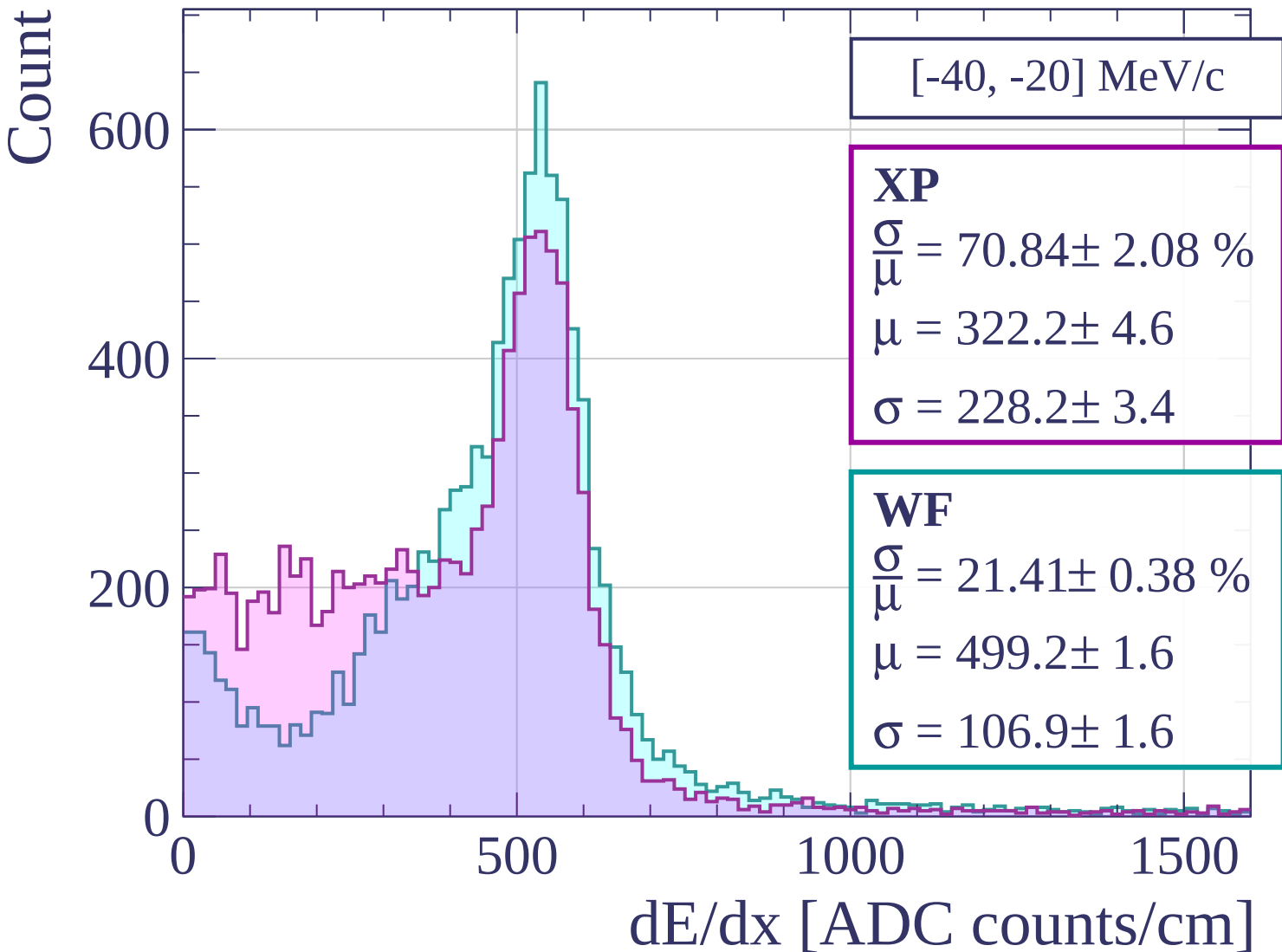
0

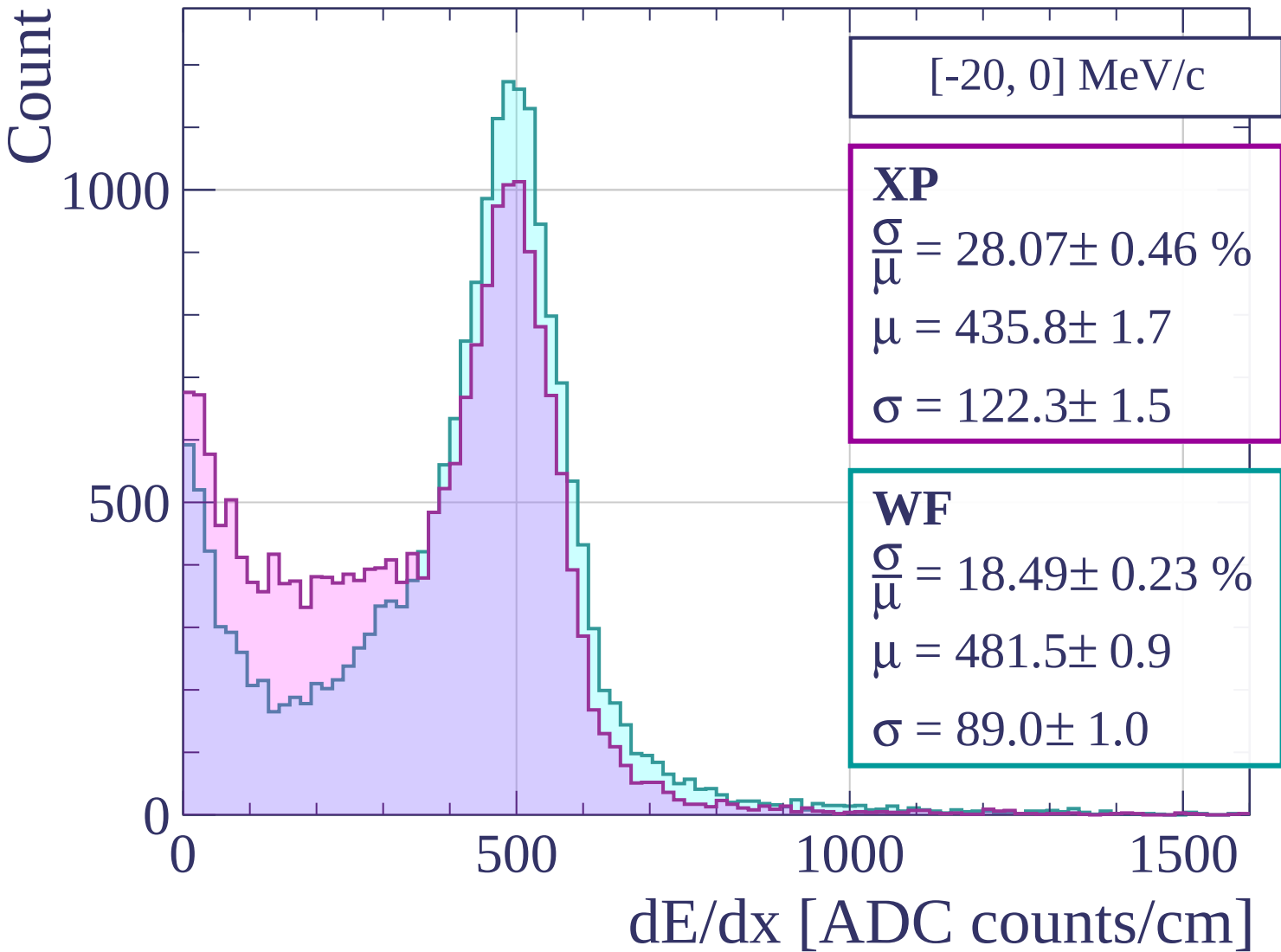
500

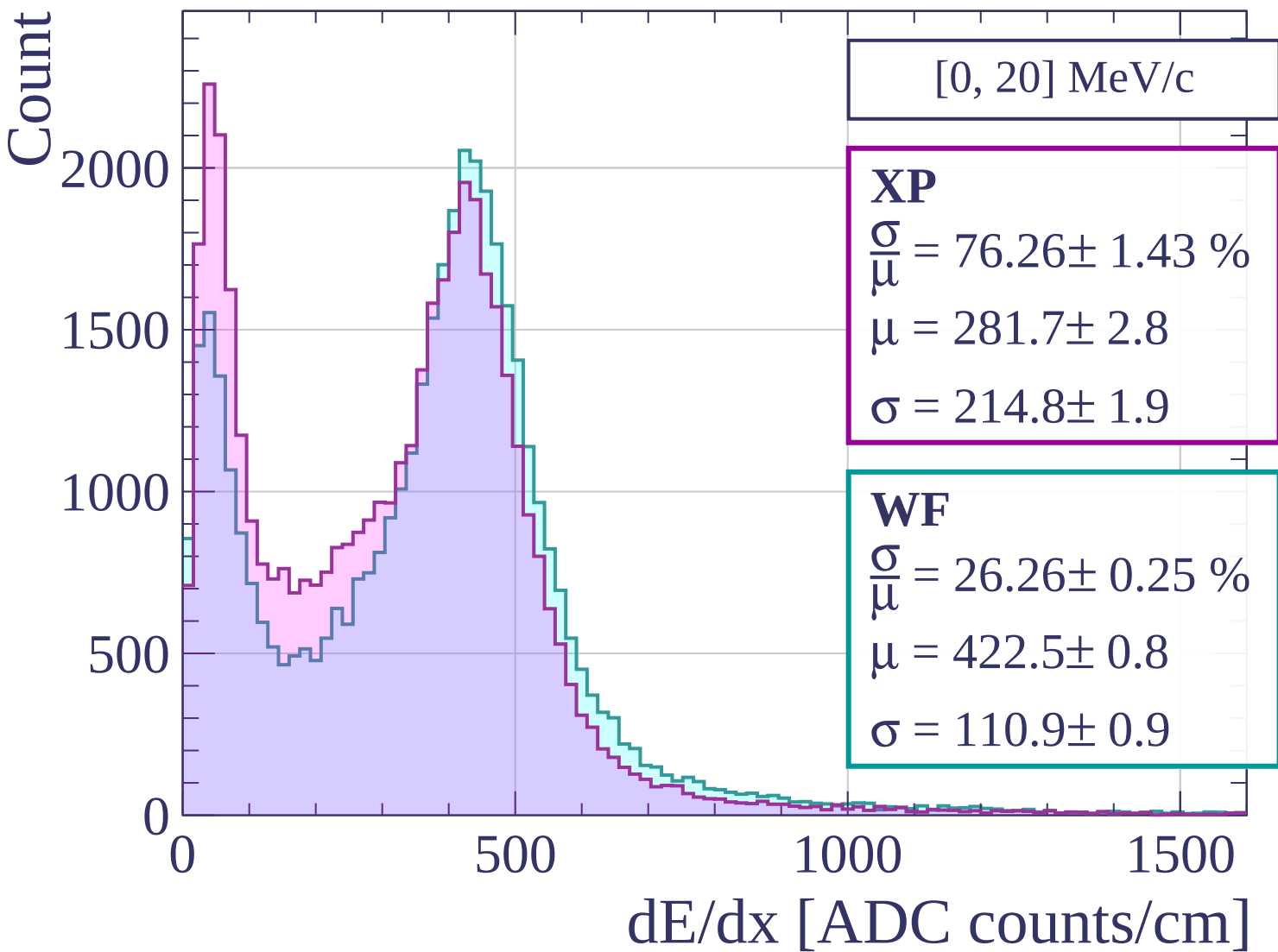
1000

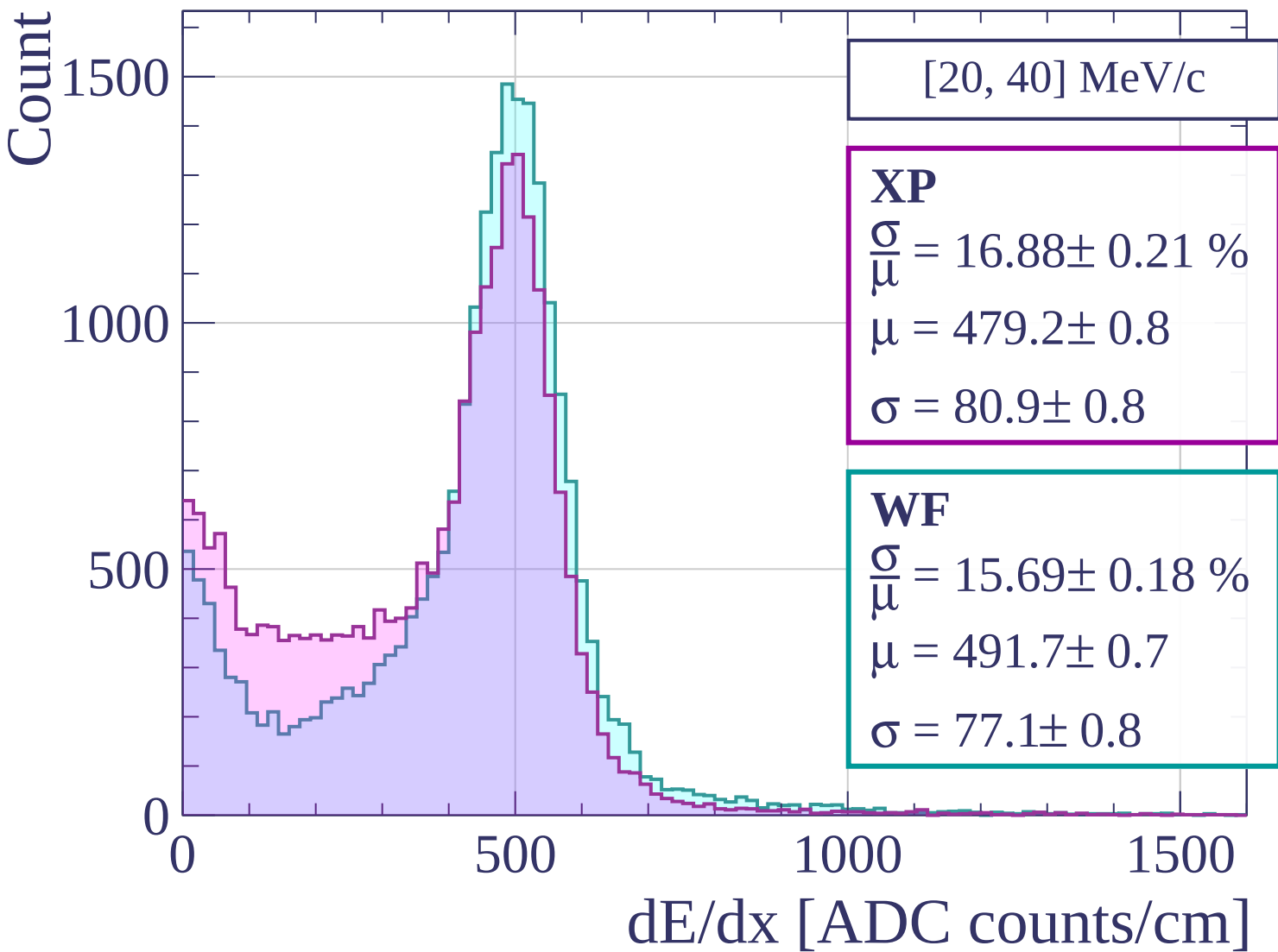
1500

dE/dx [ADC counts/cm]

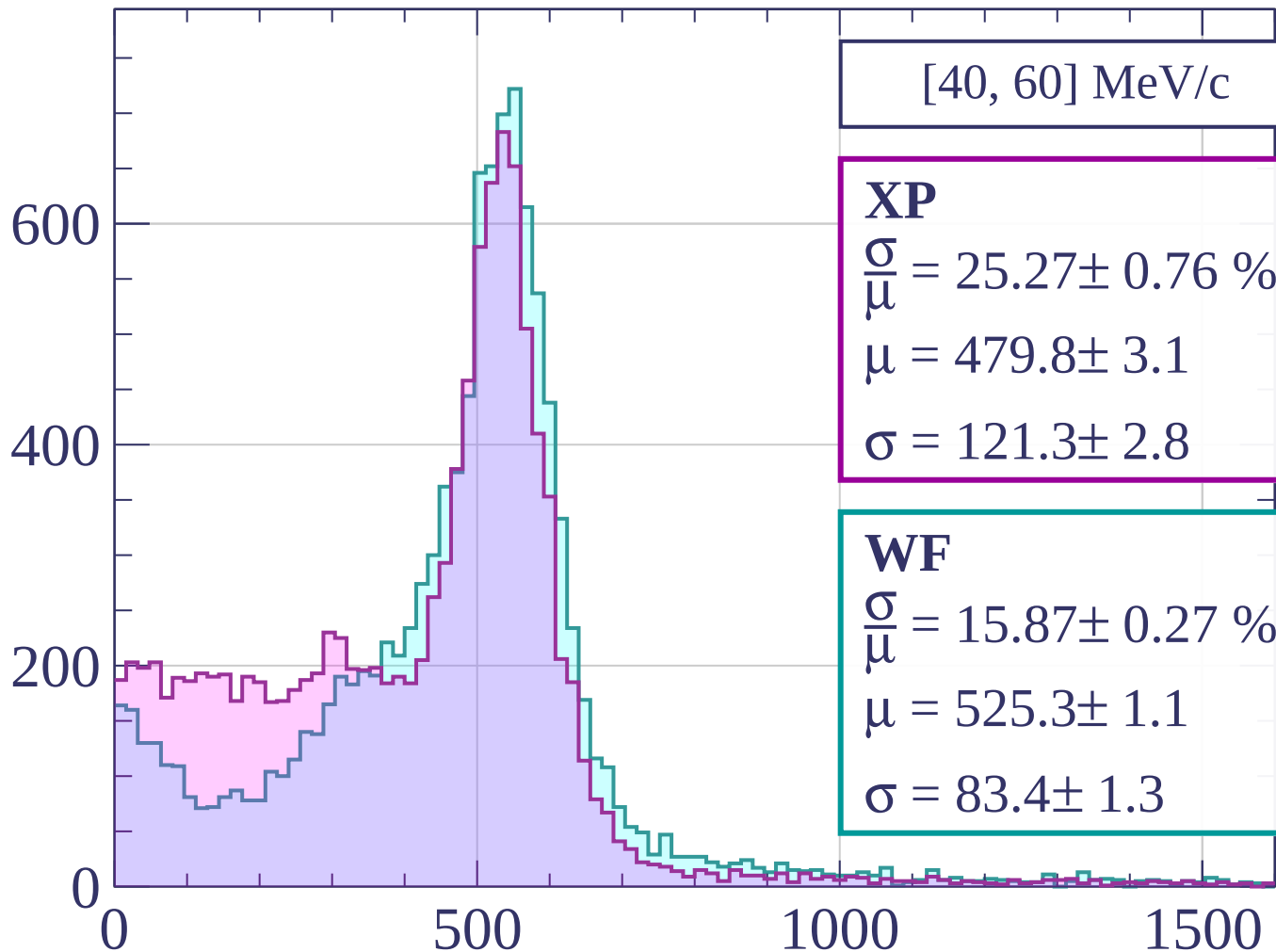




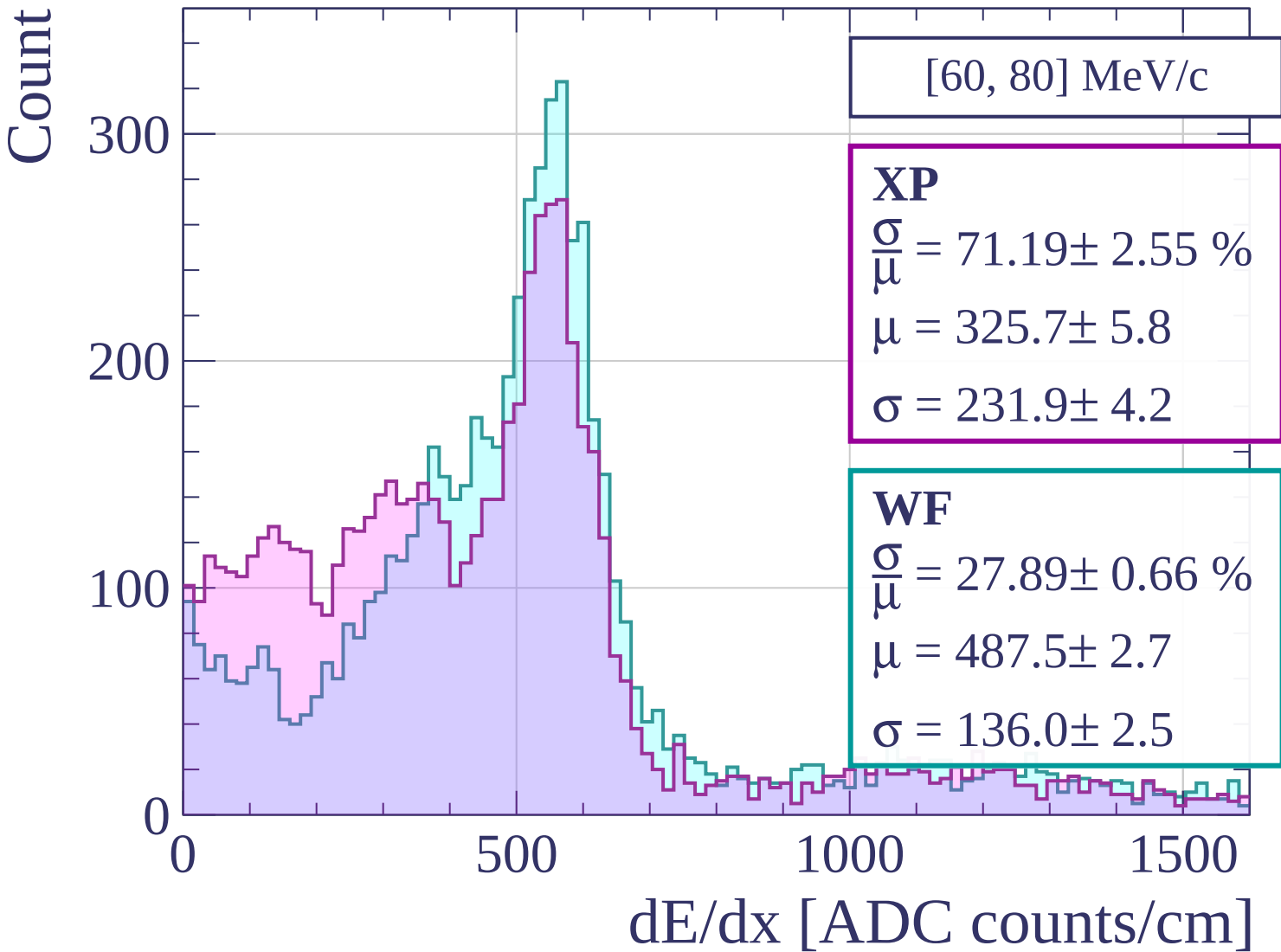




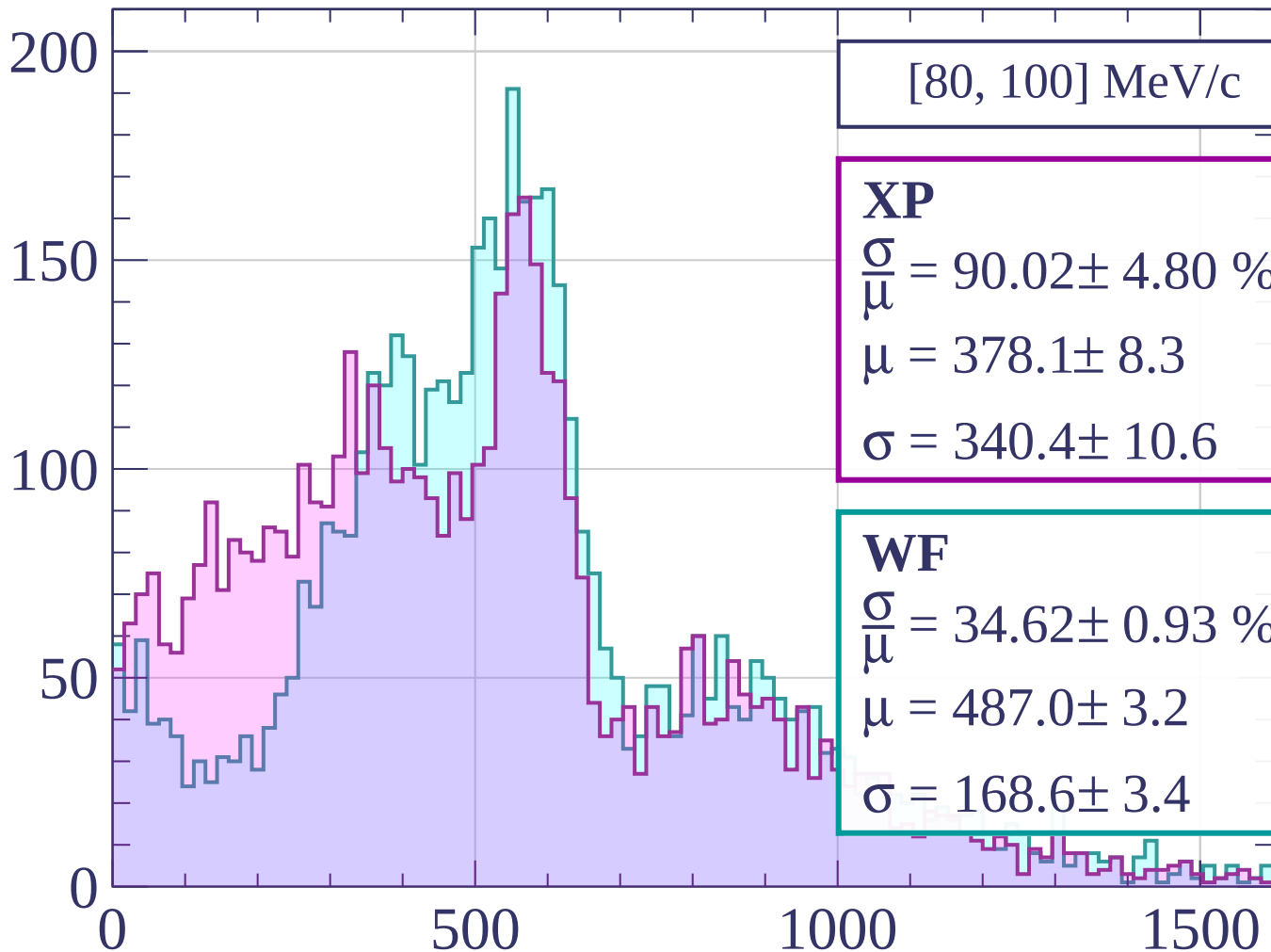
Count



dE/dx [ADC counts/cm]



Count



[80, 100] MeV/c

XP

$$\frac{\sigma}{\mu} = 90.02 \pm 4.80 \%$$

$$\mu = 378.1 \pm 8.3$$

$$\sigma = 340.4 \pm 10.6$$

WF

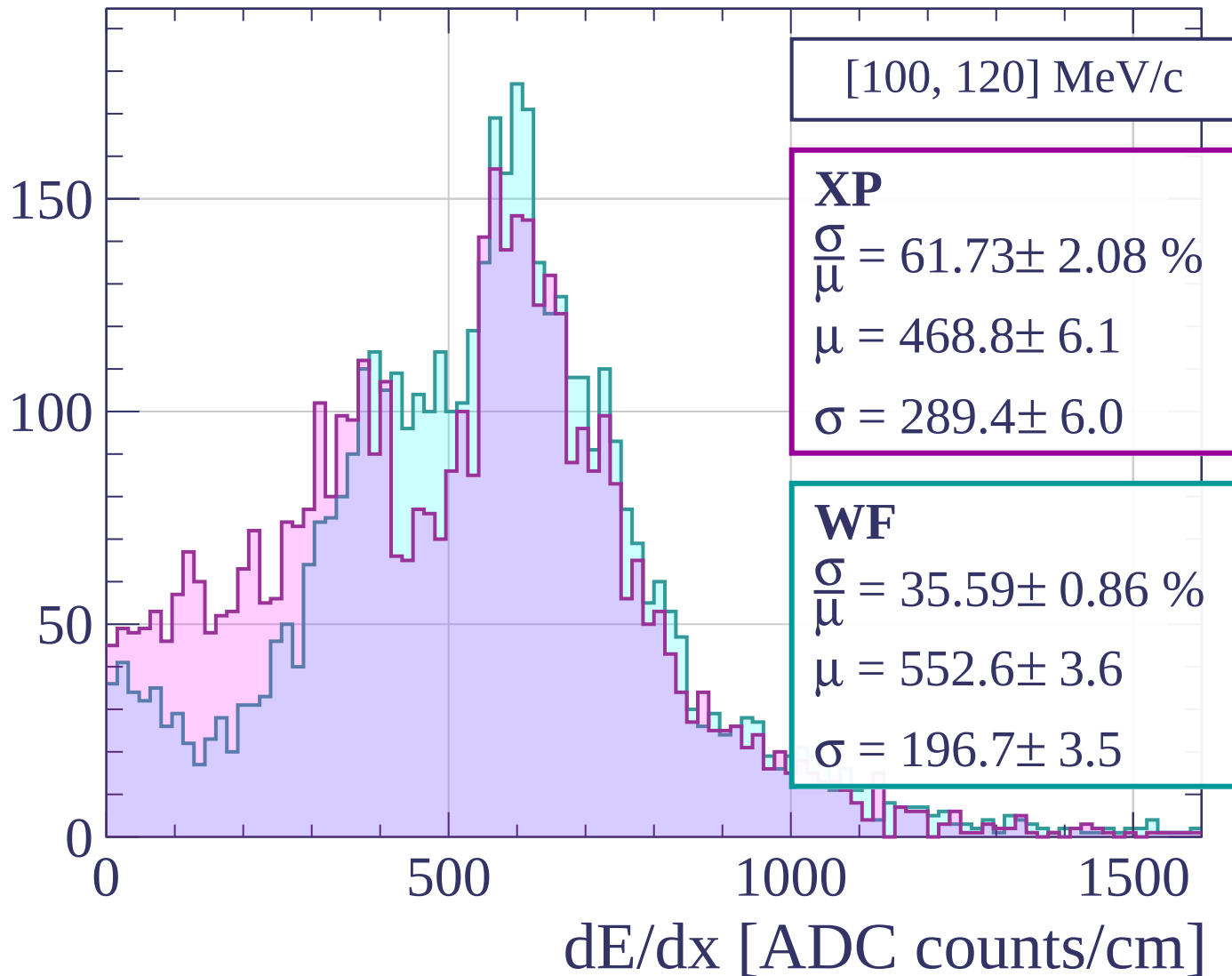
$$\frac{\sigma}{\mu} = 34.62 \pm 0.93 \%$$

$$\mu = 487.0 \pm 3.2$$

$$\sigma = 168.6 \pm 3.4$$

dE/dx [ADC counts/cm]

Count



Count

[120, 140] MeV/c

XP

$$\frac{\sigma}{\mu} = 32.08 \pm 1.03 \%$$

$$\mu = 514.3 \pm 3.9$$

$$\sigma = 165.0 \pm 4.1$$

WF

$$\frac{\sigma}{\mu} = 24.98 \pm 0.75 \%$$

$$\mu = 545.0 \pm 2.9$$

$$\sigma = 136.1 \pm 3.4$$

200

150

100

50

0

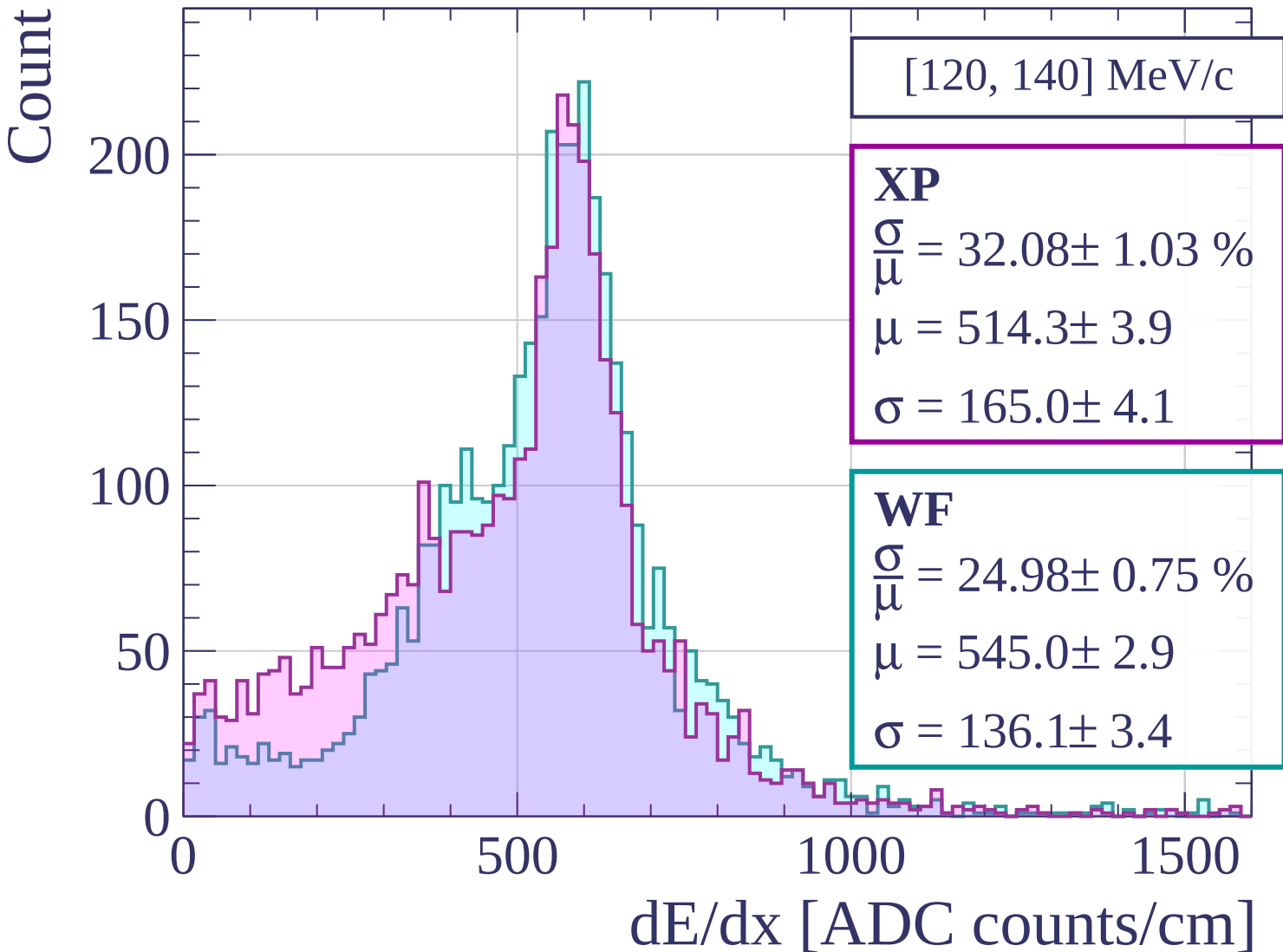
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[140, 160] MeV/c

XP

$$\frac{\sigma}{\mu} = 21.25 \pm 0.66 \%$$

$$\mu = 517.9 \pm 2.5$$

$$\sigma = 110.1 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 20.77 \pm 0.60 \%$$

$$\mu = 525.6 \pm 2.4$$

$$\sigma = 109.2 \pm 2.6$$

0

500

1000

1500

dE/dx [ADC counts/cm]

200

100

Count

[160, 180] MeV/c

XP

$$\frac{\sigma}{\mu} = 19.95 \pm 0.61 \%$$

$$\mu = 490.0 \pm 2.2$$

$$\sigma = 97.8 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 19.55 \pm 0.56 \%$$

$$\mu = 500.7 \pm 2.2$$

$$\sigma = 97.9 \pm 2.4$$

200

100

0

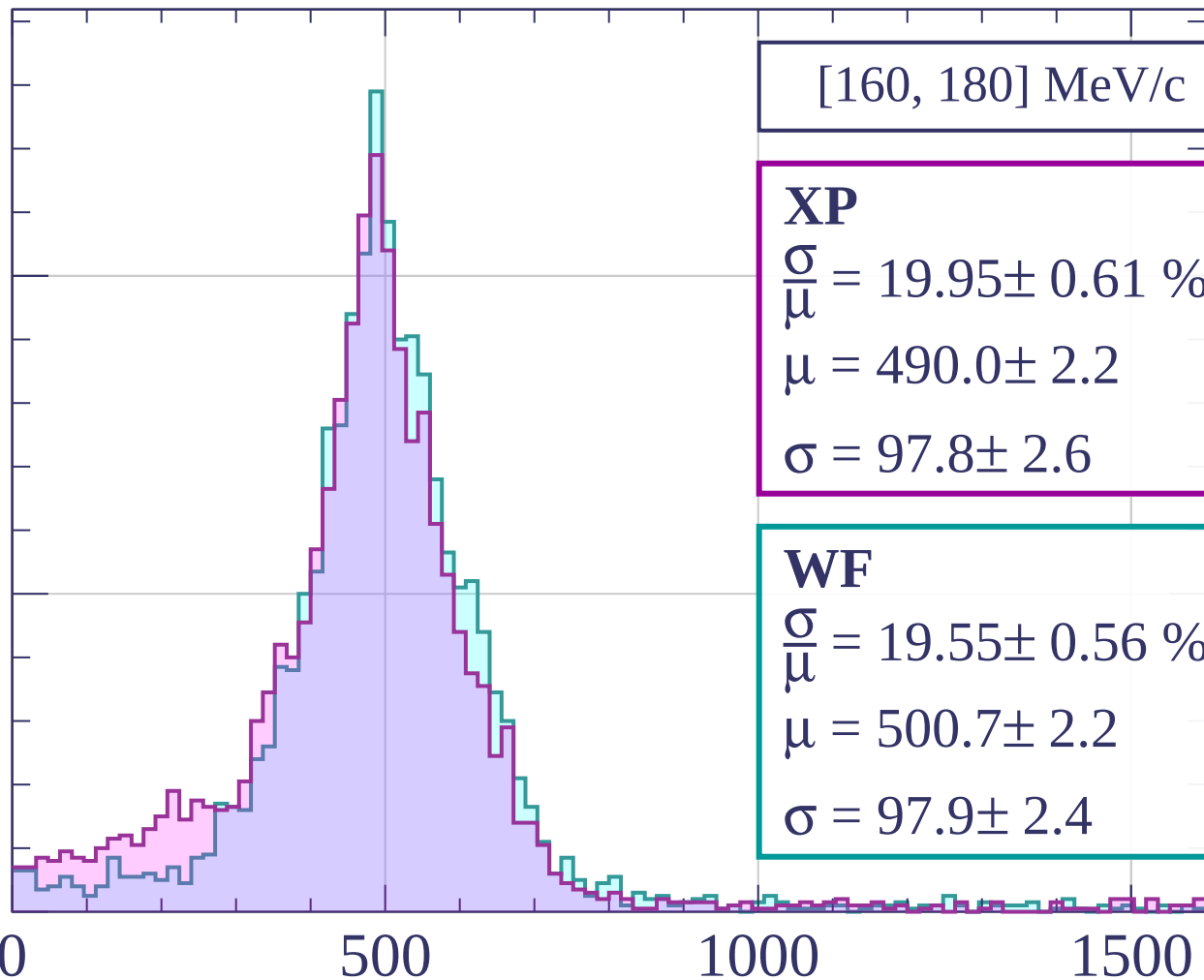
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[180, 200] MeV/c

XP

$$\frac{\sigma}{\mu} = 15.14 \pm 0.48 \%$$

$$\mu = 466.9 \pm 1.7$$

$$\sigma = 70.7 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 15.78 \pm 0.51 \%$$

$$\mu = 472.8 \pm 1.8$$

$$\sigma = 74.6 \pm 2.1$$

200

100

0

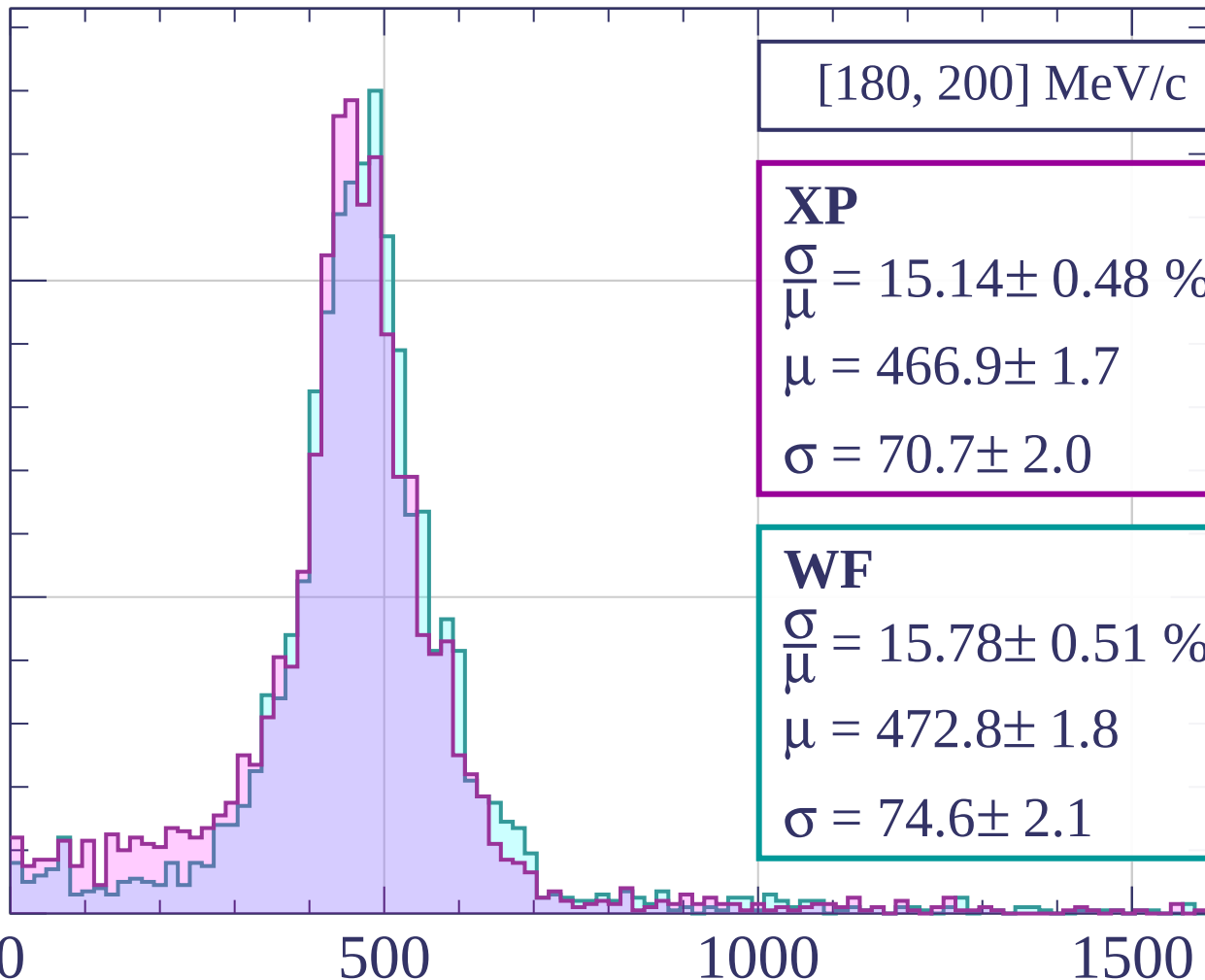
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[200, 220] MeV/c

XP

$$\frac{\sigma}{\mu} = 13.96 \pm 0.41 \%$$

$$\mu = 443.6 \pm 1.6$$

$$\sigma = 61.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 13.99 \pm 0.43 \%$$

$$\mu = 453.5 \pm 1.6$$

$$\sigma = 63.4 \pm 1.7$$

200

100

0

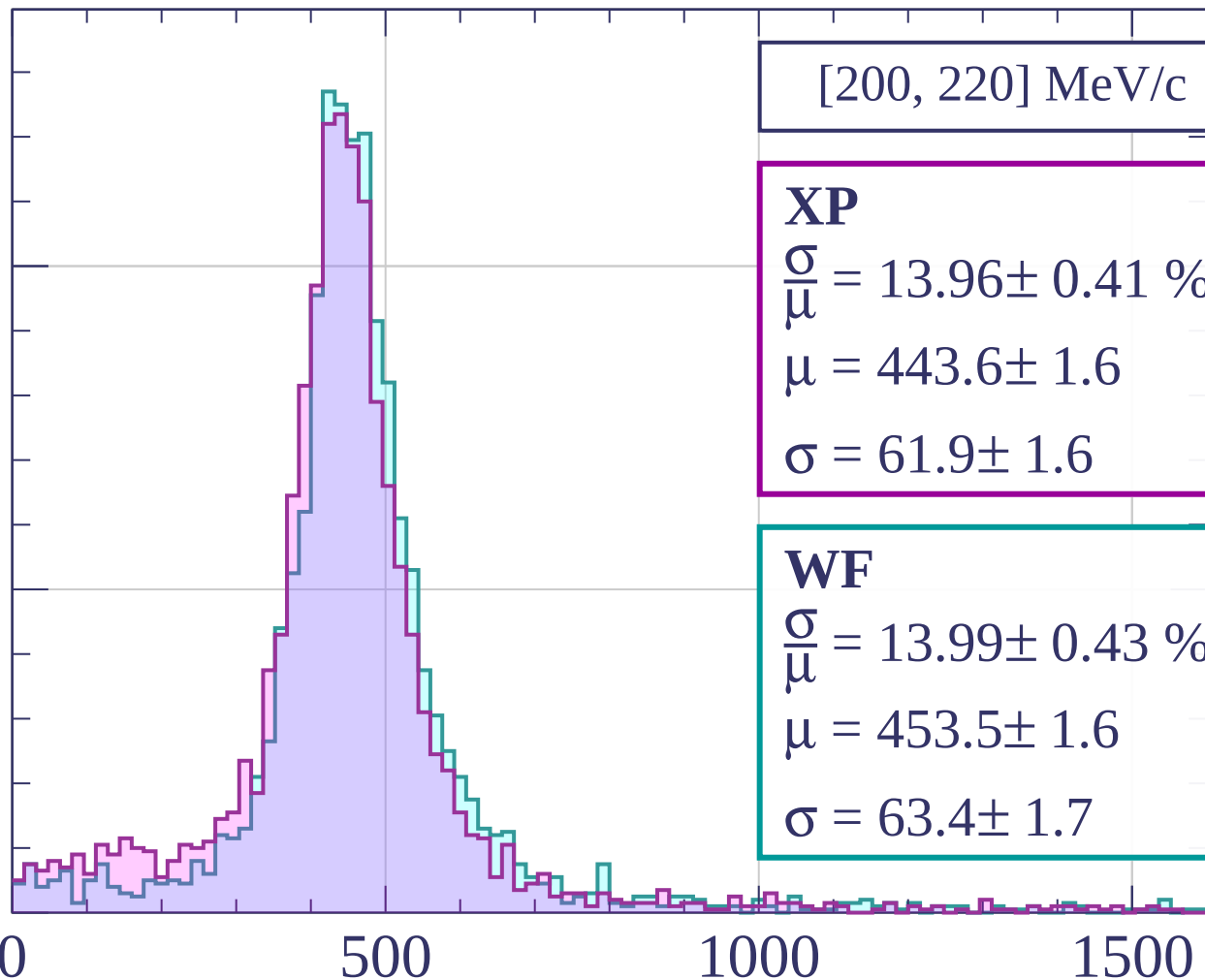
0

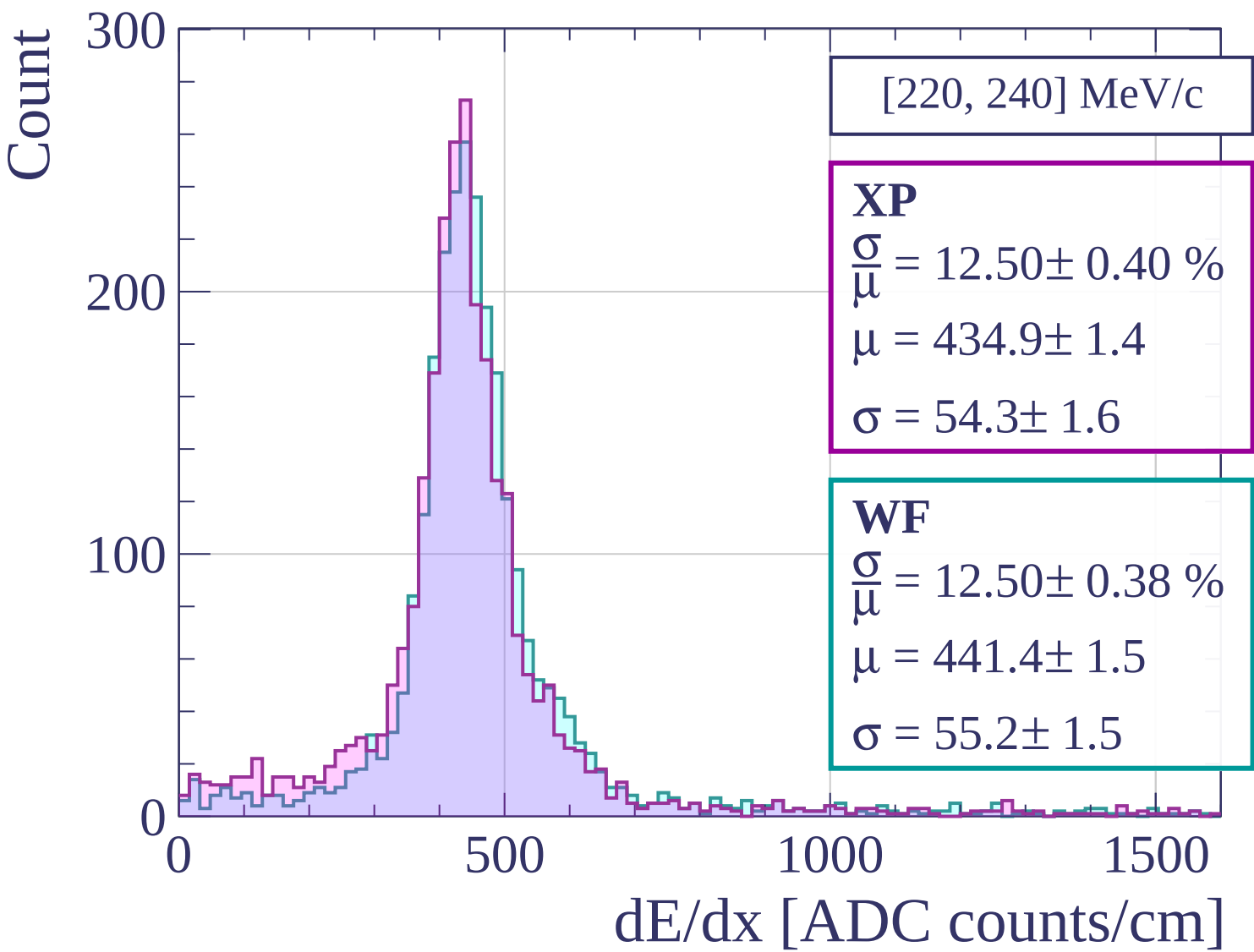
500

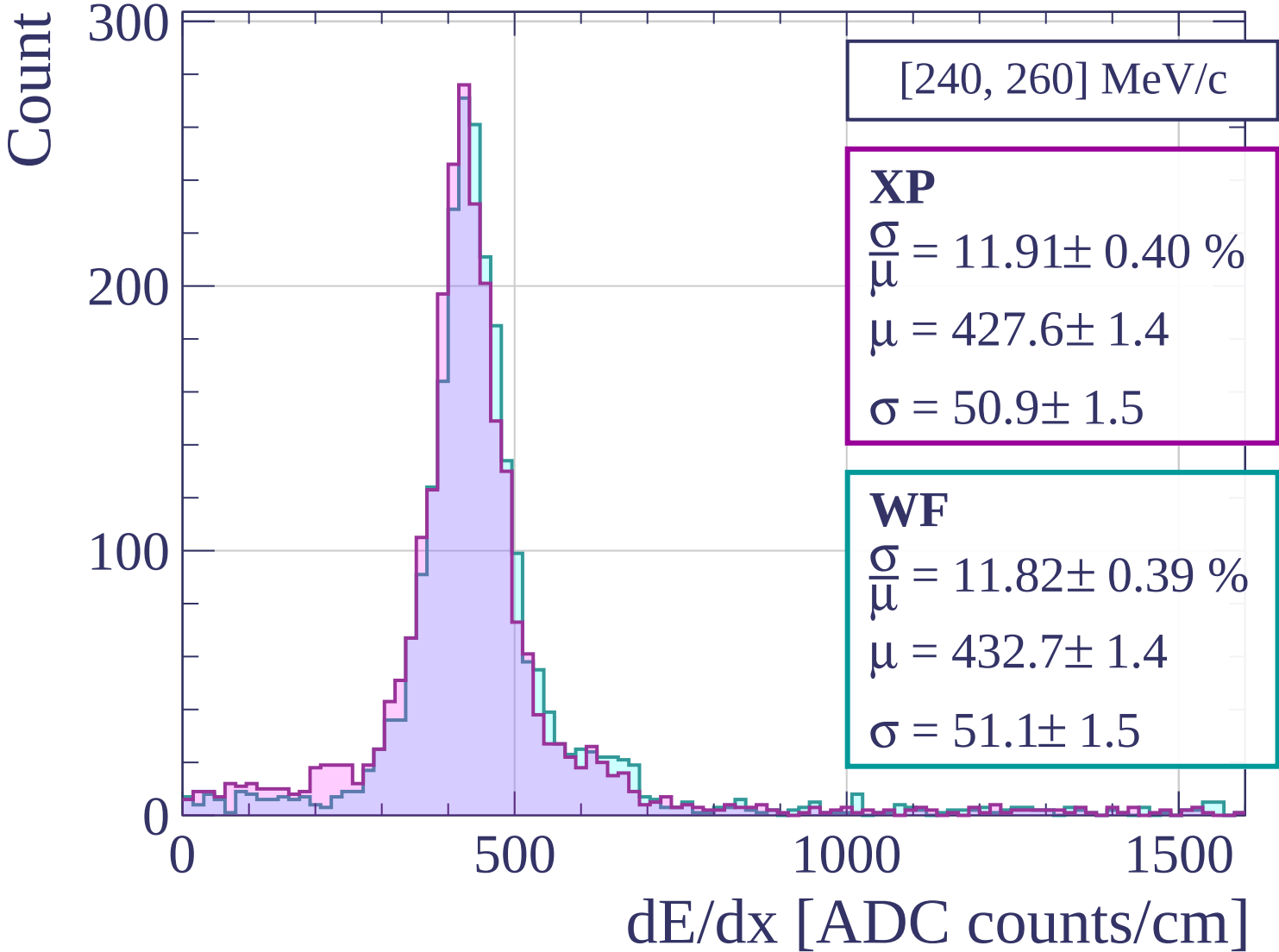
1000

1500

dE/dx [ADC counts/cm]







Count

[260, 280] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.79 \pm 0.36 \%$$

$$\mu = 418.3 \pm 1.3$$

$$\sigma = 45.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.13 \pm 0.40 \%$$

$$\mu = 423.6 \pm 1.4$$

$$\sigma = 47.2 \pm 1.5$$

200

100

0

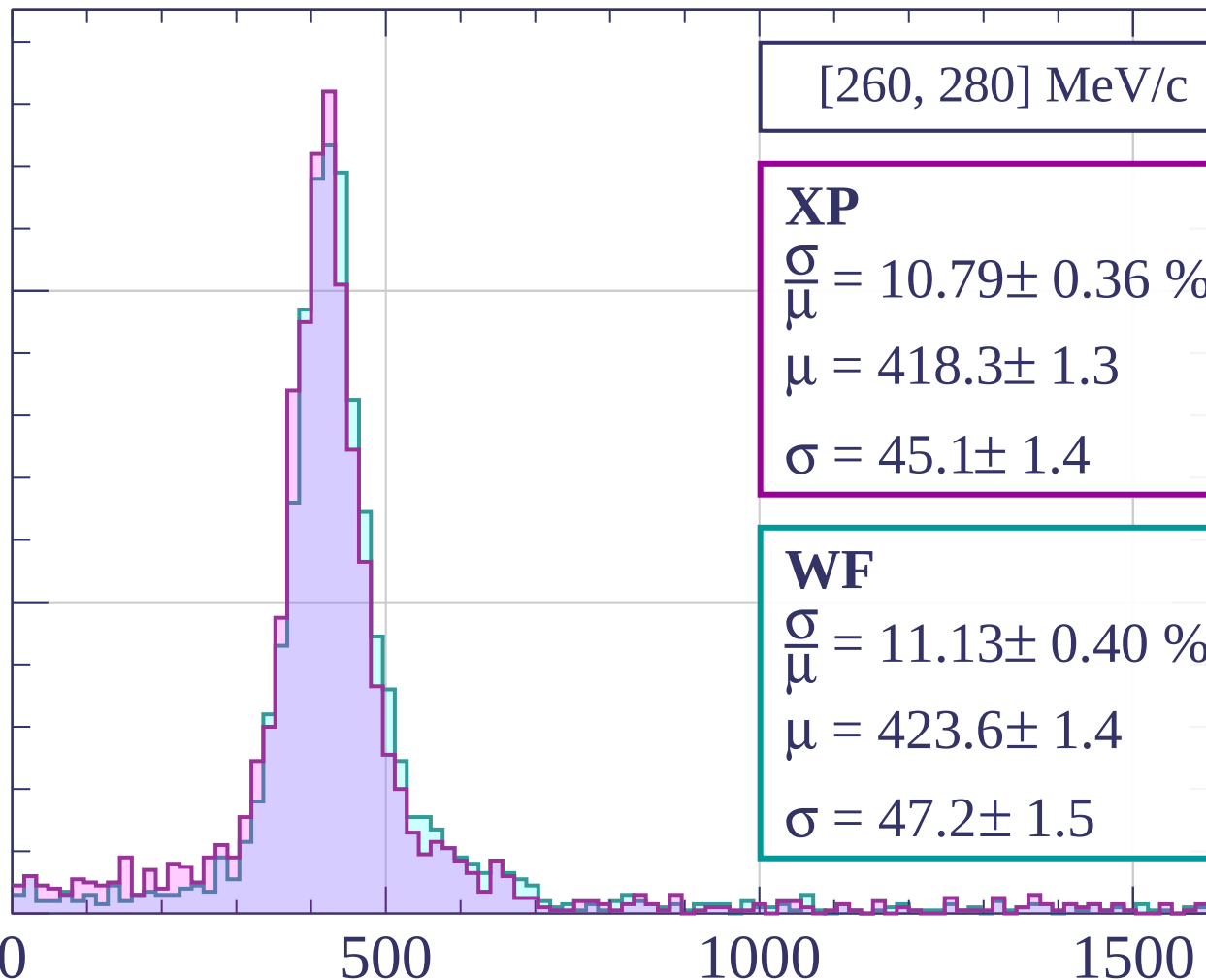
0

500

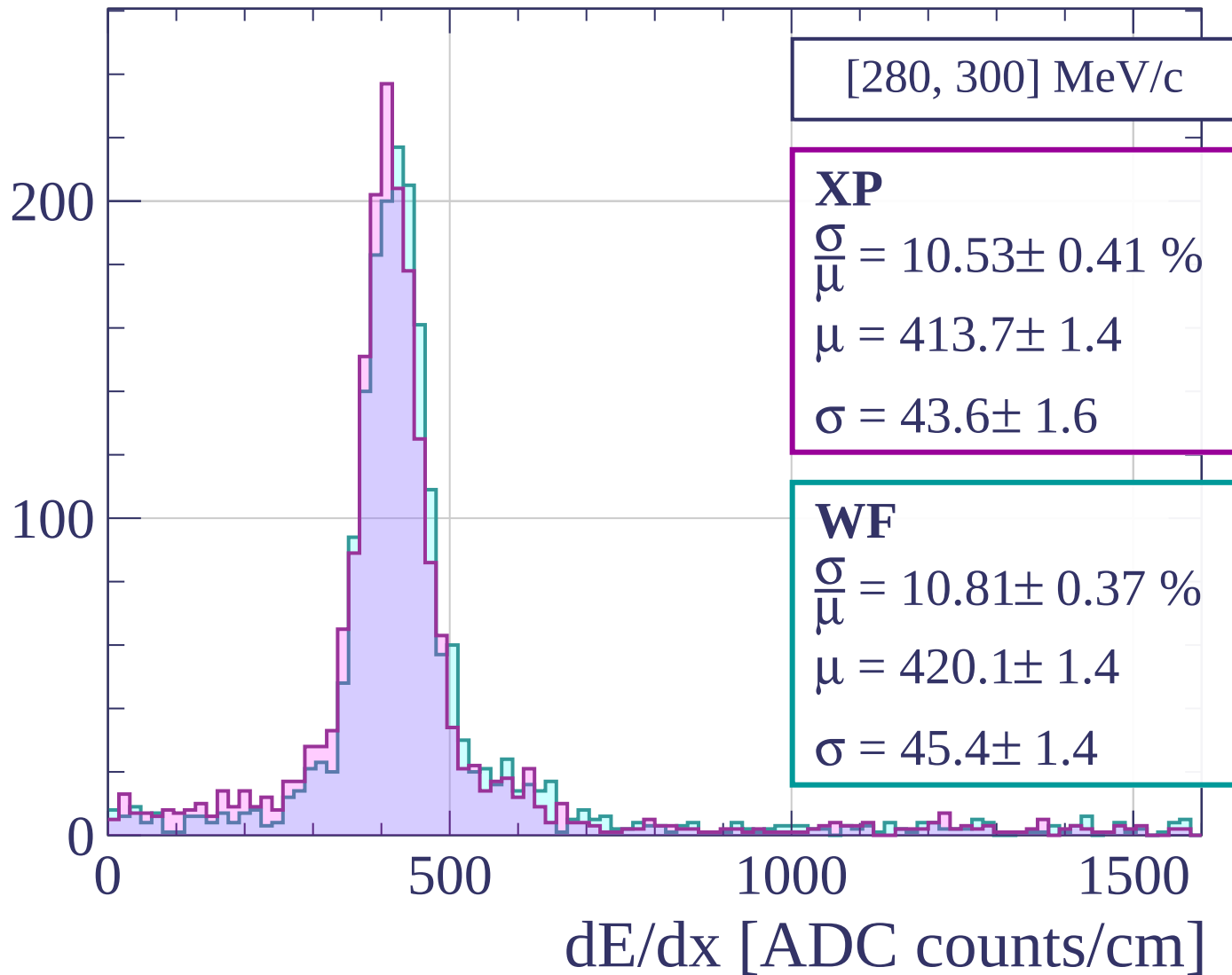
1000

1500

dE/dx [ADC counts/cm]



Count



Count

[300, 320] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.97 \pm 0.37 \%$$

$$\mu = 412.6 \pm 1.3$$

$$\sigma = 41.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.99 \pm 0.37 \%$$

$$\mu = 416.9 \pm 1.3$$

$$\sigma = 41.6 \pm 1.4$$

200

100

0

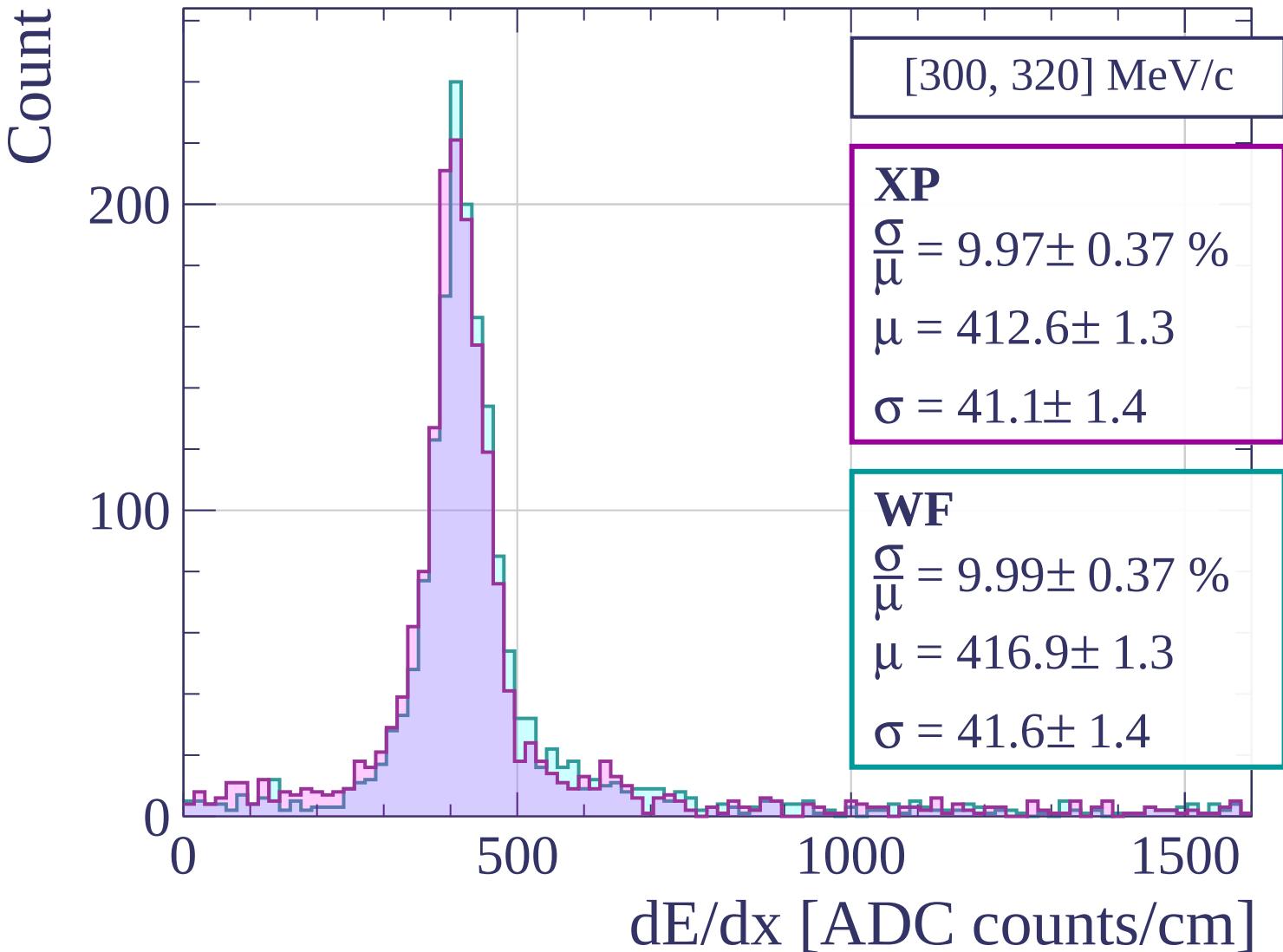
0

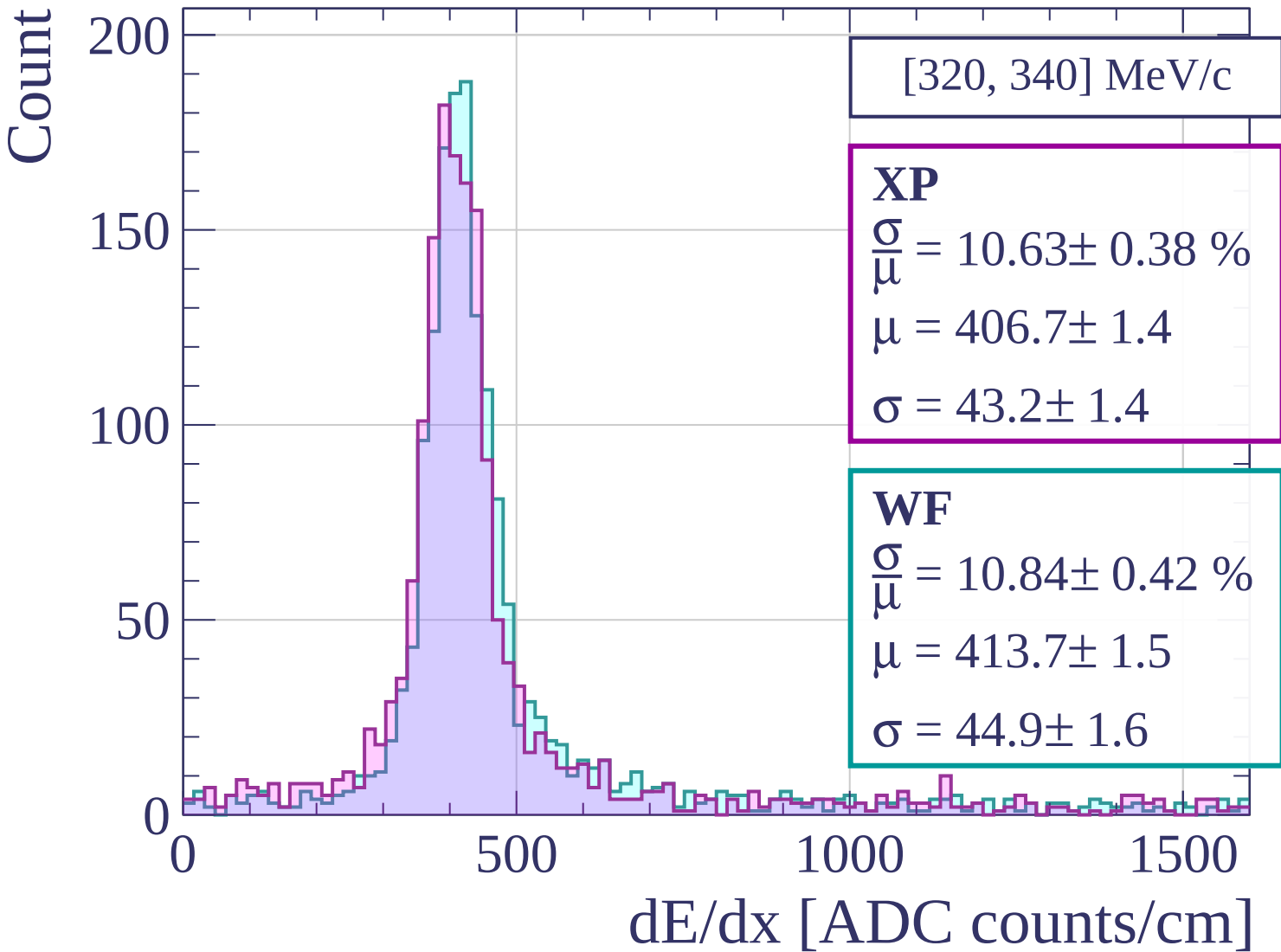
500

1000

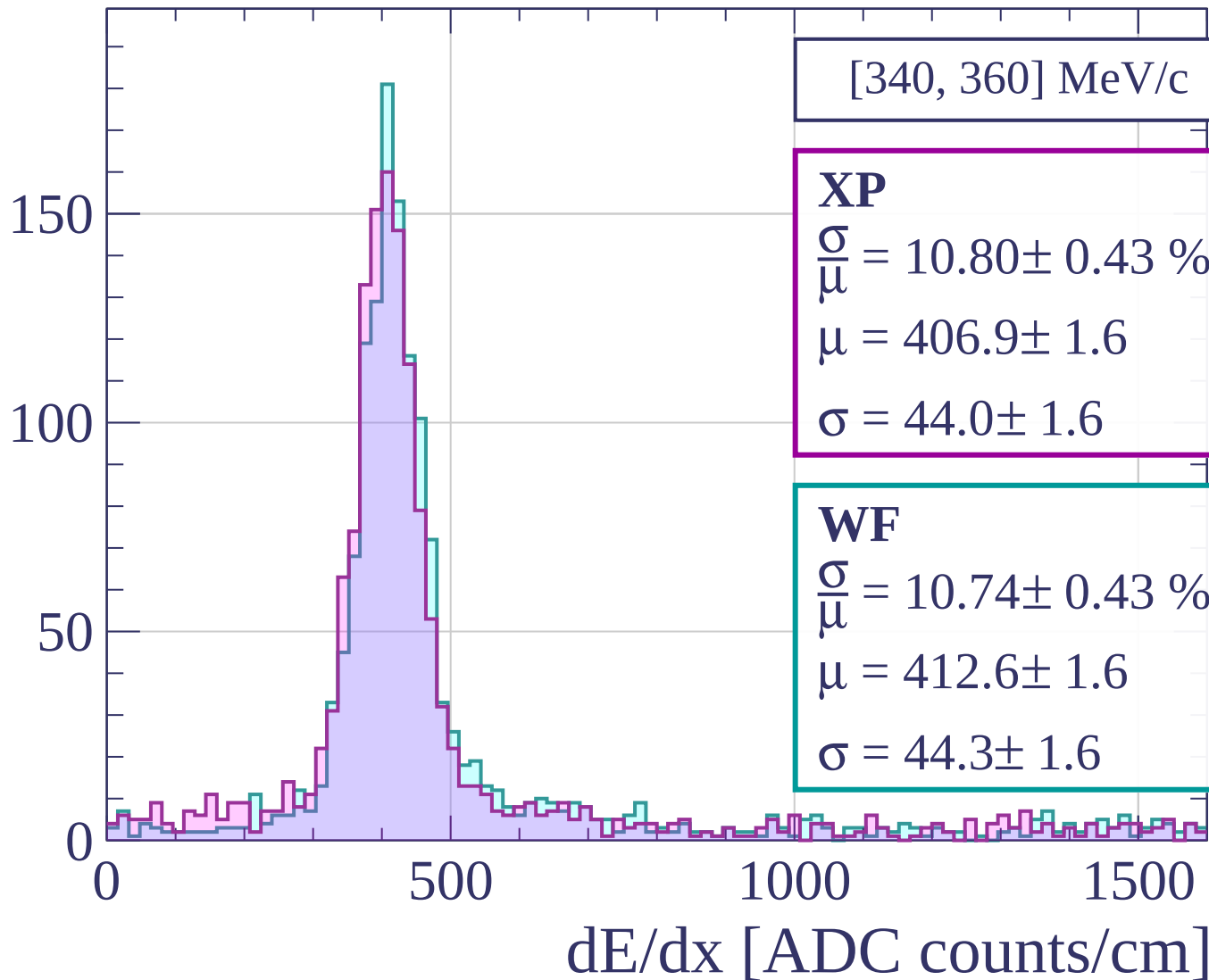
1500

dE/dx [ADC counts/cm]

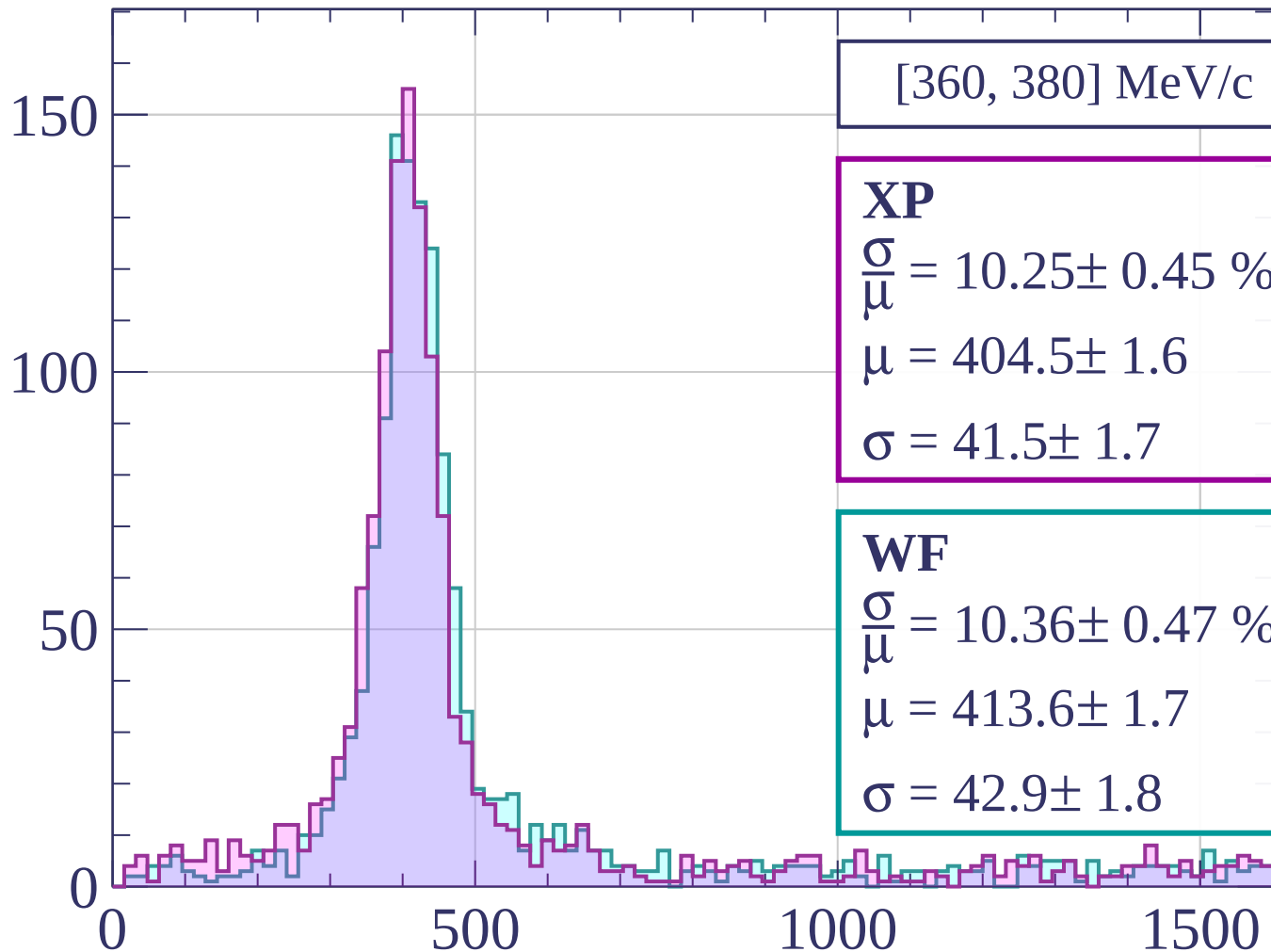




Count

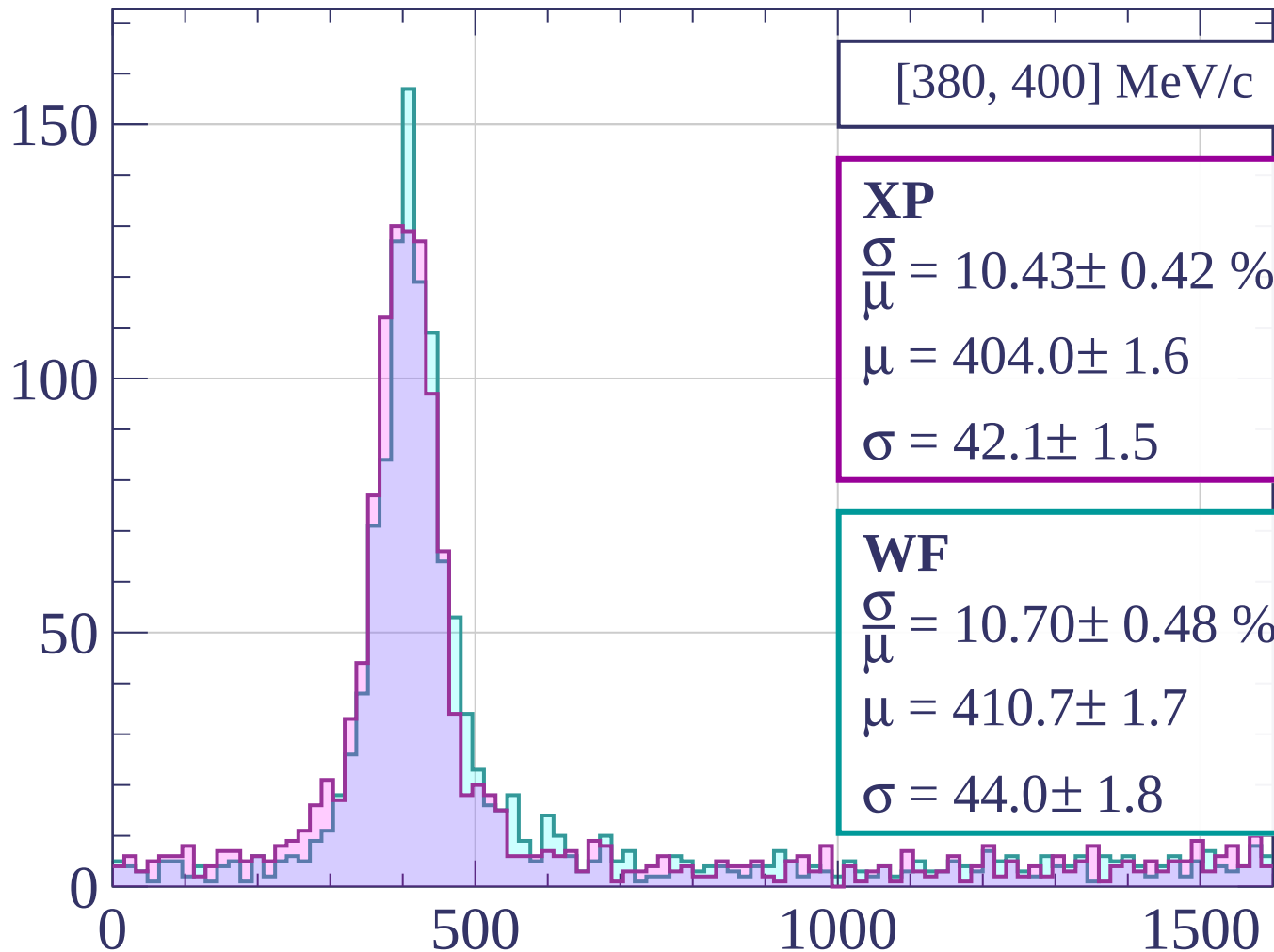


Count

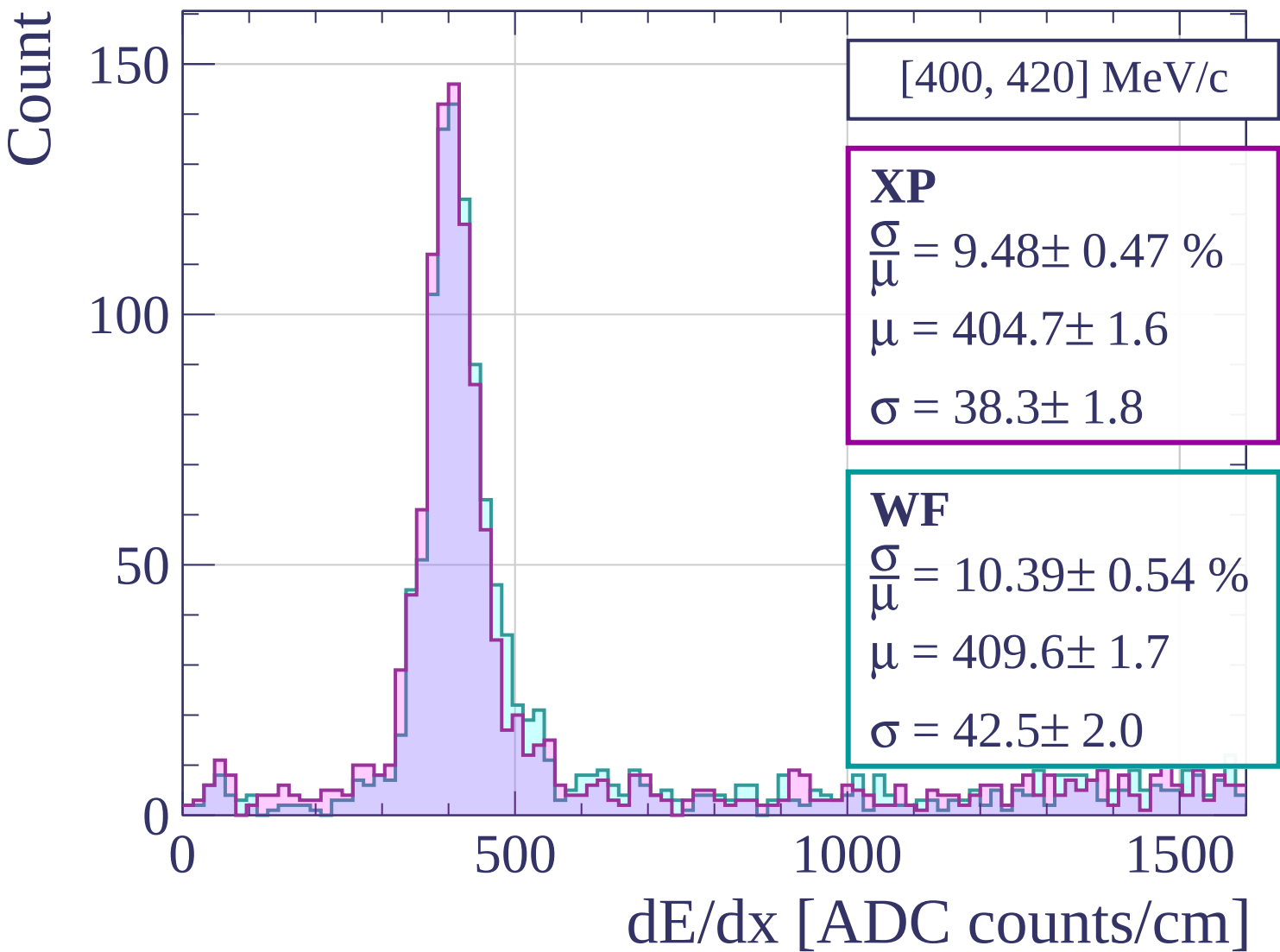


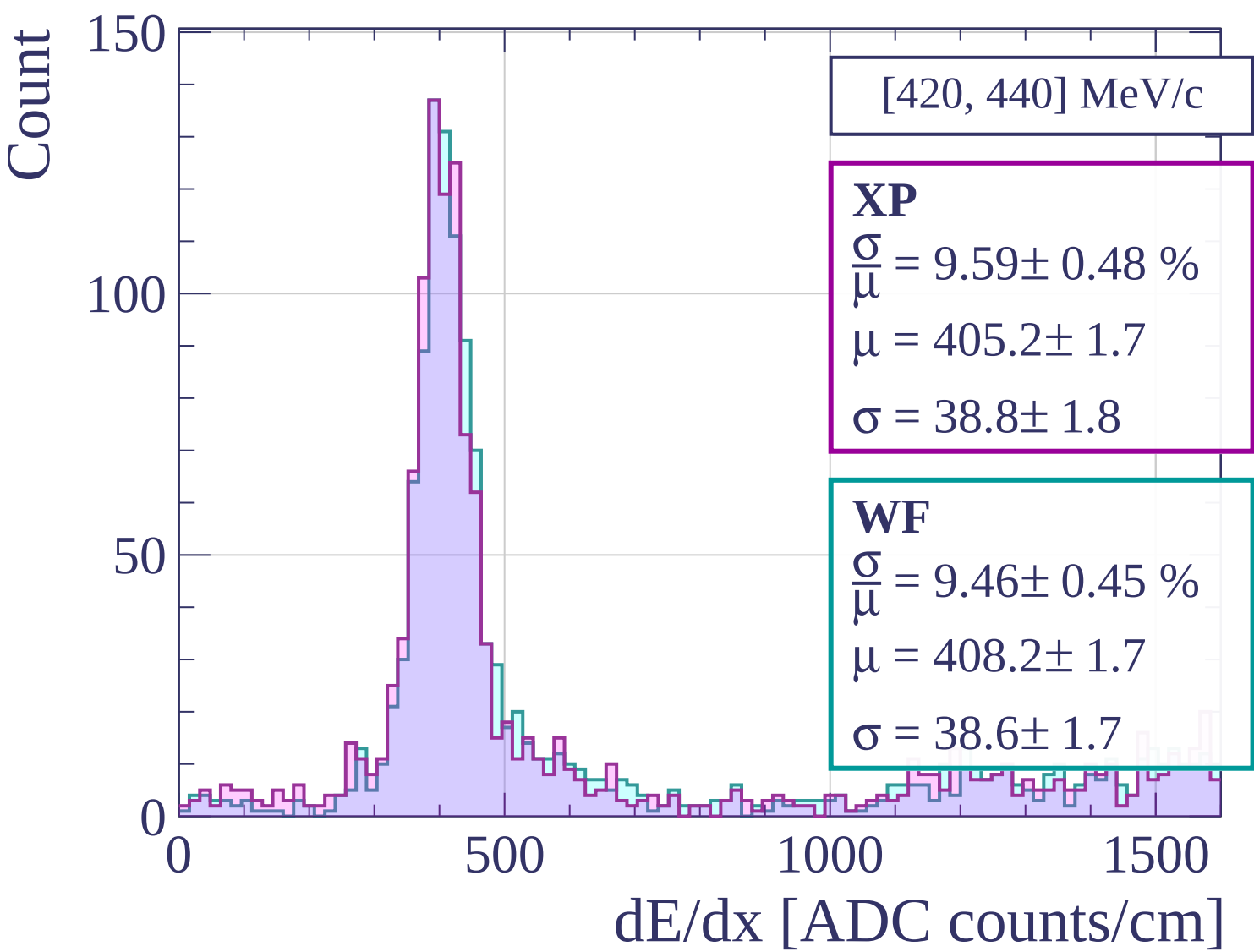
dE/dx [ADC counts/cm]

Count

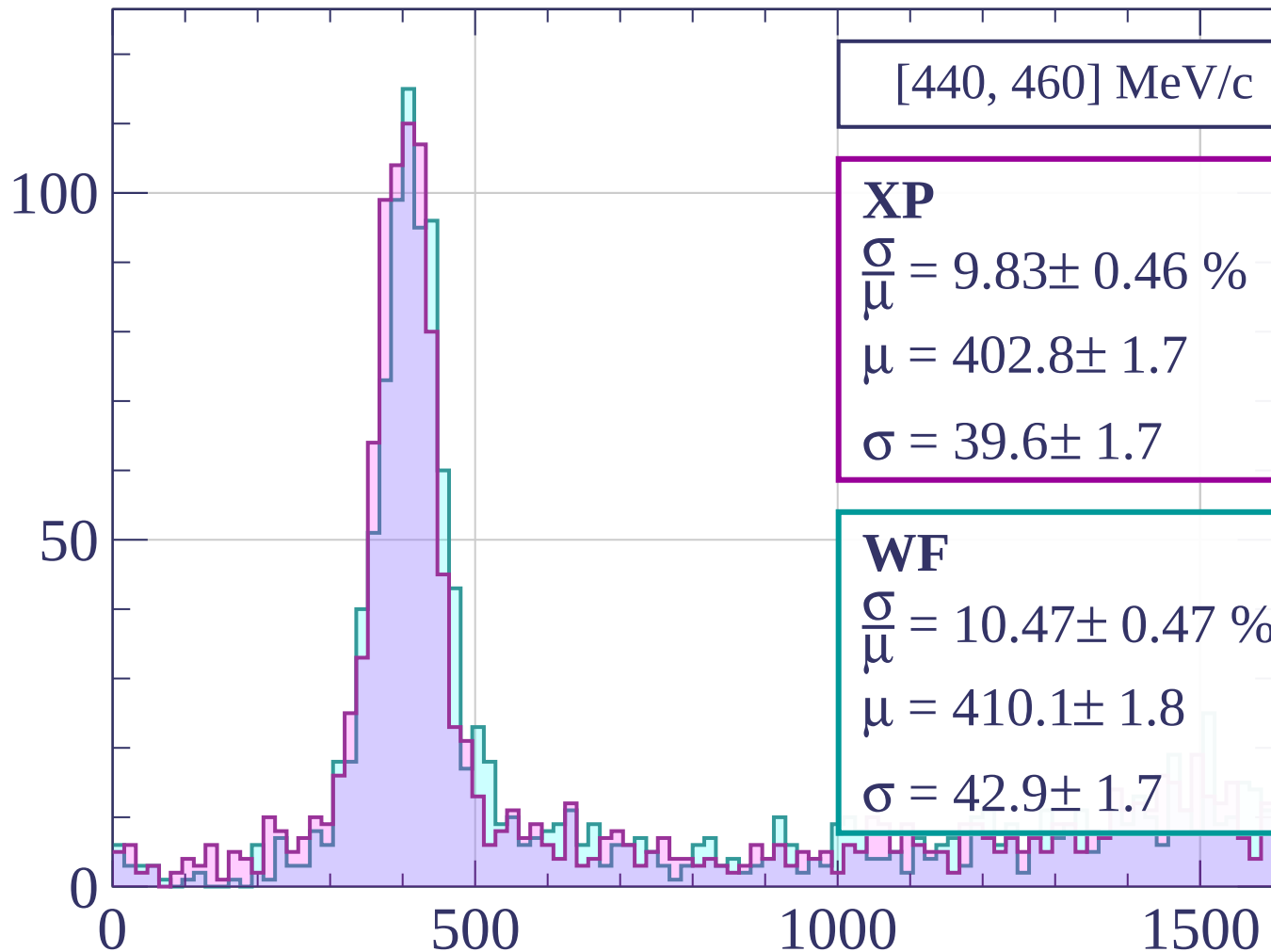


dE/dx [ADC counts/cm]





Count



[440, 460] MeV/c

XP

$$\frac{\sigma}{\mu} = 9.83 \pm 0.46 \%$$

$$\mu = 402.8 \pm 1.7$$

$$\sigma = 39.6 \pm 1.7$$

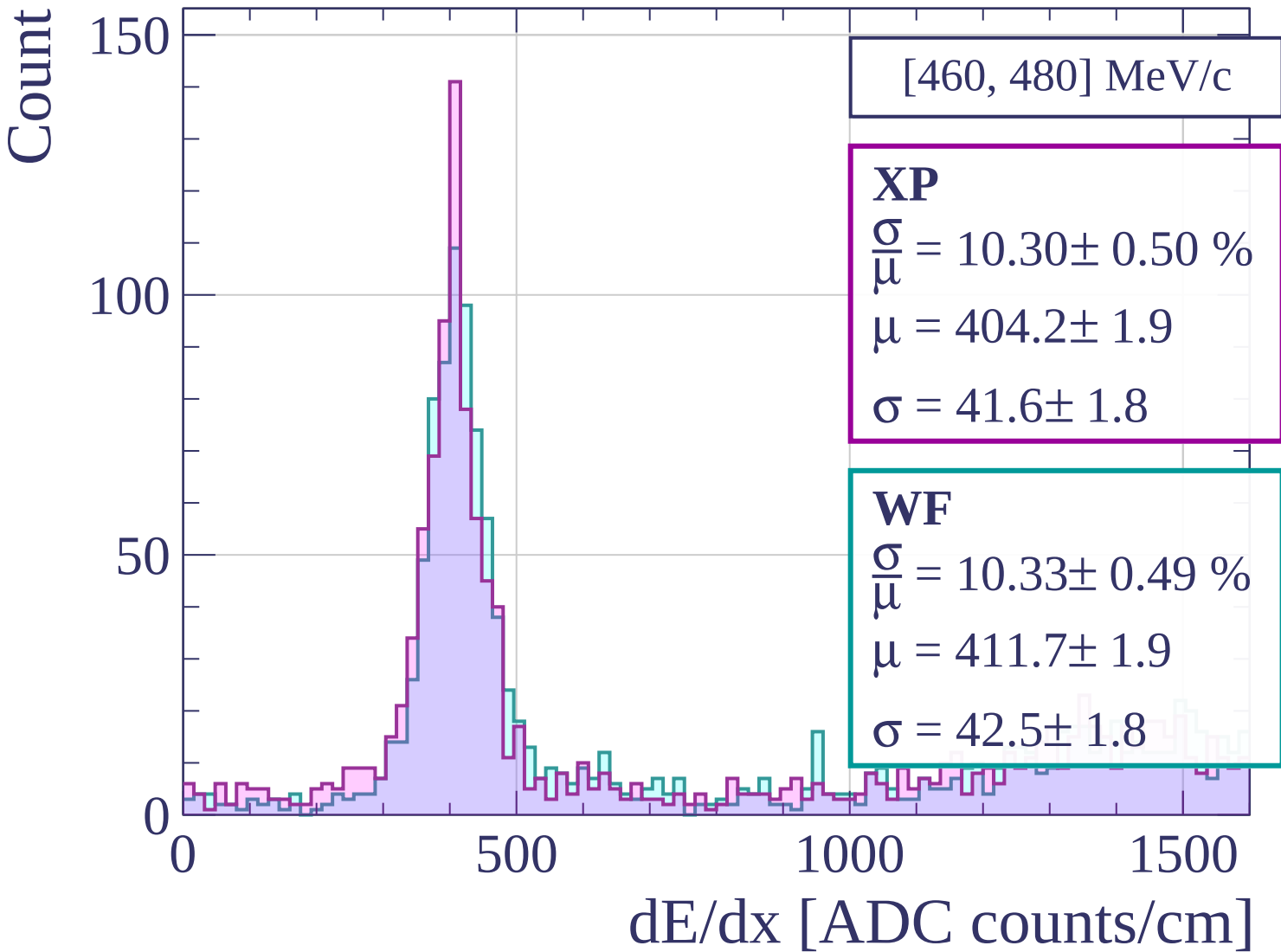
WF

$$\frac{\sigma}{\mu} = 10.47 \pm 0.47 \%$$

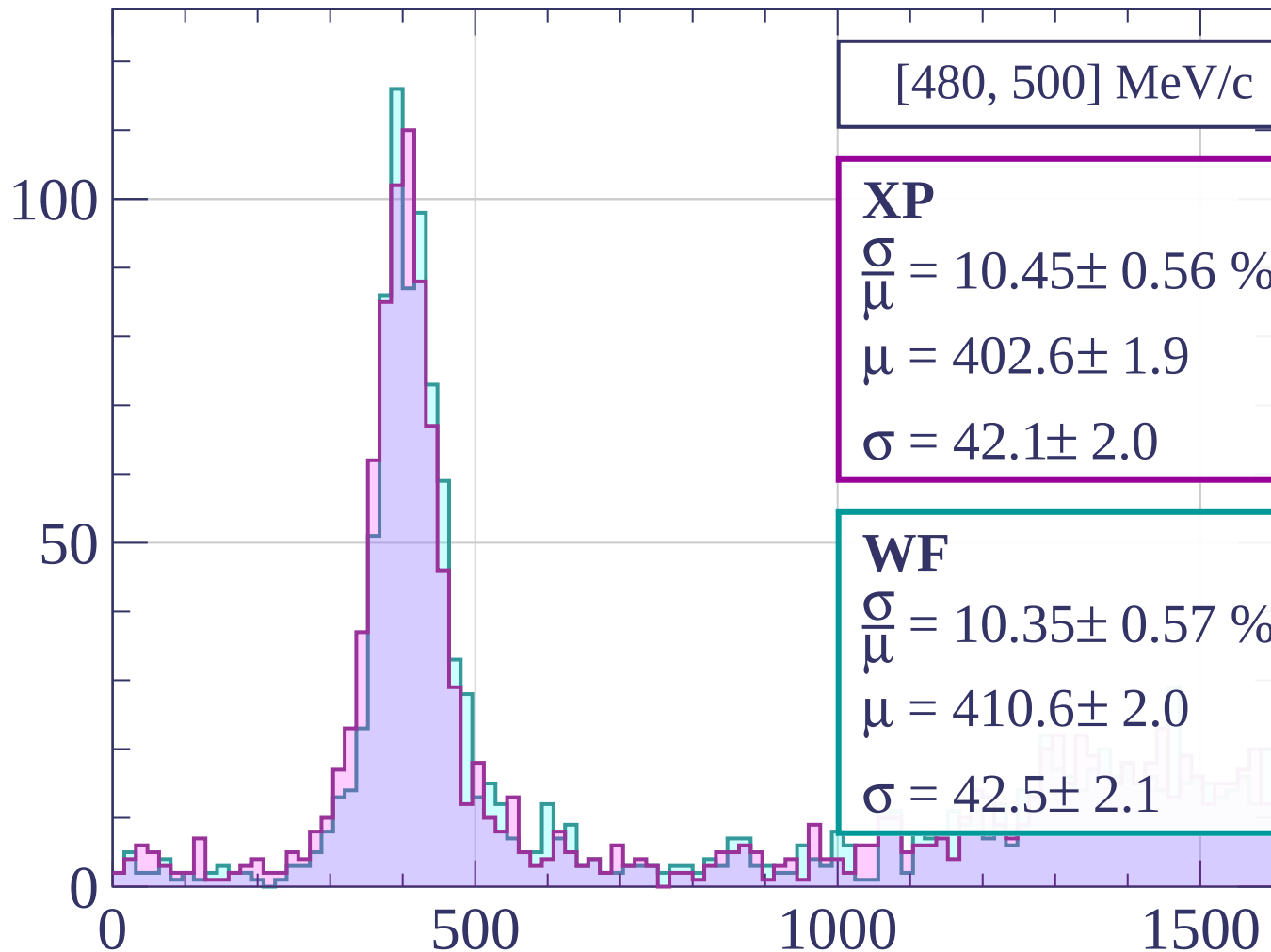
$$\mu = 410.1 \pm 1.8$$

$$\sigma = 42.9 \pm 1.7$$

dE/dx [ADC counts/cm]



Count



[480, 500] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.45 \pm 0.56 \%$$

$$\mu = 402.6 \pm 1.9$$

$$\sigma = 42.1 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 10.35 \pm 0.57 \%$$

$$\mu = 410.6 \pm 2.0$$

$$\sigma = 42.5 \pm 2.1$$

dE/dx [ADC counts/cm]

Count

[500, 520] MeV/c

XP

$$\frac{\sigma}{\mu} = 14.97 \pm 1.79 \%$$

$$\mu = 1344.6 \pm 13.7$$

$$\sigma = 201.2 \pm 22.0$$

WF

$$\frac{\sigma}{\mu} = 26.51 \pm 4.21 \%$$

$$\mu = 10811.5 \pm 1038.8$$

$$\sigma = 2866.3 \pm 179.8$$

0

500

1000

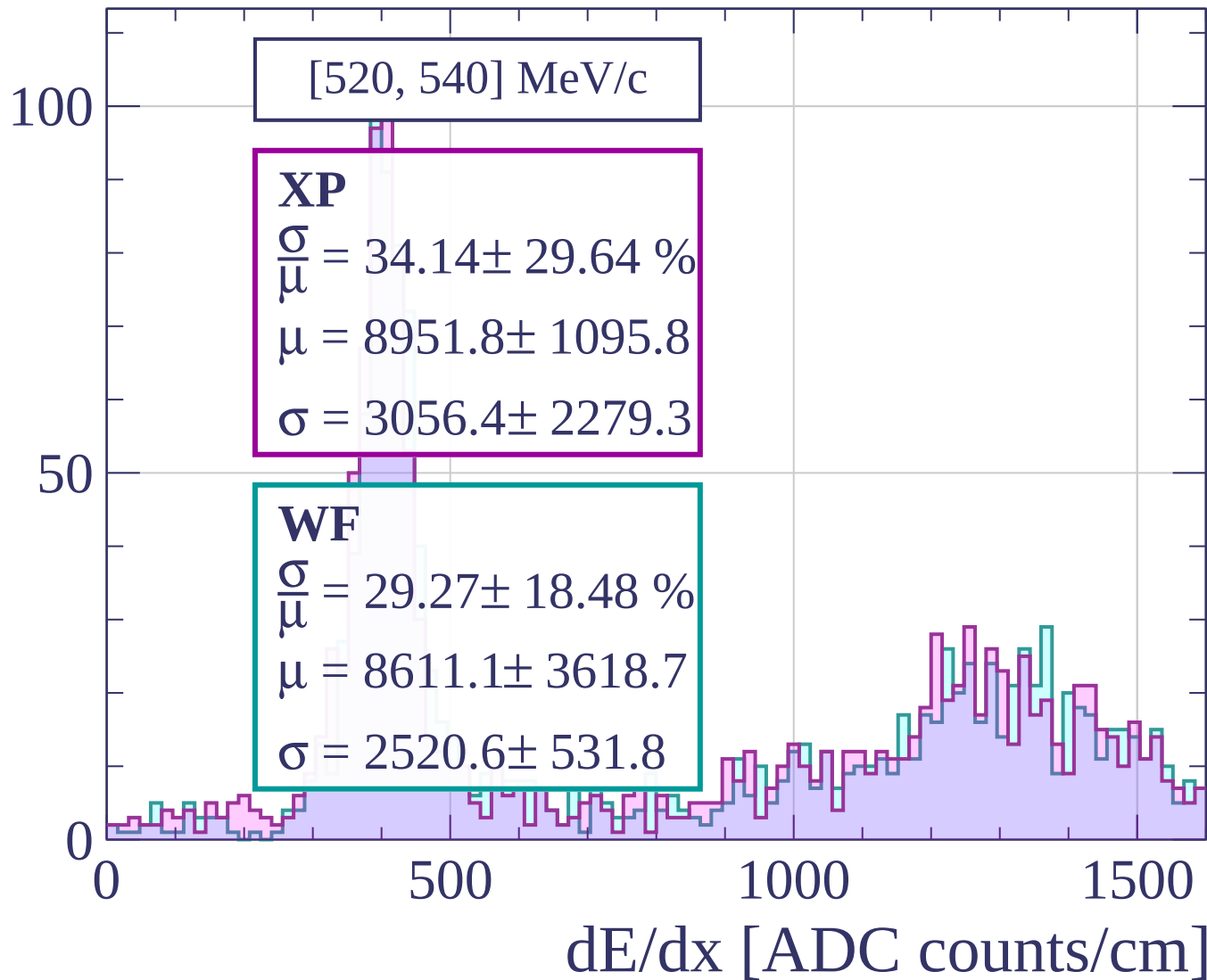
1500

dE/dx [ADC counts/cm]

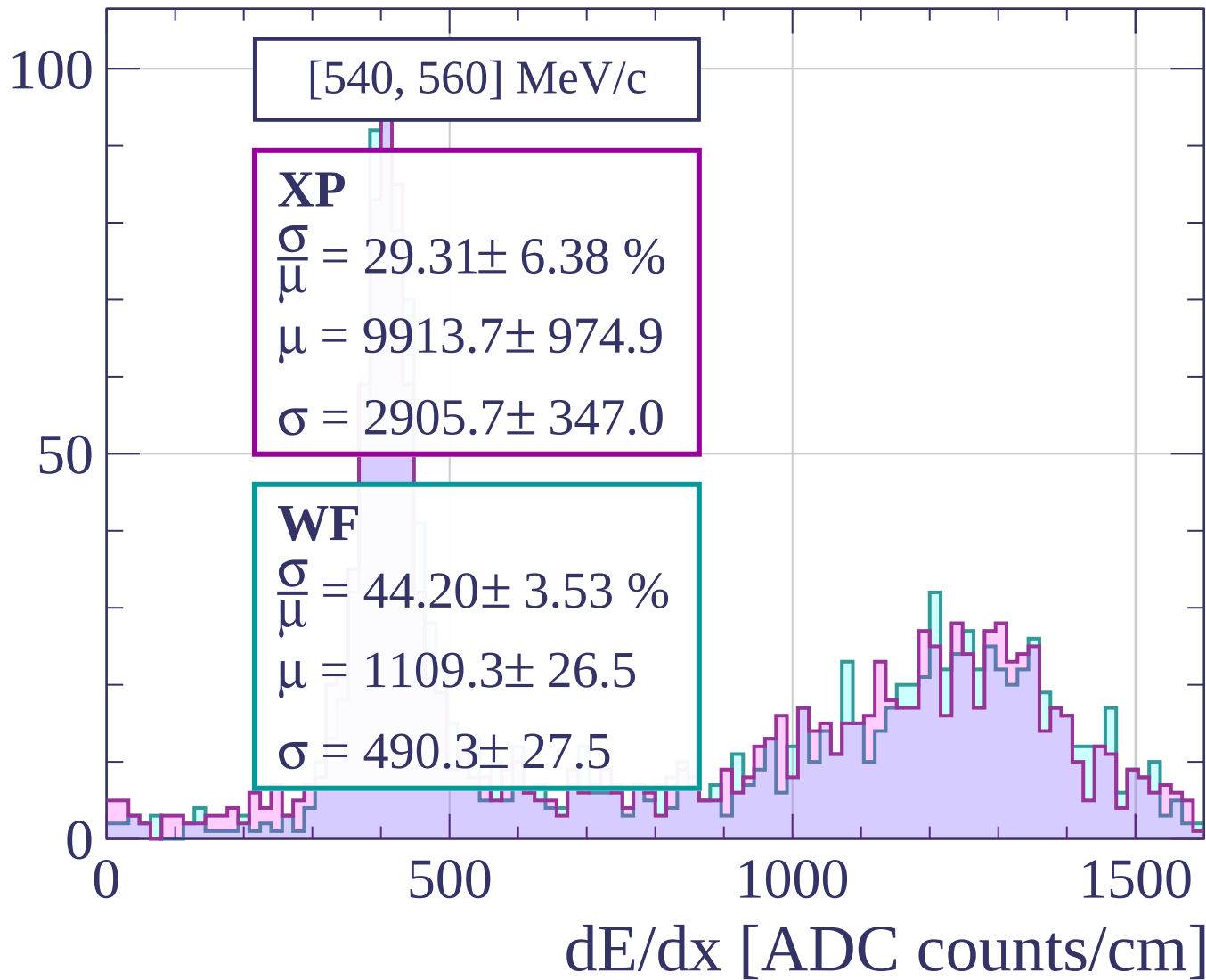
100

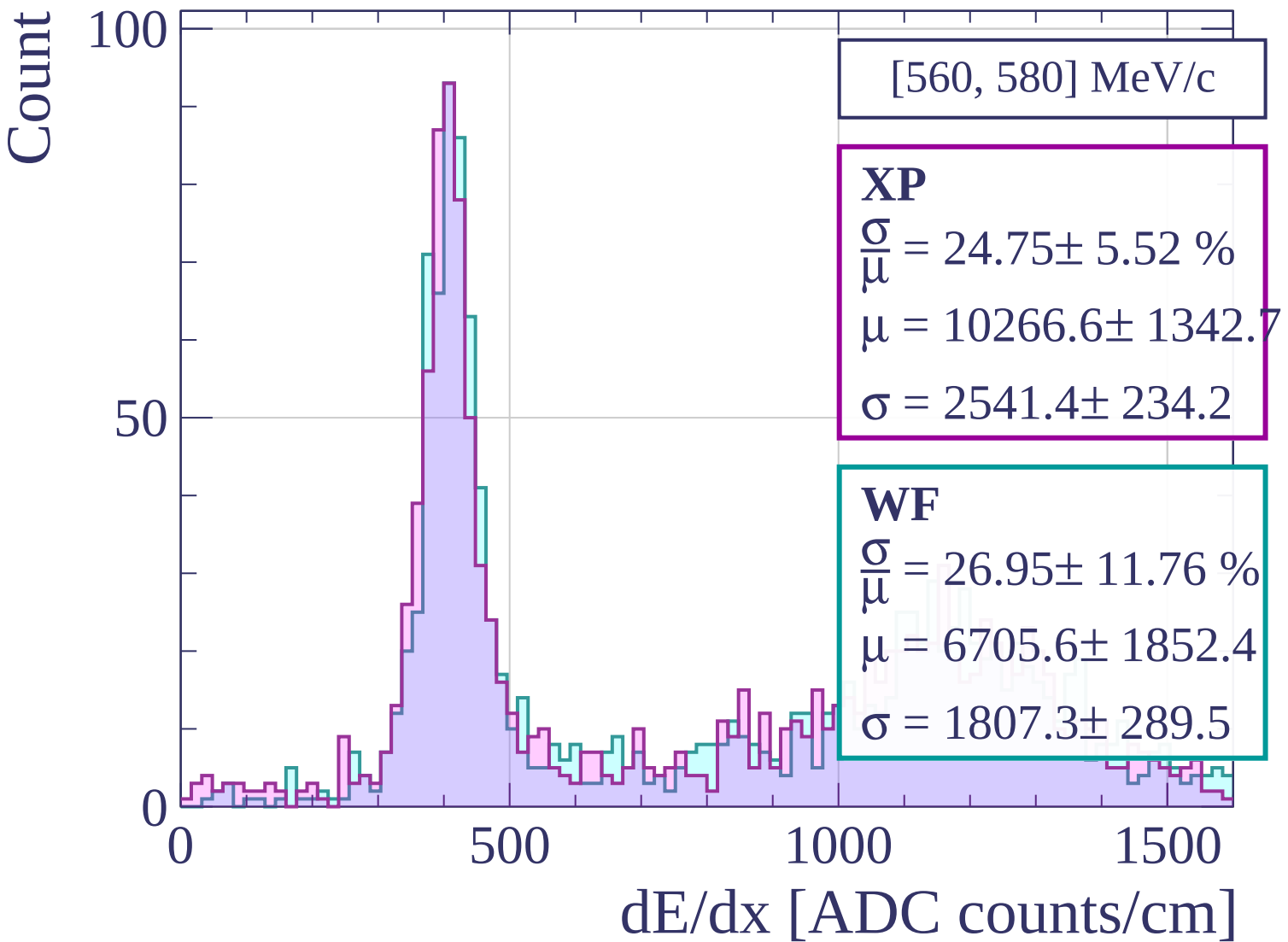
50

Count

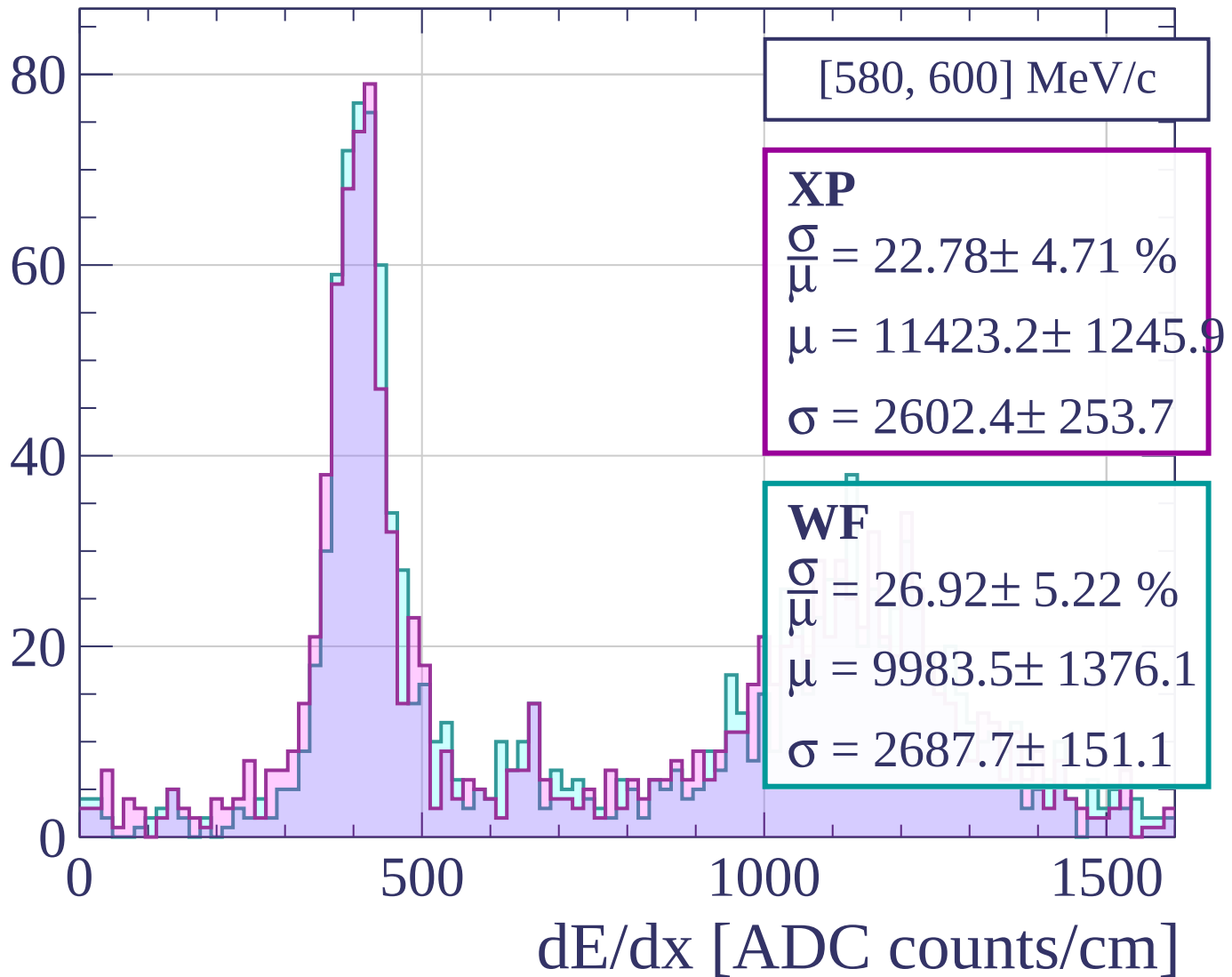


Count

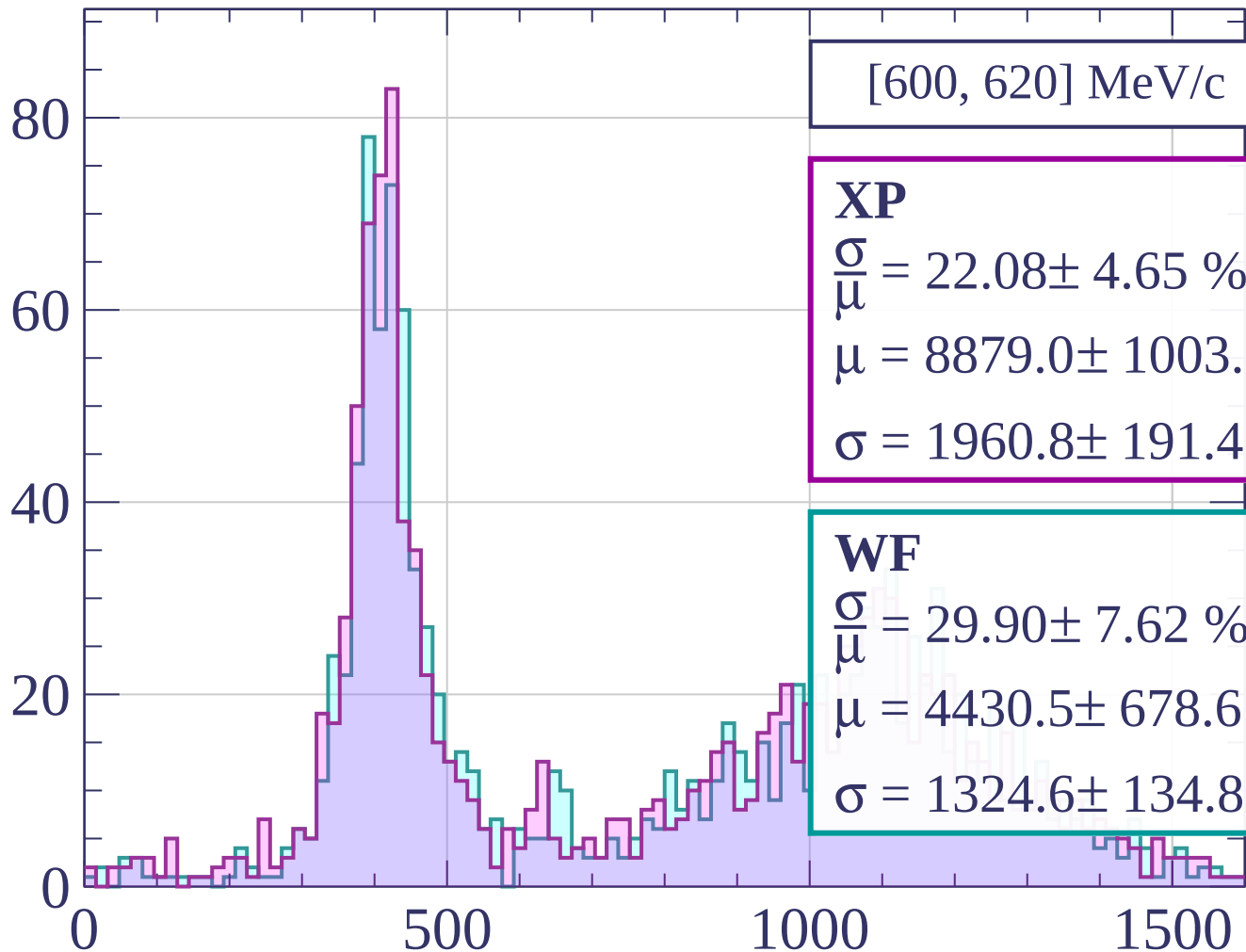




Count

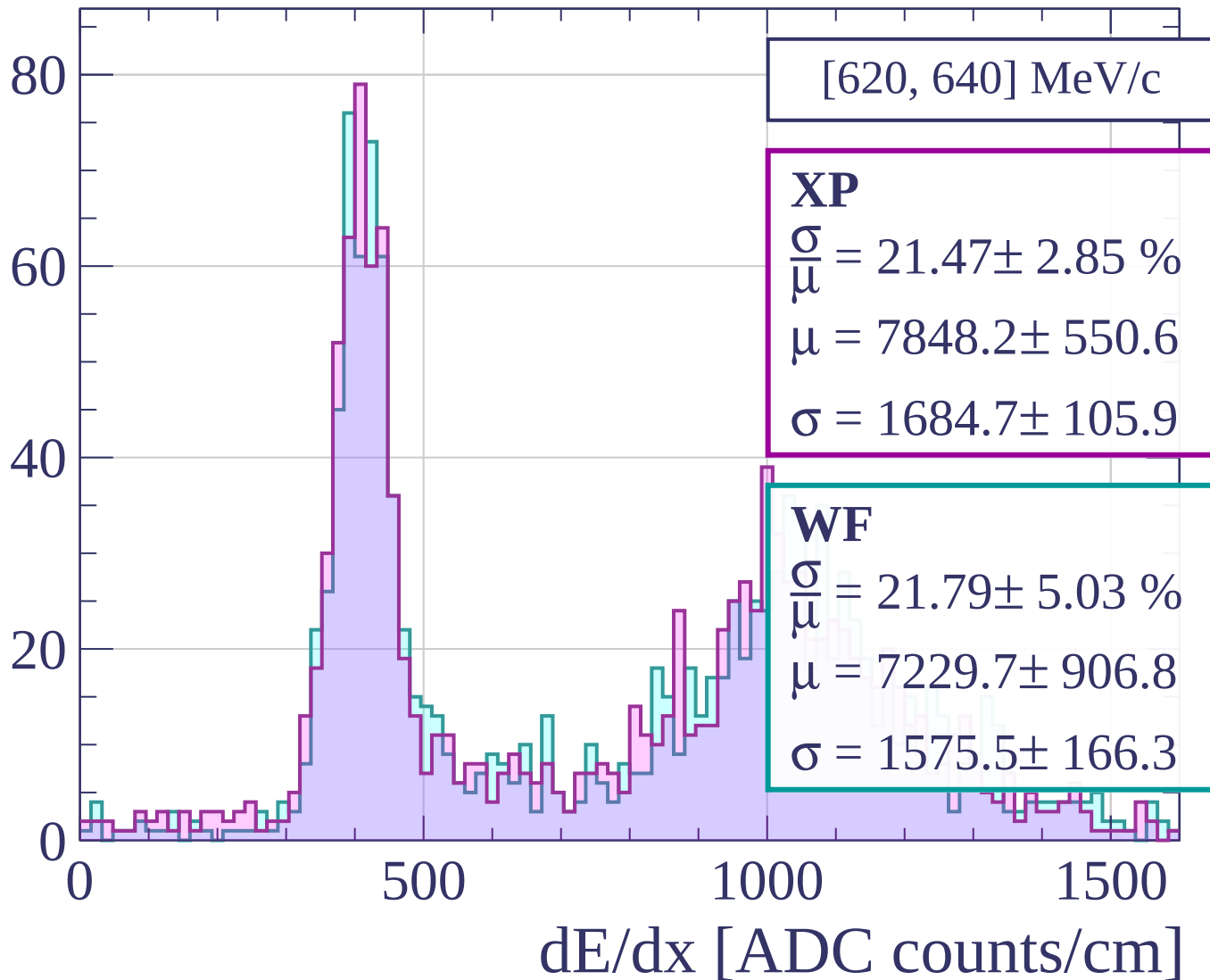


Count

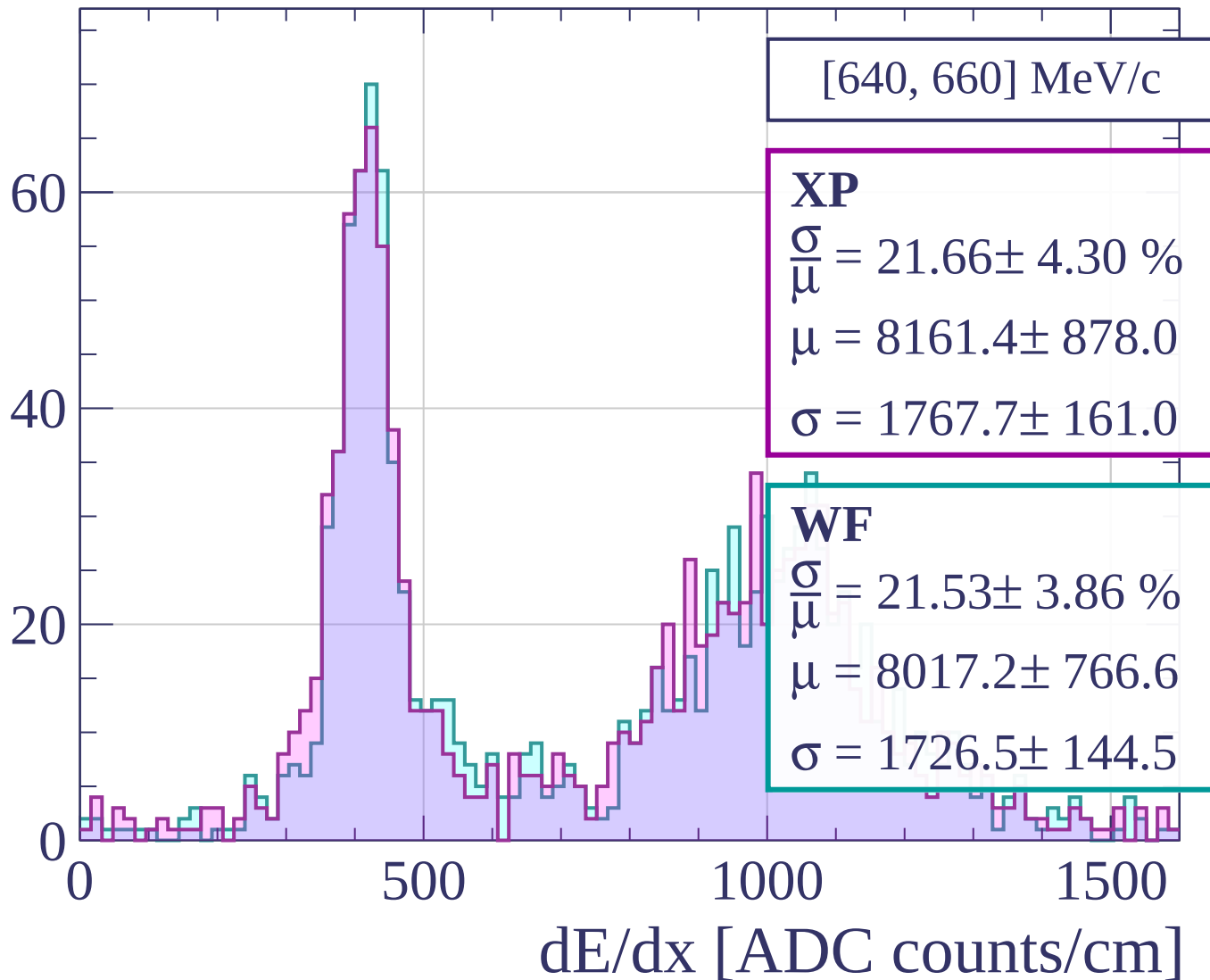


dE/dx [ADC counts/cm]

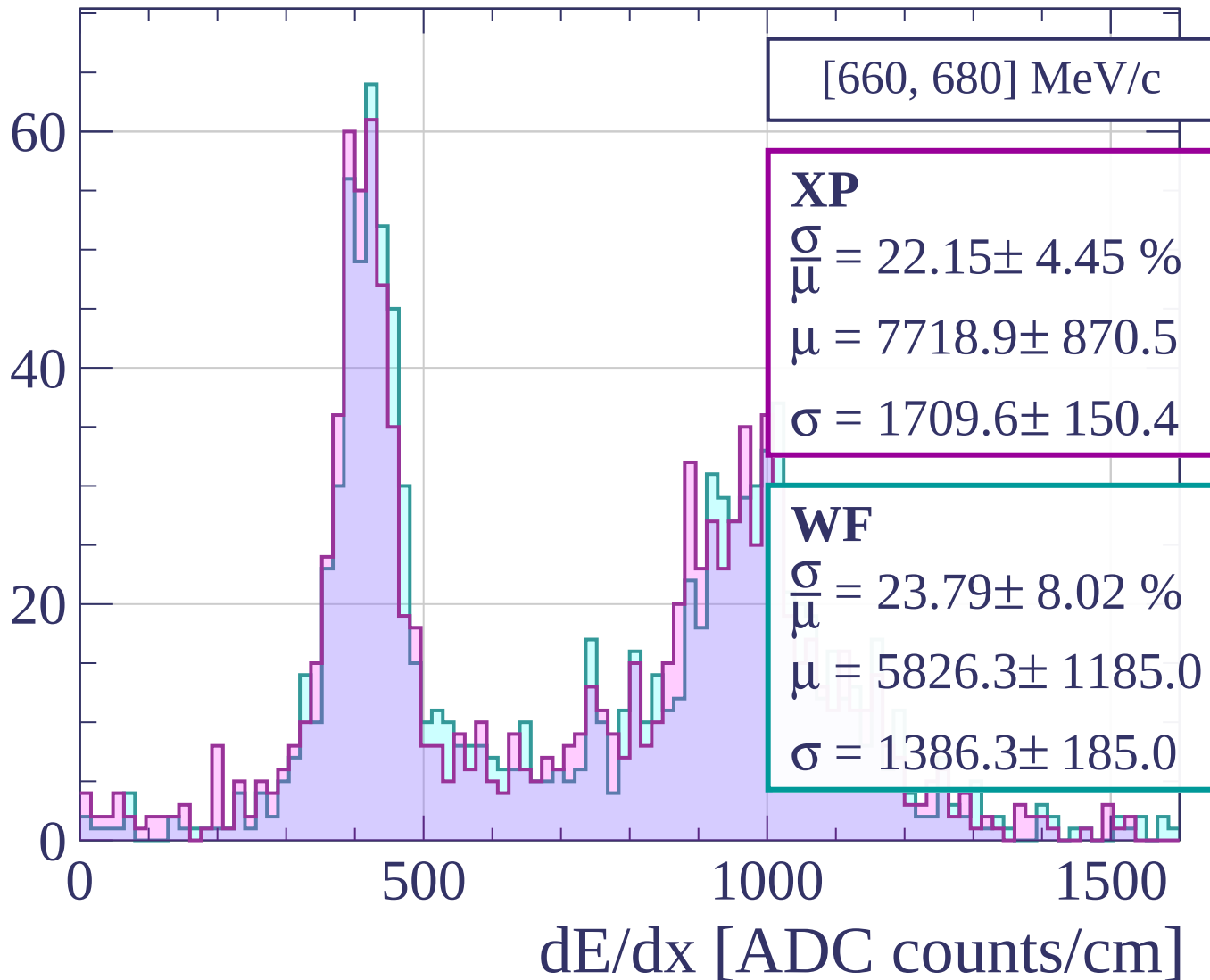
Count



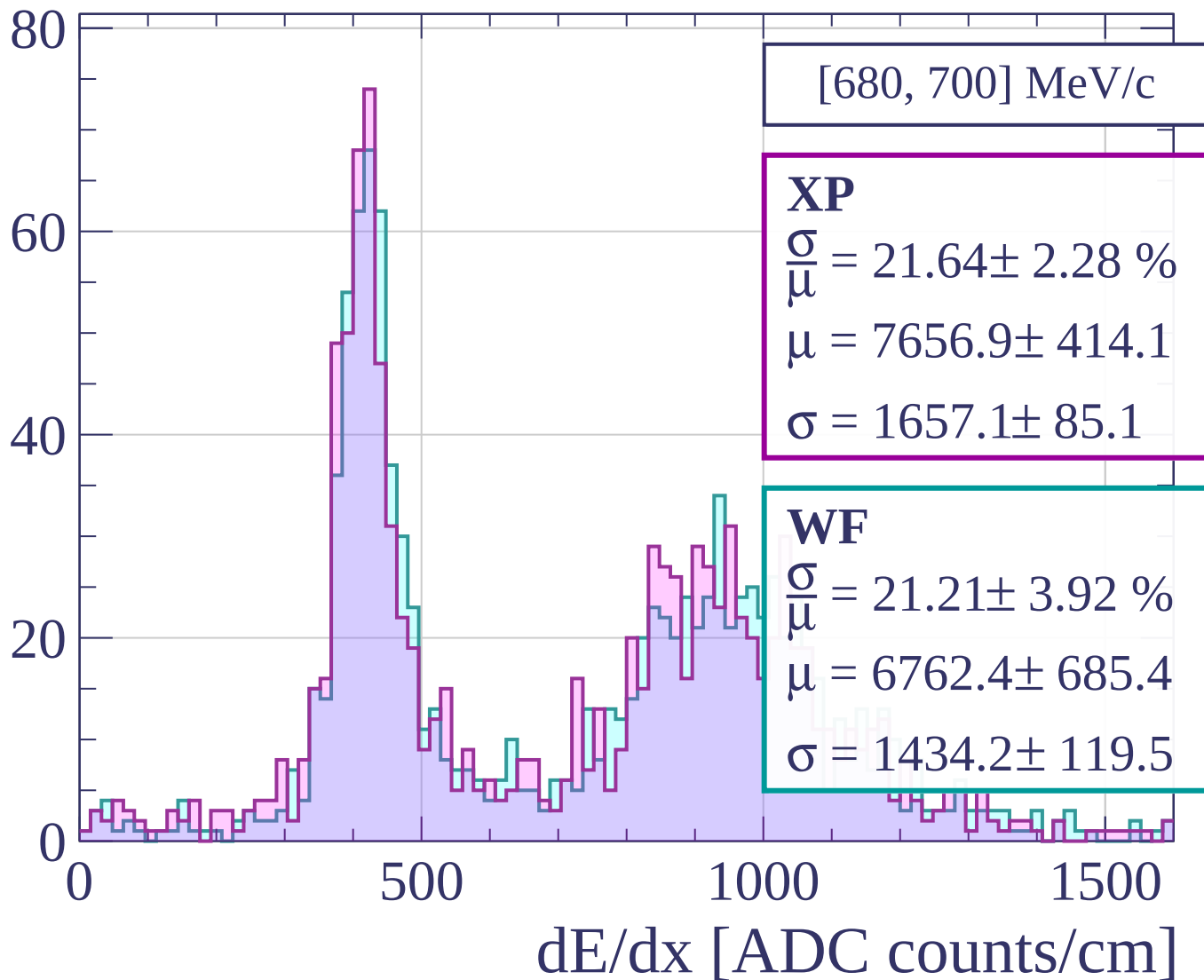
Count



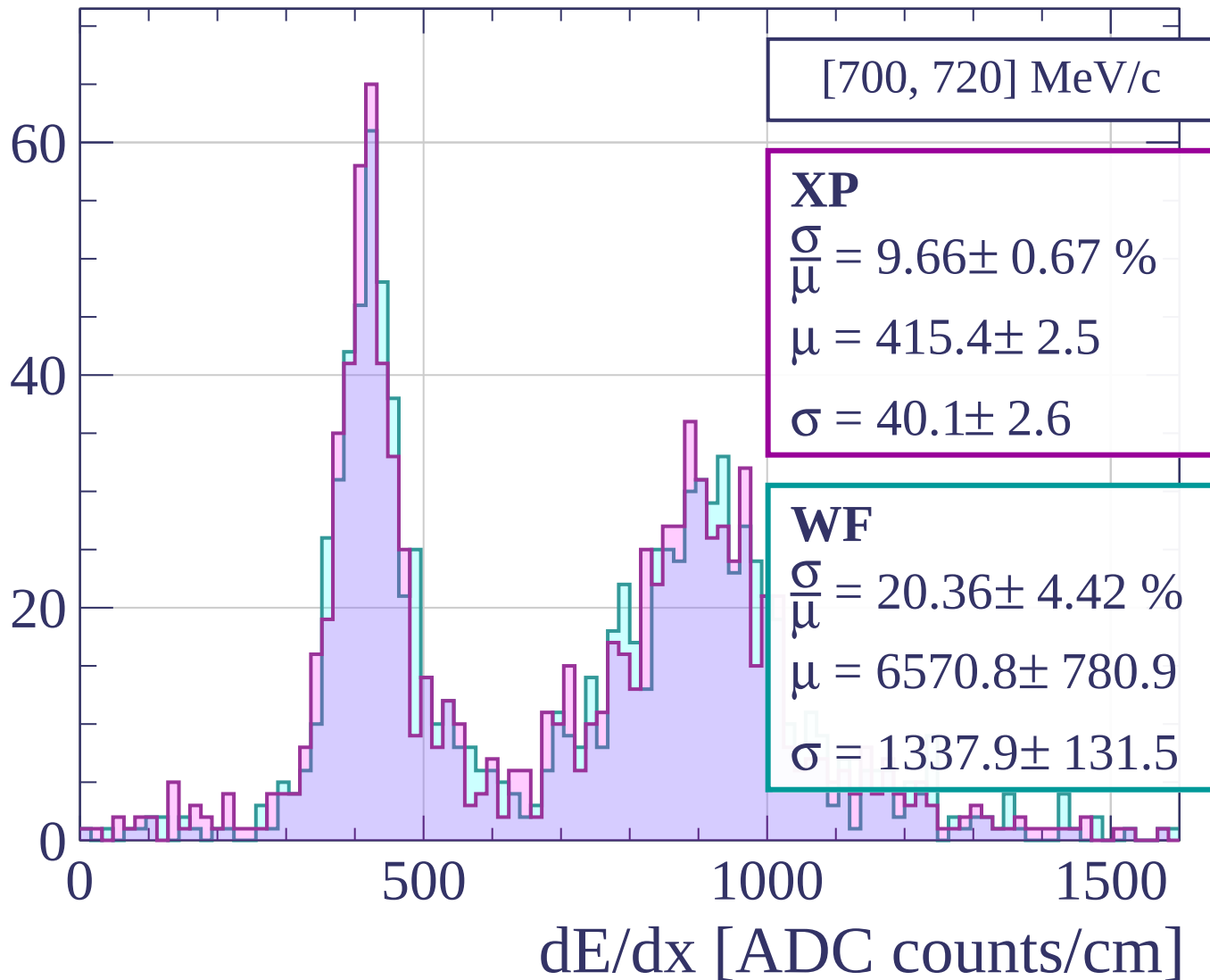
Count



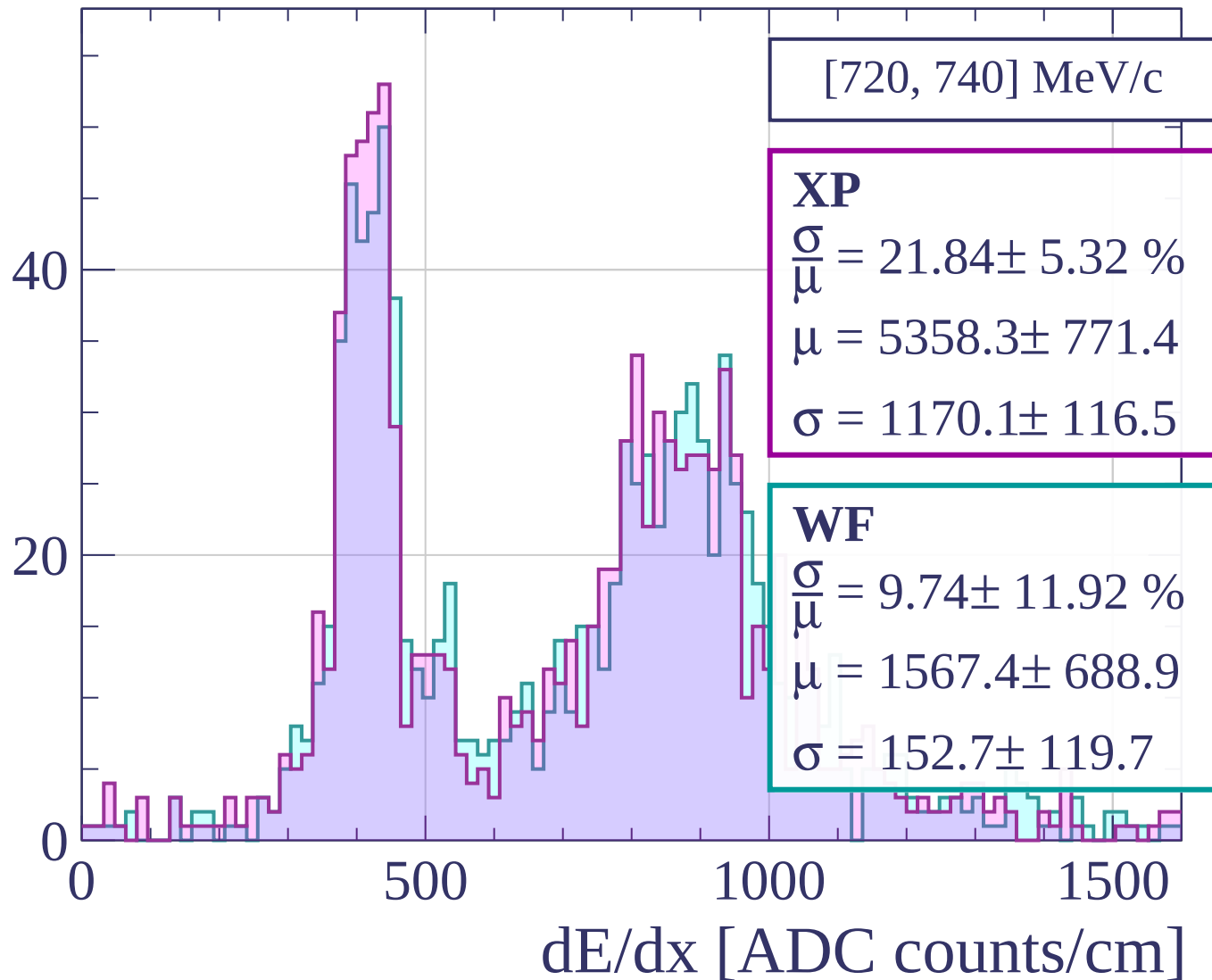
Count



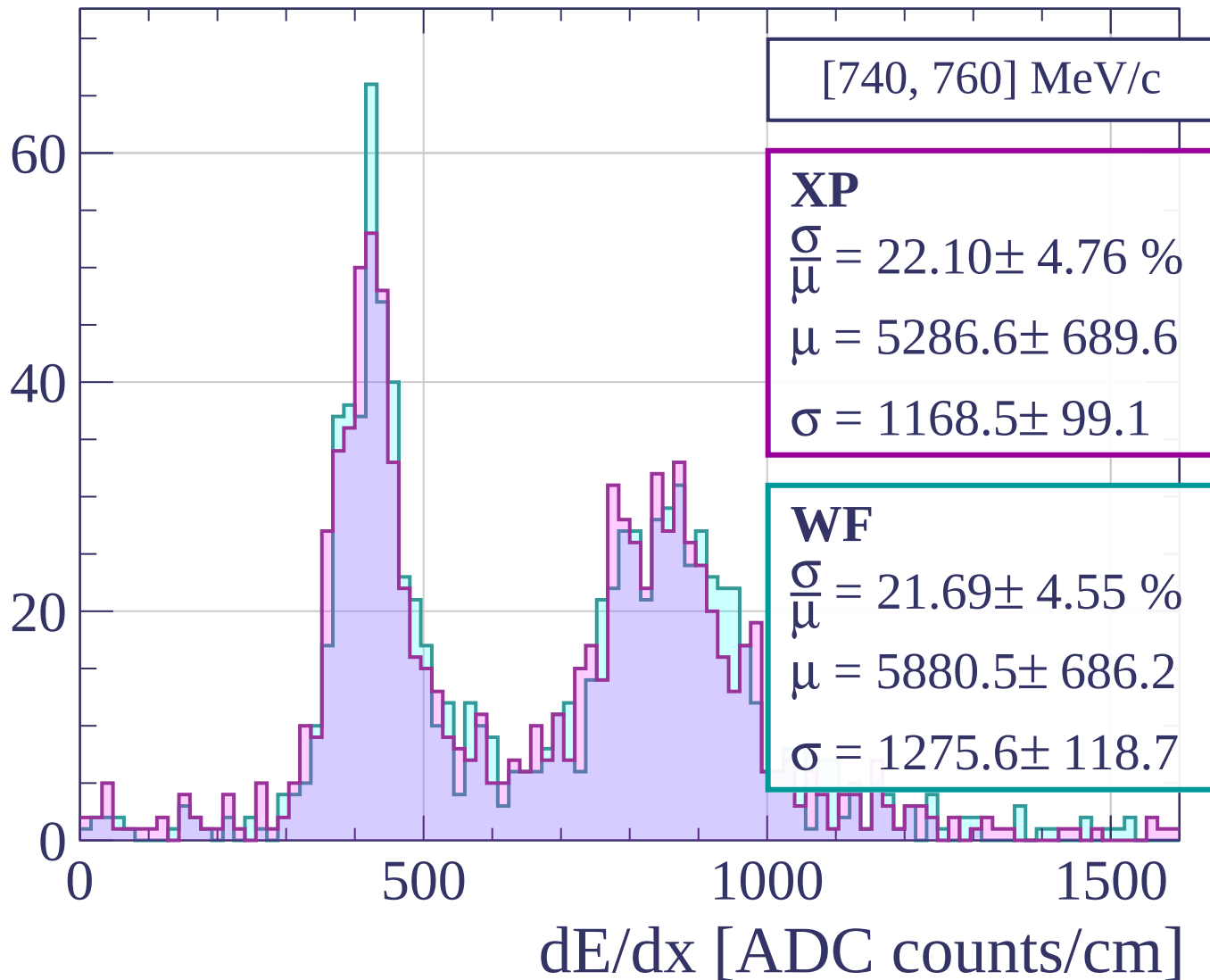
Count



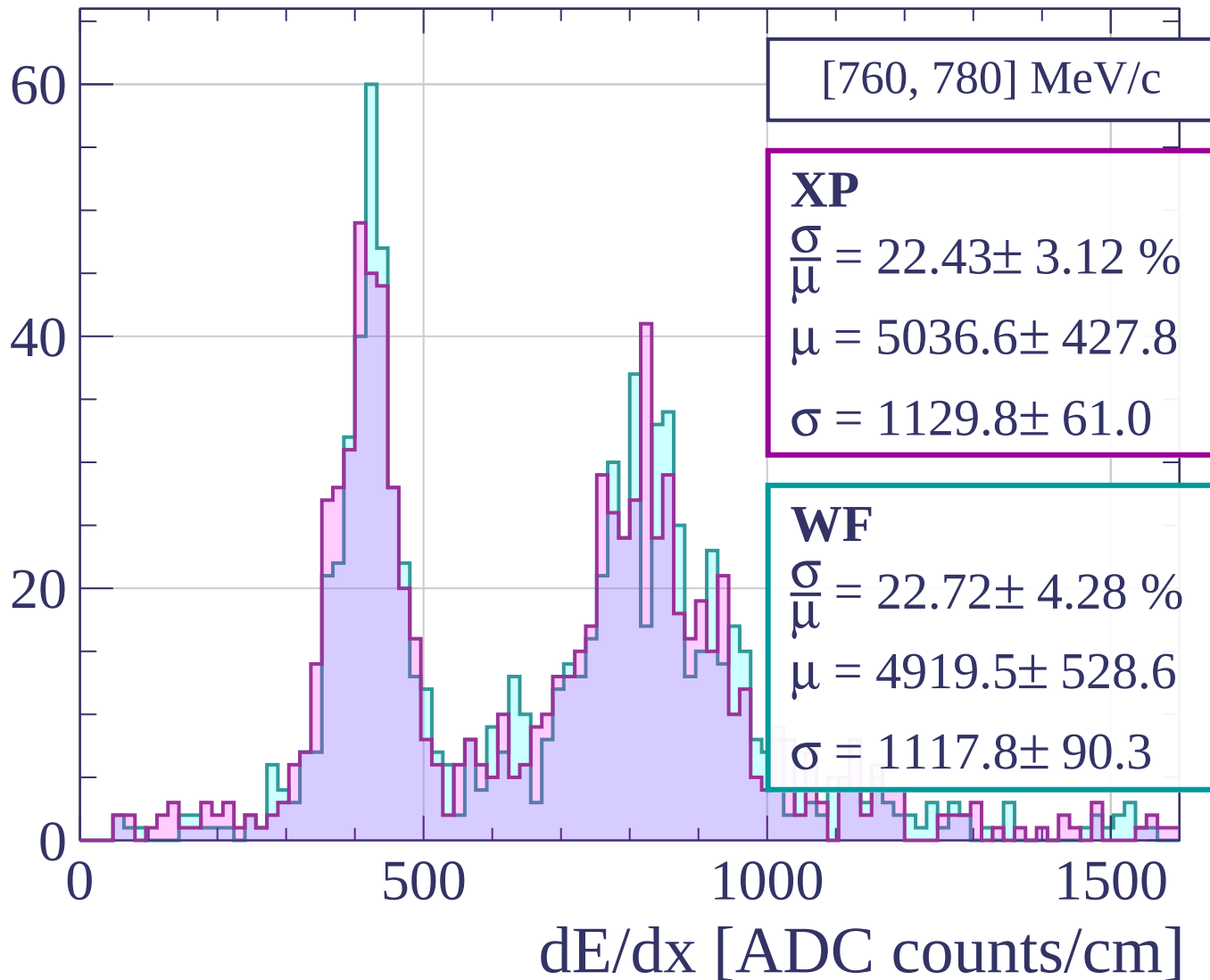
Count

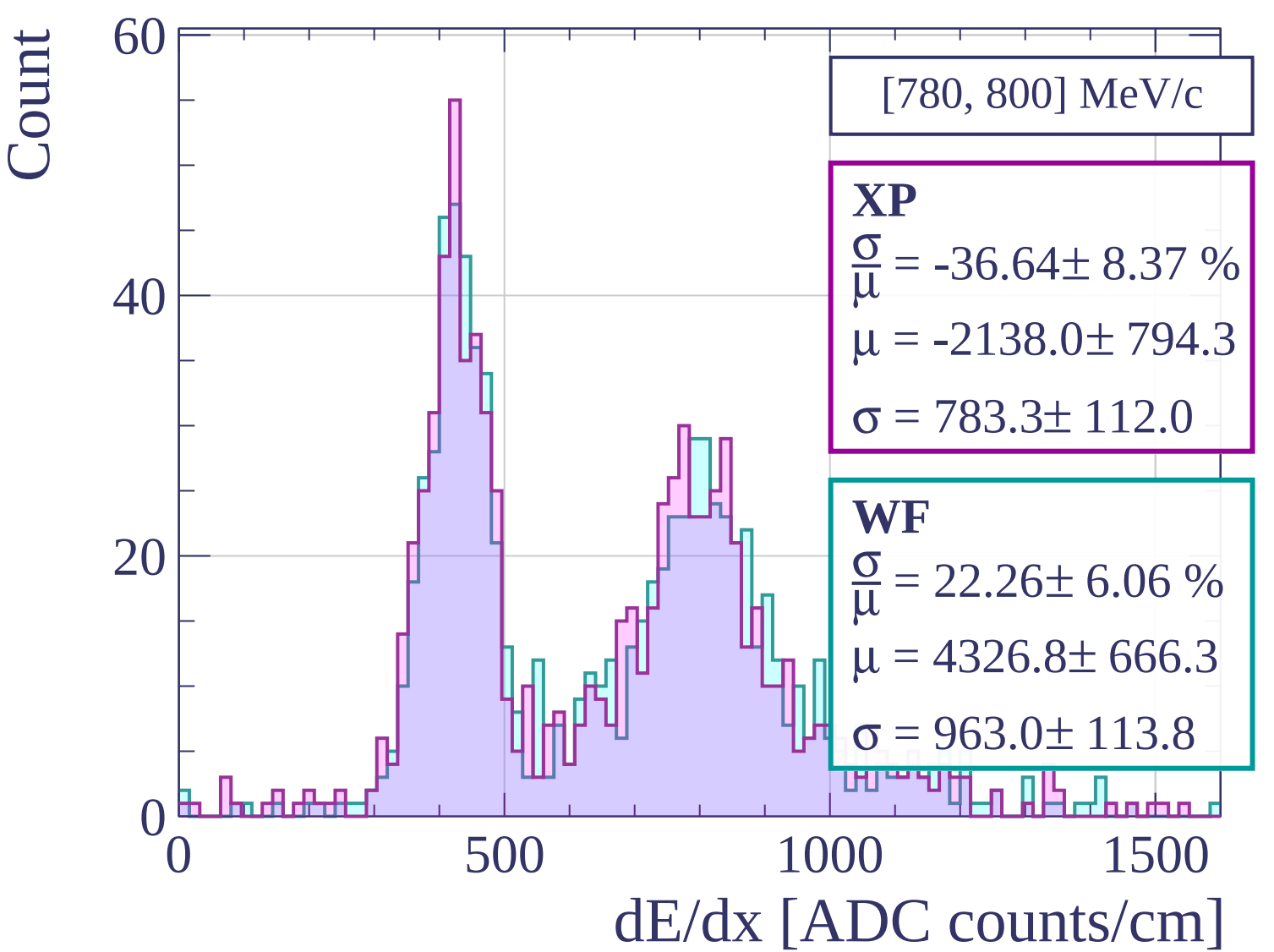


Count

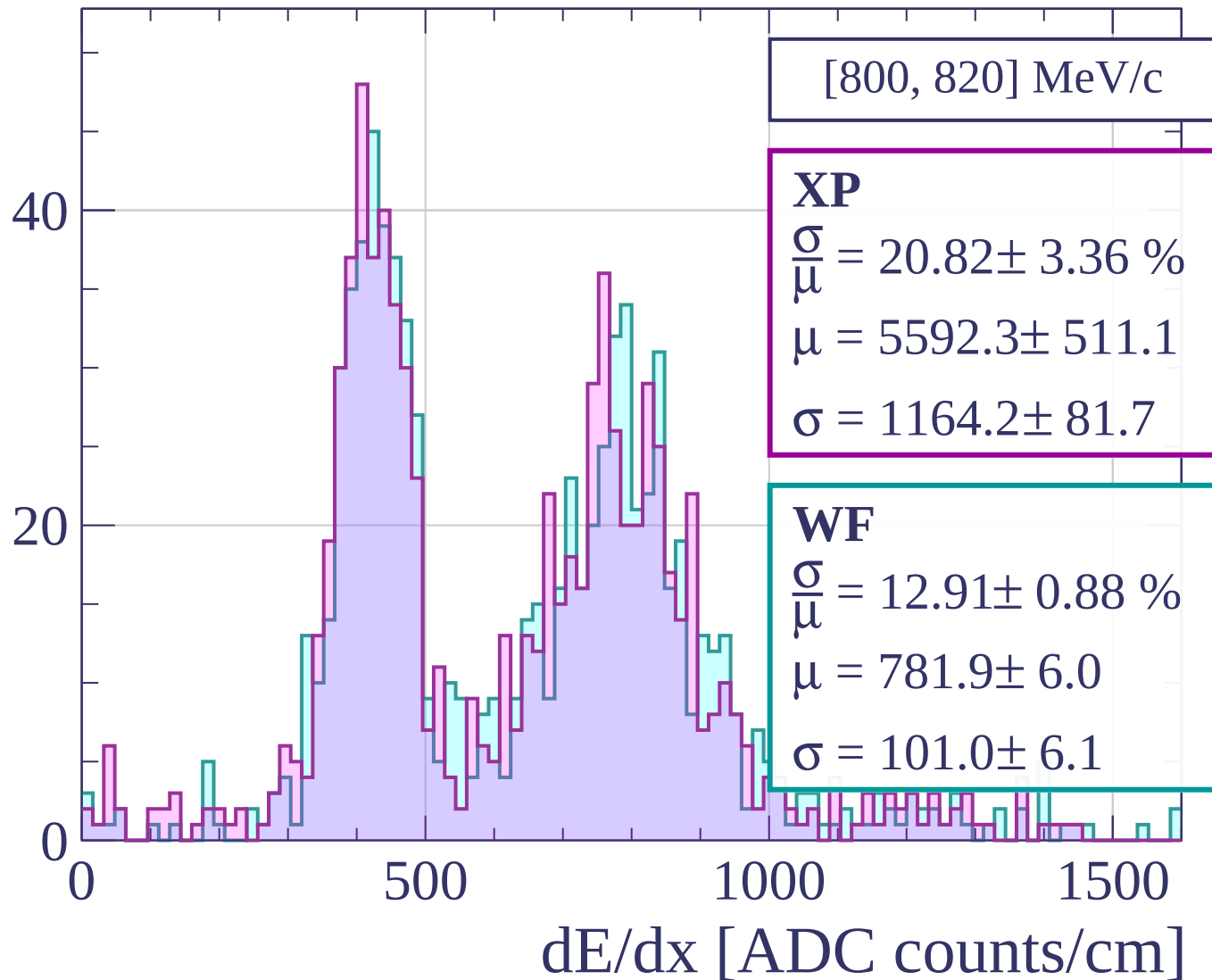


Count

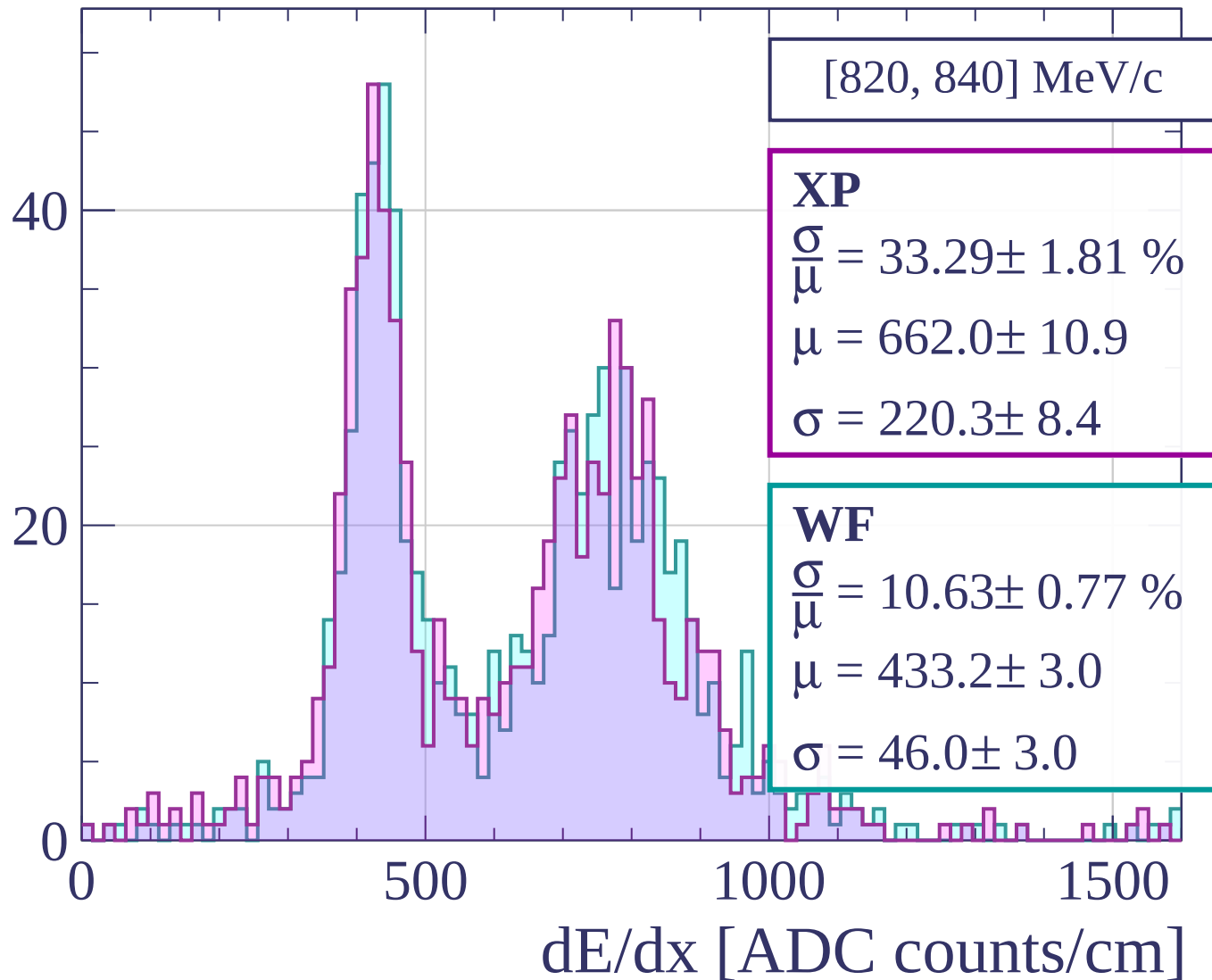




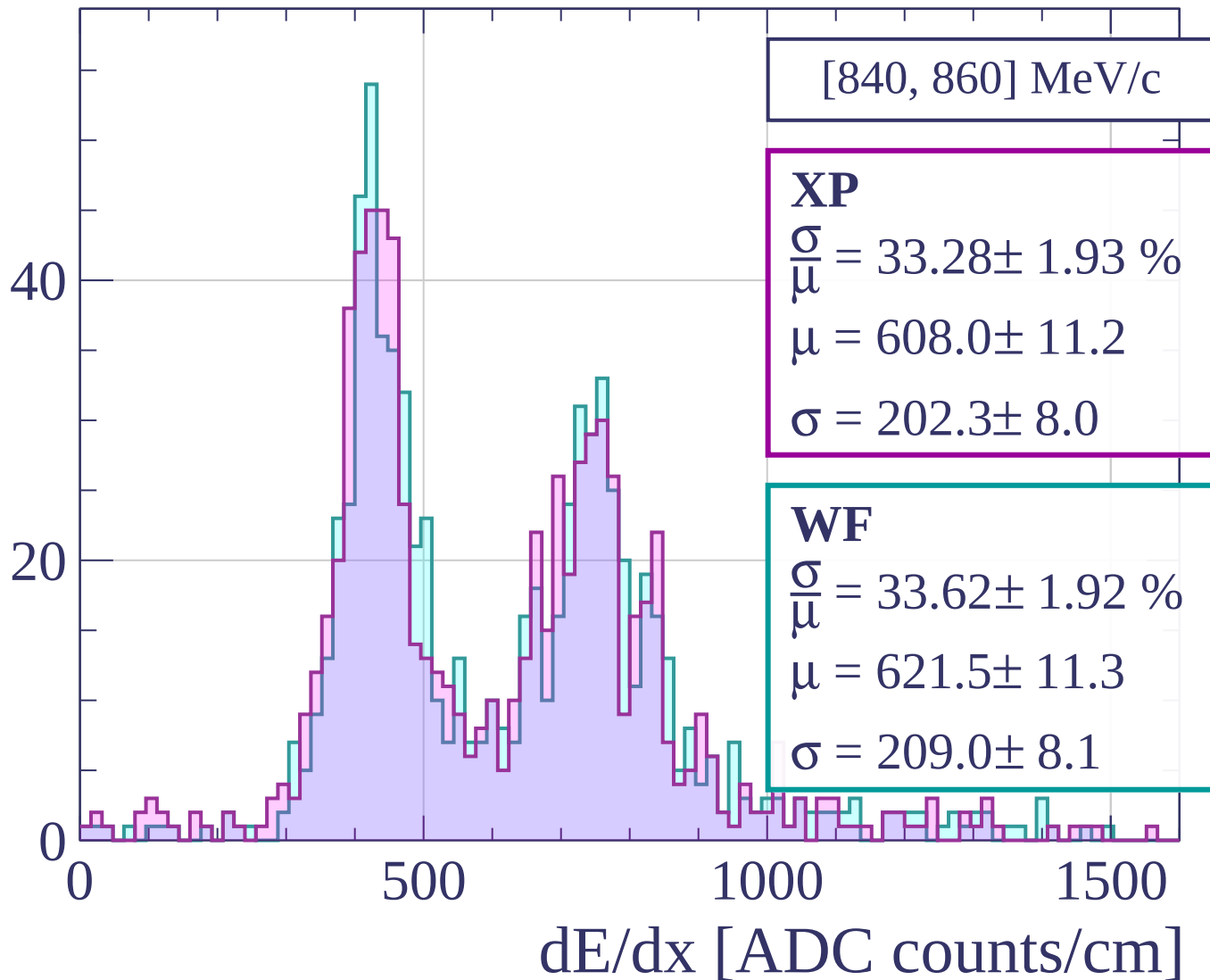
Count



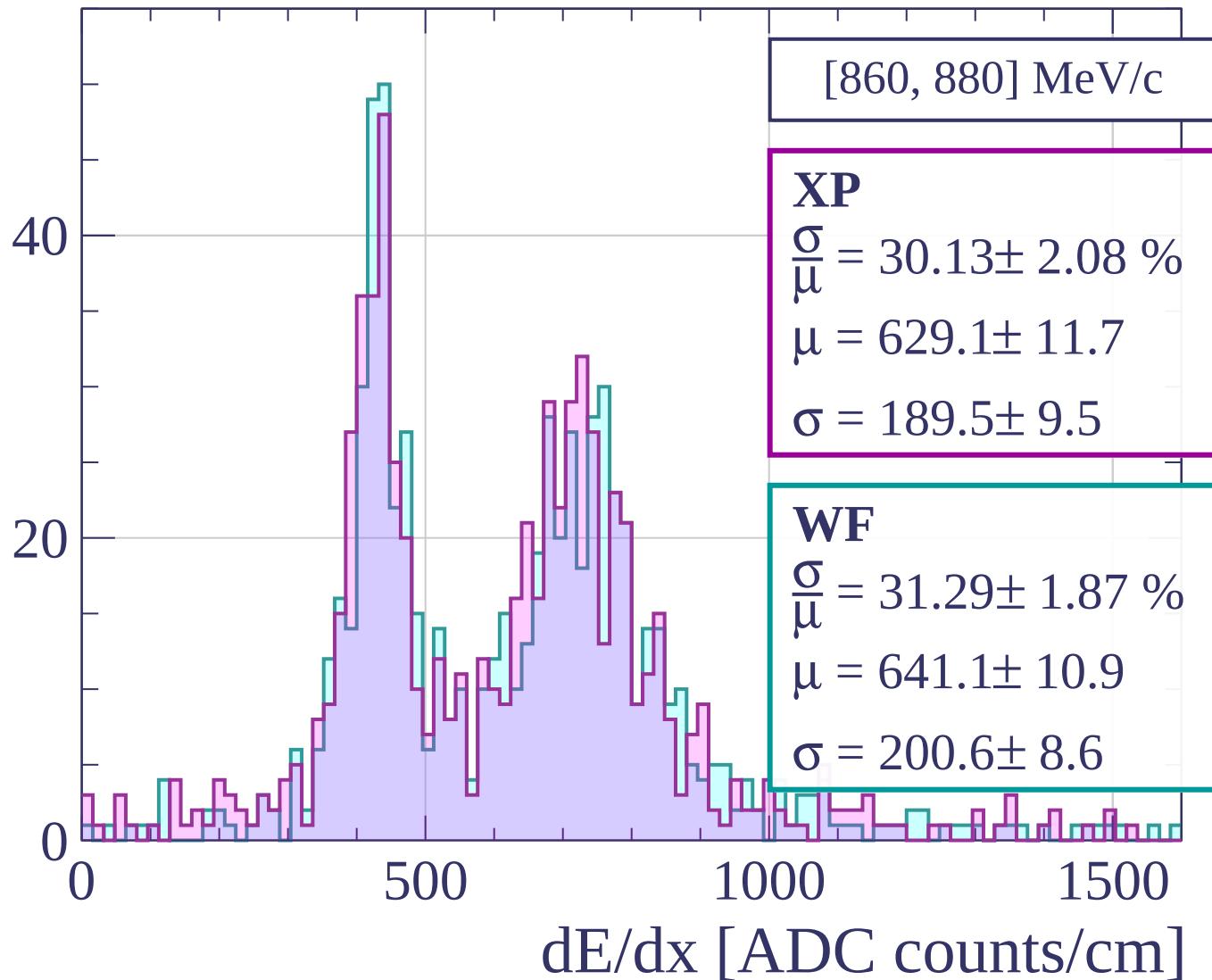
Count



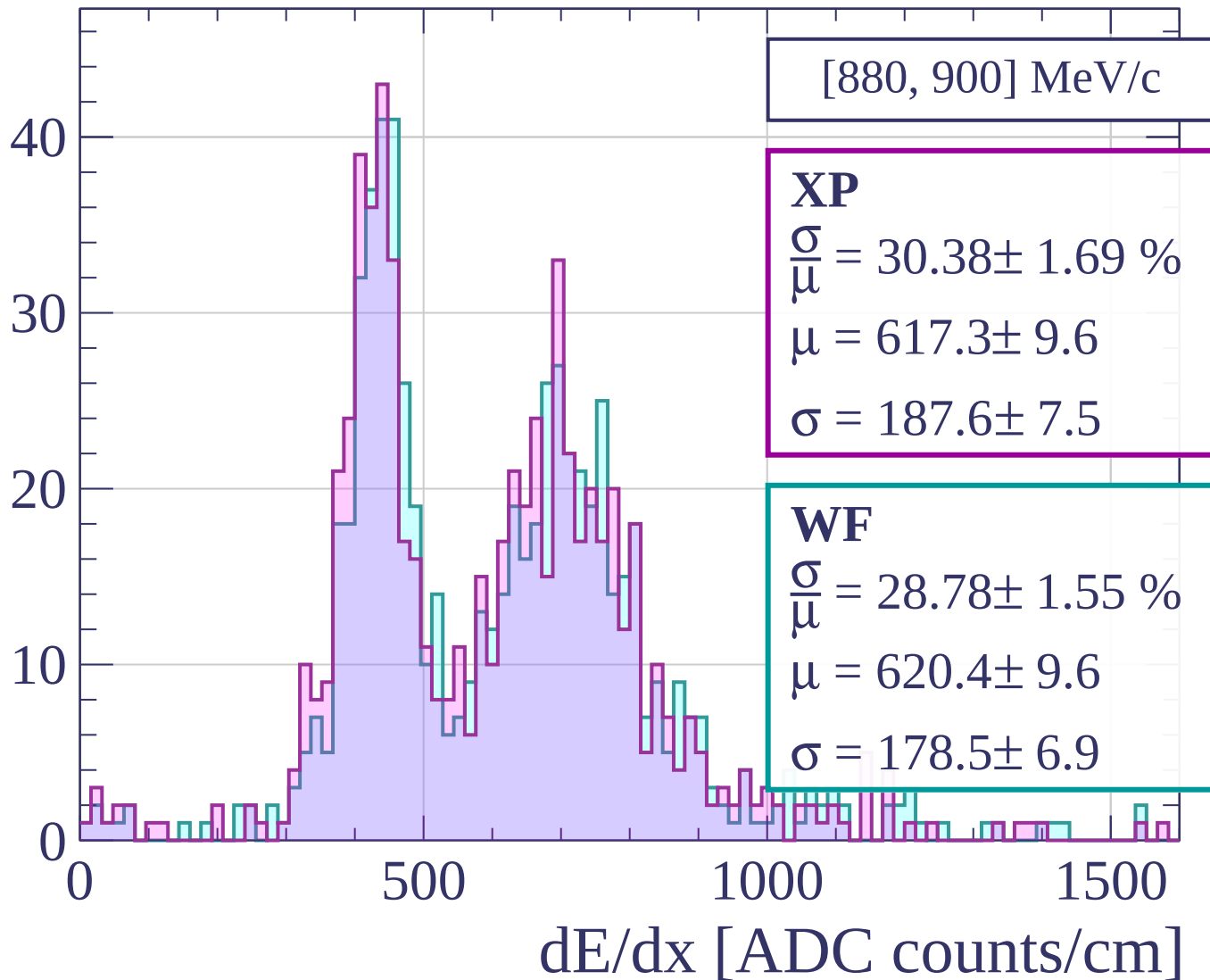
Count



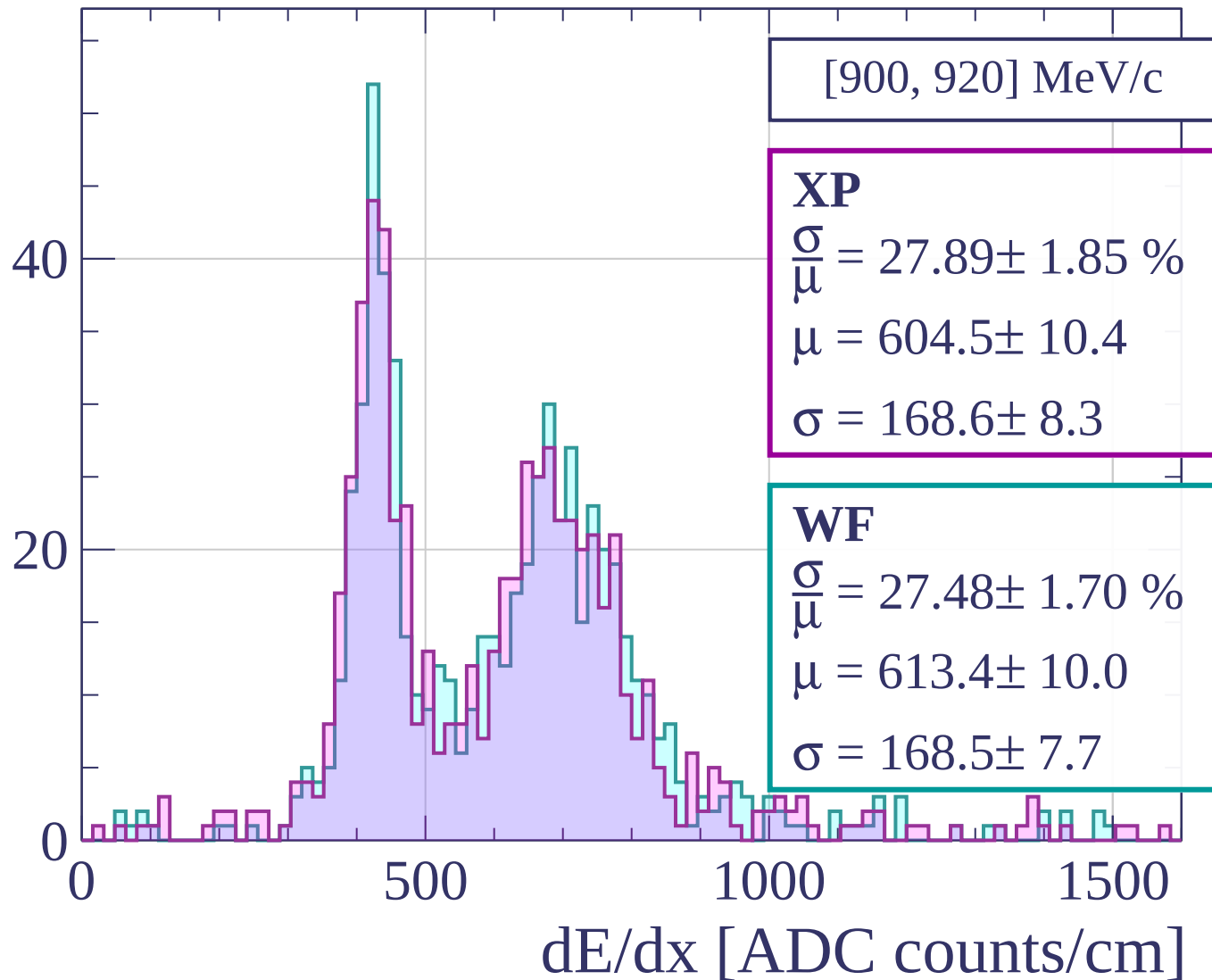
Count



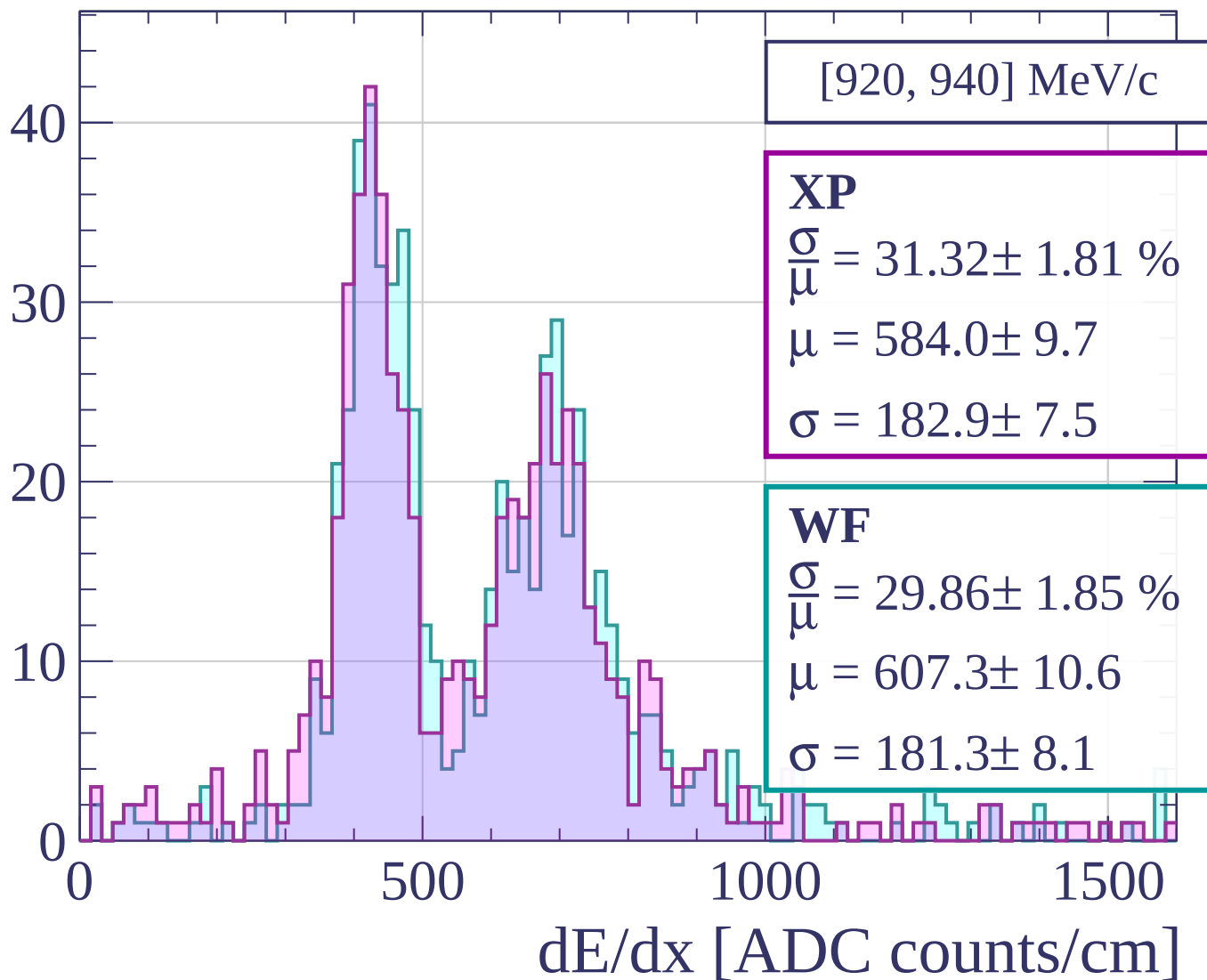
Count



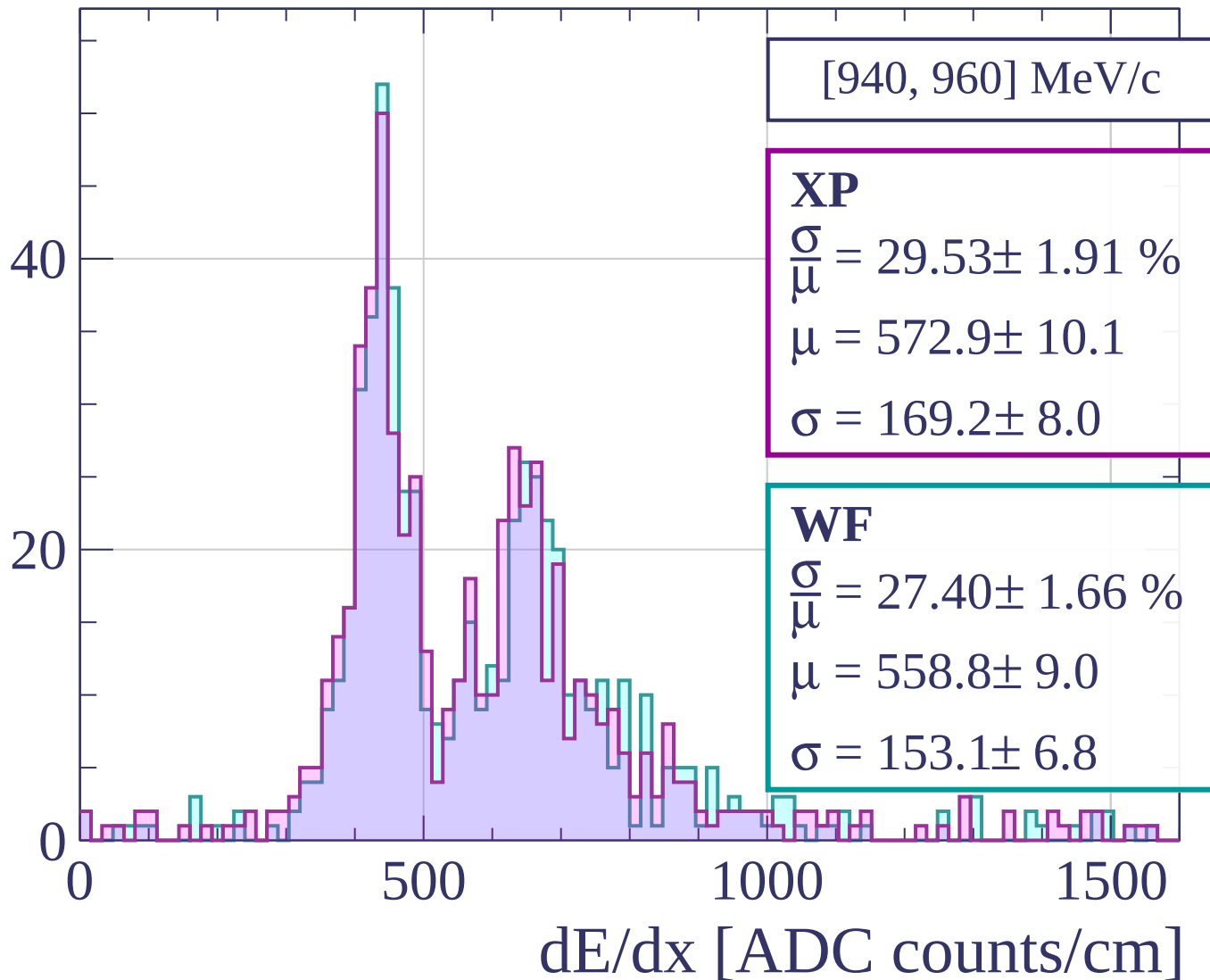
Count



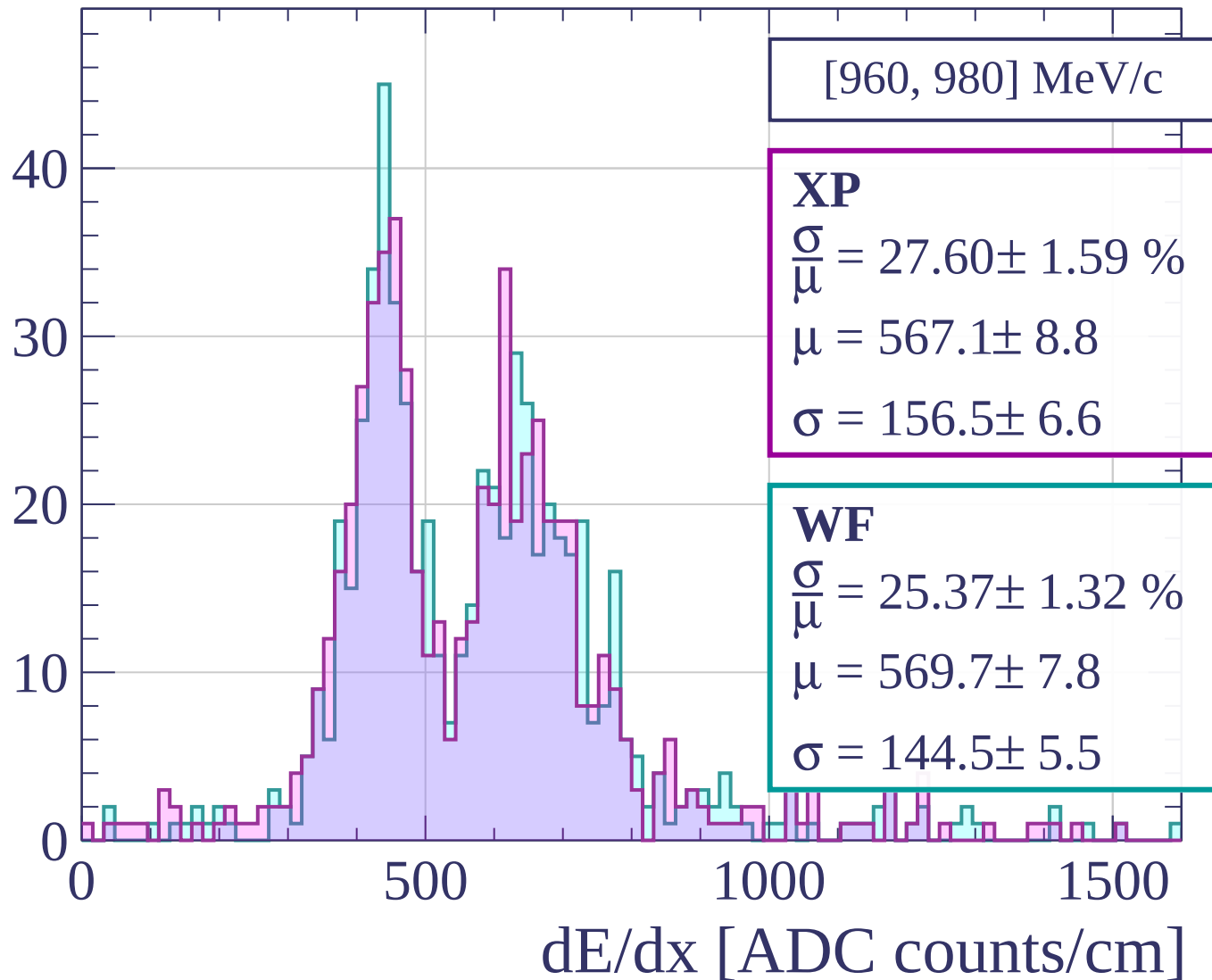
Count



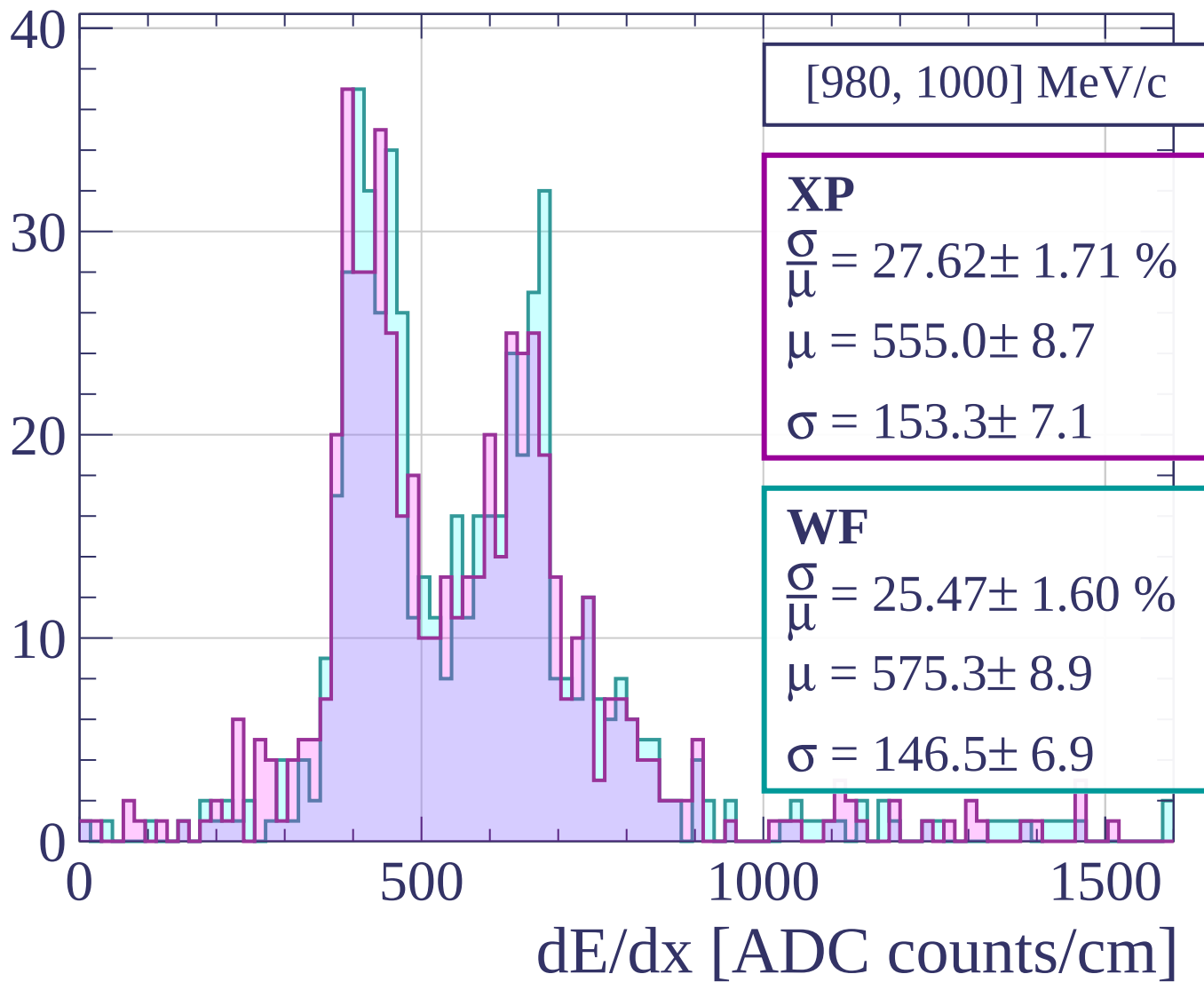
Count



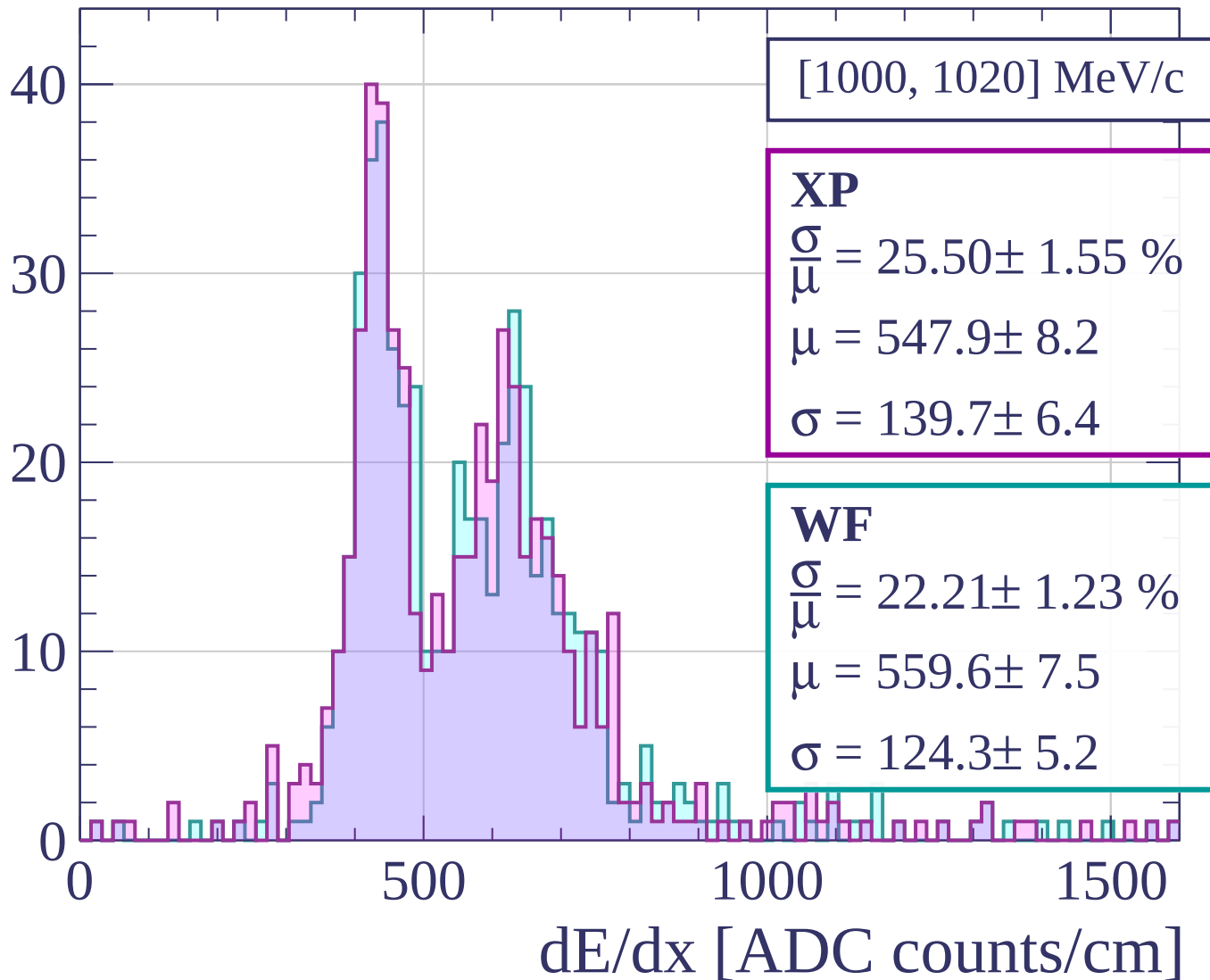
Count



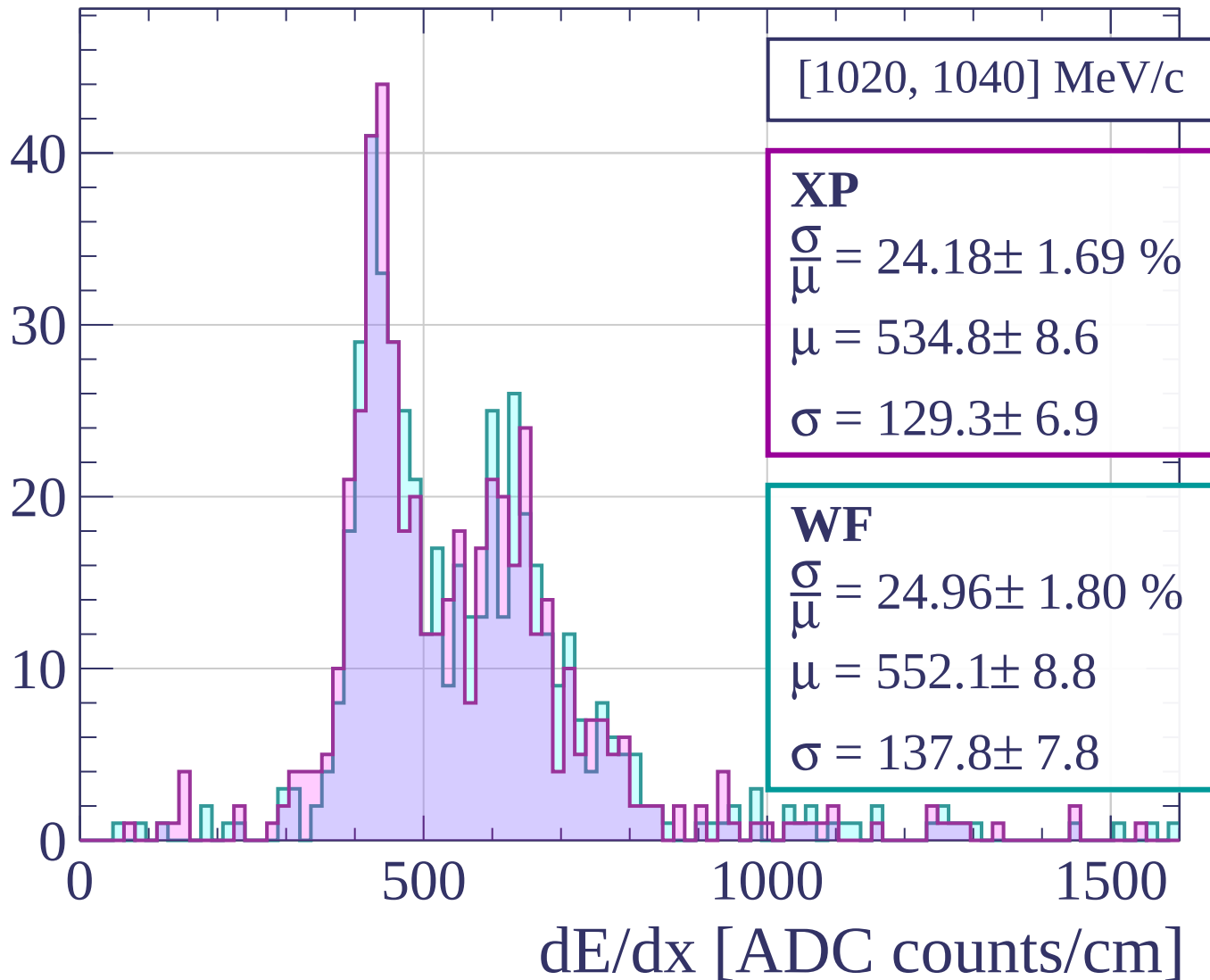
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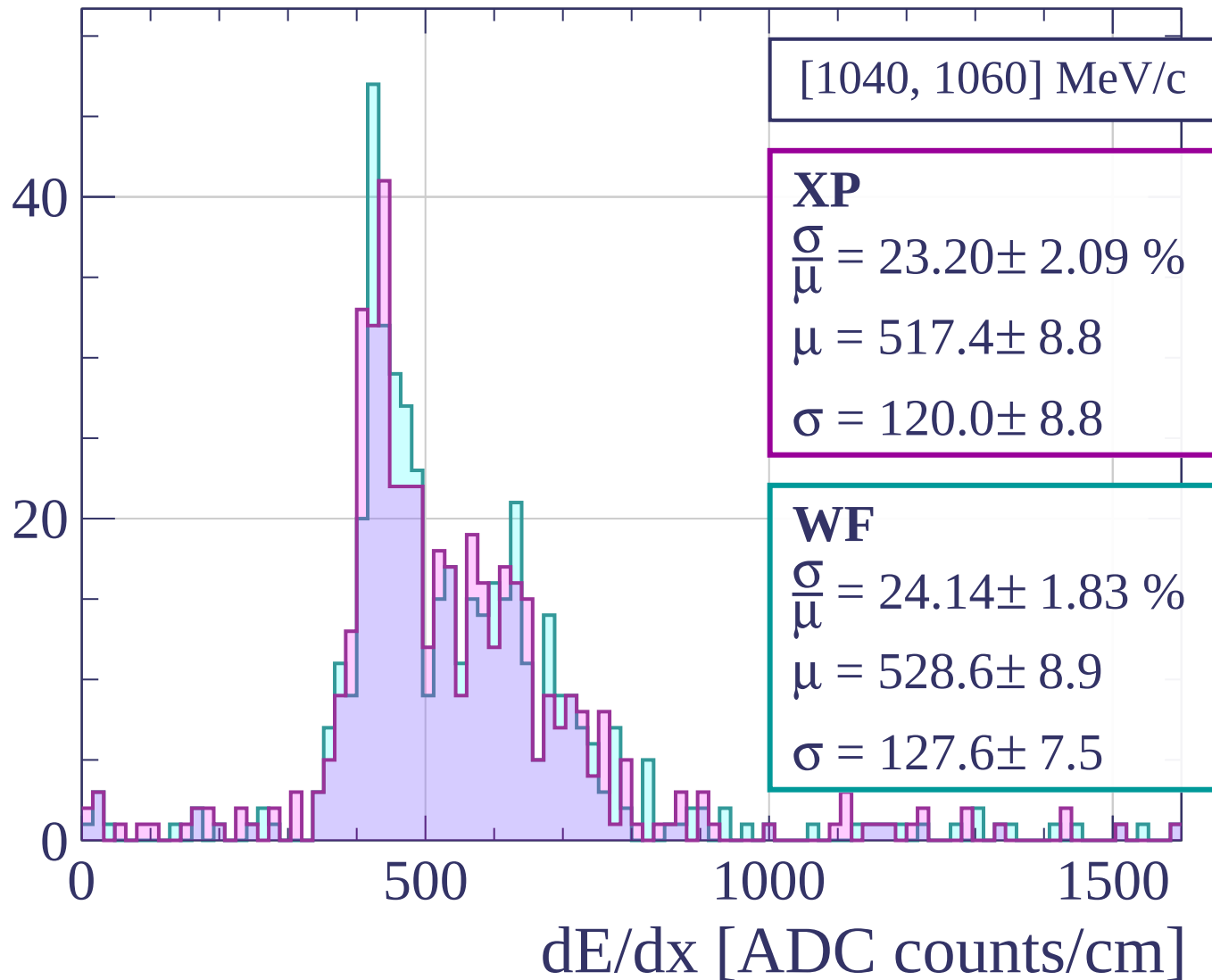
Count



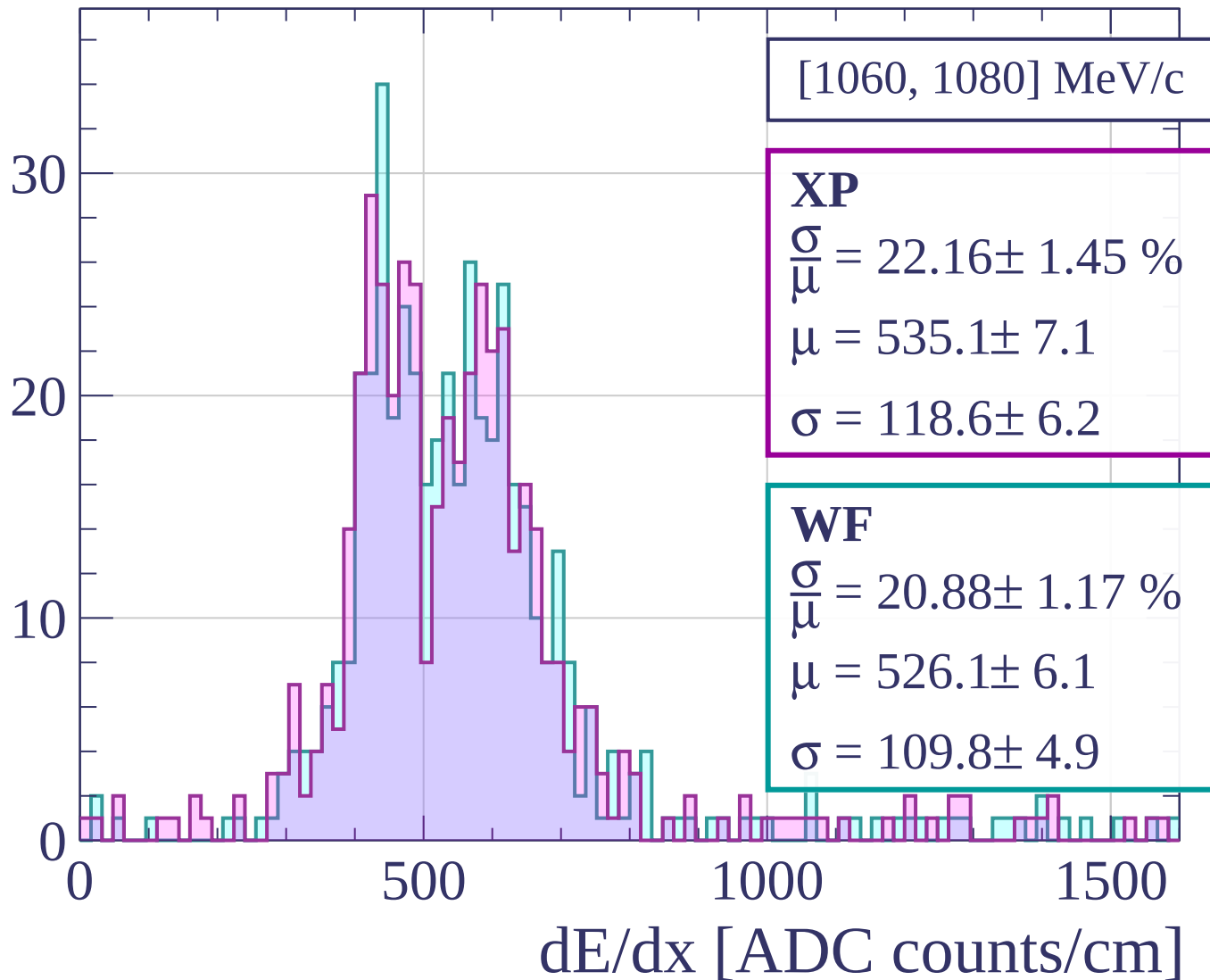
Count



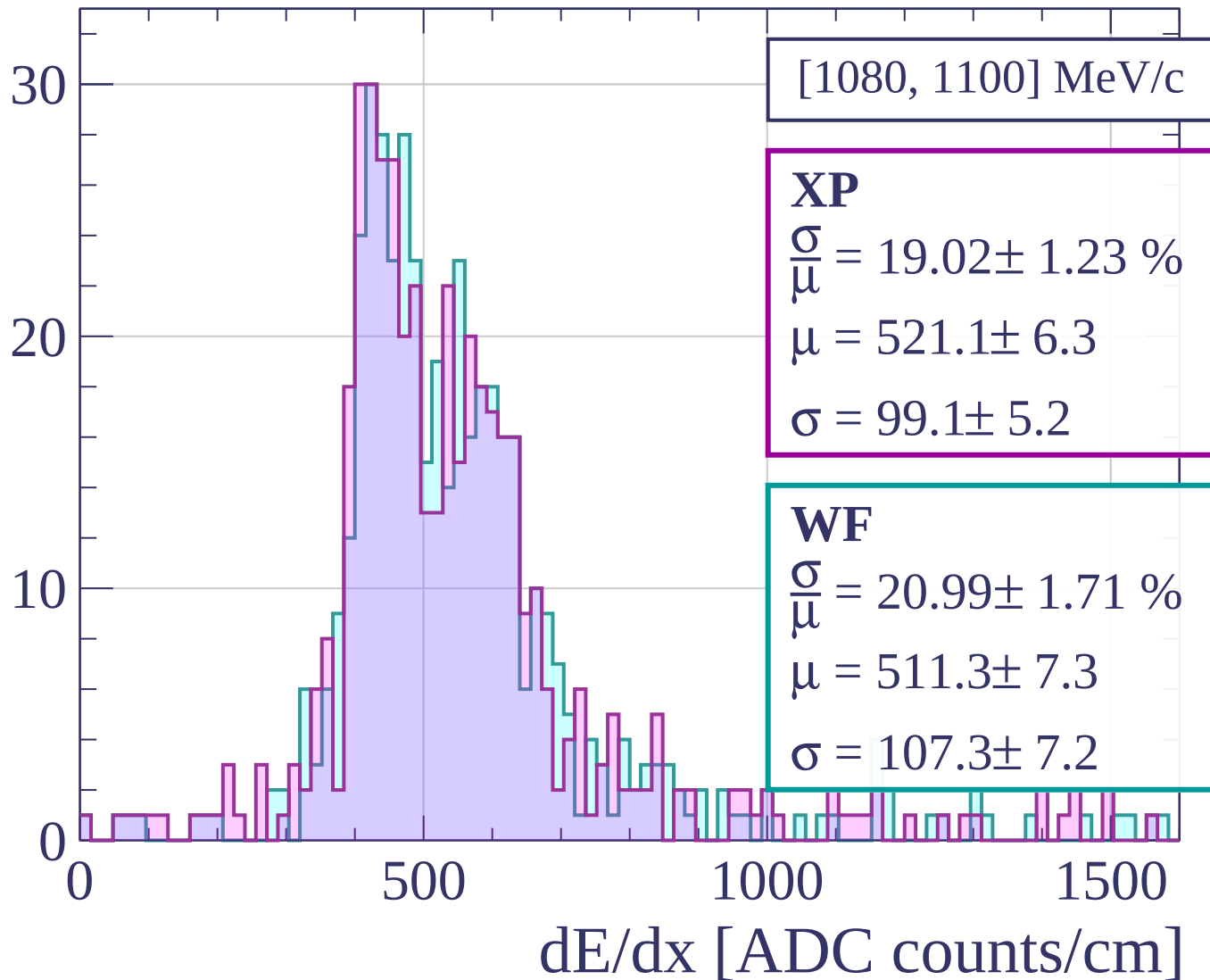
Count



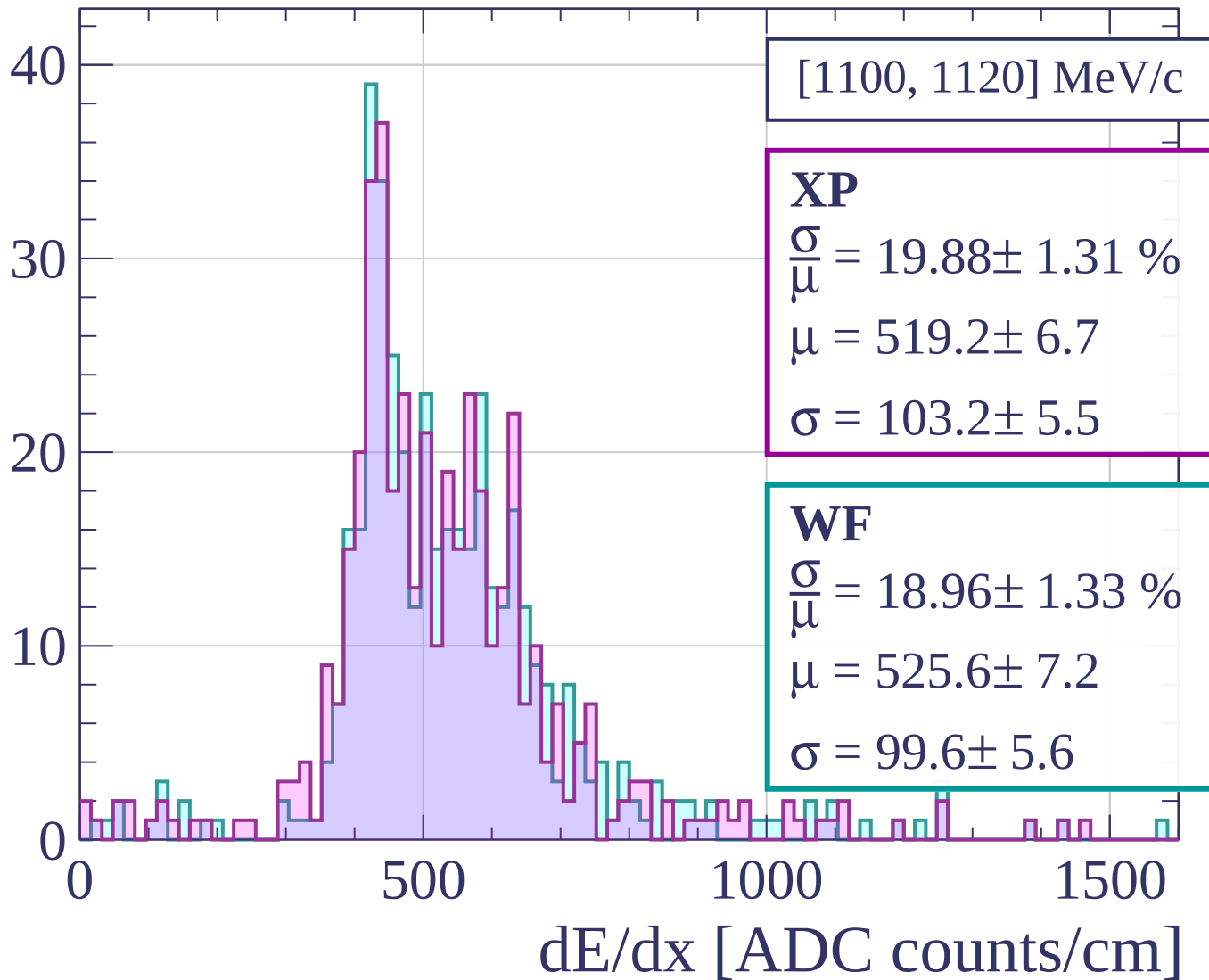
Count

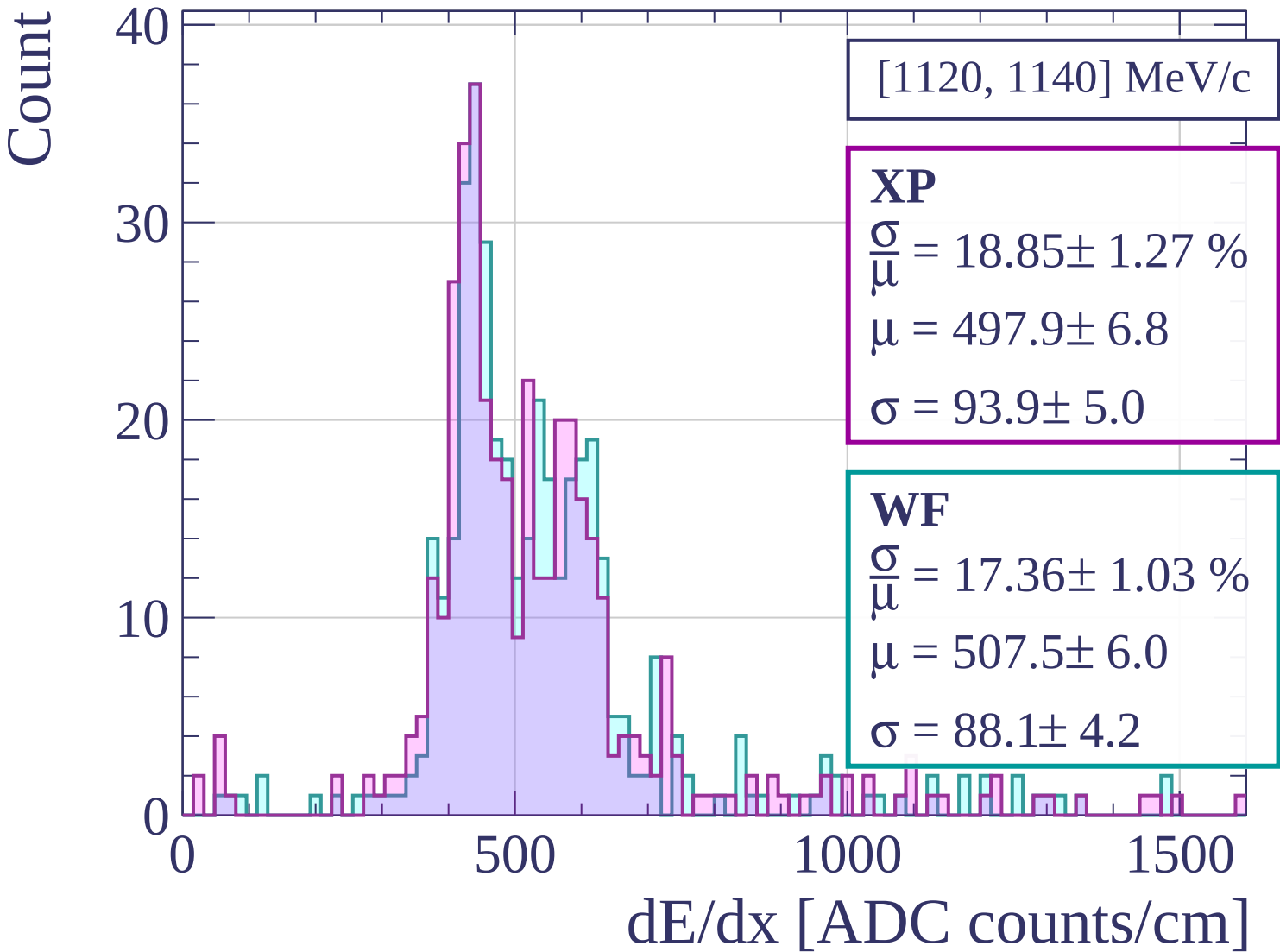


Count

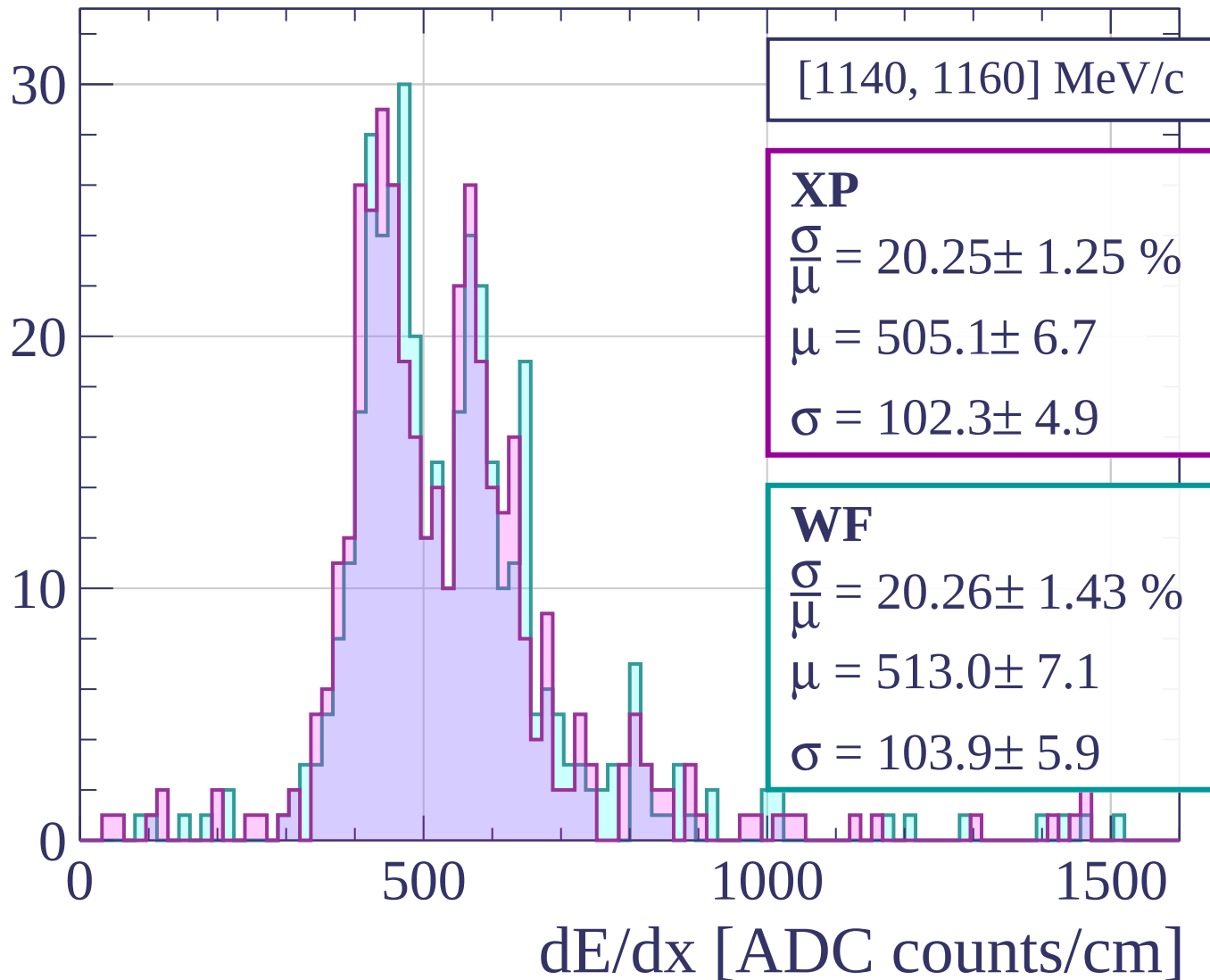


Count

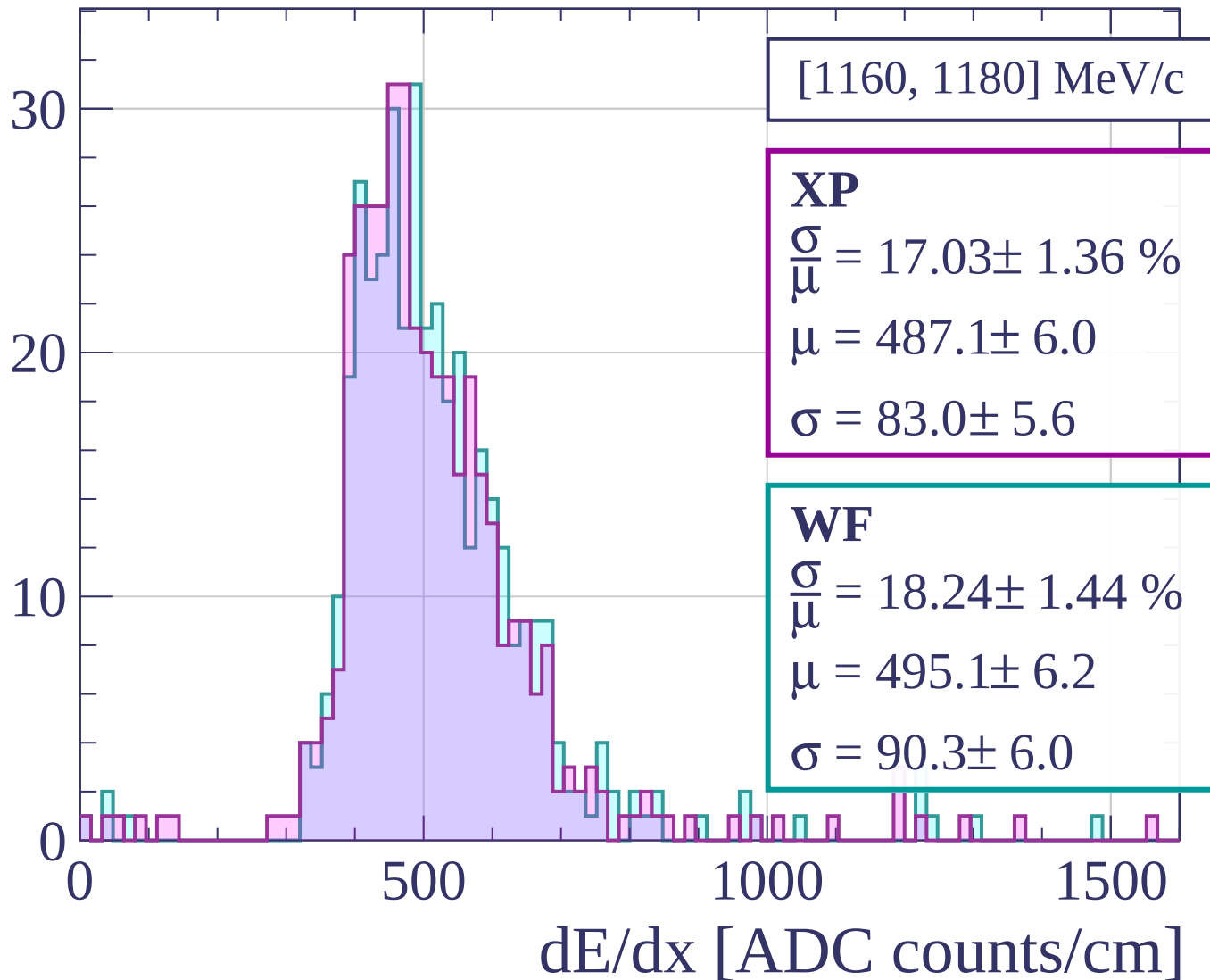




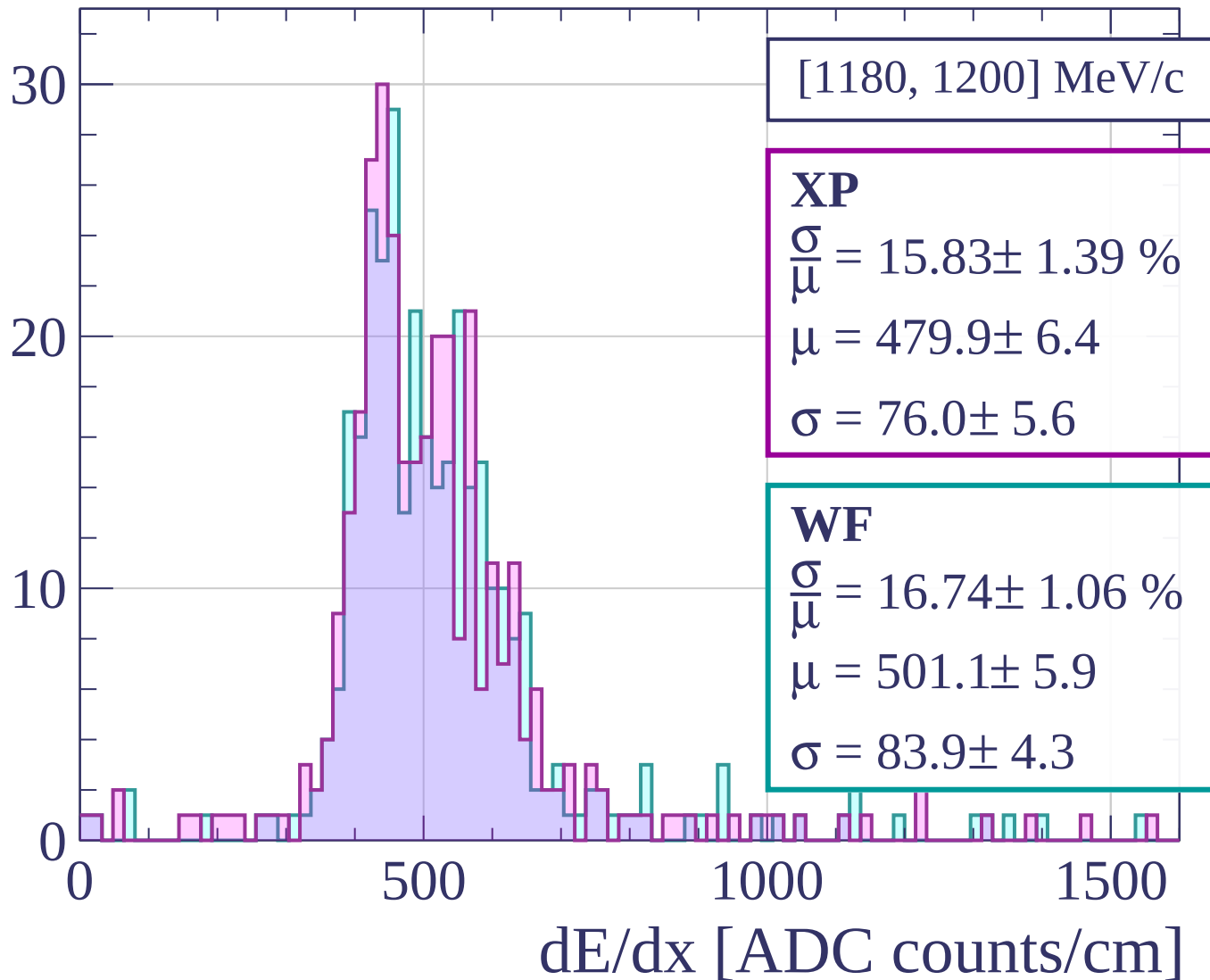
Count



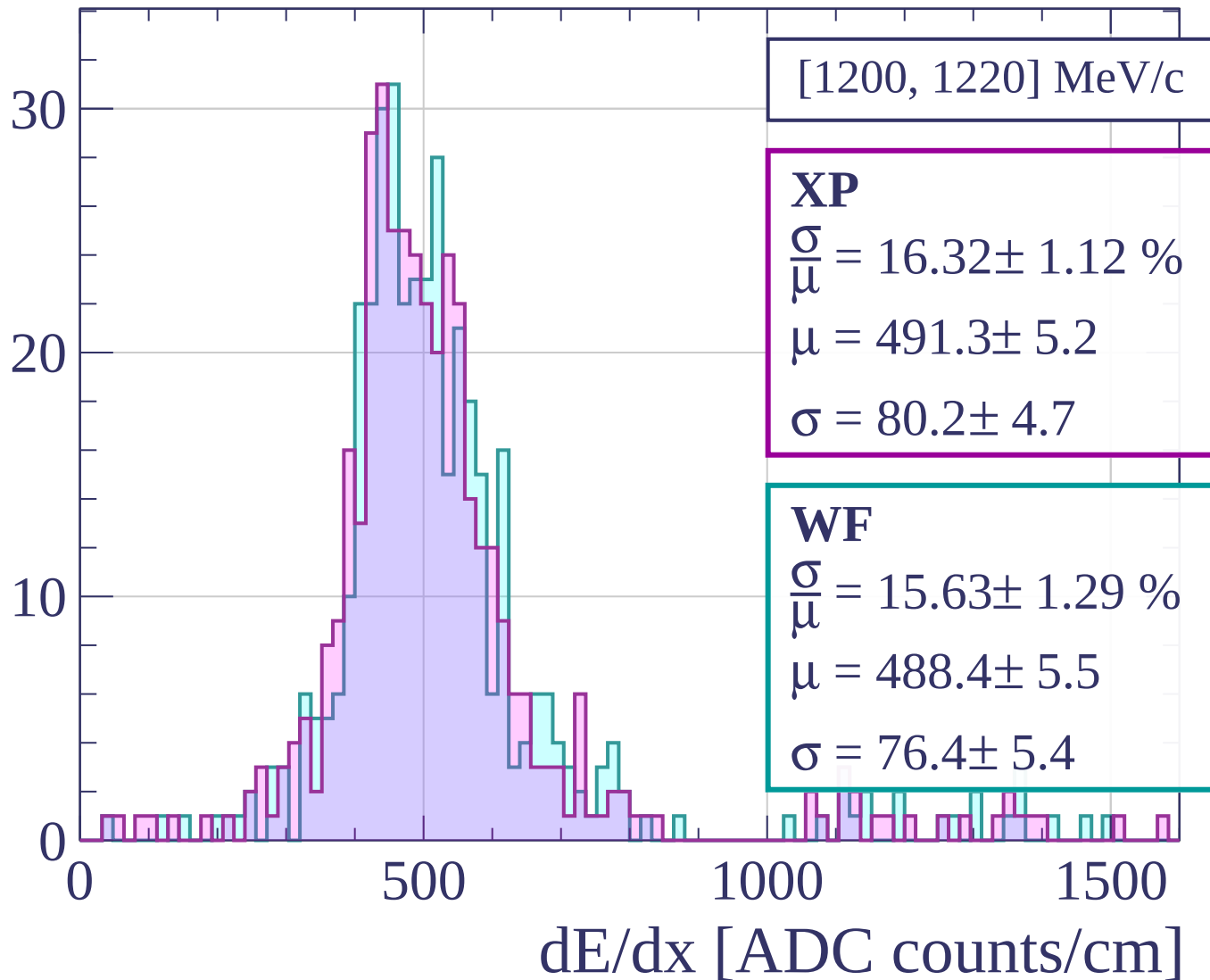
Count



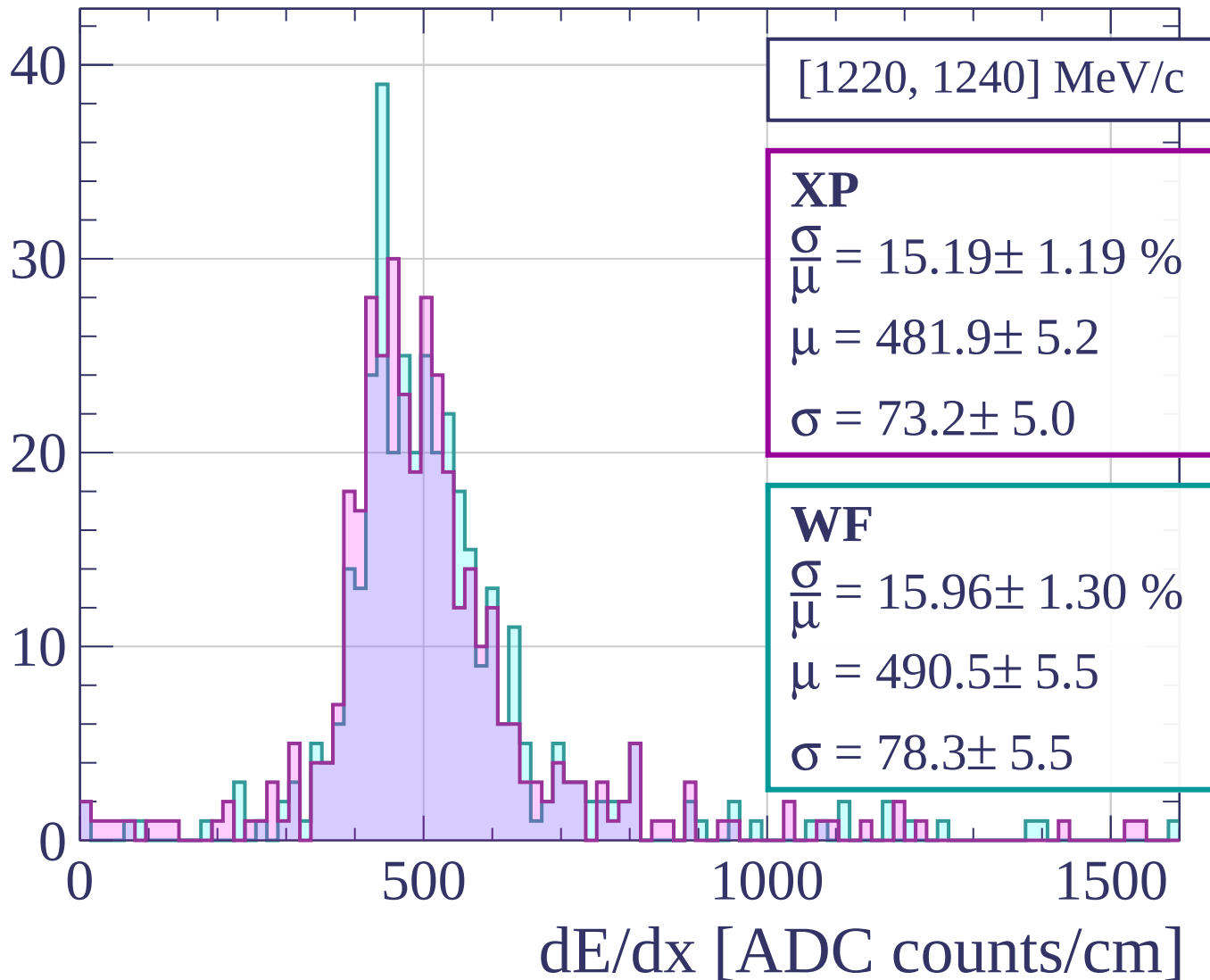
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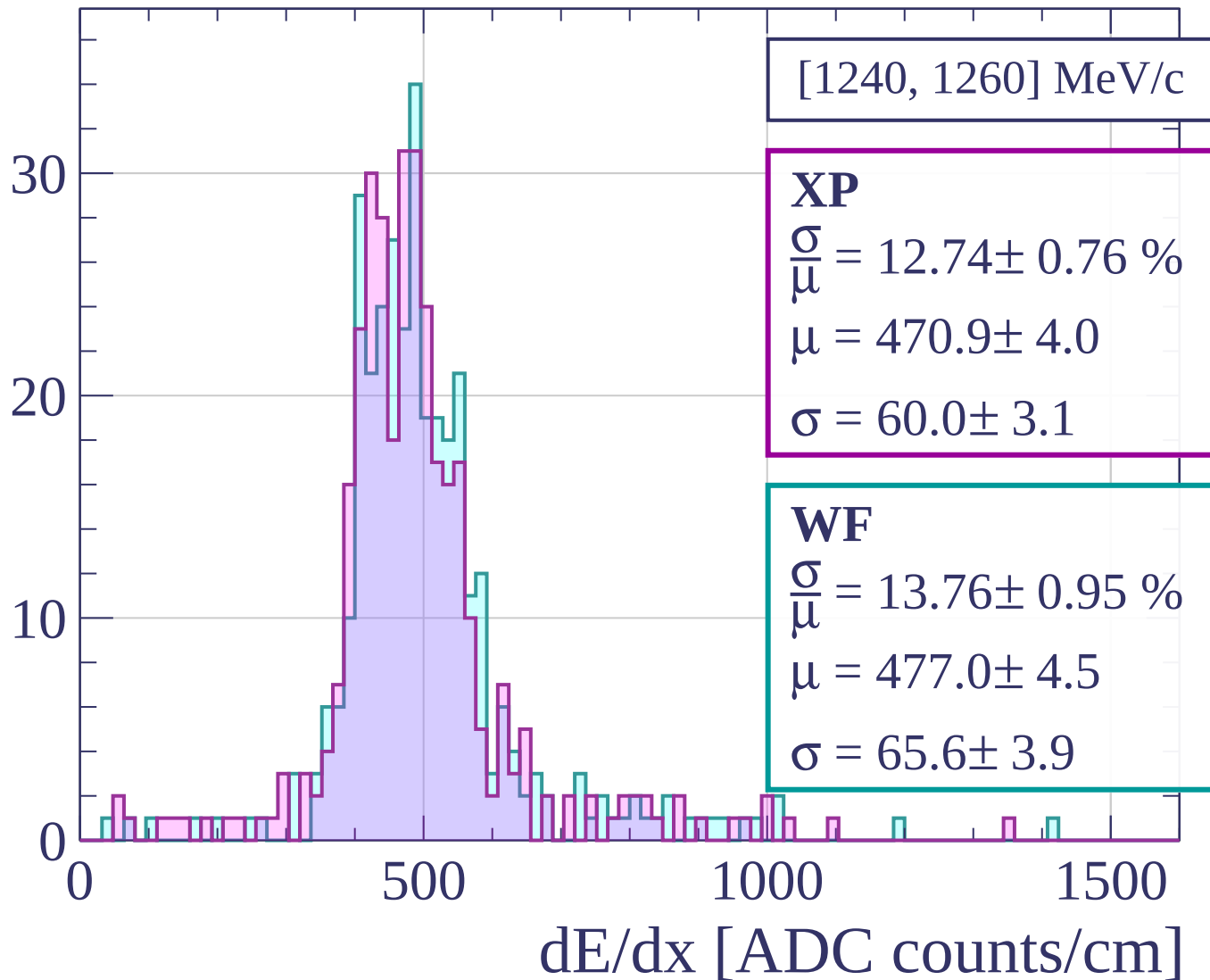
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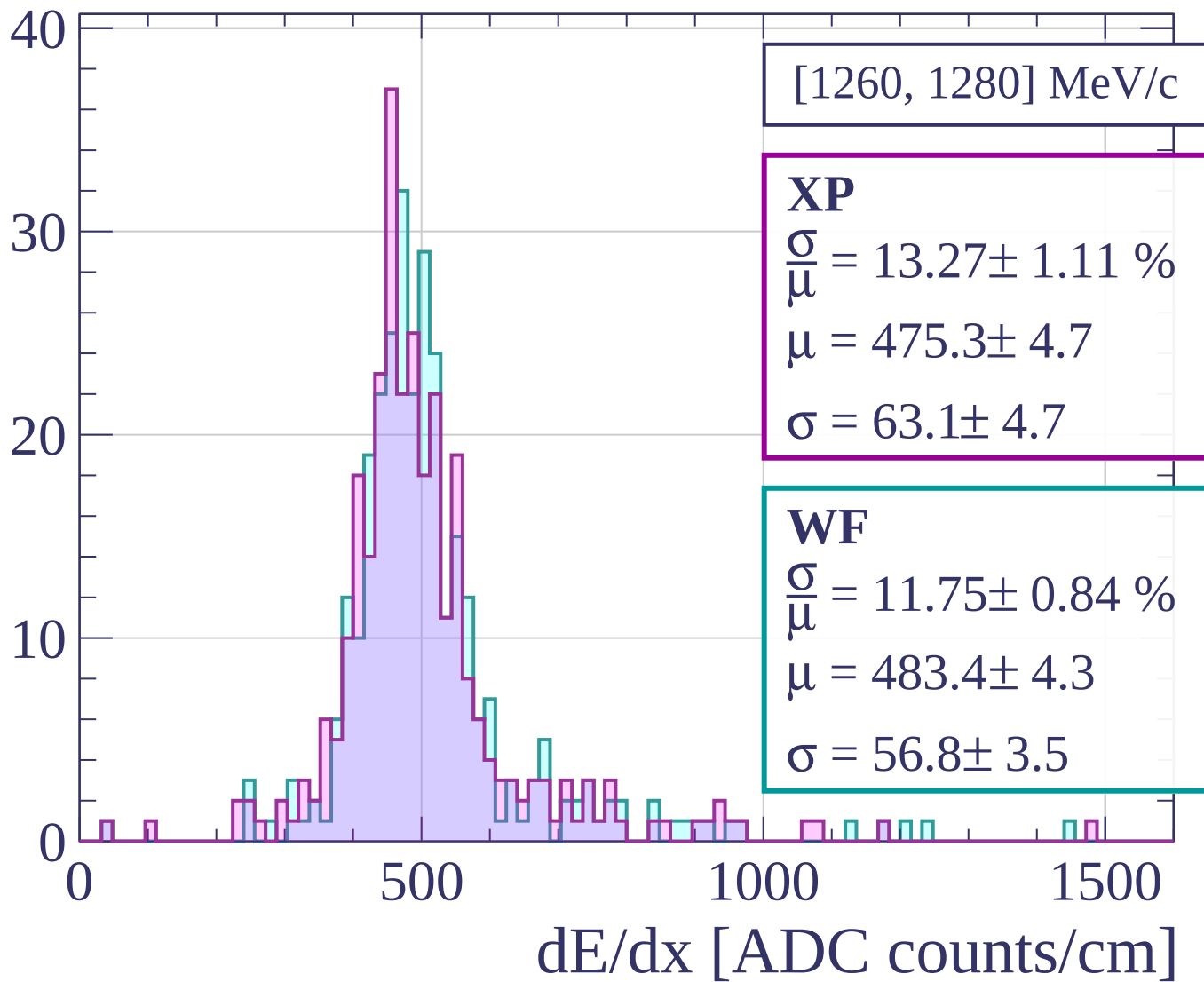
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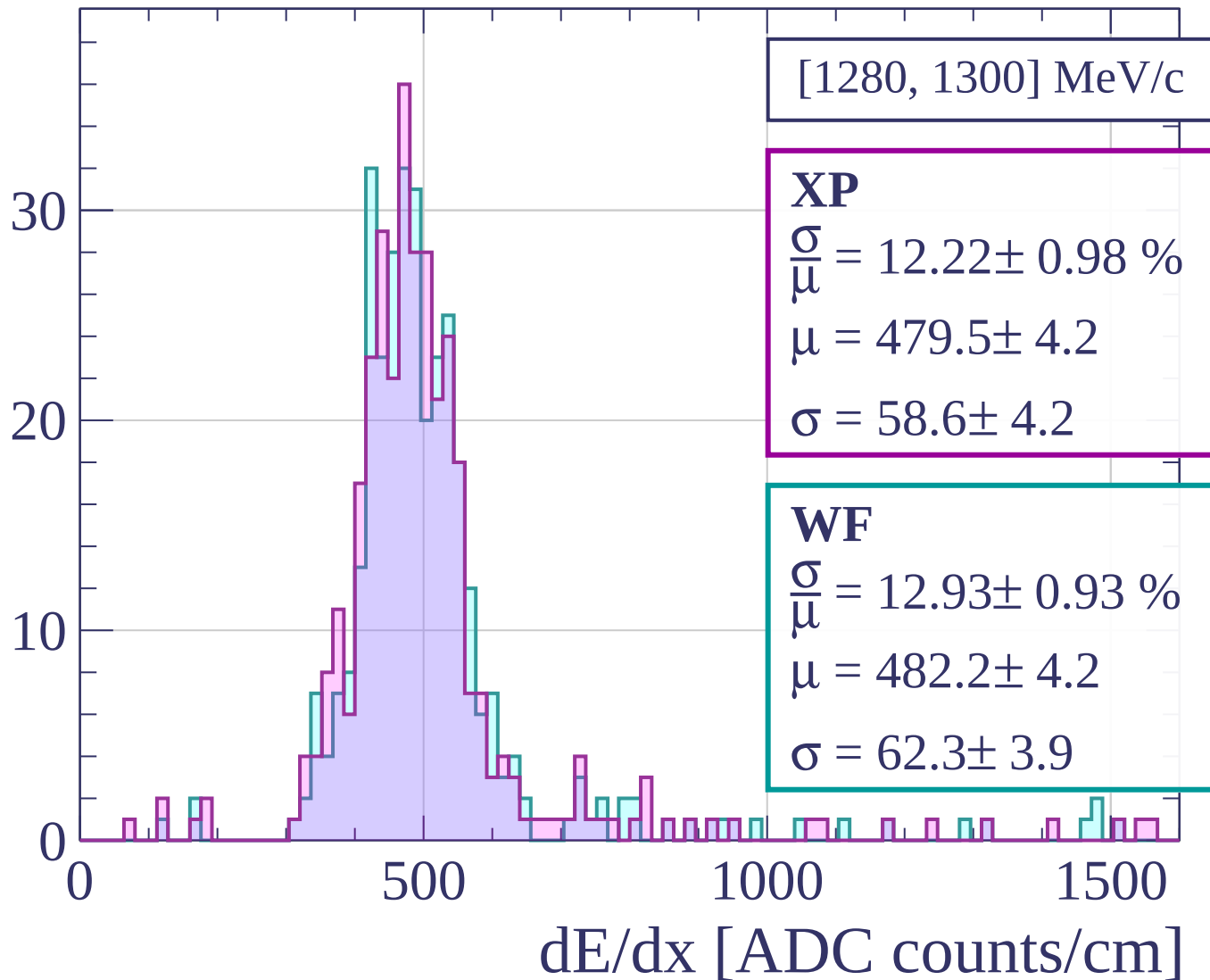
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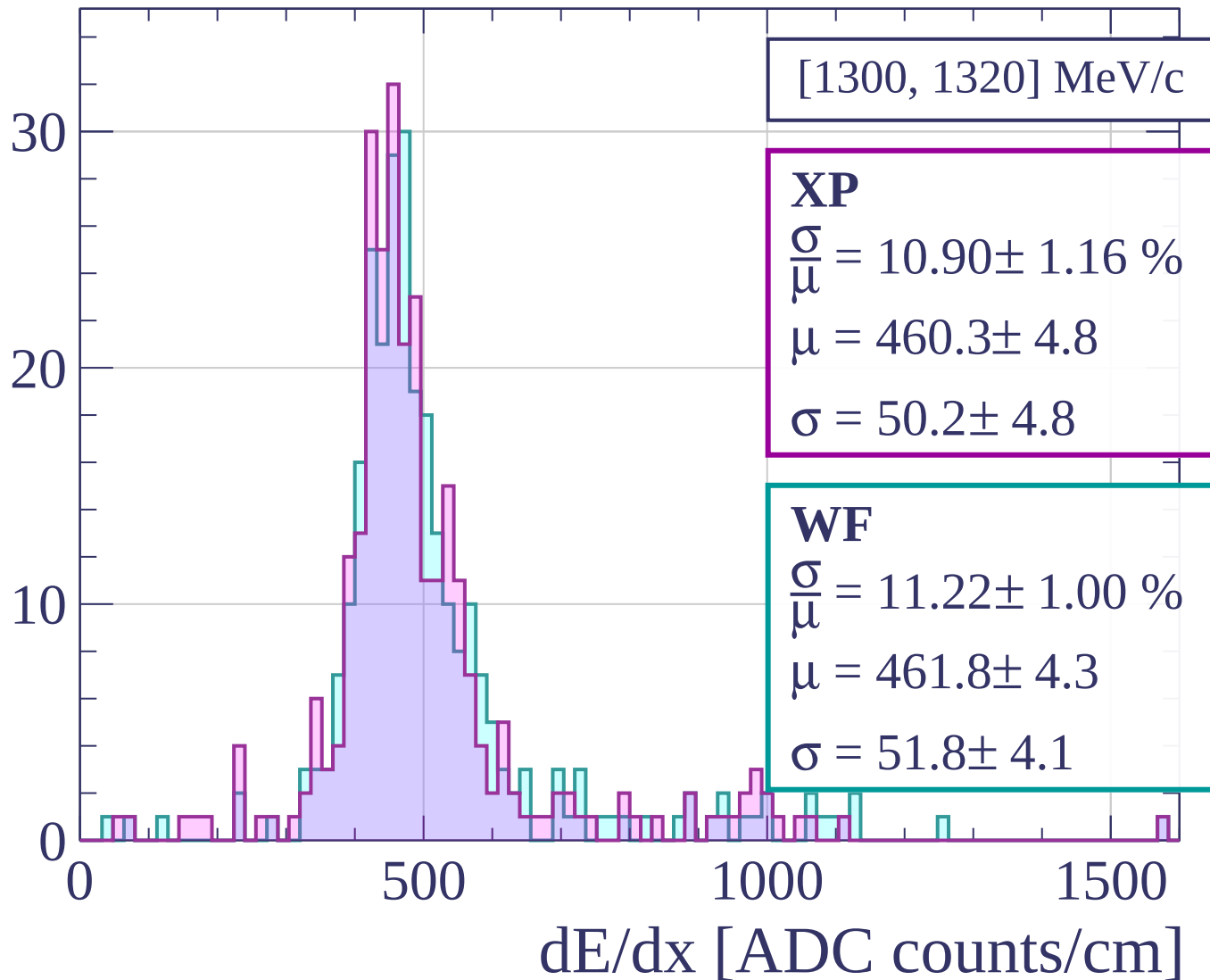
Count



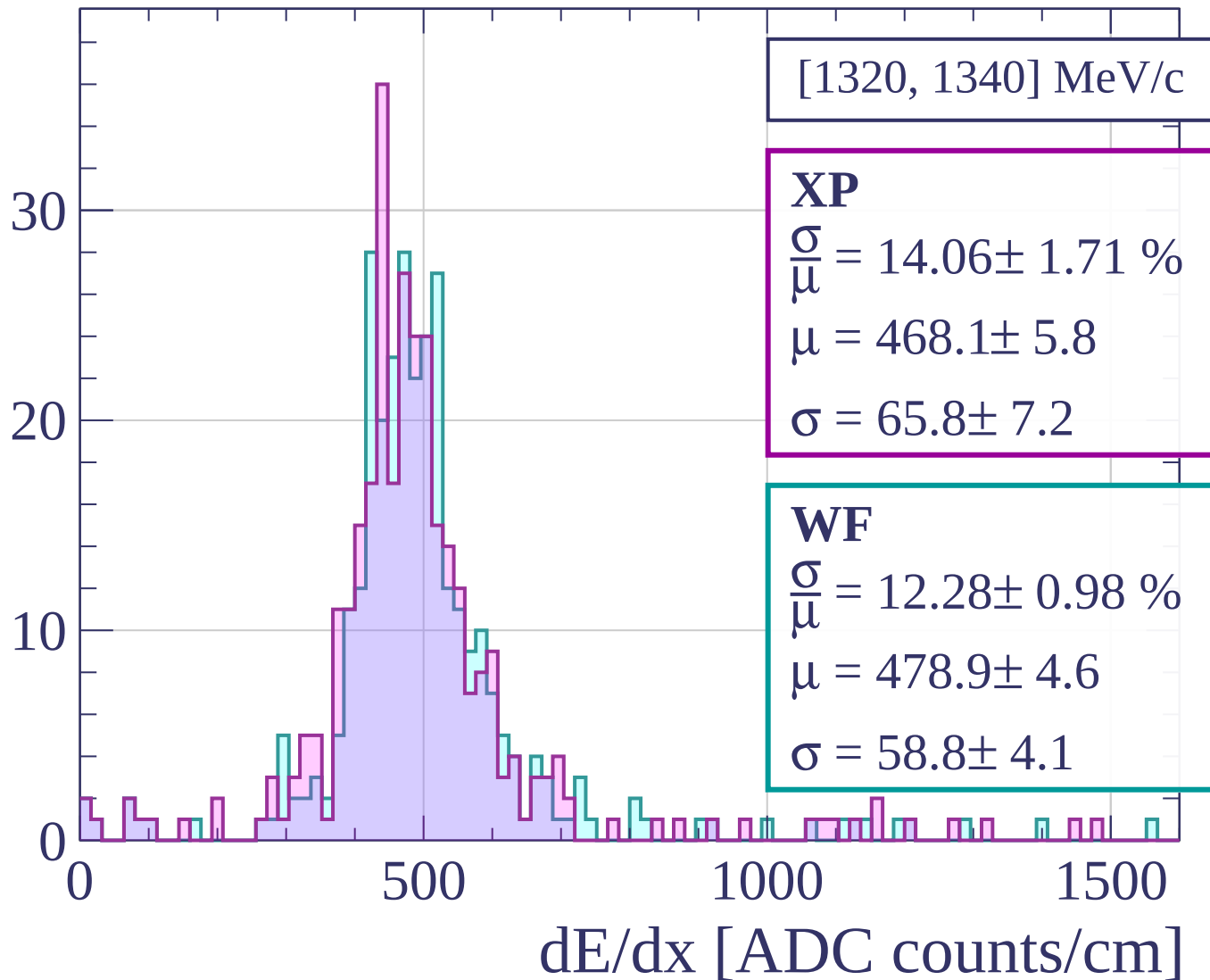
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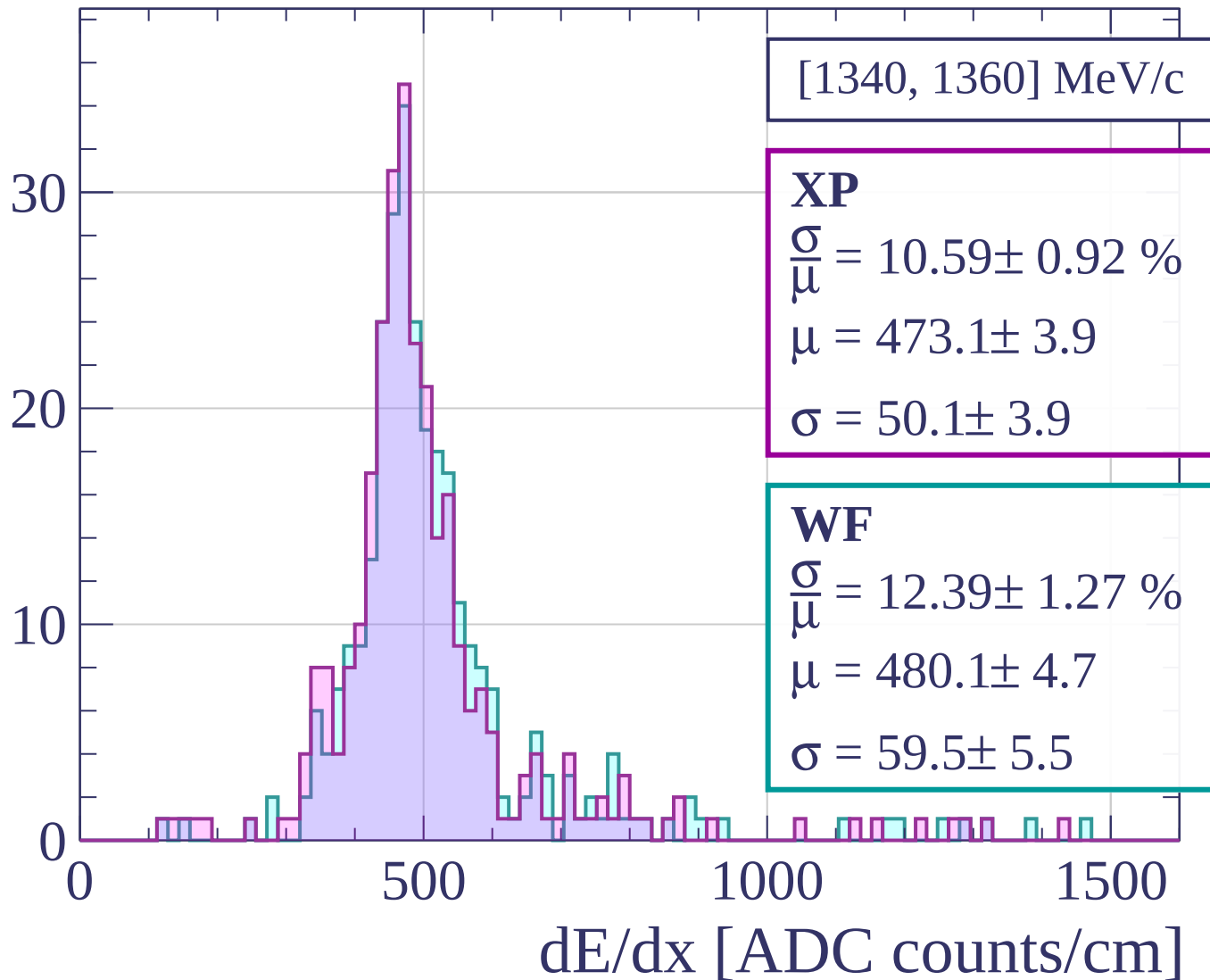
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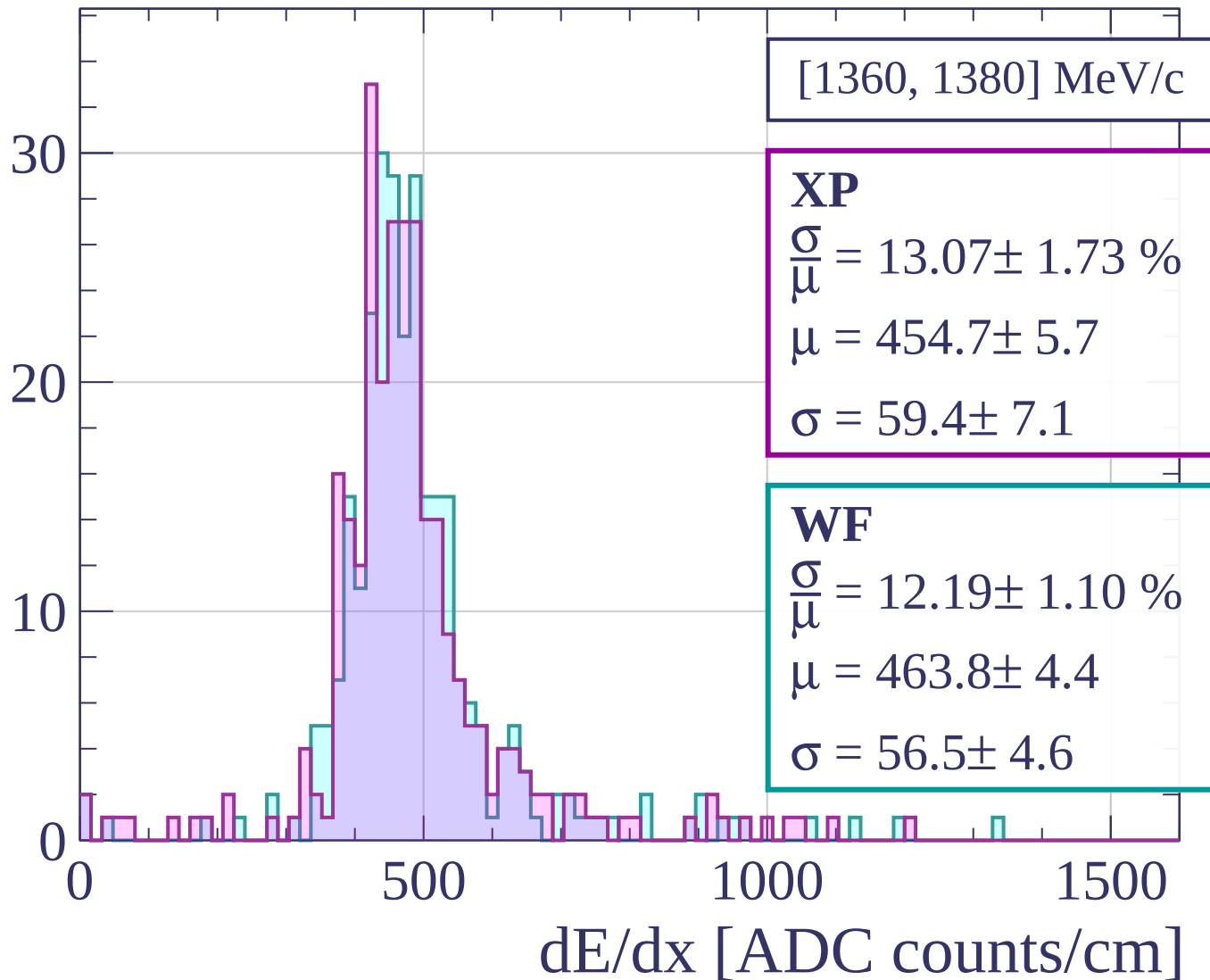
Count



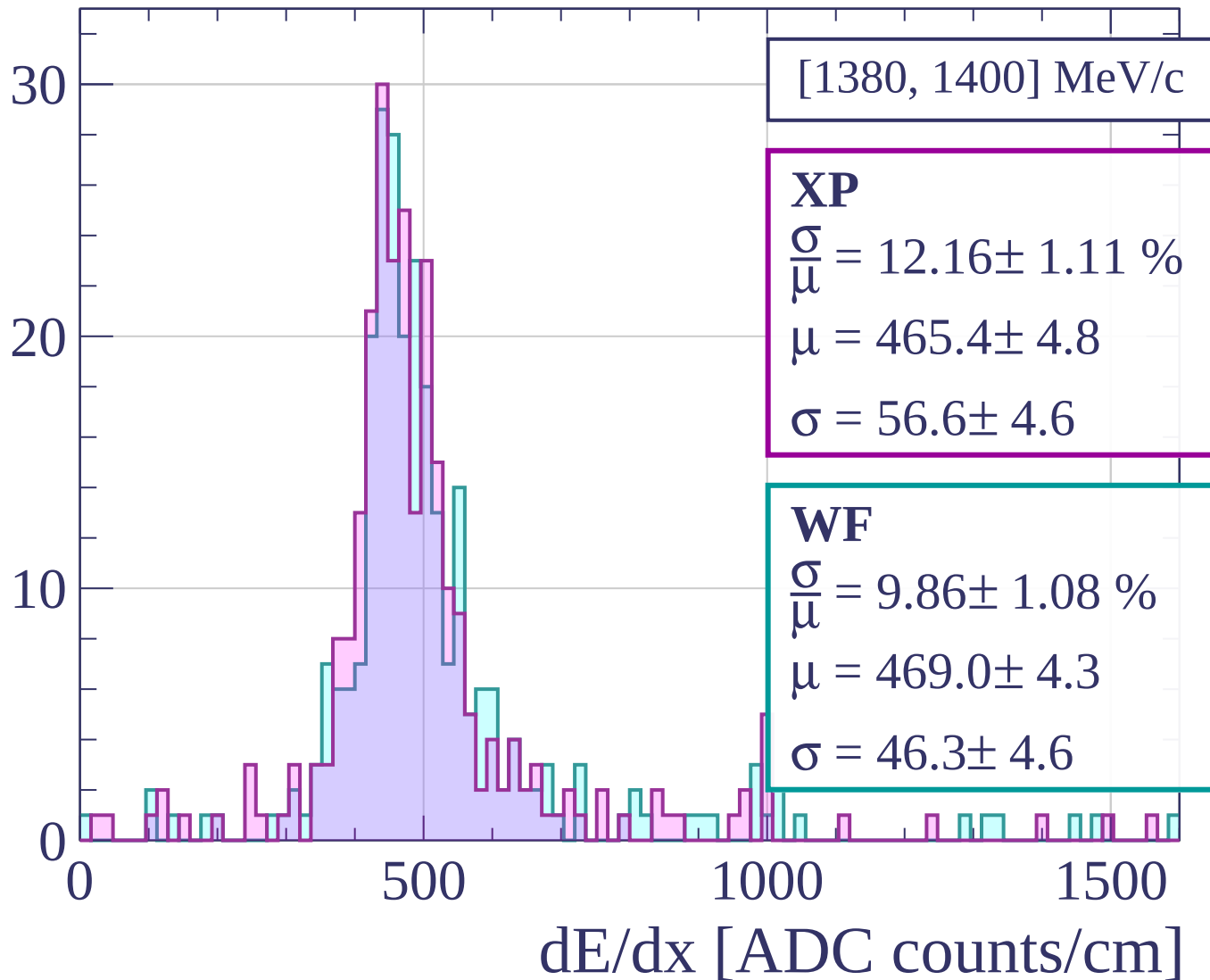
Count



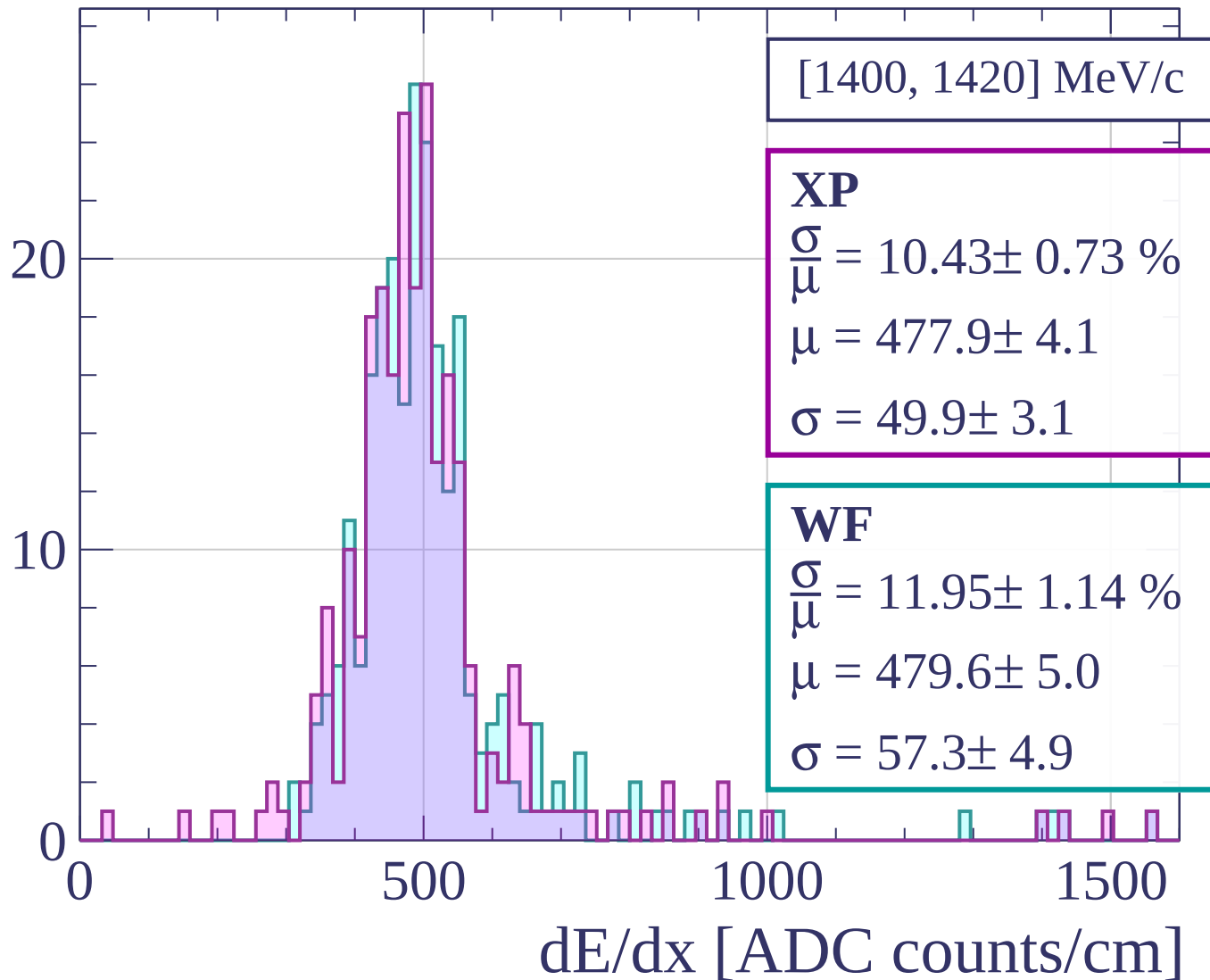
Count



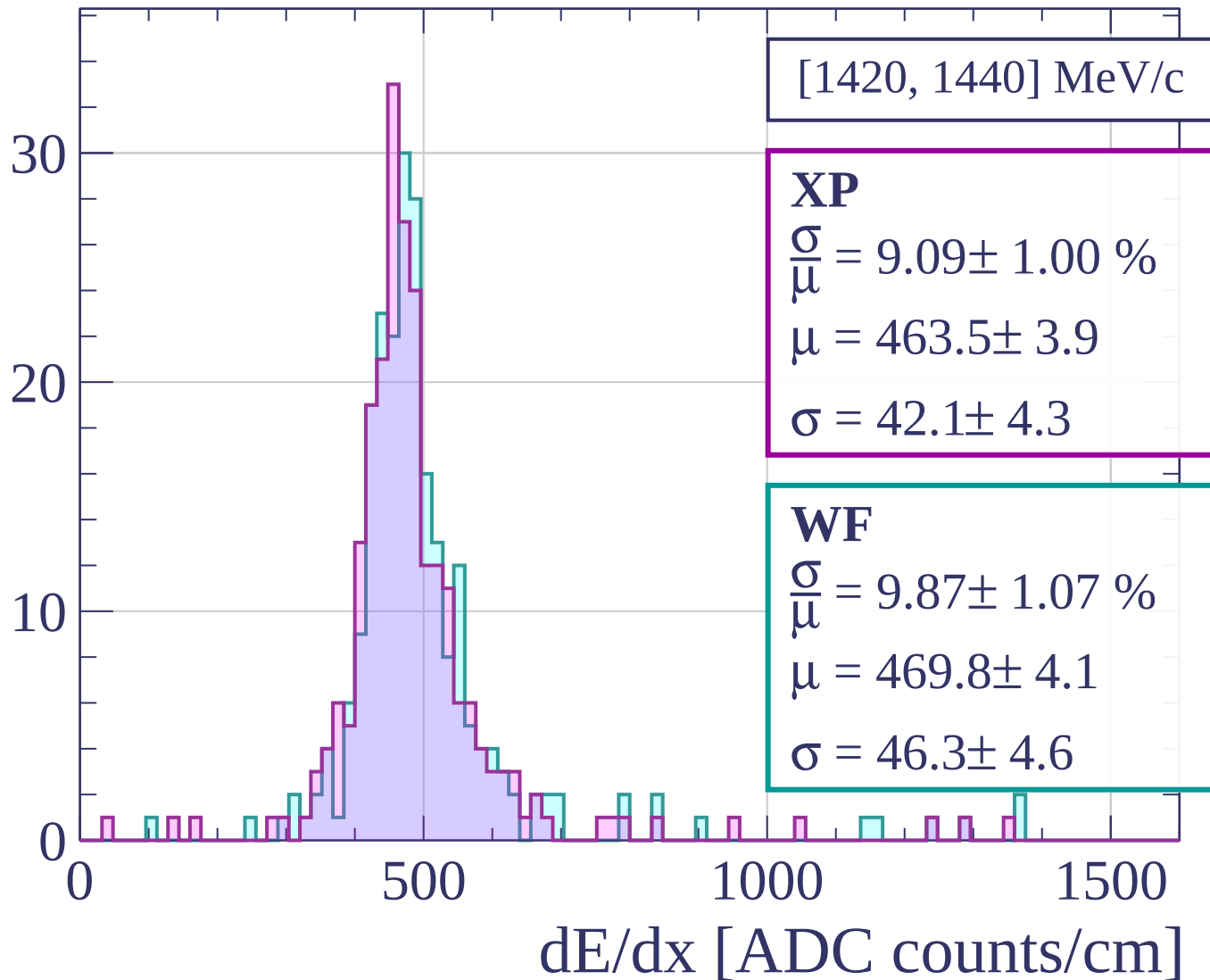
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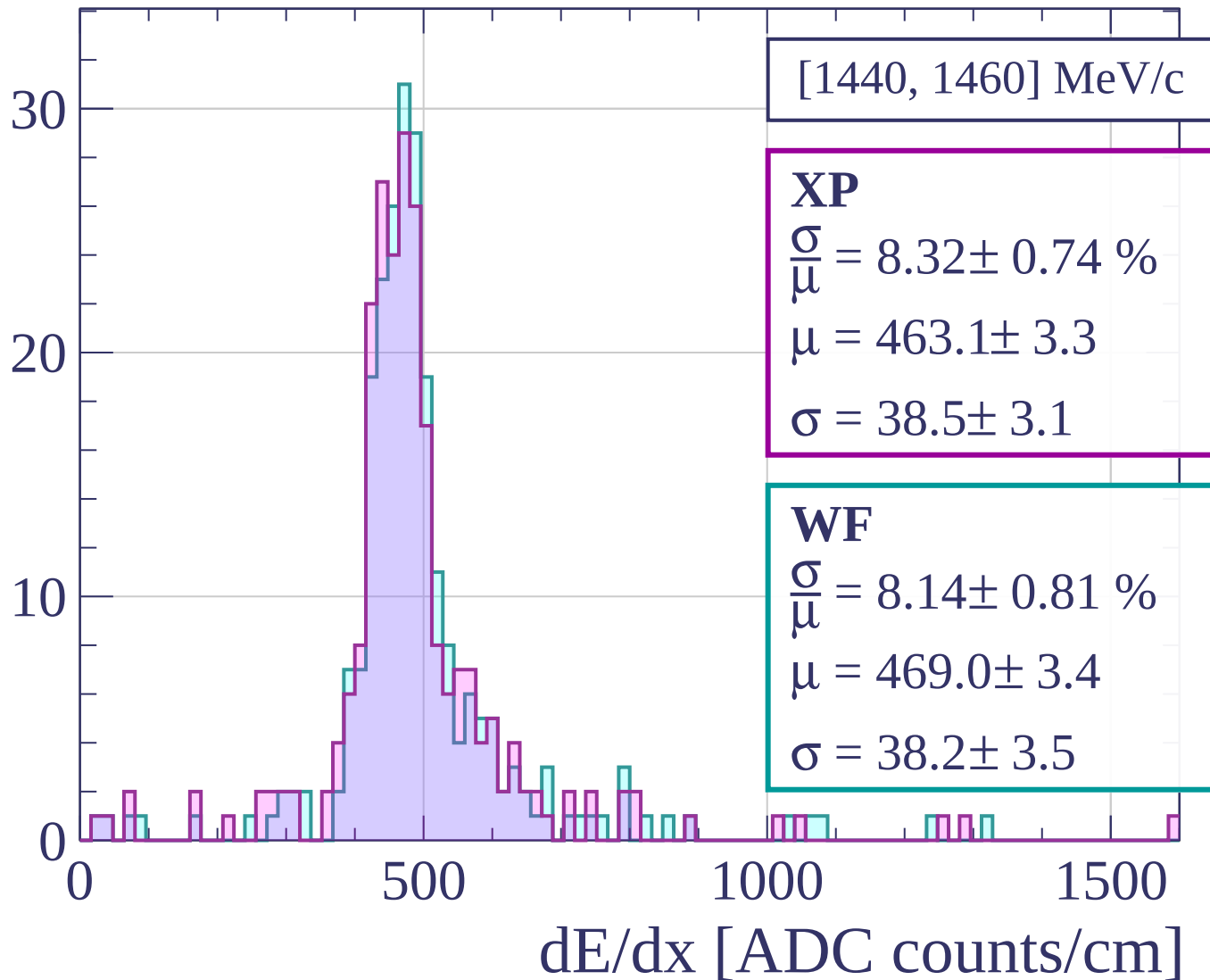
Count



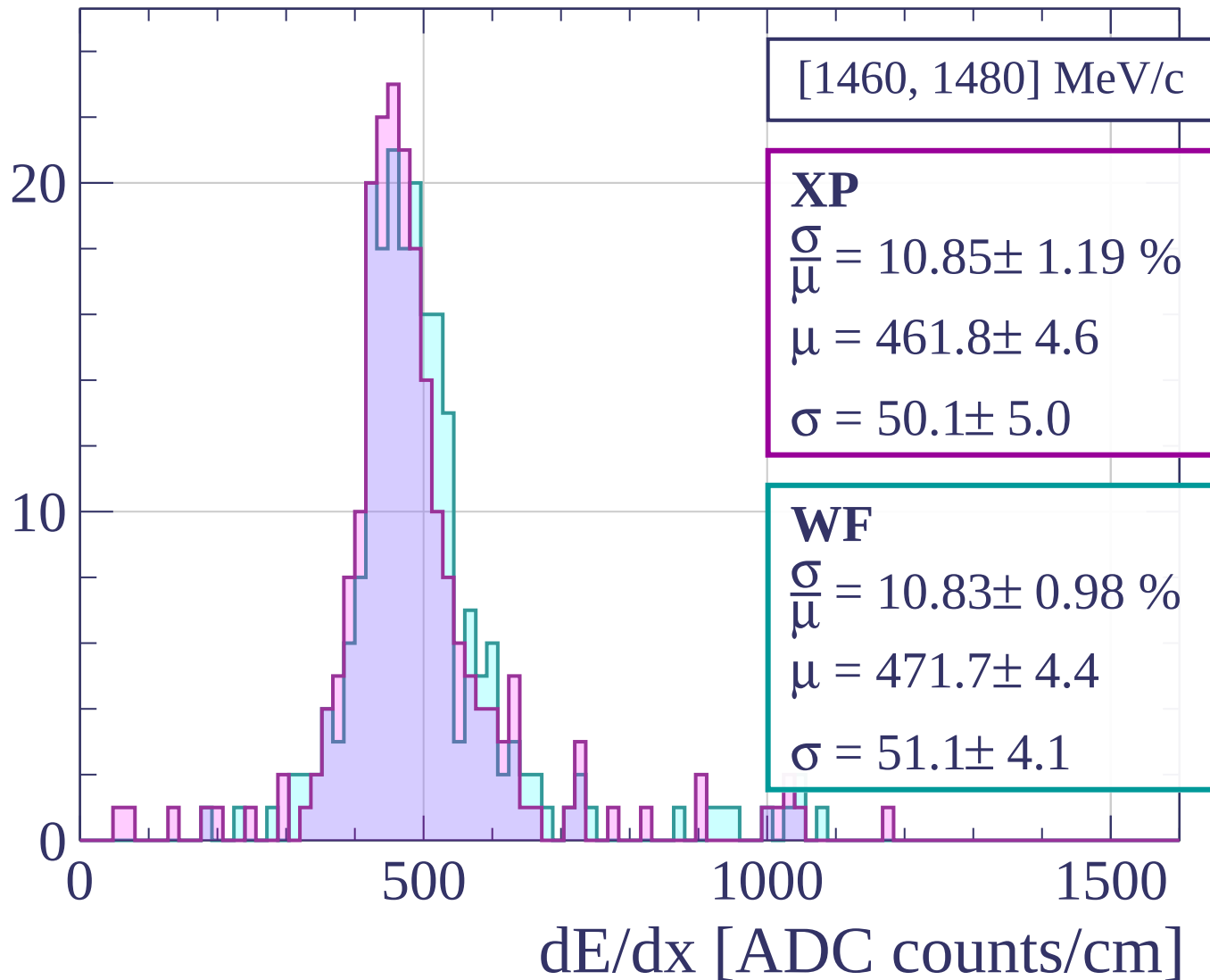
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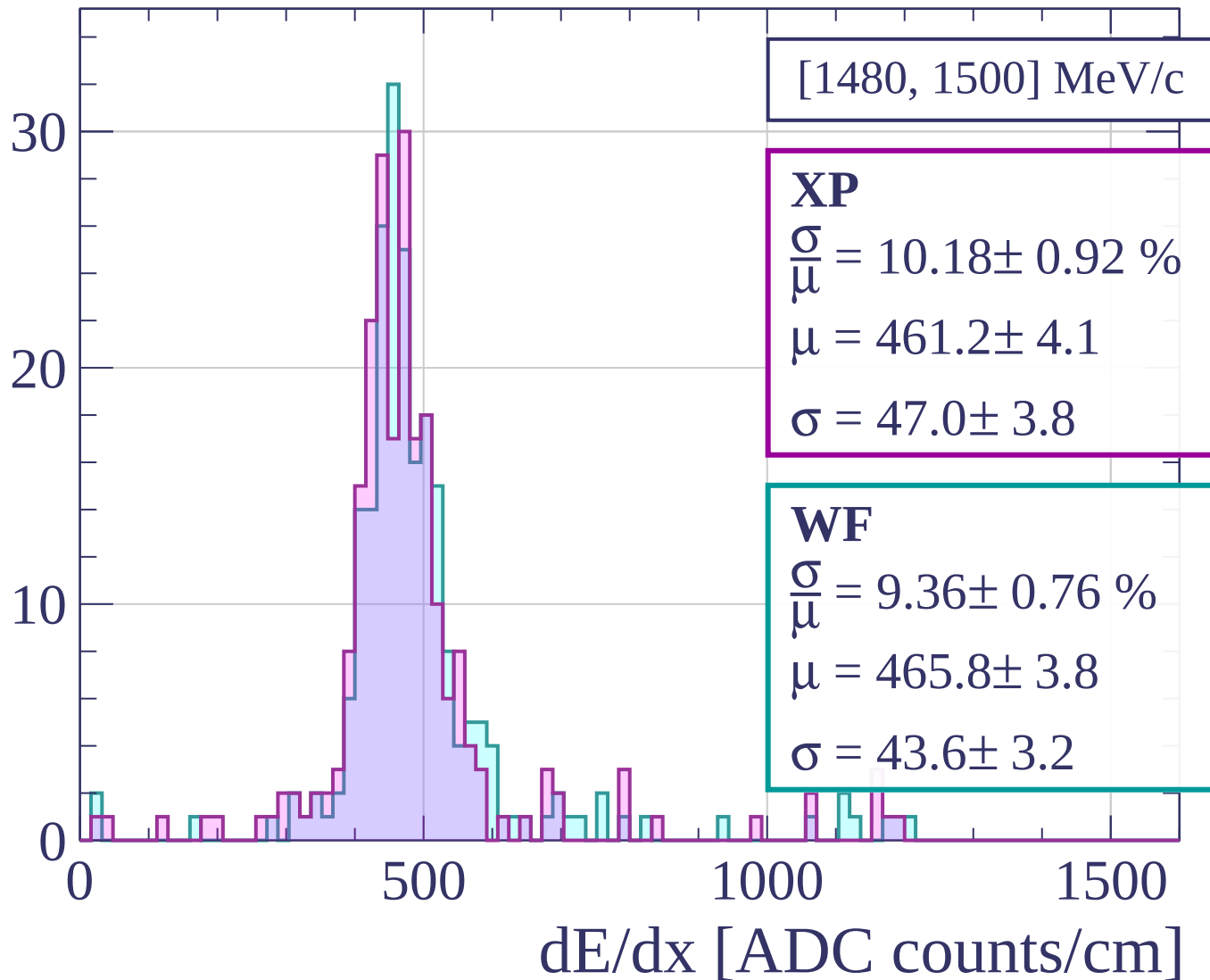
Count



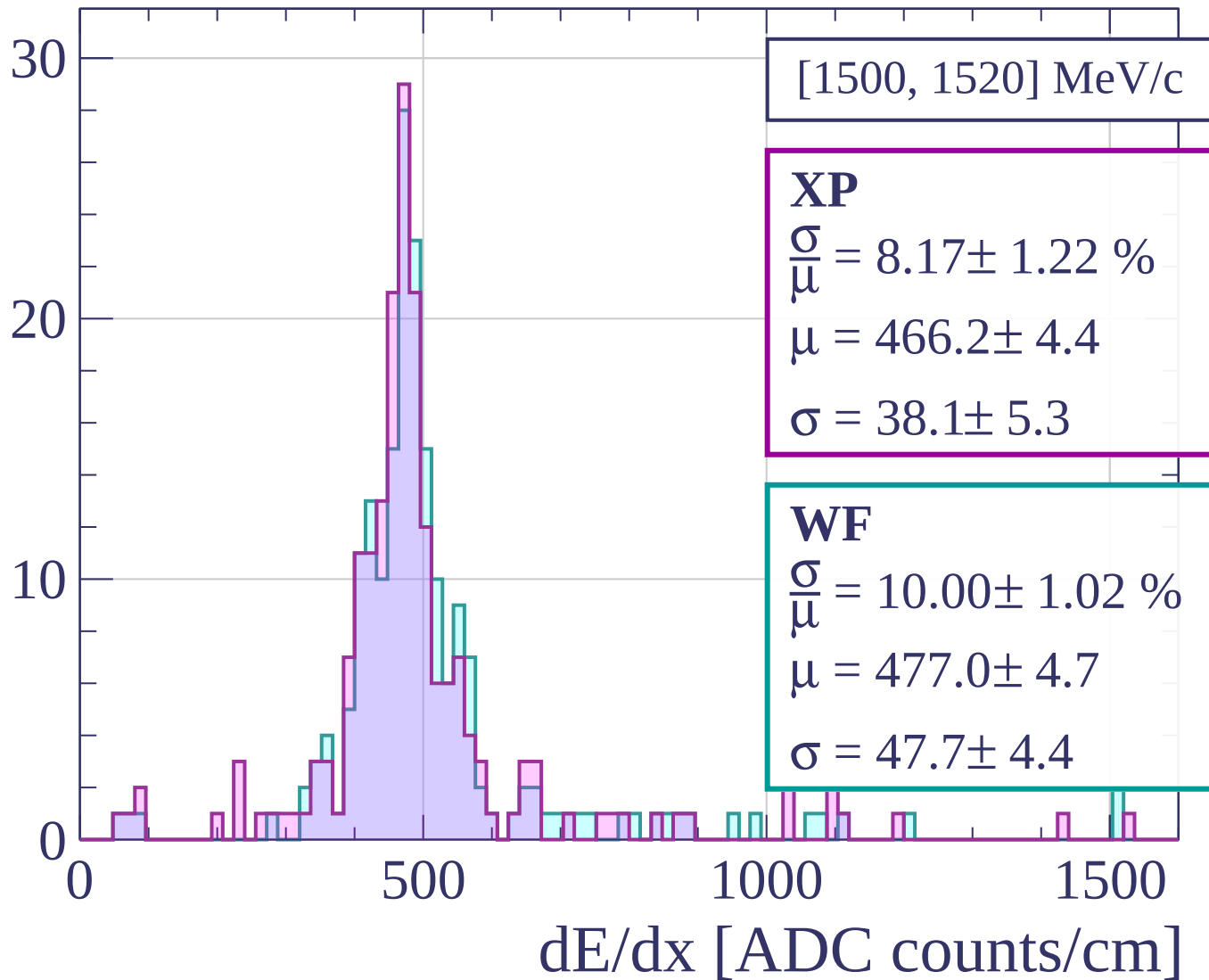
Count



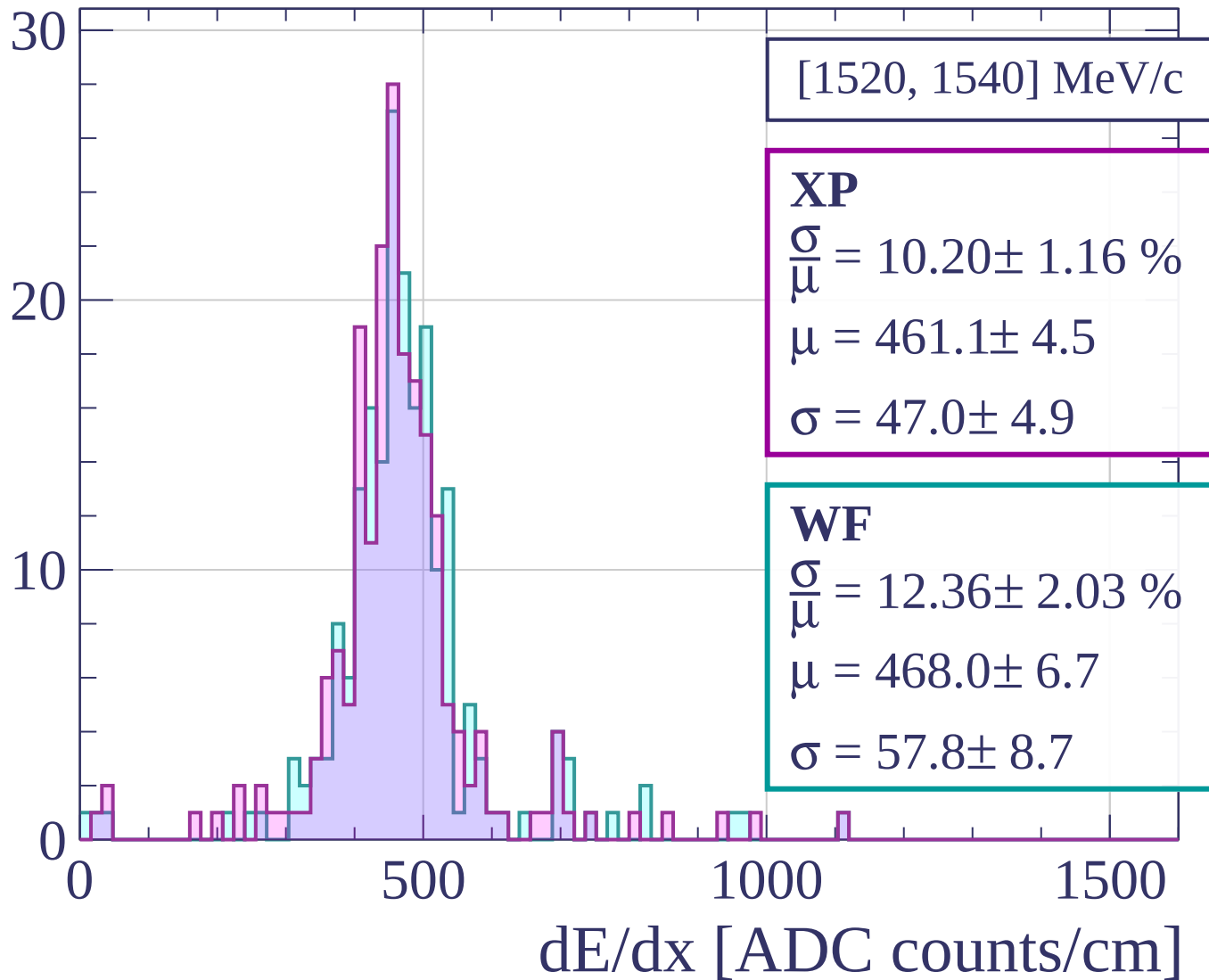
Count



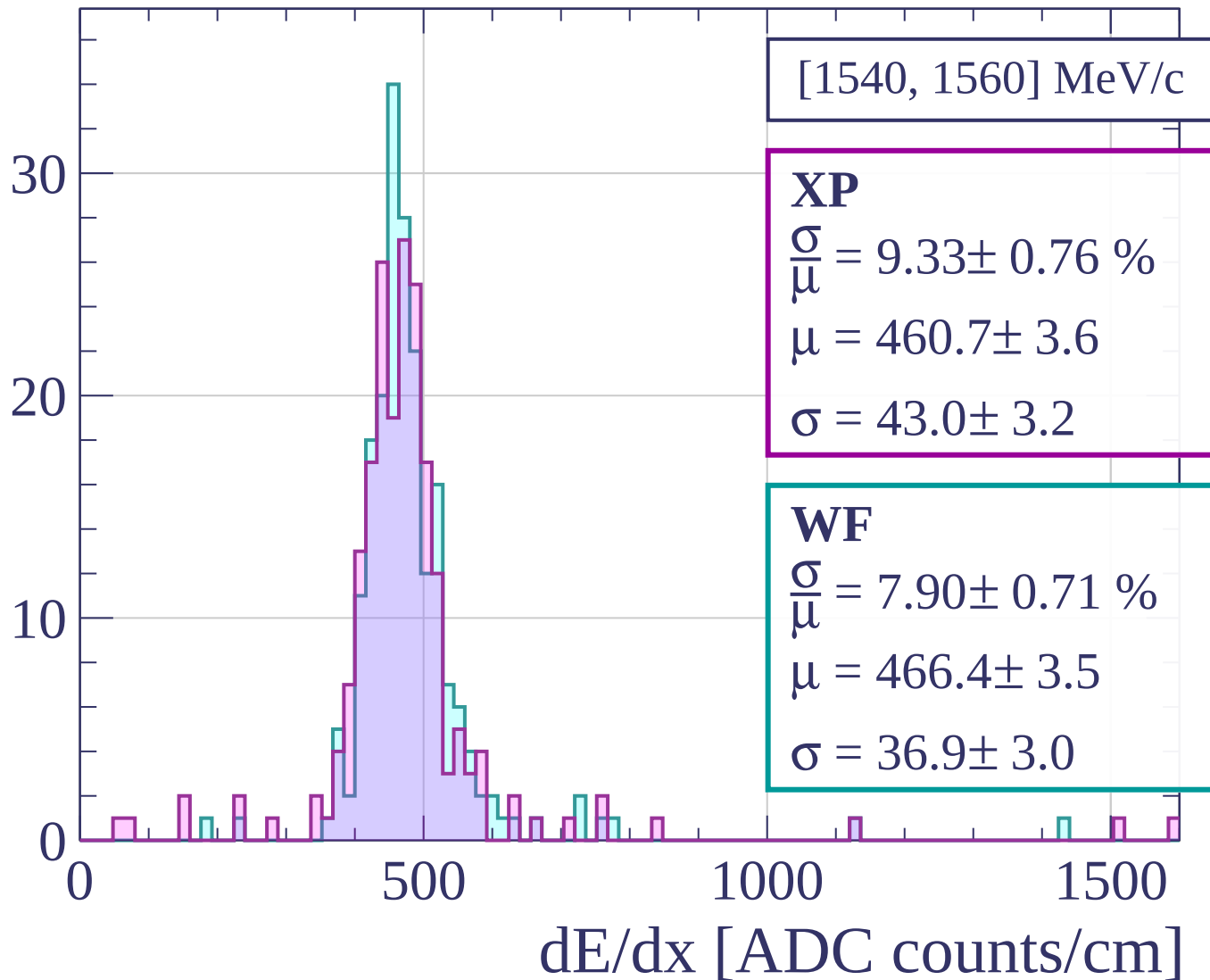
Count



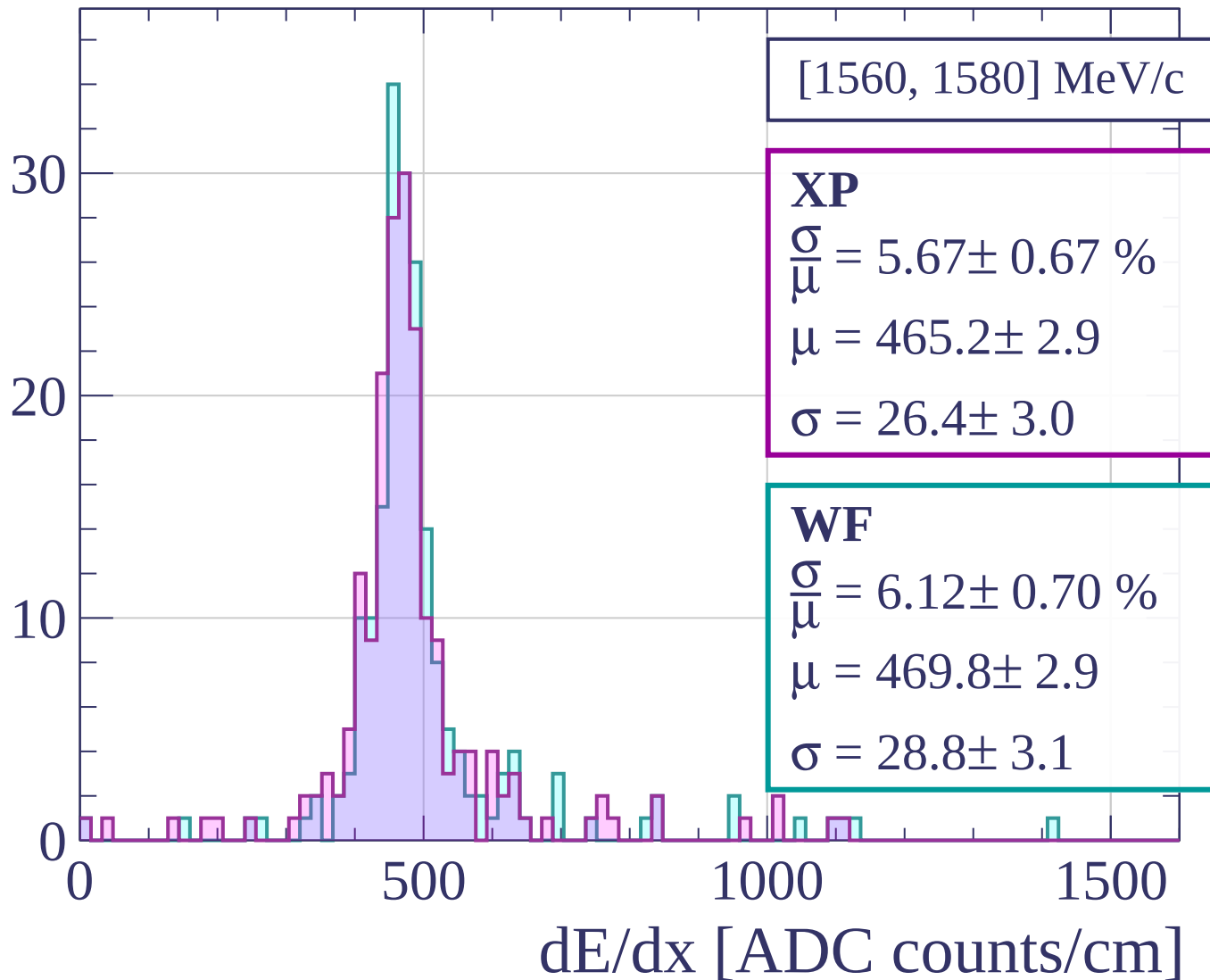
Count



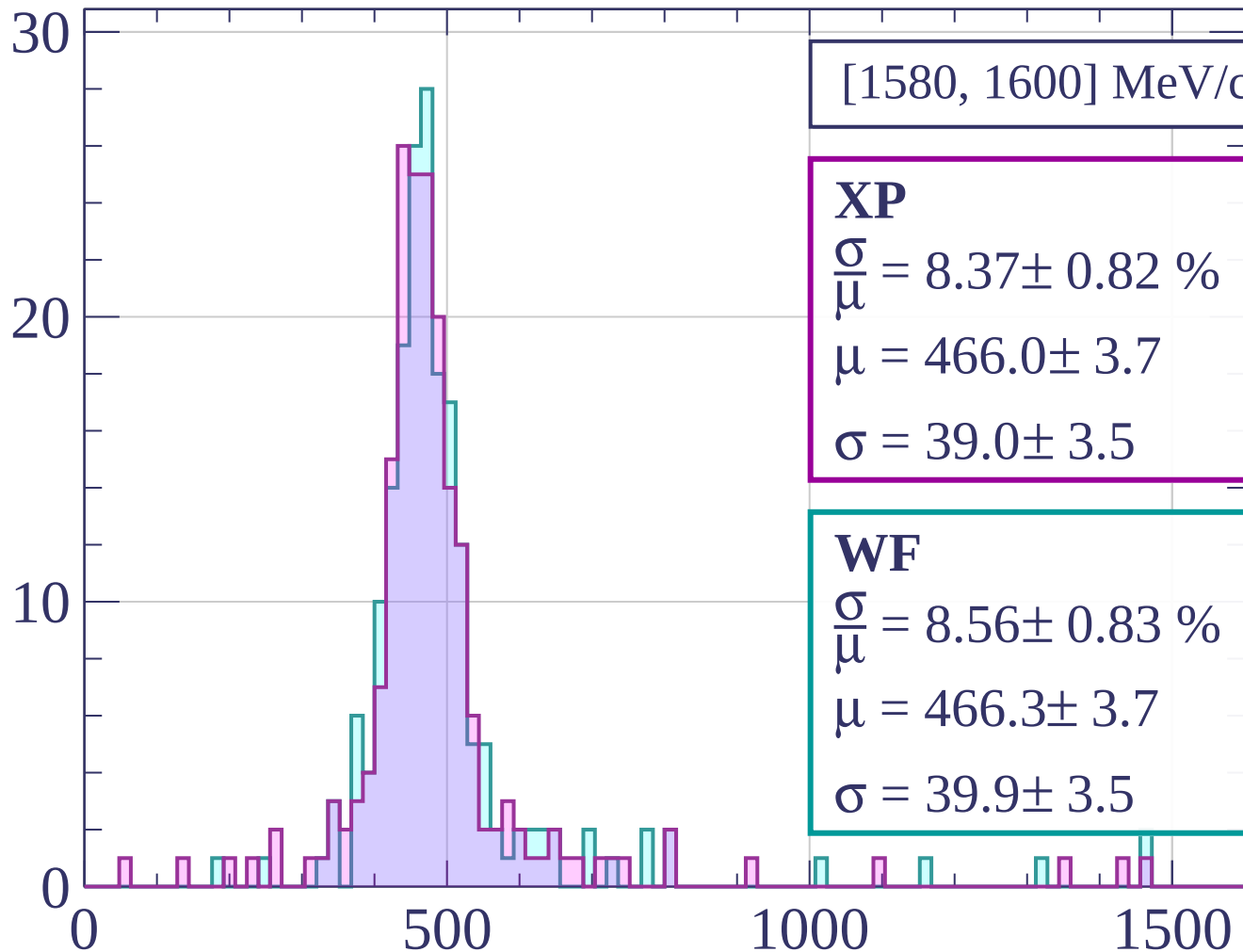
Count



Count

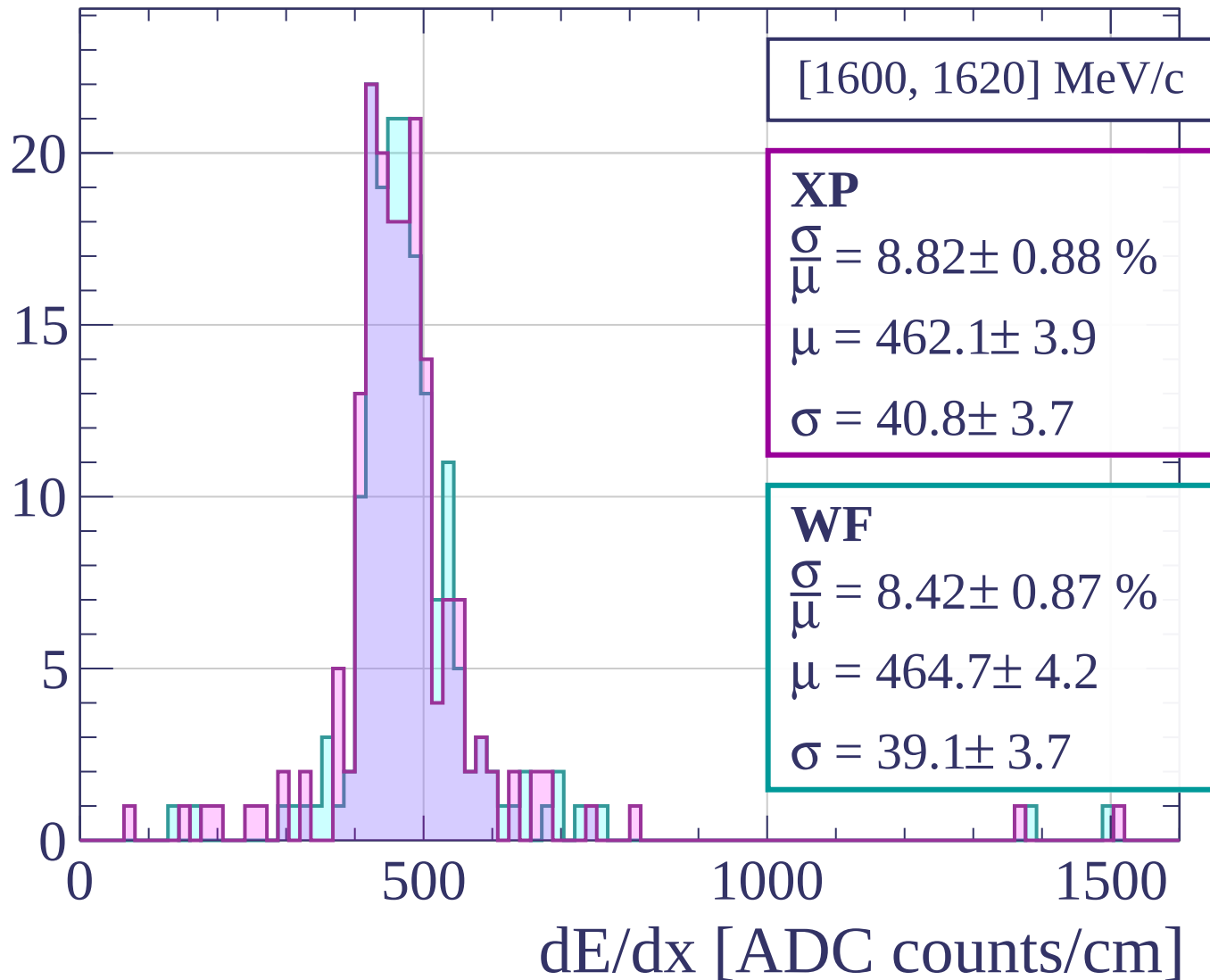


Count

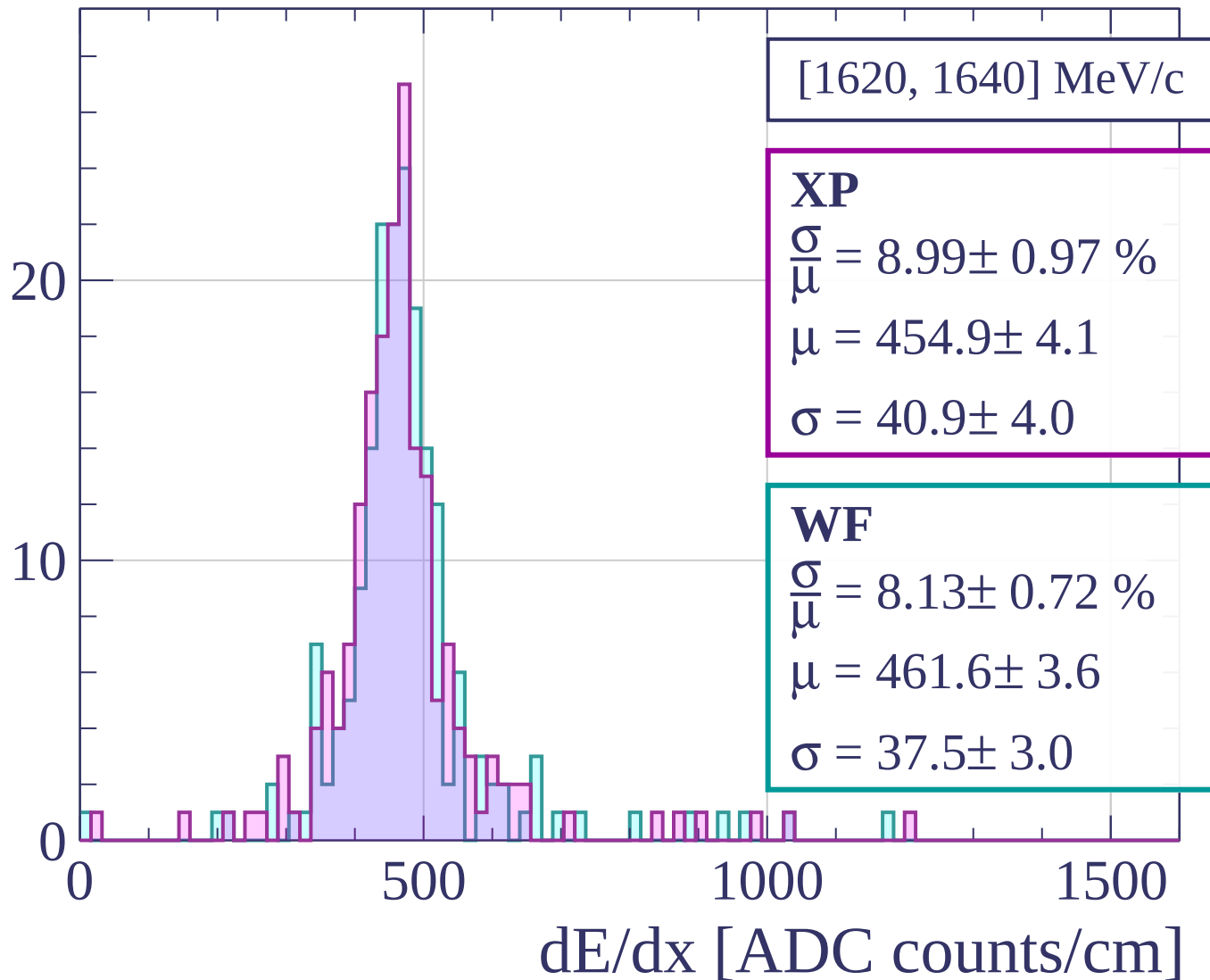


dE/dx [ADC counts/cm]

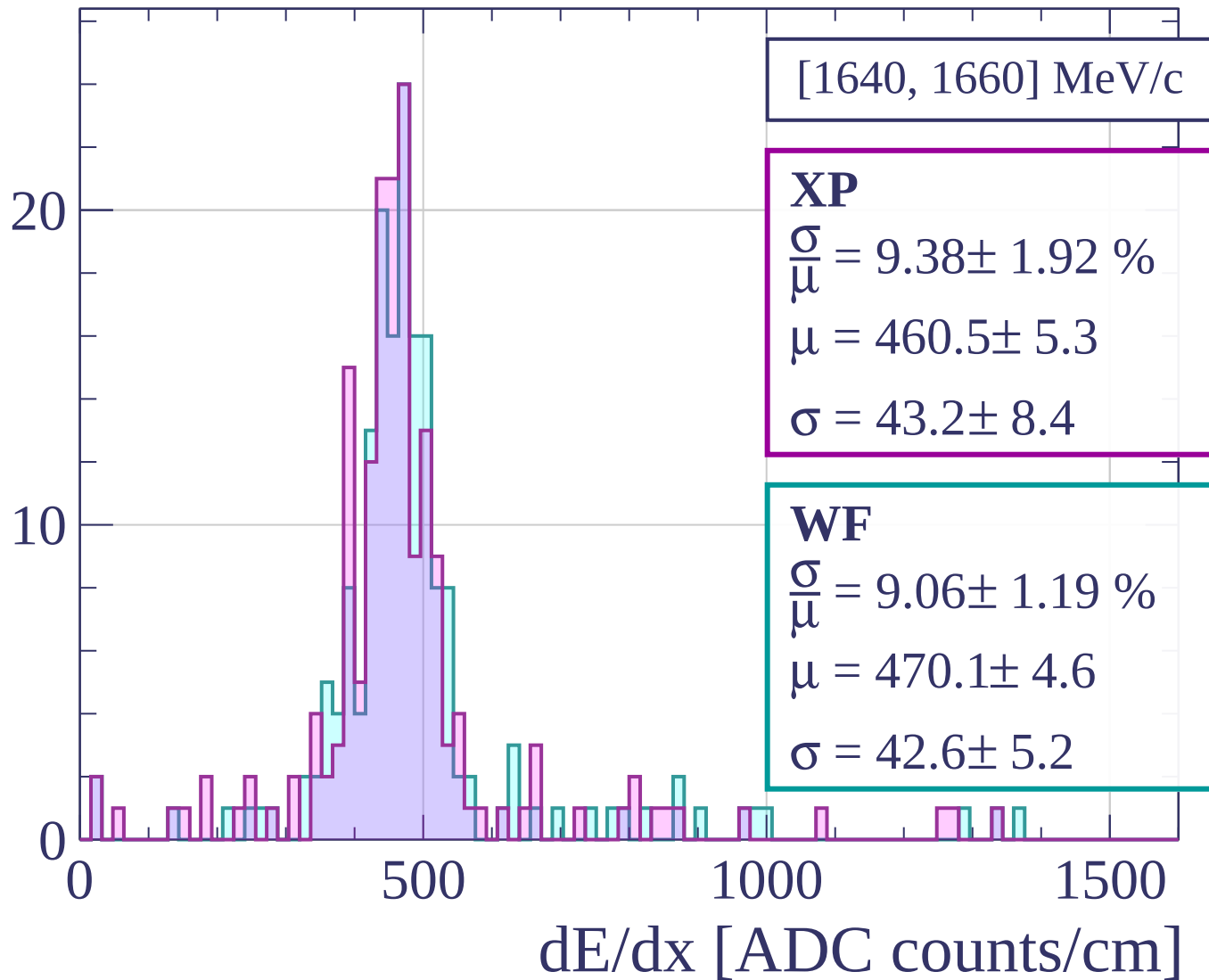
Count



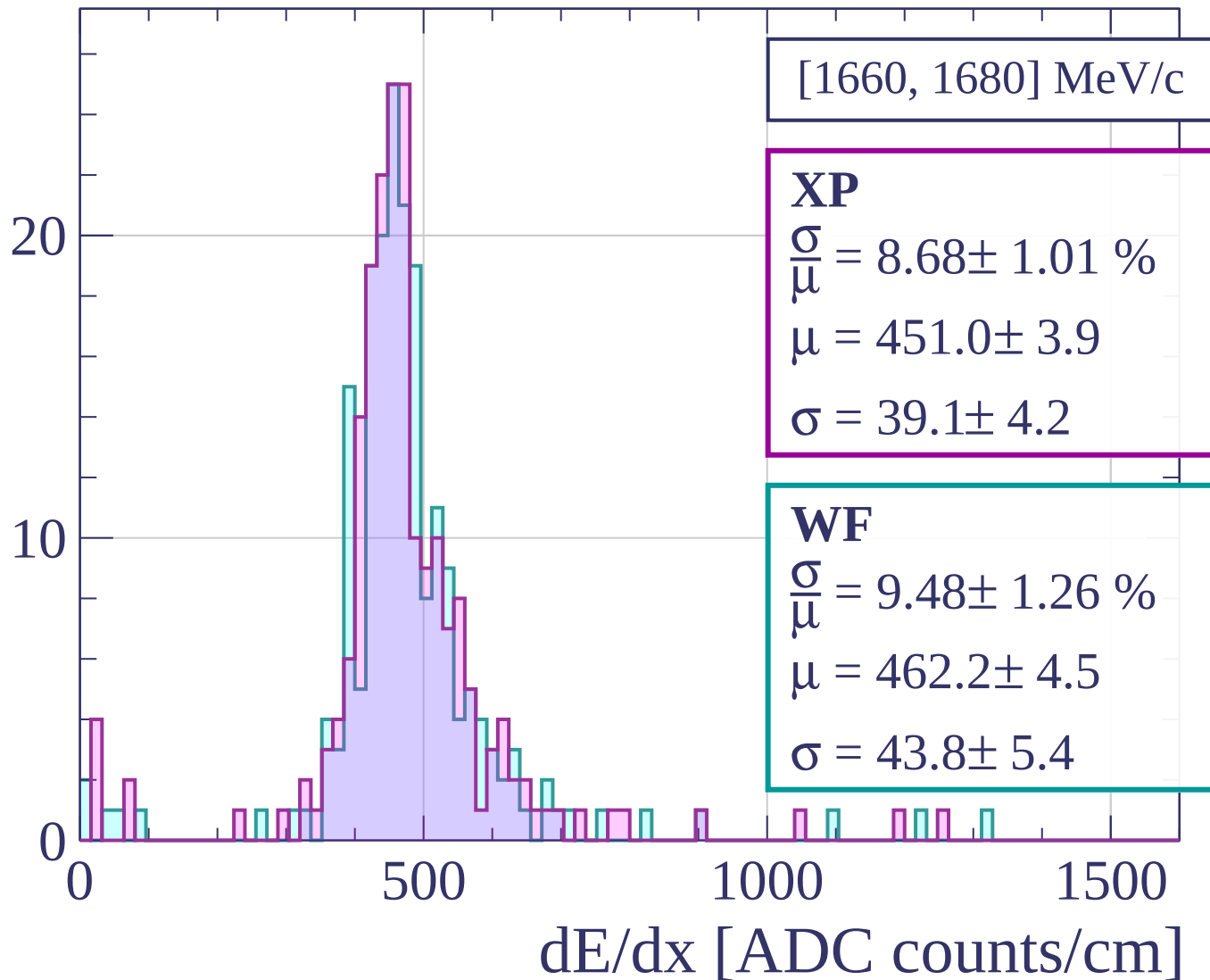
Count



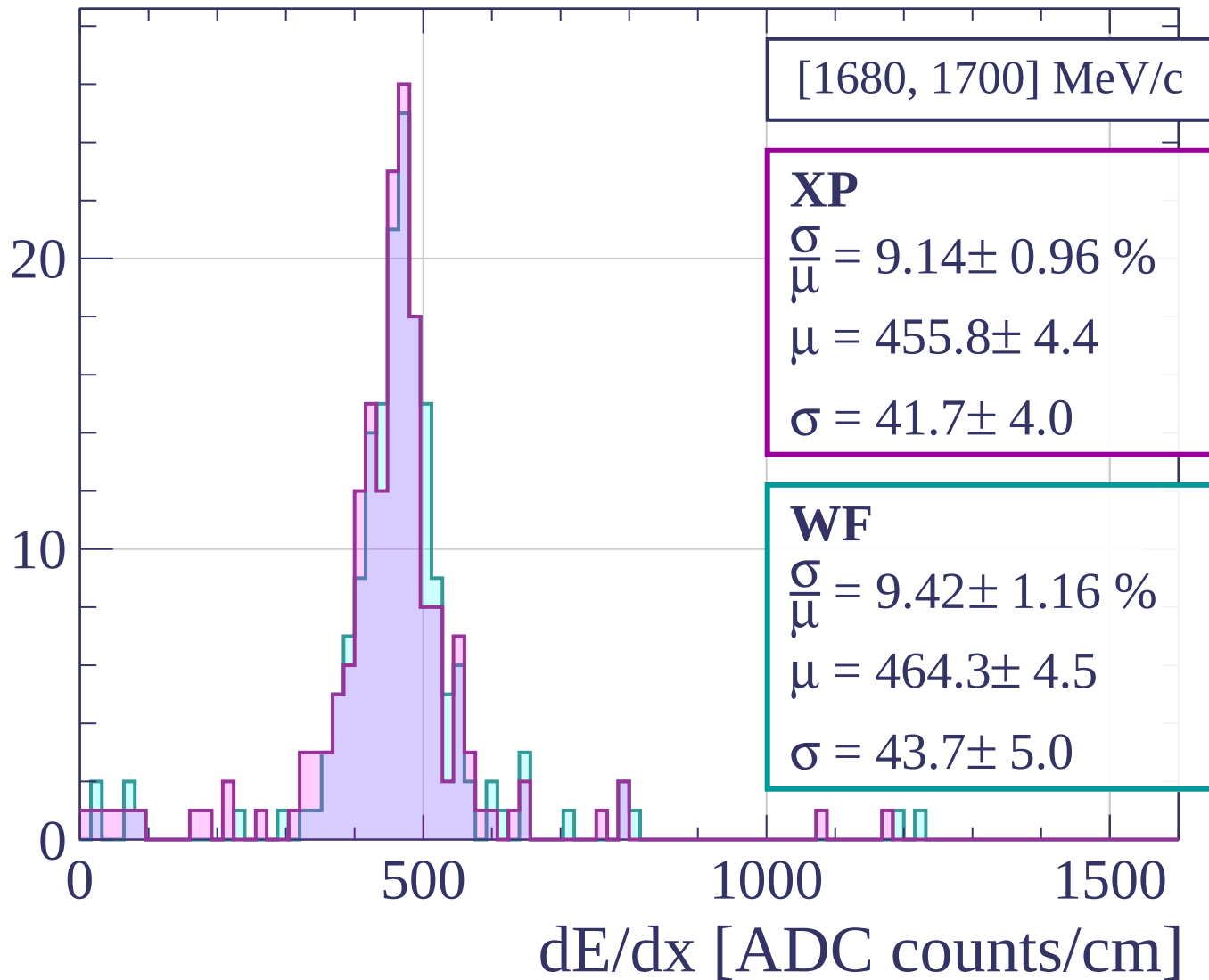
Count



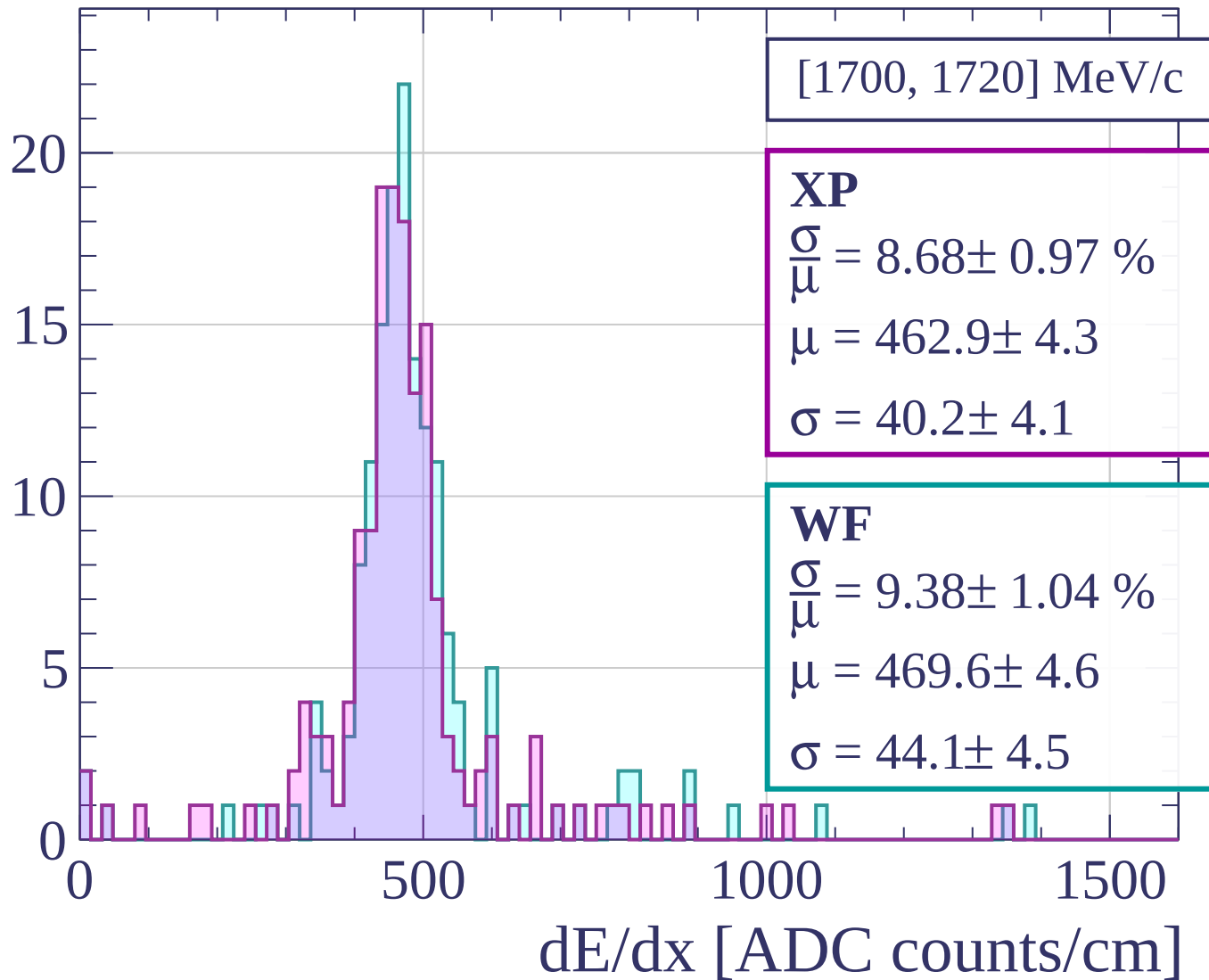
Count



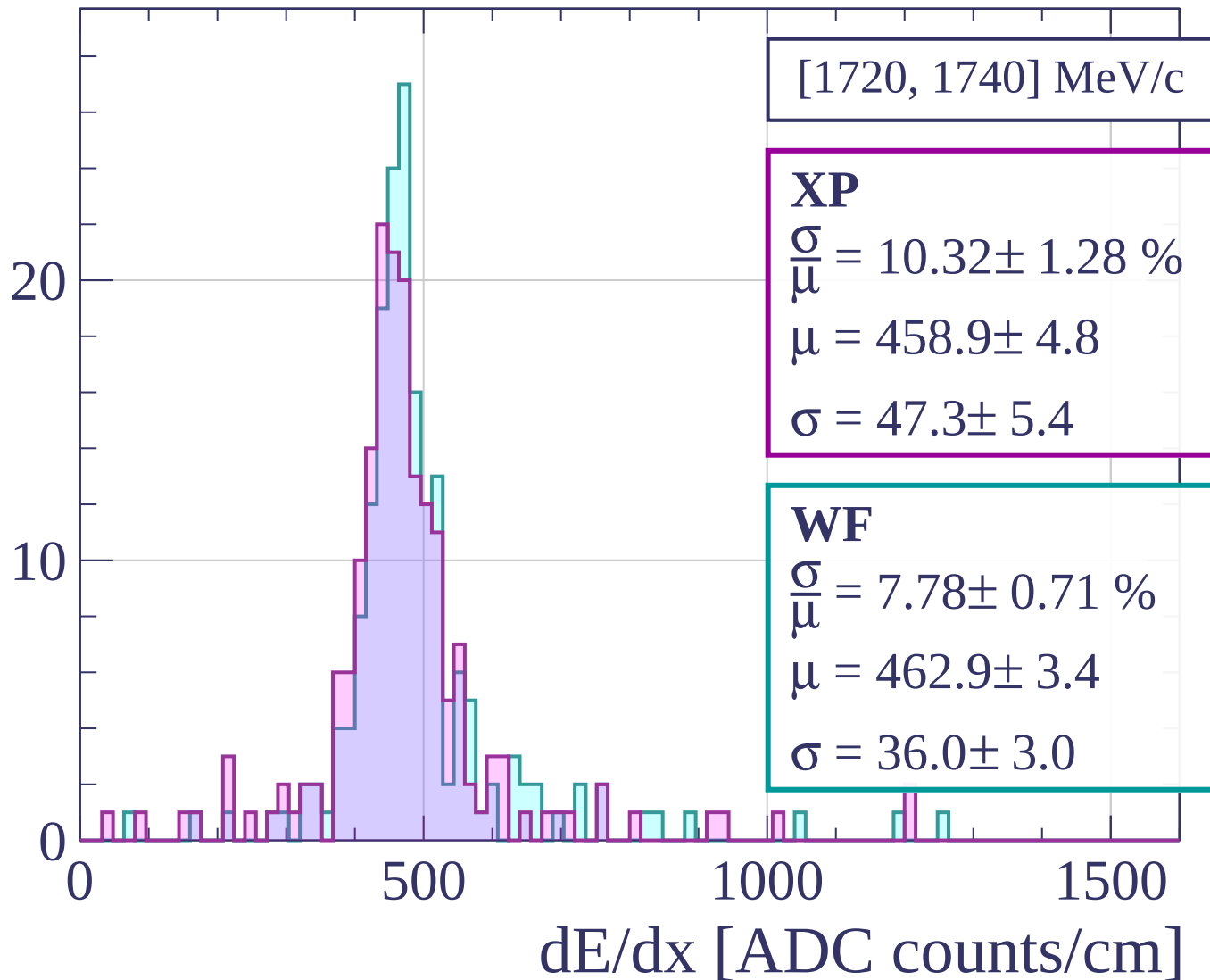
Count



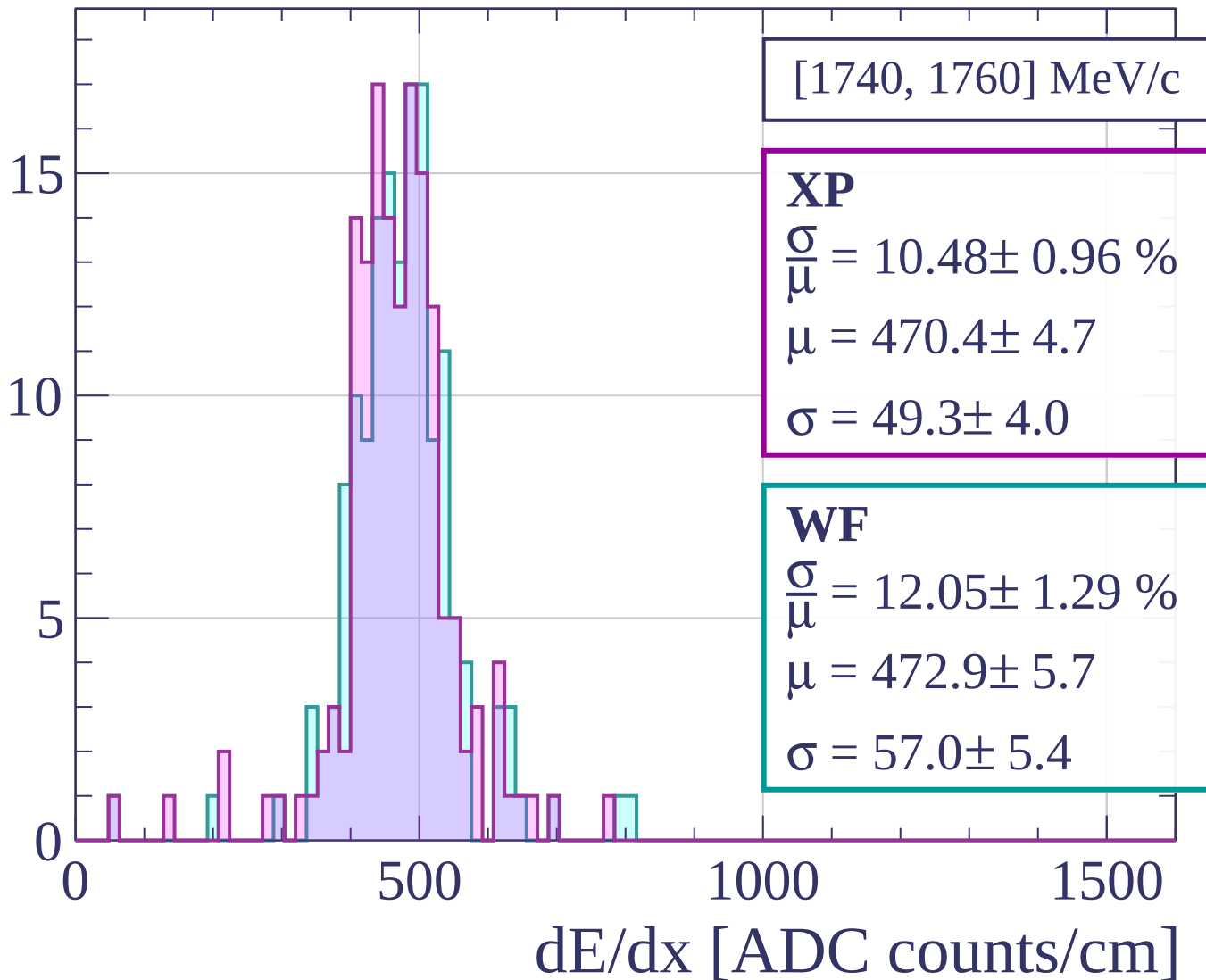
Count



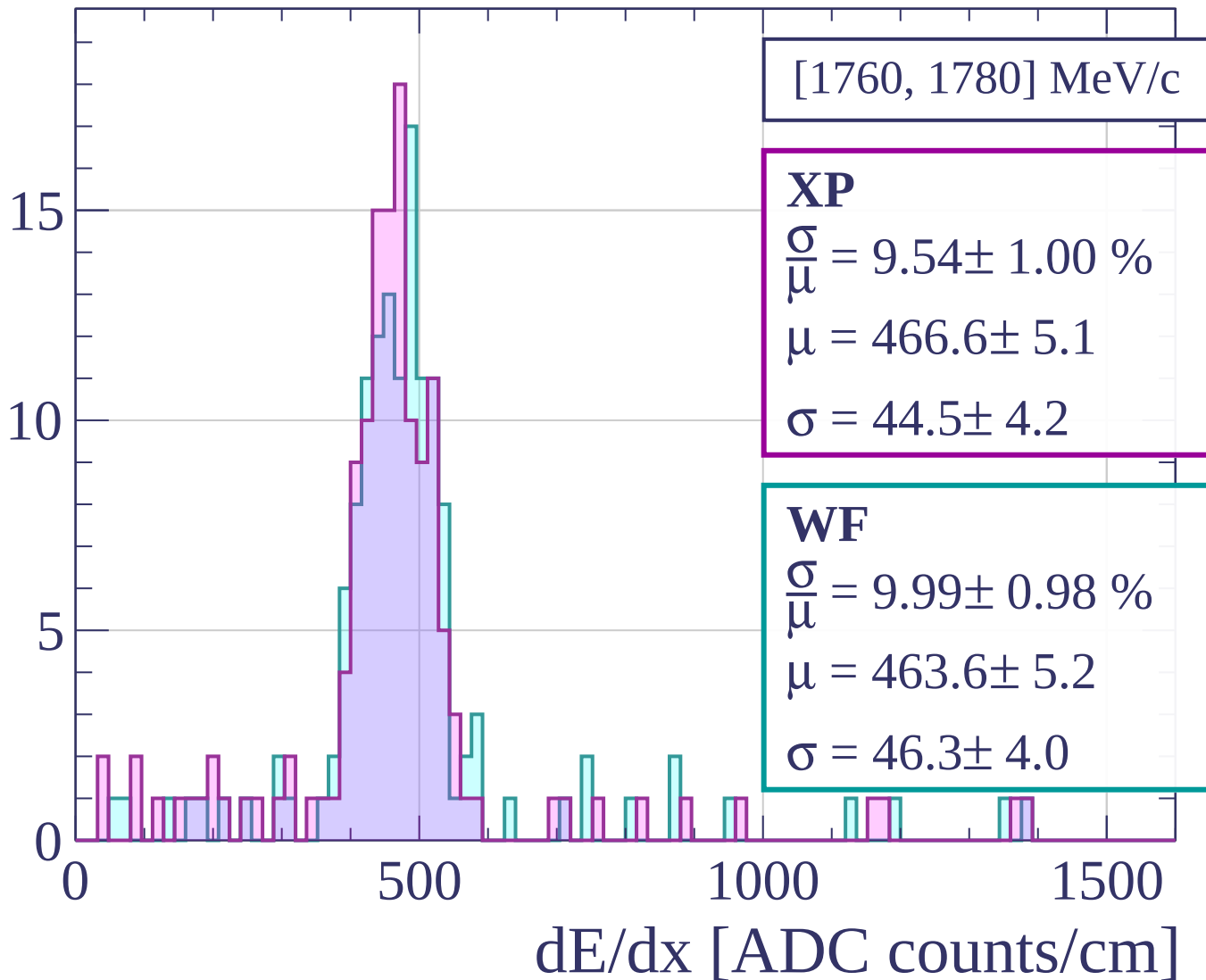
Count



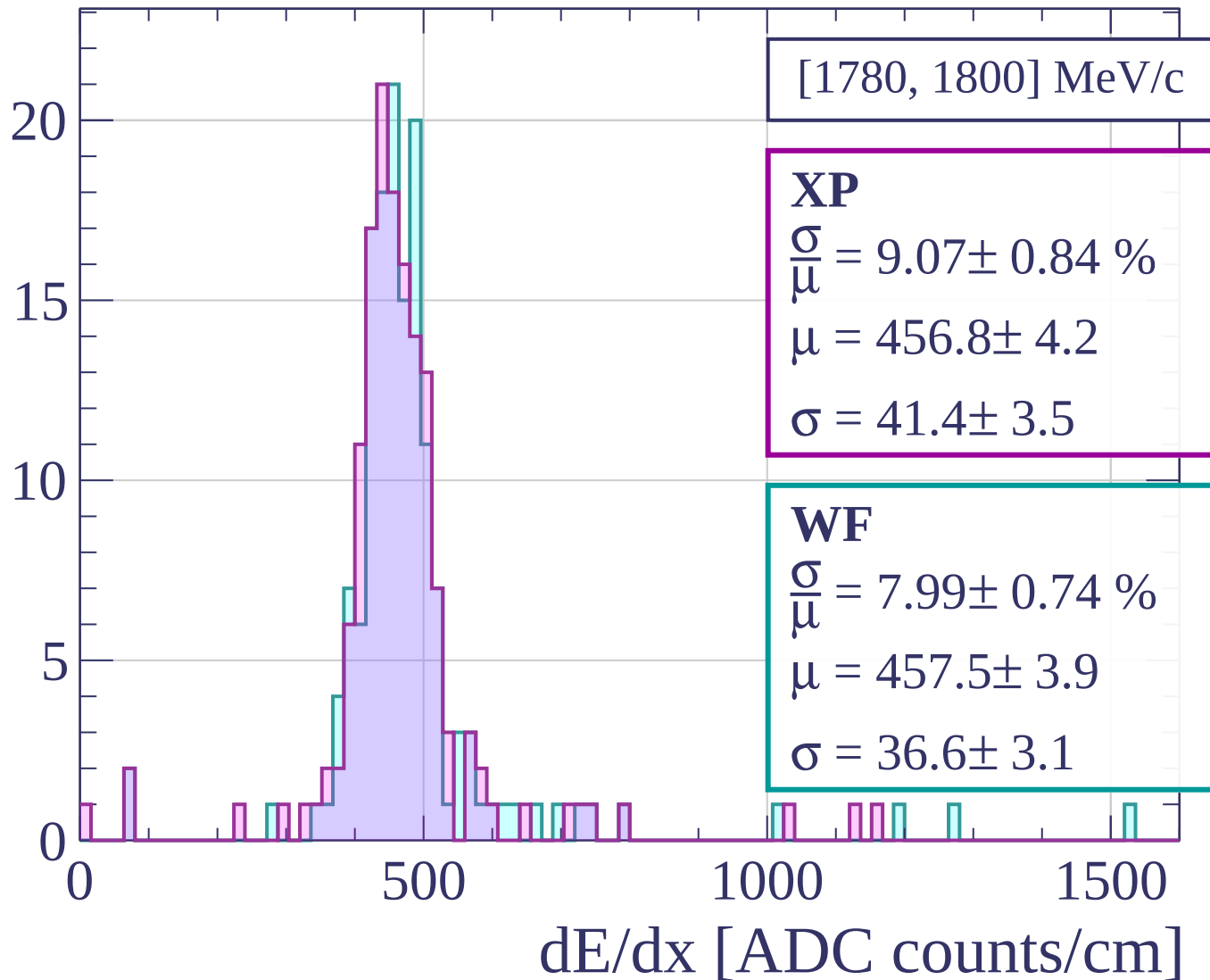
Count



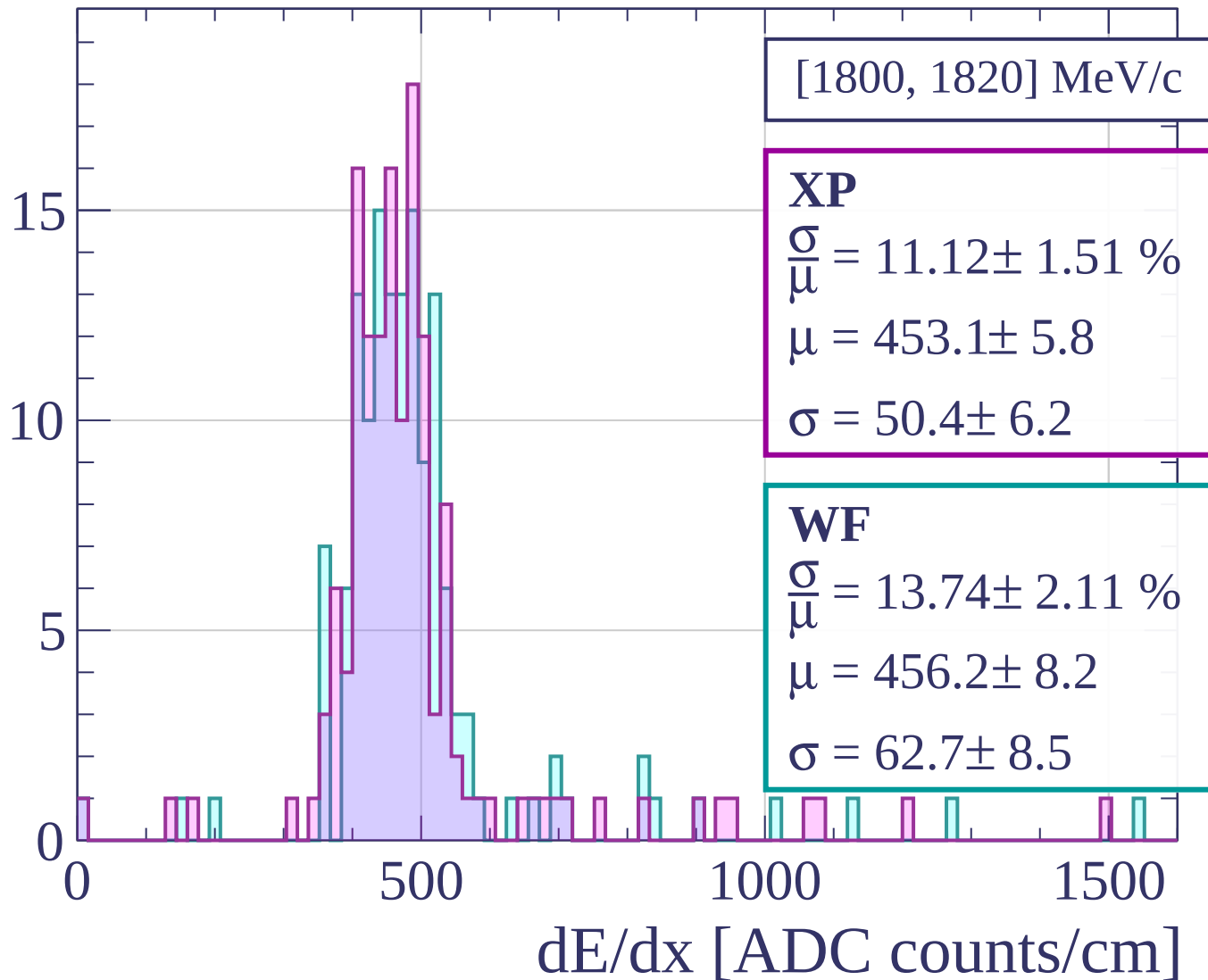
Count



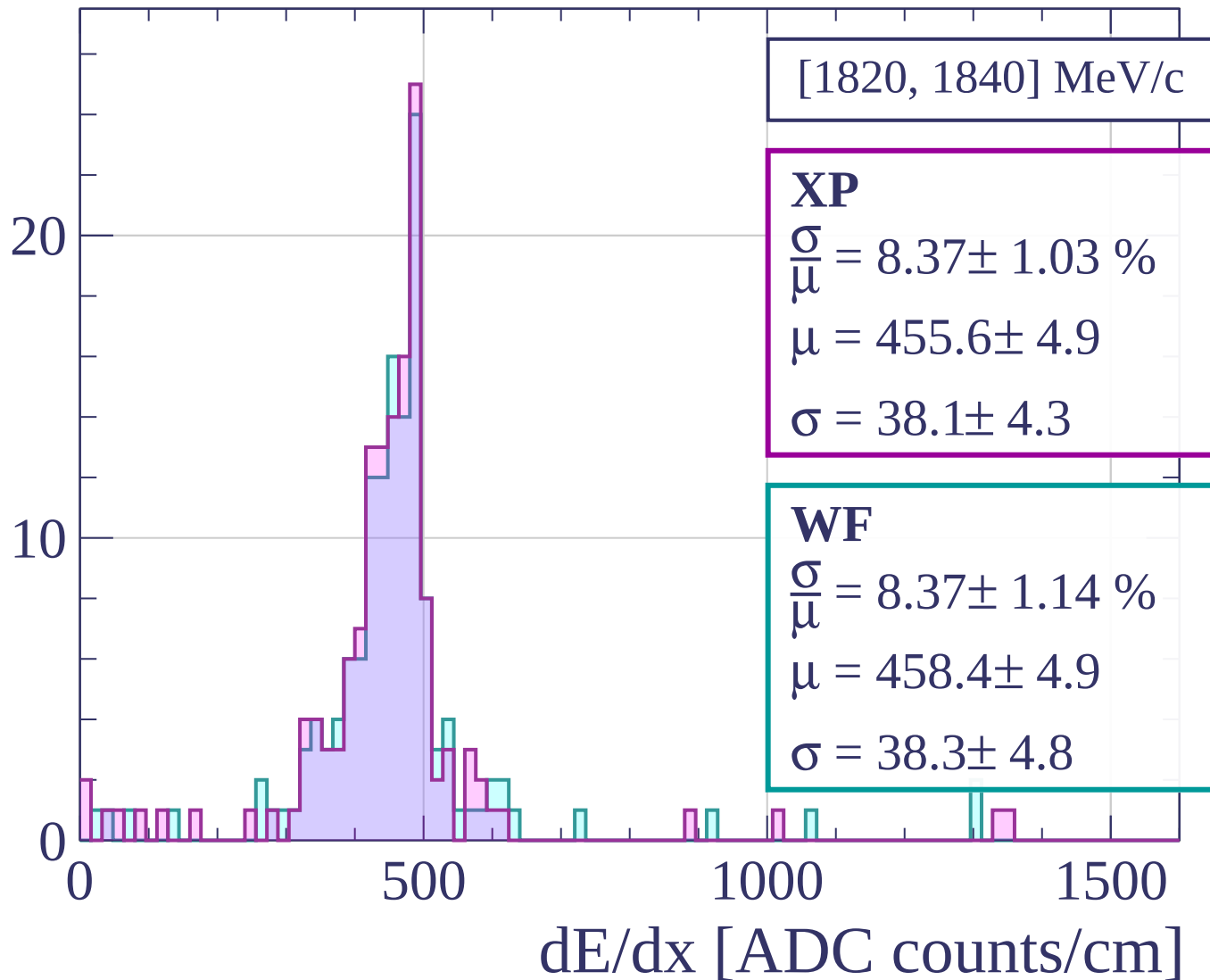
Count



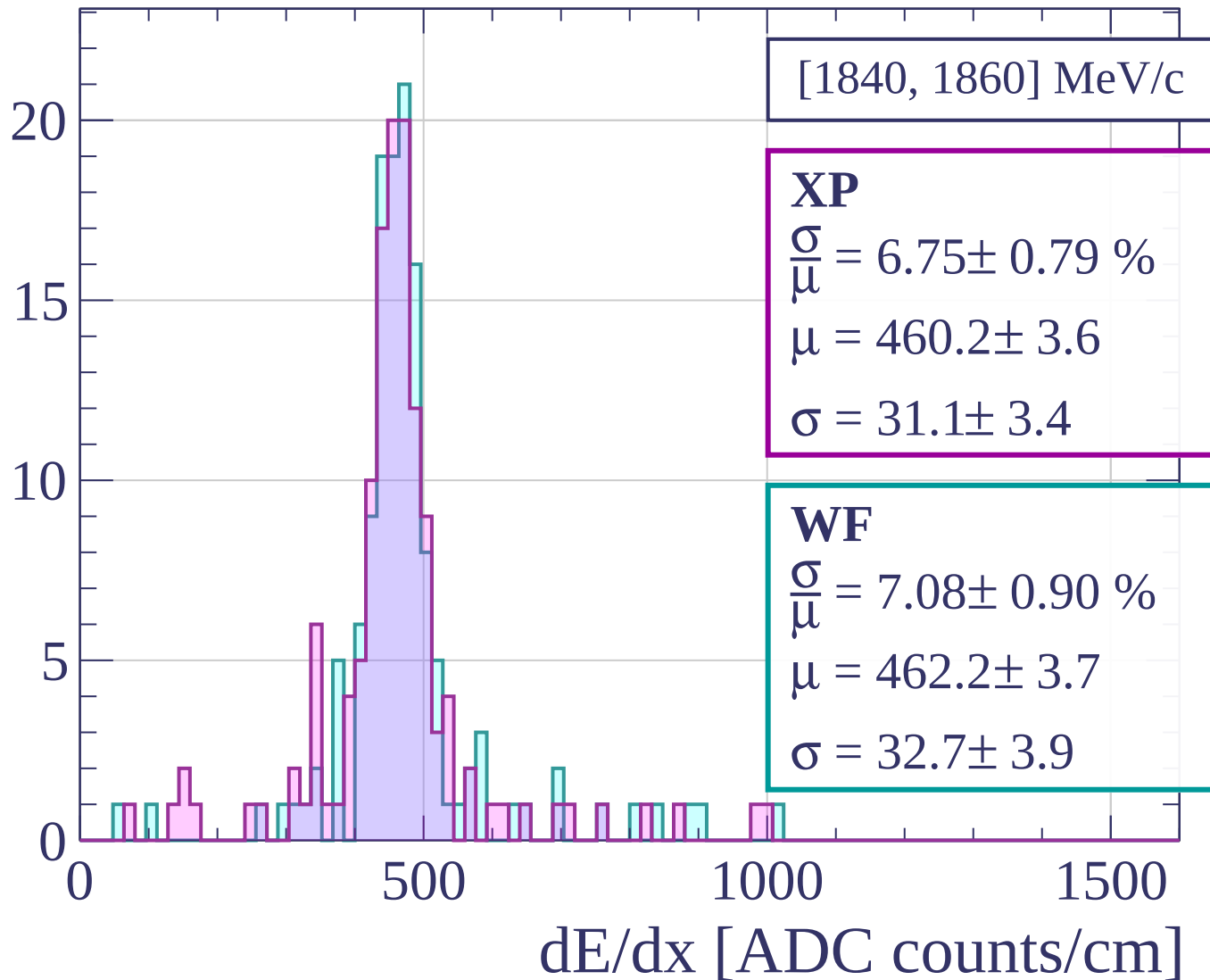
Count



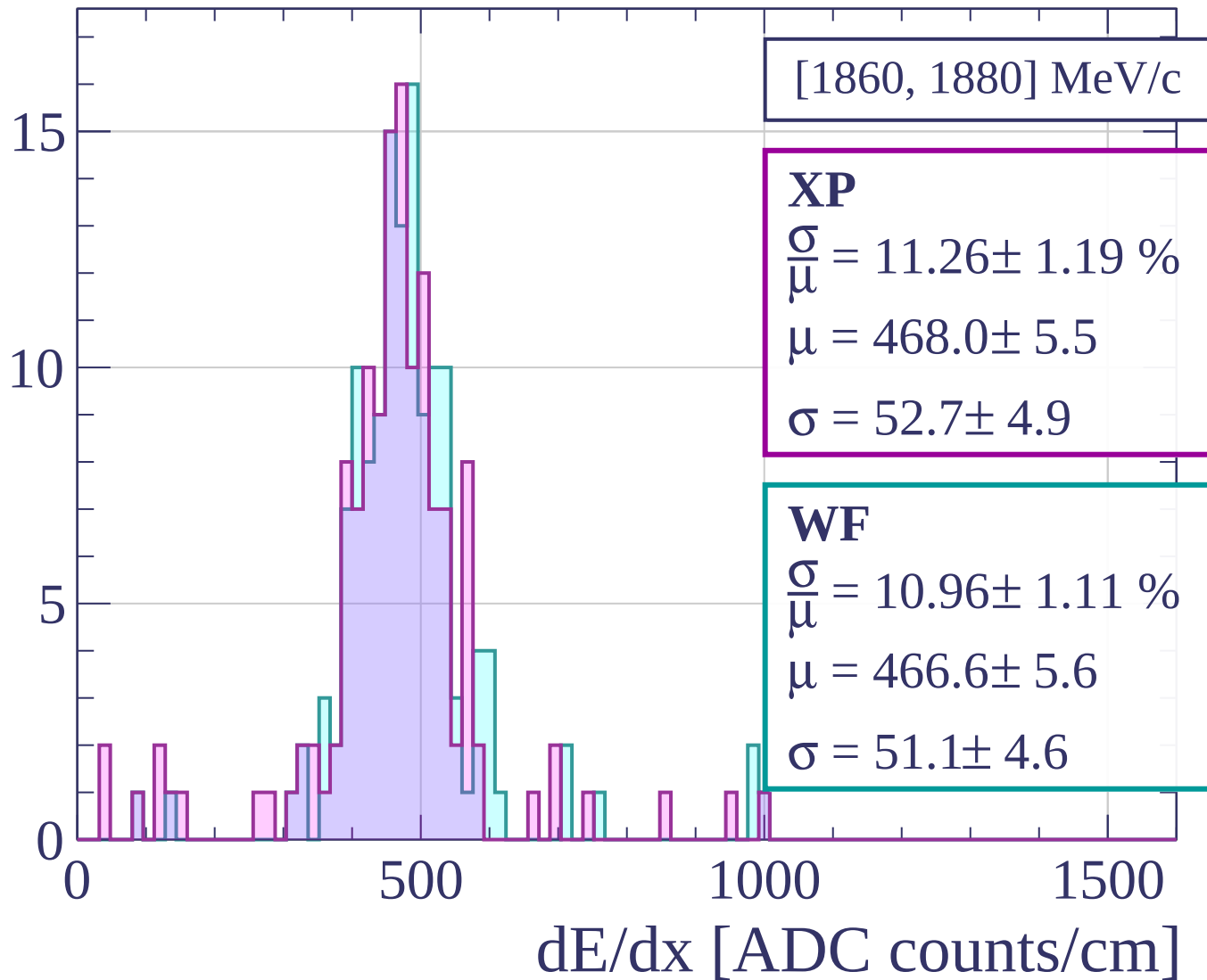
Count



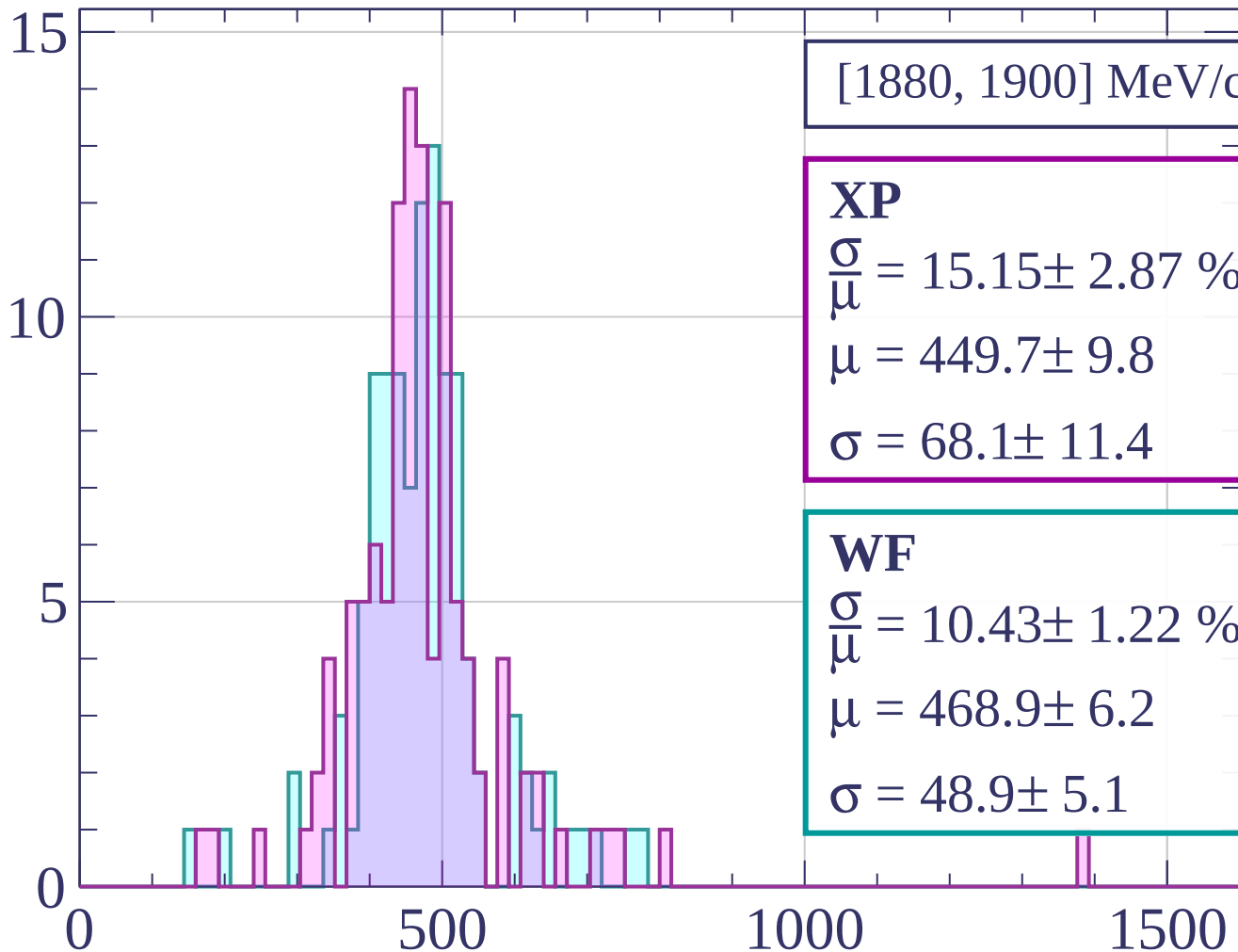
Count



Count

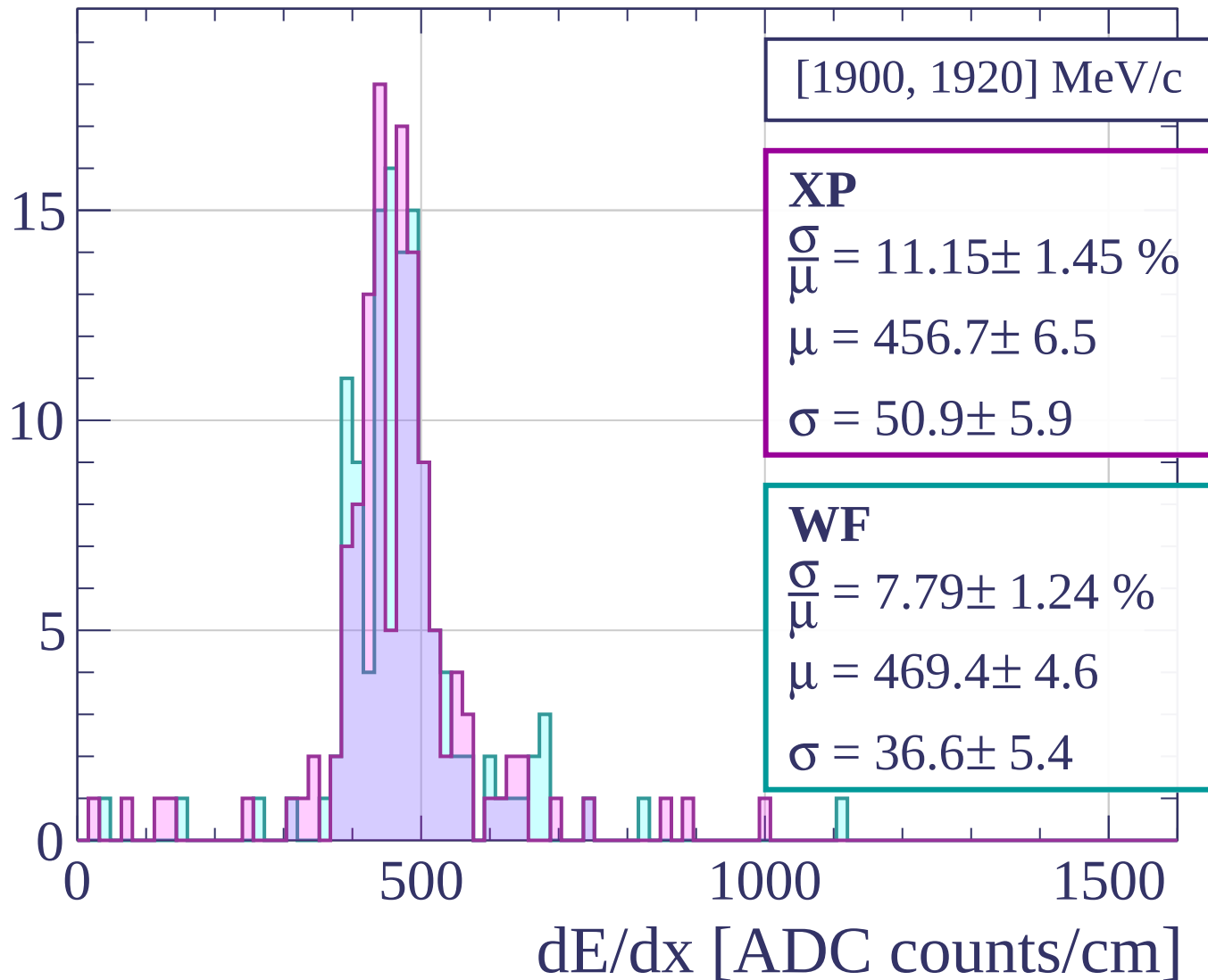


Count

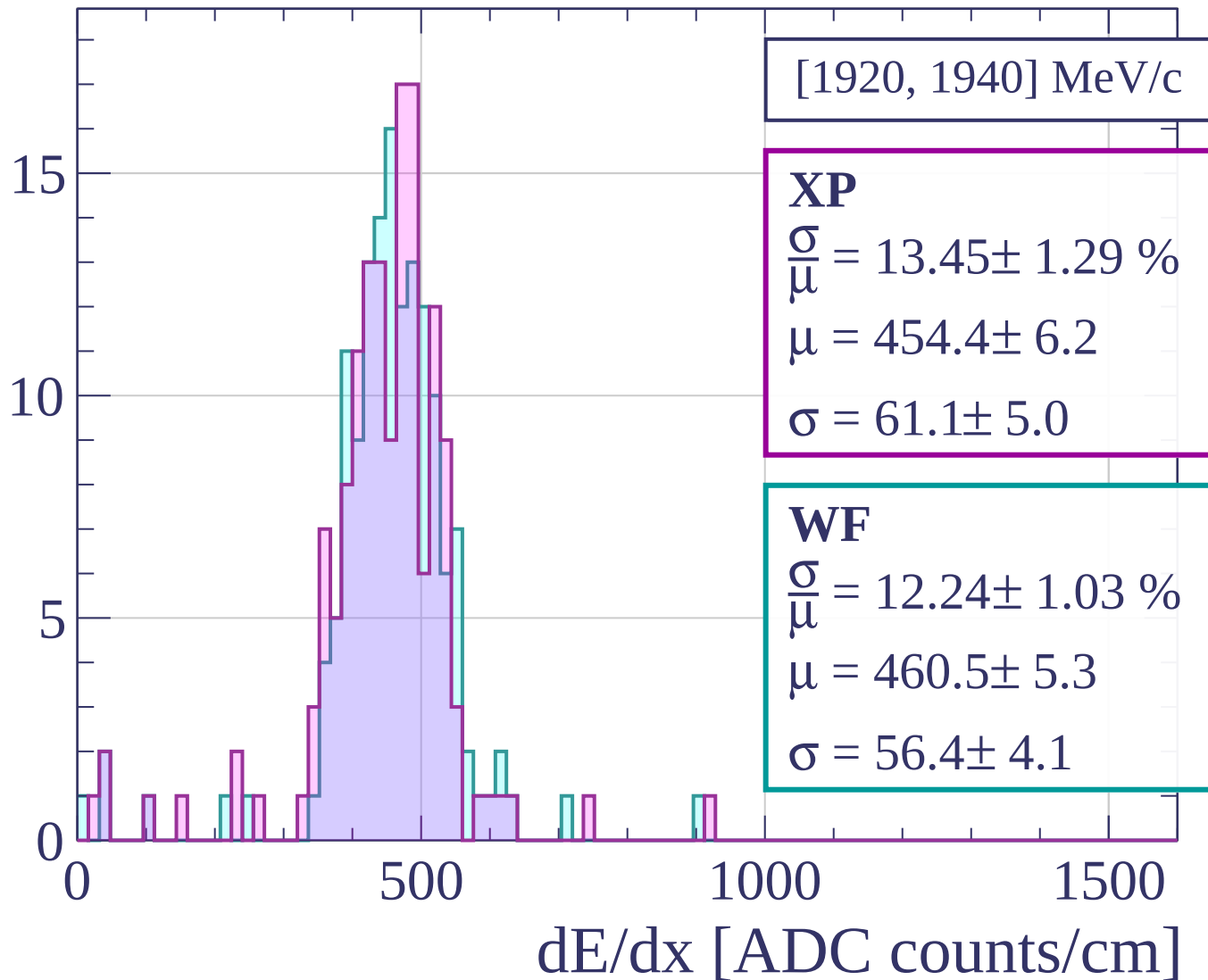


dE/dx [ADC counts/cm]

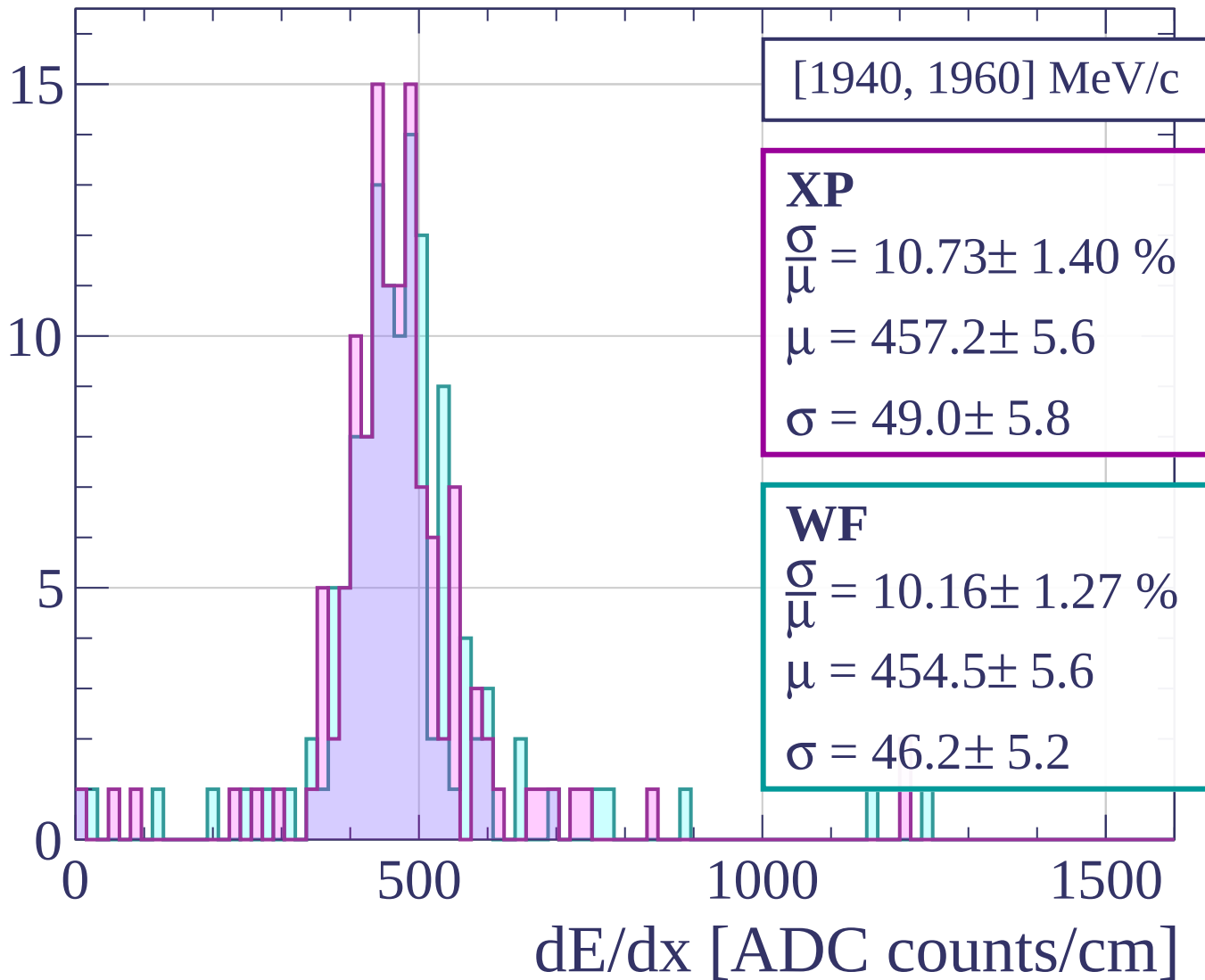
Count



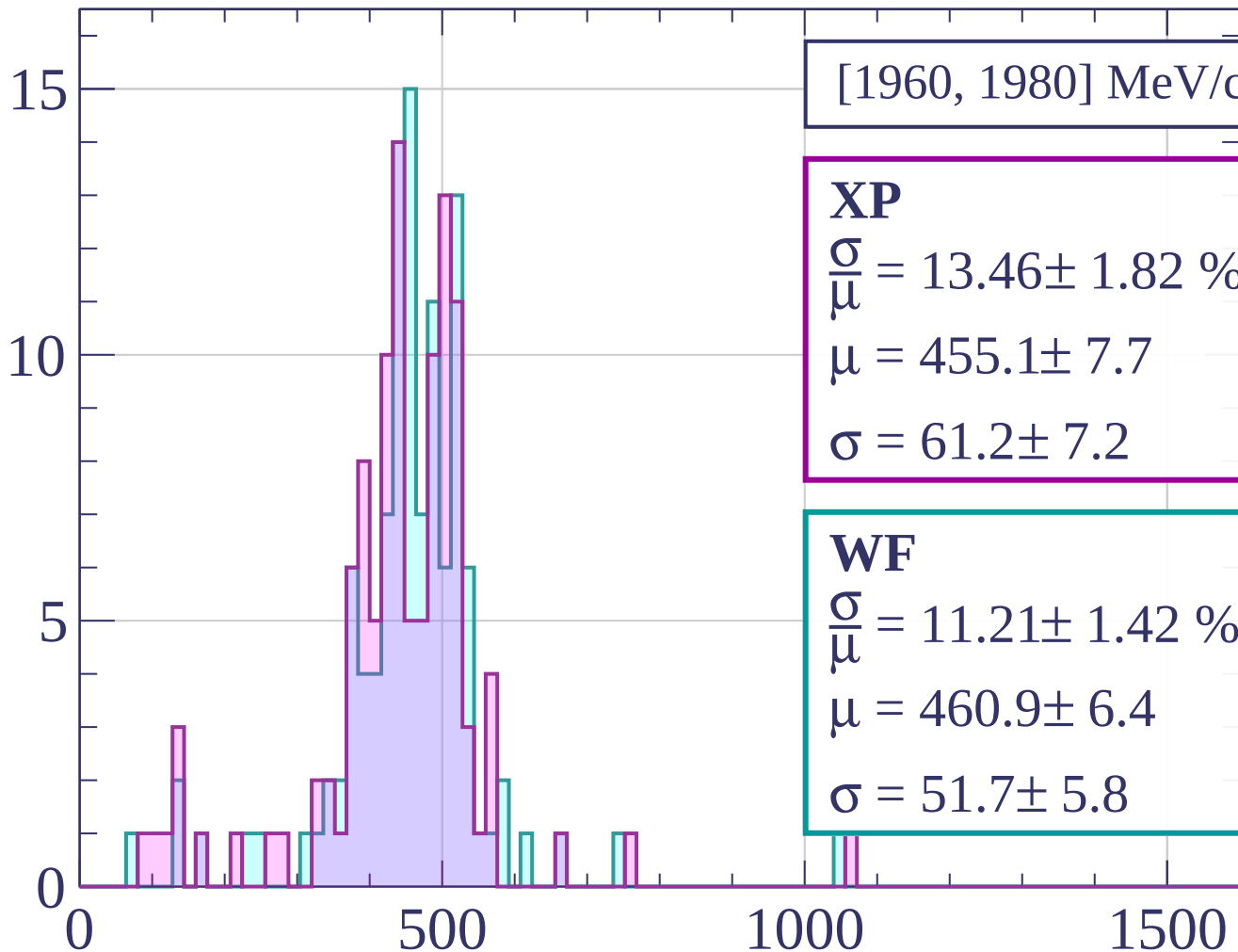
Count



Count

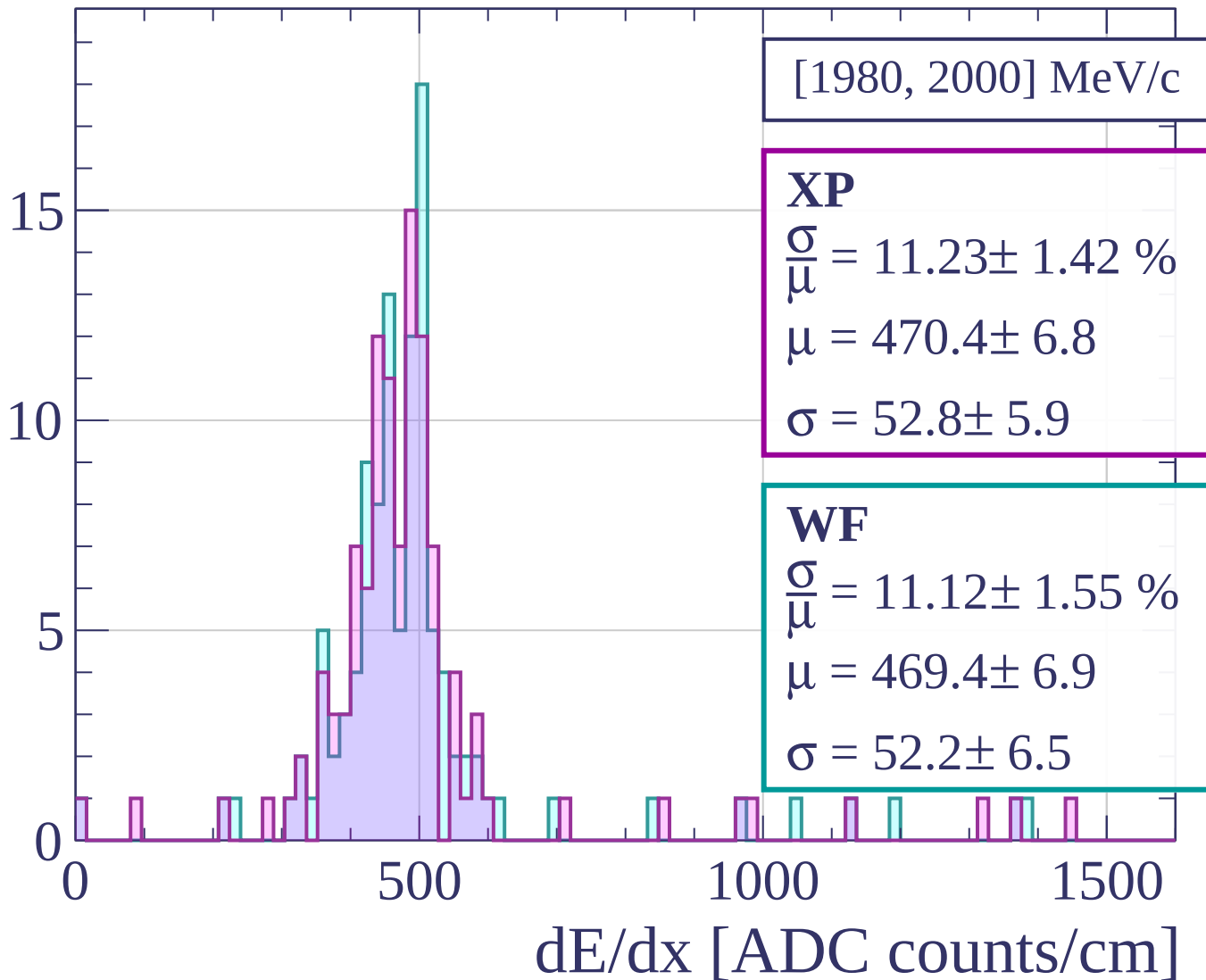


Count

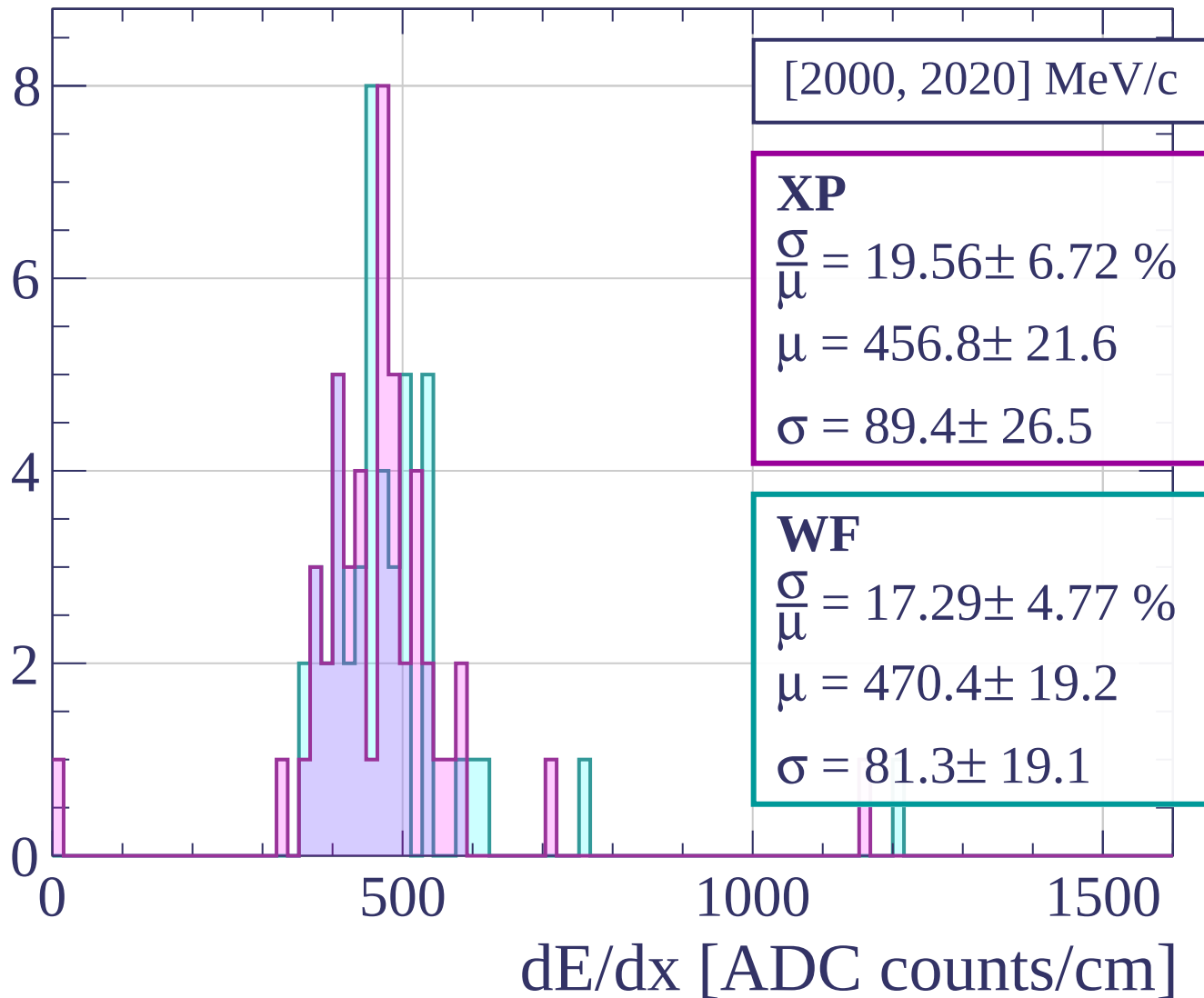


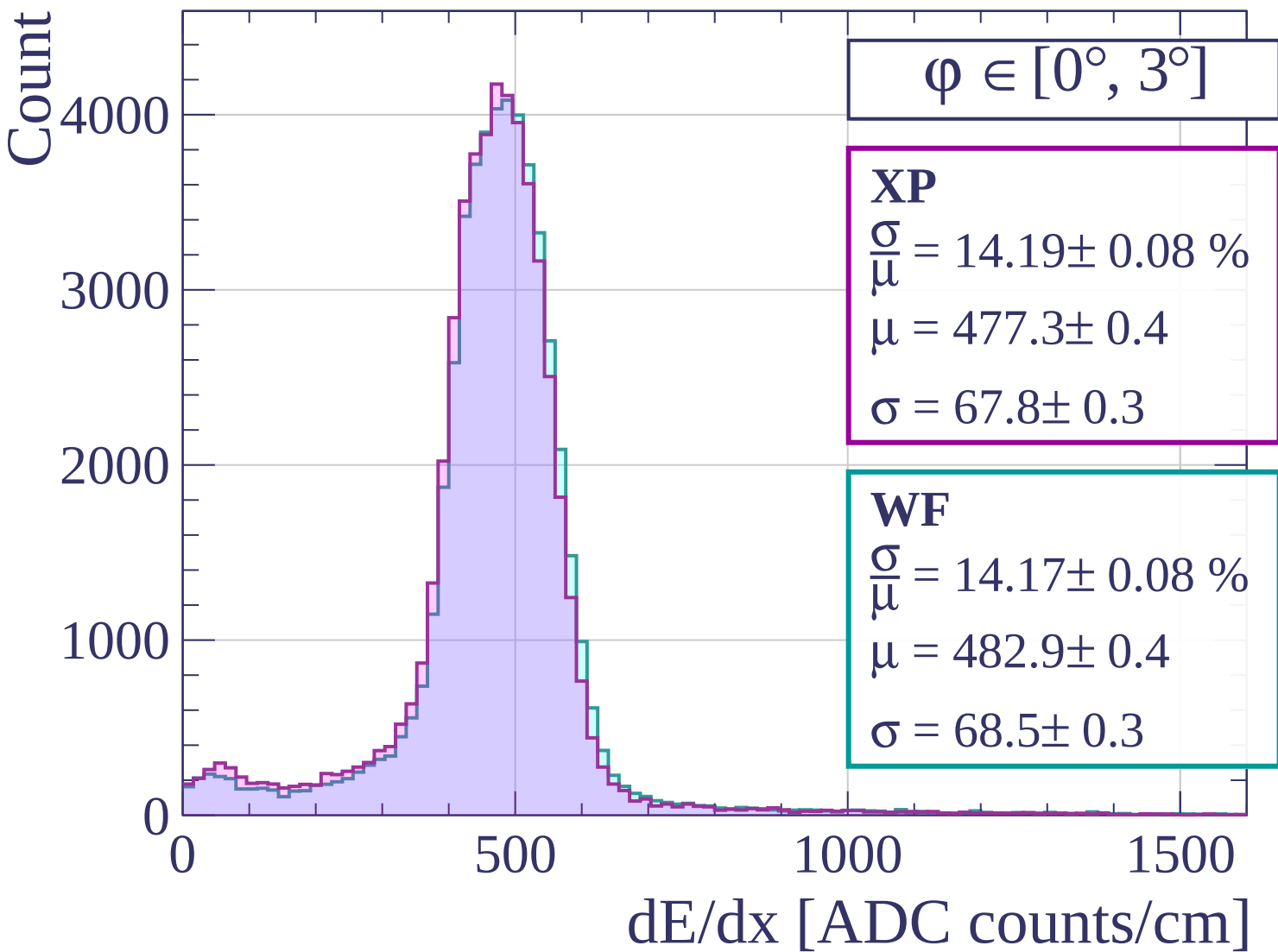
dE/dx [ADC counts/cm]

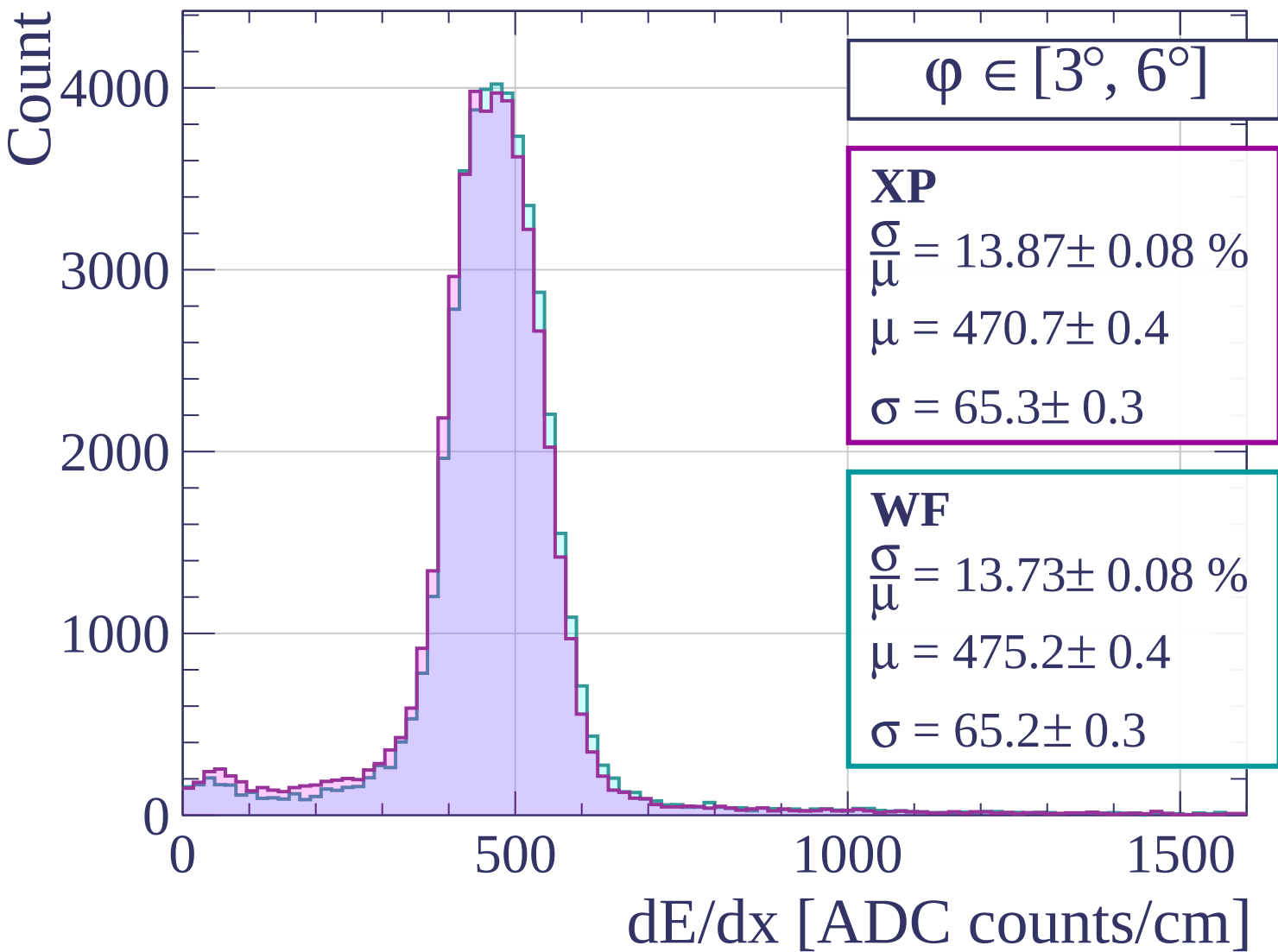
Count

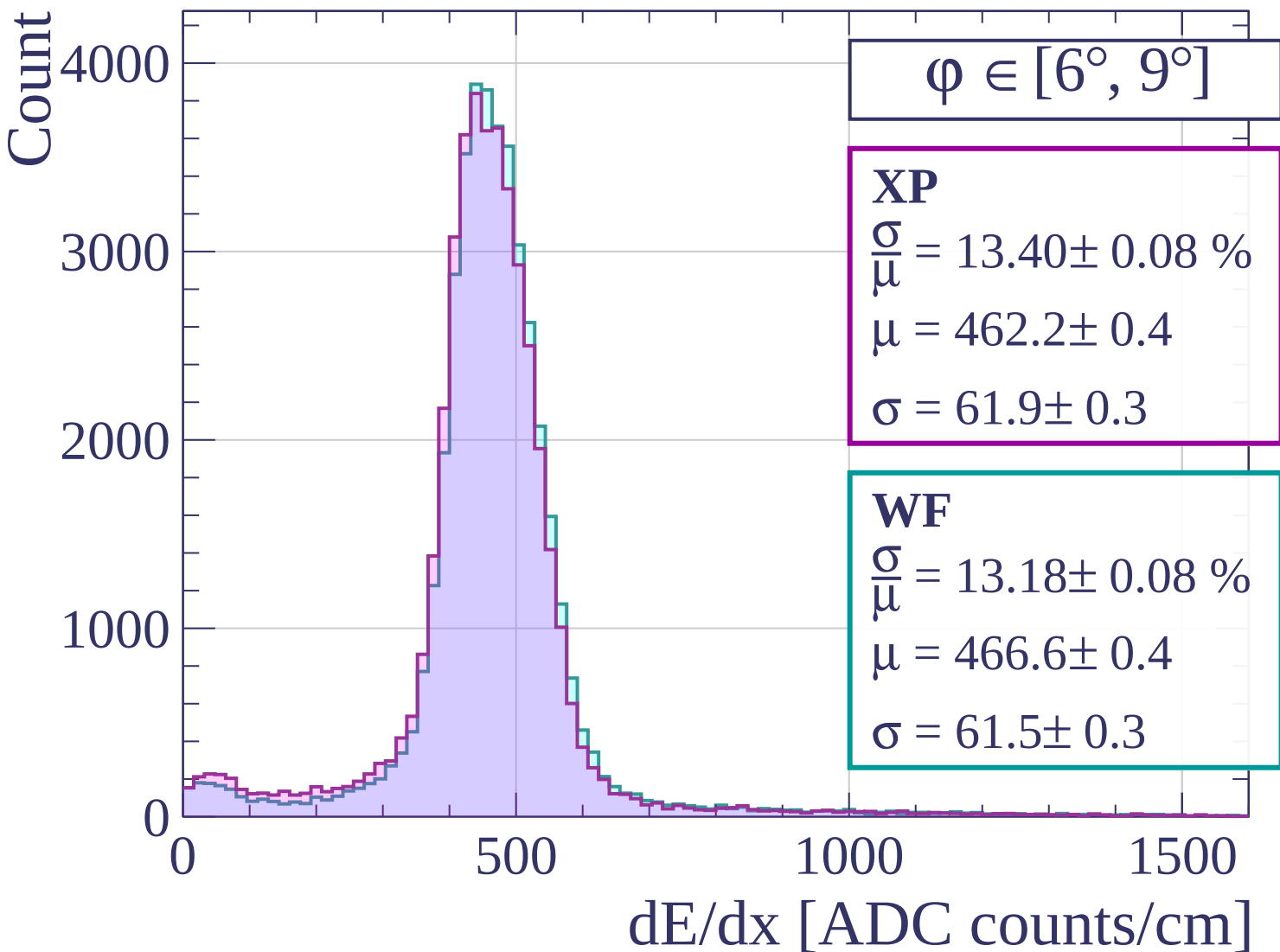


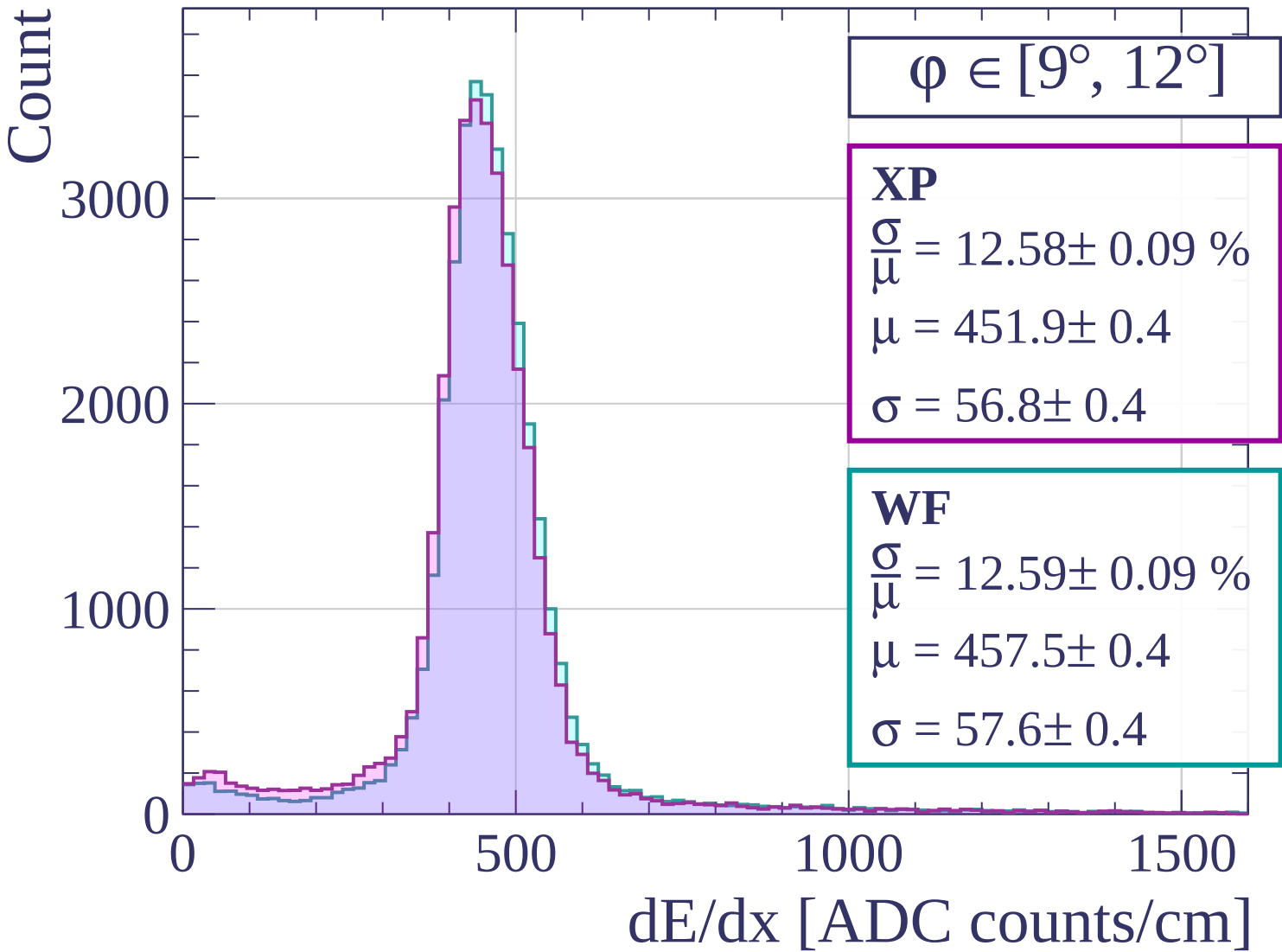
Count

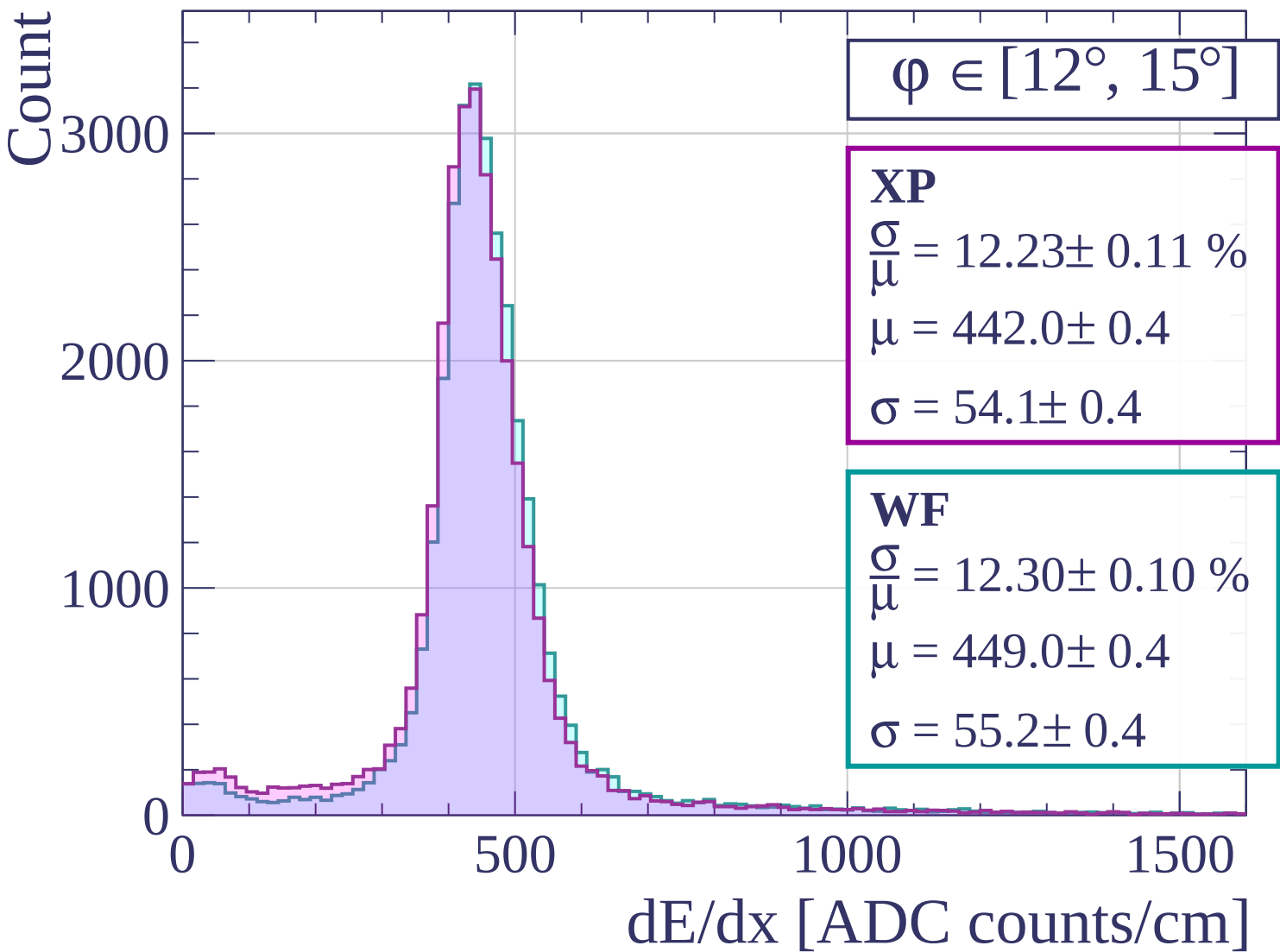


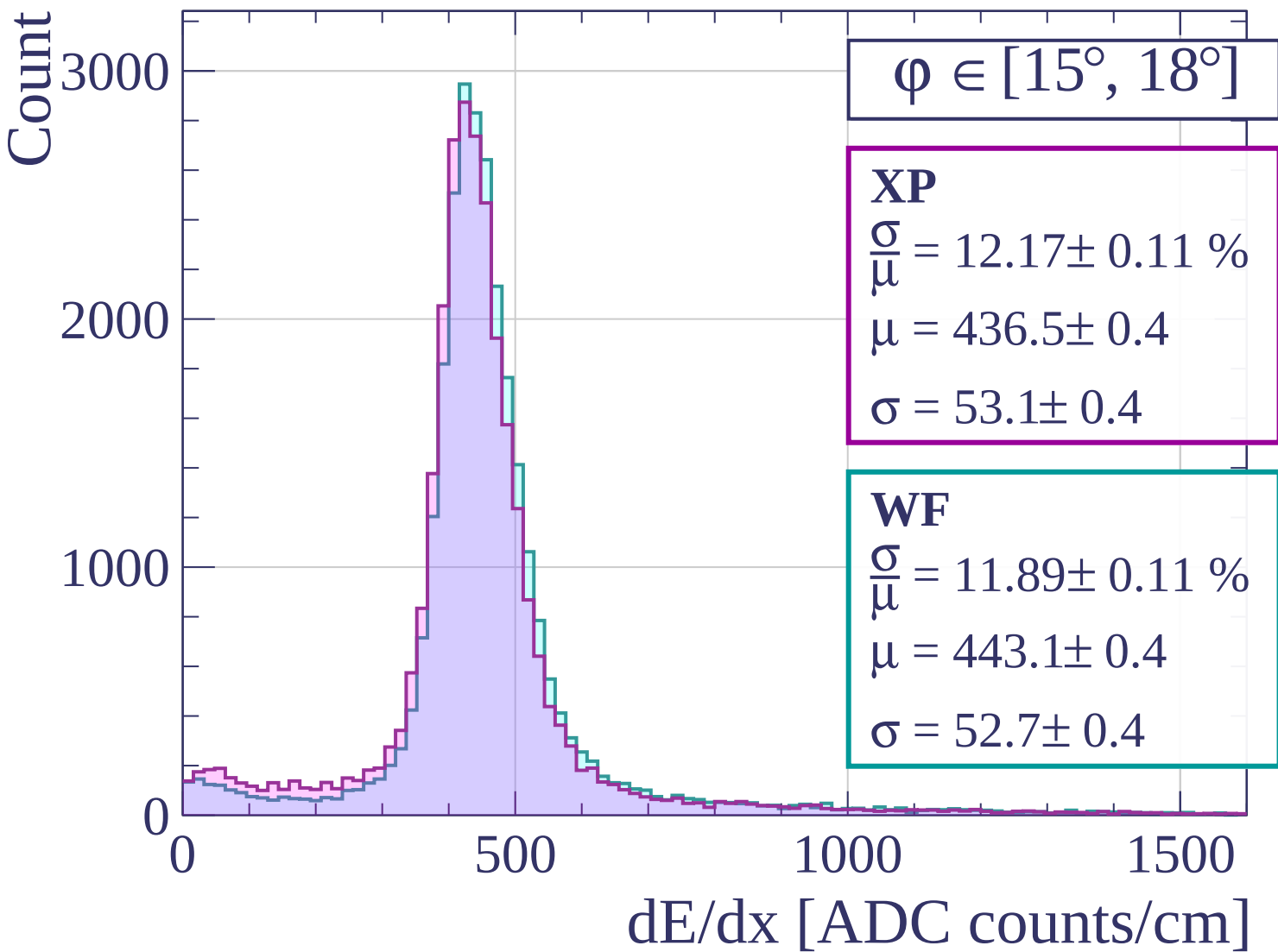


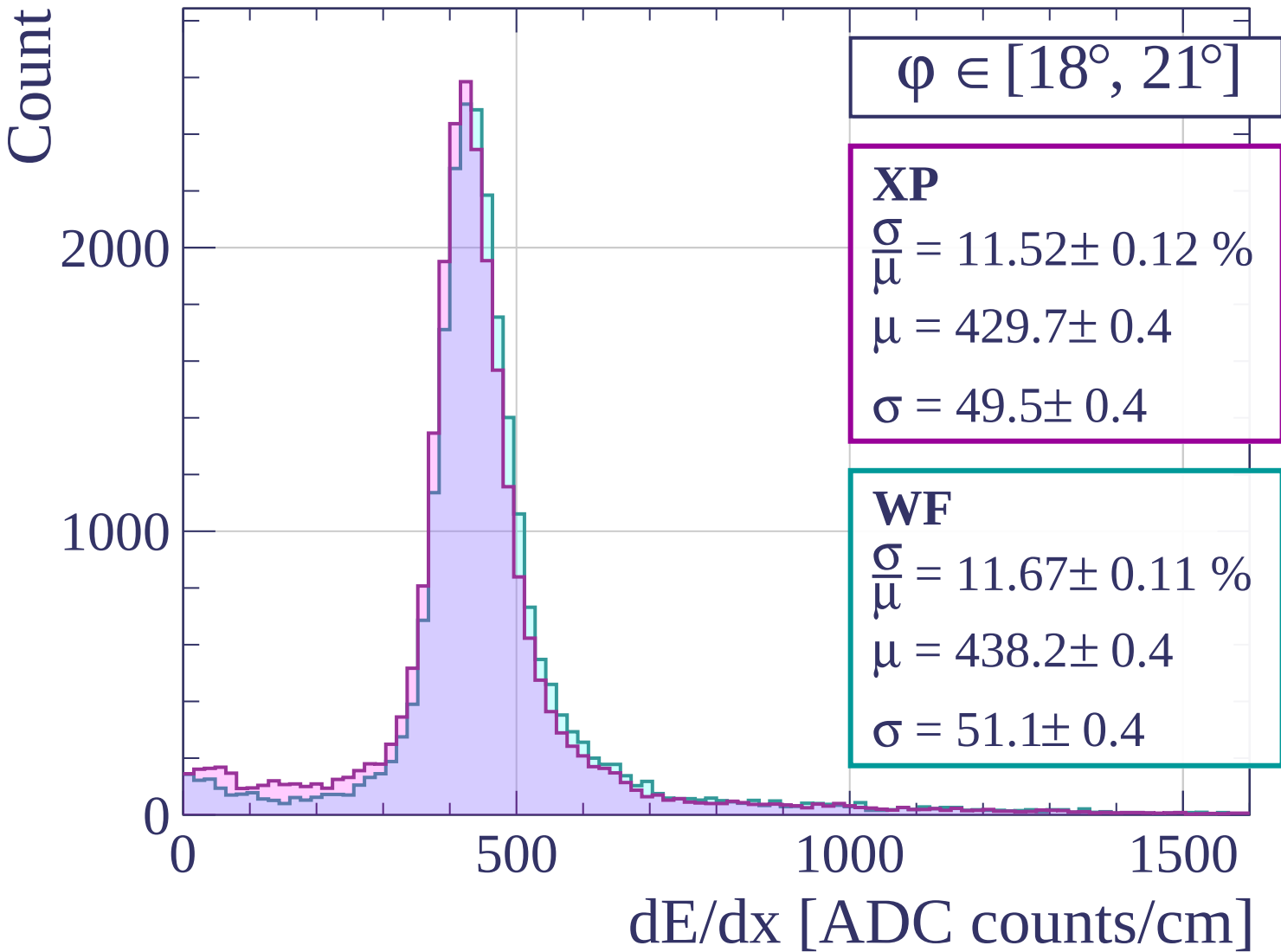


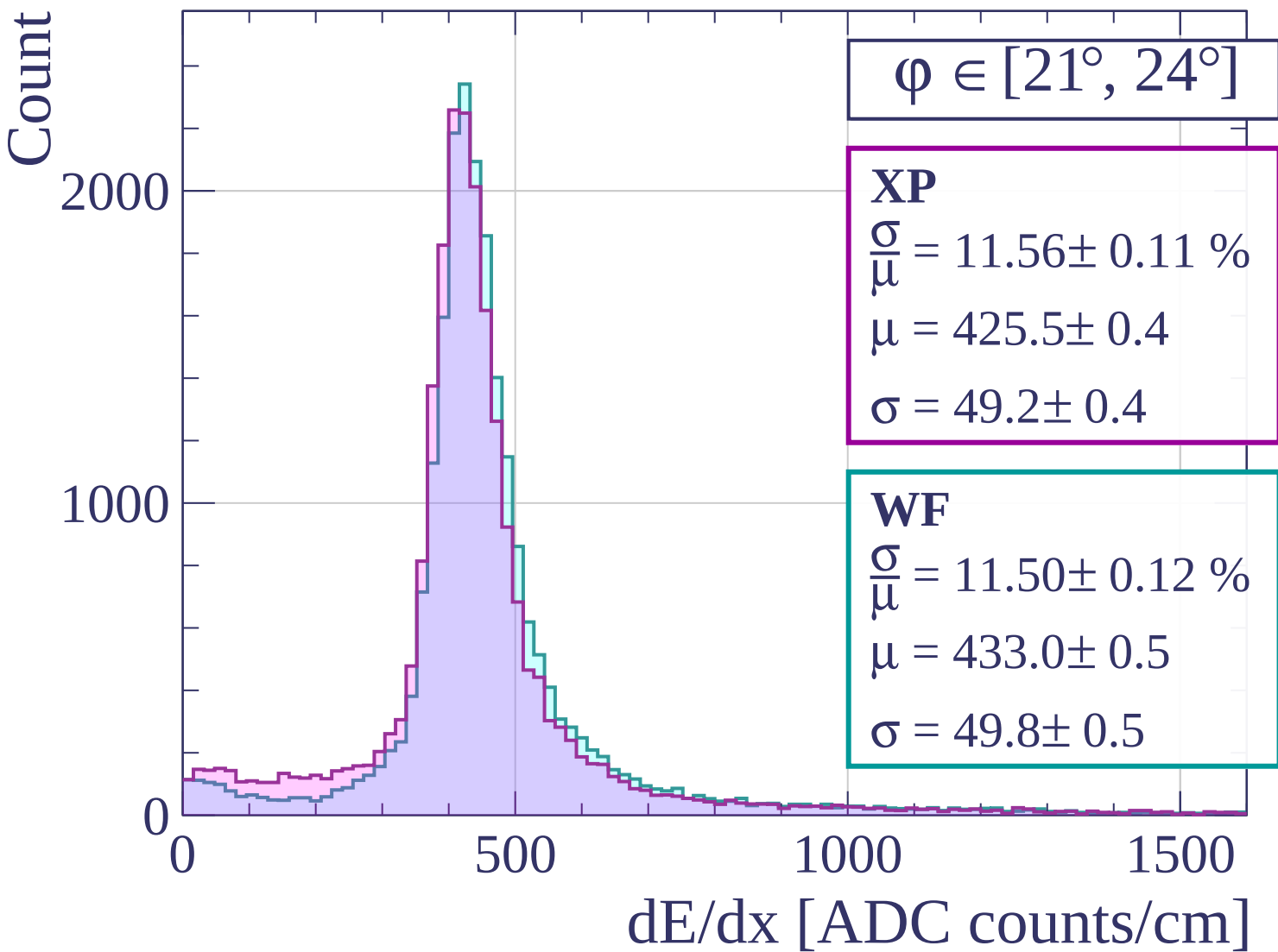


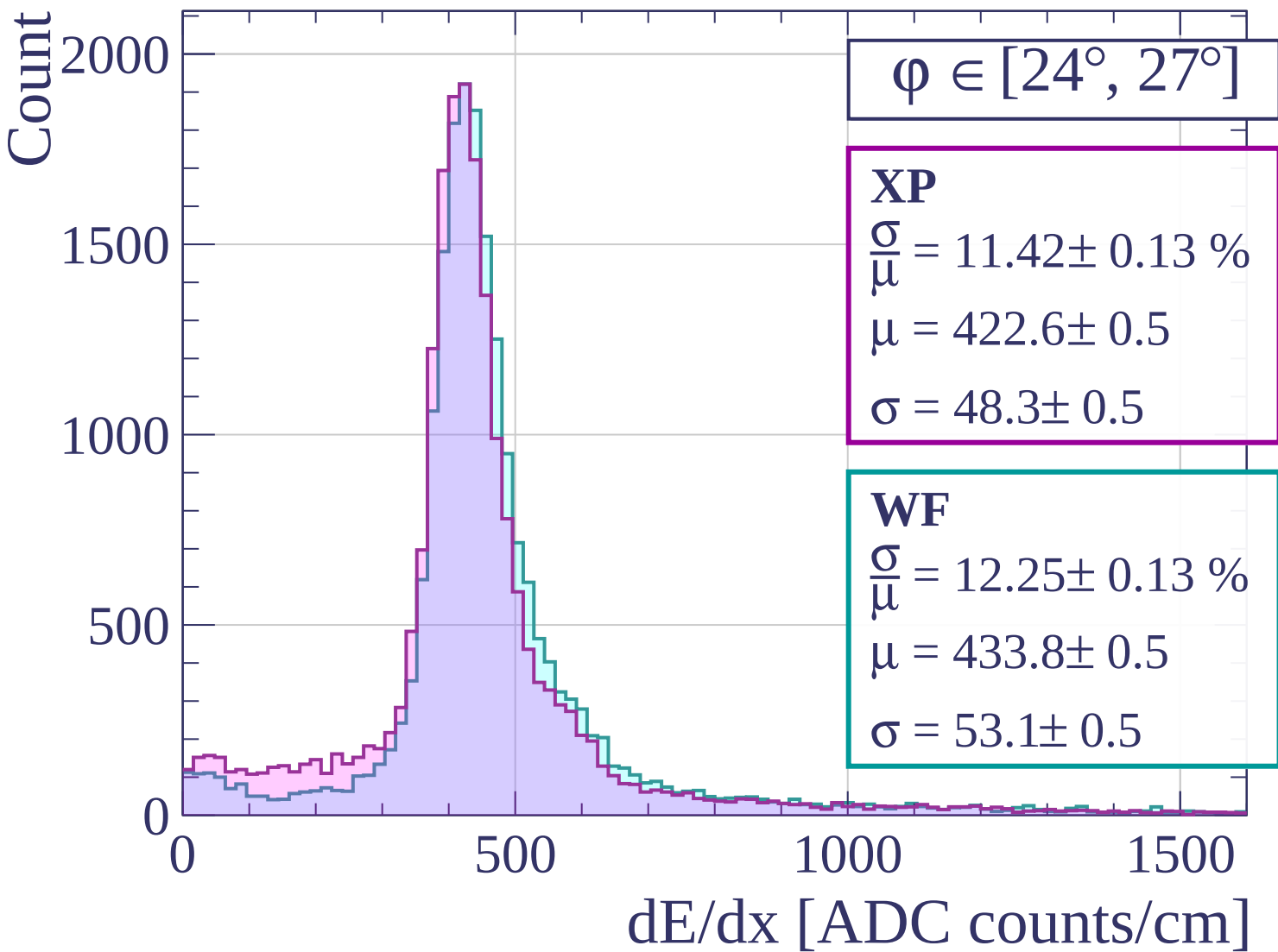


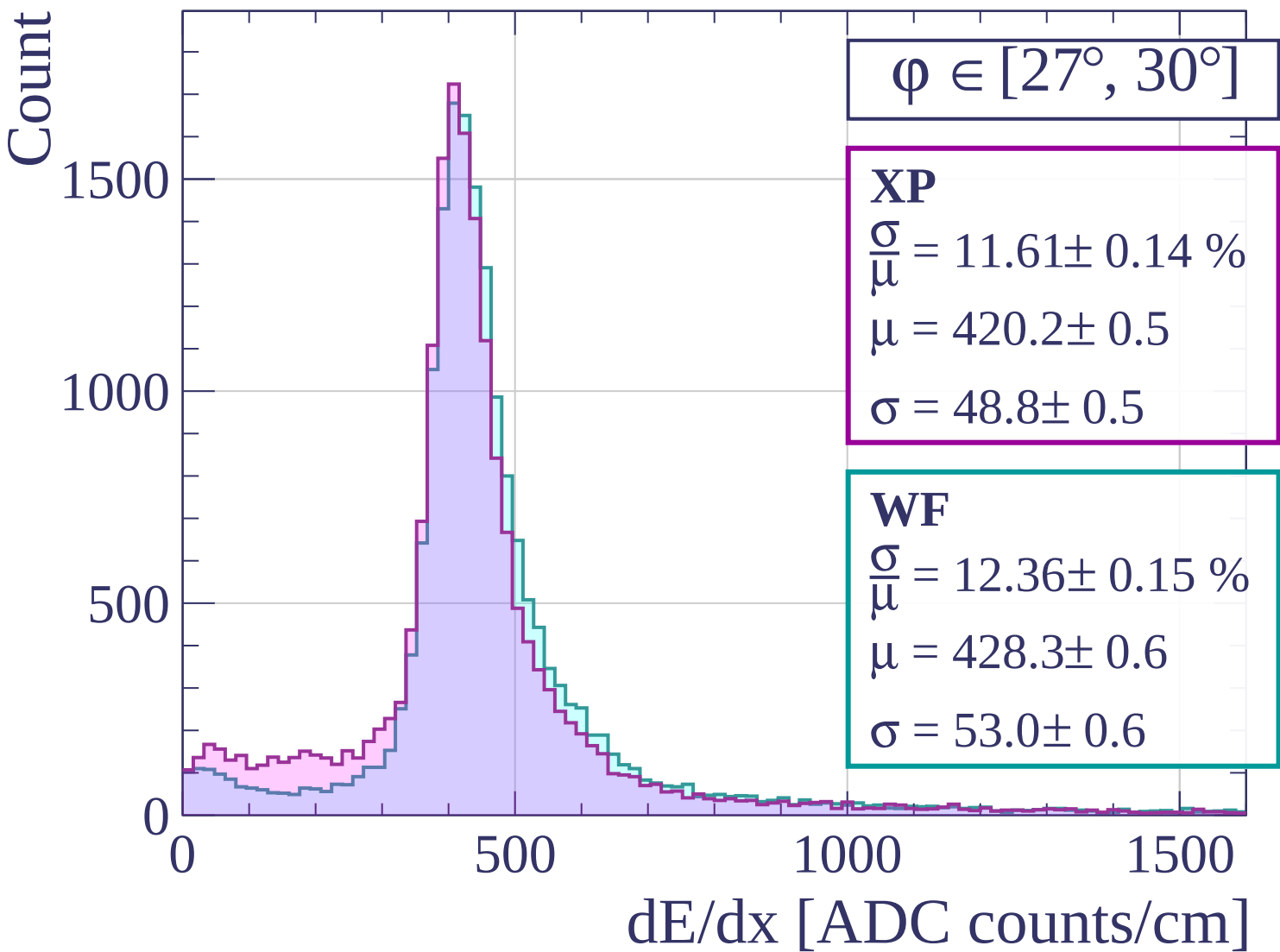


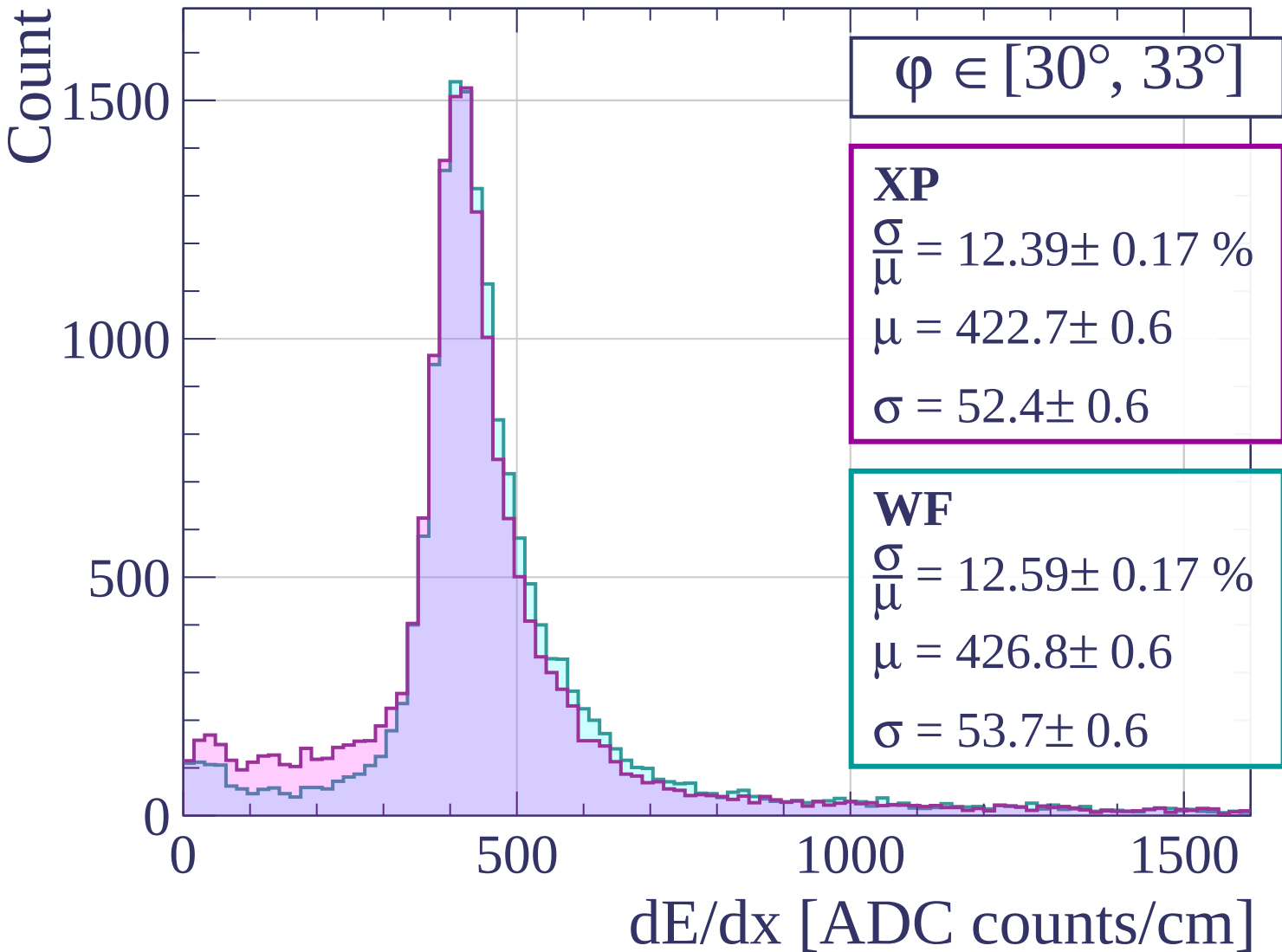


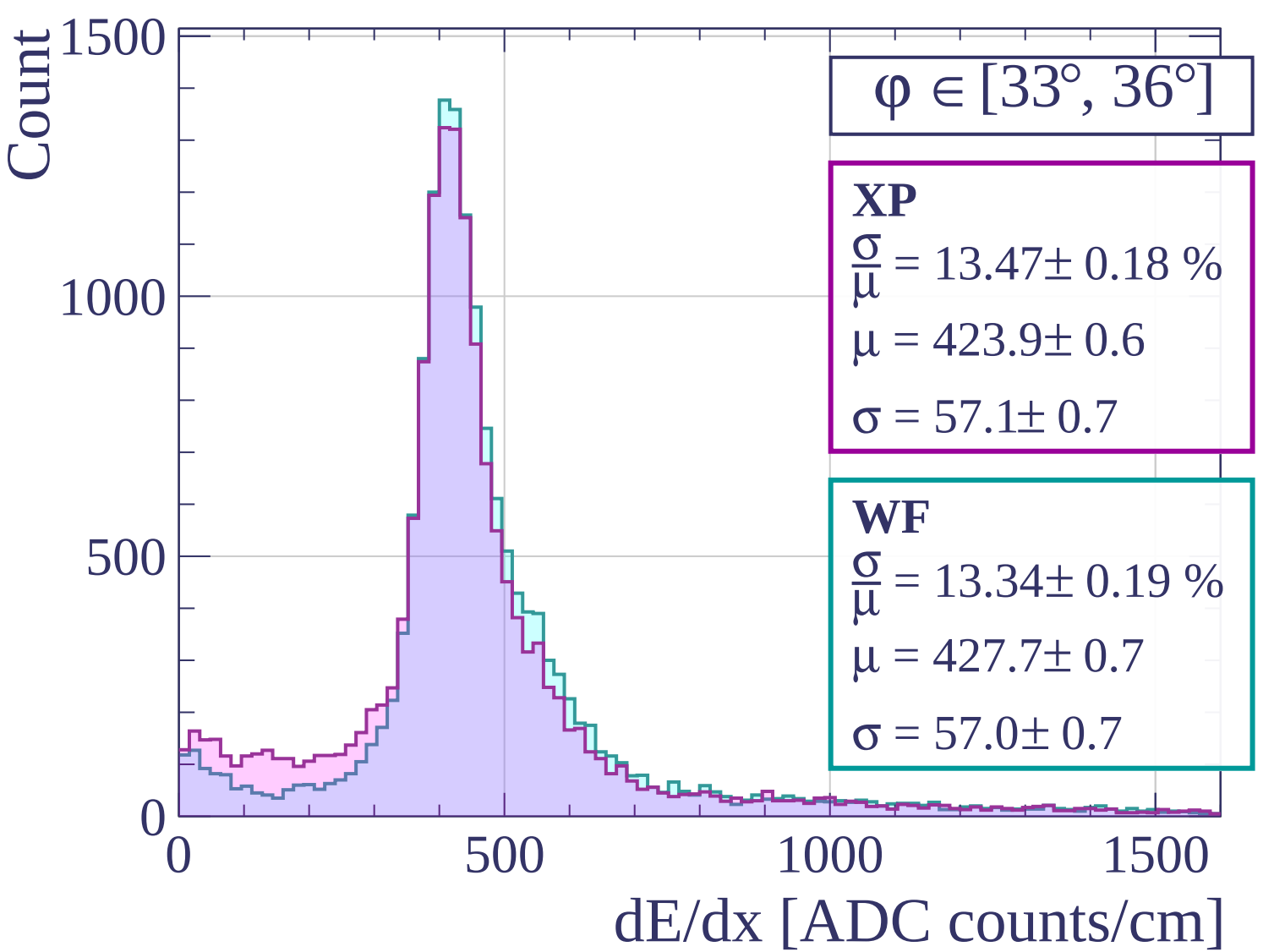


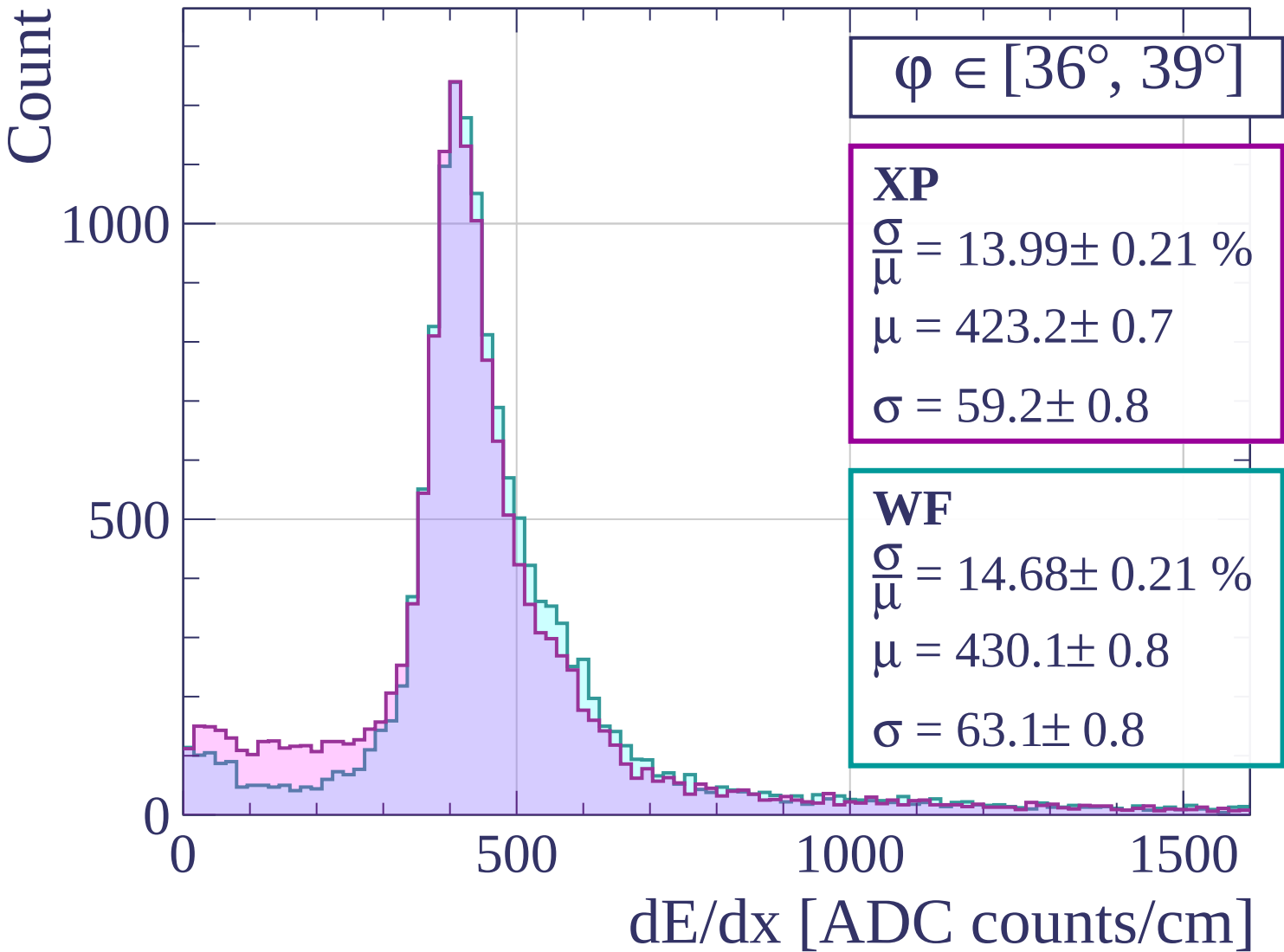


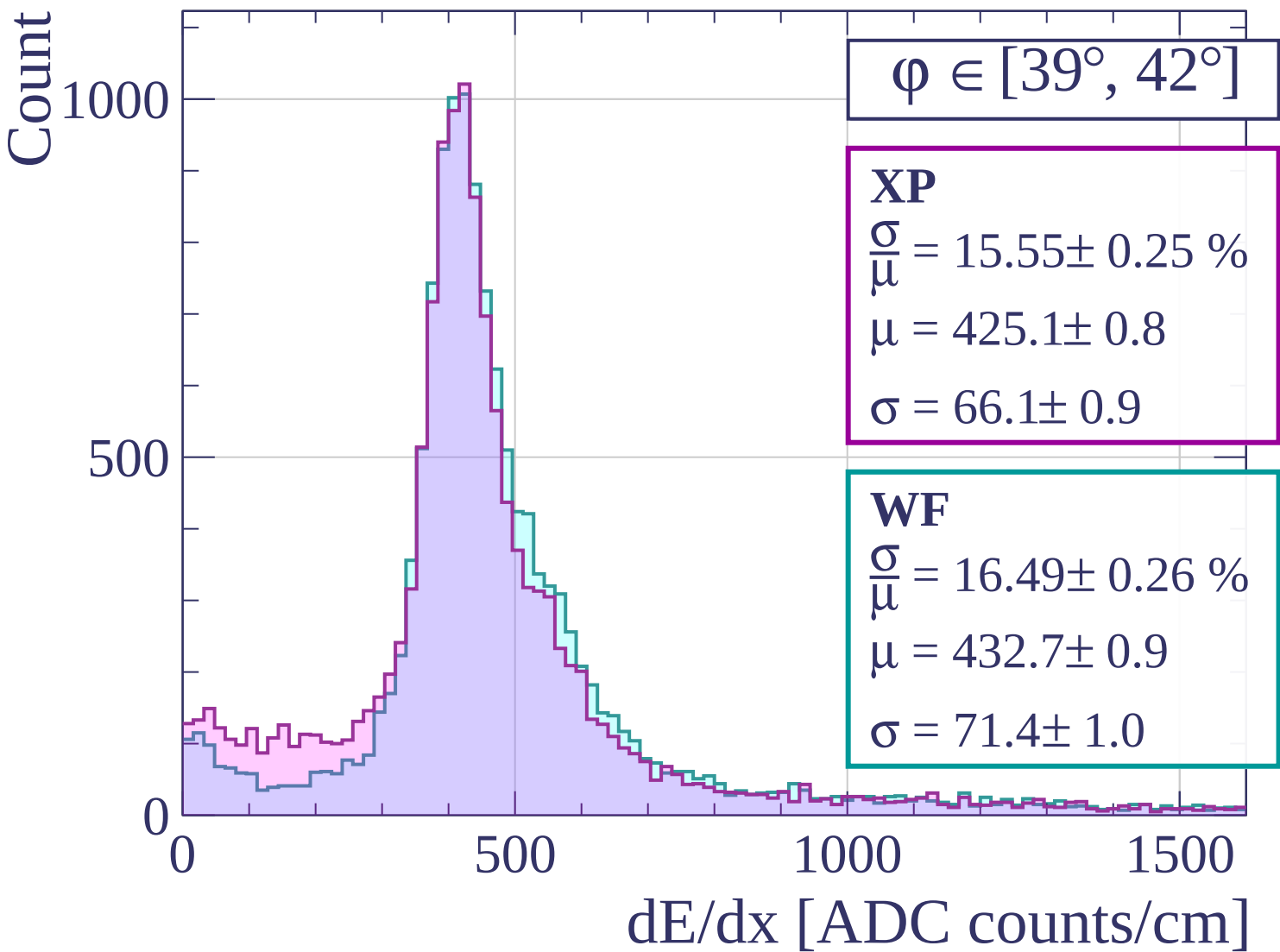


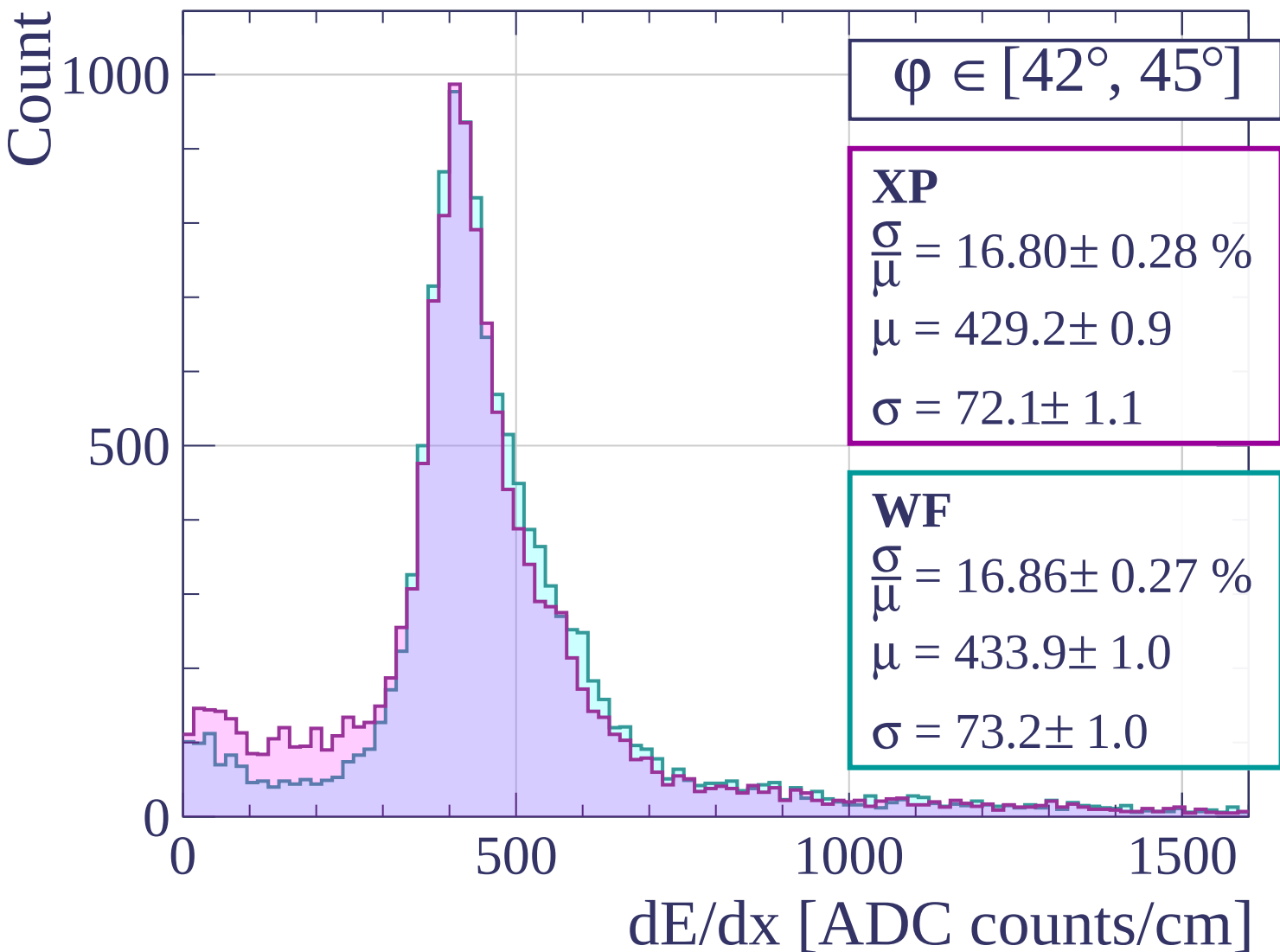












Count

$\varphi \in [45^\circ, 48^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.86 \pm 0.32 \%$$

$$\mu = 429.0 \pm 1.0$$

$$\sigma = 76.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 18.13 \pm 0.32 \%$$

$$\mu = 437.2 \pm 1.1$$

$$\sigma = 79.3 \pm 1.2$$

800

600

400

200

0

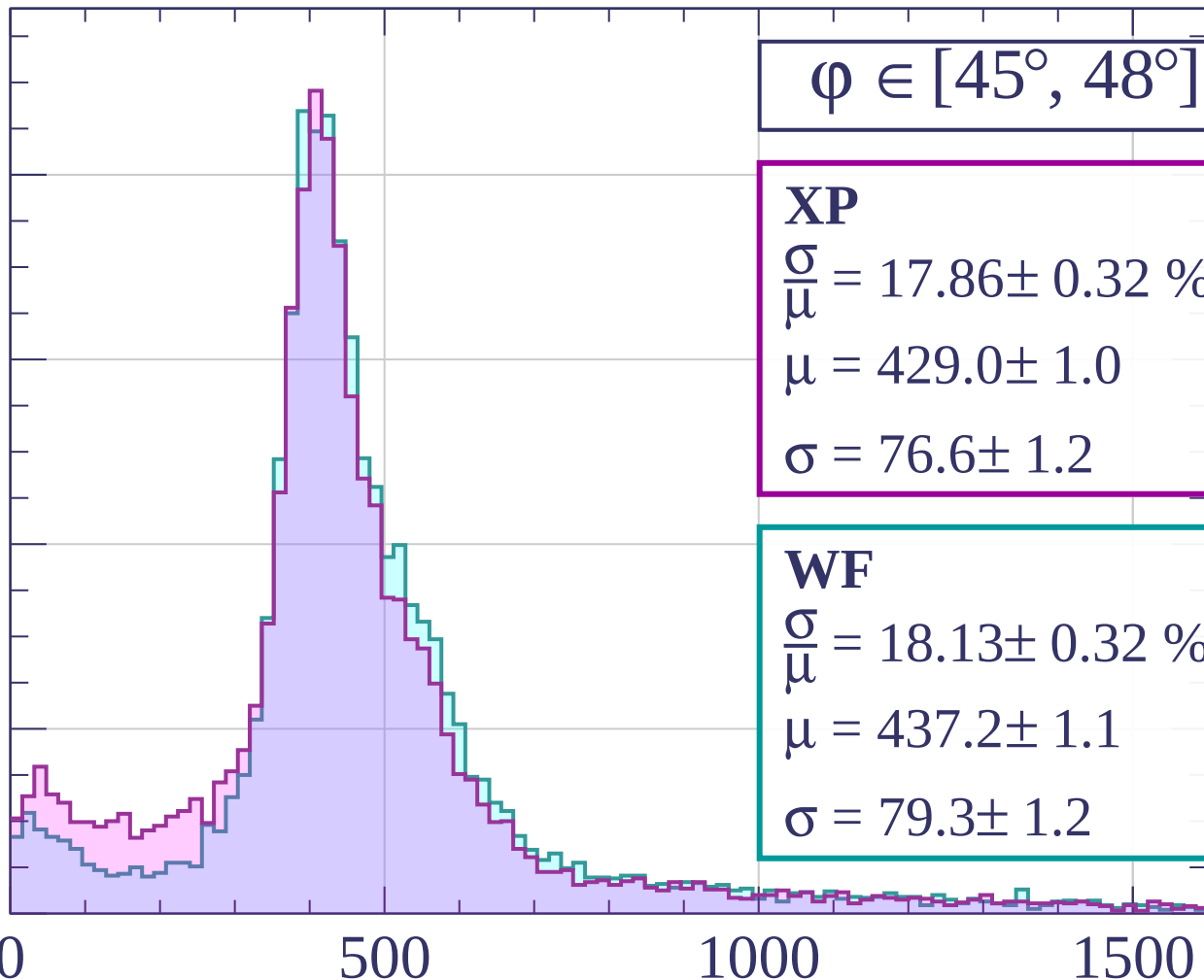
0

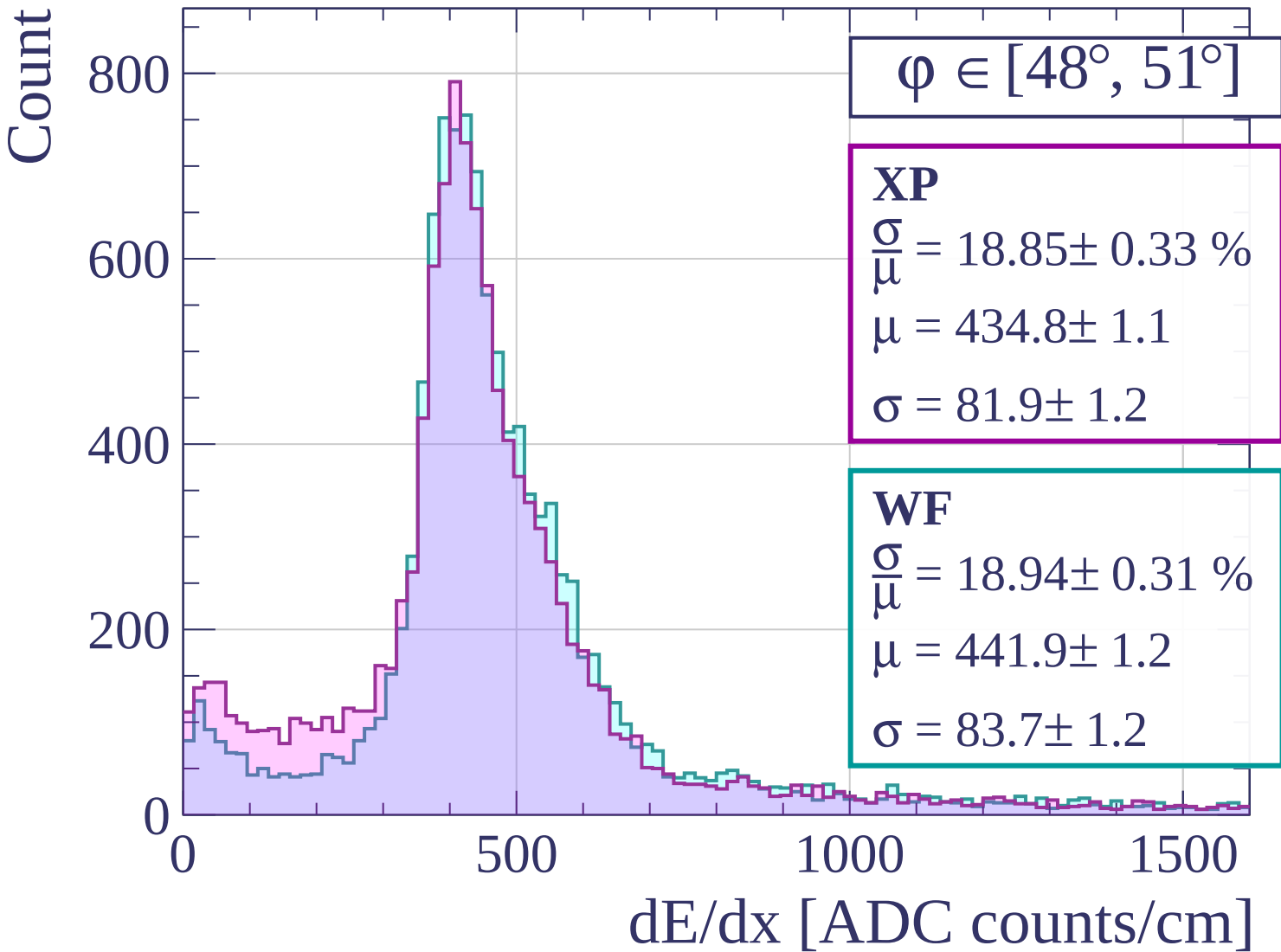
500

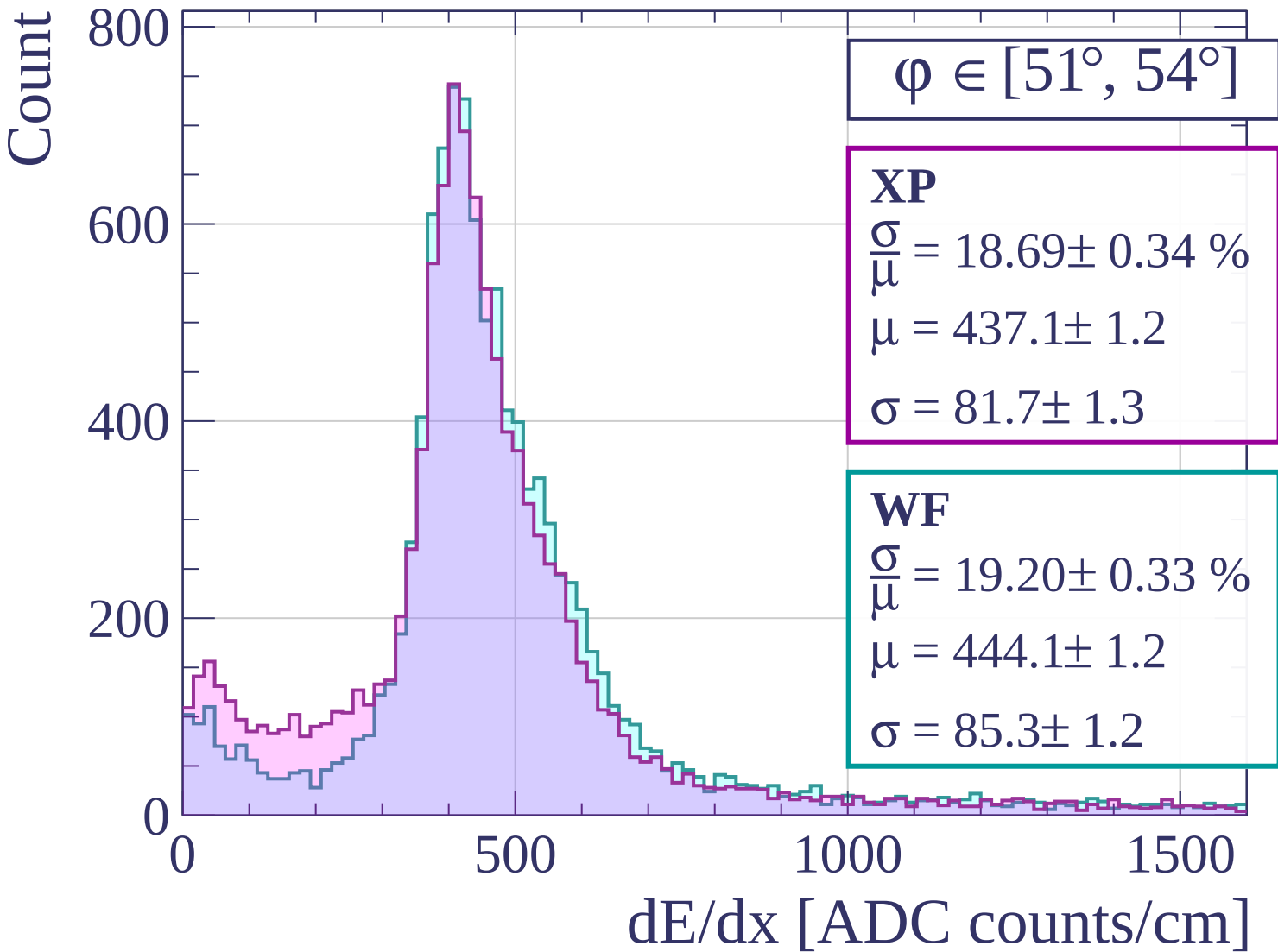
1000

1500

dE/dx [ADC counts/cm]







Count

$\phi \in [54^\circ, 57^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.05 \pm 0.37 \%$$

$$\mu = 440.1 \pm 1.3$$

$$\sigma = 88.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 19.83 \pm 0.33 \%$$

$$\mu = 449.0 \pm 1.3$$

$$\sigma = 89.0 \pm 1.2$$

600

400

200

0

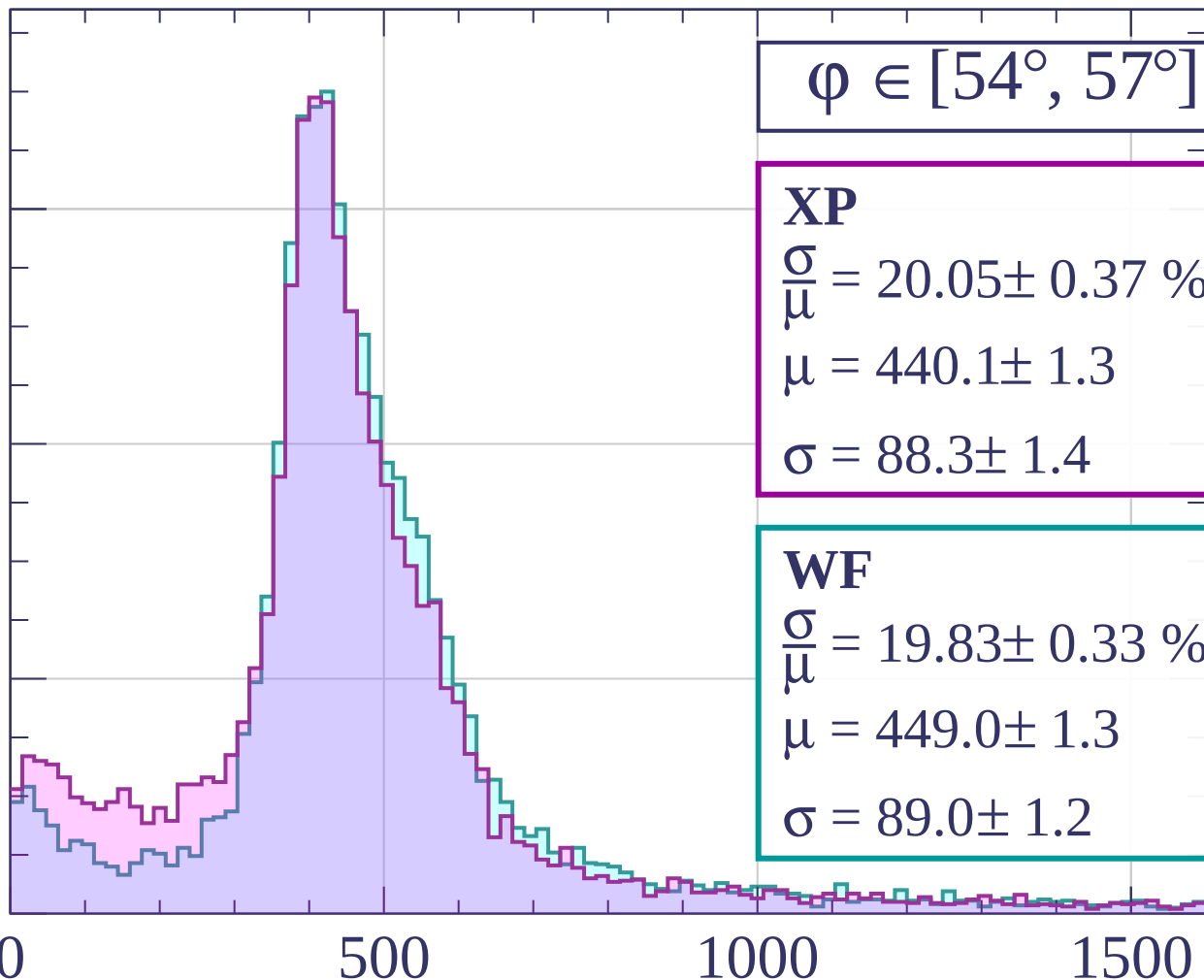
0

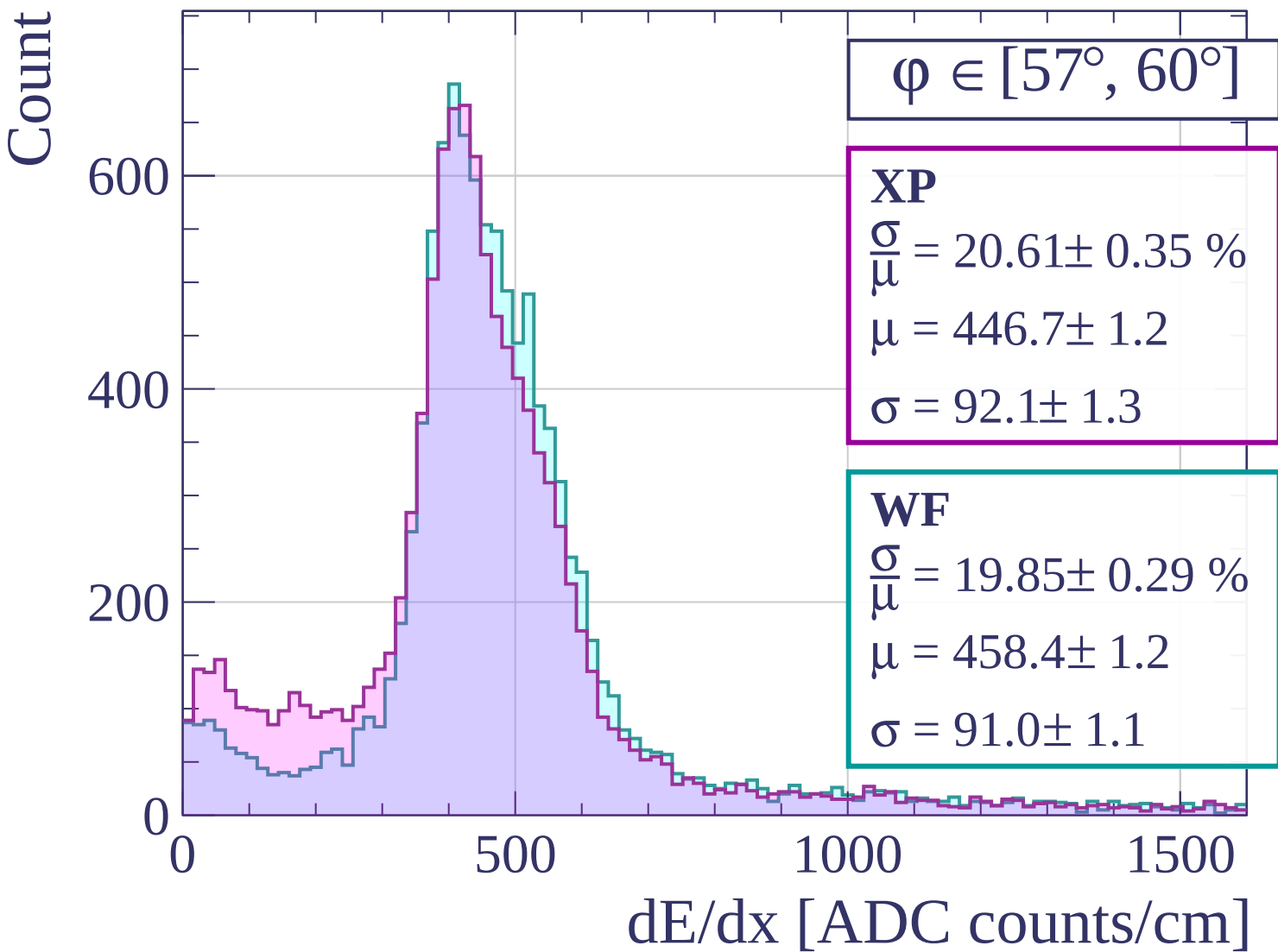
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\phi \in [60^\circ, 63^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.45 \pm 0.39 \%$$

$$\mu = 446.0 \pm 1.4$$

$$\sigma = 91.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 19.75 \pm 0.35 \%$$

$$\mu = 453.9 \pm 1.3$$

$$\sigma = 89.6 \pm 1.3$$

600

400

200

0

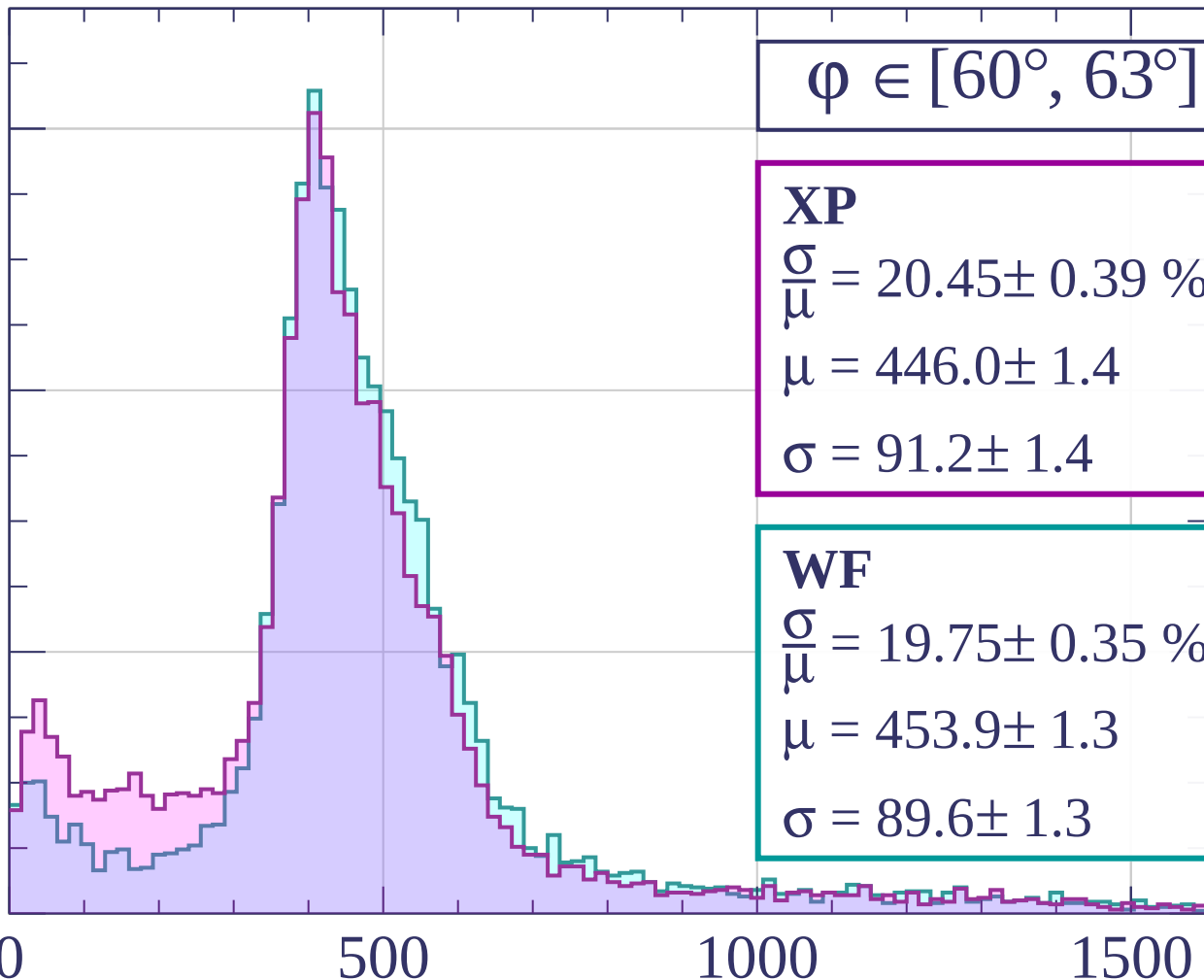
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [63^\circ, 66^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.81 \pm 0.43 \%$$

$$\mu = 447.1 \pm 1.5$$

$$\sigma = 93.1 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 19.56 \pm 0.37 \%$$

$$\mu = 455.5 \pm 1.4$$

$$\sigma = 89.1 \pm 1.4$$

400

200

0

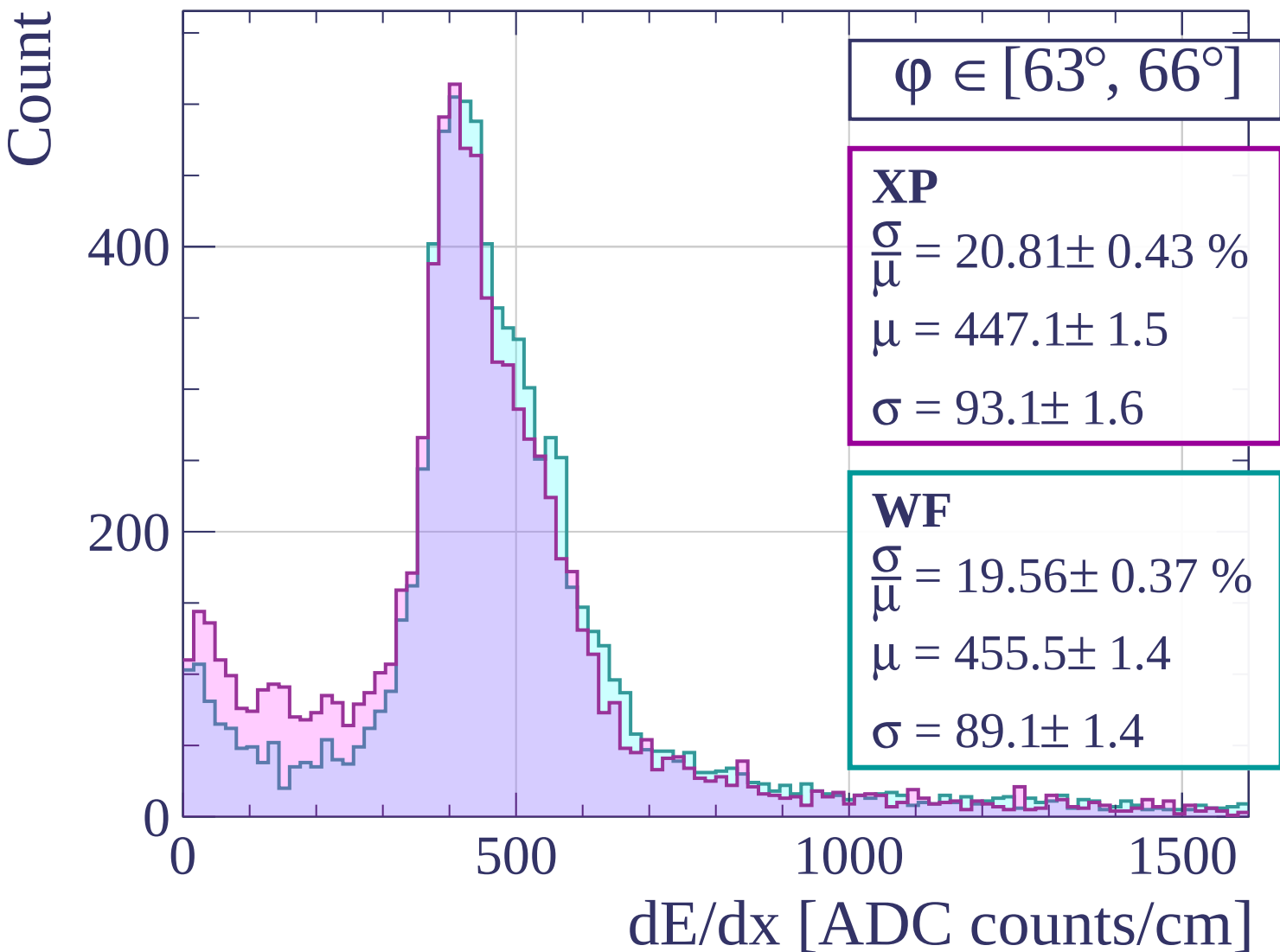
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [66^\circ, 69^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.53 \pm 0.43 \%$$

$$\mu = 448.1 \pm 1.5$$

$$\sigma = 92.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 20.07 \pm 0.37 \%$$

$$\mu = 459.4 \pm 1.5$$

$$\sigma = 92.2 \pm 1.4$$

400

200

0

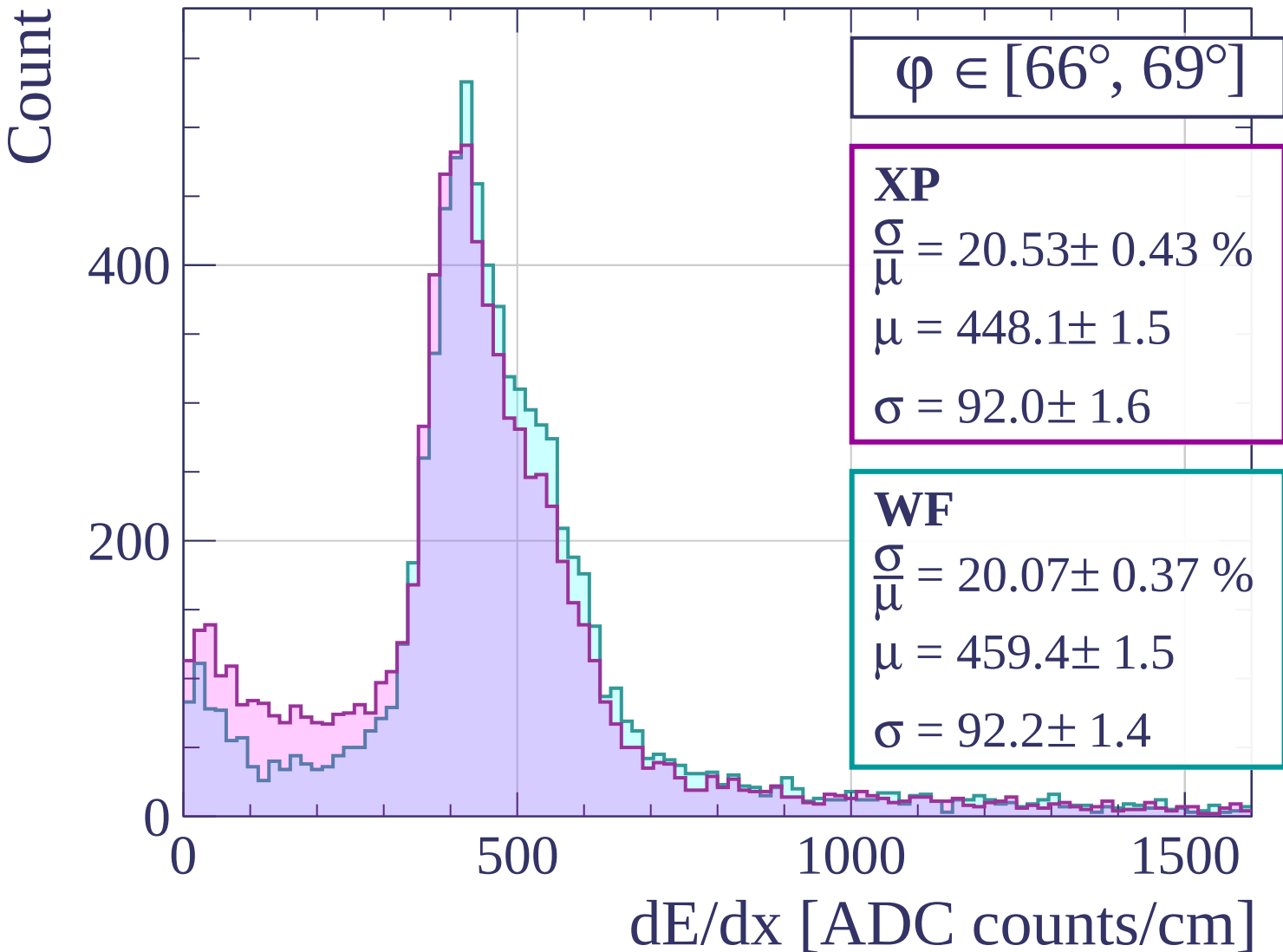
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [69^\circ, 72^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.79 \pm 0.44 \%$$

$$\mu = 449.6 \pm 1.6$$

$$\sigma = 93.5 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 19.83 \pm 0.37 \%$$

$$\mu = 462.4 \pm 1.5$$

$$\sigma = 91.7 \pm 1.4$$

400

200

0

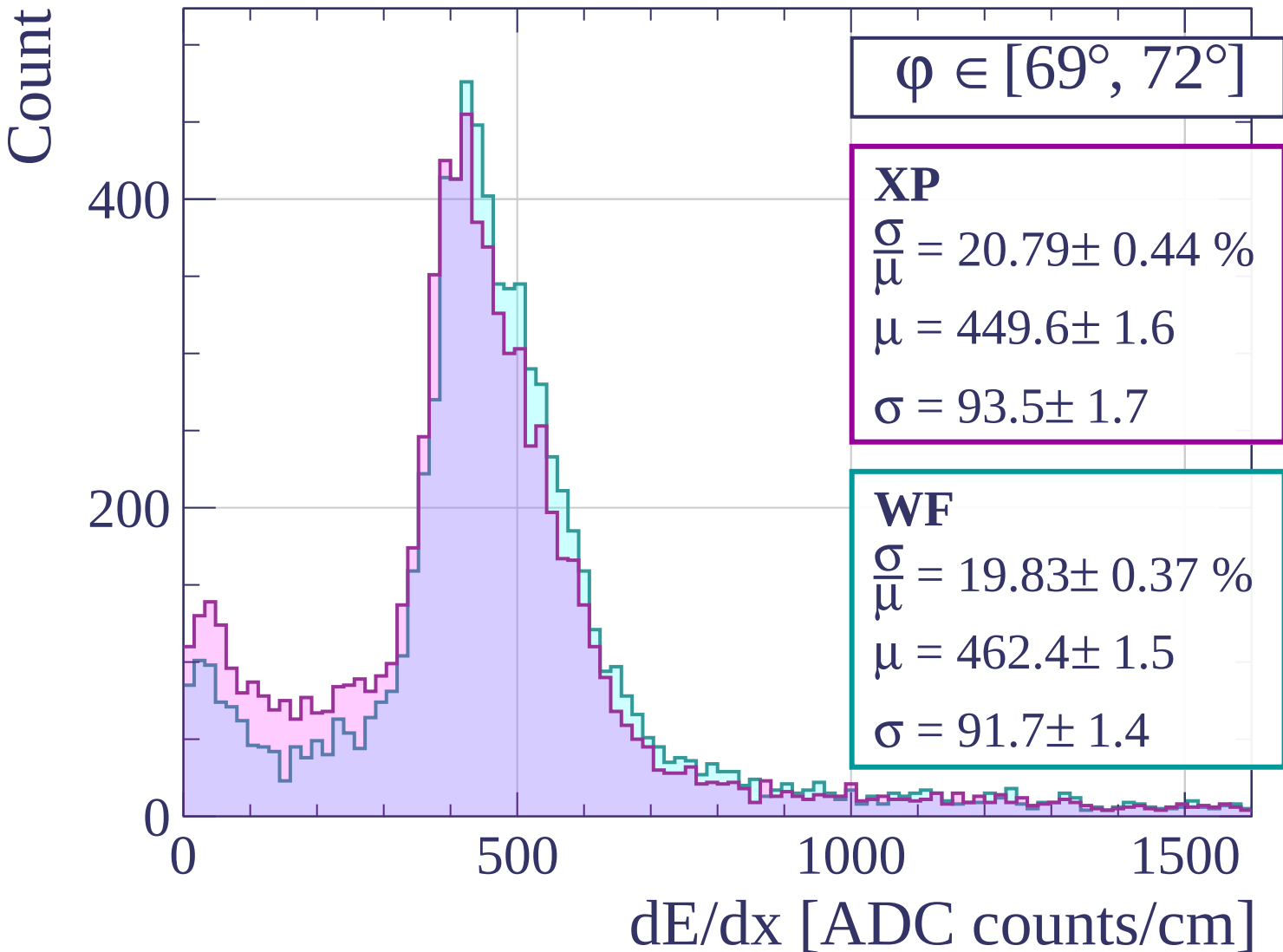
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [72^\circ, 75^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.32 \pm 0.43 \%$$

$$\mu = 452.0 \pm 1.6$$

$$\sigma = 91.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 19.46 \pm 0.37 \%$$

$$\mu = 466.8 \pm 1.5$$

$$\sigma = 90.8 \pm 1.4$$

400

300

200

100

0

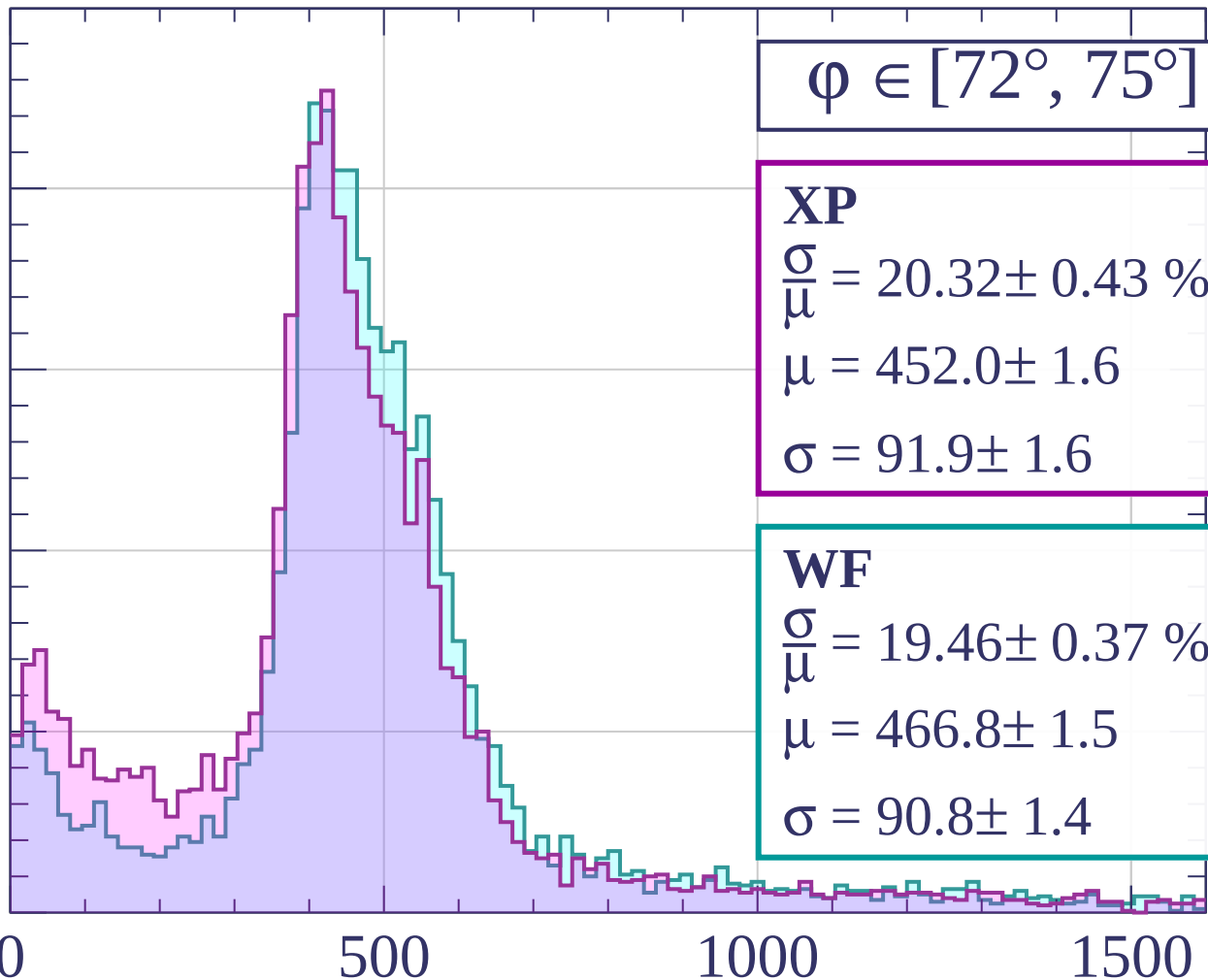
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [75^\circ, 78^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.68 \pm 0.40 \%$$

$$\mu = 455.0 \pm 1.5$$

$$\sigma = 94.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 20.09 \pm 0.36 \%$$

$$\mu = 468.6 \pm 1.5$$

$$\sigma = 94.1 \pm 1.4$$

400

300

200

100

0

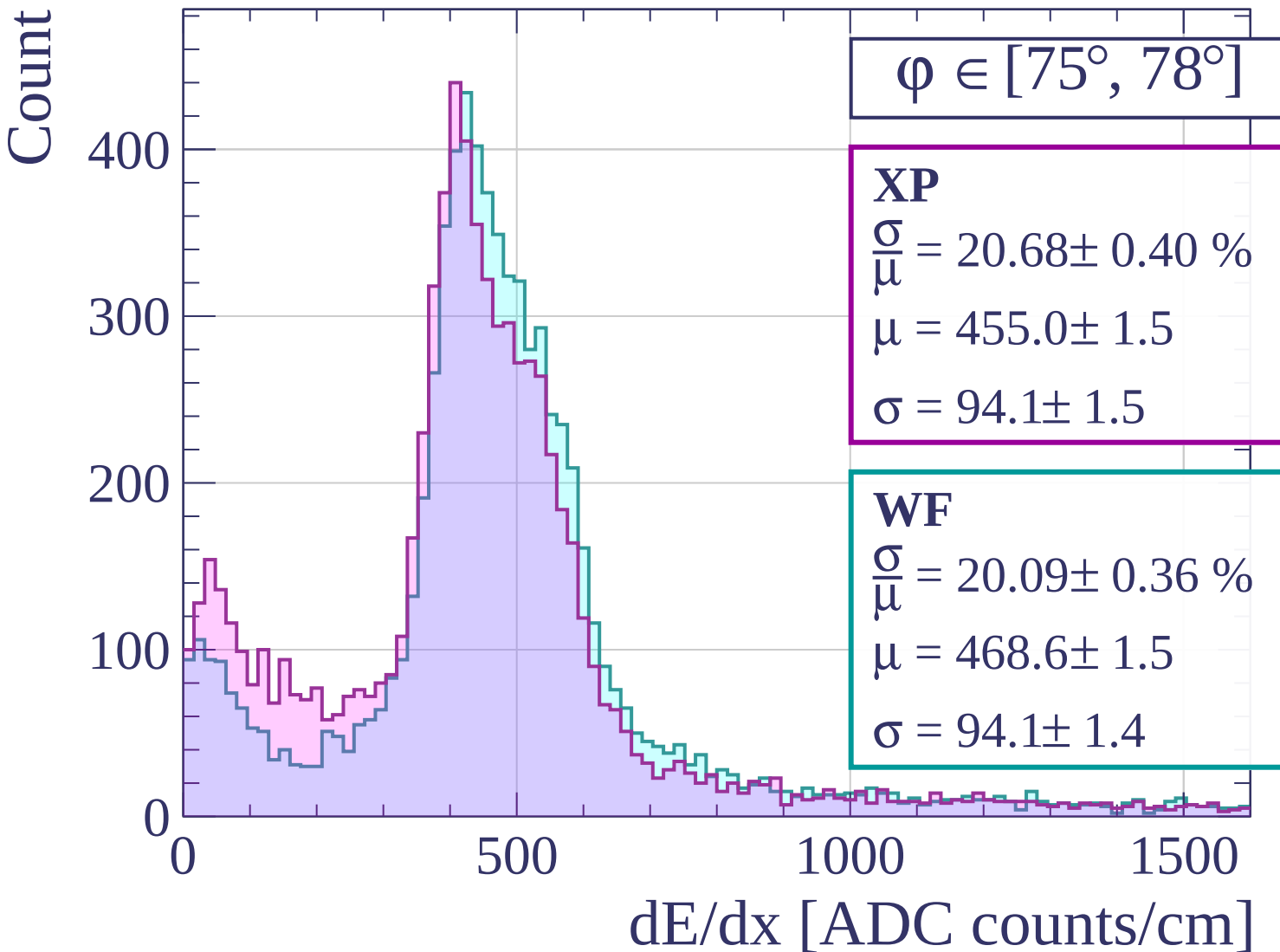
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [78^\circ, 81^\circ]$

XP

$$\frac{\sigma}{\mu} = 20.90 \pm 0.42 \%$$

$$\mu = 453.1 \pm 1.5$$

$$\sigma = 94.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 20.07 \pm 0.40 \%$$

$$\mu = 469.1 \pm 1.6$$

$$\sigma = 94.2 \pm 1.6$$

400

300

200

100

0

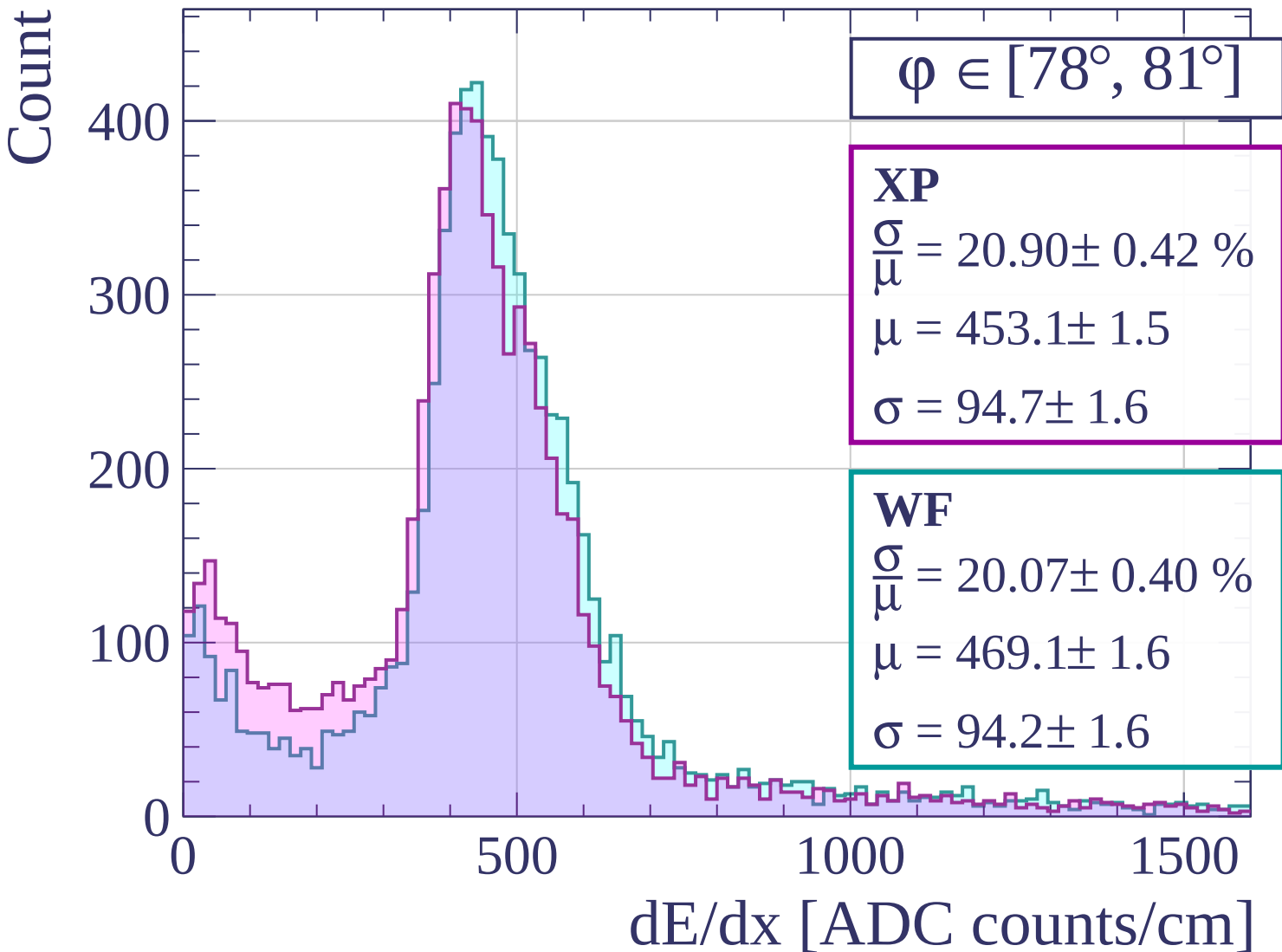
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [81^\circ, 84^\circ]$

XP

$$\frac{\sigma}{\mu} = 21.98 \pm 0.43 \%$$

$$\mu = 455.2 \pm 1.6$$

$$\sigma = 100.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 20.74 \pm 0.40 \%$$

$$\mu = 470.7 \pm 1.6$$

$$\sigma = 97.6 \pm 1.6$$

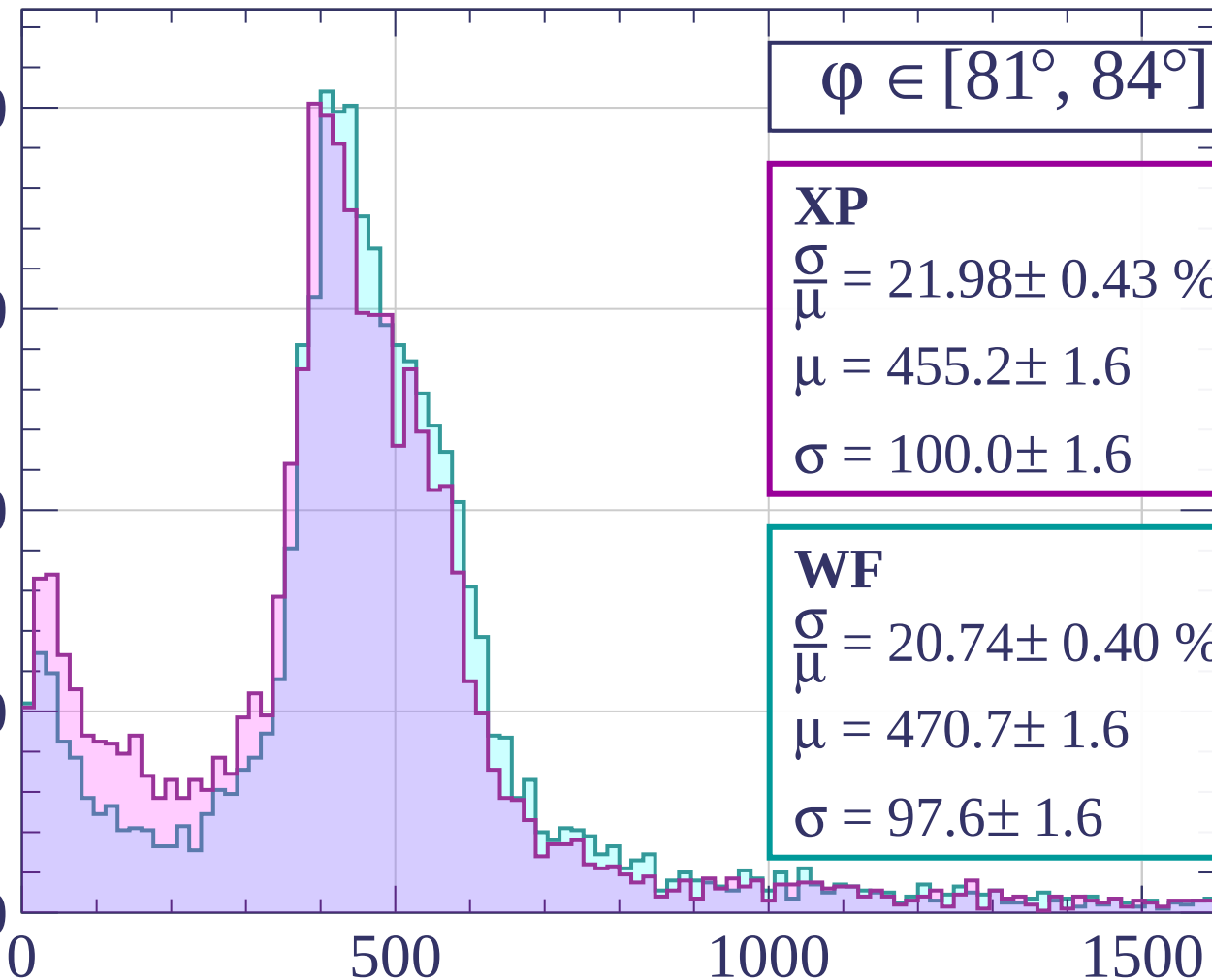
400

300

200

100

0

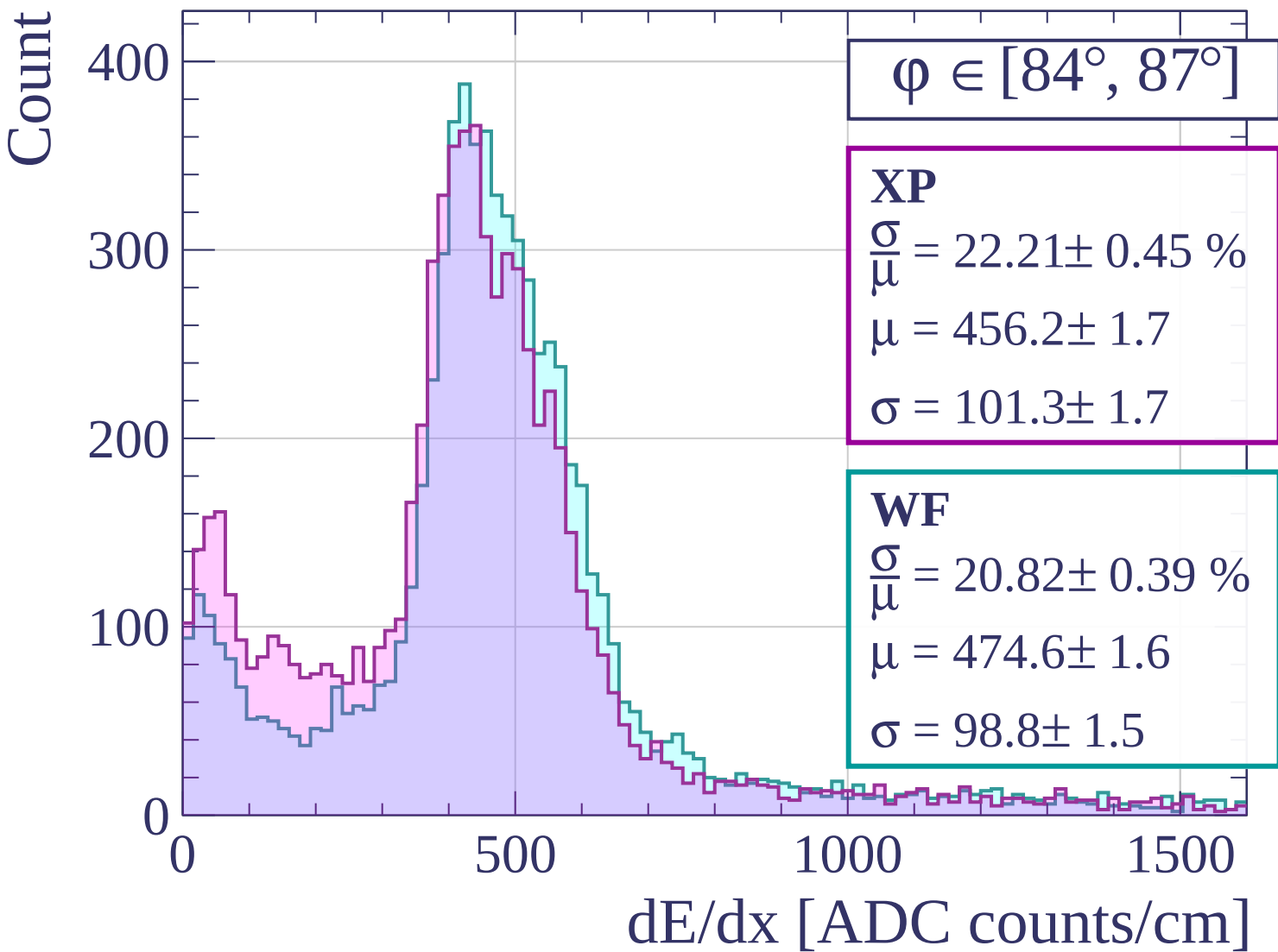


500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [87^\circ, 90^\circ]$

XP

$$\frac{\sigma}{\mu} = 21.22 \pm 0.45 \%$$

$$\mu = 455.2 \pm 1.6$$

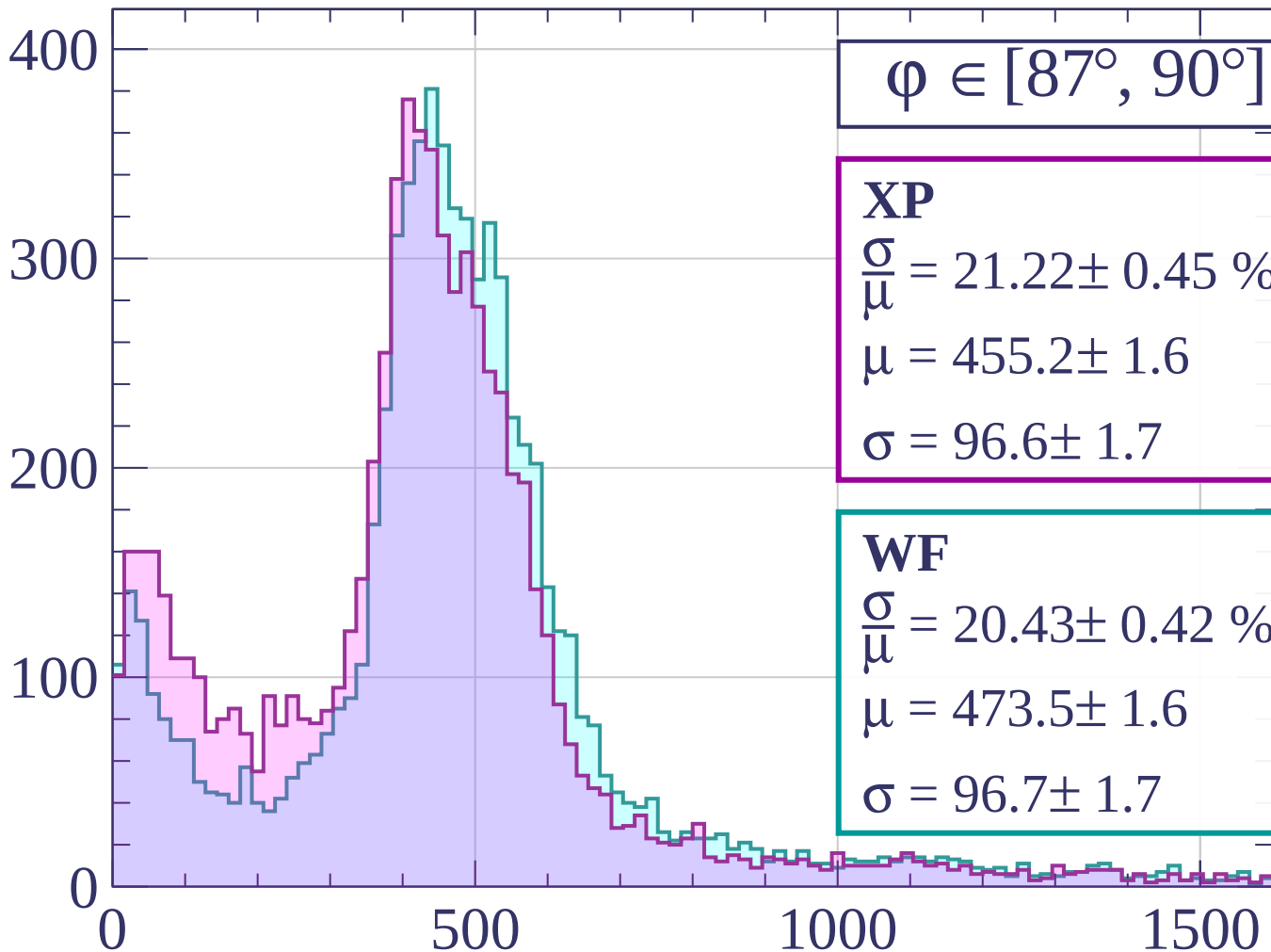
$$\sigma = 96.6 \pm 1.7$$

WF

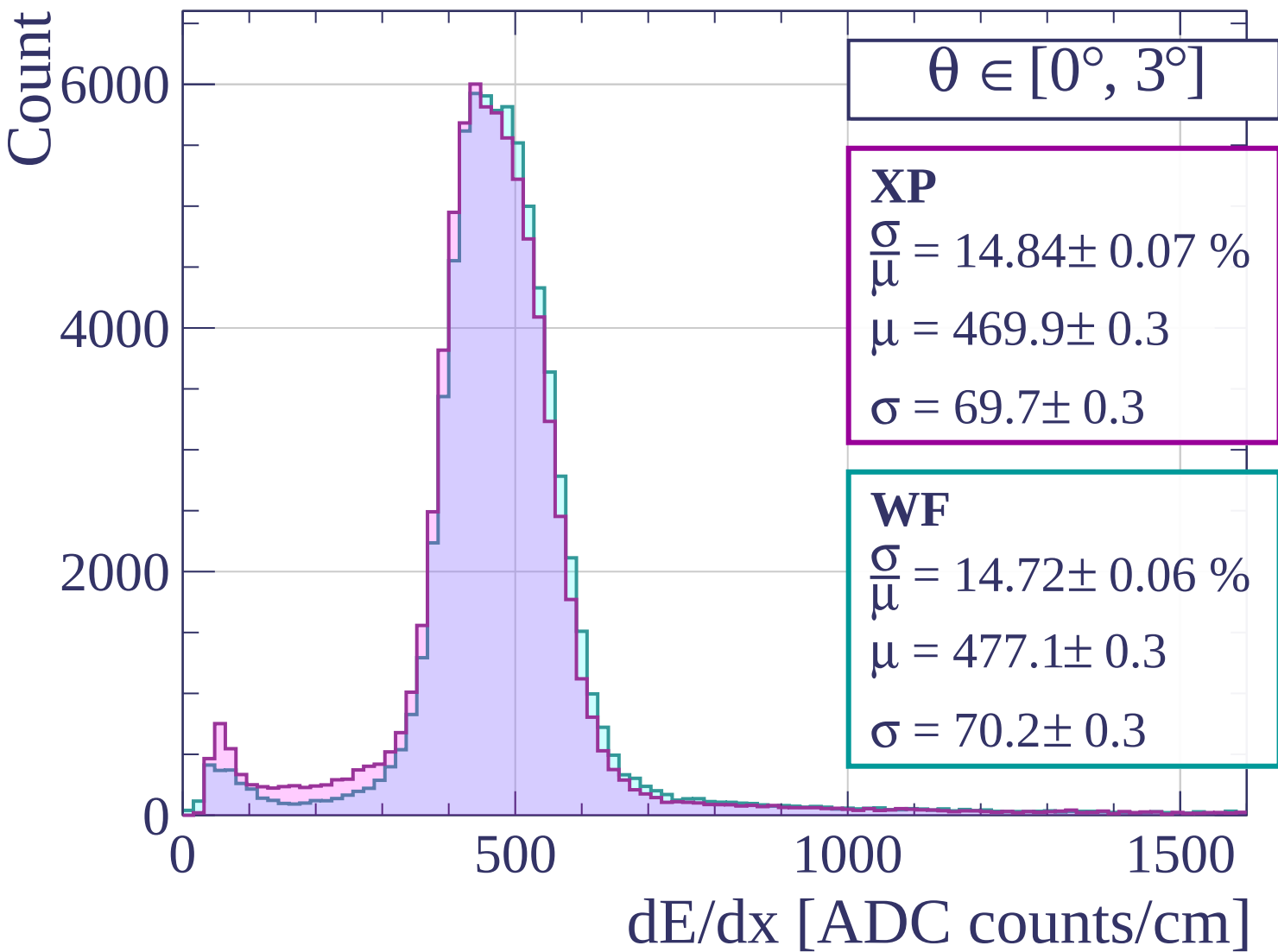
$$\frac{\sigma}{\mu} = 20.43 \pm 0.42 \%$$

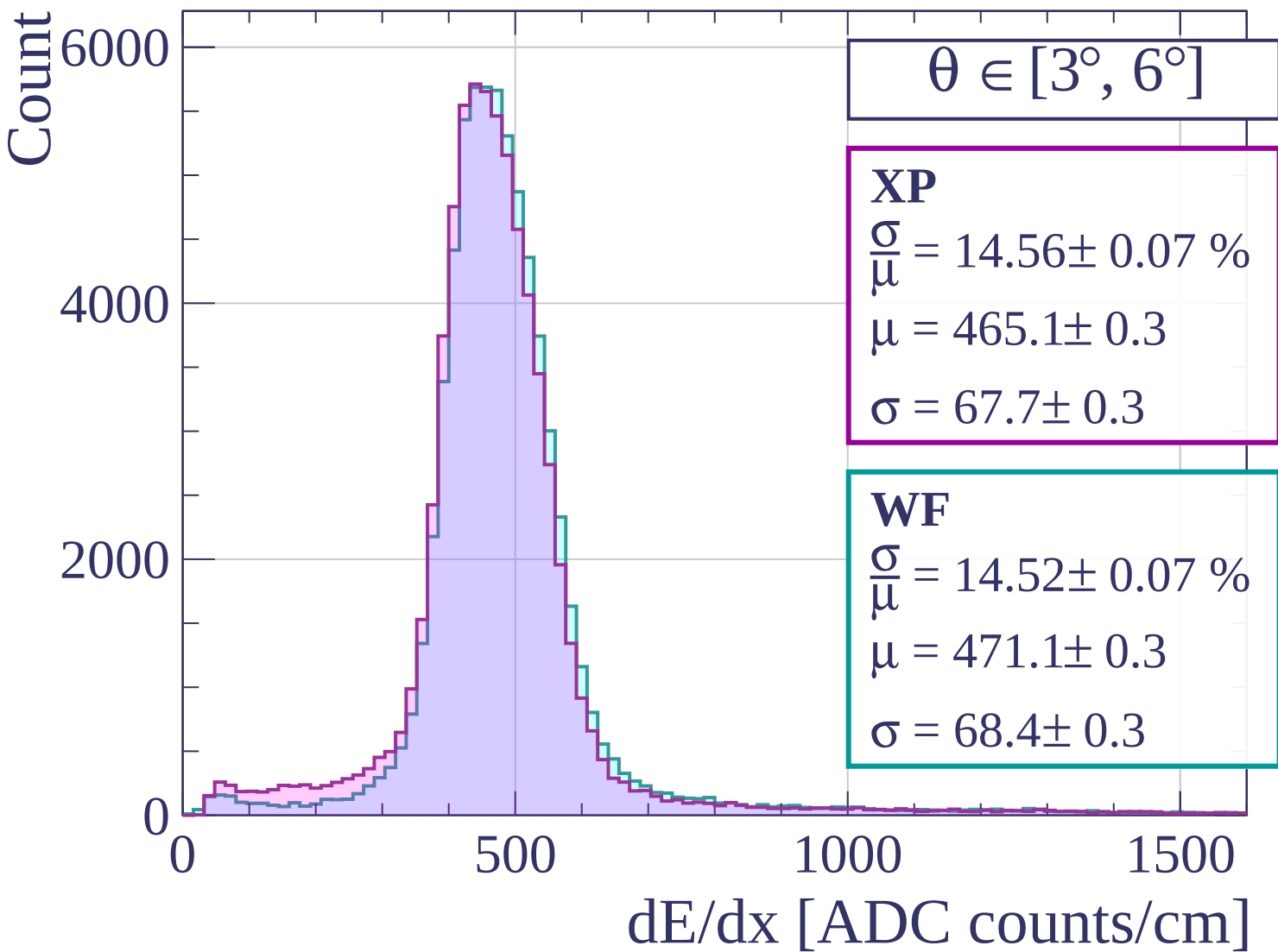
$$\mu = 473.5 \pm 1.6$$

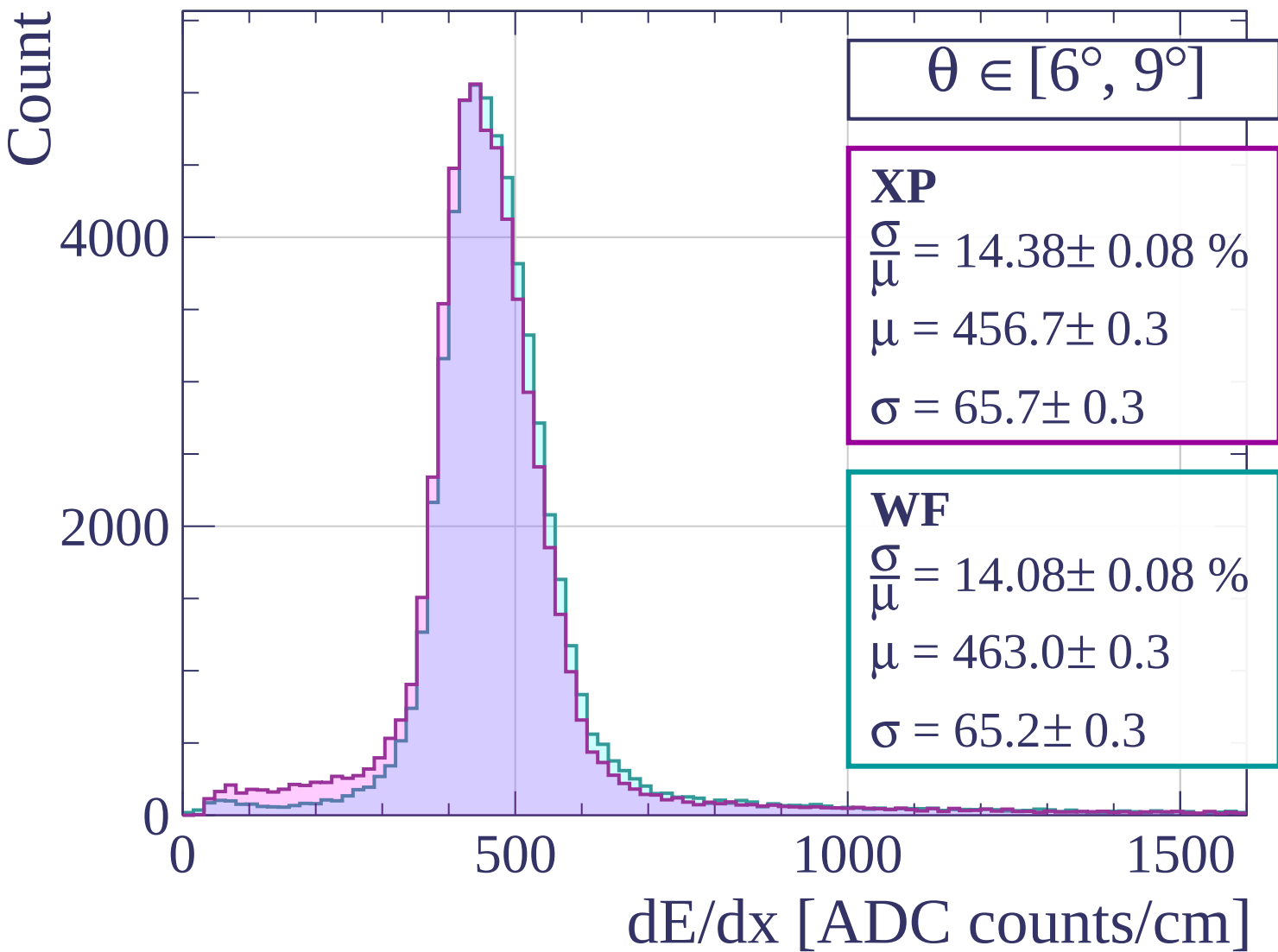
$$\sigma = 96.7 \pm 1.7$$

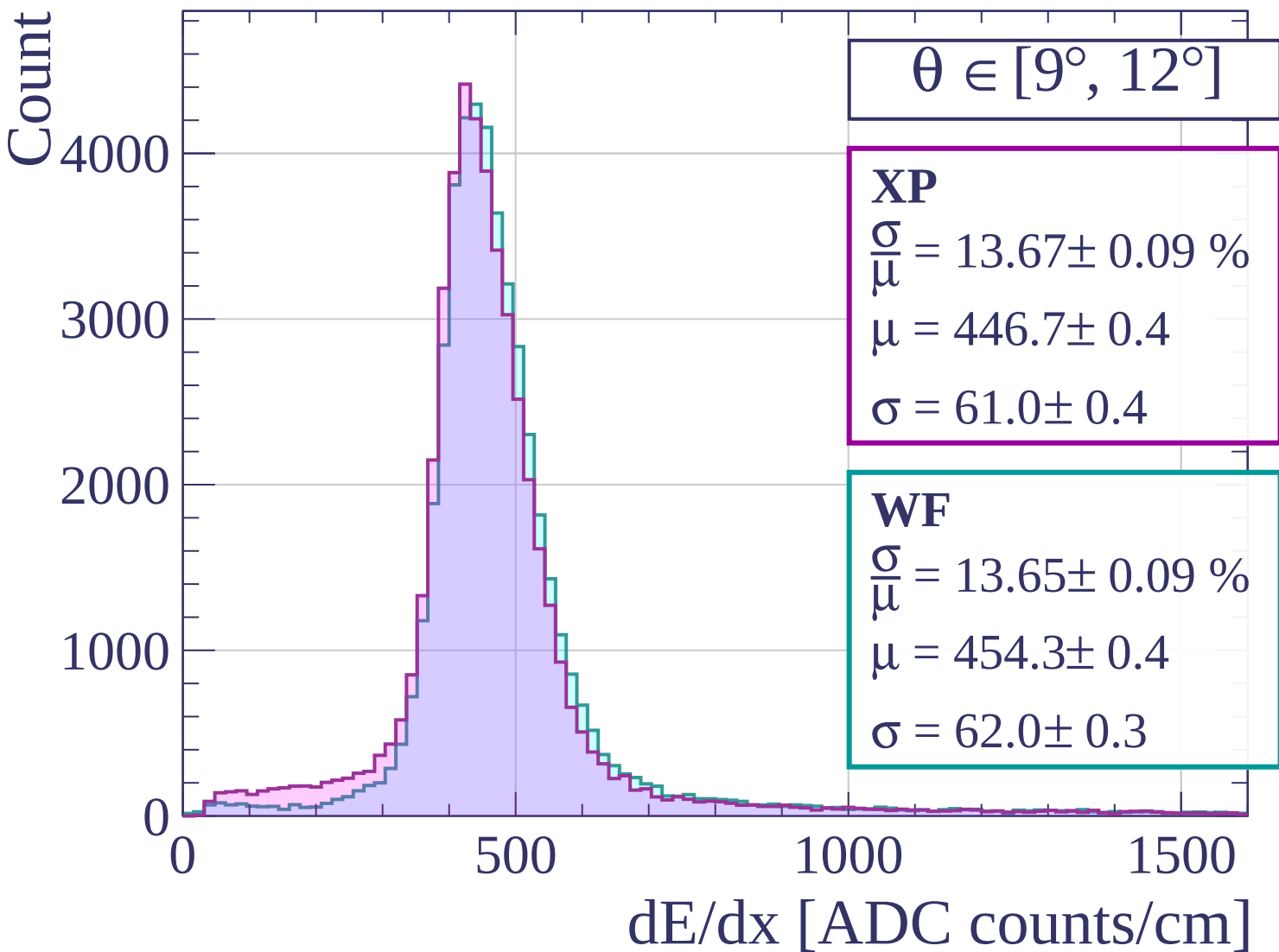


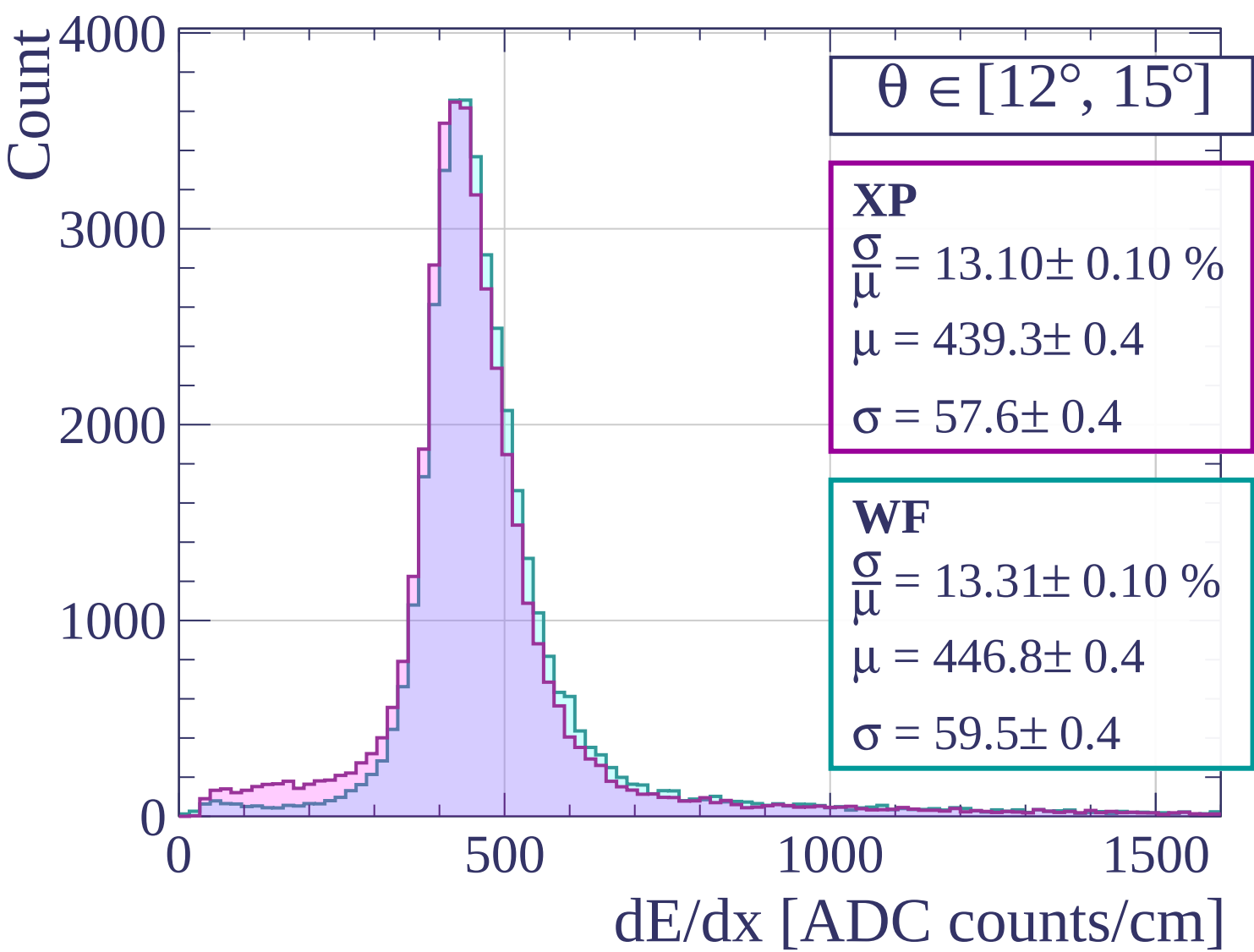
dE/dx [ADC counts/cm]

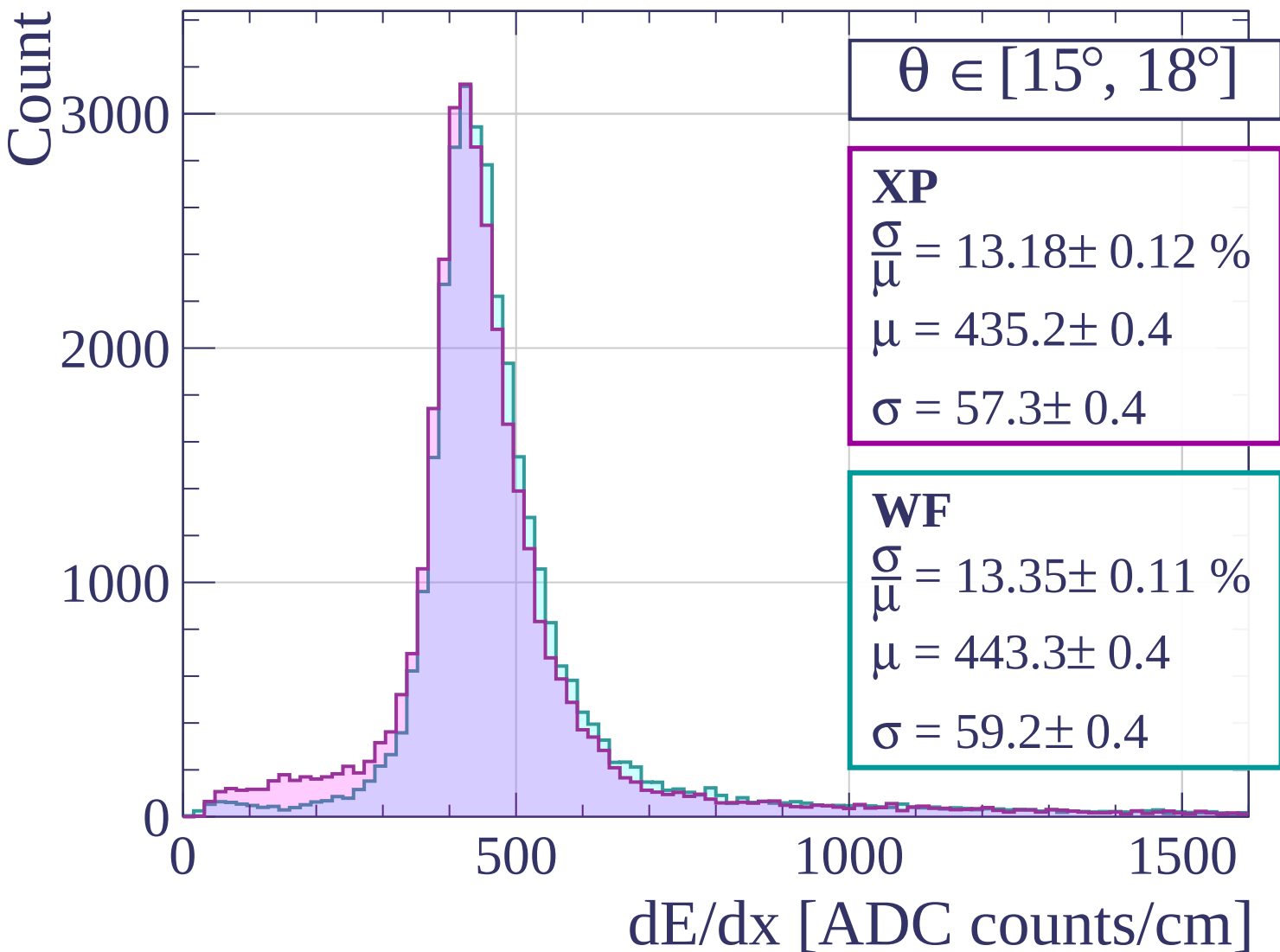




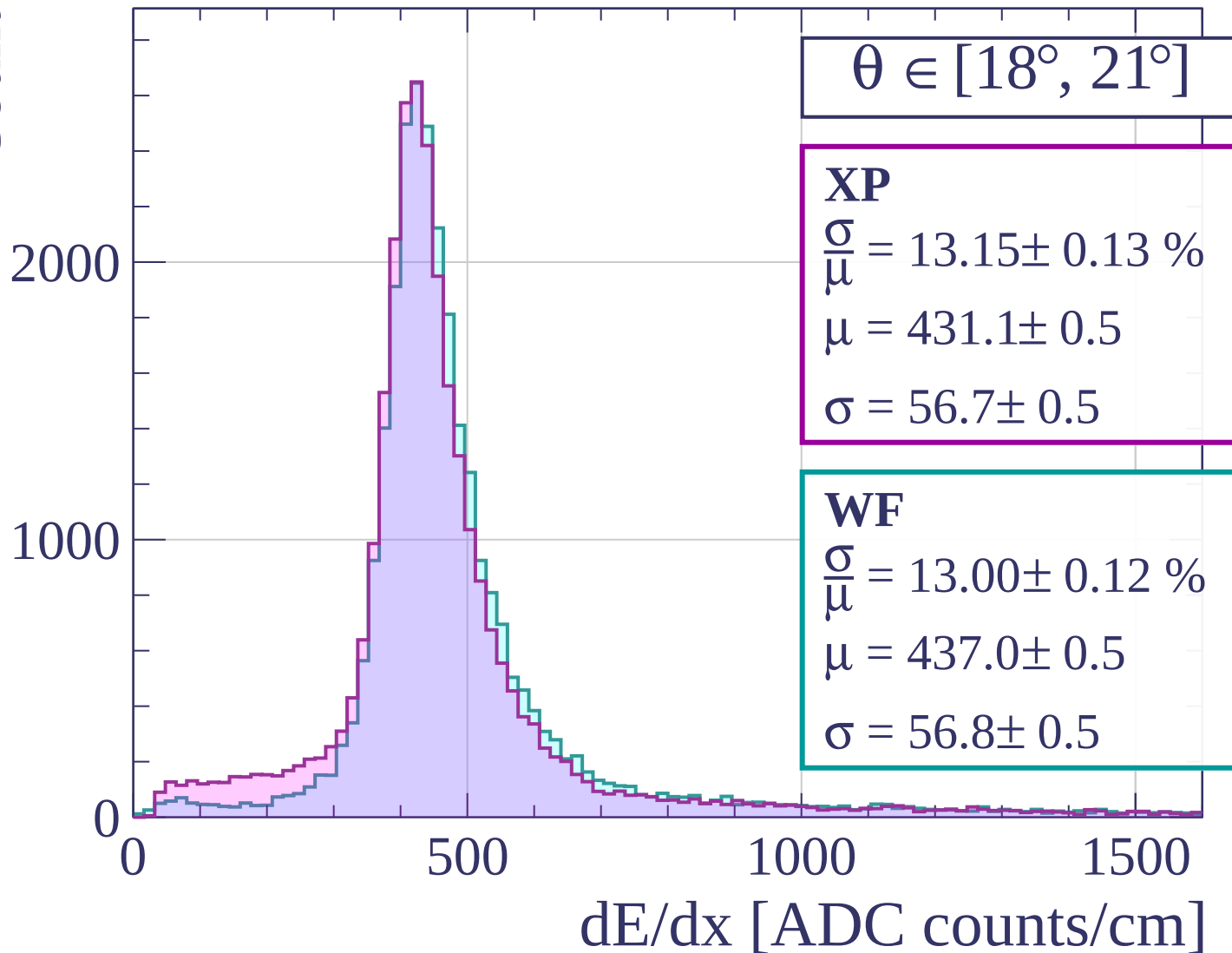


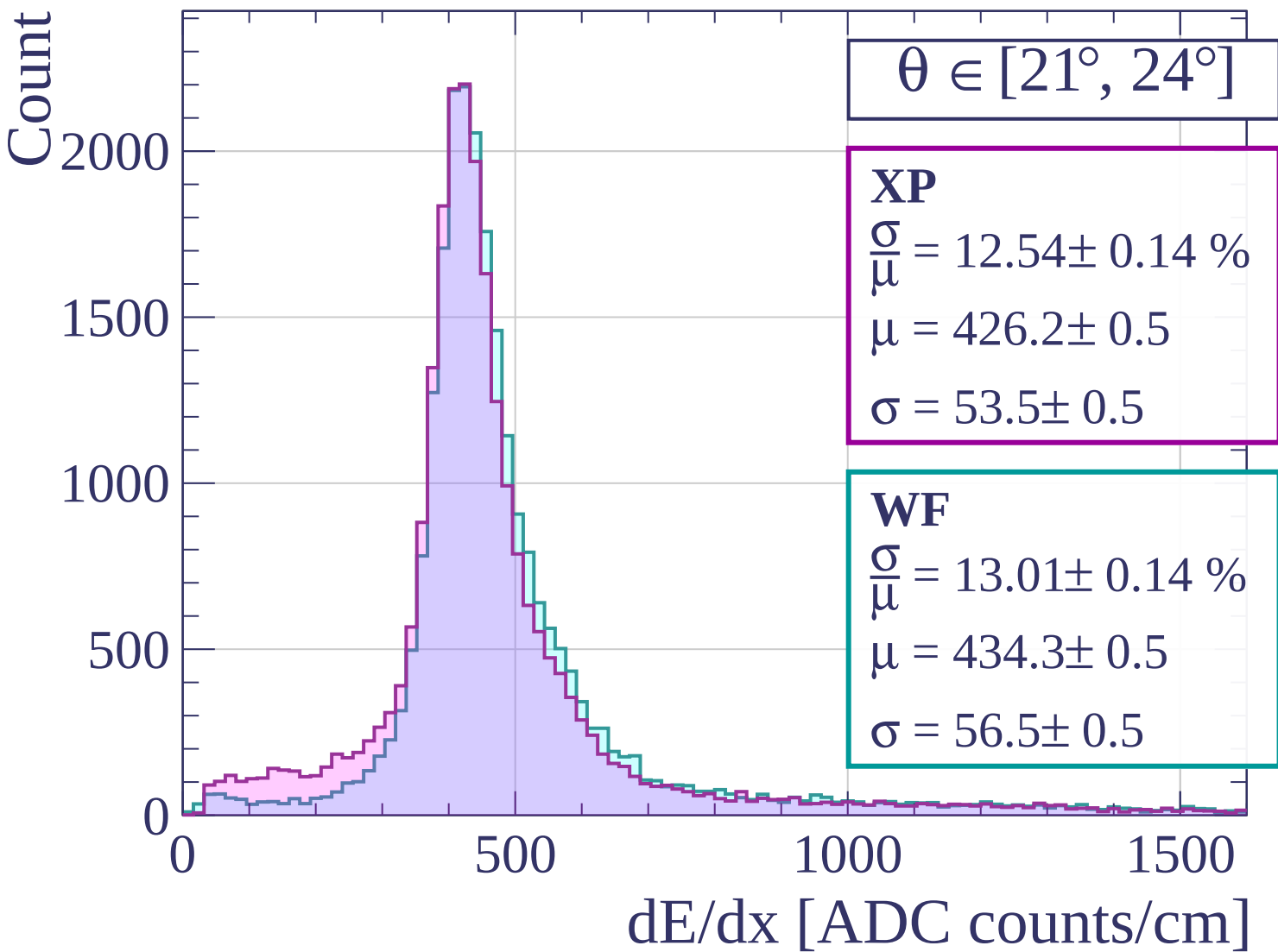


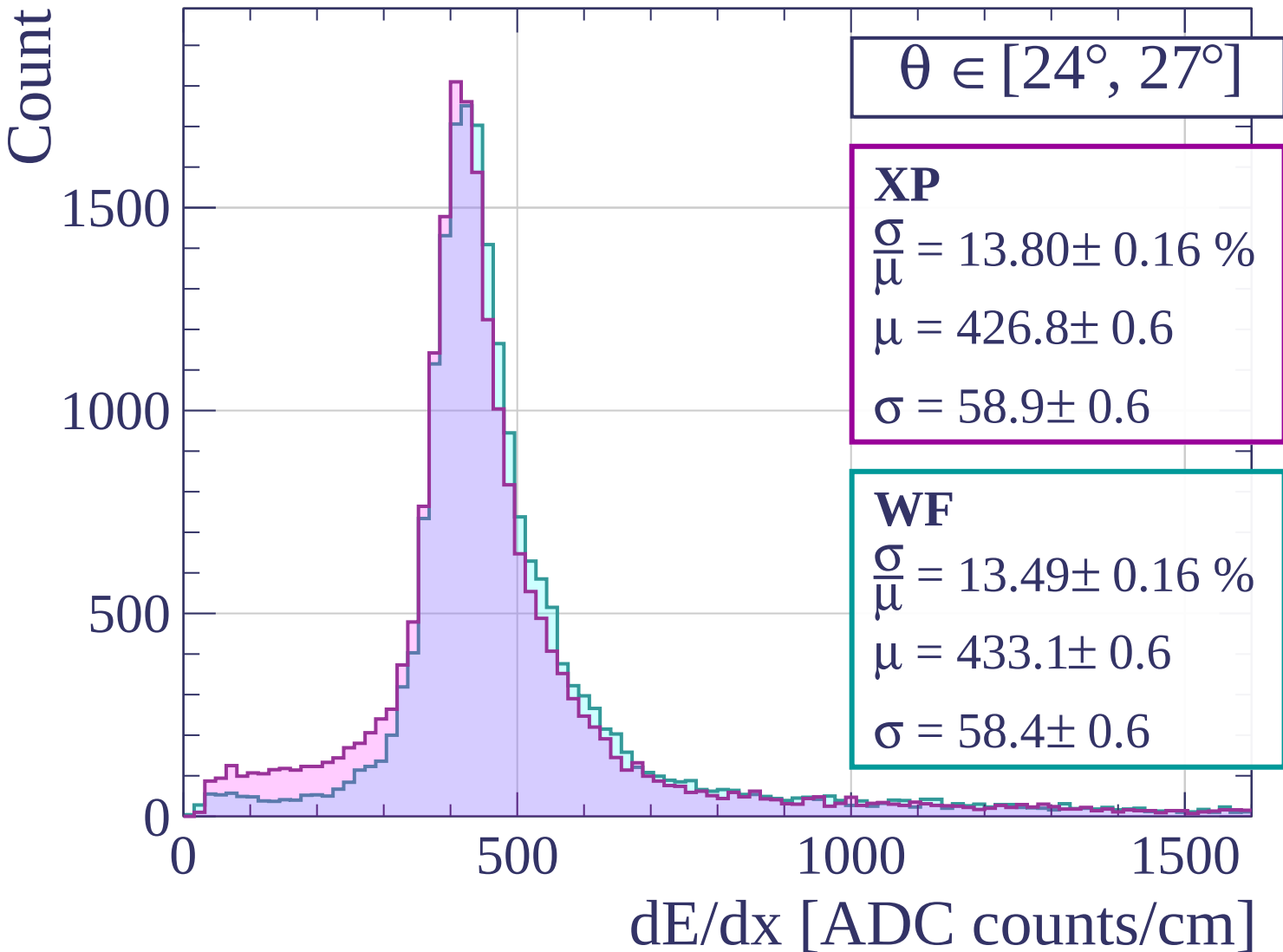


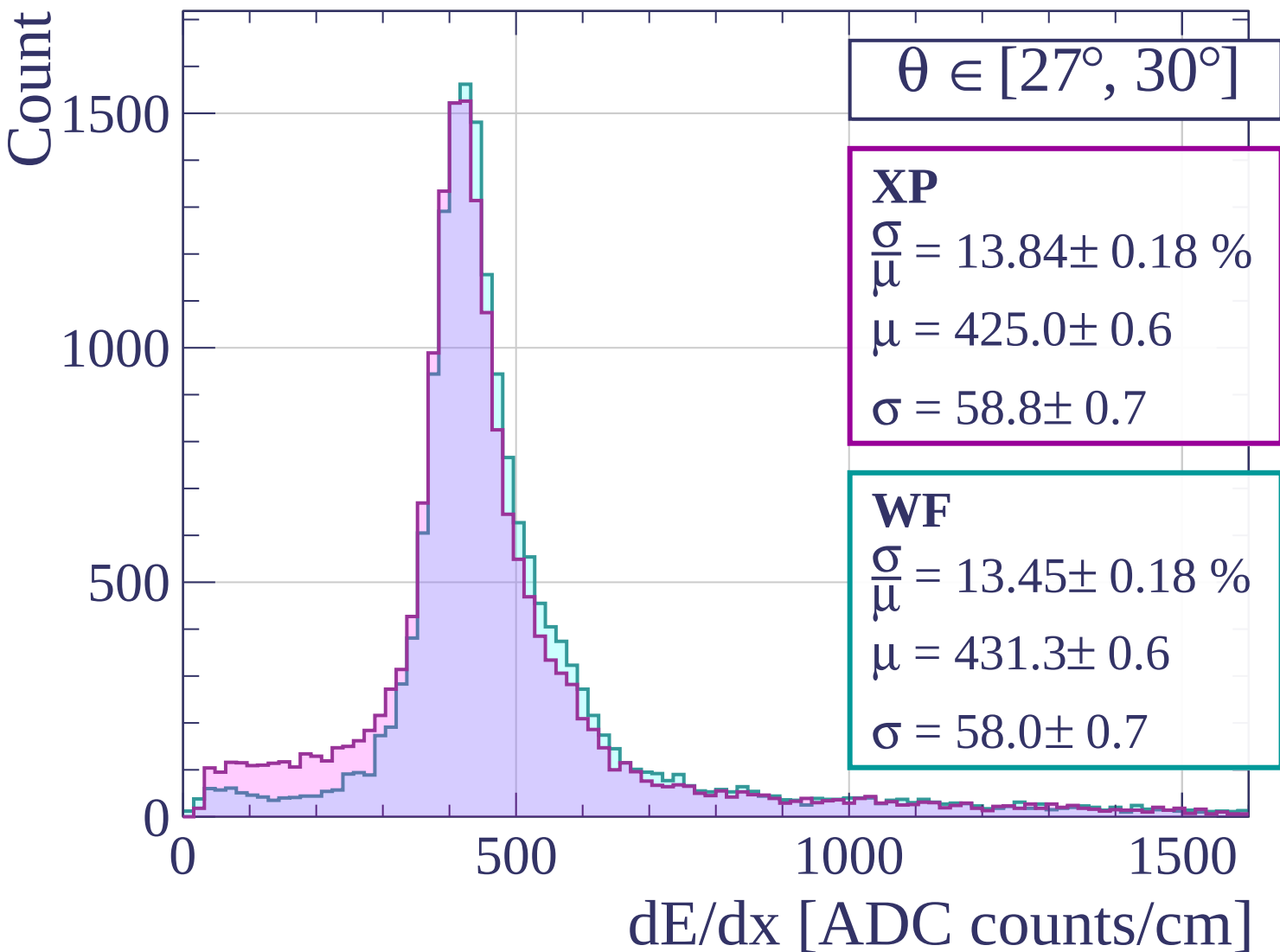


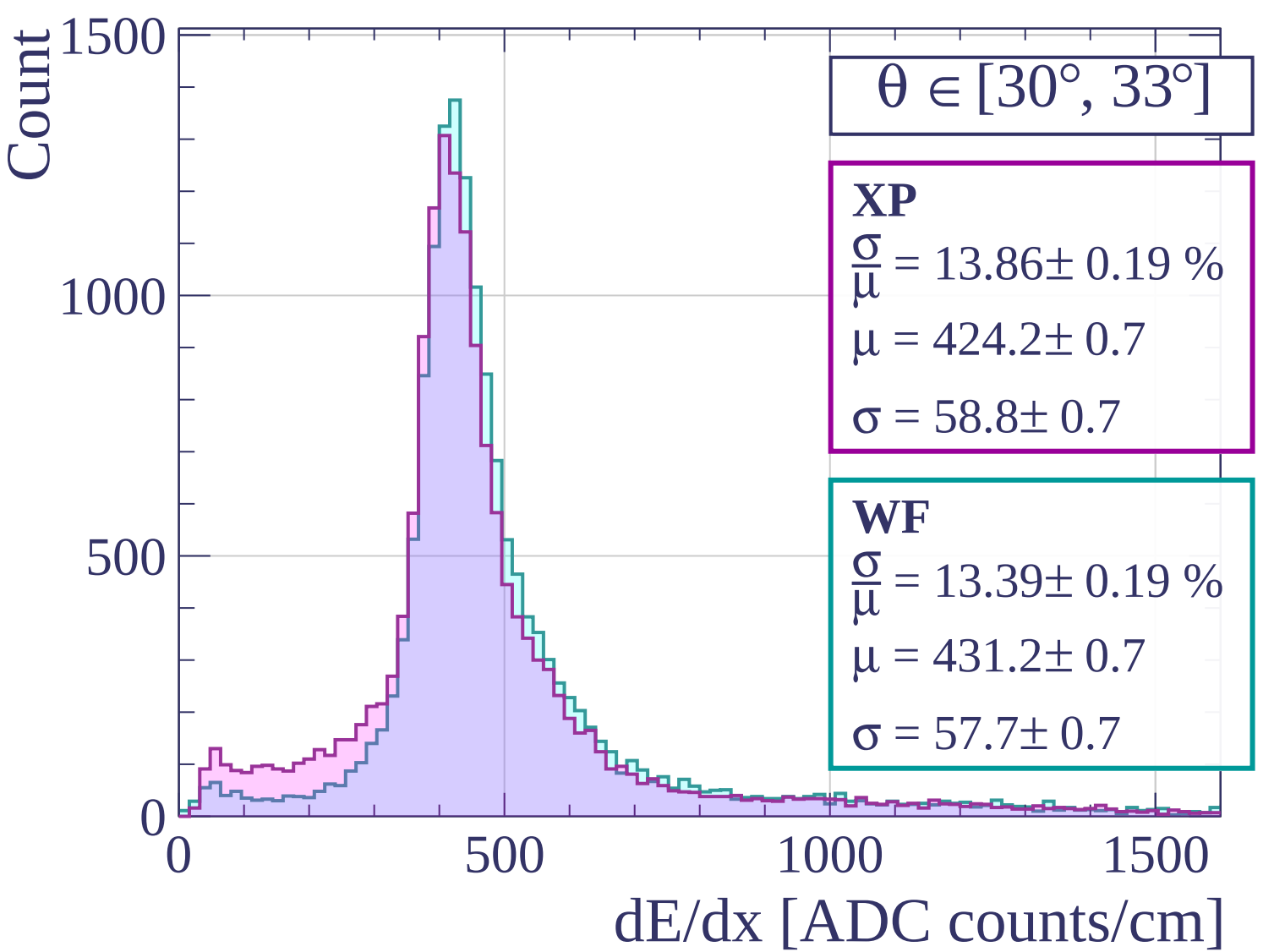
Count

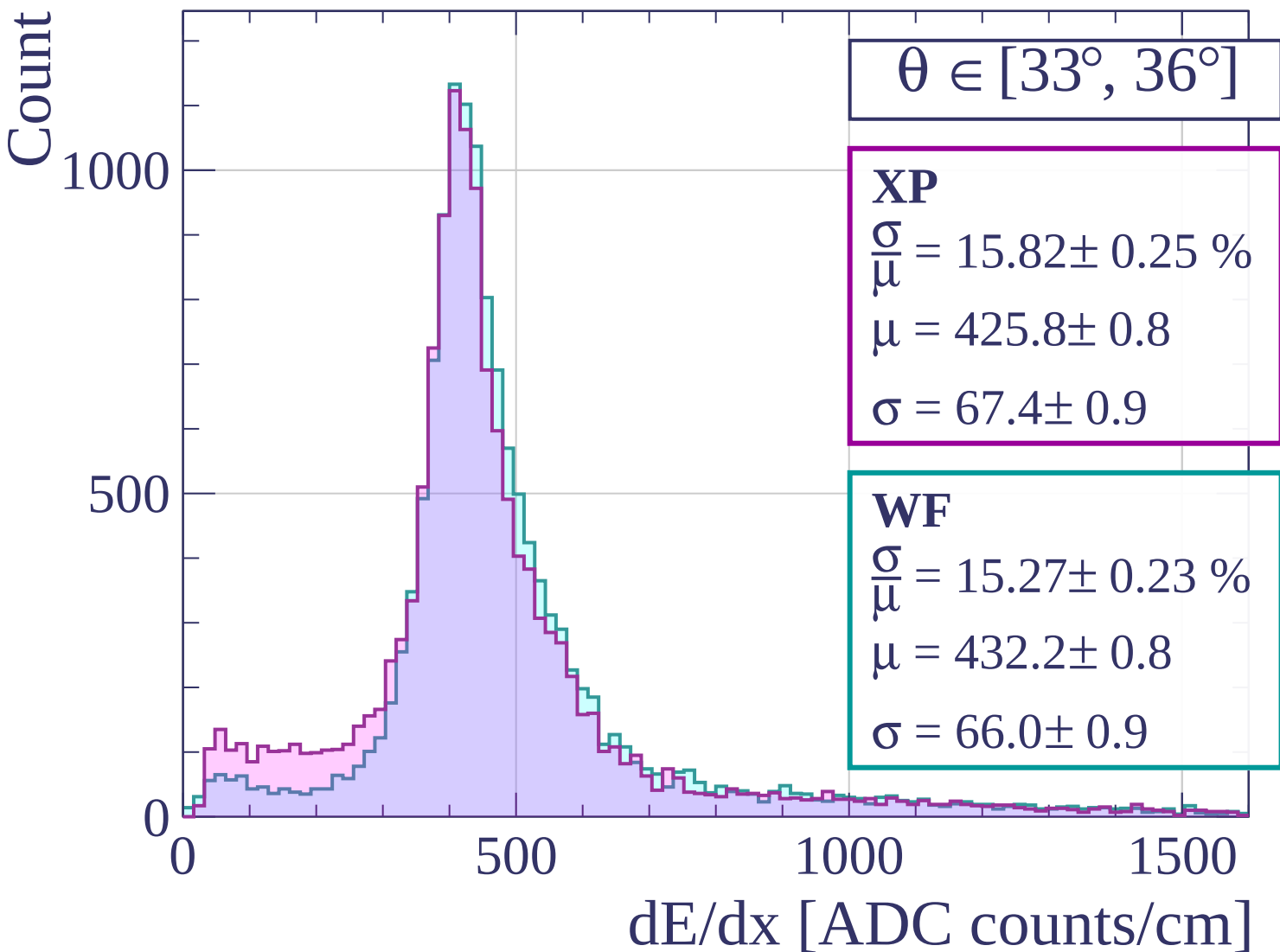












Count

$\theta \in [36^\circ, 39^\circ]$

XP

$$\frac{\sigma}{\mu} = 15.92 \pm 0.27 \%$$

$$\mu = 423.2 \pm 0.9$$

$$\sigma = 67.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 15.88 \pm 0.26 \%$$

$$\mu = 428.4 \pm 0.9$$

$$\sigma = 68.0 \pm 1.0$$

800

600

400

200

0

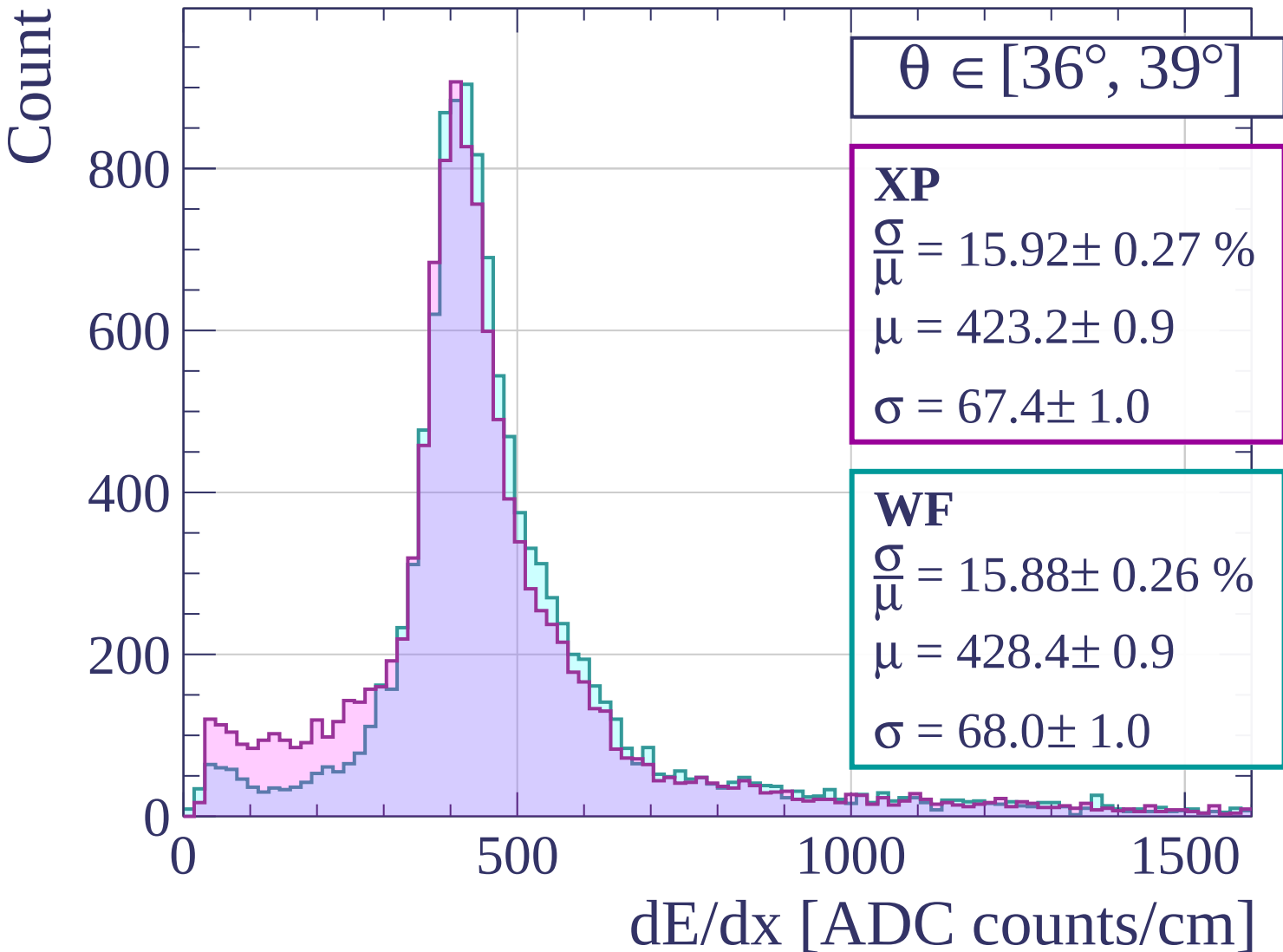
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [39^\circ, 42^\circ]$

XP

$$\frac{\sigma}{\mu} = 17.25 \pm 0.32 \%$$

$$\mu = 424.0 \pm 1.0$$

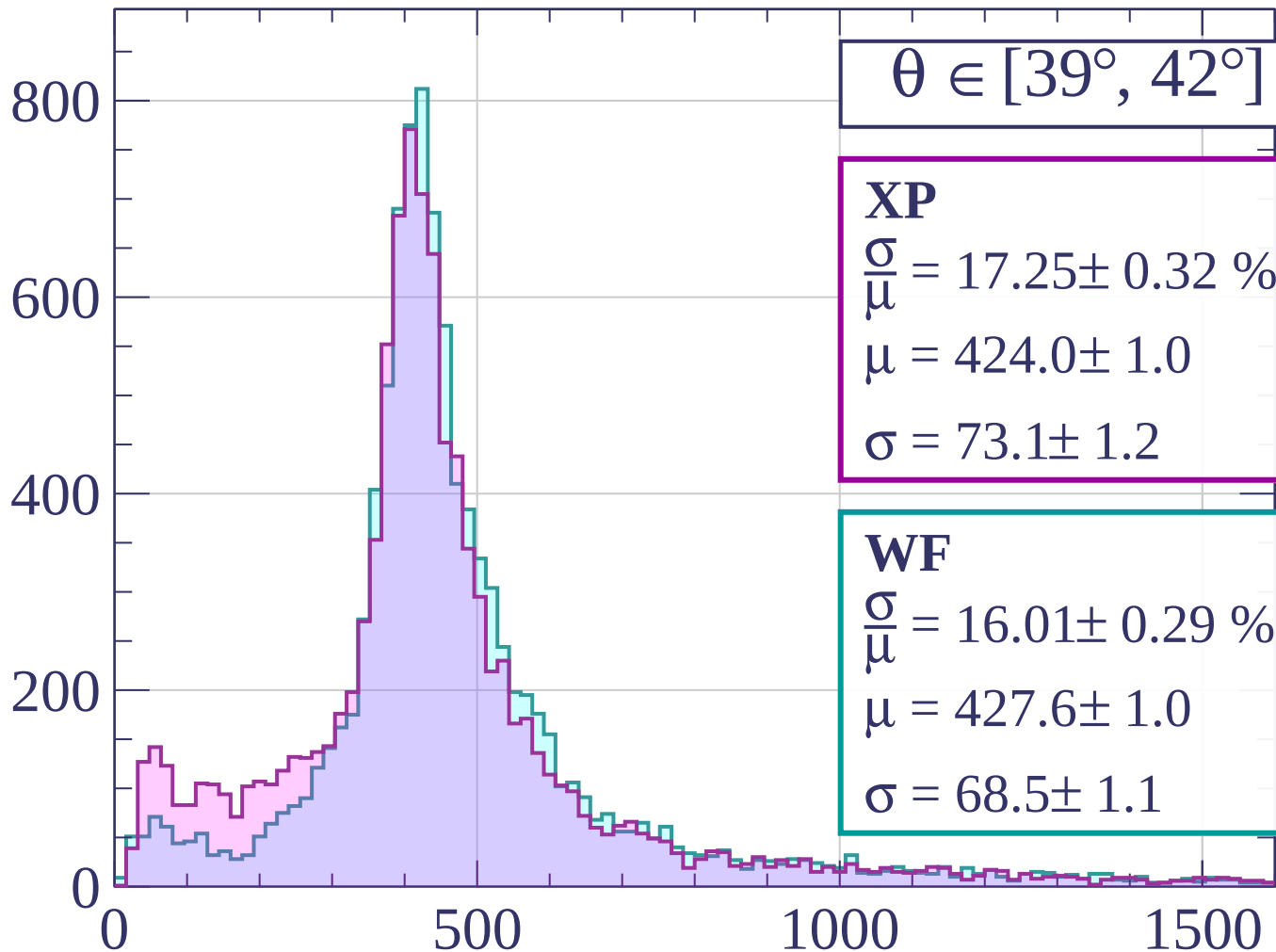
$$\sigma = 73.1 \pm 1.2$$

WF

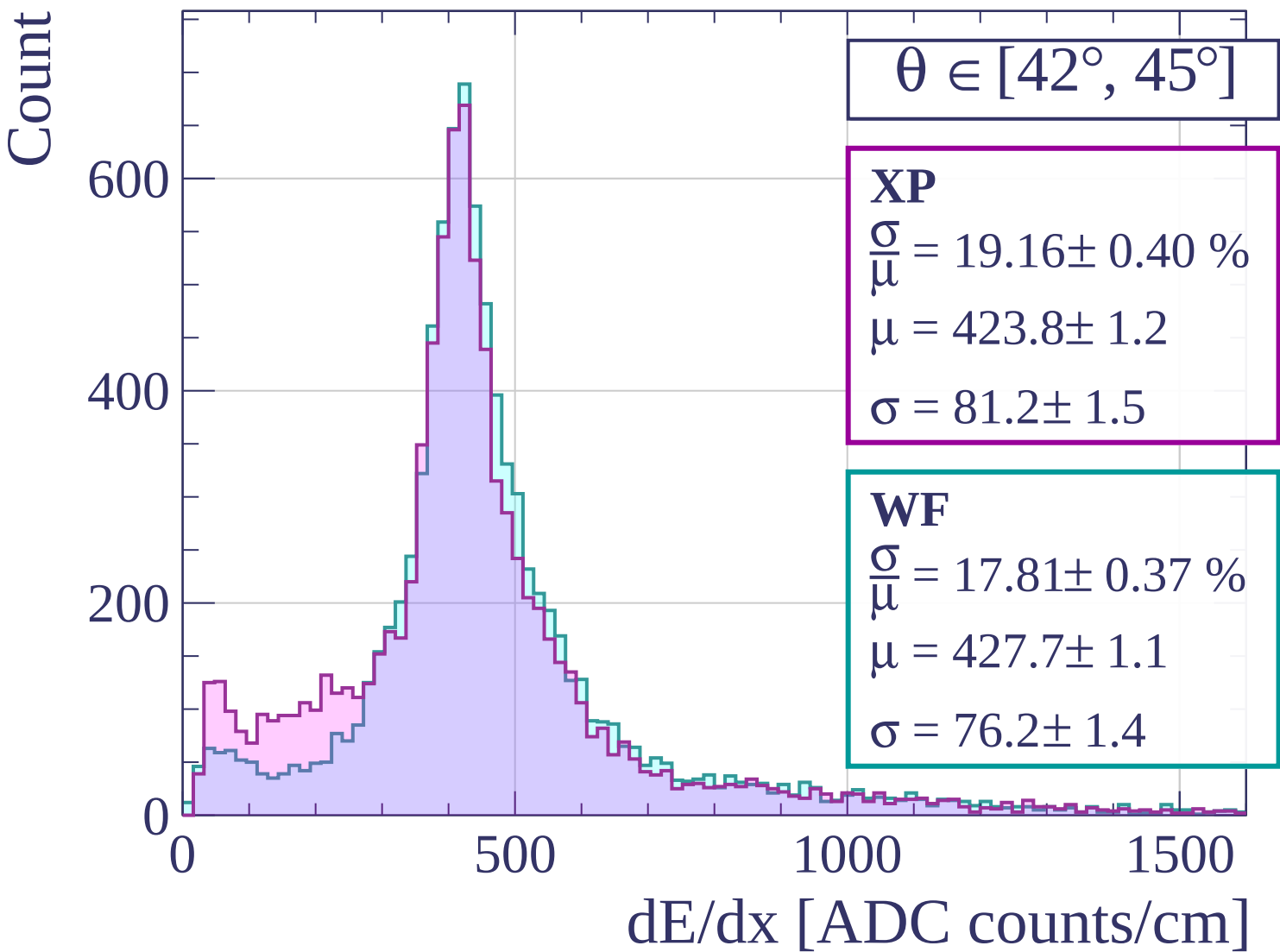
$$\frac{\sigma}{\mu} = 16.01 \pm 0.29 \%$$

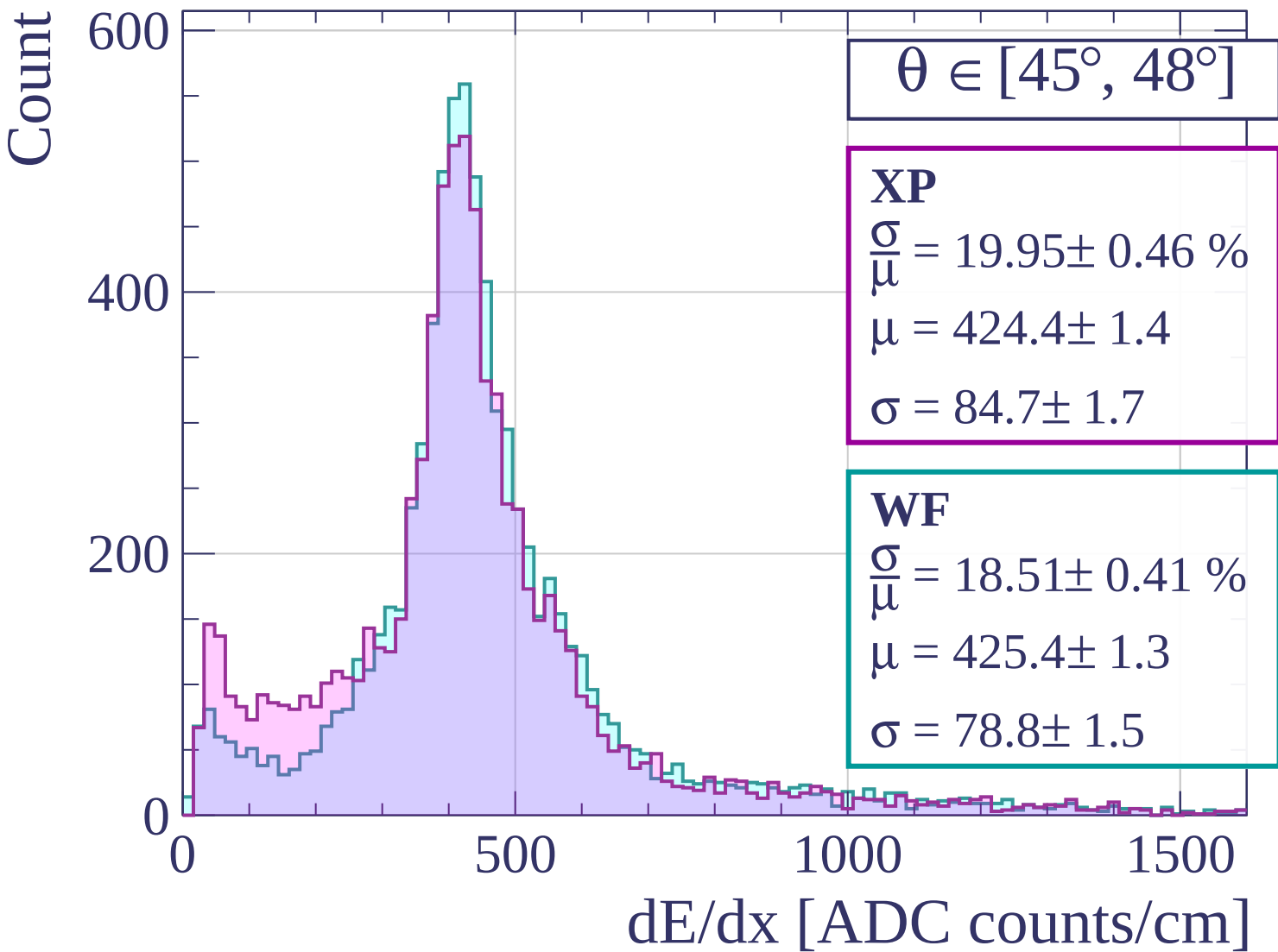
$$\mu = 427.6 \pm 1.0$$

$$\sigma = 68.5 \pm 1.1$$



dE/dx [ADC counts/cm]





Count

$\theta \in [48^\circ, 51^\circ]$

XP

$$\frac{\sigma}{\mu} = 22.21 \pm 0.52 \%$$

$$\mu = 421.9 \pm 1.5$$

$$\sigma = 93.7 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 20.93 \pm 0.50 \%$$

$$\mu = 424.2 \pm 1.5$$

$$\sigma = 88.8 \pm 1.8$$

0

500

1000

1500

dE/dx [ADC counts/cm]

400

200

Count

$$\theta \in [51^\circ, 54^\circ]$$

XP

$$\frac{\sigma}{\mu} = 31.60 \pm 1.01 \%$$

$$\mu = 407.3 \pm 2.5$$

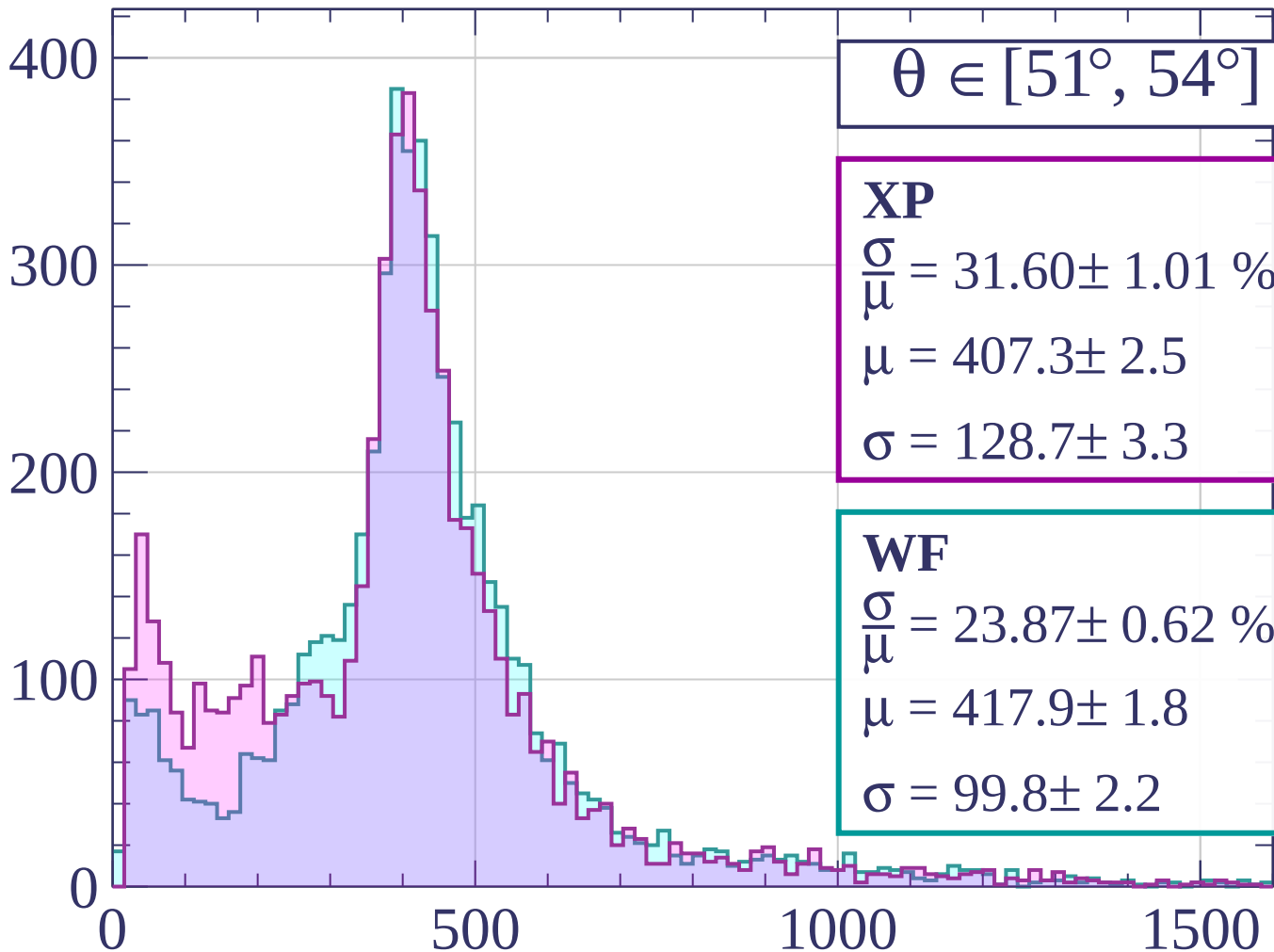
$$\sigma = 128.7 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 23.87 \pm 0.62 \%$$

$$\mu = 417.9 \pm 1.8$$

$$\sigma = 99.8 \pm 2.2$$



dE/dx [ADC counts/cm]

Count

$$\theta \in [54^\circ, 57^\circ]$$

XP

$$\frac{\sigma}{\mu} = 74.65 \pm 4.18 \%$$

$$\mu = 313.9 \pm 8.5$$

$$\sigma = 234.3 \pm 6.8$$

WF

$$\frac{\sigma}{\mu} = 27.65 \pm 0.78 \%$$

$$\mu = 414.5 \pm 2.2$$

$$\sigma = 114.6 \pm 2.6$$

300

200

100

0

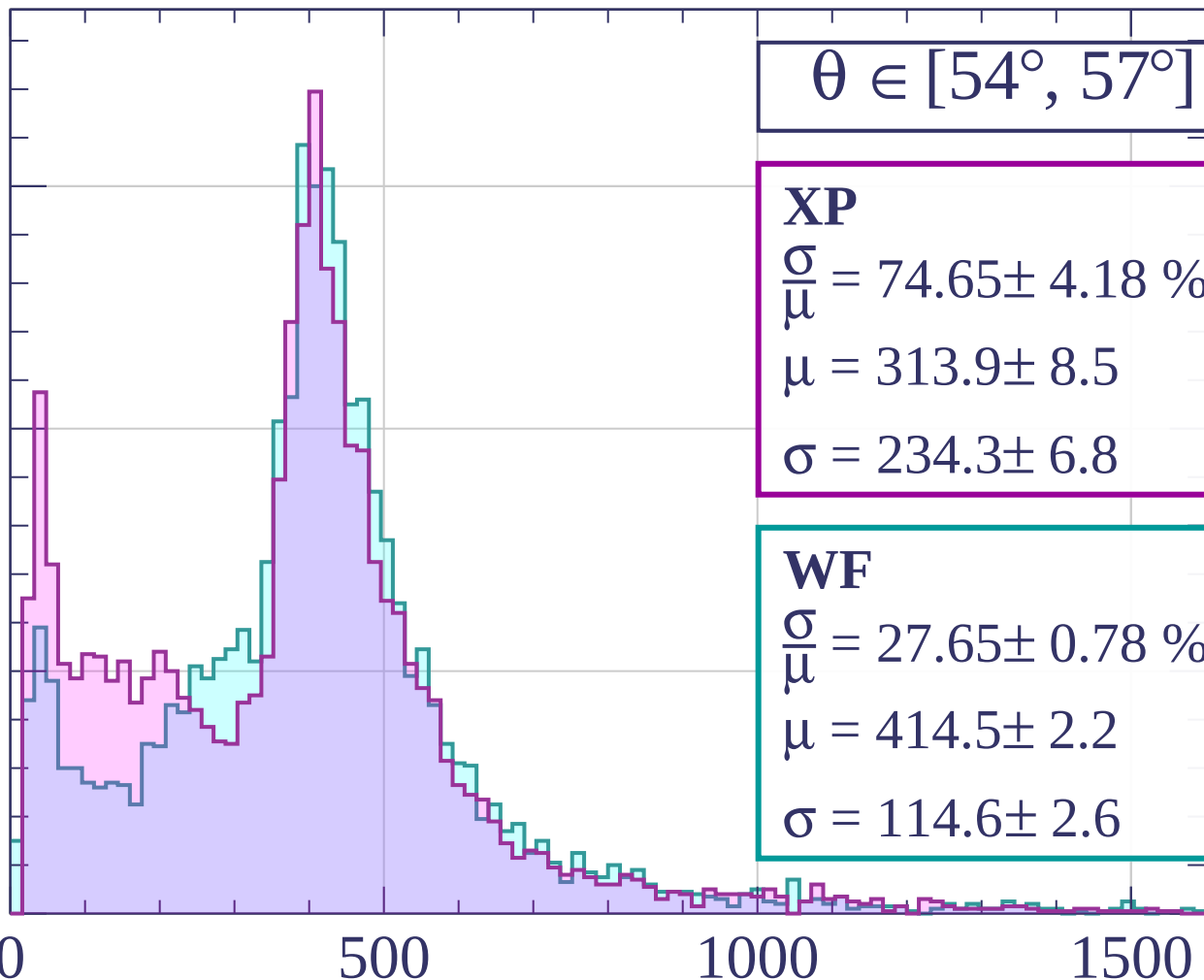
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$$\theta \in [57^\circ, 60^\circ]$$

XP

$$\frac{\sigma}{\mu} = 48.77 \pm 1.26 \%$$

$$\mu = 355.8 \pm 3.5$$

$$\sigma = 173.5 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 36.98 \pm 1.17 \%$$

$$\mu = 384.3 \pm 3.2$$

$$\sigma = 142.1 \pm 3.3$$

200

100

0

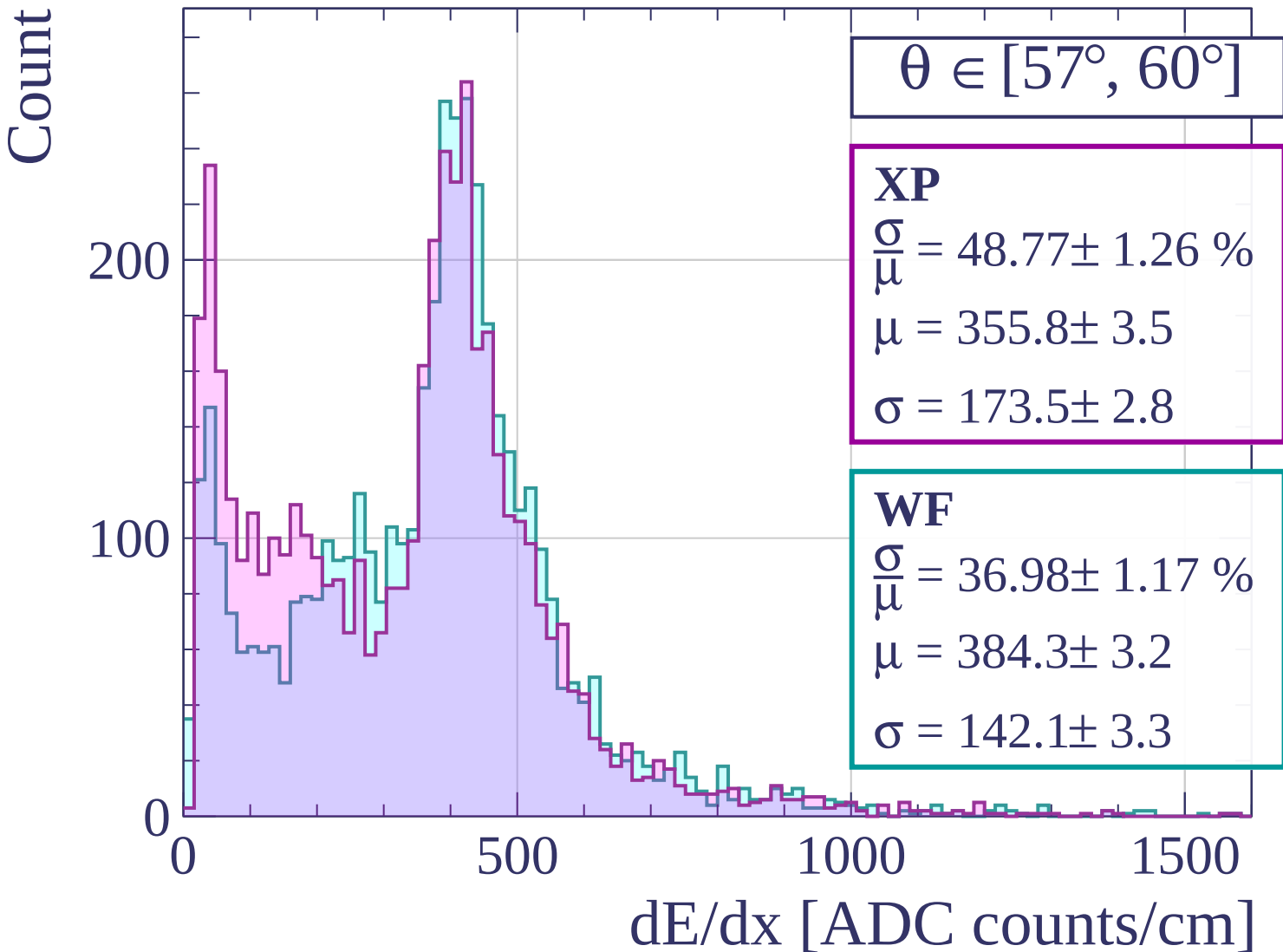
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [60^\circ, 63^\circ]$

XP

$$\frac{\sigma}{\mu} = 62.32 \pm 2.19 \%$$

$$\mu = 315.4 \pm 4.9$$

$$\sigma = 196.5 \pm 3.9$$

WF

$$\frac{\sigma}{\mu} = 57.93 \pm 2.56 \%$$

$$\mu = 327.8 \pm 5.3$$

$$\sigma = 189.9 \pm 5.3$$

200

100

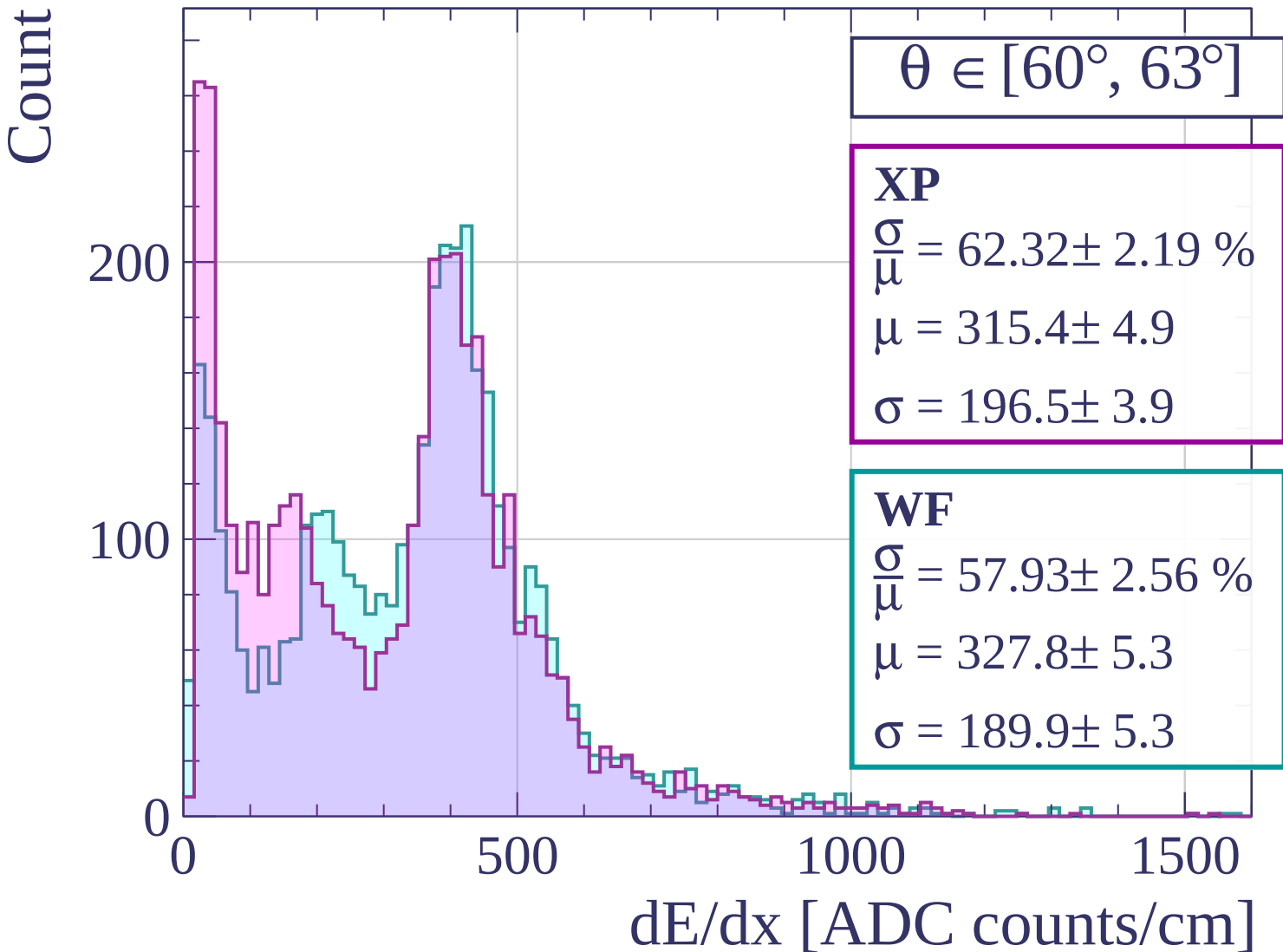
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [63^\circ, 66^\circ]$

XP

$$\frac{\sigma}{\mu} = 153.49 \pm 18.13 \%$$

$$\mu = 176.1 \pm 14.2$$

$$\sigma = 270.3 \pm 10.1$$

WF

$$\frac{\sigma}{\mu} = 89.18 \pm 5.41 \%$$

$$\mu = 246.5 \pm 8.8$$

$$\sigma = 219.8 \pm 5.5$$

300

200

100

0

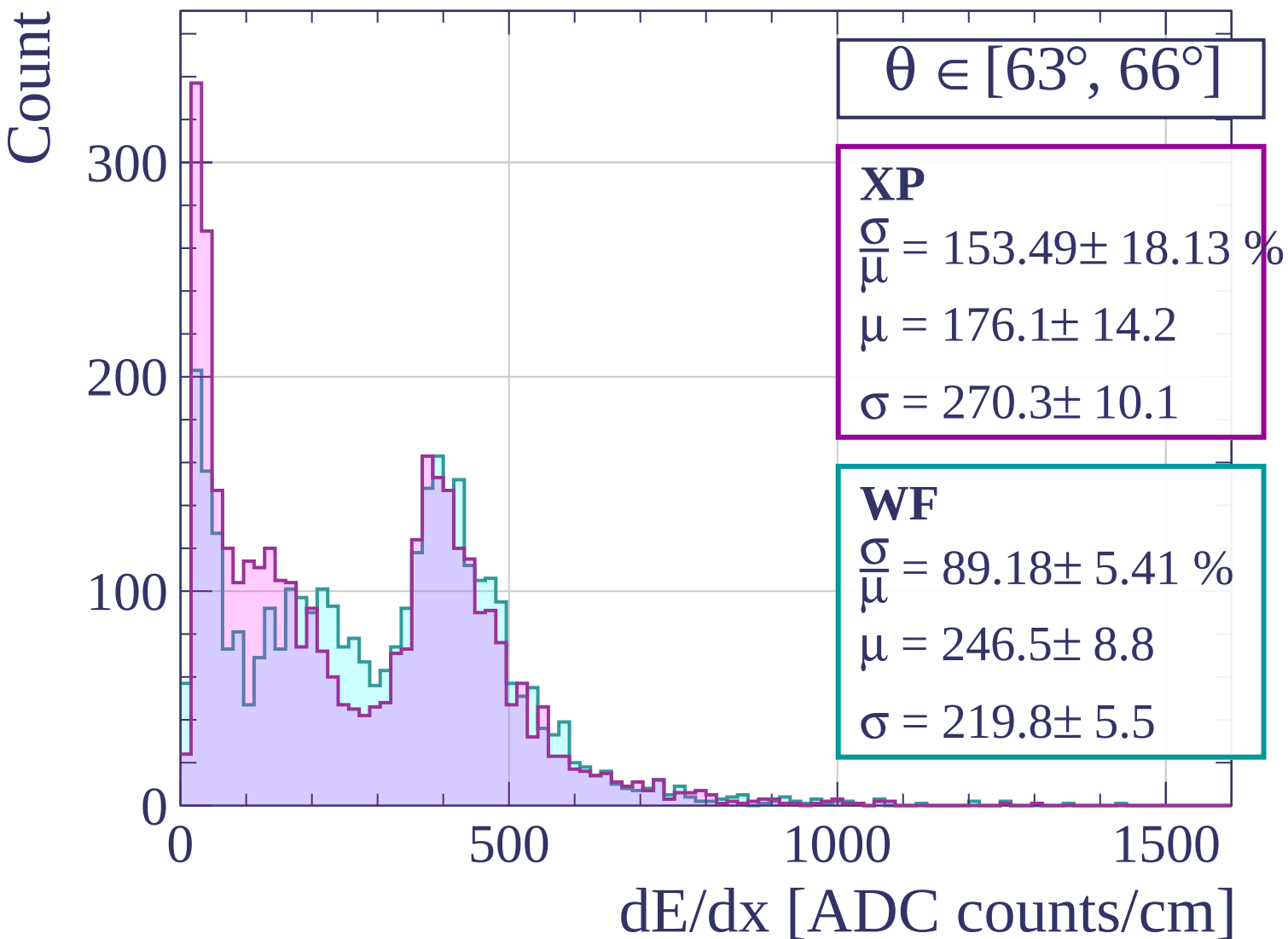
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [66^\circ, 69^\circ]$

XP

$$\frac{\sigma}{\mu} = 120.39 \pm 12.72 \%$$

$$\mu = 71.2 \pm 4.0$$

$$\sigma = 85.7 \pm 4.3$$

WF

$$\frac{\sigma}{\mu} = -32.80 \pm 1.61 \%$$

$$\mu = -3022.3 \pm 290.9$$

$$\sigma = 991.3 \pm 46.7$$

400

300

200

100

0

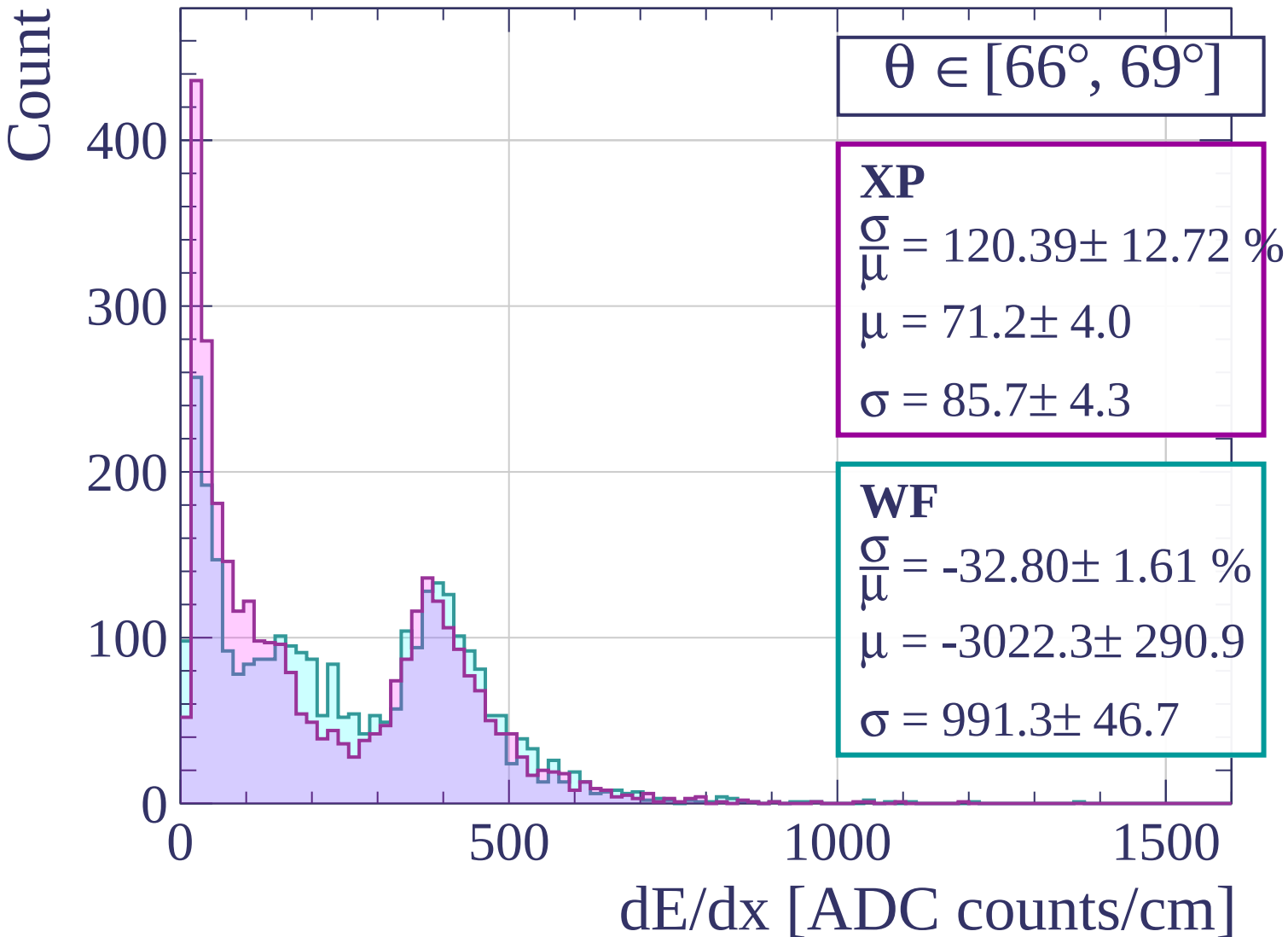
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [69^\circ, 72^\circ]$

XP

$$\frac{\sigma}{\mu} = 464.08 \pm 299.96 \%$$

$$\mu = 19.9 \pm 11.0$$

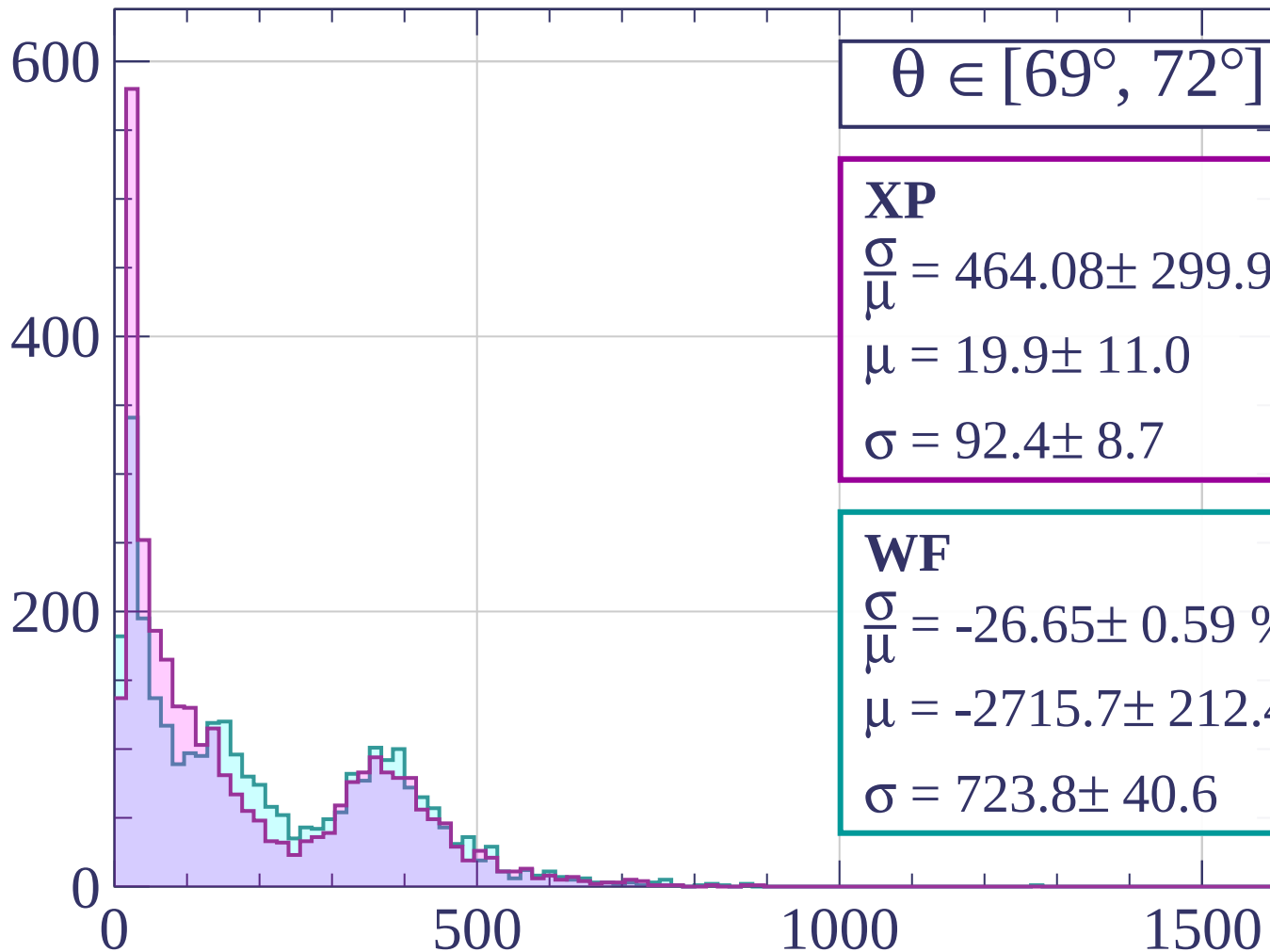
$$\sigma = 92.4 \pm 8.7$$

WF

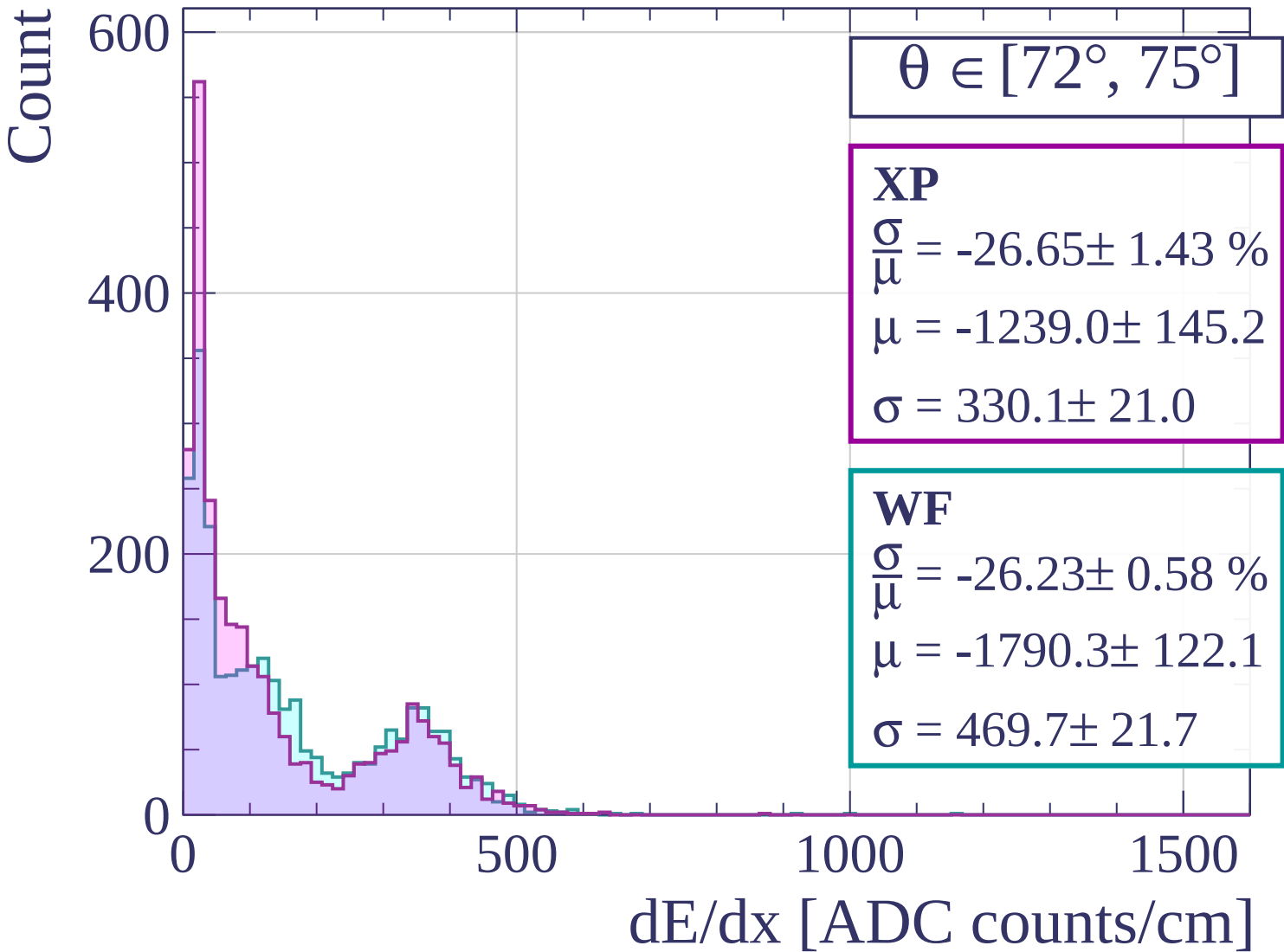
$$\frac{\sigma}{\mu} = -26.65 \pm 0.59 \%$$

$$\mu = -2715.7 \pm 212.4$$

$$\sigma = 723.8 \pm 40.6$$



dE/dx [ADC counts/cm]



Count

$\theta \in [75^\circ, 78^\circ]$

XP

$$\frac{\sigma}{\mu} = -24.68 \pm 0.67 \%$$

$$\mu = -978.6 \pm 69.2$$

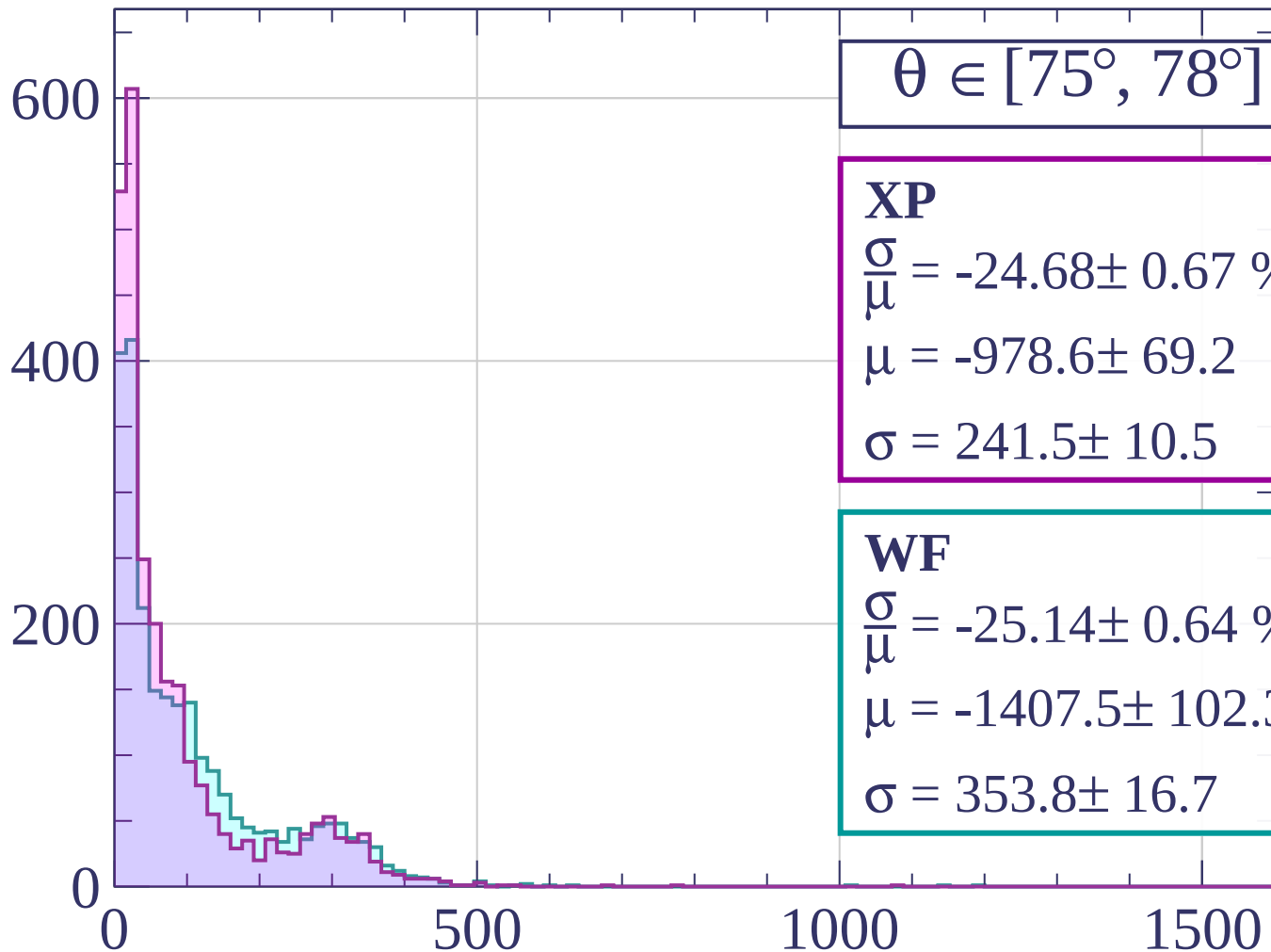
$$\sigma = 241.5 \pm 10.5$$

WF

$$\frac{\sigma}{\mu} = -25.14 \pm 0.64 \%$$

$$\mu = -1407.5 \pm 102.3$$

$$\sigma = 353.8 \pm 16.7$$



dE/dx [ADC counts/cm]

Count

$\theta \in [78^\circ, 81^\circ]$

XP

$$\frac{\sigma}{\mu} = -23.14 \pm 0.10 \%$$

$$\mu = -716.8 \pm 22.6$$

$$\sigma = 165.9 \pm 4.5$$

WF

$$\frac{\sigma}{\mu} = -23.29 \pm 0.32 \%$$

$$\mu = -979.0 \pm 53.8$$

$$\sigma = 228.0 \pm 9.3$$

800

600

400

200

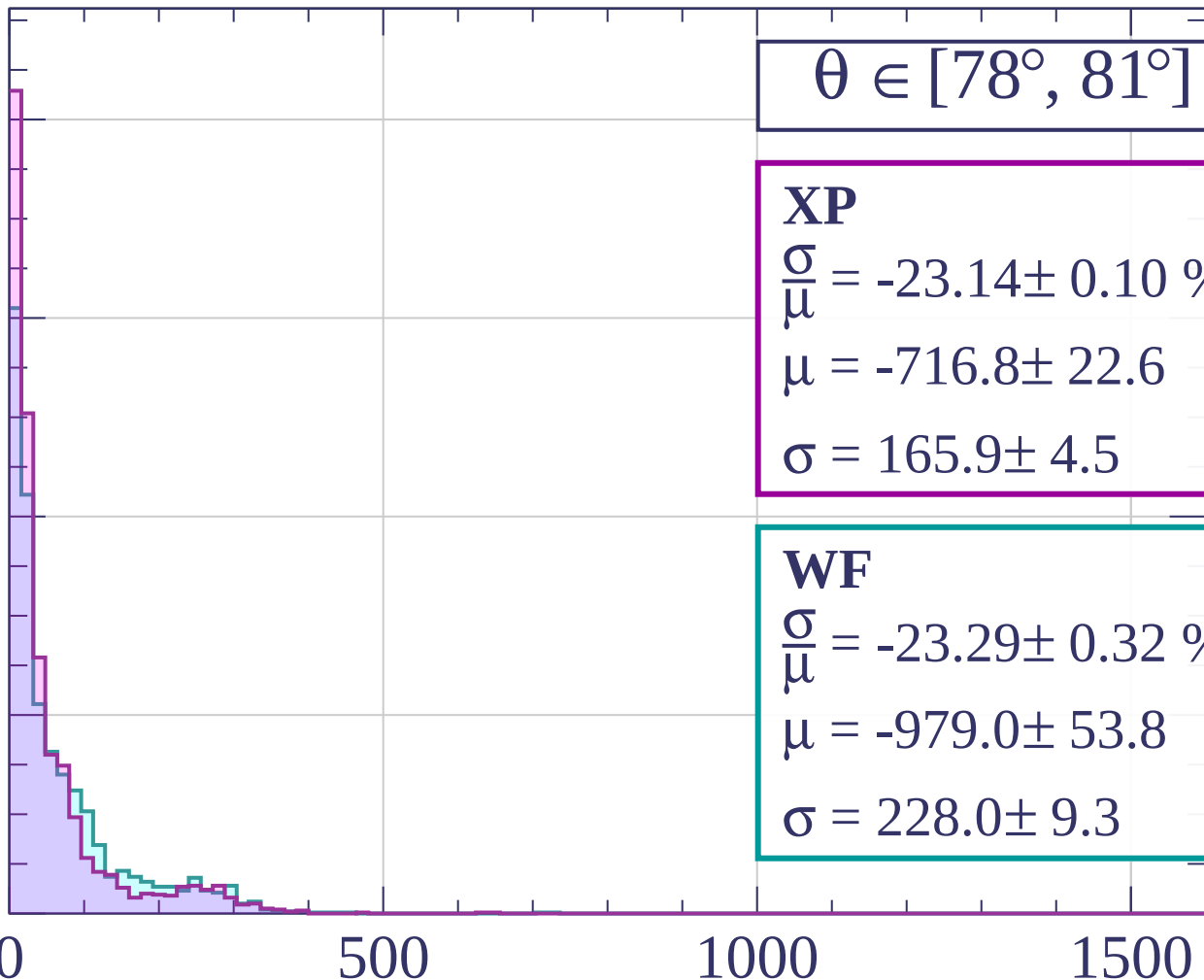
0

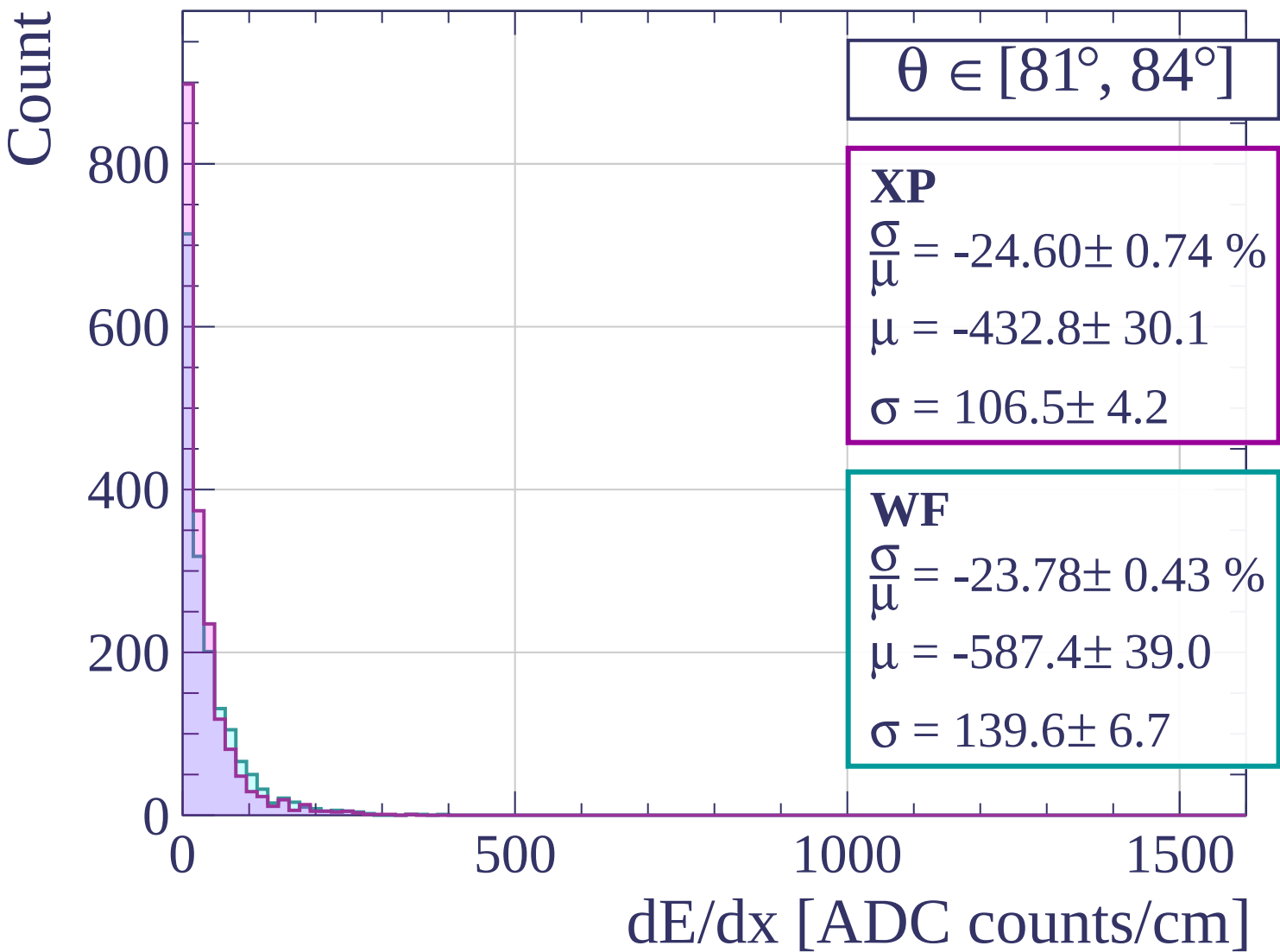
500

1000

1500

dE/dx [ADC counts/cm]





Count

$$\theta \in [84^\circ, 87^\circ]$$

XP

$$\frac{\sigma}{\mu} = -25.87 \pm 0.70 \%$$

$$\mu = -193.6 \pm 16.1$$

$$\sigma = 50.1 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = -24.57 \pm 0.81 \%$$

$$\mu = -255.9 \pm 26.6$$

$$\sigma = 62.9 \pm 4.5$$

600

400

200

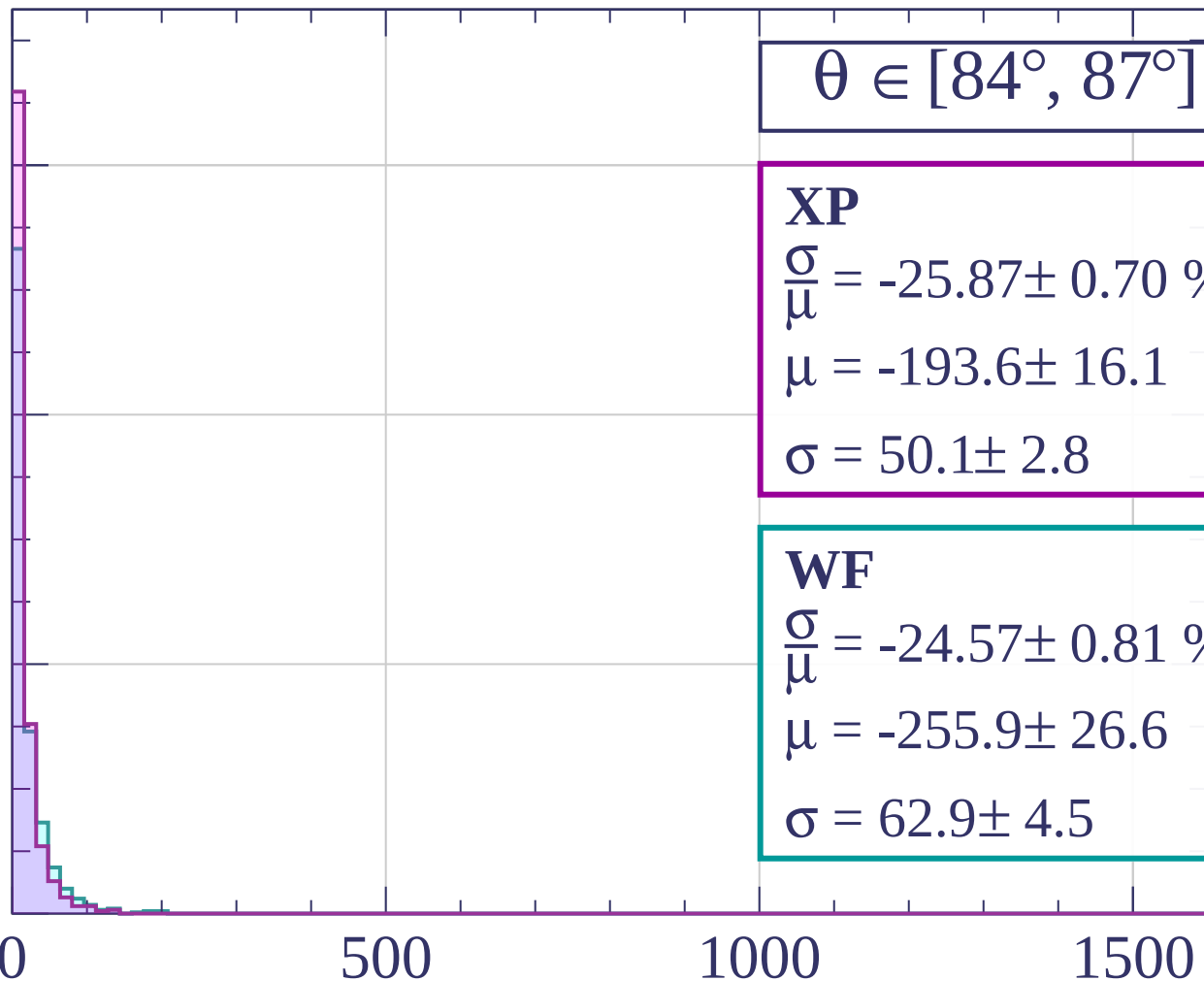
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [87^\circ, 90^\circ]$

XP

$$\frac{\sigma}{\mu} = 3.42 \pm 5.73 \%$$

$$\mu = 7.7 \pm 0.3$$

$$\sigma = 0.3 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 6.12 \pm 40.65 \%$$

$$\mu = 8.0 \pm 47.1$$

$$\sigma = 0.5 \pm 0.4$$

100

50

0

0

500

1000

1500

dE/dx [ADC counts/cm]

